MSC Corrective Maintenance
Engineering Report

USS HERSHEL WILLIAMS
(ESB 4)

N68171-28-C-XXXX
FY 2027 MTA

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USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0001	CATEGORY "N"	2026-01-30
ESB-SWI-SCOPE GENERAL REQUIREMENTS AND DEFINITIONS		Manovich, David

1.0 ABSTRACT

This item describes the intent, scope, general requirements, and definitions that apply to this work package.

2.0 REFERENCES

2.1 MSC Drawing 803-7081122, Military Sealift Command (MSC) General Technical Requirements (GTRs)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Aspects of the ship and shore support facilities as described in the work package.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES: None

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Scope

7.1.1 This work package, the accompanying contract, and contract guidance drawings define the scope of work to be performed.

7.1.2 In case of a difference or conflict between the referenced contract guidance drawings and the work items, the work items shall govern. Any omissions from the drawings, or from this work package, of details or work that is necessary to fulfill the intent of this work package or that is customarily performed in the industry or could have been obtained from a ship check (if offered), do not relieve the contractor from his/her obligation to perform such work. Notify the Administrative Contracting Officer (ACO)/OMT REP of any such issues as soon as the work package is reviewed, or such conflict is identified.

7.1.3 Provide labor and materials necessary to complete work items in the work package. Tag-out, (rearrange for order of activity) rig and unrig, connect and disconnect stage and un-stage, and remove, replace, and relocate any interference necessary to accomplish each work item in the work package. Unless specifically identified as government-furnished equipment or material (GFE/GFM), provide all equipment, material, and labor necessary to perform the contract.

7.1.4 Perform inspections, tests, and trials to demonstrate total compliance with the work package requirements.

7.2 General Requirements

7.2.1 Within the scope of this work package, complete each work item in accordance with appropriate GTRs as found in Reference 2.1.

7.2.2 Where a work item has special requirements that differ from the GTRs, the requirements of the work item shall govern. Where there are no special requirements invoked, then the GTRs shall govern.

7.2.3 Work items and contract guidance drawings provided by the Government to the contractor do not fully describe all work required to meet regulatory requirements. Ensure that work complies with regulatory requirements in accordance with Reference 2.1, GTR No. 1. The cost of meeting regulatory requirements shall be borne by the contractor. Provide the scheduling of American Bureau of Shipping (ABS) surveys and the cost and scheduling of all other regulatory services. The Government shall bear the costs of ABS ship surveys. Where a work item requires the contractor to develop working drawings and obtain ABS review and approval of those working drawings, the contractor shall bear these costs.

7.2.4 Where a work item does not cite any particular GTR, identify and perform in accordance with relevant GTRs. If any question arises as to the applicability, notify the on-site OMT REP and request further guidance.

7.2.5 Provide all necessary compliance and approval certificates and documents and submit them to the OMT REP. This includes but is not limited to:

a. Original stamped documents of each approved drawing for work performed during the availability.

b. Original stamped documents of each regulatory test and inspection performed during the availability.

c. A chronological file of all correspondence between the contractor (or sub-contractor) and all regulatory bodies.

d. Prepare and submit reports, drawings, and other documents required per the above paragraphs in accordance with the Contract Data Requirements List (CDRL).

7.3 Definitions: The following terms in alphabetical order shall have meanings as indicated below:

7.3.1 Administrative Contracting Officer (ACO) – Means the MSC designated contracting officer.

7.3.2 Article – means a separately lettered part of a work item of the work package. Articles in different items may bear the same number; hence, to identify an article completely, the item of which it is a part must be specified.

7.3.3 As-designed – means a condition meeting the original system and manufacturer's design.

7.3.4 As-found – means the condition which was found upon close inspection by the contractor. The resulting report will discuss all major components and sub-components as functional or degraded and needing specified repairs. When reporting an as-found condition, the information shall be stated clearly, concisely, and with enough detail so as to allow for preparation of a work statement and estimate. Include annotated sketches, graphs, and photographs when necessary to make a report clearly understandable to the OMT REP. Identify actual readings/dimensions taken. When a work item does not require a report, and one is determined to be necessary in order to produce a reliable or complete repair, submit one legible copy, in hard copy or electronic media, of a report with supporting data as early as possible in the contract period, so as to have the additional work completed within the original contract period.

7.3.5 As-released – means the condition of the item that exists at the time it is returned to the government. Include annotated sketches, graphs, and photographs when necessary to make a report clearly understandable to the OMT REP. Identify actual readings/dimensions taken. It shall include a summary of corrective actions made by the contractor and list any known remaining degraded conditions.

7.3.6 Category A – means a work item for which a firm fixed price has been established in the base contract, which will be executed during the period of performance.

7.3.7 Category B – means a work item which identifies additional optional repairs, labor, and materials or for which a firm fixed price has been established in the base contract, which may, or may not be invoked by the ACO during the period of performance.

7.3.8 Contractor furnished equipment (CFE) and contractor furnished material (CFM) – means the contractor provides the specified equipment or material used in the work item. Anything not identified as GFE/GFM shall be considered contractor furnished. This includes, for example, items alternatively described as installing activity furnished or vendor furnished.

7.3.9 COMSC – means the Commander, Military Sealift Command.

7.3.10 Contracting Officer's Representative (COR) – is an individual designated in accordance with DFARS subsection 201.602-2 and authorized in writing by the Contracting Officer to perform specific technical or administrative functions.

7.3.11 Contracting Officer – identifies the Contracting Officer as identified in the award or as otherwise designated by the government or MSC.

7.3.12 Contractor – identifies the shipyard holding the prime contract for the work specified in this work package.

7.3.13 Delivery – means the time when the contractor accepts the ship into their custody. The point where the ship is delivered is stated in the "DELIVERY AND REDELIVERY" Standard Work Item - 018.

7.3.14 Detach or disconnect – means to disconnect all attachments to the unit prior to moving it. Attachment points shall be tagged, identified, blanked, and protected to facilitate reinstallation. Work items do not necessarily identify interferences, and the contractor is responsible for the identification and resolution of interferences affecting a detachment and subsequent movement.

7.3.15 Drawing, As-Built (ABD)- ABDs are MSC drawings that document the final, ship specific, as-built configuration changes made by a TRANSALT. ABDs are derived from Contract Guidance Drawings (CGDs)and assigned unique MSC drawing numbers.

7.3.16 Drawing, Contract Guidance (CGD)- CGDs are MSC Drawings provided for guidance in accomplishing specific work. Together with the work item that references them, they contain adequate information and detail for industrial activities to estimate, plan and execute work requirements. They do not necessarily describe all work required to meet regulatory body requirements. CGDs are typically applicable to multiple ships in a class but based on the ship check of a single ship. Therefore a ship’s as-found condition could require the industrial activity to tailor the work.

7.3.17 Drawing, Selected Record (SRD)- SRDs are MSC drawings that document important features and systems. In most cases, they were initially developed by the shipyard that built the ship. Examples include general arrangements and system diagrams.

7.3.18 Drawing, Selected Record Markup (SRD-M)- SRD-Ms highlight additions, deletions, and configuration changes resulting from alterations. They are typically based on a tabletop review of ABDs and/or a ship check. They are retained to be incorporated into formal SRD revisions at a later time.

7.3.19 Extend – See install definition.

7.3.20 General Technical Requirements (GTRs) – defines the standards of performance for work being performed by contractors, see Reference 2.1.

7.3.21 Government furnished equipment (GFE) and Government furnished materials (GFM) – means the government provides the specified equipment or material used in the work item.

7.3.22 Hazwaste – means hazardous wastes that the

Conservation and Recovery Act (RCRA) identifies in one of four hazardous waste lists (F-list – 40 Code of Federal Regulations (CFR) 261.31; K-list – 40 CFR 261.32; P-list 40 CFR 262.32; or U-list 40 CFR 261.32; or exhibit at least one of four characteristics – ignitability, corrosivity, reactivity or toxicity, as defined in 40 CFR Part 261, Subpart C.

7.3.23 Hot work – defined per 29 CFR 1915.11(b).

7.3.24 Inspect – means to examine and assess the condition of a particular item or system and provide report with recommendation for repair and material.

7.3.25 Install, extend, and modify – means provide the piece of equipment, material, or system to be installed, extended or modified and provide the materials, structural supports, and labor to attach, connect, and test the equipment or system to effect a finished fully operational installation. In addition:

a. When new material or equipment is not specified by type, the material or equipment shall be identical to the existing material or equipment. When install is used with reference to GFE, conditions of the above definition are applicable, except the requirement to provide the piece of equipment.

b. Work items do not necessarily identify interferences and the contractor shall identify and resolve interferences affecting the installation by temporarily removing, reinstalling, or relocating interferences upon approval of OMT REP. Temporarily remove, relocate, alter, and reroute all interferences, including but not limited to, ductwork, piping, wire-ways, fixtures, insulation, joiner linings, equipment, furniture, etc., to facilitate finished fully operational installations and modifications covered by this work package. In the event that piping, ductwork, equipment, linings, etc., must be temporarily removed to facilitate installation of new or modified work, subsequently reinstall same in an as original condition. This includes but is not limited to; preservation systems, restoring floor coverings and joiner work, replacing items and equipment removed for access, plugging wiring and pipe penetrations, and restoring all other disturbed areas to match the surrounding areas. Disturbed coating shall be repaired.

7.3.26 Interference – means any object(s), equipment(s), system(s), or component(s) that must be removed and reinstalled, relocated, modified or designed around to facilitate the installation of new, repaired or modified equipment or systems.

7.3.27 Labor and Materials – means labor, materials, parts, plant facilities, supervision, services, tools, equipment, and any other resources required to accomplish the specified work.

7.3.28 Manifests – are the official shipping document forms originated and signed by the generators, transporters, and operators of the hazardous waste disposal facility, as required by federal, state, or local regulations.

7.3.29 MSC representative (MSCREP) – means the principal port engineer (PPE) for the period of performance and may be delegated to an assistant port engineer (APE) or a ship’s officer assigned to work with the ACO.

7.3.30 OMT REP approval – means OMT REP written concurrence with the general method of construction and detailing as presented by the contractor and does not relieve contractor responsibility for any error.

7.3.31 OMT REP, OMTREP – For MSC ships where ship is run by an Operating Company (OPCO), contractual authority with the contractor yard is with the operating company KO. The OMT REP denotes the OPCO Senior Management Team consisting of OPCO Program Manager, OPCO Port Engineer, OPCO Contracting Officer, OPCO Engineering Manager and OPCO Life Cycle Manager. The MSC program management team serves as owner's representative and provides the OPCO direction per the OPCO contract with MSC. MSC government representatives to include the Government Port Engineer (GPE) assigned are full time MSC representative whose function is to provide MSC Program Manager (PM) and N7 Technical director with real time status of availability progress, and firsthand information on issues, change order requirements, work quality and work completion. The GPE does not have contractual authority, but they are the primary liaison between OPCO and MSC PM and N7 whose purpose is to expedite decision making in the SY environment. For Ships operated by United States Government Employees, OMT REP is synonymous with MSC representative (MSCREP), see definition for MSC representative.

7.3.32 Non-destructive testing (NDT) - means analyzing the properties and condition of a material, component or system without causing damage and is employed to determine acceptability. Testing shall utilize currently calibrated equipment and certified testing personnel and procedures in accordance with applicable classification society and regulatory bodies requirements.

7.3.33 None additional or none - means that there are no additional requirements beyond those contained in the GTRs and work items that make up the work package. Where specific requirements are set forth in articles 6.0, 7.0 or 8.0 of a work item, such requirements are in addition to those requirements listed in the GTRs and work items that make up the work package.

7.3.34 Not separately priced (NSP) - means a work item which defines the general requirements, standards and conditions the contractor shall meet during the period of performance. NSP work items are not biddable items and do not represent a cost to the government.

7.3.35 Or equal – means that components, materials or equipment shall be equal or equivalent in terms of quality, performance (flow rate, pressure, heat transfer characteristics, etc.), services required (power, cooling water, HVAC, etc.), compatibility with interrelated systems and arrangements, and supportability over the service life of the components, material or equipment. In the case of components, materials, or equipment substitution for those items noted in the contract guidance drawings or work package, the contractor shall submit a written request delineating the design and performance data on both the specified and substituted piece of equipment for OMT REP approval, and if approved, the contractor shall take full contractual and technical responsibility for ensuring installation of the components, materials or equipment are compatible with the interrelated systems.

7.3.36 Qualified welders – means welders, whether contractor or subcontractor provided, shall hold qualification documentation from ABS and USCG for the material to be welded. Welding procedures shall be approved by the ABS Surveyor for the work specified in the work items. Qualifications and procedures shall be provided to the OMT REP upon request.

7.3.37 Redelivery – means the time that the OMT REP accepts the ship back from the contractor. The point where the ship is redelivered is stated in the "DELIVERY AND REDELIVERY" Standard Work Item – 018.

7.3.38 Refurbish – means to temporarily remove, disassemble, clean, inspect, a unit, part, equipment, or system, and then submit an as-found condition report to the OMT REP if repair work or parts are recommended beyond the scope of the work item. The OMT Rep will determine if these additional repairs or replacement parts are necessary. Upon completion the contractor shall lubricate, reassemble and reinstall the unit, equipment or system, set and adjust in accordance with manufacturer's specifications, test the unit, equipment, or system to demonstrate proper function to the manufacturer's specifications, and submit an as-released condition report to the OMT REP. The reinstalled refurbished unit, equipment, or system shall be fully operational and complete in all aspects. New screws, bolts, nuts, washers, gaskets, packing bearings, O-rings, and shims shall be used when reassembling and reinstalling the unit, equipment, or system. Additional information concerning the location of equipment refurbishment, special processes to be used in the refurbishment, and additional parts to be replaced during refurbishment shall be as specified in the individual work items.

7.3.39 Reinstall - means that the contractor shall provide labor and material to install a unit, part, equipment, material, or system after the unit, part, equipment, material, or system was temporarily removed, relocated, modified, or refurbished. The requirements of an installation as set forth in paragraphs 7.3.25, 7.3.25.a, and 7.3.25.b above with regard to completeness, operation, testing, the identification and resolution of interferences, etc. also these requirements apply to a reinstallation.

7.3.40 Relocate - means to provide labor and materials to detach the unit, part, equipment, or system and to reinstall the same unit, equipment, or system at a new or modified location. The requirements of an installation as set forth in paragraphs 7.3.25, 7.3.25.a, and 7.3.25.b above with regard to completeness, operation, testing, the identification and resolution of interferences, etc. also apply to relocation.

7.3.41 Remove or rip out - means to provide labor and materials to disconnect, detach, and transfer the unit, part, equipment, materials, or system in its entirety off the ship as specified in the GTR. Supports, stuffing tubes, collars, and other appurtenances shall be removed by burning, chipping, or cutting when a work item requires machinery, piping, wiring, ducting, structure, outfitting, joiner, or equipment removal. Areas affected by removals/rip outs shall be restored as set forth in paragraph 7.3.25.b:

a. When the work package requires the removal of machinery, piping, wiring, ducting, structure, outfitting, joiner, or equipment removal, supports, stuffing tubes, collars, and other appurtenances shall be removed by burning, chipping, or cutting, stubs, rough spots, and other surface irregularities shall be ground smooth and flush with adjacent surfaces. Once tasked work in the affected area is complete, the areas affected by the removal shall be restored to their original condition. This includes, but is not limited to, preservation system, restoring floor coverings and joiner work, painting, replacing items and equipment removed for access, plugging wiring and pipe penetrations, and restoring all other disturbed areas to match the surrounding areas. Disturbed coatings shall be repaired to match the surrounding area.

b. After removing the machinery, piping, wiring, ducting, structure, outfitting, joiner, or equipment removal, supports, stuffing tubes, collars, or other appurtenances, watertight or fire zone boundaries shall be reestablished. Holes in oil tight, watertight, and gas tight boundaries and structural bulkheads shall be blanked with flush inserts providing structural integrity and tightness equal to that which they replace. Holes in non-structural steel and steel joiner bulkheads may be blanked with lapped plates.

c. When the electrical wiring requiring removal joins a circuit that is not being removed (i.e. at a junction box), the electrical wiring or cable shall be entirely removed to the junction point. All other attachments including, but not limited to, fasteners, supports, and brackets shall be entirely removed. When electric cable is designated for removal, it shall be physically and electrically traced to the power source and the circuit de-energized before removing the cable. When cable is removed from multiple cable runs, the remaining cables shall be bound and strapped.

7.3.42 Replace or renew - means to remove the unit, equipment, or systems including interferences, and to install a new unit, equipment, or system which is either identical to or equal to that which was removed; the installation shall include, but not be limited to, any hook-ups, supports, and adapters, which are required to effect a finished fully operational installation complete in all aspects.

7.3.43 Restore - means to perform those processes (including, but not limited to, weld build-up, chrome plate, machine, grind, lap, NDT, etc.) which are required to return a component to the manufacturer’s specifications regarding dimensions, tolerances, angles, surface finish and clearances.

7.3.44 Section - means a major part of the work package and shall include a group of related work items.

7.3.45 Specification - means either a single work item or a complete set of work items.

7.3.46 Tag out - means a procedure to both notify personnel that tagged-out equipment, components, or systems are either isolated or not in a normal operating condition and to prevent injury to personnel, improper operation, or damage to the tagged-out equipment, components, or systems.

7.3.47 Temporary installation or temporarily install - means to provide labor and materials to install the unit, material, equipment or system to completely provide the function described in the individual work item, and to remove the same unit, material, equipment or system prior to redelivery of vessel or to satisfactorily meet the requirements of the individual work item(s). All the requirements of an installation and removal set forth in paragraphs 7.3.25, 7.3.25.a, 7.3.25.b and 7.3.40 with regard to completeness, operation, testing, the identification and resolution of interferences, disconnection, detachment and transfer of the unit, material, equipment, or system in its entirety off the ship as specified in the GTR apply to a temporary installation and subsequent removal.

7.3.48 Temporary removal or temporarily remove – means to provide labor and material to disconnect and move the unit, equipment, or system from its initial location, and to reinstall the same unit, equipment, or system either in the same location or elsewhere on the ship as described in the work package. All the requirements of an installation as set forth in paragraphs 7.3.25, 7.3.25.a, and 7.3.25.b with regard to completeness, operation, testing, the identification and resolution of interferences, etc. also apply to a temporary removal and reinstallation.

7.3.49 Work item or item – means a separately numbered part of the work package describing a discrete portion of the work to be accomplished.

7.3.50 Work package – means the entire written portion of the contract including the Master Ship Repair Agreement, the contract provisions, work items, and the GTRs.

8.0 GENERAL CONDITIONS

8.1 The following conditions apply to each work item in this work package and are intended to outline the minimum acceptable quality of material, workmanship, and practice.

8.1.1 Where laws, regulations, requirements, or commercial standards are referred to herein, the latest revision that is in effect on the date of the solicitation shall be applicable.

8.1.2 The contractor shall be held responsible for the protection of existing and newly installed equipment and materials and the storage and preservation of GFM/GFE. Any equipment that is damaged by the contractor and determined unusable by the OMT REP due to improper storage and preservation shall be replaced, repaired, or restored at the contractor's expense as directed by the OMT REP.

8.1.3 Materials and workmanship shall be subject to the OMT REP's inspection and approval at all times. Workmanship and materials found to be defective or not in conformity with regulatory requirements, or this work package and its associated contract guidance drawings shall be cause for rejection and removal at the contractor's expense.

8.1.4 Prepare and submit reports, As-Built Drawings, Selected Record Drawing Markups, and other documents required by the work package in accordance with the CDRL.

8.1.5 Equipment, furnishings, and materials removed, including scrap, except those specified for relocation or otherwise designated by OMT REP, are to be disposed of by the contractor at the contractor's expense, except for material requiring disposal as a hazardous waste, which shall be disposed of in accordance with work item titled, Hazardous Waste Disposal at a Contractor's Facility.

8.1.6 Install suitable protective devices on equipment, which presents a hazard to personnel. These may be carrier guards or other approved devices and shall preclude personnel injury. Examples of equipment which require protective devices are those which contain exposed rotating parts, fan blades, belts, pulleys, flywheels, wire rope reels, etc.

8.1.7 Design, workmanship, equipment, and materials must conform to commercial maritime standards such as USCG regulations, the International Regulations for Preventing Collisions as Sea (COLREGS), Navigation and Vessel Inspection Circulars (NVIC), and associated documents incorporated by reference in the aforementioned for marine inspection, and rules set forth by the ABS, U.S. Public Health Service (USPHS) regulations, International Maritime Organization (IMO), Safety and Life at Sea (SOLAS), Institute of Electrical and Electronics Engineers (IEEE) 45 and American National Standards Institute (ANSI) A17.1 (for elevators) and other applicable standards. The references required for conformance are either publicly or commercially available. Contractors are expected to have on hand any required commercially available reference which cannot be provided due to copyright restrictions.

8.1.8 Unless specifically approved by the OMT REP, contractor furnished equipment and materials shall be new. New materials and contractor furnished equipment must have traceable material documentation. The costs of ABS ship survey will be borne by the Government. The cost of other regulatory service expenses shall be borne by the contractor.

8.1.9 Conduct megger insulation resistance test (phase to phase and phase to ground) on new power cable installations and on existing disconnected and reused power cables.

8.1.10 Bond/ground equipment which is modified or installed by this work package. All bonding and bonding straps shall be in compliance with procedures and practices delineated by regulatory and professional agencies and organizations to include ABS Rules for Shipbuilding, and IEEE requirements.

8.1.11 Where rip outs leave openings in structures, decks, or shell plating, suitable insert plates or reinforced structures of equal material shall be installed to close the openings.

8.2 Hazardous Material

8.2.1 All material shall be asbestos and PCB free. If material that contains asbestos or PCB is inadvertently specified on a contract guidance drawing or other document, the contractor shall substitute an equivalent non-asbestos or non-PCB material and notify the OMT REP in writing prior to making any changes. This will not result in any adjustments to the contract.

8.2.2 All paint and coatings shall be lead free. Coal tar and coatings containing organic tin or zinc chromate shall not be used. If these materials have been inadvertently specified, the contractor shall substitute an equivalent lead free, tar free, etc. paint or coating and notify the OMT REP without any adjustment of the contract.

8.2.3 The contractor may find materials aboard the ship which are hazardous to health. These include, but are not limited to, asbestos, coatings containing lead compounds and zinc chromate, and ventilation gaskets and electrical cable containing PCBs. In the event that hazardous materials are identified, the contractor shall notify the OMT REP immediately and in writing.

8.3 Asbestos/Lead Paint/Hex Chromate Abatement and Removal requirements.

8.3.1 During the performance of this contract the contractor and subcontractors may be required to perform work which involves the removal or disturbance of asbestos or asbestos-containing products, lead paint, or hex-chromate contaminated paint. This requirement applies to each instance of removal or disturbance.

8.3.2 The contractor shall comply with the precautions required in 29 CFR 1915.1001, 40 CFR Part 61, and all other applicable Federal, state, and local restrictions. The contractor shall forward a copy of all required notices, licenses and permits to the Contracting Officer immediately upon issuance or receipt.

8.3.3 The latest change to the Federal, state, and local regulations in effect at the time of issuance of the contract shall govern. Compliance with these regulations is mandatory and is necessary to protect the employees of the contractor and Naval personnel from exposure to asbestos fibers in excess of the OSHA Action level airborne concentration (currently 0.1 f/cc of air).

8.3.4 During removal or disturbance, the contractor shall control airborne concentrations outside the removal boundary to less than 0.1 f/cc at all times.

8.3.5 After removal or disturbance is complete, the areas within the removal boundary shall not be released for re-occupancy until clearance air sampling demonstrates these spaces have concentrations of less than 0.1 f/cc.

8.3.6 In all respects, the performance of air sampling and analysis shall be performed IAW the OSHA Reference Method (Appendix A of 29 CFR 1915.1001), with the following additional specifications:

a. Aggressive clearance sampling shall be performed on 25 mm cassettes at 2.0 liters per minute for a minimum of four hours.

b. In performing the clearance sampling, the pump shall be placed within the compartment where the removal or disturbance occurred. When this operation is conducted in a multilevel space, at least one pump shall be placed on each level.

c. Air sampling shall be performed by a person competent in sampling procedures and overseen by a CIH by the ABIH.

d. Laboratory analysis of samples shall be performed by a participant in the AIHA PAT Program rated proficient for contaminants and air.

8.3.7 Personal sampling shall be conducted using breathing zone air samples which are representative of the 8-hour TWA exposure of each individual. Samples shall be collected and analyzed using the OSHA Reference Method contained in 29 CFR 1915.1001 (as amended).

8.3.8 The contractor agrees to indemnify MSC for any fines assessed by Federal, state, or local agencies, for the contractor's failure to properly follow applicable regulations.

8.3.9 The contractor shall insert this instruction in all subcontracts entered into under this contract.

1.0 ABSTRACT

This item sets general requirements for Technical and Manufacturer's Representatives.

2.0 REFERENCES: None

3.0 ITEM LOCATION/DESCRIPTION: None

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors regardless of tier shall consult the General Technical Requirements (GTR) item 024, to determine applicability to this item. In performance of work items in this package, the contractor and all subcontractors regardless of tier must comply with the requirements of GTR 024, where specified in an individual work item.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 The work item(s) specification(s) contained in the solicitation may require the contractor to furnish technical representatives or may indicate or may not indicate that Government furnished technical representative(s) will be on site. When any such instances are indicated in the work item(s) specification(s), the following shall apply:

7.2 Contractor furnished technical representatives:

7.2.1 The contractor shall furnish the services of a qualified technical representative that is:

a. An employee of the original equipment manufacturer (OEM)

b. Authorized by the OEM; or

c. Approved by MSC

7.2.2 The contractor, in order to obtain MSC approval for an OEM authorized technical representative shall provide a copy of a valid manufacturer's authorization letter, with date of issue, name, address, and telephone number for verification.

7.2.3 The technical representative shall be on site at all times whenever the following evolutions are performed to specified equipment or systems: disconnection; removal; opening; disassembly inspection; taking of clearance readings; static and dynamic balancing tests and adjustments; reassembly; shop tests; reinstallation; pre-repair and post-repair operating tests; dock trials; and sea trials.

a. The technical representative shall witness all of the above identified tasks related to the specific work hired for and shall prepare all reports of design, as-found, and final reassembly dimensions and clearances, shop test, operational test, dock trial, and sea trial results.

b. The technical representative shall witness and sign all reports attesting to his presence and witnessing of the test and concurring with the results for all tests performed by the contractor or other sub-contractors.

7.2.4 The contractor shall be prepared to submit a list of all technical representatives (by manufacturer name), intended to be employed during the contract performance period to MSC as part of the proposal submitted in response to this solicitation.

7.3 Government furnished technical representatives:

7.3.1 The Government may furnish the services of a technical representative to be on site at the ship repair facility at required times during repairs to the specified equipment of systems.

7.3.2 In the event that the contractor indicates either through written correspondence or the contractor’s production schedule that the Government’s technical representative is required to be on-site to support a production element, and subsequently fails to start the subject work item on time, the OMT REP will issue a Production Schedule Delinquent Progress Notice (PSDPN) in accordance with work item 013 of this work package. PSDPNs found to be the result of contractor delinquencies that affect the presence of a Government-furnished technical representative will result in actions to recoup the additional costs for the subject Government-furnished technical representative.

7.3.3 The Contractor shall coordinate and reiterate the production schedule dates for when the Government technical representative is to be on site. Such coordination shall be in writing to the OMT REP seven calendar days in advance of the work item's start.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

This item describes the requirements for the contractor to provide for a safe and navigable approach route to the contractor's facility; a safe and secure berth; and provide safe and sufficient mooring of the vessel.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 MIL. STD. MIL-C-28628D, Camels

2.1.2 MIL. STD. MIL-F-29248, Fenders, Marine, Foam Filled, Netless

2.1.3 MSC Drawing No. 801-8499371, General Arrangement Drawing

2.1.4 NAVSEA Drawing No. 582-8615624, Mooring Arrangement

2.1.5 DOD Uniform Facilities Criteria, UFC 4-159-03; Moorings

2.2 Enclosures

2.2.1 National Oceanographic and Atmospheric Administration (NOAA) Definition of Sounding Datum and Diagram of Charted Depths

2.2.2 MSC Ship Notional Arrival Conditions

3.0 ITEM LOCATION/DESCRIPTION

3.1 Enclosure 2.2.2 provides vessel particulars and notional arrival conditions. This information should be used for the solicitation/proposal process. Actual arrival conditions shall be as agreed between awarded contractor and OMT REP and shall be verified by the ship prior to arrival.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with all applicable GTR requirements.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract to determine their effect on the work required under this work item.

5.3 The definitions of many terms used in this work item are found in Work Item 001.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 The ship must be provided a safe approach, secure berth, and safe mooring at all times. References 2.1.1 through 2.1.5 are provided for guidance.

7.2 Approach

7.2.1 The ship shall be provided an unobstructed approach to the contractor’s facility (prime or otherwise proposed rented, leased, subcontracted location) where the vessel will be berthed and moored, for any part of the performance period. There must be no danger of the ship grounding or colliding with any overhead obstructions during transit. This applies for delivery to the contractor, redelivery to the Government, and any inter/intra facility transits.

7.2.2 During any ship transit or movement, and consistent with ship’s draft condition at the time; the ship shall remain afloat without any danger of grounding and avoid colliding with any overhead obstructions.

a. A minimum air draft clearance of not less than ten (10) feet shall be afforded between the highest vertical projection of the vessel at mean high water, all bridges and any cables passing above/across the marked channel along the entire route.

7.2.3 The following minimum clearances, with respect to vessel draft, shall be maintained to minimize probability of grounding and reduce silt buildup in piping. The datum for charted depth shall be mean lower low water (MLLW) per enclosure 2.2.1. The contractor shall include official NOAA or other regulatory agency information on tidal ranges, periodicity and length of tide ranges, including a discussion on how contractor’s proposed use of tide heights to satisfy transit requirements may affect the shipyard availability arrival/departure times and the complete scope of work:

a. Underway in confined waters (harbors) – 3 feet of charted depth.

b. Underway in inland waters – 2 feet plus squat of charted depth. (Note: Squat values only apply to MSC tanker vessels like the T-AO class ships. T-AO squat tables are available for planning.)

c. Underway in open water – 2 feet plus squat of charted depth. (Note: Squat values only apply to MSC tanker vessels like the T-AO class ships. T-AO squat tables are available for planning.)

d. During dead stick maneuvering (pilot aboard and assisted by tugs) – 2 feet of charted depth.

e. For dry-docking – 12 inches over the high block, sill, or highest projection including the effects of list, trim, and hog/sag.

f. Pierside at a contractor’s (or subcontractor’s) facility – 3 feet of facility sounding survey depth. Tidal ranges shall be included to demonstrate adequate clearance under all conditions.

7.2.4 A navigational meeting/brief shall be held 24 hours prior to arrival/departure of the vessel.

a. The conference shall be attended by ships force and OMT REP project office personnel and the contractor’s supervisory personnel concerned with the arrival/departure evolution.

NOTE: The arrival conference shall be held via phonecon so that the ship can participate in the conference remotely prior to arrival at the shipyard. Provide conference call phone number details after award for the remote conference.

b. Navigational brief shall cover arrival and departure date/time/tides/weather/ water depths/ tug and pilot services/line handlers, areas of concern, etc.

7.2.5 Contractors must submit with their proposals copies of all the latest NOAA or US Army Corps of Engineers or nationally recognized authority waterway approach charts to their facility. Charts are to show the entire route of transit from the sea buoy to the contractor's facility, or for proposed inter/intra facility transits. Charts must provide adequate demonstration that the contractor has positively determined the route(s) providing safe vessel transit and adequate water depths to meet the requirements of 7.2.3 above, including applicable tidal ranges.

7.2.6 Contractors must submit an approach chart with their proposals. The approach chart shall be equal to the berthing chart in dimensions and contain the following data:

a. Contractors shall furnish current depth soundings for the approach to the pier and/or dry-dock from the main channel.

b. Soundings must indicate the depths over the full length of the approach, from the point in the navigable channel where the contractor takes delivery of the vessel to the pier and/or drydock at a maximum of one hundred (100) foot intervals. Adequate breadth of soundings shall be provided to demonstrate sufficient underwater clearance for vessel arrival, departure and any required turning evolutions. Approach soundings shall be considered current if taken within three hundred and sixty-five (365) days prior to the date of issue of the solicitation. NOAA charts or excerpts of such documents will not be accepted.

1. Approach soundings taken by Government agencies of the contractor's home country, shall be considered current if taken within the Government agencies' current reporting period.

c. Official data on tidal ranges for the area covered by the approach charts shall also be submitted with the contractor's proposal.

d. All information must be readily available to clearly chart the progress of the ship from arrival at the port to safe mooring location or dry-dock. The ship must be provided safe access and egress to and from the contractor's proposed pier and/or dry-dock.

7.3 Berth

7.3.1 Consistent with ship's anticipated arrival drafts, the ship shall remain afloat (except when dry-docked), with no danger of grounding during the entire performance period. The minimum amount of water under the deepest/lowest projections from keel baseline at all times while at the pier and all tides shall be no less than three (3) feet of water except when weather conditions exist per work items 006 and 007. Article 7.3.3 of this work item requires the contractor to show adequate water in their berthing chart.

7.3.2 Contractors must submit with their proposal, and resubmit to the OMT REP after award, all information relative to berthing facilities. Information must include the following characteristics of the pier:

a. Location

b. Pier number and/or name

c. Street address, city, state, and zip code

d. Geographic reference to a current navigational chart using longitude and latitude coordinates

e. Length

f. Width

7.3.3 Each contractor shall draw the ship to scale, in the exact location it will occupy during the performance period, using the ship characteristics given in this solicitation on the berthing chart. For the berth chart the ship is to be superimposed over the water depth soundings grid, the entire grid to be clearly legible. The contractor shall prepare a berth chart. The berth chart shall contain the following data:

a. It shall furnish current soundings taken at mean lower low water of the berthing area. Official data on tidal ranges for the berth area shall also be submitted.

b. The soundings must list depth in a ten (10) foot grid pattern from the pier string piece. The grid pattern for the intended ship berth shall extend for the entire length of the pier and outward from the pier for a minimum of one hundred (100) feet.

c. Soundings shall be taken by a qualified waterways surveyor and certified by same. Soundings shall be considered current if taken within three hundred and sixty-five (365) days prior to the date of issue of the solicitation and should indicate the date they were taken.

d. The contractor’s berthing chart shall be approximately two (2) feet by three (3) feet minimum with a geographic reference point of latitude and longitude and a geographic indicator of north.

7.3.4 The individuals/firms preparing the berthing and approach charts shall be qualified by possession of valid licenses (or certificates of competency), as issued by the state or federal or national agency tasked within the geographic locality of the contractor with this licensing responsibility, and be stamped/illustrated on the charts, and shall indicate the date prepared.

7.4 Mooring

7.4.1 The ship shall be provided sufficient safe and secure mooring, capable of withstanding the force of winds at a speed of forty (40) miles per hour received perpendicularly on the beam at drafts consistent with minimum acceptable dry-docking displacement or the light load condition, (whichever is greater as stated in the solicitation). Reference 2.1.4 provides the ship’s mooring arrangement. Not less than seventy five percent (75%) of the ship shall be alongside the pier. Should any portion of the ship extend beyond the pier, a minimum of three (3) separate additional mooring points shall be provided for each extension. These shall be in addition to those on the pier. The additional mooring points shall be distributed on both the port and starboard sides of the ship.

7.4.2 Contractors must submit with their proposal all information relative to mooring capabilities at the proposed facility berth location.

7.4.3 Information must include the following characteristics of the mooring arrangement:

a. Submit mooring calculations for the worst anticipated loading condition during the availability using reference 2.1.5 as guidance. Determine the combined loading due to wind load from each direction and both peak flood and ebb current loads at low and high tides. Calculations may require re-submittal during the course of the repair availability if significant changes occur from the original estimate on which the calculations were based.

b. Location, pier number and/or name, street address, city, and state with geographic reference to current navigational chart using longitude and latitude

c. Length

d. Width

e. Locations, type, design and proof test loads for all mooring fittings, e.g., bollards, bits, cleats, deadmen, etc.

f. The arrangement of all mooring lines between the ship and mooring securements. All mooring lines shall be provided by the contractor. Mooring lines shall be identified by manufacturer, model/brand, size and breaking strength. Use of mixed lines is not permitted.

g. The angle of the mooring lines between the ship and each mooring securement point.

h. Quantity, size, locations and securement method of all camels and fenders.

i. General construction characteristics including: type, location, quantity of pilings; surface type and condition; location and sizes of drains; buildings and structures (sheds); width of apron and access limits for vehicles.

j. Location of and distances to nearest fire alarm call boxes.

7.4.4 Each contractor shall draw the ship to scale, in the exact location it will occupy during the performance period, using the ship characteristics given in this solicitation on the mooring chart.

7.4.5 Contractors shall furnish a chart approximately two (2) feet by three (3) feet minimum with a geographic reference point of latitude and longitude and a geographic indicator of North. The chart shall be prepared by a qualified surveyor and certified by name. The chart shall be current. The chart shall illustrate to scale, all information required above in sub-articles of 7.4.3.

7.4.6 The mooring chart as it reflects the physical characteristics of the pier as identified in article 7.4.3 shall be prepared by a licensed professional engineer. Certification shall be by the affixing to the mooring chart (original and copies), a stamp or statement attesting to the accuracy of the chart by the preparer(s). The certification shall indicate the name(s) if the individual(s) and firm(s) which prepared the chart, the date(s) the survey was conducted and the dates(s) the chart was prepared (drawn). It shall be signed and dated by the preparer(s).

7.4.7 Documentation of design load calculations for heavy weather mooring configuration shall be furnished. Heavy weather mooring configuration and calculations shall demonstrate design and construction to be capable of withstanding wind speed values anywhere within the range of seventy three (73) to one hundred thirty six (136) miles per hour, (at drafts consistent with minimum acceptable dry-docking displacement or the light load condition, whichever condition in the solicitation creates the greater sail area) using reference 2.1.5 for guidance for Type III heavy weather mooring configurations.

a. Analyze the worst-case wind directions including frontal, broadside and quartering.

b. Provide the maximum safe wind speed for mooring strength. Include strength of pier, pier fittings, mooring lines, and shipboard fittings.

c. Maximum safe surge for mooring.

d. Maximum safe elongation of mooring lines.

1. Include size and type of mooring line; percent elongation of mooring line at failure; tattletale-free length and length between attachments.

e. Sketch showing the size, type, and location (vertical and horizontal angles) of all securing devices including fenders bumpers and camels.

f. In those instances where the documentation of the design loads for securement appurtenances cannot be provided, it shall be required to have furnished copies of the most recent test certificates.

g. Test certificates shall be considered recent if tests were conducted within five (5) years of the date of issue of the solicitation, and the pier has not been subject to potential collision or weather-related damage. NOAA charts or excerpts of such documents will not be accepted.

h. All mooring appurtenances regardless of design type, i.e., bollards, bitts, or cleats, must be in an upright position and have; bases or pedestals in sound condition without any broken securement bolt holes in the base mounting flanges. All securements (bolts, washers, nuts, etc.) installed, tightly in place, and in good condition. All arms and horns in place with none missing, broken, bent, misshaped or distorted. Concrete foundations shall be in sound condition without excessive spalling and cracks or chipped and damaged concrete. Damage due to spalling, cracking, and chipping over ten percent of a foundation’s area is not acceptable. Wood and pier mooring appurtenance foundations, stringpiece mountings and support structures shall be in sound condition without cracks, rips, splits, or missing portions. Damage due to cracks, rips, splits, or missing portions of wood on the appurtenance foundations, stringpiece mountings and support structures are not acceptable.

i. All information must be readily available to clearly ascertain the safety afforded by the mooring arrangement offered the ship.

7.4.8 If the contractor is unable to show that their facility and proposed arrangement can meet the mooring criteria of 7.4.7, the contractor shall provide the heavy weather mooring configuration that would be used during the conditions specified in 7.4.7; supporting calculations in accordance with the criteria specified in 7.4.7.a through 7.4.7.e using reference 2.1.5 for guidance; and meet the requirements of 7.4.7.f through 7.4.7.i.

a. The contractor shall provide a written statement for review that includes the vessel name, maximum limits for: wind speed (knot); current (knot); storm surge (Ft) and direction of each that the facility can achieve for review by the OMT REP.

b. The contractor shall provide, both as part of SWI 006 Heavy Weather Plan and as part of this submittal, the contractor recommended contingencies necessary to protect the vessel when weather conditions exceed the mooring limits of the facility as described in 7.4.8.a above.

7.4.9 Camels and/or fenders

a. Camels and/or fenders shall be used at all times to breast out (separate) the ship from the pier. Camels shall be single or multiple, of hardwood in accordance with or equal to reference 2.1.1.

b. Fenders shall be of the suspended foam filled design in accordance with or equal to reference 2.1.2, or solid rubber dock suspended type.

c. Floating camels and/or suspended fenders shall be a minimum of four (4) feet wide/diameter and six (6) feet long.

d. Dock mounted fenders shall provide for a minimum clearance of four (4) feet.

e. The quantity and frequency of placement for camels/fenders shall be no less than one camel/fender for each two hundred (200) feet of the ship’s length overall, and adequate to protect the ship form contacting the side of the pier.

7.4.10 In instances where it is necessary to breast the ship out from the pier by use of a barge to facilitate achieving adequate depth of water for berthing, or when a floating crane is brought alongside in performing a work item(s), the barge or crane shall be fended off from the side of the ship by use of camels or fenders. Truck or heavy construction machinery tires are not acceptable.

a. The requirements for fenders and camels between the barge and the ship shall be equal to those stated in 7.4.9 above.

b. Where barges are used to breast the ship for an extended period of time, barges shall provide support coverage of at least 75% of the length of the ship in accordance with article 7.4.1 above.

c. If barges are used for an extended time period to breast out the ship, the contractor shall provide a safe and secure means to access the ship from the fixed pier. Where it is necessary for personnel to traverse the breasting barge(s), there shall be a defined safe access route with life rails and non-skid decking installed.

d. Design load calculations for moorings shall take into account breasting barge(s) arrangement when applicable.

7.4.11 During the course of this repair/overhaul period, no other ship will be docked alongside of or tied up to this vessel, nor shall this ship be docked alongside of or tied up to any floating dry-dock.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - National Oceanographic and Atmospheric Administration (NOAA) Definition of Sounding Datum and Diagram of Charted Depths

Definition of Sounding Datum

All depths indicated on nautical charts are reckoned from a selected level of the water called the sounding datum (sometimes referred to as the reference plane). For most NOAA charts of the U.S. in coastal areas, the sounding datum is Mean Lower Low Water. (MLLW). In the Great Lakes, the Plane of Reference is the International Great Lakes Datum (1985).

Depths shown on charts are the least depths to be expected under average conditions. Since the chart datum is generally a computed mean or average height at some state of the tide, the depth of water at any particular moment may be less than shown on the chart. For example, if the chart datum is MLLW, the depth of water at lower low water will be less than the charted depth as often as it is greater.

SOUNDINGS IN FEET

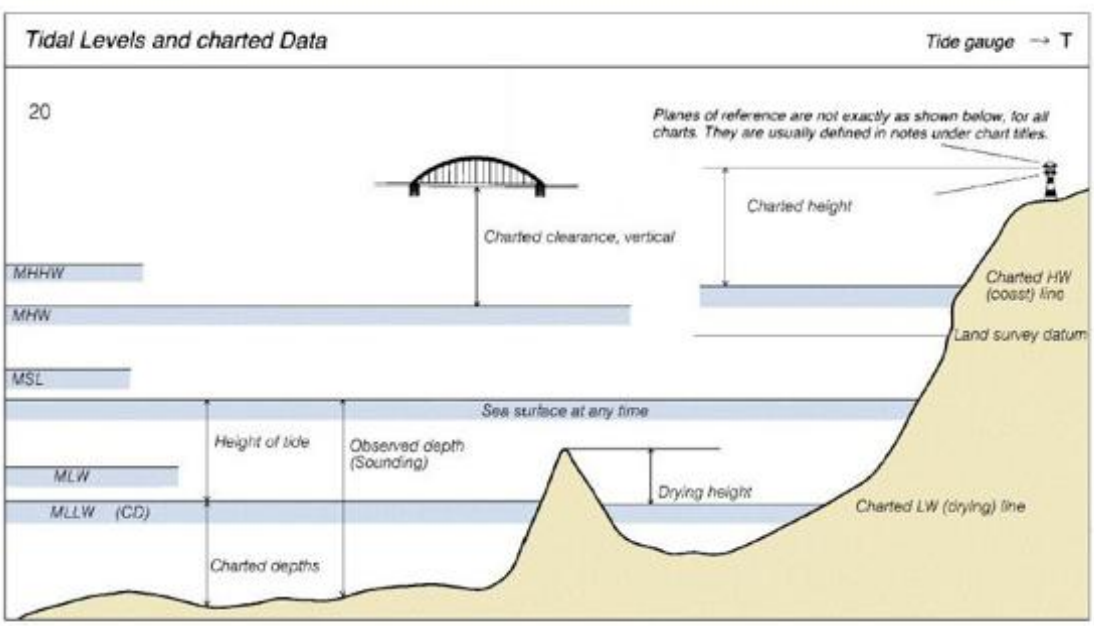
AT MEAN LOWER LOW WATER

Additional information can be obtained at nauticalcharts.noaa.gov.

TIDAL INFORMATION

Place		Height referred to datum of soundings (MLLW)			
Name	(LAT/LONG)	Mean Higher High Water	Mean High Water	Mean Low Water	Extreme Low Water
		feet	feet	feet	feet
Salem	(42°31'N/70°53'W)	9.5	9.1	0.3	-.-
Nahant	(42°25'N/70°55'W)	9.7	9.3	0.3	-3.5
Lynn Harbor	(42°27'N/70°58'W)	9.9	9.5	0.3	-3.5
Manchester Harbor	(42°34'N/70°47'W)	9.5	9.1	0.3	-3.5

(Jun 2003)



ENCLOSURE 2.2.2

MSC Ship Notional Arrival Conditions

Ship's Characteristics:

Length Overall: 239.325 m
Length between Perpendiculars: 233.215 m
Breadth Molded (Max.): 15.468 m
Depth Molded at side

(to 01 level amidships): 50 m
Displacement Full Load: 71,954 tonnes
Draft Full Load: 10.5 m
Displacement Light Ship: 35087 tonnes
Arrival Draft (ABL) Fwd Mast Top: 30 feet - 0 inches
Arrival Draft (ABL) Aft Mast Top:
Estimated Arrival Conditions:
Forward Draft: 31 feet - 1 inches
Aft Draft: 31 feet - 1 inches
Displacement: Unknown Long Tons

1.0 ABSTRACT

This item describes the requirements for the contractor to have in place an effective quality assurance program and that such program be implemented in the planning and performance of this contract.

2.0 REFERENCES: None

3.0 ITEM LOCATION/DESCRIPTION

3.1 Performance of quality assurance at the contractor's facility.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The Government may at any time without prior notification to the contractor elect to obtain the services of Government furnished technical representative(s) who will be on site at the contractor's facility acting as the Government quality assurance observer and consultant. The presence of Government furnished technical representative(s) does not relieve the contractor from performing any part of this work item.

5.2 The contractor and all subcontractors regardless of tier shall consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs (1) through (7).

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 The Government reserves the right to perform a pre-award survey of the apparently successful contractor’s facility, during which survey an inspection of the in-place quality assurance system's performance attributes in all areas may be subject to examination and review.

6.2 The successful contractor is herewith notified that regulatory body and MSC inspections are independent of the contractor's function of quality assurance. The contractor is not to rely upon MSC or regulatory bodies or their agents to perform its quality assurance inspections or tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Contractor inspection system requirements

7.1.1 Quality assurance plan/manual

a. The contractor shall document his inspection system, identifying the method of implementing the requirements contained herein. Implementation of the contractor's system shall be through a quality assurance plan or manual. The quality assurance plan or manual will be approved for use by the senior shipyard official and shall be available for review by the Government prior to the initiation of productive work. It shall meet the following minimum requirements:

1. A functional organization chart showing overall company management

2. The quality assurance organization

3. A description of the contractor's quality control system

4. The assignment of specific responsibility for the following elements of quality control including written procedures specifying the methods of implementation;

(i) Performance and witnessing of tests and inspections

(ii) Preparation and maintenance of records

(iii) Control of non-conforming material

(iv) Corrective action system

(v) Receipt inspection

(vi) Subcontractor control

7.1.2 Test and inspections

a. Inspection personnel shall be qualified to perform the duties assigned and authorized to make an accept/reject determination for the contractor.

1. Submit a list of contractor personnel authorized to witness and accept the sign off inspection and tests listed on contractor’s test and inspection records (TIR's). Include personnel title/position in corporate structure.

2. Submit a list of subcontractor firms which are authorized to independently witness, accept and sign off inspections and tests listed on TIR's on behalf of the contractor. Identify each authorized subcontractor firm by name and address, by work item number, paragraph(s), and sub paragraph(s) which the subcontractors are to accomplish. The identification shall be as broken down in the statement of work for that work item and the production planning document prepared for the contractor's offer and this contract.

3. Lists of contractor and subcontractor personnel shall be submitted as part of the offered solicitation's technical proposal. Lists shall be amended as changes in personnel or subcontractors occur. Changes shall be

subject to approval of the Contracting Officer (PCO prior to award or ship delivery/ACO (COR) after delivery) in accordance the Clause H-11 "substitution of personnel."

4. The contractor shall verify that all tests, inspections and work conform to contract requirements prior to presentation to the OMT REP for acceptance. In the event that specific criteria are not provided in the work item, the contractor shall verify that the tests, inspections or work meet regulatory body requirements (ABS, USCG etc.,) prior to presentation to the OMT REP for acceptance.

b. Final determination of acceptability shall be made by the OMT REP.

7.2 Documentation

7.2.1 Contractor inspection records

a. The contractor shall prepare a test and inspection record (TIR) for each specification item in the contract which requires productive work. TIR's shall be developed for each specification work item or change prior to productive work being accomplished for that item. TIR's at a minimum shall, include the following:

1. Identification by solicitation/contract number, ship name and Government work item number

2. Identification of each unit to be inspected by name, number, and location (e.g. Number 2 SSTG, Port Condenser, Cargo Winch Number 12, etc.). Where multiple units are contained within a work item, an entry on the TIR shall be made for each unit.

3. The listing of each specific inspection attribute, method of inspection or test and the acceptance/rejection criteria

4. Acceptability or rejection of each inspection attribute shall be indicated and shall be signed and dated by authorized personnel.

5. All TIR's shall be updated as work progresses and maintained current to within twenty-four (24) hours.

b. Test and inspection records (TIR's) are also required for all work associated with:

1. All Pre-priced option items

2. All change orders issued

3. The price for all QA required by option items shall be contained within the lowest line item level prices

4. The price for all QA required by the clause entitled "Additional Requirements" shall be considered, to be contained within the hourly labor price

5. Inspection attributes: The contractor shall list, on the TIR, all tests and inspections contained in each work item. In addition, the contractor shall include the following types of inspection required by the work item:

(i) All pertinent tolerances, clearances and conditions found upon disassembly/opening of equipment, machinery components or spaces

(ii) All in process verification, including refit/final tolerances, clearances, alignment, tests, cleanliness and internal condition prior to closure, fit-up and welding

(iii) All verification of item completion including prerequisite for operational testing (i.e., lubrication, continuity checks, rotation) and final supporting data

7.2.2 Maintenance of records

a. The contractor shall maintain records of completed TIR's, including inspection records generated by all subcontractors for a period of ninety (90) days after contract completion.

b. The records shall document the specific inspection actions or tests and provide objective quality evidence of completed work and form the basis for acceptance of the work.

c. The records shall include documentation of contractor rejected work, corrective action taken and objective quality evidence which the contractor offers as assurance of the conformance to the work item specifications.

d. The records shall include documentation of contractor inspection/test failures, corrective action taken and objective quality evidence which the contractor offers as assurance of the conformance to the work item specifications.

e. Records provided by subcontractors may be incorporated in the inspection and test documentation, or if maintained as separate documentation, shall be referenced therein and be readily available for review by the Government.

f. The inspection record shall, for each work item, reference any quality deficiency reports (QDRs) issued by the OMT REP. All QDRs require a written response from the contractor to the Government.

g. During the performance period of the contract, records shall be maintained at a location accessible to the site of the work. The records shall be available for review by the Government during weekday and weekend hours consistent with regular day shift, (approximately 0630-1800). Records shall also be available for Government's review during the 90 days following contract completion.

h. Contractor's inspection records are an integral part of the work. Therefore, the ACO and OMT REP will consider the work item incomplete if the contractor's documentation and records are not complete. Records shall include all TIR's subcontractor inspection reports, material certificates and receipt inspection records.

7.3 Corrective action

7.3.1 The contractor's corrective action shall provide for prompt correction of defects whether found by the Government or by the contractor's or subcontractor's personnel. When such defects are reported by the Government and require a written response, the contractor shall provide prompt written response indicating corrective action taken or the date by which a specific corrective action will be complete. The contractor's corrective action shall provide for the correction of the cause of defects in work or procedures reported by Government to contractor by quality deficiency report (QDRs) or by letter.

7.4 Subcontractor control and evaluation

7.4.1 The contractor shall establish procedures for selection of subcontractors and control of subcontractor quality, including:

a. Procedures for selection of qualified subcontractors, based on evaluation and assessment of the subcontractor's quality control, facilities and available resources, as appropriate, to perform the specific type of work.

b. Procedures for transmittal of all technical, design and quality requirements to the subcontractor. Quality requirements shall include the inspection and tests required by the work item as well as contractor inspection at the subcontractor's plant when such action is necessary to assure product conformance.

7.4.2 Subcontractors listed within the contractor's technical proposal as having been selected to perform individual work items or parts thereof, shall be considered key personnel as defined in Clause for "Substitution of Personnel." As key personnel they are considered during the technical evaluation process. As such any changes in subcontractors shall be subject to approval of the OMT REP.

7.5 Regulatory inspections

7.5.1 The contractor shall ensure all regulatory body mandated inspections are arranged for and conducted.

a. The contractor shall reproduce and provide copies of the contract specifications to the cognizant American Bureau of Shipping (ABS) and United States Coast Guard (USCG) area representatives.

b. The contractor shall prepare a written request for review of the specifications to identify the specific attributes for which regulatory body inspection is deemed mandatory. The contractor shall incorporate these inspections in the TIR's.

c. During the performance of the contract the contractor shall be responsible for notifying both ABS and USCG and coordinating and scheduling their inspections. Timely arrangements for the attendance of regulatory body representatives are a mandatory requirement of this item. The contractor shall be responsible for maintaining weekly contact with regulatory agencies, obtaining a written prediction of their weekly workload, their advance notification requirements, keep abreast of notification requirement changes to ensure timely contact. Copies shall be furnished to the OMT REP.

7.6 MSC inspections

7.6.1 The contractor shall provide for notification to the OMT REP or his duly authorized representative for all tests requiring MSC attendance.

a. When specification items require that a OMT REP or his representative (i.e., Port Engineer, Chief Engineer and others that may be designated by the item) accept and/or witness in process work, completed work, testing or inspections, the contractor shall provide notification to the designated personnel at least four (4) but not more than twenty four (24) hours in advance of the work sequence to be witnessed or approved.

b. When work to be witnessed or approved is to occur after normal day shift working hours the OMT REP shall be notified at least four (4) hours before the end of the last preceding regular work shift.

c. When work to be witnessed or approved is to occur on a weekend or holiday, the OMT REP shall be notified at least four (4) hours before the end of the last preceding regular work shift.

d. Notification of tests/inspections is a mandatory requirement. Identification of tests/inspections on the contractor's production schedule(s), etc., does not relieve the contractor of the requirement.

e. The OMT REP may designate checkpoints, in addition to those identified by the specification item, that are to be observed or inspected by the OMT REP or regulatory body. The notification requirements of above shall also apply to those designated checkpoints.

f. Regulatory body inspectors may also elect to designate additional checkpoint(s) tests/inspections which will require their attendance. (These are in addition to those incorporated into the TIR after specification review per 7.5 above.) This is beyond the ability or control of the OMT REP. The contractor shall provide adequate timely notification to ensure the requiring inspector's attendance at these added tests/inspections.

g. MSC inspection is an independent function of the Government and does not relieve the contractor of the responsibility for performing the tests and inspections required by the specification or those considered necessary to ensure product conformance. All inspections specified within the specification work item are mandatory inspections. The contractor shall not proceed beyond this point without prior approval from the OMT REP.

7.7 Submission of records

7.7.1 A copy of all records as defined by 7.2 shall be submitted to the Contracting Officer within ten (10) days of contract completion. The contractor shall maintain original copies in accordance with 7.2.

7.8 Quality deficiency reports (QDRs)

7.8.1 During the course of the contract it may be determined by the OMT REP that the contractor's workmanship is defective or that the quality assurance (QA) program (inspectors or inspection processes) has failed to perform in a manner consistent with the contractor's proposal as evaluated prior to award. Should such occur, the OMT REP may issue to the contractor a quality deficiency report (QDR).

7.8.2 The contractor is required to address the specific subject of the QDR and to advise the Government of how it proposes to perform permanent corrective action of the defective work or process which caused the QDR to be issued.

7.8.3 For a QDR issued in connection with a work item, the contractor is explicitly required to provide a satisfactory answer within the time frame indicated thereon for a reply. Normally, this is within two (2) calendar days from the date of issue. The contractor shall reply in the allotted time indicated on the QDR.

7.8.4 Additional reply time may be requested if needed by the contractor. A request shall be submitted in writing to the Contracting Officer for an extension to the reply date. The request letter shall give a specific, valid justification and date for submittal. In no case shall the extension requested be longer than a total of five (5) calendar additional days.

7.8.5 When the contractor's failure to submit a timely reply, (or a written request for an extension to the Contracting Officer), necessitates issuance of a letter addressing the lateness of the response, (or extension request), the contractor shall lose all further work progress for payment purposes upon the work item the QDR is issued against. This may be for the duration of the contract, or until such time as an acceptable response is furnished.

7.9 Schedule for key inspection events

7.9.1 Contractors shall prepare and submit within two (2) weeks after award a Schedule for key inspection events. The schedule for key inspection events shall show the dates for key inspections as contained in the production schedule, and the parties whose required attendance is necessary to conduct and witness a successful test. At a minimum the following inspections are to be identified:

a. Preliminary test for each final acceptance test.

b. Final acceptance tests.

c. Regulatory agency (ABS/USCG) required tests.

d. Tests indicated by work items as requiring mandatory OMT REP presence.

7.9.2 The schedule for key inspection events shall be updated weekly to show tests that have been added, deleted, completed, and all test failures. It is to reflect changes to the production plan resulting from new work items added or work items canceled. All test changes are to be reflected in the updated production schedule and test and inspection record.

7.9.3 Changes to the schedule for key inspection events for items not impacted by added/deleted work must be requested via written notification to the Contracting Officer, giving justification for the proposed change. Contracting Officer acknowledgment and concurrence is required before the change can be made. Notifications should be submitted at the weekly production meeting. For exceptional cases this may be later but not less than twenty-four (24) hours prior to the test's scheduled performance.

7.10 Inspection unpreparedness/unnecessary inspections

7.10.1 The contractor is responsible to control the number of unnecessary inspection call outs issued due to unpreparedness. The contractor may choose to accomplish pre-inspections to assure readiness. Unnecessary inspection call outs due to unpreparedness will be considered an indication of an ineffective quality assurance program and system and will be the subject of a QDR.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0005	**CATEGORY "N"**	**2026-01-30**
ESB-SWI-HIGH VOLTAGE ELECTRICAL SAFETY PROCEDURE REQUIREMENTS		**Manovich, David**

1.0 ABSTRACT

This work item establishes shipboard electrical safety requirements for both low voltage and high voltage applications. These requirements and procedures shall be followed by all contractors and subcontractors in the accomplishment of all work requirements.

2.0 REFERENCES

2.1 High Voltage Awareness Instruction Guide Powerpoint Presentation

2.2 High Voltage Awareness Powerpoint Presentation

2.3 MSC Safety Management System Procedure 2.1-004-ALL, Lock-out/Tag-out

2.4 MSC Safety Management System Procedure 2.1-016, High Voltage Systems

2.5 29CFR 1910.335 Electrical Safety Related Work Practices; Safeguards for Personnel Protection

2.6 NFPA 70E Standard for Electrical Safety Requirements for Employee Workplaces

3.0 ITEM LOCATION/DESCRIPTION

3.1 LOCATION: Throughout the ship.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None.

5.0 NOTES

5.1 Comply will all MSC General Technical Requirements (GTRs) as applicable to this work item.

5.2 Review other work items under this contract to determine their effect on the work required by this work item.

5.3 Review Work Item 001 for definitions of terms used in this work item.

5.4 These Work Item requirements apply to all Contractor and sub-Contractor personnel regardless of tier.

5.5 T-ESB class ships use a 6600 volt AC main bus distributed throughout the ship to various load centers and equipment. High Voltage areas require special awareness and access control. “High Voltage” (per ABS Rules 4-8-5 paragraph 3) is defined as 1000 volts and above.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional.

7.0 STATEMENT OF WORK REQUIRED

7.1 Electrical Safety - Lock-out/Tag-out Procedures

7.1.1 Prior to ship’s arrival, all contractor and subcontractor personnel are to receive High Voltage Awareness Training. Using Reference 2.1 for guidance, the Shipyard or Subcontractor Safety Officer is to present Reference 2.2 in a classroom setting. Prior to ship’s arrival, signed letters from Safety Officers listing personnel that took the High Voltage Training are to be submitted via condition report.

7.1.2 The Contractor shall have an established electrical safety lock-out/tag-out program in place to provide adequate protection against electrical hazards in the accomplishment of the work specified in this work package. A written description of the contractor’s electrical safety lock-out/tag-out procedures shall be provided to the OMT REP, ship’s Master and Chief Engineer upon vessel arrival at the contractor’s facility, along with a discussion on the procedures.

7.1.3 Ship's force shall maintain their existing lock-out/tag-out procedures, in accordance with Reference 2.3 for low voltage applications and Reference 2.4 for high voltage applications, in addition to the contractor's procedures, to provide a redundant safety measure for these systems. The contractor shall co-ordinate electrical safety lock-out/tag-out procedures with the OMT REP and the ship's Chief Engineer to ensure the plans complement each other and provide maximum electrical safety for ship's crew and contractor personnel.

7.1.4 Do not accomplish work on any piece of mechanical, electrical or electronic equipment, circuit, or system that has not been isolated from its energy source (electrical, mechanical, hydraulic, pneumatic, chemical, or thermal) and tagged-out in accordance with the shipyard's electrical safety plan and References 2.3 and 2.4.

7.1.5 Do not attempt to operate any piece of mechanical, electrical or electronic equipment, circuit, or system that is tagged-out by either the contractor, ship or both. Tags must be properly cleared through both the contractor's procedures and the ship's procedures before systems and equipment can be safely energized and operated.

7.1.6 All contractor and subcontractor personnel intended to work on electrical circuits and equipment onboard MSC vessels shall provide and utilize proper personnel protective equipment (PPE) and tools in accordance with References 2.5 and 2.6. Contractor and subcontractor personnel will not be permitted to perform electrical systems work onboard if they are found to not possess the proper tools and PPE.

7.2 High Voltage Systems (T-ESB 6600V Systems)

7.2.1 Ensure all Contractor and sub-Contractor personnel comply with the ship's High Voltage System procedures

(T-ESB Vessels) of Reference 2.4, in addition to the contractor's internal procedures.

7.2.2 Obey all High Voltage (HV) signs and verbal directions of the ship's force "Competent Person In-charge" (CPIC).

7.2.3 Ensure all Contractor personnel possess awareness and exercise vigilance regarding HV safety, watch for all signs indicating HV, and take prudent action to preclude unauthorized entry into HV areas.

a. Provide formal briefing to all Contractor and sub-Contractor personnel of HV concerns.

b. Submit a report to the OMT REP listing all personnel formally briefed on HV safety no less than five (5) days prior to start of the period of performance.

7.2.4 Do not disturb, cut, move or damage HV cables in any fashion.

7.2.5 Do not paint HV cables.

a. Temporarily mask, cover or protect HV cables when painting in spaces that contain HV cables.

7.2.6 Do not work on or test HV equipment:

a. For which Contractor personnel do not possess a ship's force issued "High Voltage Permit to Work" (HVPTW) or "High Voltage Sanction for Test" (HVSFT) permit.

b. Ensure only Contractor and sub-Contractor personnel designated as "qualified" work on HV equipment.

c. Prepare a list of all Contractor and sub-Contractor personnel identified as "qualified" to:

1. Work in HV spaces

2. Work on HV equipment.

d. Include on the list the specific type of formal qualification training possessed by each individual.

e. Submit copies of the list to the OMT REP and the ship's Chief Engineer before any individual commences HV permit request.

8.0 GENERAL REQUIREMENTS: None Additional.

USSHershel Williams			
(ESB 4)			
GENERAL SERVICES AND REQUIREMENTS		CONTRACT NO. N68171-26-RN-003	
ITEM NO. 0006	CATEGORY "N"		2026-01-30
ESB-SWI-HEAVY WEATHER PLAN		Manovich, David	

1.0 ABSTRACT

This item describes the requirements for the contractor to provide a heavy weather plan.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 MSC Drawing No. 835-8495718 USS Hershel Williams Trim and Stability Booklet.

2.2 Enclosures

2.2.1 Beaufort Wind Scale

3.0 ITEM LOCATION/DESCRIPTION

3.1 LOCATION: Contractor's facility

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 In order to ensure vessel safety in extreme circumstances, the contractor's Heavy Weather Plan may contain references to ballasting and deballasting the vessel. In such an instance the safety requirements of reference 2.1 shall not be exceeded. Minimum draft clearance shall be two feet beneath the keel at mean lower low water (MLLW).

5.2 The contractor shall not be permitted to alter the ballast conditions of the vessel in any manner without having first submitted detailed trim and stability calculations to the OMT REP, and having obtained written approval of same from the ship's Master and the OMT REP.

5.3 In order to ensure vessel safety in extreme circumstances, the contractor's heavy weather plan may contain references to shifting the ship to another berth. In such an instance, all safety precautions shall be taken to ensure the ship's transit route maintains the minimum draft clearances as stated herein, an adequate quantity of tugs having sufficient horsepower ratings to maintain control of the ship are provided, and the ship is properly secured to the tugs at all times during the shifting evolution.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Arrangements/Outfitting

7.1.1 The contractor shall provide a heavy weather plan as part of the technical proposal. The plan shall include a description of the protection arrangements available and the preventive measures that will be taken to ensure the safety of the ship from heavy weather conditions for a Beaufort Wind Scale, enclosure 2.2.1, Number 7 and higher. The heavy weather plan should at a minimum address the following:

a. Daily monitoring and tracking operations of heavy weather conditions reported by the National Weather Service (NWS) and National Hurricane Tracking Center (NHTC).

b. Maintenance of communications with the local offices of the NWS, NHTC, and Federal Emergency Management Agency (FEMA).

c. A timetable for making decisions including identification of the parties participating in the decision-making process.

d. A timetable for initiating contractor actions relative to deteriorating weather conditions for Beaufort Wind Scale (enclosure 2.2.1) Number 7 and higher, based on the approaching geographic position of the reported heavy weather.

e. Describe the personnel recall plan to accomplish each level of action including:

1. Facility power loss protection and precautions

2. Facility damage protection and precautions

3. Personnel safety protection and precautions

4. Plant equipment protection and precautions

5. Resources available for continued contractor's communications with the appropriate decision-making company officials

6. Resources available for continued contractor's communications with OMT REP and the ship's Master

f. Describe the actions taken for each level of the Beaufort Wind Scale (enclosure 2.2.1) deterioration in preparation to protect the vessel from damage including:

1. Vessel mooring protection and precautions

2. Vessel flooding protection and precautions

3. Vessel ballasting and deballasting as a protection and precaution

4. Trim and stability calculation for ballasting/deballasting

5. Vessel equipment protection and precautions

6. Precautions taken if the vessel is on dry dock

7. Movement of the vessel to a different

temporary lay berth for safety

8. Interim milestones for notification of OMT REP and ship's Master at each appropriate time of the decision-making process

9. Level of ship's force participation (if any) for each required action

7.1.2 The contractor is responsible for the costs associated with the planning aspects of the heavy weather plan that deal with the safety and protection of the contractor’s facility and the Government’s assets located in that facility. Extraordinary measures required to protect the vessel and its crew shall be presented to the OMT REP as a milestone event in the subject heavy weather plan. The OMT REP shall approve these extraordinary measures to prepare for heavy weather conditions via a change order.

a. Extraordinary measures may include such actions as ballasting the vessel, moving the vessel to a safe haven, and/or temporarily installing protective measures for sensitive and vulnerable ship’s equipment.

b. Non-extraordinary protective measures considered to be the responsibility of the contractor include such actions as adding additional mooring lines, temporarily disconnecting ship services, and vessel flooding protection.

7.2 Structural: None

7.3 Mechanical/Fluids: None

7.4 Electrical: None

7.5 Electronics: None

7.6 Preparation of Drawings/Documentation: None

7.7 Inspection/Test: None

7.8 Painting: None

7.9 Marking: None

7.10 Manufacturer’s Representative: None

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - BEAUFORT WIND SCALE

Beaufort Wind Scale

Developed in 1805 by Sir Francis Beaufort of England

Force	Wind (Knots)	WMO Classification	Appearance of Wind Effects	
			On the Water	On Land
0	Less than 1	Calm	Sea surface smooth and mirror-like	Calm, smoke rises vertically
1	1-3	Light Air	Scaly ripples, no foam crests	Smoke drift indicates wind direction, still wind vanes
2	4-6	Light Breeze	Small wavelets, crests glassy, no breaking	Wind felt on face, leaves rustle, vanes begin to move
3	7-10	Gentle Breeze	Large wavelets, crests begin to break, scattered whitecaps	Leaves and small twigs constantly moving, light flags extended
4	11-16	Moderate Breeze	Small waves 1-4 ft. becoming longer, numerous whitecaps	Dust, leaves, and loose paper lifted, small tree branches move
5	17-21	Fresh Breeze	Moderate waves 4-8 ft taking longer form, many whitecaps, some spray	Small trees in leaf begin to sway
6	22-27	Strong Breeze	Larger waves 8-13 ft, whitecaps common, more spray	Larger tree branches moving, whistling in wires
7	28-33	Near Gale	Sea heaps up, waves 13-20 ft, white foam streaks off breakers	Whole trees moving, resistance felt walking against wind
8	34-40	Gale	Moderately high (13-20 ft) waves of greater length, edges of crests begin to break into spindrift, foam blown in streaks	Whole trees in motion, resistance felt walking against wind
9	41-47	Strong Gale	High waves (20 ft), sea begins to roll, dense streaks of foam, spray may reduce visibility	Slight structural damage occurs, slate blows off roofs
10	48-55	Storm	Very high waves (20-30 ft) with overhanging crests, sea white with densely blown foam, heavy rolling, lowered visibility	Seldom experienced on land, trees broken or uprooted, "considerable structural damage"
11	56-63	Violent Storm	Exceptionally high (30-45 ft) waves, foam patches cover sea, visibility more reduced	
12	64+	Hurricane	Air filled with foam, waves over 45 ft, sea completely white with driving spray, visibility greatly reduced	

USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0007	CATEGORY "N"	2026-01-30
ESB-SWI-COLD WEATHER PLAN		Manovich, David

1.0 ABSTRACT

This item describes the requirements for the contractor to provide a cold weather plan which documents the planning and actions to be taken by the contractor to prevent damage to the vessel or injury to crew members in the event of significant cold weather. Applicable only to geographic areas where temperatures may drop below 32°F for periods greater than 24 hours.

2.0 REFERENCES

2.1 MSC Drawing No. 835-8495718, USS Hershel Williams Trim and Stability Booklet

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: Contractor’s facility

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 In order to ensure against cold weather damage to the vessel in extreme circumstances, the contractor's cold weather plan may contain References to ballasting and deballasting the vessel. In such an instance the safety requirements of Reference 2.1 shall not be exceeded. Minimum draft clearance shall be two feet beneath the keel at mean lower low water (MLLW).

5.2 The contractor shall not be permitted to alter the ballast conditions of the vessel in any manner without having first submitted detailed Trim and Stability calculations to the OMT REP, and having obtained written approval of same from the Ship's Master and the OMT REP.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Arrangements/Outfitting

7.1.1 The contractor shall provide, with their proposal, a cold weather plan if the facility is located in a geographic location where temperatures are likely to drop below 32°F for periods in excess of 24 hours. The plan shall include a description of the cold weather precautions and protection arrangements available and the preventive measures that will be taken to ensure the safety of the vessel, prevent damage to the ship and protect ship's force from injury during cold weather. The cold weather plan should at a minimum address the following:

a. Daily monitoring and tracking operations of cold weather conditions reported by the National Weather Service (NWS)

b. Maintenance of communications with the local offices of the NWS and Federal Emergency Management Agency (FEMA)

c. A timetable for making decisions including identification of the parties participating in the decision-making process.

d. A timetable for initiating contractor actions relative to; changing weather conditions as the cold weather season approaches, unusual cold spells for the geographic location of the contractor's facility, and winter precipitation patterns for snow, sleet and ice conditions.

e. Personnel resources to accomplish each level of plan actions.

f. Protection and precautions taken to prevent; facility power loss, damage to plant, equipment and services, personnel injury.

g. Protection and precautions taken to prevent access/travel restrictions inside the facility.

h. Actions taken to protect contractor material at outside storage areas.

i. Resources available for snow removal.

j. Resources available to maintain a heating supply for shops, warehouses, offices, etc. in the event of plant power loss.

k. Resources available for continued contractor's communications with the appropriate decision-making company officials.

l. Resources available for continued contractor's communications with OMT REP and the ship's Master.

m. Provisions for adequate lighting inside the contractor's facilities, all walkways to the ship, overhaul management team's office, and parking lots. Lighting illumination hours for normal circumstances and during inclement weather, i.e., rain and/or snowstorms.

7.1.2 Describe the actions the contractor shall plan for and execute for each increased level of protection and precautions in preparation to protect the vessel and ship's force due to the approaching cold weather season, unusual cold spells for the geographic location of the contractor's facility, and winter precipitation patterns for snow, sleet and ice conditions including:

a. Snow and ice removal from weather decks, ladders, gangways, superstructure, and masts.

b. Vessel equipment protection and precautions.

c. Precautions taken if the vessel is on dry dock.

d. Trim and stability calculations for ballasting/deballasting.

e. Interim milestones for notification of OMT REP and ship's Master at each appropriate time of the decision-making process.

f. Level of ship's force participation (if any) for each required action.

g. Covering and removal of the ship's smokestack(s).

h. Draining and maintenance of a dry condition for all pipelines, including fixtures, traps and internal tanks, throughout the ship to avoid freeze damage.

i. Record keeping and monitoring of removed pipe drainage plugs.

j. Installation of system bleed lines or recirculating lines on all temporary water, steam and air hoses/pipes to prevent freeze damage and maintain the systems in service.

k. If it is not practical to keep the pipelines/systems or fixtures drained; provide explanation of how to safely maintain the ship heated to a minimum temperature capable of preventing overnight freeze damage. Prepare and submit calculations giving the types, sizes and number of temporary heaters which might be installed in all spaces to provide the heat. The calculations for each compartment must demonstrate the quantities of heaters given are adequate to maintain each space no less than 40° Fahrenheit with the ventilation system secured.

l. Should the ship’s heating system be interrupted by any other work item of the work package; prepare and submit calculations giving the size and number of temporary heaters to be installed in all spaces for general heating. The calculations for each compartment shall demonstrate the quantity of heaters given are adequate to maintain each space no less than 60° Fahrenheit with the ventilation system secured.

m. Freeze protection shall be provided for all tanks and their associated piping systems without regard to the contractor being required to work in these tanks by a work item.

1. Freeze protection in tanks shall be by a positive action means of preventing tank bulkhead or piping ruptures.

2. The protection may be provided by deballasting and/or pumping of water tanks, opening manhole covers and inserting discharging air lines to continuously give a bubbling action, etc.

n. How deck drains will be kept clear, and in proper working condition open to drain. Actions to clear clogged and blocked drains.

o. Provisions for adequate lighting aboard the ship on all deck levels during inclement weather, i.e., rain and/or snowstorms.

7.1.3 Describe the planning and execution of actions to prevent and protect from weather impact upon the production schedule. Provide local data for the following during the past three years:

a. Normally expected number of rainfall days.

b. Normally expected amount of rainfall (in inches).

c. Normally expected number of snow/sleet fall days.

d. Normally expected amount of snow/sleet fall (in inches).

e. Normally expected number of days with temperatures below freezing.

f. Normally expected periods with temperatures below freezing in quantity and durations in days.

g. Normally expected water temperatures in degrees Fahrenheit.

h. Normally expected wind speeds.

i. Normally expected number of days when the wind speeds exceed this figure.

7.1.4 The contractor is responsible for the costs associated with the planning aspects of the cold weather plan that deal with the safety and protection of the contractor's facility and the Government's assets located in that facility. Extraordinary measures required to protect the vessel and its crew shall be presented to the OMT REP as a milestone event in the subject cold weather plan. The OMT REP shall approve these extraordinary measures to prepare for cold weather conditions via a change order.

7.1.5 Extraordinary measures may include such actions as ballasting the vessel, dealing with significant deep freeze and/or exceptionally heavy snow or ice events.

7.1.6 Non-extraordinary protective measures considered to be the responsibility of the contractor include such actions as adding additional mooring lines, temporarily disconnecting ship services, snow and ice removal, freeze protection on and in the vessel.

7.2 Structural: None

7.3 Mechanical/Fluids: None

7.4 Electrical: None

7.5 Electronics: None

7.6 Preparation of Drawings/Documentation: None

7.7 Inspection/Test: None

7.8 Painting: None

7.9 Marking: None

7.10 Manufacturer’s Representative: None

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams			
(ESB 4)			
GENERAL SERVICES AND REQUIREMENTS		CONTRACT NO. N68171-26-RN-003	
ITEM NO. 0008	CATEGORY "N"		2026-01-30
ESB-SWI-PROCUREMENT AND USE OF LUBRICANTS		Manovich, David	

1.0 ABSTRACT

This item describes the procurement and use of lubricants.

2.0 REFERENCES/ENCLOSURES

2.1 Reference: None

2.2 Enclosure:

2.2.1 USS Hershel Williams Lube Chart

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: Throughout the ship

3.2 Quantity: As required

3.3 Description/Manufacturer's Data: Per Enclosure 2.2.1

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM):

4.2.1 As required during the work scope of this work package in accordance with Enclosure 2.2.1

4.3 Government Furnished Services (GFS): None

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier shall consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier shall comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 28, and 29.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract to determine their effect on the work required under this work item. Many of the definitions relating to performance of this work item are found in Work Item 0001.

5.3 MSC has a contract with Exxon Mobil to provide lubricants for USNS vessels.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Perform procurement of lubricants, timely notification for procurement of lubricants, installation of the lubricants and disposal of waste lubricants remains as a contractor responsibility.

7.2 The lubricants listed in Enclosure 2.2.1 are available for use on all equipment and/or systems installed or worked on during the performance of this work package at no additional charge to the contractor.

7.3 Utilize lubricants identified in Enclosure 2.2.1 for all new installations of MSC furnished or contractor furnished equipment and/or systems as recommended by the original manufacturer.

7.4 Notify the OMT REP no later than 10 days after the start of the vessel's availability if an original equipment manufacturer (OEM) requires a lubricant that is not listed in Enclosure 2.2.1, for any MSC furnished or contractor furnished equipment and/or systems. If MSC is not able to provide a lubricant that is approved by the OEM, a change order will be written for the contractor to provide the approved lubricant.

7.5 The required lubricants must be ordered from the OMT REP at least 14 days prior to the required delivery date. The order must identify the type of lubricant, required use, quantity required, and type of packaging, either 55 gallon drums or 5 gallon pails. A required delivery date and a contractor point of contact (including phone # and address) must be identified for every lubricant order.

7.6 The contractor is responsible for any delays and subsequent costs incurred as a result of not placing the subject lubricant orders at least 14 days prior to required delivery date.

7.7 All lube oil fill and filtering procedures shall be approved by the Port Engineer or ship’s Chief Engineer.

7.8 All excess lubricants shall be turned over to the ship’s force upon redelivery of the vessel.

8.0 GENERAL REQUIREMENTS: None additional

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

LO Supplier: EXXONMOBIL
Author: MSC
First Issued: 10/31/18
Last Amended: **11/14/19**

1. MAIN PROPULSION				
Equipment	Make	Model	Application	EXXONMOBIL
MAIN DIESEL ENGINES (4)	MAN B&W	48/60L6	CRANKCASE SYSTEM	MOBILGARD ADL 40 /MOBIL DELVAC 1640
			TURNING GEAR SUMP	MOBIL SHC 634
			GOVERNOR	DTE HEAVY MEDIUM
GENERATOR BEARINGS	RENK	SCWFE36-280K/ERNL Q22-250	DE/NDE MOTOR BEARINGS	MOBILGARD ADL 40 /MOBIL DELVAC 1640
PROPULSION MOTOR (P&S)	RENK	SCNFQ55-560	DRIVEN END BEARING	MOBILGARD ADL 40 /MOBIL DELVAC 1640
			NON DRIVEN END BEARING	MOBILGARD ADL 40 /MOBIL DELVAC 1640
MAIN THRUST BEARING (P/STBD)	WAUKESHA	LAXRT50-515	BEARINGS	MOBILGARD ADL 40 /MOBIL DELVAC 1640
LINE SHAFT BEARINGS (P/STBD)	WAUKESHA	LRZPW52-470	BEARINGS	MOBILGARD ADL 40 /MOBIL DELVAC 1640
TURNING GEAR MAIN ENGINE			GEAR BOX	MOBILGEAR 600XP 460
			OPEN GEARS	MOBILTAC 375
			GREASE POINTS	MOBIL POLYREX EM

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

TURNING GEAR MOTOR			GEARBOX `	MOBILGEAR 600XP 460
AZIMIUTH BOW THRUSTER	ROLLS ROYCE	UL 255FP	HYDRAULIC MEDIUM	MOBIL DTE 10 EXCEL 46
			LUBRICATING	MOBILGEAR 600 XP 100 (See note A)
			GRAVITY TANK	VELOCITE OIL NO.4
BOW THRUSTER MOTOR	ALCONZA	QD630 M-8W	MOTOR BEARINGS	MOBILGREASE XHP 222
STEERING GEAR (P/STBD)	ROLLS-ROYCE NAVAL MARINE INC	L3MF 70/112	HYDRAULIC MEDIUM	MOBIL DTE 10 EXCEL 68
			ACTUATOR	MOBIL DTE 10 EXCEL 68
			RUDDER NUT COVER	MOBILGREASE 28
			HPU MOTOR BEARINGS	MOBIL POLYREX EM
2. AUXILIARY ENGINES				
Equipment	Make	Model	Application	EXXONMOBIL
EMERGENCY GENERATOR	CUMMINS	QSK50	CRANKCASE SYSTEM	MOBIL DELVAC 1300 SUPER 15W40 /MOBILGARD HSD+ 15W40
			HYDRAULIC MEDIUM	MOBIL DTE 10 EXCEL 46 (NOTE C)
			BEARINGS	MOBIL POLYREX EM
			RADIATOR FAN BEARINGS	MOBILGREASE XHP 222
			HYDRAULIC STARTING UNIT	MOBIL DTE 10 EXCEL 46 (NOTE C)
3. PURIFIERS				
Equipment	Make	Model	Application	EXXONMOBIL

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

FUEL OIL PURIFIER (2)	ALFA LAVAL	P635	DRIVE CHAMBER	MOBIL SHC RARUS 68 /MOBIL RARUS SHC 1026
			MOTOR BEARINGS	MOBILGREASE XHP 222
LUBE OIL PURIFIER (4)	ALFA-LAVAL	P635	DRIVE CHAMBER	MOBIL SHC RARUS 68 /MOBIL RARUS SHC 1026
			MOTOR BEARINGS	MOBILGREASE XHP 222
4. REFRIGERANT AND AIR CONDITIONING PLANT				
Equipment	Make	Model	Application	EXXONMOBIL
SHIP SERVICE REFRIGERATION (2)	CARRIER	18-00091-105	CRANKCASE SYSTEM	MOBIL EAL ARCTIC 68
			GREASE POINTS	MOBILGREASE XHP 222
			MOTOR BEARINGS	MOBIL POLYREX EM
ENGINE ROOM AND WORKSHOP A/C SYSTEM	CARRIER		CRANKCASE SYSTEM	MOBIL EAL ARCTIC 68
A/C PLANT COMPRESSORS (4)	CARRIER		CRANKCASE SYSTEM	NU-CALGON EMKARATE RL220 XL
5. AIR COMPRESSORS				
Equipment	Make	Model	Application	EXXONMOBIL
STARTING AIR COMPRESSOR (4)	SPERRE	HL2/105	CRANKCASE SYSTEM	MOBIL SHC RARUS 68 /MOBIL RARUS SHC 1026
GENERAL SERVICE AIR COMPRESSOR (P/STBD)	SPERRE	LL2/120	CRANKCASE SYSTEM	MOBIL SHC RARUS 68 /MOBIL RARUS SHC 1026
ROTARY GENERAL SERVICE AIR COMPRESSOR (2)			CRANKCASE SYSTEM	MOBIL SHC RARUS 68 /MOBIL RARUS SHC 1026
SCBA AIR COMPRESSORS			CRANKCASE SYSTEM	BAUER OIL-0024 (SYNTHETIC)
6. CENTRIFUGAL/ROTARY/SCREW PUMPS - FANS & ELECTRIC MOTORS				
Equipment	Make	Model	Application	EXXONMOBIL

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

FO PURIFIER SUPPLY PUMP (2)	ALFA LAVAL	ACP 045N7 NV	GEARBOX/SPINDLE	MOBIL SHC RARUS 68 /MOBIL RARUS SHC 1026
LO PURIFIER SUPPLY PUMP (4)	ALFA LAVAL	ACP 038KS 1764503	GEARBOX/SPINDLE	MOBIL SHC RARUS 68 /MOBIL RARUS SHC 1026
MAIN FO TRANSFER PUMP	LEISTRITZ CORP	L2NG 62/104 ASOG	MOTOR BEARINGS	MOBILGREASE XHP 222
FWD GENERAL SERVICE PUMP			PUMP BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM
STERN TUBE SW SUPPLY PUMP			PUMP BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM
MISSION DISTILLED WATER TRANSFER PUMP			BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM
POTABLE WATER PUMP			PUMP BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM
MSD AERATION BLOWER			ENCLOSED GEARS	MOBIL DTE HEAVY
SEWAGE TRANSFER PUMP	RED FOX ENVIRONMENTAL SERVICES INC	TTP-3		MOBIL SHC 626
			PUMP MOTOR	MOBILGREASE XHP 222
SEWAGE HOLDING TANK AERATION BLOWER	TUTHILL CORP	3205-AAR 30V	DE/NDE BEARINGS	MOBILGREASE XHP 222
			BLOWER	MOBIL RARUS 827
			MOTOR BEARINGS	MOBIL POLYREX EM
GREY WATER TANK AERATION BLOWER	MCKENNA LABORATORIES CO		DE/NDE BEARINGS	MOBIL DTE HEAVY
			MOTOR BEARINGS	MOBIL POLYREX EM
SEA WATER COOLING/DEWATERING PUMP			PUMP BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

BOW THRUSTER LIFTING LOWERING HYDRAULIC PUMP			MOTOR BEARINGS	MOBILGREASE XHP 222
FIRE & GENERAL SERVICE PUMP			PUMP BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM
HOT WATER CIRC PUMP	BUFFALO PUMPS	1N CRE	PUMP BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM
CENTRAL LOW TEMP FW COOLING WATER PUMP (4)	BUFFALO PUMPS	NA2038	PUMP BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM
CENTRAL SEA WATER COOLING PUMP	BUFFALO PUMPS	NA2040-PMPEND	PUMP BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM
HIGH TEMP FW COOLING PUMP (4)	BUFFALO PUMPS	DV4050L	PUMP BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM
CHILLED WATER CIRC PUMP			PUMP BEARINGS	MOBILGREASE 28
			MOTOR BEARINGS	MOBIL POLYREX EM
DEEPWELL BILGE PUMP (P/STBD)	FLOWAY PUMPS	11JKM-2	MOTOR BEARINGS	MOBILITH SHC 100
DISTILLER BRINE BOOSTER PUMP			BEARINGS	MOBILITH SHC 100
DILTILLER EJECTOR/FEED SW PUMP			BEARINGS	MOBILITH SHC 100
ECR A/C UNIT FAN			SHAFT BEARINGS	MOBILITH SHC 100
RO SW FEED PUMP			PUMP MOTOR	MOBILGREASE XHP 222
RO HIGH PRESSURE PUMP			PUMP MOTOR	MOBIL POLYREX EM
JP-5 SERVICE PUMP	LEISTRITZ CORP	L4NG 126/78 ASOG	GEARBOX	MOBILGEAR 600 XP 150
			UPPER SHAFT SEALS	MOBILGEAR 600 XP 150
JP-5 TRANSFER PUMP	LEISTRITZ CORP	L4NG 126/78 ASOG	GEARBOX	MOBILGEAR 600 XP 150
			UPPER SHAFT SEALS	MOBILGEAR 600 XP 150

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

OWS PROCESS PUMP	GOULDS PUMPS INC	9BF/CL	MOTOR BEARINGS	MOBIL POLYREX EM
VENT FAN (Fans M13, M24, M25, M27, M28, M17, M10, M12, M16, M11, M29, M9)			MOTOR BEARINGS	MOBILGREASE XHP 222
PORTABLE FIRE PUMPS (2)		FP-100	ENGINE	MOBIL DELVAC 1300 SUPER 15W40 /MOBILGARD HSD+ 15W40
BALLAST PUMP (3)	OSTERNECK CO INC	3N150	BEARINGS	MOBIL POLYREX EM
			MOTOR BEARINGS	MOBILGREASE XHP 222
PORTABLE VACUUM PUMPS			VACUUM PUMP OIL	MOBIL POLYREX EM
7. DECK MACHINERY AND EQUIPMENT				
Equipment	Make	Model	Application	EXXONMOBIL
ANCHOR WINDLASS/MOORING WINCH (2)	TTS MARINE GMBH	1909/5530	GEARBOX	MOBILGEAR 600 XP 150
			HYDRAULICS	MOBIL DTE 10 EXCEL 46 (NOTE C)
			GREASE POINTS	MOBILGREASE XHP 222
			OPEN GEARS	MOBILTAC 375
MOORING WINCH (7)	TTS MARINE GMBH	55303	GEARBOX	MOBILGEAR 600 XP 150
			HYDRAULICS	MOBIL DTE 10 EXCEL 46 (NOTE C)
			GREASE POINTS	MOBILGREASE XHP 222
FAS STATION	IMECO	GYPSY WINCH	GEARBOX	MOBIL SHC 634
			BEARINGS	MOBILGREASE 28
			MOTOR BEARING	MOBILGREASE 28
			MOTOR COUPLING	MOBILGREASE 28
		DOUBLE PROBE RECEIVER	HINGE PIN	MOBIL DTE OIL HEAVY

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

			SWIVEL ARM	MOBIL DTE OIL HEAVY
			RELEASE LEVER	MOBIL DTE OIL HEAVY
			RELEASE LEVER PINS	MOBIL DTE OIL HEAVY
ACCOMMODATION LADDER	SCHOELLHO RN-ALBRECHT		BEARINGS AND PIVOT POINTS	MOBILGREASE XHP 222
			WIRE ROPE	MOBILARMA 798
			REDUCTION GEAR	MOBIL SHC RARUS 46
WATER TIGHT DOORS			HYDRAULICS	MOBIL DTE 10 EXCEL 32
			GREASE POINTS	MOBILGREASE XHP 222
CHAIN STOPPERS (P/STBD)			HYDRAULICS	MOBIL DTE 10 EXCEL 46 (NOTE C)
			GREASE POINTS	MOBILITH SHC 100
EMERGENCY TOWING CHAIN STOPPER			GREASE POINTS	MOBILITH SHC 100
GYROCOMPASS (2)				ANSCHUTZ SUPPORTING FLUID
INLINE AIR LUBRICATORS (SHIPWIDE)				MOBIL DTE 10 EXCEL 46 (NOTE C)
PALLET JACK			HYDRAULIC MEDIUM	MOBIL DTE 10 EXCEL 46 (NOTE C)
S-BAND RADAR PEDESTAL			GEARBOX	MOBIL SHC 634
X-BAND RADAR ANTENNA			GEARBOX	MOBIL SHC 634
RADAR			GREASE POINTS	SILICON GREASE
AFFF CONCENTRATE PUMP			BEARINGS	MOBILGREASE XHP 222
DIESEL PRESSURE WASHER ENGINE			ENGINE	Delvac 1300 Super 15W40/ Mobilgard HSD+ 15W40
			PUMP	Mobil DTE Heavy

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

INLINE AIR LUBRICATORS (SHIP WIDE)			LUBRICATORS	Mobil DTE 10 Excel 32
JP-5 HOSE REEL BEARINGS			BEARINGS	Mobil DTE 10 Excel 46
PALLET JACK HYDRAULIC OIL			HYDRAULICS	Mobil DTE 10 Excel 32
8. SMALL BOATS				
Equipment	Make	Model	Application	EXXONMOBIL
WORK BOAT ENGINE	BUKH STEYR	M0144M38	ENGINE	MOBIL DELVAC SUPER 1300 15W40 /MOBILGARD HSD+ 15W40
			GEARBOX	MOBIL ATF DM /MOBIL ATF 320 /MOBIL MULTI-PUROSE ATF
			STEERING	MOBIL DTE 10 EXCEL 15
			WATER JET STEERING HYDRAULICS	MOBIL ATF DM /MOBIL ATF 320 /MOBIL MULTI-PUROSE ATF
WORK BOAT DAVIT WINCH		RHS 26/5.0	Davit Winch Enclosed Gears	MOBILGEAR 600 XP 68
			Davit Winch Freewheeling Clutch	OEM LIFE TIME LUBE
			Davit Winch Hydraulics	MOBIL DTE 10 EXCEL 46 (NOTE C)
			Swing Drive No. 1 Enclosed Gears	MOBIL DTE 10 EXCEL 46 (NOTE C)
			Swing Drive No. 2 Enclosed Gears	MOBIL DTE 10 EXCEL 46 (NOTE C)
9. RESCUE & LIFEBOATS				
Equipment	Make	Model	Application	EXXONMOBIL
LIFE BOAT DAVIT AND WINCH	MARINE EQUIPMENT INC	GSP.FP.90/ 6.25	DAVIT WINCH GEARBOX	MOBILGEAR 600 XP 68
			GREASE POINTS	MOBILGREASE XHP 222

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

			WIRE ROPES	MOBILARMA 798
			Davit Winch Freewheeling Clutch	OEM LIFE TIME LUBE
LIFE BOAT (#1 & #2)	SABB	L3.139LB	ENGINE	MOBIL DELVAC SUPER 1300 15W40 /MOBILGARD HSD+ 15W40
			GEARBOX	MOBIL ATF DM /MOBIL ATF 320 /MOBIL MULTI-PUROSE ATF
			STERN TUBE ABD RUDDER BEARING	MOBILGREASE XHP 222
			STEERING WHEEL PUMP	MOBIL DTE 10 EXCEL 46 (NOTE C)
LIFE RAFT DAVIT	SURVITEC		WIRE ROPES	MOBILARMA 798
			GREASE POINTS	MOBILGREASE XHP 222
			LIFE SAVING WINCH	MOBILGEAR 600XP 220
RESCUE BOAT	BUKH STEYR	M0144M38	ENGINE	MOBIL DELVAC SUPER 1300 15W40 /MOBILGARD HSD+ 15W40
			JET DRIVE	MOBIL ATF DM /MOBIL ATF 320 /MOBIL MULTI-PUROSE ATF
RESCUE BOAT DAVIT AND WINCH	MARINE EQUIPMENT INC	RHS 26/5.0	Davit Winch Enclosed Gears	MOBILGEAR 600 XP 68
			Davit Winch Freewheeling Clutch	OEM LIFE TIME LUBE
			Davit Winch Hydraulics	MOBIL DTE 10 EXCEL 46 (NOTE C)
			Swing Drive No. 1 Enclosed Gears	MOBIL DTE 10 EXCEL 46 (NOTE C)
			Swing Drive No. 2 Enclosed Gears	MOBIL DTE 10 EXCEL 46 (NOTE C)

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USS Hershel Williams Lubrication Chart

10. WORKSHOP				
Equipment	Make	Model	Application	EXXONMOBIL
LATHE	JET EQUIPMENT & TOOLS	GH-1660ZX	GEARBOX	MOBIL DTE HEAVY MEDIUM
			HEADSTOCK	MOBIL DTE HEAVY MEDIUM
			WAYS, TAIL STOCK, LEAD SCREW,	MOBIL DTE HEAVY MEDIUM
			APRON	MOBIL DTE HEAVY MEDIUM
			CUTTING FLUID	OEM Grade
BENCH DRILL PRESS	JET EQUIPMENT & TOOLS	GHD-20	GREASE POINTS	MOBILGREASE XHP 222
PEDESTAL DRILL PRESS	JET EQUIPMENT & TOOLS	J-2530	HEADSTOCK	MOBIL DTE 10 EXCEL 46 (NOTE C)
			CUTTING FLUID	OEM Grade
			GREASE POINTS	MOBILGREASE XHP 222
BAND SAW	JET EQUIPMENT & TOOLS	HVBS-7MW	GEARBOX	MOBILUBE HD 80W90
			CUTTING FLUID	OEM Grade
			GREASE POINTS	MOBILGREASE XHP 222
HYDRAULIC PIPE BENDER			HYDRAULIC SYSTEM	MOBIL DTE 10 EXCEL 46 (NOTE C)
PIPE THREADING MACHINE			CUTTING FLUID	OEM Grade
11. CARGO HANDLING GEAR & SYSTEM				
Equipment	Make	Model	Application	EXXONMOBIL
WEATHERTIGHT DOORS (6)			HYDRAULIC SYSTEM	MOBIL DTE 10 EXCEL 46 (NOTE C)
			GREASE POINTS	MOBILGREASE XHP 222
			WEATHER DECK VENT GASKETS	SILICON GREASE

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

Scooting Winch			GEARBOX	Mobilgear 600 XP 220
Tagline Winch			GEARBOX	Mobilgear 600 XP 220
12. DECK AND CARGO EQUIPMENT				
Equipment	Make	Model	Application	EXXONMOBIL
No. 1 Forklift	-	-	ENGINE	MOBIL DELVAC SUPER 1300 15W40 /MOBILGARD HSD+ 15W40
	-	-	HYDRAULICS	MOBIL DTE 10 EXCEL 46
	-	-	TRANSMISSION	MOBILFLUID 424
	-	-	DIFFERENTIAL	MOBILLUBE HD 80W90
	-	-	BRAKE	DOT-3 BRAKE FLUID
No. 2 Forklift	-	-	ENGINE	MOBIL DELVAC SUPER 1300 15W40 /MOBILGARD HSD+ 15W40
-	-	-	HYDRAULICS	MOBIL DTE 10 EXCEL 46
-	-	-	TRANSMISSION	MOBILFLUID 424
-	-	-	DIFFERENTIAL	MOBILLUBE HD 80W90
-	-	-	BRAKE	DOT-3 BRAKE FLUID
Manlift	-	-	HYDRAULICS	MOBIL DTE 10 EXCEL 46
	-	-	NO. 1 DRIVE HUB	MOBILLUBE HD 80W90
	-	-	NO. 2 DRIVE HUB	MOBILLUBE HD 80W90
13. CRANES				
Equipment	Make	Model	Application	EXXONMOBIL
HANGER BRIDGE CRANE	IMECO	6 KIP	HOIST GEAR CASE	MOBILFLUID 424
-	-	-	WIRE ROPE	MOBILARMA 798

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

-	-	-	MOTOR SLEEVE AND DRUM BEARINGS	MOBILGREASE XHP 222
-	-	-	BRIDGE AND TROLLEY GEAR MOTORS	MOBIL SHC 630
-	-	-	HOIST MOTOR BRAKE LINKAGE	MOBIL DTE HEAVY MEDIUM
-	-	-	GUIDE ROLLER AND PILLOW BLOCKS	MOBILGREASE XHP 222
ENGINE ROOM BRIDGE CRANE (P/STBD)	C S CONTROLS INC	3912-A000	FORE/AFT TRAVEL GEARBOX	MOBILGEAR 600 XP 220
			PORT/STBD TRAVEL GEARBOX	MOBILGEAR 600 XP 220
			HOIST GEARBOX	MOBILGEAR 600 XP 220
			SLEWING GEARBOX	MOBILGEAR 600 XP 220
			GREASE POINTS	MOBILITH SHC 100
			MOTOR BEARINGS	MOBILITH SHC 100
			OPEN GEARS	MOBILTAC 375 NC
			WIRE ROPE	MOBILARMA 798
FORWARD AND AFT KNUCKLE BOOM CRANE	APPLETON MARINE INC.	AMD-2310	HYDRAULIC MEDIUM	MOBIL DTE 10 EXCEL 46 (NOTE C)
			CABLE WINCH GEARBOX	MOBILGEAR 600 XP 150
			SWING DRIVE ENCLOSED GEARS	MOBILGEAR 600 XP 150
			SWING DRIVE BREAK	MOBIL DTE 10 EXCEL 46 (NOTE C)
			GREASE POINTS	MOBILGREASE XHP 222
			OPEN GEARS	MOBILTAC 375 NC

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

			WIRE ROPE	MOBILARMA 798
PROVISION CRANE (PORT/STBD)	ORIENTAL PRECISION & ENGINEERIN G	HPC140-0517	HYDRAULIC MEDIUM	MOBIL DTE 10 EXCEL 46 (NOTE C)
			GREASE POINTS	MOBILGREASE XHP 222
			WIRE ROPE	MOBILARMA 798
			CABLE WINCH GEARBOX	MOBILGEAR 600 XP 220
			CABLE WINCH BREAK	MOBIL DTE 10 EXCEL 46 (NOTE C)
			SWING DRIVE ENCLOSED GEARS	MOBILGEAR 600 XP 220
			SWING DRIVE BREAK	MOBIL DTE 10 EXCEL 46 (NOTE C)
			SWING DRIVE OPEN GEARS	MOBILTAC 375 NC
MACHINERY ROOM JIB CRANE	CS CONTROLS		GREASE POINTS	MOBIL POLYREX EM
			WIRE ROPE	MOBILARMA 798
			SLEWING AND TELESCOPING REDUCERS	MOBIL SHC 630
			AIR WINCH MOTOR	MOBIL RARUS 827
			AIR WINCH MTORO LUBRICATOR	MOBIL DTE 10 EXCEL 32
JIB CRANE	ORIENTAL PRECISION & ENGINEERIN G	JC0510-2500R5-540	GREASE POINTS	MOBILGREASE XHP 222
BOW CRANE			HYDRAULICS	MOBIL DTE 10 EXCEL 46 (NOTE C)
			SWING DRIVE NO.1	MOBILLUBE HD 80W90
			SWING DRIVE NO.2	MOBILLUBE HD 80W90
			TELESCOPING GEARBOX NO.1	MOBILLUBE HD 80W90

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

			TELESCOPING GEARBOX NO.2	MOBILLUBE HD 80W90
14. EMERGENCY EQUIPMENT				
Portable Fire Pumps			No. 1 FP-100 Engine	MOBIL DELVAC SUPER 1300 15W40 /MOBILGARD HSD+ 15W40
			No. 2 FP-100 Engine	MOBIL DELVAC SUPER 1300 15W40 /MOBILGARD HSD+ 15W40
			GREASE POINTS	MOBILGREASE XHP 222
15. MISCELLANEOUS				
GENERAL OIL LUBRICATIONS				MOBIL DTE 10 EXCEL 46 (NOTE C)
GENERAL GREASE LUBRICATION				MOBILGREASE XHP 222
WIRE ROPES FOR NON UNREP APPLICATIONS				MOBILARMA 798
WHISTLE			CON. ROD BEARINGS	OEM
			GEARS	MOBILTAC 375 NC
TRASH COMPACTOR			HYDRAULIC MEDIUM	MOBIL DTE 10 EXCEL 46 (NOTE C)
			GUIDE RODS, HINGES, LATCHES	MOBILGREASE XHP 222
INCINERATOR			EXHAUST GAS FAN MOTOR BEARINGS	MOBILITH SHC 460
			EXHAUST FAN BEARINGS	MOBIL POLYREX EM
			GREASE POINTS	MOBIL POLYREX EM

Enclosure 2.2.1

USS Hershel Williams Lubrication Chart

QCV ACTIVATION STATION (LO&FO)			HYDRAULIC MEDIUM	MOBIL DTE 10 EXCEL 46 (NOTE C)
HOBART MIXER (GALLEY)	HOBART	HL200 ML-134457	TRANSMISSION GREASE	OEM
			GEAR GREASE	OEM
			SIDEWAY & BOWL CLAMP GREASE	OEM
AIR WINCH (SHAFT ALLEY) MOTOR			ENCLOSED GEARS	MOBIL DTE HEAVY
			OUTBOARD DRUM BEARINGS	MOBILGREASE XHP 222
			WIRE ROPE	MOBILARMA 798
FIRE HOSE/FIRE STATION			THREADS	SILICON GREASE
DUMBWAITER			DOOR LATCH	MOBIL DTE 10 EXCEL 32

SURVEY NOTES (IF APPLICABLE):

A. MOBILGEAR 600XP 100 IF WATER TEMPERATUE IS 68 DEGREES OR LESS, MOBILGEAR 600XP 150 IF WATER TEMPERATURE IS GREATER THAN 68 DEGREES FAHRENHEIT.
B. OEM PRODUCT IN USE. POLYUREA GREASE NOT ON MSC CONTRACT
C. MOBIL DTE 10 EXCEL 46 (NOTE C) IS REPLACING ALL SHEL TELLUS SHELL TELLUS S2 VX 32 BECAUSE ISO 32 HYDRULIC FLUID DOES NT HAVE A FZG RATING OF 12 OR GREATER

MSC Technical Manager	-	-	-	**Contractor Program Manager**
Name: Kyle Beckett			-	Name: Chuck Rockwell
Email: kyle.beckett@navy.mil			-	Email: charles.f.rockwell@exxonmobil.com
Phone: (757) 341-5125				Phone: (908) 335-3764

1.0 ABSTRACT

This item describes the requirements for the contractor to provide a private office facility, equipment, and supplies for use by the overhaul management team (OMT) during the overhaul period.

2.0 REFERENCES: None

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity: Offices shall be land based and located near the ship. Offices shall be provided as described in article 7.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES: None

5.1 This work item is for the MSC OMT only and is not inclusive of FDRMC/SRF or and MILCREW requirements.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 The contractor-provided office facilities for the OMT use shall be separated from those used by contractor personnel. The office and services described herein shall be provided 24 hours/day, seven days/week, commencing three days before arrival of the vessel and terminating three days after vessel redelivery.

7.1.1 Offices shall be land based, air conditioned, heated and located near the ship and preferably other shipyard shops and offices.

7.1.2 The minimum acceptable size of each office workspace layout is 100 square feet, and the minimum size of each private office within the facility is 100 square feet. All office facilities shall be provided with adequate lighting for accomplishment of normal office work. See article 7.2 for office features.

7.1.3 All offices shall be close to one another. If separate facilities must be utilized (such as multiple trailers), the facilities shall be a maximum distance of 40 feet from each other. Easy walking access shall be provided between the separate facilities, with no obstructions or obstacles, such as fences, roads, etc. separating the facilities.

7.1.4 The office facility shall be configured to have a common area where the administrative assistant's desk and computer will be located.

7.1.5 At the primary entrance to the office, the contractor shall supply a touchless, automatic hand sanitation dispenser station. The station shall have a minimum volume of 1000 ml and be refillable. The sanitizer shall be alcohol based with a minimum alcohol concentration of 60%. The contractor shall provide and maintain an extra refill (at least 1000 ml) located at the station throughout the period of performance.

7.2 OMT REP Offices and Workspaces

7.2.1 Provide three private offices and ten individual workspaces in the OMT office facility. One private office shall be designated strictly for the OMT REP's use. The private offices shall be completely enclosed with floor to ceiling walls and shall have lockable doors. The total number of workspaces above includes one workspace for the administrative assistant and one workspace for the translator.

7.2.2 Provide, install, and maintain for the duration of the performance period the following office equipment in each private office and at each workspace:

a. Private Office(s)

1. One each executive desk with central drawer

2. One each swivel chair

3. One each two-drawer legal size file cabinet with lock

4. One each wastebasket

5. One each white board with dry erase markers and eraser, minimum board size 4’ × 3’

6. One each 3-month view planning wall calendar, laminated, with at least 2” × 2” date blocks

b. Workspace(s)

1. One each standard desk

2. One each swivel chair

3. One each wastebasket

7.3 Overhaul management team conference/meeting area: Provide a conference/meeting area within or adjacent to the overhaul management team’s office with the following:

7.3.1 Table large enough to seat 15 people

7.3.2 Chairs for 15 people

7.3.3 Large white board hanging from the wall that can be viewed from the conference table

7.4 Janitorial Services: The contractor shall provide janitorial services to empty the wastebaskets on daily basis, dust and pickup office facilities twice weekly, clean the bathroom daily, restock toilet paper as needed, mop the floor and/or vacuum on a weekly basis.

7.4.1 Frequently touched objects and surfaces within the office shall be disinfected twice per week utilizing damp wipe cleaning method and an EPA approved /registered disinfectant cleaning product. Objects to be cleaned in this manner include, but are not limited to the following:

a. Telephones

b. Computer keyboard and mouse

c. Desk horizontal surfaces

d. Chair arm rests

e. Door/desk/cabinet/window handles, pushbars, etc.

f. Switches (lights, climate controls, etc.)

g. Refrigerator handles

h. Microwave oven panel/handle

i. All washroom surfaces

j. Copy machine/printer surfaces

7.5 Office equipment and miscellaneous supplies: The contractor shall provide, install, and maintain the following office equipment/supplies for the duration of the performance period:

7.5.1 One each automatic drip coffee maker

7.5.2 One each refrigerator, minimum of 18 cubic feet capacity

7.5.3 One each microwave oven

7.5.4 Four each additional side chairs

7.5.5 Three each coat racks

7.5.6 Provide a source of chilled drinking water, meeting state and local health standards (lavatory sinks are not an acceptable source of drinking water)

7.5.7 Adjustable automatic climate control system sufficient to maintain 75 degrees Fahrenheit in office facilities

7.5.8 Three each wall clocks

7.5.9 Two each high capacity cross-cut office shredders with shred size ⅛ inch or smaller

7.5.10 Private washrooms (one designated as male and one designated as female) located within each office facility, including sink, mirror, and toilet

7.5.11 Provide soap, toilet paper, paper towels, and hand-sanitizer on a daily basis

7.5.12 Provide locks with 3 sets of keys for each access door leading into the main entrance of the office/common area. Provide three sets of keys to each individual private office.

7.6 Office supplies: The contractor shall provide the following office supplies for exclusive use by USS Hershel Woody Williams OMT REP for the duration of the performance period:

7.6.1 Paper for the copy machine and printers

a. 4 boxes of 8-½ × 11

b. 1 boxes for 8-½ × 14

c. 1 boxes for 11 × 17

7.6.2 (4) boxes each of blue/black/red pens, assorted markers, colored highlighters, and mechanical pencils.

7.6.3 (2) dozen note pads (8-½ × 11 size), steno notebooks and Post-it pads

7.6.4 (6) each staplers and staples

7.6.5 Two each three-hole punches

7.6.6 0 boxes hanging folders for the file cabinets

7.6.7 3 each - 3” 3-ring binders

7.6.8 3 each - 1” 3-ring binders

7.6.9 6 each scissors

7.6.10 6 each staple remover

7.6.11 12 each black sharpie pens

7.6.12 25 each CD-R’s

7.6.13 25 each DVD-R's

7.6.14 24 each all-weather notepad, 3 in. x 5 in. with polydura cover, top spiral bound on the 3 in. side

7.7 Safety equipment retained by the MSC: The contractor shall provide the following safety equipment:

7.7.1 12 each hard hats

7.7.2 12 each ULINE Heavy Duty gloves S-15355, size large, or similar.

7.7.3 12 each safety eye shields or glasses.

7.7.4 200 pack of EARsoft yellow neon corded earplugs or equivalent – NRR 33 dB.

7.7.5 Five each Pelican Super Sabrelite or other pocket-size intrinsically safe, UL rated, Class 1, Division 1 LED flashlights, with five sets of alkaline batteries or two sets of rechargeable batteries and one charger, for each flashlight. Flashlights to be MSHA approved.

7.7.6 (2) boxes of NIOSH approved N95 particulate respirators (20 per box)

7.7.7 Five each inspection mirrors

7.7.8 (6) each raincoats w/ pants, size (XL)

7.7.9 (12) each disposable Tyvek suits

7.7.10 (2) boxes of disposable latex gloves, minimum 6-mil (100 per box) size (XL)

7.7.11 (2) boxes of disposable nitrile gloves, minimum 6-mil (100 per box) size (XL)

7.7.12 4 Disinfectant wipes containers of at least 75 wipes in each

7.7.13 Four Honeywell BW™ MicroClip X3 (4-gas detection meter configuration) portable gas meters or US OSHA approved portable gas meters for Port Engineer/OMT use.

a. Meters are to be provided with charging stations, calibrated for use for the duration of the availability, calibration and in-service dates clearly defined and marked on the meters, test gas for bump test, and any applicable operation/owner’s manuals.

b. Manufacturer’s website:

https://sps.honeywell.com/us/en/products/safety/gas-and-flame- detection/portables/honeywell-bw-microclip-series

c. Custody of meters will be transferred back to contractor post availability for future use by MSC ships.

7.8 Mini specs retained by the MSC

7.8.1 Provide (15) each, bound mini specifications, approximate size of (4” × 7”), bounded left to right on the long side (book fashion) of the MSC work package; provide all copies to the Port Engineer. Work item specification format shall not be altered.

7.9 Mail delivery: Provide Federal Express or equal mail services to pick up and deliver overnight express mail to and from the OMT REP. Estimate twice a week FedEx Envelope and six FedEx 10kg boxes of documents at the end of the availability.

7.10 Parking spaces for project office personnel

7.10.1 Provide minimum of (12) independently accessible parking spaces adjacent, (within 300 feet) to the OMT REP office.

7.10.2 Mark the parking spaces with weather resistant signs that reads:

RESERVED FOR USNS “HERSHEL WILLIAMS” PROJECT OFFICE PERSONNEL ONLY

7.11 Foreign Language Interpretation and Translation Services:

7.11.1 The contractor shall provide a full-time linguist, eight hours per day, Monday – Friday from a temporary employment agency. The linguist shall be supplied with a workspace and office supplies per this work item.

7.11.2 One week before the start of the availability, the contractor shall provide a minimum of three resumes for candidates to the OMT REP. While the OMT REP may indicate a preference, the hiring decision will be made by the contractor.

7.11.3 The linguist shall:

a. Fluently read, write, and speak the English language and the local language to include both interpretation and translation.

b. Be able to interpret and translate effectively, accurately, and impartially, both receptively and expressly, to and from such language(s) and English, using any necessary specialized vocabulary, terminology and phraseology.

7.11.4 The linguist shall not work directly for the contractor performing office duties, but the contractor is required to exercise direct administrative control over the provider. The contractor shall coordinate such matters as requested leave with the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0011	**CATEGORY "A"**	**2026-01-30**
ESB-SWI-GENERAL SERVICES FOR SHIP		**Manovich, David**

1.0 ABSTRACT

This item describes the requirements for the contractor to provide general services to the ship for duration of the overhaul period.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSEA Drawing No. General Arrangement, 801-8499371

2.1.2 MSC Chemical Treatment Handbook

2.2 Enclosures

2.2.1 Heat Lamp List: Motors

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: See Reference 2.1.1 and specification article 7.0.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with all applicable GTR requirements.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract to determine their effect on the work required under this work item.

5.3 The definitions of many terms used in this work item are found in Work Item 001.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Furnish the services specified in this work item, from the date of the ship’s arrival at the Contractor's facility until the date of final departure from the Contractor's facility:

7.1.1 Electrical Shore Power

a. The contractor shall provide a reliable source of electrical power at a minimum capacity of 6000 amps at 505 Volts. Voltage is to be 505 Volts under no load conditions at the shore power cable ends to be connected to the ship's shore power receptacle. Loaded Line to line voltage under the ship’s normal electrical load at the ship’s service switchboard shall be maintained at 485 +/- 5% Volts, 60 Hz +/- 3%. Line voltage imbalance shall not exceed 2%. Shore power substations shall be enclosed in a vault or designed to be weather-proof. Substation shall have a primary isolation switch. Substation secondary shall be a three-phase ungrounded system and shall have 400 amp circuit breakers for each shore power connection.

b. 15, 400 amp shore power cables are required to be furnished by the contractor, to connect the pier side substation secondary to the ship’s shore power connection. All electric cables utilized shall be in good material condition, free of tears, cracks or poorly repaired insulation. Cables shall be sized such that the voltage drop between the substation secondary and the ship’s shore power connection does not exceed 4% at full load. All three phases shall be carried in each cable. All conductors of each cable shall be of equal cross-sectional area. Cable lengths shall be of equal length +/- 10%

of the average length from all shore power vaults to the ship’s shore power connections. Cables shall be routed such that there is sufficient slack to allow for natural movement of the ship with tide and weather, but length shall not be excessive such that the cables be allowed to fall into the water or become trapped between the ship’s hull and the pier. Cables shall avoid sharp edges and tight bends, and proper strain relief shall be provided for vertical runs. Cables shall be protected from pedestrian and vehicular traffic. In line connectors and splices shall be properly insulated, sealed and elevated such that they will not be exposed to standing water. Subsequent to installation, but prior to energizing, resistance tests (500 V, 1 megohm minimum corrected to 70 degrees F) shall be performed between conductors, and between each conductor and ground for all cables. Each cable shall be checked for proper phase rotation.

c. Contractor shall be responsible to connect and disconnect as required upon ship arrival, in-yard shifts, dock and sea trial events, and for ship’s departure. Ship’s shore power connection boxes are located per Reference 2.1.1. Contractor shall perform walkthrough with the OMT REP and the vessel’s Chief Engineer upon arrival to lay out an approved shore power cable routing.

d. Electrical Shore Power Pricing: Provide a calibrated in-line electrical power meter with which to measure the vessel’s shore power consumption. The meter shall only reflect power usage by the ship. The meter shall be installed in the immediate vicinity of the ship’s shore power receiving station and shall be readily accessible to the OMT REP and the vessel’s Chief Engineer. An initial meter reading shall be witnessed by the OMT REP and the Chief Engineer. Meter readings shall be recorded and presented to the OMT REP on a weekly basis. At the end of the availability, the final meter reading shall be recorded and witnessed by the OMT REP and the Chief Engineer. A report shall be prepared and submitted with total calculated electrical power consumption. The Contractor will only be paid for actual electrical power consumed. In the event that a generator is used, all generator fuel, upkeep, rental and maintenance shall be included in the shore power pricing per KWH. The contractor will base its offer on providing 37,000 KWH per day for 45 days, 1,665,000 KWH total.

e. In addition, contractor shall bid a unit price per kWh for adjustments to the award price for lesser or greater total consumption. Any required adjustments shall be made via a contract change order.

f. The shore power connections are located on the on the top Center Line approximately frame 23. The connections are located approximately 41 m ABL. The contractor shall provide crane services to lift shore power cables to and from the ship's connections.

g. There are 2 shore power stations. The port station is located at 1-74-2 and the starboard station is located at 1-74-1.

h. In the event of a power failure from the contractor supplied shore power, a backup generator shall be furnished that is capable of supplying all emergency loads as well as ships accommodation HVAC and chiller and freezer room units. The generator shall be onsite and supplying within 8 hours of the power failure event. The generator shall be supported on a 24 hour basis until shore power is restored and stabilized. Exceptions to this requirement may be approved by onsite Port Engineer. Contractor shall provide pricing for mobilization/demobilization of the generator as well as a per day cost for use.

7.1.2 Potable Water

a. Provide potable water, 60 psig at approximately 100 GPM. Provide and install all valves, fittings and hoses required to connect to the ship’s shore potable water connection. The contractor shall provide a regulating or relief valve on the water source to prevent water pressure from exceeding 75 psig during periods of low consumption. All equipment and installations used shall meet United States Public Health Service or equivalent international public health service standards for drinking water intended for human consumption. It is strictly prohibited to connect anything other than potable water to the potable water manifold. The potable water connection shall be fully dedicated and direct connection to a potable water source. No split lines or additional headers allowed. Connect and disconnect potable water upon ship arrival, at each in-yard shift, for tests and trials and for ship’s departure. Potable water connections are located per Reference 2.1.1.

b. Potable Water Pricing: Provide a calibrated in-line water meter with which to measure the vessel’s shore potable water consumption. The meter shall be installed in the immediate vicinity of the ship’s shore potable water receiving station(s) and shall be readily accessible to the OMT REP and the vessel’s Chief Engineer. An initial meter reading shall be witnessed by the OMT REP and the Chief Engineer. Meter readings in gallons shall be recorded and presented to the OMT REP on a weekly and cumulative basis. At the end of the availability, the final meter reading shall be recorded and witnessed by the OMT REP and the Chief Engineer. A report shall be prepared and submitted with total calculated potable water consumption. The Contractor will only be paid for actual potable water consumed. The contractor shall provide a bid price based on an estimated potable water consumption of 400,000 gallons. In addition, contractor shall bid a unit price per gallon for adjustments to the award price for lesser or greater total consumption. Any required adjustments shall be made via a contract change order.

c. The potable water connections are located on the Main Deck approximately Frame 58.

1. Port valve PWC-V-027

2. Stbd valve PWC-V-028

3. Valve designation shall be confirmed by the Chief Engineer prior to connection.

d. Once the ships potable water system is secured the KTR shall provide a source of chilled drinking water that meets or exceeds the United States Public Health Service or equivalent international public health service standards for drinking water intended for human consumption. Dispenser shall be a bubbler type dispenser (UL listed) that can accept the ships electrical power receptacles (110v). Provide disposable paper cups and trash receptacle. Dispenser will be placed at a location designated by the MSCREP. Replacement/refill jugs to be provided weekly and delivered to the dispenser location. Provide a sign in close proximity to the dispenser designating use “only for ship’s personnel” in both English and the local language as required.

7.1.3 De-ionized Water

a. Provide de-ionized water for vessel MPDE and SSDG engine cooling. This water shall meet the requirements in Reference 2.1.1. Provide and install all valves, fittings and hoses required to connect and fill the ship’s MPDE and SSDG cooling systems. If a pierside demineralizer is used, a Y type strainer with 100 mesh shall be fit to prevent resin contamination of the ship’s system.

b. De-ionized Water Pricing: Provide a calibrated in-line water meter with which to measure the amount of de-ionized water provided to the vessel. The meter shall be installed in the immediate vicinity of the tank fill line and shall be readily accessible to the OMT REP and the vessel's Chief Engineer. An initial meter reading shall be witnessed by the OMT REP and the Chief Engineer. Upon completion of filling, the final meter reading shall be recorded and witnessed by the OMT REP and the Chief Engineer. A weekly and cumulative report shall be prepared and submitted with total calculated de-ionized water in gallons provided. The Contractor will only be paid for actual de-ionized water received. The contractor shall provide a bid price based on an estimated de-ionized water consumption of 50,000 gallons. In addition, contractor shall bid a unit price per gallon for adjustments to the award price for lesser or greater total consumption. Any required adjustments above or below this total volume will be made via contract change order using the prorated unit price per gallon based on the contractor's initial bid price.

7.1.4 Heat Exchanger Saltwater Cooling

a. Provide saltwater cooling water supply and return for air conditioning and ship's reefer machinery. Ship's air conditioning condenser shall be supplied with, 100 psig at approximately 600 GPM. Ship's refrigeration machinery shall be supplied with, 100 psig at approximately 600 GPM. Provide and install all valves, fittings and hoses required to connect to the ship's saltwater cooling connection(s). Fittings shall include relief valves to protect vessel systems at indicated pressure settings. Connect and disconnect saltwater cooling lines as required upon ship arrival, at each in-yard shift, for tests and trials and for ship's departure.

7.1.5 Compressed Air

a. Provide 125 psig compressed air at 300 CFM at each of three (3) manifolds for ship's force use. Manifold locations shall be designated by the OMT REP. Air shall be clean, dry, and oil free. Provide all necessary hose and fittings and make all hook ups. Connect and disconnect compressed air lines upon ship arrival, at each in-yard shift, for tests and trials and for ship’s departure.

b. Provide 125 psig compressed ships service air at a minimum capacity of 300 CFM. Air shall be clean, dry, and oil free. The air line shall be fit with an air dehydrator, air filter and oil separator in the vicinity of the connection to the ship’s compressed air system. Provide necessary hose, fittings, and make all hook ups. Contact Chief Engineer prior to hook up. Connect and disconnect compressed air lines upon ship arrival, at each in-yard shift, for tests and trials and for ship’s departure.

7.1.6 Fire Protection

a. Ship’s fire protection requirements are provided under Work Item 016 of the work package.

7.1.7 Temporary Lighting and Ventilation

a. Provide and maintain adequate temporary lighting and ventilation for the performance of work required by this specification in all areas where ship's lighting and/or ventilation is disrupted or inadequate.

7.1.8 Temporary Space Cooling

a. Provide temporary space cooling by means of portable air conditioning units to maintain an ambient temperature of 75 degrees F for the entire performance period for the following locations:

1. Helicopter Control, A8-87-0
2. Communications Ctr, A8-94-0
3. Intel Ctr, A8-93-2
4. Tactical Ops Ctr, A8-91-0
5. Training Room, A8-89-0
6. Communications Module, E-48-0
7. Port(2-56-2)and Starboard (2-56-1) ECR
8. Aft House Accommodation Spaces and Offices
9. Fwd House Accommodation Spaces and Offices
10. Aft Crew Mess A-45-3
11. Bridge E-52-4
12. Aft SVCE Trunk (server room) 1-52-3

7.1.9 Sewage Disposal

a. Provide all pumping arrangements, hoses, connections and shore-side disposal facilities necessary to remove and dispose of sewage from the ship for the duration of the performance period, in accordance with all federal, state and local (CONUS) or international (OCONUS) requirements. Ensure all equipment is in good material condition and that all connections are 100% leak-free. Connect and disconnect sewage upon ship arrival, at each in-yard shift, for tests and trials and for vessel departure.

b. Approximately 3,500 gallons of Sewage effluent are expected to be generated per day when reduced crew is berthing onboard the vessel.

7.1.10 Heat Lamps

a. Provide, install, and maintain infrared heating lamps/strips on all motors, generators, and exciters, listed on enclosure 2.2.1 and additional to quantity of 73 EA. Lamps shall be installed immediately upon plant shutdown. Heat lamps shall be separately powered from a pierside power source. The ship's electrical systems shall not be used to power heat lamps.

7.1.11 Debris and Garbage Removal

a. Clean the entire ship on a daily basis, Monday – Saturday, except federal holidays. All decks, passageways, and work areas shall be swept clean and debris free on a daily basis. Clean and disinfect frequently touched objects and surfaces using regular household cleaning spray and wipe. All paint containers, chips, shavings, dirt, paper, boxes, and other work item debris shall be collected and placed in metal containers which shall be removed daily from the ship and disposed of in accordance with state and local regulations (CONUS) and international and local regulations (OCONUS). Oil, refuse, hazardous wastes, and scrap materials shall also be removed on a daily basis and disposed of in a proper manner and in accordance with federal, state and local laws and regulations (CONUS) and international requirements (OCONUS). Hazardous wastes shall be removed and disposed of in accordance with specification Work Item No. 023. No debris or garbage shall be left onboard the vessel overnight.

b. All garbage that has been generated by the ship's crew and shipyard personnel shall be collected and removed on a daily basis. The contractor shall provide metal containers at appropriate locations for garbage deposit. Minimum ¾" thick plywood deck protection shall be placed under all garbage containers. Garbage containers shall be maintained in a sanitary condition, free of insects and rodents.

7.1.12 Interior Bulkhead and Deck Protection

a. Install 1/8” minimum thickness plywood, pressed fiberboard, herculite, or suitable covering acceptable to the OMT REP, on all interior decks, bulkheads, passageways, stair towers, entrance to interior decks, bulkheads, passageways, stair towers, and other areas that will be traversed by the contractor. The protective covering shall be securely fastened to all decks and bulkheads and renewed as necessary, or at the direction of the OMT REP. Just prior to redelivery, the contractor shall remove all protective coverings and securing materials and clean all decks, bulkheads, passageways, and stair towers to a finished, ready to inhabit condition, to the satisfaction of the OMT REP.

7.1.13 Helicopter Deck Protection

a. Contractor shall NOT utilize Flight Deck or Aircraft Parking Area for lay-down, garbage containers or staging areas without written consent from the OMT REP. As authorized, contractor shall temporarily install adequate means to protect Flight Deck or Aircraft Parking Area non-skid surfaces. Blocking, plywood, and HERCULITE lay down materials shall be utilized wherever contractor lands materials and/or equipment on the Flight Deck or Aircraft Parking Area. Any damage to Flight Deck or Aircraft Parking Area surfaces shall be repaired, restored, and certified (per NAVAIR regulations) to like new condition, at contractor’s expense.

7.1.14 Toilet and Sanitary Facilities

a. Shipyard personnel shall not use shipboard toilet and sanitary facilities. The contractor shall provide and install signs to this effect. In cases where the ship’s sanitary system is not active, the contractor is to provide and maintain a minimum of four portable toilet and sanitary facilities for the shipboard personnel aboard the vessel in locations designated by the ship’s Master. One portable toilet shall be designated for female use only. Adjacent to or integral to each portable toilet a freestanding hand washing, and sanitation station shall be provided. Free standing hand wash and sanitation station shall include hot/cold water up to 100-degree F., liquid soap, hand sanitizer, paper towels, and a trash bin for disposal. Drains for the hand sanitation station shall be piped off the vessel or in a collection tank and emptied on a daily basis. Hand sanitizer shall be alcohol-based with at least 60% alcohol. The contractor shall also provide daily janitorial services to maintain cleanliness and toilet supplies. This includes the cleaning and disinfecting of frequently touched objects and surfaces using a regular household cleaning spray or wipe. Any contamination of the shipboard facilities or MSD system by the contractor shall be cleaned and restored at the contractor's expense.

1. Portable toilet facilities shall be locked with the keys delivered to the gangway watch.

7.1.15 Oil Containment Boom

a. Upon arrival of the vessel at the contractor’s facility, an oil boom shall be rigged to completely encircle the vessel and shall be maintained in good material condition throughout the entire availability. Connect and disconnect oil booms upon ship arrival, at each in-yard shift, for tests and trials and for vessel departure. In the event the availability includes a drydocking evolution, the oil boom shall be rigged around the floating drydock or the graving dock gate during the time the vessel is on drydock.

7.1.16 Gangways

a. Provide and install two safe and proper gangways for access to the ship. One gangway shall be used for routine access to the vessel. Other gangway(s) shall be designated for emergency use only. Gangway locations shall be to the satisfaction of the OMT REP and ship's Master with the gangways separated a distance greater than 1/4 distance of the vessel length.

b. Gangways shall meet the requirements of OSHA Standard 1915.74, Access to Vessels (CONUS) or applicable international occupational safety and standards (OCONUS). Gangways shall be a minimum of 20 inches wide. Gangways shall be adequately secured against shifting, shall be illuminated, and shall be properly trimmed.

c. Each side of the gangway, and turntable if used, shall have a railing with a minimum height of 33 inches, and shall be fit with a mid-rail. Rails shall be wood, pipe, chain, wire or rope and shall be kept taut at all times. The gangway shall be properly trimmed at all times.

d. Obstructions shall not be laid on or across the gangway. Means of access shall be adequately illuminated for its full length. Safety nets shall be provided and secured to the ship and to the pier edge to provide adequate coverage. The net shall extend 6 feet on all sides of the gangway.

e. The emergency use gangway shall have signs posted at each end indicating its use for emergencies only.

f. Adjacent the main gangway a freestanding hand washing, and sanitation station shall be provided. Free standing hand wash and sanitation station shall include hot/cold water up to 100-degree F, liquid soap, hand sanitizer, paper towels, and a trash bin for disposal. Drains for the hand sanitation station shall be piped off the vessel or in a collection tank and emptied on a daily basis. Hand sanitizer shall be alcohol-based with at least 60% alcohol.

7.1.17 Bilge Water Removal

a. Contractor shall provide all pumps, hoses, fittings, and personnel to safely remove all bilge water from the spaces noted in Section 3.0 on a continuous basis, such that the bilges are maintained in a dry state. Locations of spaces are indicated in Reference 2.1.1. Applicable federal, state, local or international procedures for transferring oily wastes over water shall be followed by the contractor. Ensure all equipment is in good material condition and that all connections are 100% leak-free. Connect and disconnect bilge pumping arrangements upon ship arrival, at each in-yard shift, for tests and trials and for vessel departure.

b. Provide a calibrated in-line water meter with which to measure the amount of bilge water removed from the vessel. The meter shall be installed in a location readily accessible to the OMT REP and the vessel's Chief Engineer. An initial meter reading shall be witnessed by the OMT REP and the Chief Engineer. Weekly readings shall be provided to the OMT REP via a condition report. A weekly and final report shall be prepared and submitted with weekly and cumulative readings in gallons calculated. The Contractor will only be paid for the actual quantity of bilge water removed from the vessel.

c. The contractor shall provide a bid price for bilge water disposal based on an estimated quantity of 50,000 gallons. In addition, contractor shall bid a unit price per gallon for adjustments to the award price for lesser or greater total gallons removed. The actual quantity of bilge water disposal shall not be greater than the actual quantity of bilge water removed from the vessel. Any required adjustments above or below this total volume will be made via contract change order using the prorated unit price per gallon based on the contractor’s initial bid price.

7.1.18 Fuel and Oil Transfer

a. All internal fuel and oil transfers aboard the ship shall be conducted by the ship’s force. All transfers to or from the ship will be a coordinated effort with the contractor and the ship’s force and shall be in accordance with all USCG and international regulations concerning fuel transfers. The ship’s fuel transfer systems shall be used to off load or receive fuel via the designated deck riser.

b. In transferring the fuel and other oils, the contractor shall comply with U. S. Coast Guard regulations 33 CFR section 156.100 and all applicable local regulations and requirements. The contractor shall provide all labor, equipment and materials necessary to safely accomplish the transfer of fuel and other oils to and from the vessel.

c. In the event that hot work is required adjacent to a fuel tank or if a fuel tank is required to be temporarily emptied allow work in this package to be completed, the contractor will assist the crew in either transferring or offloading and loading such fuel and provide temporary storage for the time period required.

7.1.19 Compartment Access and Security

a. All compartments shall remain locked unless the contractor is required to accomplish daily work.

b. Ship’s force shall be notified in advance of required entry into any particular compartment.

c. The contractor shall permit free movement of properly identified employees and materials of the government to all points of the ship at all times during the contract period. The contractor shall also permit free movement of properly identified government employees and subcontractors through the yard and to the ship. The contractor shall allow private contractors, hired by the government to perform certain installation work aboard the ship. The Contractor shall grant passage through its facility to the ship to all properly identified technicians or other workmen hired by the government. A driver’s license or similar ID and a copy of a government work order or purchase order shall constitute proper identification.

d. Contractor and subcontractor personnel shall not be permitted to eat in the ship’s mess. Ship’s mess decks are off limits to contractor and subcontractor personnel.

e. Contractor and subcontractor personnel shall not use the ship’s 1MC or telephone systems. The contractor shall provide a separate, independent means of communications for contractor and subcontractor personnel.

f. Contractor shall not use ship’s cargo elevators.

g. The contractor shall take appropriate measures to ensure that all contractor and subcontractor personnel comply with all posted notices onboard.

7.1.20 Cold Weather Protection

a. In the event that this ship availability is accomplished at a time and location susceptible to cold weather conditions, contractor shall be responsible to protect the vessel from the effects of these cold weather conditions.

b. Contractor shall adhere to the Cold Weather Plan provided in accordance with work item 007 of this work package, to provide adequate protection to the ship, and to provide a safe working environment for ship’s crew, OMT staff and contractor personnel.

7.1.21 Care of the Ship

a. Provide a safe berth for the vessel at a pier where the ship will be afloat with a minimum of 3 feet of water under the keel/lowest projection at all times and during all tides, as specified in work item 003 of this work package. The ship shall be securely moored to the pier with the contractor’s mooring lines, sufficient camels and/or fenders between the ship and pier to prevent any structural and coating system damage. Mooring facilities and locations of protective camels and/or fenders shall be presented for OMT REP approval per work item 003 prior to vessel’s arrival at the contractor’s facility.

b. Upon arrival of the vessel at the contractor's facility, conduct a joint survey with OMT REP to ascertain the operational status of all equipment and systems onboard the vessel, and the cleanliness of all spaces onboard the vessel. The ship's crew will be made available for up to 72 hours after arrival to assist in this survey.

1. Submit an as found condition report of the arrival survey to the OMT REP within five days after arrival of the vessel at the contractor's facility.

c. The contractor shall be responsible for the entire ship from the time of delivery to the contractor until the OMT REP accepts it for redelivery. All ship's parts, equipment, systems, etc., damaged or destroyed by the contractor shall be repaired or replaced, as original, at contractor's expense. Such consideration includes, but is not limited to, piping, electric cables, machinery, insulated/refrigerated storerooms, compartments, bulkheads, decks, and ducting damaged by neglect, weather, personnel, dirt, foreign substances, etc., or lost through theft.

d. All paint and flammable liquids that are temporarily stored onboard the ship by the contractor shall be stored in a suitable container designed in accordance with international, federal, state, and local regulations for flammable liquid storage.

e. Rigid control of welding and grounding shall be maintained for the protection of the hull, and hull appendages. Care shall be taken to ensure that welding polarity and ground connections of welding machines used aboard the ship or other ships in the immediate vicinity, on the pier, or on the wharf to which the ship is moored, shall not damage any part of the ship.

1. LOCATION OF GROUND RETURN CABLES. When systems such as piping, pressure vessels, or machinery are being welded, the ground return cable connection should be located as close to the work as possible. This ensures that welding current does not flow through bearings, threaded joints, and other areas where arcing could occur. Ground return cable connection should be no farther than 10 feet from work.

2. Proper setup will reduce potential corrosion due to the welding current. Underwater hull and shaft corrosion is, in large part, directly attributable to improper hookup of welding leads while work is being performed on ships which are waterborne. Corrosion resulting from improper weld lead hookup is induced through electrolytic action by stray electrical currents.

f. The contractor shall adequately protect the underwater portion of the hull from the time of delivery until the OMT REP accepts the ship for redelivery. If, at any time prior to redelivery, circumstances exist for believing that the underwater portion of the hull has been damaged or impaired, the contractor shall make arrangements at its expense for an underwater survey. If damage has in fact occurred, the contractor shall place the ship in drydock and completely repair, clean, and paint the damaged areas.

g. The contractor shall survey all deck drains in way of work areas upon vessel arrival and submit a condition report on whether subject drains are free and clear or clogged. Drains found clogged subsequent to this initial survey shall be kept clear and in proper working condition, unless previously identified as being clogged.

h. The contractor shall maintain all machinery space bilges in a clean and dry condition for the entire shipyard availability period.

7.1.22 Stack Cover

a. Template from the smokepipes, provide and install temporary stack covers. Covers shall be fit and installed over each individual stack exhaust. Remove covers prior to any engine operation.

1. Each individual stack exhaust

7.1.23 External Vessel Contamination Protection

a. Prevent contamination of ship’s ladder ways, decks, deck coverings, equipment, materials, furnishings, and spaces at all times throughout the performance period of this contract. This shall include, but not be limited to operations during abrasive and/or hydro blasting, vacuum blasting, scaling, surface cleaning, sanding, painting, welding, grinding, pumping/transferring of water/oils/liquids, and any other source of contamination by work accomplished in this specification package. Contractor shall be responsible for protecting against contamination from neighboring and/or adjacent facilities and/or other work projects within contractor facility, that are capable of emitting contaminants onto any/all areas of the vessel.

b. Accomplish below specified requirements as found required in support of this work specification package and as found required to protect vessel from outside sources of contamination.

1. Temporarily install protective coverings on the deck, deck equipment, machinery, sheaves, wire ropes,

cranes, boats, stern ramp spectra rope, and any other materials and/or equipment in way of possible contamination. Contractor shall use heavy duty canvas, then cover the canvas covering with heavy duty PVC sheets. Protection and covering material shall be fastened securely to prevent protective coverings from being removed.

2. Temporarily plug, blank, wrap, cover or mask openings and portlights to prevent entry of contaminants into machinery, equipment, systems, electronic equipment, valves, vents, and other openings, when not in use.

3. Temporarily install new industrial foam filter material on the intake and exhaust end of the ventilation systems. Renew filtering material when air flow becomes restricted on exhaust and intake of ventilation systems. Change out and renew filter material when required, for the entire contract performance period.

c. Removal of any protective measures specified above shall not be accomplished without permission of the OMT REP. Where removal of preventative measures specified above become necessary to accomplish other related work items, the contractor shall remain responsible for ensuring no contamination of the areas are involved, and preventive measures are restored.

d. Once approved by the OMT REP the contractor shall remove protective measures and sweep, clean, and washdown area, and equipment impacted.

7.1.24 Final Vessel Cleaning

a. Four (4) days prior to the crew move-aboard milestone if applicable, or approximately 9 days prior to contract performance period end date, the contractor shall accomplish the following cleaning requirements onboard the vessel.

1. All temporary deck protective covering shall

be removed. All tiled decks shall be swept and mopped.

2. All bulkheads and overheads which are soiled shall be washed with soap and water/cleaning solvent and wiped dry.

3. All exposed wireways and pipe shall be vacuumed free of all loose dust and grit. All horizontal surfaces shall be examined and likewise vacuumed of loose dust and grit.

4. All galley(s), pantry(s), scullery(s) and other food preparation, serving, messing and food consumption spaces and equipment shall be cleaned and sanitized.

5. Upon completion of all work within each machinery space the contractor shall conduct a close out inspection with the OMT REP. As acceptance occurs the contractor shall perform a final bilge cleaning. The contractor shall perform the final bilge cleaning using an environmentally safe biodegradable bilge cleaning product and a high pressure water wash down. All bilge drainage limber holes and snipes, and all bilge wells shall have all trash, garbage, dirt and industrial debris therein removed. All bilge drain well covers shall be left open with the well pumped dry until accepted as clean. All fuel and lube oils, grease, or petroleum based materials, etc. shall be cleaned and removed from the bilges and other surfaces.

7.1.25 Crane, Forklift and Transportation Services

a. Provide crane, rigging, forklift and transportation service at the request of the OMT REP to accomplish vessel material and supplies movement. Estimate 300 crane lifts, inclusive of all associated man-hours and equipment to support these lifts, to accomplish vessel material and supplies movement.

1. These lifts shall be in addition to any

crane service required by other items in this work package.

2. Contractor shall establish a chit system to track usage of these lifts. For each crane lift, a usage chit shall be prepared documenting the lift event. The chit shall be signed by the OMT REP. Contractor shall maintain a running log of lifts used, and shall provide a weekly summary to the OMT REP. The Contractor will only be paid for actual lifts used. Provide a unit price per lift for adjustments to the award price for lesser or greater total usage. Any required adjustments shall be made via a contract change order.

b. Provide transportation (on an as needed basis) for the OMT REP and the Chief Engineer for inspection visits to subcontractor or vendors facilities outside the contractor’s facility.

7.1.26 Megger Readings

a. Within the first ten days after delivery, the contractor shall take and record megger readings of all motors listed in enclosure 2.2.1 and supply a Conditions Found Report of the readings to the OMT REP. Any low megger readings shall be reported to the OMT REP immediately so corrective actions can be taken.

b. Twenty days prior to redelivery, the contractor shall re-take and record megger readings of all motors listed in enclosure 2.2.1 and supply a Conditions Found Report of the readings to the OMT REP. Any low megger readings shall be reported to the OMT REP immediately so corrective actions can be taken.

7.1.27 Provide 10 parking spaces for ship’s officers and official guests. All parking spaces shall be adjacent to the vessel and properly marked as USNS “HERSHEL WILLIAMS” OFFICER'S PARKING ONLY.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - HEAT LAMP LIST: MOTORS

MOTOR	MEL	MFR	DE BRG	ODE BRG	TYPE	SIZE	FRAME	RPM
AFT STEERING MOTORS PORT							2-16-2, Steering Gear Room	
AFT STEERING MOTORS PORT							2-16-2, Steering Gear Room	
AFT STEERING MOTORS STBD							2-16-1, Steering Gear Room	
AFT STEERING MOTORS STBD							2-16-1, Steering Gear Room	
MDG PRE-LUBE PUMP 2S							MMR 1, 3rd Deck	
LO XFER PUMP STBD							MMR 1, 3rd Deck	
MDG PRE-LUBE PUMP 1S							MMR 1, 3rd Deck	
SUPPLY PUMP 1S LO PUR MOTOR							MMR 1, 3rd Deck	
SUPPLY PUMP 2S LO PUR MOTOR							MMR 1, 3rd Deck	
LO XFR PUMP PORT							MMR 2, 3rd Deck	
MDG PRE-LUBE PUMP 1P							MMR 2, 3rd Deck	
MDG PRE-LUBE PUMP 2P							MMR 2, 3rd Deck	
SUPPLY PUMP LO PUR 1P							MMR 2, 3rd Deck	

SUPPLY PUMP LO PUR 2P							MMR 2, 3rd Deck	
LUBE OIL PUR 1P							Purifier Room, 3-56-2	
LUBE OIL PUR 2P							Purifier Room, 3-56-2	
LUBE OIL PUR 3P							Purifier Room, 3-56-2	
LUBE OIL PUR 4P							Purifier Room, 3-56-2	
F.O. PUR #1							Purifier Room, 3-56-2	
F.O. PUR#2							Purifier Room, 3-56-2	
F.O. SERVICE PUMP 1P							FO Service Room, 3-56-4	
F.O. SERVICE PUMP 2P							FO Service Room, 3-56-4	
HT FW COOLING PUMP 1P							MMR 2, 4th Deck	
HT FW COOLING PUMP 2P							MMR 2, 4th Deck	
HT FW COOLING PUMP 2S							MMR 1, 4th Deck	
HT FW							MMR 1,	

COOLING PUMP 1S							4th Deck	
CTL LT FW COOLING PUMP 2S- 2 STRIPS							MMR 1, 4th Deck	
CTL LT FW COOLING PUMP 2S- 2 STRIPS							MMR 1, 4th Deck	
CTL LT FW COOLING PUMP 1S-2 STRIPS							MMR 1, 4th Deck	
CTL LT FW COOLING PUMP 1S-2 STRIPS							MMR 1, 4th Deck	
MAIN FUEL XFR PUMP							MMR 1, 4th Deck	
CTL LT FW COOLING PUMP 2P- 2 STRIPS							MMR 2, 4th Deck	
CTL LT FW COOLING PUMP 2P- 2 STRIPS							MMR 2, 4th Deck	
CTL LT FW COOLING PUMP 1P-2 STRIPS							MMR 2, 4th Deck	
CTL LT FW COOLING PUMP 1P-2 STRIPS							MMR 2, 4th Deck	
CTL SW COOLING PUMP 2P- 2 STRIPS							MMR 2, 5th Deck	

CTL SW COOLING PUMP 2P- 2 STRIPS							MMR 2, 5th Deck	
CTL SW COOLING PUMP 1P - 2 STRIPS							MMR 2, 5th Deck	
CTL SW COOLING PUMP 1P - 2 STRIPS							MMR 2, 5th Deck	
AUX FUEL XFER PUMP PORT							MMR 2, 4th Deck	
FIRE GEN SVCE PUMP PORT							MMR 2, 5th Deck	
DSTL EJECTOR PUMP PORT AND STBD							MMR 1 and 2, Tank Top	
STERN TUBE SW SUPPLY PUMP PORT							MMR 2, 5th Deck	
MISSION DW SUPPLY TRANSFER PUMP							MMR 2, 5th Deck	
CTL SW COOLING PUMP 1S -2 SRTIPS							MMR 1, 5th Deck	
CTL SW COOLING PUMP 1S -2 SRTIPS							MMR 1, 5th Deck	
DEEPWELL BILGE PUMP MOTOR							MMR 1 and 2, 5th Deck	

MISSION MODULE SW COOLING PUMP#1							MMR 1, 5th Deck	
MISSION MODULE SW COOLING PUMP#2							MMR 1, 5th Deck	
FM PUMP-1 FIRE & GEN SVCE PUMP STBD- 2 STRIPS							MMR 1, 5th Deck	
FM PUMP-1 FIRE & GEN SVCE PUMP STBD- 2 STRIPS							MMR 1, 5th Deck	
STERN TUBE SW SUPPLY PUMP STBD							MMR 1, 5th Deck	
STBY STERN TUBE SW SUPPLY PUMP STBD							MMR 1, 5th Deck	
PROP SHAFT THRUST BRG LUBRICATION MOTOR PORT AND STBD							MMR 1 and 2, Shaft Alley	
FUEL SUPPLY PUMP 1S							FO Service Room, 3-56-5	
AC/CHILL WATER PUMP#1							Hotel Services Room, 2-48-1	
AC/CHILL WATER							Hotel Services	

PUMP#2							Room, 2-48-1	
FUEL SUPPLY PUMP 2S							FO Service Room, 3-56-5	
FWD CW PUMP#1							Support Services, A1-96-0	
FWD CW PUMP#2							Support Services, A1-96-0	
R.O UNIT MOTOR FWD							AMR, 3rd Deck	
BOW THRUSTER MOTOR							AMR, 5th Deck	
BOW THRUSTER LT FW COOLING PUMP#1							AMR, 4th Deck	
BOW THRUSTER LT FW COOLING PUMP#2							AMR, 4th Deck	
FWD POTW HYDROPHORE PUMP MOTORS 1&2							AMR, 4th Deck	
BOW THRUSTER SW COOLING PUMP MOTORS 1&2							AMR, 5th Deck	
JP5-MISSION XFER PUMP							JP-5 Pump Room, 5-108-1	
JP5-							JP-5 Pump	

MISSION SVC PUMP							Room, 5-108-2	
MISSION DECK CRANE MOTOR							Mission Deck, Port Side	
STBD/PORT STORES CRANES							B Deck, Aft Casing, P/S	
APPLETON CRANE MOTOR							Upper Deck, Port Side	
FIREMAIN CHARGING PUMP MOTOR							PUMP ROOM 5-57-0	
HPSW PUMP MOTOR							PUMP ROOM 5-57-0	

USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0012	CATEGORY "A"	2026-01-30
ESB-SWI-INFORMATION TECHNOLOGY SERVICES		Manovich, David

1.0 ABSTRACT

This item describes the requirements for the contractor to information technology equipment, supplies, and services to the ship and in the OMT's Office for the duration of the overhaul period.

2.0 REFERENCES: None

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 One telephone at the gangway watch station with unlimited local phone service

3.1.2 Provide (2) direct phone lines connected to the ship's shore telephone PBX connection box with unlimited local phone service.

3.1.3 Provide (5) telephones and direct lines in the OMT's office with unlimited local and unlimited international calling capability (OCONUS SY locations). Telephone for the administrative assistant's use shall be capable of picking up and transferring to all other phones within the office.

3.1.4 Shipboard high-speed internet connection: Provide a T3 type or equivalent high-speed internet connection to the ship, with adequate capacity to support a minimum of 20 users onboard simultaneously.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor is reminded that since direct lines may be difficult to obtain, arrangements should be made in advance of the vessel's arrival at the contractor's facility.

5.2 All phones and phone lines shall be maintained in operating condition, 24-hours a day, seven days a week, throughout the availability period.

5.3 The definitions of many terms used in this work item are found in Work Item 001.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 Provide dedicated/continuous telephone and internet connectivity.

7.0 STATEMENT OF WORK

7.1 Communication equipment and services:

7.1.1 Provide, install and maintain telephone service and ship’s high-speed internet service as described in article 3.1. Contractor shall connect phone lines to the ship’s PBX connection box. Contractor shall connect high speed internet line to the ship’s internet connection box.

7.1.2 Telephone lines shall be direct access lines (separate lines and not through shipyard switchboard) for 24-hour/day, seven day/week use, throughout the shipyard availability period. Telephone services for the OMT’s office shall be provided commencing three (3) days before arrival of the vessel and terminating three days after redelivery of the vessel.

7.1.3 OMT office telephones shall be touch-tone type, with the following features:

a. Hold button

b. Speaker phone

c. 3-way calling

d. Ability to switch between calls

7.1.4 The gangway phone is for unlimited local phone service only. Phone is for official or emergency use only. However, to ensure that unauthorized personnel will not make phone calls, the contractor shall provide and install a phenolic sign as shown below (black letters on a red background) at the gangway station(s):

NOTICE

PHONE IS FOR U.S. GOVERNMENT OFFICIAL, OR EMERGENCY USE ONLY,

NOT FOR NON-EMERGENCY USE BY CONTRACTOR PERSONNEL

7.2 Computer equipment and servicing:

7.2.1 Office and workstation numbers are delineated in Work Item 010 article 7.2.1. Each private office and workstation shall be equipped with:

a. A minimum of one LAN drop/high speed internet connection shall be provided at each workstation and private office.

b. In addition to the one LAN drop at each computer, each office and workstation shall have an additional internet connection (Cat 5) cable to permit connection of personnel’s laptop computers to the internet. Any installed firewalls must not restrict Navy Marine Corp Intranet (NMCI) virtual private network (VPN) functions.

7.2.2 Provide (**4**) computers for the OMT’s use and provide two (2) computers for the administrative assistants. One computer shall be outfitted for each of the office and workstation described in paragraph 7.2.1 of WI 010.

7.2.3 Provide and maintain High speed Wifi with coverage for the entire OMT office. Wifi shall meet the additional following requirements:

a. Be minimum 100Mbps download speed and minimum 10Mbps upload speed.

b. Have both 2.4Ghz and 5Ghz networks.

c. Be capable of being password protected with the password set by the OMT Rep.

d. Have unlimited and unrestricted data transfer

7.2.4 Provide a service contract for the entire performance period that will provide computer equipment technical support, trouble shooting and repair during period of 0800 to 1730 local time, six days a week. The service will cover all equipment and be at no additional cost. Technical support will be available by phone. Troubleshooting and repair will be performed on site within four (4) hours of call to service agent.

7.2.5 Provide, install, and maintain the following computer hardware and software for the duration of the performance period:

a. A server which networks all of the computers together and meets the following:

b. Computers provided must meet the following specifications:

1. All computers connected to unrestricted high-speed internet (T3 type or equivalent high-speed internet connection)

2. Each computer shall have two monitors that are, at minimum, 19” LCD flat panel color monitor with 1920 ×

1080 resolution

3. Minimum Intel I5 processor

4. Minimum speed 3.0 Ghz

5. Minimum 8 gigabyte of memory/RAM

6. Minimum 250 gigabyte hard drive

7. Ethernet connection

8. DVD/CD ROM drive/burner 16× or higher, capable of reading and burning DVD's and CD's including +R, -R, +RW, and -RW

9. Minimum of 4 USB ports (easily accessible) on each computer

10. Optical mouse and a keyboard

11. MSC to maintain administrative rights to all computers and network

7.2.6 Contractor shall provide the following software on all computers, temporarily provide the manufacturer's newest/latest versions of the following software for the overhaul duration:

a. Microsoft Windows operating system (Windows 2010 or later)

b. Microsoft Office (Microsoft Office 2016 or later)

c. Microsoft Edge or Google Chrome

d. Microsoft Project

e. Adobe Acrobat Standard DC

f. AutoCAD Lite latest revision

g. Software to read and write to CDs and DVDs

7.2.7 Printers/Copiers/Scanners:

a. One HP Color LaserJet Enterprise MFP M680 or equal multi-function device connected to office network. To be considered “or equal,” the printer must meet the following salient characteristics:

1. Ability to copy and print 8-½” × 11”, 11” × 17”, and 8-½” × 14”

2. Digital flatbed scanner with a duplexing automatic document feeder

3. Capable of scanning up to 11 × 17 size paper

4. 600 × 600 dpi minimum copying and scanning resolution for color and black and white

5. 9600 × 600 dpi minimum printing resolution for color and black and white

6. High speed (40 copies per minute minimum black and white)

7. Automatic feed

8. Zoom

9. Collating capability

b. Three each HP Color LaserJet Pro M277dw or equal connected to each private office’s computer (One located in the Administrative Contracting Officer's office; one in Project Engineer's office and one located in the Principal Port Engineer's office). To be considered "or equal," the printer must meet the following salient characteristics:

1. Ability to copy and print 8-1/2" × 11" and 8-1/2" × 14"

2. Digital flatbed scanner with a duplexing automatic document feeder

3. 300 × 420 dpi minimum copying and scanning resolution for color and black and white

4. 600 × 600 dpi minimum printing resolution for color and black and white

5. Average speed (18 copies per minute minimum black and white)

6. Provide and maintain spare toner for copier and printers in paragraph 7.2.7

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0013	**CATEGORY "A"**	**2026-01-30**
ESB-SWI-INTEGRATED PROJECT PLANNING AND PRODUCTION STATUS MONITORING REPORTS		**Manovich, David**

1.0 ABSTRACT

Project planning and management are of utmost importance to accomplish all required work items within the defined Period of Performance (PoP). The OMT may have 3rd party contractor teams assigned to accomplish specific work requirements onboard the vessel. This item describes the requirements for the contractor to establish and utilize a robust integrated project management program and to provide production charts, progress reports, and manning charts to support overhaul management team monitoring of project status on a weekly basis.

2.0 REFERENCES/ENCLOSURES

2.1 References: None

2.2 Enclosures:

2.2.1 MSC QMS N7.75.3120.1-Q, MSC Engineering (N7) Port Engineering Manual, Appendix C11 – Sample, MSC Post Award Meeting Template

3.0 ITEM LOCATION/DESCRIPTION

3.1 Quantity/Description: Throughout the period of performance

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors regardless of tier shall consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs (1) through (7).

5.2 The contractor and all subcontractors regardless of tier are advised to review other work items under this contract, to determine their effect on the work required under this work item. Many of the definitions relating to performance of this work item are found in Work Item 0001.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 The contractor shall utilize project planner(s) to accomplish integrated production planning, manpower/resource allocation, production tracking, and production schedule updates based upon inputs from the ship superintendent/project manager and OMT REP, and actual progress to fulfill the requirements of this work item.

7.1.1 This project planning function should be independent of ship superintendent/project managers, but planner(s) can be supporting multiple projects.

7.1.2 The contractor’s assigned Project Planner shall attend the weekly progress meeting.

7.2 Production chart

7.2.1 The integrated production chart is the timetable for the use of resources and processes required by the contractor to complete the project in the contractual PoP. Upon award of the ship repair contract, the OMT REP will advise the contractor of any 3rd party OMT furnished contractor teams planned to accomplish work on the vessel during the scheduled PoP. The Contractor’s Project Planner shall integrate the work of the OMT Furnished contractors into the integrated production chart, and shall work with the OMT REP to de-conflict issues that may arise between OMT Furnished contractors and shipyard work requirements and milestones. These integration efforts are intended to initiate and provide cross-communication between the ship repair contractor, the 3rd party contractor and the OMT REP to minimize impacts to the production plan.

7.2.2 OMT REP will provide to the contractor within five days after award, a listing of 3rd party OMT Furnished contractors in the following format:

a. Company name of OMT Furnished contractor

b. OMT Furnished contractor POC

c. A brief description of work to be accomplished by that contractor.

d. Listing of any OMT Furnished contractor support work items included in the work package.

e. Specific milestones included in the OMT Furnished contractor work requirements that should be included in the integrated production plan.

7.2.3 Within ten days after receiving the information about OMT Furnished contractors in 7.2.2 above, the contractor will schedule a post award conference call with MSC OMT and other parties as agreed to by MSC. No less than one day prior to the meeting the contractor must prepare and submit to the OMT REP an electronic copy in portable document format (PDF), of a Gantt or similar (bar type) integrated production chart clearly indicating planned start date, planned completion date, and planned manning for each item of the specification for contractor and subcontractors (manning for OMT Furnished contractor not required). Subcontracted work shall be assigned as discrete activities. Estimated manning shall be assigned to each prime and subcontractor activity. Additional agenda items will be provided per Enclosure 2.2.1, Appendix C11 – Sample, MSC Post Award Meeting Template.

7.2.4 Each work item shall be subdivided into its major components, i.e., working plans, material delivery, equipment removal, shop repair, reinstallation, testing, regulatory approvals. The Y-axis of the production chart will be divided by each work item title. Critical path work items indicated by asterisk next to the work item or positively identified by other means. Significant milestones, including those provided for OMT Furnished contractor work, shall also be identified on the chart. The X-axis of the production chart shall be subdivided by days.

7.2.5 The integrated production chart shall be amended weekly, until the completion of the performance period, to incorporate all added and deleted work under contract change orders (CCOs) and each bar shall be legibly marked to indicate current status of the work item. OMT REP will provide status of OMT Furnished contractor work and any adjustments to milestones. Production chart revisions are an integral part of project management to reflect project elements that were simpler or more complex than originally planned. Revised dates for sequence start, stop, milestone date completion along with revised manning and work-shift projections shall be highlighted on the updated production charts. An electronic copy in portable document format (PDF), and a second electronic copy of underlying *.csv or *.xml data needed to produce production chart shall be submitted to the OMT REP 24 hours prior to the start of each of the weekly progress meetings, the submitted chart shall be produced to current schedule and shall highlight changes in each work item, CCO, or milestone schedules.

7.2.6 The production chart shall have an appropriate title block indicating job identification number, vessel name, chart date, amendment letter and approval authority.

7.2.7 All production charts shall clearly indicate both original and new work items that are critical to the completion of the total work package with an asterisk.

7.2.8 The contractor's production chart is the tool that both the Government and the contractor will use to measure, track and adjust performance in order to maintain contract end dates. It is incumbent on the contractor to constantly monitor and adjust the production schedule, particularly for events that are taking longer to progress than anticipated. Likewise, the OMT REP will monitor the integrated production plan and collaboratively work with the Ship Superintendent and Production Planner to deconflict any issues between OMT Furnished contractor milestones and shipyard milestones and production elements. In the event that contractor's performance does not substantially support its planned work effort and changes have not been reflected on the production chart, the contractor shall be required to respond to requests issued by the OMT REP via production schedule delinquent progress notification (PSDPN) form. Each PSDPN issued against work performance on specific work items requires the contractor to submit his recovery plan in writing within 48 hours of receiving the notification. This recovery plan shall include the reason for progress delay and corrective action to be implemented to assure timely contract completion. PSDPNs may be issued whenever work item physical progress and/or milestones lag at least three days behind projected progress per the production schedule.

7.2.9 Milestones: The contractor shall incorporate the following milestones, at a minimum, into the production planning schedules and reports.

a. ABS Inspection of all tanks: The contractor shall schedule and complete checkpoints for ABS inspection of all tanks that are open/gas free within five days of safe for entry.

b. Structural inspections: The contractor shall schedule and complete all structural inspection by 20% of the availability. This is to include: WI 101 Structural Inspections and WI 0111, 0112, 0113 ABS CBP Annual/Intermediate/5 year Hull Inspection.

c. The following milestones for critical path items shall be completed within the listed percent period of performance as follows:

1. Tank inspections post coating system blasting: 75%

d. Annual inspections for SOLAS related and other items: Shall be completed within the first 30% of the period of performance. These inspections shall include, but are not limited to work items pertaining to the following: Annual Lifeboat and Davit Inspection, Annual Rescue Boat and Davit Inspection, Liferafts, Associated Liferaft and Lifeboat Equipment (Survival Kits, Signaling Devices, Immersion Suits, Sea Anchor, Repair Kit, Oars, Hooks, Bailers, Survival Manual), Lifejackets, Buoyancy Apparatus, Marine Evacuation Systems, SCBA Annual Inspection, Fire Detection Systems inspections and tests, All Fixed Firefighting Systems Inspections and Tests, Fire Extinguishers, Navigation Equipment (including Magnetic Compass and Pelorus, VHF Radiotelephone, Search and Rescue Transponder (SART), NAVTEX Receiver), Annual Inspection of Alarms including Emergency Alarm Systems, Public Address Systems. Provide a CFR for any deficiency found in these systems within three days of discovery per 7.6.2.

e. 50% Progress Review: As Determined by OMT

f. Habitability, galley and mess turn-over (for work packages where shipboard work requirements necessitated crew living off-ship): Shall be scheduled to occur four days prior to crew move aboard milestone. This milestone signifies that all contractor related work in spaces and on ship systems affecting berthing, lavatories, common spaces, galley, and mess areas, hotel services and any other system required to support crew living onboard, shall be completed. All fire detection systems and all fire extinguishing features must be fully active and operable in the habitability, galley and mess areas for this milestone.

g. Crew move onboard (for work packages where shipboard work requirements necessitated crew living off-ship): Shall be scheduled to occur two days prior to dock trial milestone. This milestone signifies the day that crew members will move back onboard the vessel. All fire detection systems and all fire extinguishing features must be fully active and operable in the habitability areas for this milestone.

h. Machinery space turn-over: Shall be scheduled to occur at least five days prior to the dock trial milestone. This milestone signifies that all contractor related work on ship systems and equipment within all machinery spaces is complete and equipment is ready for start-up. All tag-outs shall be cleared by shipyard personnel in concert with ship’s force for equipment worked on during the availability. Ship’s crew shall utilize the time between machinery space turn-over and dock trials to complete MSC mandated ship start-up and readiness assessment checklists. The contractor shall assist the OMT REP and Chief Engineer to assemble information and reports necessary to complete ship start-up readiness assessments for machinery and equipment worked on by the contractor or their subcontractors during the repair availability.

1. Machinery spaces are considered to be any spaces onboard the ship containing machinery and equipment including but not limited to main and auxiliary engines, generators, switchboards and load centers, pumps and motors, air

compressors, water-makers, hydraulic power units, thrusters, water-jet machinery, steering gear, windlass machinery, propulsion chain gears and bearings. Only those spaces where machinery, equipment and systems were the subject of work package maintenance and repair requirements are covered under this milestone.

i. Pre start-up meeting: Shall be scheduled to occur at least three days prior to the dock trial. This milestone marks the requirement for the contractor, all subcontractors and technical representatives involved in the installation, maintenance and repair of shipboard equipment and systems to meet with the OMT REP, Chief Engineer and Master to review written reports from the contractor, subcontractors and technical representatives asserting the readiness of subject equipment for start-up, commissioning and operational testing. Subject reports shall also include recommended start-up, commissioning, and testing procedures to prove proper operating conditions of subject equipment.

1. Subcontractors and technical representatives may participate in the pre start-up meeting by teleconference providing the following has been satisfied:

2. Subcontractor and/or technical representative written completion reports were provided to the OMT REP and ship’s Chief Engineer prior to their departure from the local area and conducted an outbrief with the ship superintendent, OMT REP and Chief Engineer.

3. Subcontractor and/or technical representative report includes a statement that the equipment is ready for start-up and contains a comprehensive start-up, commissioning and operational testing procedure.

j. Bridge turn-over: Shall be scheduled to occur at least three days prior to the dock trial milestone. This milestone signifies that all contractor related work on ship systems and equipment within the bridge space such as but not limited to navigation, ship propulsion control, alarms and indicators, communications equipment, and associated topside antennas and rotating equipment is complete. Ship’s crew will utilize this time to complete MSC mandated Bridge Resource Management training.

k. Dock trial: Shall be scheduled to occur at least two days prior to the sea trial milestone. This milestone marks the requirement to conduct static, pier-side testing of all equipment and systems worked on during the availability. Any discrepancies found during dock trials must be corrected by the contractor and verified by the OMT REP prior to accomplishment of vessel sea trials. Any shipboard systems and equipment to which maintenance, repair or upgrade was accomplished per the work package and per additional contract modifications, for which manufacturers or qualified technical representatives were used to complete that work, shall have those technical representatives present during equipment start-up and dock trial testing. Manufacturers and authorized technical representatives must provide written reports asserting the readiness of the equipment for start-up and testing, as well as assessments of the testing and equipment performance in comparison to acceptance criteria in accordance with WI 022 Paragraph 7.2.

l. Sea trial: Shall be scheduled to occur one day prior to the contract performance period end date. This milestone marks the requirement to conduct open-water underway operational testing of equipment and systems worked on during the availability. Any shipboard systems and equipment to which maintenance, repair or upgrade was accomplished per the work package and per additional contract modifications, for which manufacturers or qualified technical representatives were used to complete that work, shall have those technical representatives present during equipment start-up and sea trial testing. Manufacturers and authorized technical representatives must provide written reports asserting the readiness of the equipment for start-up and testing, as well as assessments of the testing and equipment performance in comparison to acceptance criteria in accordance with WI 0022 Paragraph 7.3.

7.2.10 Category B work items have been identified as possible additional work if funding or material become available. The contractor will provide in their bid proposal the last calendar date for which activation of each Category "B" work item will not affect original scheduled completion date of the availability.

7.3 Daily production meetings and plan of the day (POD):

7.3.1 The contractor shall conduct daily production meetings with the OMT REP (and others as needed) for the purpose of discussing the shipyard's plan of the day (POD). The daily production meeting shall be held each morning at a mutually agreed time and location. OMT REP will ensure representatives from each OMT Furnished contractor are present to discuss their work plans for the day and to coordinate those actions with the shipyard. To the maximum extent possible OMT REP will ensure OMT Furnished contractors provide a copy of their POD to the ship Superintendent at the end of the previous day.

7.3.2 The contractor shall provide hardcopies of the POD each morning to the OMT REP detailing the work planned for the shipyard and their subcontractors for that day. At a minimum the POD shall include safety concerns (immediate or predictable), specific work items being worked on, areas of concern such as cleanliness of vessel, areas of hot-work, check points, call-outs, major crane lifts, ABS/USCG inspections and any other items that may require situational awareness and/or assistance from the OMT or ship's force.

7.4 Progress report

7.4.1 Within five days after start of the contract the contractor shall prepare and submit a progress report, listing specification item titles and numbers with appropriate blocks to record weekly percentages completion for each item over the period of the contract.

a. This report shall include contractor's estimate of percentage completion of each work item and the previously reported progress.

b. Authorized change order modifications shall also be included with completion estimates. Final weekly completion estimates shall be negotiated during the weekly progress meeting.

c. A current status report of all change orders and condition found reports shall be provided with the progress report for review and discussion.

7.4.2 The progress report shall be subdivided by original items and added items and amended weekly until the completion of the contract. Added changes shall be listed under the parent work that they pertain to when applicable. An electronic copy in portable document format (PDF), of the progress report shall be submitted weekly to the OMT REP 24 hours prior to the start of the weekly progress meetings, a second electronic copy of the report containing the data needed to create the PDF report shall be provided in *.csv or *.xml format. The weekly progress meeting shall be held on Wednesdays at 1300 hours for the duration of the contract period.

7.5 Manning chart

7.5.1 One week before the start of the contract, the contractor shall prepare and submit to the OMT REP a manning chart, clearly indicating the planned contractor and subcontractor manning used on a daily basis, including weekends, throughout the contract performance period. The manning chart shall be updated weekly, until the completion of the contract period to reflect increased or decreased planned manning as a result of production status and CCOs issued by the government, cancellation/increase and/or contractor initiated manning changes. Note that the production chart required under 7.2 requires manning by work item to be indicated on that chart. There should be a direct correlation between the manning numbers represented on these two charts.

a. The contractor shall provide to the OMT REP at each daily meeting a list of major trades or work centers indicating the manpower associated with the respective major trade or work center for that day. This shall include all prime and subcontractor manpower. This shall be provided on the POD required per 7.3.

7.5.2 An electronic copy in portable document format (PDF), of the updated planned manning chart shall be submitted to the OMT REP, 24 hours prior to the start of the weekly progress meetings, a second electronic copy of the report containing the data needed to create the PDF report shall be provided in *.csv or *.xml format.

7.6 Condition found reports (CFRs)

7.6.1 Shipyard shall provide an electronic copy in PDF of all CFRs to the OMT REP.

7.6.2 CFRs shall be submitted with a sequential numbering convention for MSC serialization purposes starting with 001. That is to say, the first CFR shall be serialized 001 and following shall be sequential (002, 003, 004...). All CFRs shall be submitted through the shipyard project manager for all trades and subcontractors within three days of discovery. CFRs will not be accepted from individual trade supervisors nor subcontractors.

7.6.3 The OMT REP will provide a written response to each CFR indicating the decision regarding the CFR. The response may indicate a CCO will be issued if additional work or a credit is required, or it may simply acknowledge receipt of the CFR with no further action required or authorized. The response to the CFR does not constitute authorization to proceed with any additional work or a change order.

7.6.4 All project reports called out in 7.2, 7.4 and 7.5 must be assigned a CFR number and submitted as discussed applicable section.

7.7 Contract change order (CCO)

7.7.1 The OMT REP (via the ACO for GOGO vessels) will provide electronic copies (PDF) of each CCO to the shipyard /contract manager in response to a CFR if additional work is required or desired to correct the condition described by a given CFR. Not every CCO will have a related CFR as some represent new work requirements. The shipyard shall respond to the CCOs with a quotation for the additional work requested within three days of receipt of the CCO. Once the costs and PoP have been negotiated and agreed by both the shipyard contract manager and the ACO and officially signed, the CCO will be considered an effective change order to the contract.

7.7.2 Contractor response to CCOs shall include an analysis of the requirements against the production chart and shall include the following additional information:

a. A firm statement concerning impacts to the production chart: This statement will either confirm that the additional work can be accomplished within the current PoP, or it will state that the additional work will affect the PoP completion date. In this case, the response shall note the primary and any secondary work items that would be affected, including an excerpt of the contractor’s planning production chart showing the affects to the project.

b. The contractor shall also provide an analysis and alternative proposal to maintain existing PoP, including such things as alternative procedures, accelerated and/or premium labor efforts, or any other ideas contractor may present to avoid contract PoP extensions.

7.8 Overhaul Lessons Learned

7.8.1 As part of the availability closeout process, key members of the shipyard project management team and/or the program manager shall attend an Overhaul Lessons Learned Conference, hosted by the OMT. The purpose of the overhaul lessons learned conference is to engage with the shipyard and other invitees sharing lessons learned from all parties, such as positive or negative impacts to the budget and the schedule.

8.0 GENERAL REQUIREMENTS: None additional

Enclosure 2.2.1 MSC QMS N7.75.3120.1-Q, Appendix C11 – Sample, MSC Post Award Meeting Template

<u>Vessel Name ROH-DD MSC Post-Award Meeting</u>

Date/ Time

- Greetings and Introductions (All principal participants; Shipyard, Sub Contractors and MSC)
- Purpose
- WI Index Review
  - o Any potential issues (GFM/CFM, changes planned)
  - o Critical Path, Risks
  - o Specific work item risk assessments (from MSC or Shipyard point of view)
  - o Specific GFM or CFM issues
- Service Order Contractor Access
- ABS Inspections
- Contracting
  - o CFR/CCO process
  - o CDRLS
- Safety/ Security
  - o Safety and Security Standard Work Items
  - o Shipyard procedures/standards, if add'l info
  - o Force Protection
  - o Additional Shipyard Safety/ Security
- Ships force working hours and support for:
  - o Lock-out/Tag-outs (and ins)
  - o Hotwork
  - o Crane & Forktruck Lifts
  - o Checkpoints
  - o Space Close-out

  - Any Contractor Comments
- Pre-arrival meeting with SY & mooring line transfer
- SY Meetings:
  - Daily Production
  - Weekly Status
  - Remote Communications, if necessary
- Ships force input:
  - Timeline/schedule for crew move off
  - Ship's force work list
- General Expectations/ Other
  - Communication Plan
- Last call and action Item/ due date review
- ADJOURN

1.0 ABSTRACT

This item describes the requirements for the contractor to provide a weight report listing all the weight and moment changes to the ship resultant of all work accomplished in this availability.

2.0 REFERENCES: None

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: Throughout the ship

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with all applicable GTR requirements.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract to determine their effect on the work required under this work item.

5.3 The definitions of many terms used in this work item are found in Work Item 001.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Arrangements/Outfitting: None

7.2 Structural: None

7.3 Mechanical/Fluids: None

7.4 Electrical: None

7.5 Electronics: None

7.6 Preparation of Drawings/Documentation

7.6.1 The contractor shall provide a detailed weight report listing all the weight additions, removals and relocations to the ship including the associated centers of gravity, as a result of the work performed in the course of this contract period of performance. Changes in buoyancy must also be reported.

7.6.2 The report shall account for all weight changes.

a. Small weights (less than 0.1 long tons (224 lbs.)) may be combined and estimated.

b. Weights shall be documented either by the manufacturer, by estimating, actually weighing, or calculating.

c. Account for items at the component level. For systems normally containing liquids (piping, machinery with sumps, etc.) report the weight of liquid in the system separate from the dry weight of the system.

7.6.3 The weight format shall be in the tabular format and, at a minimum, shall contain the following information: The tabulation shall be grouped according to work items of the work package.

a. Item, number and description

b. Weight, (long tons)

c. Vertical lever (VCG), feet above baseline

d. Vertical moment, foot-tons

e. Longitudinal lever (LCG), feet. aft or forward of the forward perpendicular

f. Longitudinal moment, foot-tons

g. Transverse lever (TCG), feet off centerline port/starboard

h. Transverse moment, foot-tons port/starboard

i. Resultant weight, long tons

j. Resultant VCG, feet above baseline

k. Resultant LCG, feet aft of forward perpendicular

l. Resultant TCG, feet off centerline port/starboard

1. Weight (tons) and distance (ft.) shall be rounded off to two places to the right of the decimal point. Moments shall be rounded to the whole foot-ton.

2. Dimensions aft of the forward perpendicular, above baseline and to port shall always be considered positive.

7.6.4 The contractor shall also provide a preliminary report five days prior to docking if any weights have already been removed or added to the vessel. Two copies of the preliminary weight report shall be forwarded for review five days prior to re-floating the ship from the dry-dock; a revised copy of the weight report one week before the end of the shipyard period; and two copies of the final weight report two weeks after the shipyard period. The final report shall incorporate any comments provided about the preliminary report. Submit each report to the MSCREP.

7.6.5 Deliver to the OMT REP two hard copies of a preliminary weight report for review, one week prior to sea trials.

a. Retain all field notes taken and make available to the OMT REP upon request.

7.6.6 Provide the final weight report (incorporating, or responding to, comments on the preliminary weight report) as two hard copies and one electronic copy in Microsoft Excel format.

a. Ensure the report is prepared and signed by a degreed engineer or licensed professional engineer with annotation of the individual’s qualifying credential(s) (e.g., BS Naval Architecture, state PE license no., etc.). It shall contain the following certification in the lower right corner:

1. Undersigned certifies that the weight report herein is a true representation of the weight effects of all changes made to the vessel.

7.6.7 One additional electronic copy, on CD, shall be mailed via FedEx or similar courier service to the MSC Technical Library at the following address:

MSC Technical Library - USNS HERSHEL WILLIAMS,
Final Weight & Moment Report
9276 3rd Ave., Bldg LP-26
Norfolk, VA 23511
Tel: (757) 443-2595

7.7 Inspection/Test: None

7.8 Painting: None

7.9 Marking: None

7.10 Manufacturer’s Representative: None

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0015	**CATEGORY "A"**	**2026-01-30**
ESB-SWI-ILS-GFM		**Manovich, David**

1.0 ABSTRACT

This item contains the Integrated Logistics Support Requirements (ILS), Government Furnished Equipment (GFE) and Government Furnished Material (GFM) support requirements for all the work items contained in this specification package.

2.0 REFERENCES

2.1 Department of Defense Security Manual for Safeguarding Classified Information, DOD Manual 5220.22-M

3.0 ITEM LOCATION/DESCRIPTION: None

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with all applicable GTR requirements.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract to determine their effect on the work required under this work item. The definitions of many terms used in this work item are found in Work Item 001.

5.3 Requirements of this item apply to the base contract and all modifications inclusive. Additional ILS requirements resulting from change orders shall be priced and negotiated on that change order and/or modification.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Logistics support requirements: Logistics support requirements establish and maintain life cycle support for equipment procured by the contractor in support of work item (WI) requirements in this work package (WP). The logistics documentation required by the contract is in the form of: an equipment technical manual (TM); technical support data (TSD); repair parts support; purchase orders; drawings; and residual asset file.

7.1.1 TM requirements: Equipment procured under this contract shall be delivered with three hard copies and one electronic copy of the supporting TM. An additional electronic copy shall be mailed to the MSC Tech Library. Address is provided in article 7.4. The equipment TM shall be permanently imprinted by mechanical means and the cover shall be durable to withstand frequent handling and exposure to oil and water. The binding will permit adding and removing pages. All TM shall include the following, as applicable (commensurate with the complexity of the equipment):

a. Cover and title page

b. Manufacturer’s name and address

c. Equipment name and application

d. Table of contents (including a list of drawings and tables)

e. Safety precautions (cautions, warnings, and notes)

f. General theory of operation

1. Complete functional description of equipment based on a block diagram

2. Complete explanation of mechanical features using block diagrams or cutaway drawings

3. Major assemblies broken into individual circuits, accompanied by complete circuit analysis keyed to a simplified schematic

4. Brief descriptions of complex and unusual circuits

5. Voltage waveforms at significant point in the circuit

6. Memory maps and a description of microprocessor functions

g. Preparation for use, installation, and initial adjustment instructions

h. Operational instructions

i. Maintenance instructions (preventive and corrective)

j. Cleaning and lubrication instructions

k. Performance verification and test features

l. Frequency of adjustment/test equipment

m. Troubleshooting instructions

n. Disassembly, repair, replacement, and re-assembly instructions

o. Installation instructions

p. Diagrams, illustrations, and schematics

q. Parts list data - The parts list will identify all parts necessary to provide for 100% bill of material. The following requirements apply to the parts list:

1. Identification of OEM and OEM’s part number, OEM CAGE code, if available, or identification of OEM including address and telephone number

2. All parts shall be keyed (using index numbers) to exploded view drawings

3. Parts in the listings shall be grouped by assemblies, subassemblies, and modules. Parts shall be identified in the assembly in which they are components.

4. Parts listed in the TM will match the TSD parts list as required by article 7.1.2.c. of this item.

r. All data shall be provided in hard-copy and CD-ROM format.

7.1.2 Technical support data:

a. The contractor shall provide complete and accurate data with delivery of the equipment. All data will be submitted in the English language only. The contractor shall submit a revision whenever engineering changes and/or modifications occur which add to, delete from, or modify previously submitted TSD (including changes to manufacturer’s part numbers).

b. When TSD documents are prepared by a contractor other than the original equipment manufacturer (OEM), the preparer shall be identified by their company’s name, address, telephone number and point of contact.

c. The TSD documentation consists of a bill of materials/list of all repair parts, assemblies and subassemblies, special tools and test equipment required to maintain, repair, or overhaul the equipment/components as specified by an illustrated parts breakdown. The documentation shall include at a minimum the following technical data for each individual part, assembly, and subassembly for the equipment/components specified:

1. OEM's part number
2. OEM's CAGE (if no CAGE is available, provide OEM address, description and the telephone number)
3. OEM's drawing that identifies the part (illustrated parts breakdown)
4. Characteristic/name plate data/certification data for the end item and all equipment
5. Item name
6. Reference Symbol Number (for electronic provisioning only)
7. Production lead time (if known)
8. Unit of issue
9. Unit price
10. Unit of measure

d. TSD or parts of it, containing classified information shall be protected and marked in accordance with Reference 2.1.

e. Alternate submission of TSD: The contractor may substitute the requirements of TSD documentation with an equipment TM as required in article 7.1.1, providing the TM contains the minimum data requirements set forth in article 7.1.2.c. The contractor is responsible to provide any additional TSD documentation not contained in the TM in order to meet the minimum requirements of article 7.1.2.

7.1.3 Statement of prior submission (SPS): The contractor may submit SPS documentation in lieu of the requirement in article 7.1.2, as applicable.

a. Repair parts support: The contractor is required to provide the OMT REP a pre-priced recommended list of spare parts with delivery of equipment. Listing shall be on CD-ROM. The OEM/vendor recommended spares list shall be of sufficient range and depth to provide one year of preventive maintenance and 10 years of corrective maintenance support for equipment purchased in support of the contract. The Government may exercise the option to purchase any or all parts recommended by the contractor. In determining the necessary spares support, the contractor shall consider the equipment’s maintenance requirements (preventive and corrective), component criticality, and historic failure rates. The contractor recommended spares list shall include at a minimum:

1. Part number
2. Part nomenclature/description
3. Part quantity per component
4. Contractor recommended quantity
5. Unit of issue
6. Unit price
7. Total price

7.1.4 Purchase orders: The contractor shall provide copies of all purchase orders for all work items where contractor-furnished materials and/or equipment were procured. This data is not required for repairs to the hull (doors, bulkheads, tanks, etc.) or for new installations of piping, wiring, bulkheads, etc. Purchase orders will identify the following:

a. Material

1. Work item
2. Original manufacturer’s part number
3. Identification of manufacturer
4. Quantity
5. Unit price
6. Total price

b. Equipment

1. OEM’s part number
2. Identification of manufacturer
3. Equipment nameplate
4. Equipment characteristics
5. Equipment Serial number
6. Unit price
7. Quantity
8. Total price

7.2 Drawings: The contractor shall provide specific drawings as called out in the individual Work Items. Drawings shall be IAW MSC GTR No.5.

7.2.1 Any drawings that were provided to the contractor in AutoCAD format for modification shall be returned in AutoCAD format.

7.2.2 Any newly developed drawings shall be developed in AutoCAD format.

7.2.3 If drawings were provided in a format other than AutoCAD for red-lining or other modification, that drawing may be returned in AutoCAD, PDF, or the format in which the drawing was originally provided.

7.2.4 If the contractor is required to obtain drawing/design approval from regulatory bodies such as USCG and ABS, any approval letters for the drawings shall be provided as a separate file in PDF format. A hard copy of the approval letter shall accompany the hard copy of the stamp approved drawing.

7.2.5 An additional electronic copy shall be mailed to the MSC Tech Library. Address is provided in article 7.4.

7.3 Contractor support

7.3.1 Receipt of GFM

a. The contractor shall receive all GFM listed in this WP, whether on board the ship or delivered to the contractor by common carrier or other means. The contractor shall accept GFM beginning 44 days prior to the start of the overhaul.

b. The contractor shall inspect the GFM at the time of receipt and verify that the material received is the same as that indicated on the GFM listing. Note and describe any physical/visible damage or defects to the equipment/packaging on the carrier’s bill of lading or delivery confirmation, to include a printed name and signature of the individual transferring custody indicating acknowledgement of condition at time of delivery. The contractor shall immediately notify the OMT REP of the condition and schedule a joint survey with the OMT REP.

c. The contractor receiving report which notes discrepancies of documented and observed damage to the GFM will only be signed by the OMT REP if the OMT REP concurs that the discrepancy/damage was not the result of contractor negligence.

7.3.2 Storage: The contractor shall provide storage facilities as specified below. Access to storage areas in all cases shall be limited to authorized personnel only. The OMT REP or any individual specified by the OMT REP shall have access to all GFM/GFE storage facilities for the duration of the period of performance (POP).

a. Outside storage: Outside storage shall be limited to material not subject to environmental degradation. Protection against physical abuse and theft is required. Designated outside storage shall require prior approval of the OMT REP.

b. Inside storage: Inside storage shall be dry, protected, ventilated, and fork truck accessible. A minimum of 3000 square feet of inside storage area is required and shall be adequately lighted with an integral structure of floor(s), walls, roof, and capable of being locked. The warehouse shall also have multi-level industrial storage racks (ISR) capable of supporting 30 individual positions of single stacked, fully loaded standard sized (48 × 40 × 36), heavy duty rated tri-walls or similarly palletized loads. Note: In lieu of multi-level ISR, additional floor space equivalence will be accepted.

c. Temperature controlled storage: Temperature controlled storage shall consist of a minimum of 200 square feet of inside storage area in which the temperature is maintained in a range of 65 to 85 degrees Fahrenheit year-round. The 200 square foot area shall be segregated from other areas in the same building and be capable of being locked.

d. Environmentally controlled storage: Environmentally controlled storage areas shall consist of a minimum of 200 square feet of inside storage area in which the relative humidity does not exceed 50 percent and the temperature is maintained at 70 degrees Fahrenheit. The 200 square foot area shall be segregated from other areas in the building and be capable of being locked.

e. Locked: The term ‘locked’ shall mean that the storage area has a physical separation from all other areas. The physical separation shall be by means of durable walls or expanded metal and shall be of such construction and arrangement so that a normally locked door controls entry to the area. The contractor shall, in the case of locked storage, provide security as indicated below. If contractor shipyard security is used for such duties, so note.

f. The contractor shall be responsible for the security of all classes of storage. Security measures shall prevent unauthorized entry to the storage areas. Means to accomplish this shall include high quality security locks and roving security guards on a 24-hour basis.

1. The roving security guards shall maintain a log of each inspection. The log shall contain the ship’s name, security contractor’s firm, contract number, and entry blocks for each inspection including the following:

2. Start and end date of the period clocked

3. Signature of the first and last shift’s security watch

4. Signature of the shipyard security head

5. Signature of the shipyard project manager

6. At the end of the 24-hour period the log shall be signed by the head of shipyard security and the shipyard Project Manager and delivered to the OMT REP, as a

contract deliverable document.

7.3.3 Protection: The contractor shall at all times protect GFM/GFE from physical or other types of abuse. Protective dust covers and other appropriate means of protecting stored equipment shall be used. The contractor shall use care in handling, moving, transporting, and rigging all GFM. Damage occurring to the GFM, subsequent to receipt, shall be corrected at the contractor's expense.

7.3.4 Handling

a. The contractor shall provide all the services necessary to safeguard, inventory, record, document, administer, control, and document issue and install of the GFE/GFM received at the shipyard.

b. The contractor shall take delivery and maintain documented accountability of all GFM whether stored on board the ship or delivered to the contractor's designated location. The contractor shall maintain accountability over the material at all times, to include documentation of receipt, release, current location, issuance, installation, or approved disposal.

c. It is the contractor's responsibility to handle, care, protect, and provide security for all GFE/GFM as delineated by this WI. The contractor shall provide all labor and equipment necessary to properly handle and deliver all GFE/GFM to the job site when it is scheduled for installation.

d. All unused GFE/GFM shall be returned to the Government in the same condition as received. All equipment that requires special preparation such as being properly drained, cleaned and/or special packaging will be organized for safe transportation in accordance with state and federal regulations by the contractor.

e. Material and equipment designated for turn over to the Government shall be packed, crated, and prepared for shipment by the contractor. Each WI or specification item will be prepared for shipment in its own unique container. Containers may be consolidated into larger shipping crates, but each crate must be clearly marked, and copies of documents enclosed as noted below. An accurate material inventory including nomenclature, part number, and quantity shall be prepared for each container/crate. Place one copy of the material inventory on the inside of the crate/container and affix another copy to the outside of the crate/container. Enclose both copies in a suitable weather resistant packing envelope. Two additional copies shall be provided to the Government onsite ILS Representative.

f. All crating shall be IAW the International Standards for Phytosanitary Measures (ISPM): Guidelines for Regulating Wood Packaging Material in International Trade (ISPM Publication No. 15) to include certifying that the wood packaging material bears the mark indicating it has been subjected to an approved measure.

g. Markings on outside of box/crate/container shall consist of minimum two-inch stenciled letters with ship’s name, hull number, job number, WI number, and contract number. Any previous markings that contradict current material inventory shall be painted over/removed.

h. Provide certified weights and dimensions of materials prepared for shipment to the Government onsite ILS representative. The Government will arrange transportation and freight.

7.3.5 Reports

a. GFM/GFE/CFM summary list: The contractor shall prepare and update weekly a summary list of GFM/GFE and contractor furnished material (CFM) received at commencement of the contract until contract completion. The list shall describe the material, shipment date and tracking number (where applicable), receiving date, storage location, installation date, condition, etc. The summary report shall also indicate any GFM/GFE and CFM material which was not used in the course of the contract. These items shall be identified as residual assets.

b. Damage report: Any damage to the GFM/GFE, while in the custody of the contractor, shall be reported immediately to the OMT REP so that arrangements for joint survey can be made. The contractor shall prepare a damage report outlining the damage, conditions that led to the damage, and proposed method of repair. Concurrence of the OMT REP is required prior to initiation of recommended repairs.

c. Scrap report: Material and equipment permanently removed from the vessel as a result of work accomplished in this WP which is not identified in this WP for turnover to the Government and is determined by the OMT REP to be unfit for reuse, shall be classified as scrap. The contractor shall prepare and submit a scrap report for all material and equipment permanently removed from the vessel and identified as scrap. The scrap report shall include the specification work item and paragraph under which the contractor was required to remove the material, scrap description, weight, and current scrap value. All scrap shall be separated by material composition and weighed separately. The contractor shall obtain current scrap market prices for the separately identified and weighed variations of scrap prior to hauling and disposing of it. The contractor shall prepare and submit two copies of the scrap report, including price quotes, to the OMT REP. Scrap value will be the subject of a negotiated credit change order.

d. Residual asset report: The contractor shall provide a list of GFM and CFM that are residual assets at completion of the availability. The report will list part number, nomenclature, quantity, associated work item, condition code, weight, and dimension. Data shall be submitted in electronic MS Excel spreadsheet format on a CD-ROM. This report shall be submitted to the OMT REP no later than the last day of the contract POP.

7.4 As required, technical documentation shall be mailed via FedEx or similar courier service to the MSC Technical Library at the following address:

MSC Technical Library - USS HERSHEL WILLIAMS
9276 3rd Ave., Bldg. LP-26
Norfolk, VA 23511
Tel: (757) 443-2595
E-mail: MSC_N7_TechLibrary@us.navy.mil

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

This item describes the requirements to provide and install fire and safety protection capabilities, establish and manage fire and safety programs and inspections, and provide for fire watch services and oversight onboard the ship.

2.0 REFERENCES

2.1 National Fire Protection Association 312, Standard for Fire Protection of Vessels During Construction, Repair and Lay-up (Available Commercially)

2.2 National Fire Protection Association 51B, Standard for Fire Prevention in Use of Cutting and Welding Processes (Available Commercially)

2.3 OSHA Fire Protection while in a shipyard, 29 Code of Federal Regulations (CFR) Part 1915.506, titled "Hazards of fixed extinguishing systems on board vessels and vessel sections"

2.4 OSHA Confined Space Entry Program, 29 Code of Federal Regulations (CFR) Part 1910.146, titled "Permit-required confined spaces"

2.5 MSC Drawing No. 830-8496378, Fire Control Plan

2.6 NAVSEA Drawing No. 521-8499368, Firemain System Diagram

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: Various

4.0 GOVERNMENT FURNISHED MATERIAL/EQUIPMENT/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors regardless of tier shall consult the General Technical Requirements (GTR) to determine applicability to this work item. In performing this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs (1) through (7).

5.2 The contractor and all subcontractors regardless of tier are advised to review other work items under this contract, to determine their effect on the work required under this work item. Many of the definitions relating to performance of this work item are found in Work Item 001.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Submit the following plan as part of the contractors' initial proposal. Amendments to the initial Contractor's written Fire and Safety Plan must be submitted in writing to the OMT REP by submitting a condition report.

7.1.1 The Contractor must develop and implement a written Fire and Safety Plan that details placement of temporary firefighting appliances and actions that must be taken to ensure shipboard fire safety as detailed in this work item. The Contractor must include the following in the Fire and Safety Plan:

a. Contractor's responsibilities as detailed in 29 CFR 1915.502(a)-(b).

b. Contractor's required written policy that complies with the local municipal response as required in 29 CFR 1915.505(b)(2) which details the following:

i. The types of fire suppression incidents to which the fire response organization is expected to respond at the employer's facility or worksite;

ii. The liaisons between the Contractor and the local municipal response organization; and

iii. A plan for fire response functions that:

(a) Addresses procedures for obtaining assistance from the local municipal response organization;

(b) Familiarize the outside fire response organization with the layout of the Contractors facility and worksite; including access routes to controlled areas, and site-specific operations, occupancies, vessels or vessel sections; and hazards; and,

(c) Sets forth how hose coupling connection threads are to be made compatible and includes where the adapter couplings are kept; or

(d) States that the Contractor will not allow the use of incompatible hose connections

c. Description of the ship repair facility fire protection and flooding protection measures/procedures;

d. Agreements with local municipal response organizations;

e. Firefighting response plan;

f. Flooding response plan;

g. Rescue response plan;

h. Medical emergency response plan;

i. Dewatering equipment available;

j. Alternative firefighting provisions in the case of disabled fixed firefighting systems;

k. Preventive measures that will be taken to ensure ship's safety during fire or flooding emergencies;

l. Procedures for notifying; methods to account for; and off ship muster point locations for the ship’s crew, Government/Operating Company personnel, Government/Operating Company contractors, and visitors located at the shipyard of a fire emergency;

m. Names and contact information of shipyard personnel that can be contacted in case of an emergency;

n. Specific mitigation actions for any ship fire detection systems or firefighting system required to be disabled for maintenance, repair, and/or modifications;

o. Point of contact for conducting daily joint fire safety walkthrough of the ship, including point of contact for fire safety concerns on night shift for 24/7 coverage;

p. Incident command location;

q. Hot work procedures;

r. Planned off ship firefighting support capabilities, including peirside firemain size and capacities to include pressure and gallons per minute, all firemain riser connection locations and sizes around the pier/dock, and other shipyard firefighting resources;

s. Shipyard firefighting systems dedicated to the ship availability, including temporary shipyard firefighting systems to be deployed on the ship, interfaces/shore-side water supply to ship’s firemain system, fire and smoke spreading mitigation plan, compartment closure plan, quick disconnect locations at fire/smoke boundaries as required by this work item;

t. Planned access cuts.

7.1.2 The Fire and Safety Plan must address the following measures to be provided by the contractor to ensure adequate protection is provided:

a. The plan must include a copy of the Fire Control Plan, , Reference 2.5, where ingress and egress routes are clearly marked, along with locations of contractor placed temporary firefighting manifolds, contractor connections to vessel’s fixed firemain and contractor provided portable fire extinguishers. A copy of this temporary Fire Control Plan shall be posted in common passageways near machinery space ingress/egress locations and at the gangways.

b. The contractor's submitted Fire and Safety Plan must indicate if ship's firemain can be pressurized and utilized (with Shipyard confirmation that no work or isolation of shipboard firemain is planned), and Shipyard facility connections to the firemain including isolation valve locations.

c. All temporary fire suppression measures shall be located to not present any kind of hazard or blockage to normal or emergency safe passage. All temporary equipment shall be staged and marked for easy recognition by both contractor and shipboard personnel.

7.1.3 Shipyard to provide condition report that requirements of approved fire and safety plan are met prior to commencing hot work.

7.2 Place additional contractor provided portable fire extinguishers (PFE) for spaces that are protected by a gaseous fixed fire suppression system(s). The contractor must provide these PFEs regardless if the fixed system is being disabled and must be installed prior to the system being disabled. These PFEs are in addition to those already installed as detailed in Reference 2.5. Each contractor provided PFE must have a current inspection and not require an internal cylinder inspection for the duration of the overhaul. Each PFE must be located adjacent to an entrance/exit or evenly distributed among the primary and secondary egress routes. The contractor must clearly state these requirements in their Fire and Safety Plan documentation.

Location	10# ABC	20# Fast Flow PKP	15# CO2	2.5 Gal K
Main Space	2/deck			
Aux Machinery Space	2/deck			
Steering Gear	2/deck			
Emergency Diesel		4/space	2/space	

Location	10# ABC	20# Fast Flow PKP	15# CO2	2.5 Gal K
Generator				
Pump Room	2/deck			
Locker(Paint, Lube Oil, Grease, Gas Cylinder)-	1/space			
Motor Room	1/space		1/space	
Storerooms(Cargo, Flammable Liquid)	2/space			
Galley	1[1]			1/space[2]
Incinerator	1/space			
Bow/Stern Thruster	2/space			
Purifier Room	2/space			
Medical Space			1/space	
All Other Spaces	1/space			

[1]One 10# ABC per Galley Exit
[2]One 10# K per space containing a Deep Fat Fryer

7.3 In the event that the contractor requires the disabling of fixed fire suppression systems, using Reference 2.3 as guidance, immediately upon vessel's arrival at a shipyard and/or before any work is done in a shipboard space equipped with fixed fire suppression system, the contractor shall provide the services of an ABS Recognized Service Supplier for Firefighting Extinguishing System Inspection/Maintenance Firm manufacturer's representative to temporarily disable the ship's fixed fire suppression system(s) for those affected spaces. A list of ABS Recognized Service Suppliers can be found at the following URL: https://www.eagle.org/ABSEaglePrograms/es/es-search.jsp.

7.4 The contractor is responsible to determine from the work items which spaces require deactivation of fixed fire suppression systems. The contractor shall ensure the following conditions are met before isolation takes place and shall ensure these conditions remain in place while the fixed fire suppression system(s) is/are isolated. The contractor must clearly state these requirements in their Fire and Safety Plan documentation.

7.4.1 Adequate temporary firefighting system, as offered by the contractor in the Fire and Safety Plan shall be in place and verified by the OMT REP to meet the requirements of this entire work item. The contractor shall ensure that all areas of the affected space are adequately protected with the temporary system.

7.4.2 The fixed extinguishing system shall be locked out / tagged out by both Ship's Force representative and the assigned shipyard representative using requirements found in MSC Shipboard SMS Procedure 2.1-004-ALL "Lockout / Tag out" or Operating Company "Lockout / Tag out" SMS Procedure. The system shall be de-activated by removing as few components as possible. This permits the system to be quickly re-activated in case of emergency.

7.4.3 All bilges shall be free of oil or oily water and shall remain free of oil or oily water for the duration of the overhaul, as is clearly required per Work Item 011. At the beginning and end of each shift, the contractor and Chief Engineer's representative shall inspect the space to ensure the bilges remain clear of fire hazards.

7.4.4 Minimize the movement of oil, either pressurized or by gravity in ship's piping. Temporary fire extinguishing systems must be in place and operational.

7.4.5 Ensure all doors, hatches, scuttles, and other exit openings remain working and accessible for escape.

7.4.6 When physically servicing low pressure or high pressure CO2 fixed fire suppression system(s), ensure all inward opening doors, hatches, scuttles, and other potential barriers to safe exit are locked open, braced, or otherwise secured in the CO2 Storage Room and the CO2 protected space so that they remain open and accessible for escape.

7.4.7 Contractor shall provide the services of an ABS Recognized Service Supplier for Firefighting Extinguishing System Inspection/Maintenance Firm manufacturer’s rep to return the fixed fire suppression system(s) to normal condition and proven fully operational prior to Machinery Turnover milestone or when work is completed in the affected spaces.

a. Placards shall indicate that the fixed fire suppression system is disabled and shall be posted at all accesses to the affected space(s) and at the ship’s gangway.

7.5 Immediately after ship’s arrival conference, the shipyard shall provide Shipyard Safety Awareness Training (SSAT). Additionally, the shipyard shall make available a supplemental Safety Awareness Training video or hold a Safety Awareness Training class for new or visiting MSC crew or representatives prior to gaining access to the vessel.

7.5.1 Shipyard Safety Awareness Training (SSAT) shall be mandatory for all ship’s personnel, MSC personnel and any Government provided contractors. This training can be conducted onboard the vessel or in a shipyard facility and should be approximately 1 hour in length. At a minimum, SSAT shall address the following:

a. Benefits of SSAT

b. Personal Protection Equipment (PPE) while in the shipyard

c. Awareness of unsafe areas and conditions such as loading/unloading zones and crane operations

d. Observance and compliance with shipyard safety signs, alarms and markings

e. Shipyard confined space entry program including gas free requirements, entry permitting, safety considerations and restrictions, and shipyard confined space rescue procedures. The shipyard confined space entry program and training shall conform with Reference 2.4

f. Shipyard working aloft safety, to include scaffolding safety and fall protection safety

g. Shipyard vehicle traffic areas

h. Reporting accidents, near misses, unsafe conditions or emergencies

i. Contractor shall provide emergency POCs and phone numbers

j. Shipboard stainless steel welding procedure for containment and personal protection

7.5.2 Conduct a Firefighting and Fire Prevention Conference immediately after SSAT Training detailed in paragraph 7.5.1. At this conference, the contractor shall address and familiarize the ship's force with the contractor's procedures for fire prevention, firefighting and all procedures that will be implemented by municipal firefighting organizations as required by Title 29 CFR 1915.505(b)(2). Contractor in conjunction with ship's force will familiarize non-ship local municipal emergency personnel from all duty rotations with ship's layout of spaces and firefighting equipment locations, the ship arrangement, shipboard fire protection and firefighting systems, equipment and organization and familiarize all parties with the scope of work and aspects of the work or ship conditions that have significance in fire prevention and firefighting.

7.5.3 The shipyard must conduct procedural training on how to safely secure each quick disconnect fitting type found on all temporary services detailed in paragraph 7.15 during the Firefighting and Fire Prevention Conference.

a. Samples of each quick disconnect fitting must be made available to physically demonstrate how to secure and disconnect each safely.

b. Contractor must delegate two shipyard employees per shift, one primary and one alternate, and the vessel's Safety Officer will delegate two members of ship's force per shift, one primary and one alternate that will be responsible for securing quick disconnect fittings for each Class "A" boundary penetration.

7.5.4 During the Firefighting and Fire Prevention Conference conduct training on what to do upon discovering a fire.

a. Sound the alarm by activating the nearest fire alarm pull station, passing the word via phone or radio to the gangway, or by sending other personnel to pass the word.

b. Once the word is passed, secure the area by making closures to doors, hatches, scuttles and/or vents in the direct evacuation path.

c. Whenever possible, upon evacuation, pass along as much information as possible to Shipyard Management and a Vessel's Officer. Information such as the fires location, what is on fire, type of fire, size of fire, etc. Then promptly evacuate the vessel.

d. Shipyard personnel in conjunction with Ship's Force will be responsible for securing all quick disconnect fittings and making all remaining vertical and horizontal fire boundaries closures.

7.6 Temporary Firefighting Manifolds:

7.6.1 Contractor must temporarily provide and maintain a minimum of one Temporary Firefighting Manifold for every 200 feet of vessel length rounded up to the next multiple of 200 feet for the duration of the overhaul. (i.e. 805 feet equals 5 temporary firefighting manifolds onboard the ship). The contractor must clearly state these requirements in their Fire and Safety Plan documentation.

7.6.2 Contractor must utilize a sufficient number of 2½ inch, 3-inch, or 4-inch size fire hoses supplying the temporary firefighting manifolds in order to efficiently carry the required flowrate and pressure. These fire hoses must meet the requirements of Underwriters Laboratory 19(UL19) or MIL-H-24606. Each temporary firefighting manifold must include three, 1.5 inch diameter valved outlet connections and an appropriately sized inlet connection.

7.6.3 Temporary firefighting manifolds must be evenly spaced amongst the vessel and placed where each part of the vessel can be reached by at least two fire hoses feed from separate temporary firefighting manifolds.

7.6.4 Water supplying each temporary firefighting manifold must be capable of maintaining a minimum 150 psig static pressure as measured by a contractor installed, in-line, pressure gauge at each temporary firefighting manifold inlet connection.

a. All gauges must be calibrated within the last 12 months and be marked with a sticker with the current calibration date. Gauges must not require recalibration for the duration of the overhaul.

7.6.5 Two of the valved outlet connections must have a 100-ft length fire hose and a vari-nozzle pre-connected to each. Two each fire hoses and two each vari-nozzles total per temporary firefighting manifold. Neatly stow the fire hoses on a hose saddle in a ready and accessible state. Mount two spanner wrenches for each size fire hose coupling at each temporary firefighting manifold. Provide four each additional 100-ft length fire hoses and one each additional vari-nozzle at each temporary firefighting manifold. Neatly stow the hoses in a ready and accessible state.

a. Fire hoses must meet the requirements of UL19 or MIL-H-24606. Fire hoses must have 1.5 inch couplings with a hose diameter of 1.75 inches. Fire hose couplings must match the vessels hose thread, either NH or NPSH, as detailed in reference 2.6. Fire hoses must have been successfully hydrostatically tested within the last 12 months and be stenciled with the current hydrostatic testing date. Fire hoses must not require requalification by hydrostatic testing for the duration of the overhaul.

b. Vari-nozzles must be 46 CFR 162.027 USCG Type Approved or meet the requirements of MIL-N-24408, have a pistol grip, 1.5 inch couplings, have matching threads and be rated to 95 gpm at 100 psi.

7.6.6 Clearly identify each temporary firefighting manifold water hose station. Post a placard noting source of water supply and providing clear operating instructions at each station. Where weather conditions may cause freezing, provide a recirculation capability to prevent freezing at each manifold.

7.6.7 Each temporary firefighting manifold must be capable of delivering water to both connected fire hoses simultaneously. Conduct a visual inspection and operational test of each Temporary Firefighting Manifold in the presence of the OMT REP, Ship’s designated representative and Shipyard quality assurance representative. Verify that all firefighting appliances are correctly installed and in place prior to operational test. To operationally test each temporary firefighting manifold, fully layout both fire hoses so that each one fire hose reaches the farthest capable horizontally and the other reaches the highest capable vertically with a maximum 300 foot per hose. Verify minimum discharge pressure of 60 p.s.i at a 95 gpm flow rate at each nozzle while discharging both fire hoses simultaneously. Repeat this test for each remaining temporary firefighting manifold, one at a time. Submit a copy of the checkpoint results in a condition report to the OMT REP. This test shall be re-accomplished upon any vessel movement to another pier or drydock.

7.7 Fixed Firemain System:

7.7.1 Contractor must temporarily provide water connections from the shore to the vessels fixed firemain system for the duration of the overhaul. The contractor must clearly state these requirements in their Fire and Safety Plan documentation.

7.7.2 Contractor must utilize a sufficient number of 2½ inch, 3-inch, or 4-inch size fire hoses supplying the vessel's fixed firemain system in order to efficiently carry the required flowrate and pressure. Connect to at least two widely separated locations as proposed by the contractor and approved by the OMT REP. At each shipboard connection point, install a contractor supplied cut off valve.

a. These fire hoses must meet the requirements of UL19 or MIL-H-24606. Fire hoses must have been successfully hydrostatically tested within the last 12 months and be stenciled with the current hydrostatic testing date. Fire hoses must not require requalification by hydrostatic testing for the duration of the overhaul.

7.7.3 Fixed firemain system water supply must be capable of maintaining a minimum 100 psig static pressure and a maximum pressure shall not exceed the system maximum operating pressure as detailed in Reference 2.6, as measured by a contractor installed, in-line, pressure gauge at each fixed firemain system connection. Total water supplied to fixed firemain system must be a minimum of 1000 gpm.

a. Fixed firemain system supply connections must be filled with water and charged up to the shipboard connection point at the contractors supplied cutoff valve. The fixed firemain system will remain dry for dry systems as detailed in Reference 2.6, otherwise the system will remain wet.

b. All gauges must be calibrated within the last 12 months and be marked with a sticker with the current calibration date. Gauges must not require recalibration for the duration of the overhaul.

7.7.4 Clearly identify these connections to the fixed firemain system. Post a placard noting source of water supply and providing clear operating instructions at each connection. Verify unobstructed flow. Where weather conditions may cause freezing, provide a recirculation capability to prevent freezing at each connection.

7.7.5 Conduct a visual inspection and operational test of fixed firemain system in the presence of the OMT REP, Ship's designated representative and shipyards quality assurance representative. Verify that all connections are correct and in place prior to operational test. Prior to being installed, contractor supplied cut off valves must be pressure tested to at least 1.5 times the maximum working pressure of the vessel's firemain system as detailed in Reference 2.6. Allowable seat leakage, none. To operationally test the fixed firemain system, operate a 1.5 inch fire hose feed from the highest and farthest shipboard fire station from the shore connection. Verify minimum nozzle discharge pressure of 60 p.s.i and a flowrate of 95 gpm at each location. Submit a copy of the checkpoint results in a condition report to the OMT REP. This test shall be re-accomplished upon any vessel movement to another pier or drydock. Upon successful test, the firemain must be returned to normal operating condition.

7.8 When the work package requirements put any part of ship's permanent firemain in an unusable condition where fire hose coverage cannot be provided using the ship's fixed firemain and where fire hose coverage from the temporary firefighting manifolds in Paragraph 7.6 cannot provide coverage, then the contractor must submit a condition report to the OMT REP detailing the deficiency.

7.9 Assign fire watch personnel for each welding, brazing, burning or other contractor responsible hot work operations. Contractor fire watch personnel shall meet the requirements of References 2.1 and 2.2.

7.9.1 Fire watch shall be exclusive of personnel performing the actual hot work. Individual fire watch personnel shall be assigned to 'each' hot work activity. Contractor shall not split fire watch responsibilities for adjacent hot work activities.

7.9.2 Fire watch personnel shall be trained to combat various class and type of fires and shall know the regulations, procedures and facilities for sounding an alarm and shall carry them out in the event of fire.

7.9.3 Fire watch personnel shall be on location prior to the start of hot work and shall remain on location a minimum of 60 minutes after the hot work is completed.

a. Assign additional fire watch personnel on the opposite side of a structure on which welding, cutting or heating is being performed, as required (inclusive of all interconnected plating areas in way of shell, decks, overheads, bulkheads, and intersecting corners connecting more than one tank or space).

7.9.4 Fire watches shall be equipped with suitable fire extinguishing equipment to combat and extinguishing class “A”, “B”, and “C” fires. Fire watch equipment shall be maintained in state of readiness for instant use. Ship’s fire extinguishers shall not be used for fire watch. Fire watches must be able to effectively communicate with OMT Rep and Ship’s Force Personnel.

7.10 Provide written notice for each job or separate area of hot work aboard ship. Submit one copy of each notice to the OMT REP and one notice to the ship’s Chief Engineer or designated representative each morning at the morning production meeting for all hot work planned for the next 24 hours. The notice shall state the following:

7.10.1 A description of the work to be performed

7.10.2 The specific location of the hot work and compartments adjacent to decks, bulkheads and similar structure upon which work is to be accomplished

7.10.3 Time hot work will commence

7.10.4 Current gas free status of the areas, including a copy of the applicable Gas Free certificate

7.10.5 Absence or existence of combustible material in the vicinity of the operation and if combustible material exists the action taken to protect material and equipment adjacent to the hot work area from damage or fire

7.10.6 The provision and assignment of the fire watch, and affirmation that conditions at work site (ventilation, temporary lighting, and access) permit the fire watch to observe all areas where the hot work is being performed

7.10.7 The notice shall be signed by a supervisor specifically designated as responsible for coordination of hot work and fire watch requirements

7.11 In the event that a fire does occur onboard the vessel, contractor shall notify the ship’s duty officer and the OMT REP immediately. Contractor shall prepare and submit a preliminary draft report within 24 hours of the occurrence, including information on cause of the fire, actions taken to extinguish the fire, personnel involved, any damage or injuries, and corrective actions being taken by the contractor to prevent future occurrences. A final formal report shall be submitted within 5 business days after the occurrence.

7.12 Contractor shall provide communication system for contractor’s supervising personnel and workers. Contractor may use the ship’s public address system only in an emergency.

7.13 When flammable gas cylinders are required onboard ship, they shall be located on the weather decks. The number of in-use cylinders shall be limited to those which are required for work in progress and which have pressure regulators connected to the cylinder valve. Onboard reserve gas cylinders shall not exceed one-half the number of in-use cylinders and shall be located in a remote area of the weather decks.

7.13.1 When not in use, gas cylinders onboard shall have lines disconnected and pulled back, protective cover (cap) in place and shall be secured and in an upright position.

7.14 Fire exits and ship's access shall be maintained clean and clear. Contractor shall clean the ship's gangways on daily basis as required.

7.15 Rigging of hose, welding leads and temporary lights shall be kept clear of decks on temporary trees or brackets and be arranged to avoid tripping and other safety hazards. Passageway shall be kept and maintained clear of obstructions.

7.15.1 Non-Fluid Systems:

a. Quick disconnect fitting(s) must be installed within ten (10) feet of where all temporary services pass through penetration(s) designated as, Class "A-60" & "A-30" Bulkhead(s), Overhead(s) and Decks that are detailed in Reference 2.5.

7.15.2 Fluid Systems:

a. Quick disconnect fitting(s) must be installed with use of isolation valves within six feet of where all temporary services pass through penetration(s) designated as, Class "A-60" & "A-30" Bulkhead(s) , Overhead(s) and Deck(s) that are detailed in Reference 2.5. Quick disconnect fittings with isolation valves must be capable of being disconnected safely on pressurized temporary systems..

7.15.3 Indicate each Class "A-60" & "A-30" by installing a sign adjacent to each entrance. Mark each sign with international orange tape. All quick disconnect fitting must be marked with international orange tape on both sides of each quick disconnect.

7.16 Designate a safety inspector(s) responsible for ship's safety on daily basis during performance period.

7.16.1 Perform a fire safety inspection once per working shift. The inspection shall be made jointly with the OMT REP or their designated representative, a representative of the ship's Master and a representative from the shipyard. A written report of the discrepancies and three copies shall be submitted to the OMT REP within four hours after completion of the inspection.

7.16.2 As part of the inspection the following must be inspected once per shift:

a. Current condition of all fixed fire suppression and monitoring system(s)

b. Temporary fire suppression system must be fully operational and placards detailing operating instructions must be in place

c. Verify the bilge(s) are free of oil, fuel, water, dirt and debris

d. Visually inspect hotwork location(s) for a valid hot work permit; adequate deck, bulkhead and equipment protection; and location(s) have adequate fire watch personnel

e. Fire screen doors and water tight doors are closed, as required

f. Spaces that require entry are gas free.

g. Verify quick disconnect fittings are properly installed where temporary services passes through, Class "A-60" & "A-30" Bulkhead(s), Overhead(s) or Deck(s)

h. Verify the pier and vessel remain accessible and approachable by shore base fire-fighting or other emergency equipment at all times and tides

i. Discrepancies found must be corrected within 24 hours unless adjudicated upon discovery by the Safety Inspection Team. All discrepancies found during the walkthrough must be submitted in a Condition Report to the OMT REP at least once per day. Detail discrepancies that cannot be corrected within 24 hours with a corrective action plan.

7.17 Preparation of drawings: None additional

7.18 Manufacturer’s Representative

7.18.1 See requirements in Paragraphs 7.3 and 7.4.7.

8.0 GENERAL REQUIREMENTS: None additional.

USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0017A	CATEGORY "A"	2026-01-30
ESB-SWI-HANDLING SHIPS STORES - STORING		Manovich, David

1.0 ABSTRACT

Offload, store, and load chill/frozen and dry stores.

2.0 REFERENCES/ENCLOSURE: None

3.0 ITEM LOCATION/DESCRIPTION

3.1 Quantity/Description

3.1.1 Five tons frozen

3.1.2 Two tons chill

3.1.3 20 tons of miscellaneous dry stores

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 Freeze/chill boxes shall be placed as near the ship as possible.

5.2 Freeze boxes to be maintained between 0 degrees and 5 degrees Fahrenheit. Chill boxes to be maintained between 33 degrees and 35 degrees Fahrenheit.

6.0 QUALITY ASSURANCE REQUIREMENTS: None Additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide labor, equipment and crane services necessary to off load dry and general stores, medical supplies, spare parts and frozen and refrigerated stores from their stored locations aboard ship. Store same throughout the availability period and return to the vessel and load when directed by the OMT REP.

7.1.1 Provide secure storage facilities for the dry and general stores, medical supplies, and spare parts described in Paragraph 3.1. Prior to off-load all dry stores shall be jointly inventoried and inspected by the Chief Steward and a Contractor's Representative. Any damaged material shall be documented at this time. Dry stores shall be removed when directed by the OMT REP and the Chief Steward and/or Supply Officer and shall be stored within secured, weatherproof, enclosed facilities.

7.1.2 Frozen and chill stores shall be removed from the vessel when directed by the OMT REP and the Chief Steward and stored within temperature controlled and monitored frozen and chill boxes placed near the ship. Provide unit pricing per day for frozen and chill boxes. Prior to off-load, all frozen and chill stores shall be jointly inventoried and inspected by the Chief Steward and a Contractor's Representative. Any damaged, spoiled or otherwise unusable food material shall be documented at this time, and shall be disposed of by the contractor as directed by the OMT REP.

7.1.3 Frozen and chill boxes not located immediately adjacent to the vessel shall require use of contractor furnished frozen and chill trucks for transporting stores to and from the vessel.

7.1.4 Frozen and chill boxes shall have both external reading recording type and standard thermometers for monitoring box temperatures. Frozen and chill box temperatures shall be monitored by the contractor every four hours. Any malfunctions or irregularities shall immediately be corrected and reported to the attention of the OMT REP and the Contractor’s Representative. Two copies of a temperature graph for each box shall be submitted to the OMT REP weekly.

7.1.5 When directed by the OMT REP return and load all stores aboard the vessel and return to their original locations as directed by the Chief Steward and/or Supply Officer. Upon return to the vessel all stores shall again be jointly inventoried and inspected by the Chief Steward and a Contractor's Representative. Any damaged material shall be documented at this time.

a. The contractor shall be responsible for any loss or damage to ship's stores which has occurred during storage by the contractor. At the discretion of MSC, the contractor shall be required to replace the lost or damaged stores in kind, or the stores shall be replaced at the Government's expense and the value of the contract be reduced accordingly.

7.2 Preparation of Drawings: None additional

7.3 Manufacturer's Representative: None

8.0 GENERAL REQUIREMENTS: None Additional

USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0018	CATEGORY "A"	2026-01-30
ESB-SWI-DELIVERY AND REDELIVERY OF THE VESSEL		Manovich, David

1.0 ABSTRACT

This item describes the contractor's responsibilities associated with the delivery, care, and redelivery of the ship.

2.0 REFERENCES

2.1 NAVMED P-5010-6, Manual of Naval Preventive Medicine, Chapter 6, Water Supply Afloat

2.2 MSC Drawing No. 582-8615624, Mooring and Towing Arrangement

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: Entire vessel

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with all applicable GTR requirements.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract to determine their effect on the work required under this work item.

5.3 The definitions of many terms used in this work item are found in Work Item 001.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Delivery

7.1.1 The contractor shall provide tugs and pilots in accordance with USCG and local requirements to guide/escort the ship from the sea buoy on the approach from open ocean to a transfer point in the navigable channel as close as possible to the contractor's facility as agreed upon by the OMT REP and the contractor.

7.1.2 The Government/Operating Company shall deliver the ship to the contractor at a transfer point in the navigable channel as close as possible to the contractor's facility as agreed upon by the OMT REP and the contractor. The contractor shall accept custody of the ship upon delivery and assume total responsibility for its safety, security, and protection. The contractor shall furnish tugs, pilots, equipment, mooring lines, line handlers, safe berth, etc. to accept delivery of the vessel and safely berth the ship at the Contractor's facility.

a. Contractor provided mooring lines for use on the ship shall all be of same manufacture, type, size and breaking strength, and shall be free from excessive wear or defect. Use of mixed lines to moor the ship is not permitted.

b. Contractor shall provide mooring lines with rated breaking strength equivalent to the ship's lines per reference Mooring Arrangement Drawing (Reference 2.2). Maximum breaking strength of contractor provided mooring lines shall not exceed rated strength of ship's lines.

c. In the event that the ship needs to be reoriented or shifted at the berth or pier to accomplish any of the work required by this work package, the contractor shall furnish tugs, pilots, equipment, mooring lines, line handlers, safe berth, etc. at the expense of the contractor to enable the ship to safely be reoriented or shifted berths or piers at the Contractor's facility. If the ship is shifted from berth or pier all requirements of WI 0011, General Services for Ship, shall be maintained.

7.2 Re-delivery

7.2.1 Upon completion of successful testing, dock and sea trials, the contractor shall provide all line handlers, tugs and pilots to redeliver the ship to the government at a transfer point in the navigable channel in the vicinity of the contractor’s facility, as agreed upon by the OMT REP and the contractor. At the time of redelivery, the contractor shall ensure that the ship is ready for service in accordance with this work package. The ship shall be thoroughly cleaned, with all parts of the ship free of dirt, debris, and garbage.

7.2.2 Potable water systems and tanks, if entered or worked by the contractor, (including piping) shall be super-chlorinated, drained, replenished and certified fit for consumption by a qualified medical testing facility prior to dock trials, in accordance with the requirements of Reference 2.1.

7.2.3 The contractor shall provide tugs and pilots in accordance with USCG and local requirements to guide/escort the ship from the transfer point in the vicinity of the contractor’s facility for the transit back to open ocean. In the event that sea trials and the ship repair period conclude at the sea buoy, such that the ship does not return to the Contractor’s facility, the OMT REP will issue a credit RFP to recoup the costs for these tug and pilot services.

7.2.4 Existence of any major uncorrected deficiencies, for which the contractor is responsible that adversely affect the safe navigation and/or proper operation of the ship for its intended service, shall be sufficient cause to reject redelivery of the ship pending correction of the item(s) in question. Consequential delays in redelivery, as a result of uncorrected deficiencies or unfinished work shall be the contractor’s responsibility. All tests and inspections required by this work package shall be completed prior to redelivery of the ship. Existence of a large number of uncorrected deficiencies will be sufficient cause for rejection of redelivery.

7.2.5 To ensure that the ship is in proper condition for redelivery, a final joint acceptance survey of the ship will be made by the OMT REP and the contractor at least five days prior to the scheduled redelivery date. This acceptance survey will include a completion status review. Based on this survey, an agreement shall be reached between the contractor and the OMT REP regarding all remaining work requirements, the extent of further cleaning and correction of deficiencies that must be completed prior to the OMT REP accepting redelivery of the ship.

8.0 GENERAL REQUIREMENTS: None additional

USSHershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0019	CATEGORY "A"	2026-01-30
ESB-SWI-SHIPBOARD ACCESS AND SECURITY		Manovich, David

1.0 ABSTRACT

This item describes the requirements for security services and shipboard access.

2.0 REFERENCES

2.1 COMSCINST 5530.3 Series, Military Sealift Command (MSC) Shipboard Force Protection (FP) Program

2.2 NAVSEA Standard Item 009-72 (current revision)

2.3 MSC Drawing No. 801-8495724, Booklet of General Plans

3.0 ITEM LOCATION/DESCRIPTION

3.1 LOCATION: Entire vessel at the contractor's facility

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES: None

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Security plan: Submit security plan in accordance with Reference 2.2.

7.2 Shipboard access

7.2.1 No less than five (5) days before scheduled ship arrival, the contractor shall deliver to the OMT REP a list of all contractor and subcontractor (including contractor furnished technical representatives) personnel who will be involved in the onboard performance of the contract. The list shall contain the names, identification numbers, and if available, the security clearance of all contractor and subcontractor personnel.

7.2.2 Prior to the commencement of each weekend or unscheduled overtime shift, not identified in the contractor's production plan schedule as originally submitted with the contractor's proposal (or subsequently revised during contract performance), the contractor shall provide to the OMT REP a list of all contractor and subcontractor (including contractor furnished technical representatives) personnel scheduled to work each shift. The list shall contain the names, and if available, the security clearance of all contractor and subcontractor personnel.

7.2.3 The contractor shall have and demonstrate for approval at the time of submitting his proposal, an employee identification badge/pass and security system in accordance with References 2.1 and 2.2. The requirements of References 2.1 and 2.2 are for all direct employees to be issued badges which as a minimum have:

a. Bust photo minimum size: 1"×1¼"

b. Name of employee (typed)

c. Badge serial number (printed)

d. Employee's position/trade (typed)

e. Employee's signature

f. Validation officer's name (typed)

g. Validation officer's signature

h. Expiration date (typed)

i. Lamination of badge

7.2.4 Temporary individual badges shall be issued to all subcontractor workers and visitors. Subcontractor and visitor badges shall be clearly identifiable as such from those of full-time employees by being distinctly different in color and marking from those issued direct employees and each from the other.

7.2.5 Badge records/inventories shall be maintained:

a. An updated record (weekly) shall be maintained for all effective badges issued and surrendered.

b. All temporary (subcontractor and visitor) badges shall be numbered and inventoried daily.

c. When for any one of the individual badge types (direct employee, subcontractor, visitor), a loss of ten percent occurs, that entire series of badges shall be canceled and reissued with a new series of numbers and colors.

d. Inventories shall be made available to the OMT REP for review upon demand and shall be delivered as a contract deliverable for this work item at the end of the performance period.

7.2.6 The contractor shall obtain from the ship’s Master and OMT REP a list of crew members (and changes thereto as they occur), Government technical representatives, and other known visitors for each day. These lists shall be the basis for MSC access to the ship.

7.2.7 The OMT REP and the ship’s Master may deny access to the ship to contractor personnel if the personnel list required by Paragraph 7.2.1 is not delivered prior to the start of the performance period.

7.2.8 The OMT REP and the ship’s Master may deny access to the ship to contractor personnel, on a shift by shift basis, if the personnel list required by Paragraph 7.2.2 is not delivered prior to the start of weekend or unscheduled overtime shifts.

7.3 Shipboard Security

7.3.1 Uniformed security guards shall be posted ashore immediately adjacent to the ship's gangway(s) on a 24-hour around the clock basis. The guards shall be other than full time production workers assigned for this task on a temporary basis. The use of a shipyard installed 24-hr monitored central monitored camera system does not remove the requirement for 24-hr gangway personnel or roving patrol personnel.

7.3.2 The guards shall ensure that only authorized personnel are allowed onboard, this to include: OMT REP and associated overhaul management team staff, all ship's force, contractor workers, subcontractor workers, Government furnished technical representatives, and visitors (MSC and others when prior authorization is issued by OMT REP).

7.3.3 Verification of each individual's identity shall be required by presentation of a valid contractor issued identification card or Government issued identification card. Only these documents will be acceptable identification.

7.3.4 Unannounced visitors without contractor issued or government issued identification are not permitted onboard. The guard shall immediately notify the OMT REP, the ship's Master and the contractor’s project manager to verify the visitor's identity.

7.3.5 Visitors log

a. A daily log shall be maintained for visitors who do not possess contractor furnished or government furnished identification.

b. These visitors shall be logged in and out. The log shall contain entry blocks for the printed name, organization represented, type of identification presented (company ID, driver’s license), the ID card number, time onboard, time departed and signature of the visitor, and name of crew escort. A new page shall be started for each new day starting at 0001 hours.

c. In addition to the shipyard’s visitors pass, visitors coming onboard the ship shall be issued a shipboard visitor’s pass by the gangway watch readily identifiable from those of the shipyard and subcontractor workers by a unique color.

7.3.6 Secondary gangways rigged for emergency egress only do not require a dedicated security guard.

7.3.7 The contractor shall ensure notification of contractor and subcontractor (including contractor furnished technical representatives) personnel and all visitors, that a precondition for being granted access to the ship is that all packages, suit cases, briefcases, boxes, tool bags, and tool boxes going onboard or ashore may be subject to inspection for the following prohibited articles: Weapons, explosives, government property or property belonging to the crew. That any person refusing to submit to such inspection when boarding the ship will be refused access to the ship. That any person refusing to submit to inspection upon departing the ship may be subject to detention for civil authorities and being prohibited further access to the contractor’s facility and the ship.

a. The guard(s) shall be required to inspect all visitors’ (and randomly inspect all other persons’) packages, suitcases, briefcases, boxes, tool bags and tool boxes going onboard or ashore for: weapons, explosives, government property or property belonging to the crew. The rate of random inspections shall be mutually agreed upon and monitored by the shipyard superintendent and OMT REP so as not to impede production. An attempt, without resorting to force, shall be made to detain at the gangway anyone attempting to bring any prohibited item(s) onboard, remove prohibited property from the ship, or refusing to submit to an inspection. The shipyard security force shall be immediately notified.

b. The contractor shall ensure that painted signs with one-inch lettering giving notice of the conditions for access described in Paragraph 7.3.7 are prominently displayed at the foot of each gangway. Signs shall be in English language, and additionally in the language of the local area.

7.3.8 A roving patrol is required on the ship on a 24-hour basis. This is to be a separate and distinct individual from the gangway guard. The roving patrol shall be other than full time production workers assigned for this task on a temporary basis.

a. The roving patrol shall continuously tour all spaces, Reference 2.3 is provided for guidance. At a minimum, all spaces will be toured once every 2 hours. Spaces to be inspected shall include but not limited to:

1. The lowest level of all machinery spaces; engine room(s), auxiliary machinery room(s), pump room(s), shaft alley(s), generator room(s), boiler/fire room(s), steering gear room(s), thruster room(s), and all other main and auxiliary machinery spaces

2. All engineering, cargo hold, cargo hold cofferdams, voids, and rosebox(es), pipe tunnel(s), dry, general material, refrigerated and frozen, paint and flammable, boatswain's, sponsor, and other storerooms

3. All communication, scientific data, telemetry, sponsor data, sonar trunk(s), laboratories, work and

machine shops

4. All galley(s), pantry(s), mess rooms, lounges, hospital(s), ship's exchange(s), (P.X.), berthing areas, toilets and showers, and other spaces

5. All strong room(s) and gun locker(s)

b. The roving patrol shall be required to enter all unlocked spaces and traverse same in a manner that necessitates using an exit other than the one used for entry, except in those cases where only one entry/exit route/door exists.

c. All doors to spaces not requiring entry by contractor workers in the performance of the repair work set forth in the work items of the specifications shall be checked to ensure they are locked.

d. All such spaces will be locked by the ship's force upon arrival at the contractor's facility. The contractor’s project manager shall ensure that the head of the shipyard security force has assigned a responsible individual to perform an initial inspection of all spaces with a ship’s force officer to identify the locked spaces and prepare a list of these for use by the roving patrol. The list shall be updated daily or as frequently as necessary. Doors on the list found unlocked shall be reported immediately to the head of the shipyard security department and to the shipyard project manager. Copies of the list and each revision shall be furnished to the OMT REP.

e. The roving patrol shall punch a time clock on each inspection round of the ship.

1. The contractor shall temporarily install a punch time clock system throughout the ship to demonstrate inspection of the entire ship, has been performed on a

continuous round basis.

2. Key stations for such a system shall be installed in a manner which discourages theft or tampering.

f. Time clock cards shall be kept:

1. Time clock cards shall be kept from 1200 of the preceding day to 1200 of the succeeding day. Each day's clock card shall be delivered to the gangway watchman at 1200 for the start of the next twenty-four hour period. No extra copies shall be allowed in the possession of the watchman.

2. As verification that the ship was inspected, the clock cards shall contain entry blocks on the back for the following:

(i) Ship’s name

(ii) Contractor’s firm

(iii) Contract number

(iv) Start and end date of the period clocked

(v) Signature of the first and last shift’s security watch

(vi) Signature of the shipyard security head

(vii) Signature of the shipyard project manager

3. At the end of the 24-hour period the clock card shall be signed by the head of shipyard security and the shipyard project manager and delivered to the OMT REP, as a contract deliverable document.

g. The roving patrol shall be aware of and on the alert to the signs and conditions which indicate any of the following:

1. Flooding and rising water levels in machinery spaces and holds

2. Fire

3. Water/oil damage (from leaking or broken steam, water or oil piping)

4. Explosion dangers (from escaping welding/burning and sewage gases)

5. Live disconnected electrical lines or welding leads

6. Loose and unsecured equipment or machinery which can fall/drop from heights

7. Unauthorized persons onboard the ship during work and non-work hours

h. It shall be the contractor's responsibility to provide the required training to the roving patrol on how to identify these conditions in a shipyard work environment.

7.3.9 The roving patrol and gangway guard(s) shall have portable radio communications at all times. In addition, communications with other shipyard security forces shall also be maintained.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0020	CATEGORY "A"	2026-01-30
ESB-SWI-GAS FREE CERTIFICATES		Manovich, David

1.0 ABSTRACT

This item describes the requirements for the contractor to provide and maintain Marine Chemist gas free certificates to certify spaces and systems as gas free safe for personnel entry and safe for hotwork on the ship where work will be accomplished on a daily basis.

2.0 REFERENCES: None

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Machinery rooms, pump rooms, and tanks and spaces as noted in Enclosure 2.2.1 of Work Item 021 of this work package.

3.1.2 Areas throughout the vessel where hot work is being performed by the contractor.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with all applicable GTR requirements.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract to determine their effect on the work required under this work item.

5.3 The definitions of many terms used in this work item are found in Work Item 001.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 In the United States or its territories and possessions, a national fire protection association (NFPA) Certified Marine Chemist shall be used to certify spaces gas free safe for personnel entry and safe for hot work (29 CFR 1915, Subpart B Appendix B). Outside of the United States, a Marine Chemist or Industrial Hygienist, certified in the country or locale where the work is to be accomplished, shall be utilized.

7.1.1 Contractor actions to prepare tanks and spaces to attain gas free certification shall be accomplished in accordance with USCG, OSHA (Subparts 1915.11, 1915.12, and 1915.13) and OCONUS specific locale requirements.

7.2 Test and prove gas free "Safe for Men" and "Safe for Hot Work" as applicable for all areas as listed in 3.1, including all other spaces where work requiring gas free condition is required by the work items of these specifications:

7.2.1 Cleaning required to attain gas free status is covered under individual work items.

7.3 Furnish the OMT REP with two copies of the gas free certificates signed by a NFPA certified Marine Chemist (or equivalent OCONUS Marine Chemist or Industrial Hygienist) prior to start of any confined space entry and/or hot work. Furnish a new gas free certificate, signed by a NFPA certified Marine Chemist (or equivalent OCONUS Marine Chemist or Industrial Hygienist), daily thereafter for the duration of the particular work necessitating gas freeing of spaces safe for men and hot work. One copy of said certificate is also to the posted near the gangway(s).

7.3.1 Gas free certificates shall be maintained on a daily basis. This may be accomplished by a shipyard competent person, provided same competent person is acceptable to all regulatory agency (e.g. USCG, ABS) inspection personnel during inspections. Should regulatory agency inspectors require NFPA certified Marine Chemist (or equivalent OCONUS Marine Chemist or Industrial Hygienist) certificates, the additional costs of contracting a Marine Chemist shall be borne by the contractor.

7.3.2 Shipyard competent persons shall be trained by a NFPA certified Marine Chemist (or equivalent OCONUS Marine Chemist or Industrial Hygienist), and credentials demonstrating this training shall be provided to the OMT REP prior to any confined space entries.

7.4 Any variation of originally certified conditions such as shifting vessel, changing ballast, and transport of any other fluids, etc., must be immediately brought to the attention of the OMT REP. A new Marine Chemist gas free certificate shall also be issued.

7.5 Burning or other hot work as defined by 29 CFR 1915.11(b) shall not be done by the contractor on any part of the vessel, including piping systems unless permission to do so is specifically granted by the NFPA certified Marine Chemist (or equivalent OCONUS Marine Chemist or Industrial Hygienist), documented with issued certificates.

7.6 Preparation of Drawings: None

7.7 Manufacturer’s Representative: None

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

This item describes the requirements for the contractor to open, clean, gas-free, maintain, and close miscellaneous tanks, voids, cofferdams, and spaces. Storage of re-usable fuel and lube oils is also required.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSEA Drawing No. 801-8499371, General Arrangement, USNS Hershel Williams (ESB 4)

2.1.2 NAVSEA Drawing No. 835-8495718, USNS Hershel Williams, Trim & Stability Booklet

2.2 Enclosures

2.2.1 USNS Hershel Williams (ESB 4) List of Applicable Tanks, Voids and Cofferdams to be Opened during Current Availability

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity: See Enclosure 2.2.1

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with all applicable GTR requirements.

5.2 The contractor and all subcontractors, regardless of

tier are advised to review other work items under this contract to determine their effect on the work required under this work item.

5.3 The definitions of many terms used in this work item are found in Work Item 001.

5.4 In the event that any tank contents are deemed to be hazardous wastes in accordance with federal, state and local (CONUS) or international (OCONUS) standards, the costs for handling and disposal of those wastes shall be covered under Work Item 023 of this work package.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 Tank coating system conditions shall be identified and documented by the contractor during tank surveys. ABS surveyor shall receive a copy of contractor’s reports.

6.2 Provide unit pricing for each facility listed in 7.3.3 Storage facility must be available throughout the duration of the entire availability including any extensions.

6.3

Requirement	Unit/Quantity	Price
Clean Storage Facility for Fuel Oil	1 Lot/ XXX Barrels	
Clean Storage Facility for Lube Oil	1 Lot/ XXX Barrels	
Clean Storage Facility for JP-5	1 Lot/ XXX Barrels	

7.0 STATEMENT OF WORK REQUIRED

7.1 It is the contractor’s responsibility to analyze and properly plan tank unloading procedures to ensure that no undue stresses are imposed on the vessel structure. Ship’s force and OMT REP will assist contractor with ship loading documents and software. The contractor shall utilize the services of a naval architect or structural engineer to support this planning. The contractor shall provide their tank unloading plan to the OMT REP at least two business days prior to starting this work to permit review and approval by MSC naval architect.

7.2 Contractor and ship's force shall on arrival at the contractor's facility take, record, and verify soundings of all tanks listed in Enclosure 2.2.1, using References 2.1.1 and 2.1.2 as guidance. The Chief Engineer shall review and endorse all soundings taken/recorded. Submit copy of endorsed record to the OMT REP.

7.3 Provide a clean storage facility with a capacity of approximately XXX tons (XXXX gal) of fuel oil, XX tons (XXXX gal) of JP-5 and XX tons (XXXX gal) of lube oil. Contractor designated storage facility shall be inspected by the OMT REP for cleanliness and acceptability prior to any transferring of fuel oil and lube oil. Prior to off-loading liquids, samples shall be taken by the contractor of each of the tanks to be moved to temporary storage. These samples shall be turned over to the Chief Engineer. Each sample bottle shall be labeled with tank taken from, contents, and date taken. Any visual indications of contamination shall be brought to the attention of the OMT REP immediately. Upon completion of transferring of liquids, the contractor, in conjunction with ship's force, shall validate quantities at the storage facility.

7.3.1 In the event that contamination is suspected, a change order will be issued to the contractor to have a laboratory analysis conducted on the sample.

7.4 Removals: The ship will pump down tanks to low suction, either to other tanks onboard or to shore storage facilities. In the event that the tank contents are being moved off the ship, the contractor shall provide support to the vessel, including all hoses and manpower on the shore side to accomplish this removal. Contractor shall be responsible to provide all pumps, hoses, fittings and personnel to safely remove all remaining tank contents as identified in Enclosure 2.2.1. Applicable federal, state, local or international procedures for transferring fuel and lube oils over water shall be followed by the contractor. Contents to be temporarily stored are indicated in the enclosure. All other liquids shall be disposed of in accordance with local, state, federal and/or international regulations. In some states, the presence of oil in the wash water may deem the mixture to be hazardous waste. When pricing this item, the contractor shall assume that oil is present in the tank wash water. The cost for disposal of the mixture shall be included in the price quote regardless of whether or not, due to the presence of oil, the wash water is deemed hazardous waste or non-hazardous waste. If the mixture is determined to be hazardous waste due to some contaminant other than oil, the procedures outlined in Work Item 023 shall be used for the identification, handling and disposal. The costs for removal are still part of this work item (021). All other costs for offload and disposal shall be captured in this work item.

7.5 When directed by the OMT REP, return all fuel oil and lube oil to the vessel. Prior to returning of liquids, the contractor, together with ship's force, shall take samples of all stored liquids. These samples shall be turned over to the Chief Engineer. Each sample bottle shall be labeled with temporary storage tank taken from, contents, and date taken. Any visual indications of contamination shall be brought to the attention of the OMT REP immediately.

7.6 Accomplish the following for each tank/space listed in Enclosure 2.2.1:

7.6.1 Open manholes and erect/maintain safety guards over openings.

7.6.2 Temporarily provide blowers and ventilate tanks and spaces, inclusive of confined areas around manhole covers.

a. Ventilation of the tanks shall be accomplished so as to prevent tank fumes from exhausting in the interior compartments of the ship.

7.6.3 Temporarily provide and maintain, as required, tank lamping/lighting. Lighting shall be explosion proof, safety type lights.

7.7 Clean and Gas Free tanks and spaces listed in Enclosure 2.2.1. Cleaning shall completely remove all traces of dirt, sludge, debris and liquids. All tanks and spaces shall be presented in a clean and dry condition and shall be ready to be certified and maintained safe for men/safe for hot work. The provision of gas free certificates is covered under Work Item 020 of this work package.

7.7.1 Where a battery of tanks is designated for cleaning and gas freeing, all interior spaces in way of same, such as cofferdams, voids, bilges, bilge wells and other spaces, even if not specifically mentioned, shall be considered as being included as spaces to be cleaned and gas freed.

7.7.2 For all tanks and spaces indicated in enclosure 2.2.1, air vents, sounding tubes, filling and suction lines and other appurtenances shall be included for cleaning and gas freeing. Secondary manhole covers for each tank shall also be opened.

7.7.3 Post original gas free certification at each tank/space location. Certificate shall be mounted directly at the tank/space manhole entrance.

7.7.4 Post copy of gas free certificates at the designated gangway access to the vessel. Certifications posted at gangway shall be in a protective weather/waterproof enclosure that is accessible and visible.

7.8 Surveys and testing requirements for these tanks shall be addressed in other work items in this work package.

7.9 At the conclusion of all required work, conduct a final closeout inspection for each tank/space listed in enclosure 2.2.1 with the OMT REP. No tank shall be closed until a final inspection is accomplished with the OMT REP in attendance.

7.9.1 Contractor is responsible for repairing all tank coating damage caused by work in this solicitation work package. During close out inspection, if coatings are found damaged by contractor’s work, contractor shall repair damaged areas prior to final close out of tank/space, at contractor’s expense. Damaged coatings shall be repaired in accordance with MSC paint system and the paint representative’s recommendations.

7.9.2 Ensure all blowers, lighting, staging, tools, and miscellaneous equipment are removed prior to the final inspection. Any staging necessary for OMT REP close out inspection shall be removed just prior to final acceptance close out of tank/space.

7.9.3 Present tank in final clean condition. No dirt, dust, debris or liquids shall be allowed. Any tank or space failing the close out inspection, shall be re-cleaned until passing OMT REP inspection (repeated cleanings and inspections will be at contractor expense, and may become subject to issuance of a QDR).

7.10 Upon acceptance of final closeout inspection and by direction of the OMT REP, reinstall manhole covers using new gaskets, grommets, and wicking. Submit a condition found report for any missing or damaged fasteners or hardware.

7.11 The following tanks and spaces, included in Enclosure

2.2.1, shall be opened, cleaned and prepared for gas free certification within (5) days of the shipyard availability start:

7.11.1 Gray Water Tank A1-92-6

7.11.2 Food Waste A1-92.5-2

7.11.3 All Salt Water Ballast Tanks

7.12 Required deliverable report: Contractor shall create, maintain, update, and submit weekly during the progress meeting, a tank/space work matrix. This matrix shall contain the following information:

7.12.1 Tank/Space Name

7.12.2 Tank/Space Location Designation

7.12.3 Date Tank/Space Opened

7.12.4 Date Gas Free Cleaning Started

7.12.5 Date Gas Free Cleaning Completed

7.12.6 Date Gas Free Certificate Received

7.12.7 Date ABS Initial Structural Survey Complete

7.12.8 Date Air Test/Hydro Test Completed

7.12.9 Date Final Tank/Space Cleaning Started

7.12.10 Date Final Tank/Space Cleaning Completed

7.12.11 Date Tank/Space Final Close

7.12.12 Tank/Space Related Repair Status: List major job sequence within a tank/space followed by date start and date completed (such as below):

a. Access Cuts -> date removed/date installed

b. Blasting -> date start/date complete

c. Painting -> date start/date complete

d. Related work items internal repairs (pipe, valve, structure, etc.)

e. Related work items external repairs (welding, UT’s, zincs, other structural related; etc.)

f. Fluids added/removed from tank/space (pot water chlorination; Undocking stability; fuels/oils; etc.)

7.13 Paint all disturbed surfaces to match surroundings

8.0 GENERAL REQUIREMENTS: None additional

Enclosure 2.2.1 - USNS Hershel Williams (T-ESB 4) List of Applicable Tanks, Voids and Cofferdams to be Opened during Current Availability

Mission Liquids Tanks (JP-5)				
Description	Tank Location	95% Capacity (gals)	Area (sqft)	Arrival Condition (Ltons)
JP-5 Overflow 5-97-3	97-98		TBD	TBD

Salt Water Ballast				
Description	Tank Location	95% Capacity (gals)	Area (sqft)	Arrival Condition (Ltons)
SWB Aft Peak 4-16-0	FR 7-16	172882.4306	TBD	TBD
SWB Fore Peak 3-126-0	FR 108-126	1512520.476	TBD	TBD

AWB Aux Machinery Room 4-108-0	FR 97-108	1512520.476	TBD	TBD
SWB Mission Deck Stbd. 5-97-5	FR 89-97	1390614.267	TBD	TBD
SWB Mission Deck Port 5-97-4	FR 89-97	350332.893	TBD	TBD
SWB Engine Room Stbd. 4-57-3	FR 38-57	172882.4306	TBD	TBD
SWB Engine Room Port 4-57-4	FR 38-57	172882.4306	TBD	TBD

Miscellaneous Tanks

Description	Tank Location	95% Capacity (gals)	Area (sqft)	Arrival Condition (Ltons)
INCINERATOR SLUDGE TANK (A-29-1)	31-33	752.9723	TBD	TBD
GRAY WATER HOLDING (A1-92-6)	90-92	22865.26	TBD	TBD
FOOD WASTE (A1-92.5-2)	92-92.5	1430.647	TBD	TBD

1.0 ABSTRACT

This item describes the requirements for machinery space turn-over, dock trials, and sea trials.

2.0 REFERENCES: None

3.0 ITEM LOCATION AND DESCRIPTION

3.1 Various

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIALS/SERVICES: None

5.0 NOTES

5.1 The contractor and subcontractors must consult Military Sealift Command’s General Technical Requirements, (GTR’s), to determine applicability to this work item. The contractor and all subcontractors must comply with all applicable GTR requirements.

5.2 The contractor and subcontractors shall review other work items under this contract, to determine their effect on the work required by this item. Based on this review the contractor shall plan and schedule work to minimize conflicts between work items.

5.3 The definitions of many terms used in this work item are found in work item 001.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Machinery space turn-over

7.1.1 Machinery space turn-over signifies that all contractor related work on ship systems and equipment within the machinery spaces is complete.

a. All tag-outs shall be cleared by shipyard personnel in concert with ship’s force for machinery space equipment worked on during the availability after verifying the work is complete.

b. All temporary hoses, power cables, rigging and staging has been removed from the machinery spaces. All temporary openings have been restored.

c. All tank covers have been closed with proper gasketing and bolting and all gratings are in place and properly secured within the machinery spaces.

d. All machinery space bilges are free of trash, clean, and dry.

e. All machinery space fire detection and fire extinguishing systems are fully operational.

f. All guards and shields are in place for machinery space equipment.

7.1.2 Ship’s crew shall utilize the time between machinery space turn-over and dock trials to complete MSC mandated ship start-up and readiness assessment checklists. The contractor shall assist the OMT REP and Chief Engineer in assembling information and reports necessary to complete ship start-up readiness assessments as detailed in Work Item 013 article 7.2.8.f.

7.2 Dock Trials

7.2.1 Develop and submit to OMT REP a dock trial agenda at the 50% time point of the availability. Develop and submit to the OMT REP, four days prior to dock trials, a final version of the dock trial agenda that includes the following items. (Note that under Work Item 013, the contractor shall support a pre start-up meeting scheduled to occur at least 3 days prior to the dock trial. This milestone marks the requirement for the contractor, all subcontractors and technical representatives involved in the installation, maintenance and repair of shipboard equipment and systems to meet with the OMT REP, Chief Engineer and Master to review written reports from the contractor, subcontractors and technical representatives asserting the readiness of subject equipment for start-up, commissioning and operational testing. Subject reports shall also include recommended start-up, commissioning and testing procedures to prove proper operating conditions of subject equipment.):

a. Written certification from the Contractor or the subcontractor that all work on each piece of machinery space equipment is complete and that the equipment is fully ready for start-up and operation.

b. List of equipment to be tested

c. List of tests

d. Test performance criteria, including OEM commissioning procedures

e. Test acceptance criteria

f. Test schedule

g. Test data to be recorded

h. Test data recording format

i. Instrumentation required

j. List of shipyard personnel involved

k. List of vendor technical representatives

l. List of regulatory body representatives

7.2.2 Dock trials shall be carried out jointly with ship’s force. Representatives of contractor or subcontractor shall be present onboard to witness testing and record operational testing and performance data. Dock trial results shall be summarized by the contractor and submitted to the OMT REP prior to sea trials.

7.2.3 Conduct dock trials as scheduled per work item 013.

7.2.4 Allocate four hours minimum duration for running of each equipment/system.

7.2.5 Schedule all dock trials during the day shift.

7.2.6 Perform dock trials in accordance with the published agenda.

7.2.7 Operate and test main propulsion machinery, auxiliary machinery, and equipment that has been overhauled, repaired or otherwise worked on during the performance of the work package.

7.2.8 Turn the shaft(s) during dock trials, taking all necessary precautions to obtain wheel clearance prior to turning the shaft(s). Contractor shall provide tug services at the pier to provide positive holdback during dock trials. Contractor shall provide additional lines, make temporary removals and disconnections as necessary to support dock trial evolution.

7.2.9 Manufacturer and authorized technical representatives, used by the contractor to install, repair and/or modify any equipment or system shall be present during operational testing of that equipment or system during the dock trial event. OEM commissioning procedures shall be provided by the technical representative and shall be included in the test procedures provided by the contractor in 7.2.1.

7.2.10 Correct deficiencies attributable to contractor or subcontractor workmanship found during dock trials, prior to accomplishing sea trials.

7.2.11 At the completion of dock trials, the Contractor shall issue a report outlining the observed performance of all tested equipment. The report shall document observed performance and note acceptance/non-acceptance based upon the stated test criteria per 7.2.1. In addition, the report shall contain reports from manufacturer and authorized technical representatives stating conditions noted for commissioning/start-up and testing of subject equipment and an affirmation of the equipment operating within acceptable parameters. The report shall be issued no later than 24 hours after completion of the dock trials and shall be the basis for determination that the ship is ready to execute the sea trial event.

7.3 Sea Trials

7.3.1 Sea trials shall not be executed without prior receipt of the preliminary weight and moment report required in WI 014 article 7.6.4.

7.3.2 Develop and submit to OMT REP a sea trial agenda at the 50% time point of the availability. Develop and submit to the OMT REP, two days prior to sea trials, a final version of the sea trial agenda that includes:

a. List of equipment to be tested

b. List of tests

c. Test performance criteria

d. Test acceptance criteria

e. Test schedule

f. Test data to be recorded

g. Test data recording format

h. Instrumentation required

i. List of shipyard personnel involved

j. List of vendor technical representatives. All manufacturer and authorized technical representatives, used by the contractor to install, repair and/or modify any equipment or system shall be present during operational testing of that equipment or system during the sea trial event.

k. List of regulatory body representatives

l. List of equipment requiring special operating conditions (e.g., sea state, navigational clearance, test ranges, etc.)

m. Equipment requiring calibration and adjustment while at sea.

7.3.3 Conduct sea trials one day prior to scheduled redelivery.

7.3.4 Perform sea trials in accordance with published agenda.

7.3.5 Operate and test at equipment full power rating, all main propulsion machinery, auxiliary machinery, and equipment that has been overhauled, repaired or otherwise worked on during the performance of the Work Package.

7.3.6 Allocate six hours minimum of actual running time in open seas. Transit time shall not be included in the sea trial time requirement.

7.3.7 Manufacturer and authorized technical representatives, used by the contractor to install, repair and/or modify any equipment or system shall be present during operational testing of that equipment or system during the sea trial event. OEM commissioning procedures shall be provided by the technical representative and shall be included in the test procedures provided by the contractor in 7.3.2.

7.3.8 Contractor shall provide the following shipyard journeymen at a minimum (with respective craft tools) to ride the ship and perform final adjustments, calibrations, data collection, deficiency correction and observance of equipment/system performance during the entire sea trial:

a. One supervisor

b. Two machinists

c. Two pipe fitters

d. Two electricians

7.3.9 At the completion of sea trials, the Contractor shall issue a report to the OMT REP outlining the observed performance of all tested equipment. The report shall document observed performance and note acceptance/non-acceptance based upon the stated test criteria per 7.3.2. In addition, the report shall contain reports from manufacturer and authorized technical representatives stating conditions noted for commissioning/start-up and testing of subject equipment and an affirmation of the equipment operating within acceptable parameters. The report shall be issued no later than 24 hours after completion of the sea trials and shall be the basis for determination that all shipyard completed work is acceptable.

7.3.10 Correct deficiencies attributed to contractor or sub-contractor workmanship found during sea trials prior to redelivery.

7.3.11 Repeat the sea trial at Contractor’s expense until satisfactory results are achieved if the initial sea trial is deemed unsatisfactory by the OMT REP.

7.3.12 Upon successful completion of the sea trial, as accepted by the OMT REP, contractor shall provide sea and/or land transportation to pick up all contractor, subcontractor and government riders from the traditional pick-up point for the local launch service and return those personnel to the shipyard.

7.3.13 If a sail-away sea trial is conducted, a change order will be issued to accept redelivery of the vessel at an alternate location from that listed in Work Item 018 (Delivery and Redelivery).

7.3.14 Contractor shall provide all tugs, pilots, line-handlers, etc. to support ship departure and return from sea trials.

7.4 Dock and Sea Trial Reports

7.4.1 Prepare final dock trial and sea trial reports addressing the following. Reports must be completed and submitted to the OMT REP within 24 hours of completing the dock trial and the sea trial:

a. Equipment/systems tested

b. Test environmental conditions

c. Test date and time

d. Test acceptance criteria

e. Test results

8.0 GENERAL REQUIREMENTS: None additional

<table>
<tr><td colspan="3" align="center">USS Hershel Williams</td></tr>
<tr><td colspan="3" align="center">(ESB 4)</td></tr>
<tr><td>GENERAL SERVICES AND REQUIREMENTS</td><td colspan="2" align="right">CONTRACT NO. N68171-26-RN-003</td></tr>
<tr><td>ITEM NO. 0023</td><td>CATEGORY "A"</td><td align="right">2026-01-30</td></tr>
<tr><td>ESB-SWI-HAZARDOUS WASTE DISPOSAL AT A CONTRACTOR FACILITY</td><td colspan="2" align="right">Manovich, David</td></tr>
</table>

1.0 ABSTRACT

This item describes the requirements for the contractor to provide services to identify, test, handle, store, transport, and dispose of any hazardous wastes generated during the performance of this work package.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 Resource Conservation and Recovery Act (RCRA) as amended (Available Commercially)

2.1.2 10 U.S.C. §7311

2.1.3 Applicable Uniform Hazardous Waste Manifest Form (Available Commercially)

2.1.4 Test Methods for Evaluating Solid Waste, Physical/Chemical Methods; EPA Document, SW-846; the official compendium of analytical and sampling methods that have been evaluated and approved for use in complying with the RCRA regulations. (Available Commercially)

2.1.5 OPNAVINST M 5090.1(SERIES), ENVIRONMENTAL READINESS PROGRAM MANUAL

2.2 Enclosures

2.2.1 List of Expected Types & Quantities of Hazardous Wastes

2.2.2 Hazardous Materials Survey Requirements

3.0 ITEM LOCATION/DESCRIPTION

3.1 Throughout the ship and adjacent areas where ship related work is accomplished causing the generation of hazardous waste as stated below.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with all applicable GTR requirements.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract to determine their effect on the work required under this work item.

5.3 The definitions of many terms used in this work item are found in Work Item 001.

5.4 Hazardous Waste Defined

5.4.1 RCRA hazardous waste is waste that appears on one of the four hazardous wastes lists (F-list – 40 CFR 261.31, K-list – 40 CFR 261.32, P-list 40 CFR 261.33, or U-list 40 CFR 261.33), or exhibits at least one of four characteristics; ignitability, corrosivity, reactivity, or toxicity as defined in 40 CFR Part 261, Subpart C.

5.4.2 Any material that is classified and regulated as a hazardous waste by international, federal, state and local regulations for the location where the work is being performed shall be treated as hazardous waste in this item.

5.5 If a material is not specifically determined to be hazardous waste in the location where the work is being performed, the waste shall be considered as non-hazardous waste for the purposes of this contract and shall be properly disposed of as part of the contractor’s bid work requirements for the work item that generates the waste. The costs associated with the removal, handling and disposal of these non-hazardous wastes shall not be charged against this item.

5.6 The waste stream list identified in Enclosure 2.2.1 represent potentially hazardous wastes that may have been encountered on MSC ships in the past. Hazardous waste generated during the actual performance of the work may vary in type or amount from waste identified in Enclosure 2.2.1, as classified and regulated by international, federal, state and local regulations for the location where the work is being performed.

5.7 This item does not cover the disposal of bilge water and/or tank wash water that is deemed hazardous waste due to the presence of oil in the water, as may be the case in certain states. Oil contaminated bilge water disposal is covered under Work Item 011. Oil contaminated wash water is covered under Work Item 021.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 The proper characterization of potentially hazardous wastes shall be performed by one of the following means:

6.1.1 Chemical analysis by a third party certified testing laboratory in accordance with Reference 2.1.4 (required for all hazardous wastes with aggregate disposal costs above $5,000).

6.1.2 Reference to a material safety data sheet, (MSDS).

6.1.3 Applying knowledge and objective qualitative and quantitative data of the hazardous characteristics of the wastes based on the materials or process(es) used.

6.2 The contractor must provide a means to accurately measure and record/document the amount of potentially hazardous wastes generated in this work package.

6.3 The contractor shall provide the regulatory cite, referencing the threshold values exceeded which identify the material in question to be hazardous waste.

7.0 STATEMENT OF WORK REQUIRED

7.1 Hazardous waste management and control

7.1.1 The contractor shall have in place an effective hazardous waste management and control system for the identification, handling, storage, transportation and disposal of hazardous waste in accordance with Reference 2.1.1.

7.1.2 The contractor shall submit a hazardous waste management and control plan, signed by the contractor’s environmental manager, as part of the bidder’s proposal submittal package.

a. Additionally, the contractor shall submit up-to-date reference documentation delineating all hazardous waste regulations applicable for work performed at their facility. This document is to be in portable document format (PDF) and shall be part of the contractor’s bid proposal submittal package.

7.1.3 The contractor's control system shall be capable of limiting the amounts of hazardous waste generated in the performance of all work items and preventing the generation of all unnecessary hazardous waste materials during performance of the contract.

7.1.4 The control system shall ensure that all hazardous wastes are handled, stored, transported, and disposed of in accordance with References 2.1.1 and 2.1.2 and all applicable international, federal, state, and local rules and regulations. It shall further ensure that all processes and procedures which have the potential to generate hazardous waste contain readily identifiable language (or other markings), which indicates that hazardous waste is a residual product of the work performance, and that such wastes are to be handled, stored, transported and disposed of in accordance with References 2.1.1 and 2.1.2 and all applicable international federal, state and local rules and regulations.

7.2 Generator numbers

7.2.1 Generator numbers shall be used on all Uniform Hazardous Waste Manifest Forms.

7.2.2 When the contractor or any of its sub-contractors performs work which in any manner results in the creation of hazardous waste, documentation related to such waste shall bear the contractor’s generator number.

7.2.3 When work is performed internationally, the international and local regulations shall be followed, however, in all cases, the contractor shall be the generator of the hazardous waste.

7.2.4 Per NAVSEA doctrine (OPNAVINST M 5090.1(Series), ENVIRONMENTAL READINESS PROGRAM MANUAL, Section 22-6.1.4), hazardous wastes generated solely by the ship shall be retained onboard and shall only be disposed of to a Naval Facility. Therefore, MSC should not be claimed as the generator of hazardous waste while the ship is at the contractor’s facility.

7.3 Procedure for initiating work

7.3.1 MSC has identified potentially hazardous wastes which may be produced during performance of this contract in Enclosure 2.2.1.

7.3.2 Upon vessel arrival at the contractor's facility, the contractor shall survey all ship spaces, areas, equipment, lockers, and storerooms in the presence of the OMT REP to identify material requiring disposal which may be characterized as hazardous waste. The nature of the potentially hazardous waste shall be determined through the testing requirements of this work item listed in 6.1. The survey results shall be tabulated in a spreadsheet format per Enclosure 2.2.2 of this work item.

7.3.3 Following the arrival survey, if potentially hazardous wastes are discovered through any work associated with a work item in this work package, the contractor is required to notify the OMT REP in writing. Each notification shall be in accordance with Enclosure 2.2.2.

7.3.4 Hazardous material is only to be handled, stored, transported and disposed of when authorized by the OMT REP.

7.3.5 In the event that a hazardous waste is identified onboard the ship and is authorized for removal by the OMT REP, and the OSHA regulations require containment and special personal protective equipment and health monitoring, these requirements shall be specifically identified by the contractor, and the costs shall be the subject of a change order to the generating work item.

7.3.6 In some states, bilge water may be determined to be hazardous waste due to the presence of oil in the water. When this occurs, the disposal costs shall not be charged against this item. Oil contaminated bilge water is covered in Work Item 011. In the event the bilge water is determined to be hazardous waste due to some contaminant other than oil, the disposal charges shall be charged against this work item.

7.3.7 In some states, tank wash water may be determined to be hazardous waste due to the presence of oil in the water. When this occurs, the disposal costs shall not be charged against this item. Oil contaminated tank wash water is covered in Work Item 021. In the event the tank wash water is determined to be hazardous waste due to some contaminant other than oil, the disposal charges shall be charged against this work item.

7.3.8 The physical labor to accomplish the work requirements of a particular work item, including the removal of coverings and interferences, is covered under that work item. In the event that hazardous waste is discovered in preparation for the accomplishment of that work item, the following guidance is provided.

a. Special costs for containments, medical screening and other precautionary measures shall be in accordance with article 7.3.5 of this work item.

b. Cost associated with the testing, handling, segregation, movement off the ship, temporary storage, transportation, management and disposal shall be covered under this work item, as long as the requirements of this work item are fulfilled.

7.3.9 The contractor shall submit weekly reports via condition found report to the OMT rep itemizing all materials being handled as hazardous waste up to that time. This report shall be submitted whether hazardous wastes have been handled or not.

7.4 Hazardous Waste Disposal

7.4.1 Contractor’s offer shall include $75,000 towards the cost of hazardous waste disposal, plus the total price for contractor management and handling fees. If the actual cost of hazardous waste disposal exceeds $75,000 plus the total price for contractor management and handling fees, a change order will be issued for the difference. If the actual cost of hazardous waste disposal is less than $75,000 plus the total price for contractor management and handling fees, a change order will be issued to credit the Government for the difference. The contractor shall receive the management and handling fee on a pro-rata basis. Change orders and credits due shall be on a pro-rata basis of the waste disposed.

7.4.2 Compensation will be based on properly identified hazardous waste and quantities as shown on the applicable Uniform Hazardous Waste Manifests. This will include matching the waste with the waste description identified in Enclosure 2.2.1 or additional items as appropriate. Compensation will be paid against actual invoices submitted for the transportation and disposal of the hazardous wastes. All applicable documentation, including invoices and proof of disposal at an authorized hazardous waste disposal site and signed by an authorized representative of the authorized disposal facility shall be required for reimbursement.

7.4.3 The contractor shall include a management and handling fee, as a percentage of the total estimated disposal costs, to account for the contractor’s efforts to identify, test, manage, handle and temporarily store the hazardous wastes in the yard. “Test” above includes all testing required in 6.1. This fee shall be added as a pro-rata expense to each submitted invoice to cover the contractor’s work efforts.

7.4.4 Submit Uniform Hazardous Waste Manifest Forms to the OMT REP when disposing of hazardous waste complete with the contractor’s EPA generator number.

a. Outgoing Uniform Hazardous Waste Manifest Form: Submit three copies of the completed form signed by the generator and the transporter prior to transporting hazardous waste from the contractor’s facility.

b. Returning Uniform Hazardous Waste Manifest Form: Submit three copies of the form completed and signed by the owner or operator of an authorized disposal facility acknowledging receipt of the hazardous waste within five days of receipt.

7.5 The contractor shall ship the hazardous waste to a facility legally authorized to accept the identified waste. The contractors shall ensure that transportation of hazardous waste is accomplished only by haulers registered to perform such transportation with cognizant local, state, federal, and international requirements for the locality that the work is being performed.

7.6 The contractor must submit the following in order to receive compensation under this work item: If any of the items listed in 7.6.1 through 7.6.5 are not provided, the contractor shall not receive compensation until such time as they are provided.

7.6.1 Shipyard’s invoice for the waste disposal. The invoice shall cover the pass-through charge from the waste disposal facility and the shipyard’s pro-rata charge for the management and handling fees.

7.6.2 A document identifying the regulatory reference and method utilized under article 6.1 to determine the material was hazardous waste. When analytical testing is utilized, test results shall be included.

7.6.3 Copies of the outgoing Uniform Hazardous Waste Manifests.

7.6.4 Copies of the return Uniform Hazardous Waste Manifests.

7.6.5 Copies of invoices from the hazardous waste transfer and disposal facilities submitted for the transportation and disposal of the hazardous wastes. This invoice must be the invoice the shipyard receives from the transfer and disposal facility and show the amount the shipyard was charged for the hazardous waste disposal.

7.7 Upon completion of the work package, submit a Final Hazardous Waste Removal Summary Report to the OMT REP. The report shall provide an itemized list of the hazardous wastes removed by type, quantity, unit price, extended price, analysis method used, regulatory cite that determined the material hazardous waste, work item that generated the waste, generator assignment along with the applicable manifests for each item, and any third party chemical analysis conducted in accordance with article 6.1.

7.7.1 An electronic copy of the Final Hazardous Waste Removal Summary Report shall be sent to:

a. The OMT REP

b. The MSC Contracting Officer

c. The MSC Technical Library at email address MSC_N7_TechLibrary@us.navy.mil

7.8 Nothing contained in this work item shall relieve the contractor from complying with applicable international, federal, state, and local laws, codes, ordinances, and regulations, including the obtaining of licenses and permits, in connection with hazardous waste identification, handling, transportation, and disposal in the performance of this contract.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - LIST OF EXPECTED TYPES & QUANTITIES OF HAZARDOUS WASTES

Listing of Potentially Hazardous Wastes for Item 023				
Item	Description			
1	Asbestos			
2	PCB Contaminated Media			
3	Acid & Caustics (Sulfuric Acid)			
4	Ethlyene Glycol			
5	Diesel Engine Coolant (Maxigard)			
6	Cooling Water (Drew LiquidDEWT)			
7	Cleaning Solvents			
8	Fluorocarbons			
9	Spent Abrasive Grit (Contaminated – contaminated beyond hazardous waste thresholds)			
10	Paint Sludges (UHP Residue – contaminated beyond hazardous waste thresholds)			
11	Paint Chips (Contaminated – contaminated beyond hazardous waste thresholds)			

12	Misc. Chemicals (Flammable)			
13	Misc. Chemicals (Corrosive)			
14	Misc. Chemicals (Reactive)			
15	Misc. Chemicals (EP toxic)			
16	Paints (enamel, epoxy, non-skid, etc.)			
17	Paints (lead, cadmium, chrome)			
18	Paint Stripper & Thinners (phenols, lead, chromium)			
19	Used & Empty Paint Cans			
20	Oily Cloths			
21	Used Boiler Refractory/Magnesite Deck/Underlay & Deck Tile			
22	Electrical Cable Asbestos Insulation			
23	Batteries			
24	Lead Acid Batteries			

The items listed in the table above are examples of wastes that have, at times, been determined to be hazardous wastes based on the locality and characteristics of the substance. If an item on this list is not regarded as a hazardous waste in the locality where the work will be performed, then the contractor shall cross out these items from the list, indicating that they are not considered to be hazardous wastes. Accordingly, the disposal of these non-hazardous wastes generated in the conduct of the work items shall be covered entirely in the bid costs presented by the contractor for the generating work item and shall not be claimed as a cost against WI 023.

ENCLOSURE 2.2.2 - HAZARDOUS MATERIALS SURVEY REQUIREMENTS

Asbestos, lead paint, PCB and general survey forms shall be separate.

1. Column 1: Indicates Contractor assigned identifier number for each sample.

2. Column 2: Indicates reason sample was obtained e.g. suspected Lead Paint, suspected Asbestos, etc.

3. Column 3: Indicates the general location from which the sample was obtained; for example, Store Space 2-50-1, forward bulkhead. Where sketches are used to indicate the location, enter “See Sketch” in column 3. Sketches must be serialized and reference the contractor assigned identifier in Column 1.

4. Column 4: Indicates the specific location from which the sample was obtained. Where sketches or photographs are used to indicate a specific location, enter “see sketch” or “see photograph” in column 4. Sketches and photographs must be serialized and reference the contractor assigned identifier in Column 1.

5. Column 5: Indicates whether the sample was hazardous or not and the method used to identify the nature of the waste. In instances where tests were required to be performed in accordance with Reference 2.1.4, the test results shall be included and serialized and reference the contractor assigned identifier in Column 1.

6. Column 6: Cites the specific reference and threshold that characterizes the material as hazardous waste for the particular locality that the work is being performed.

7. Column 7: Cites the work item that generated the waste.

8. All analysis results or copies of Material Safety Data Sheets (MSDS) which indicate a material's hazardous condition must be attached to the form and serialized according to the identifier number. The identifier number shall be entered directly on the laboratory test result sheets and the MSDSs.

1.0 ABSTRACT

This item describes the requirements for the contractor to continue delivering services described in the standard work items.

2.0 REFERENCES: None

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Location: As described in parent work item

3.1.2 Quantity: As described in parent work item

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES: None

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Submit a unit price per day for 15 days for the continuation services described in parent work item 010 FURNISH OFFICE FOR OVERHAUL MANAGEMENT TEAM (OMT). The proposed daily rate will be used for the duration of the extension.

7.2 Submit a unit price per day for 15 days for the continuation services described in parent work item 011 GENERAL SERVICES FOR THE SHIP. The proposed daily rate will be used for the duration of the extension. Weekly reports shall continue and be charged against the base work item until exhausted. The daily rate will not include additional electrical power consumption, potable water consumption, distilled water provided, bilge water removed, and crane services. If additional quantities are needed, these shall be negotiated under a separate CCO.

7.3 Submit a unit price per day for 15 days for the continuation services described in parent work item 012 INFORMATION TECHNOLOGY SERVICES. The proposed daily rate will be used for the duration of the extension.

7.4 Submit a unit price per day for 15 days for the continuation services described in parent work item 013 PROJECT PLANNING AND PRODUCTION STATUS MONITORING REPORTS. The proposed daily rate will be used for the duration of the extension.

7.5 Submit a unit price per day for 15 days for the continuation services described in parent work item 016 FIRE PROTECTION AND SHIP’S SAFETY PROGRAM. The proposed daily rate will be used for the duration of the extension.

7.6 Submit a unit price per day for 15 days for the continuation services described in parent work item 017 HANDLING SHIPS STORES. The proposed daily rate will be used for the duration of the extension.

7.7 Submit a unit price per day for 15 days for the continuation services described in parent work item 019 SHIPBOARD ACCESS AND SECURITY. The proposed daily rate will be used for the duration of the extension.

7.8 Submit a unit price per day for 15 days for the continuation services described in parent work item 020 GAS FREE CERTIFICATES. The proposed daily rate will be used for the duration of the extension.

7.9 Submit a unit price per day for 15 days for the continuation services described in parent work item 040 OCONUS FORCE PROTECTION. The proposed daily rate will be used for the duration of the extension.

7.10 Work items that are not separately priced continue to remain in effect and apply throughout the period of the performance extension.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
GENERAL SERVICES AND REQUIREMENTS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0040	CATEGORY "A"	2026-01-30
ESB-SWI-OCONUS FORCE PROTECTION		Manovich, David

1.0 ABSTRACT

This item describes the minimum Force Protection standards for facilities that repair MSC ships in operating areas outside of the US and its territories.

2.0 REFERENCES/ENCLOSURES: None

3.0 ITEM LOCATION AND DESCRIPTION

3.1 Location: Various

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor and subcontractors shall review other work items under this contract, to determine their effect on the work required by this item. Based on this review the contractor shall plan and schedule work to minimize conflicts between work items.

5.3 The definitions of many terms used in this work item are found in work item 001.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 FORCE PROTECTION: The contractor shall, as a minimum, provide the following means of Force Protection while the vessel is being repaired at their facility.

7.1.1 24 hours per day/7 days per week Access control. All facility access points will be locked when not in use. When not locked, facility access points will be manned by at least two Security guards. The facility access control guards will perform the following functions while MSC ships are moored at the facility:

a. The facility guards will check the identification of all personnel at the Facility Perimeter. Using access lists or other official correspondence, the guard will verify that the person entering the facility has official business that requires access to the facility. Turnstiles and badge systems, while not mandatory, are encouraged as these will help meet this requirement.

b. The facility will inspect vehicles, hand carried items, cargo, and stores at the Facility perimeter.

7.2 The facility will employ water-filled or concrete block barriers to keep unauthorized vehicles at least 100 feet from the ship.

7.3 The facility will have designated parking areas and parking restrictions. Any vehicles will be inspected before being allowed within 100 feet of the ship.

7.4 A continuous wooden, concrete or metal fence line (at least 6 feet high) will be maintained at the Facility boundaries. Buildings with wood, concrete and/or metal construction can be used to maintain boundaries in lieu of a fence line.

7.5 Consistent with Naval Vessel Protection Zones, whenever possible, MSC ships will be berthed or dry docked so as to be no closer than 100 yards to other facilities or 100 yards to ships which are moored at other facilities.

7.6 The facility will employ lighting sufficient to illuminate the pier to which the MSC ship is moored as well as 10 feet on either side of all landside perimeters. Portable lighting can be used to meet this requirement.

7.7 All Facility Security Guards shall have at least two separate means of communicating with MSC ships moored or dry docked at their facilities. At least one of these means shall be a two-way UHF or VHF portable radio. MSC ships' portable radios can be used to meet this requirement. Exception: Portable UHF/VHF radios and cell phones will not be used whenever hostile explosives are suspected. In that case, only hard-wired or non-electronic communications will be used.

7.8 Provide waterborne patrols, 24 hours per day/7 days per week, with two security guards onboard for the entire duration the vessel is at the contractor's facility.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

This item describes the addition of mooring chocks and bitts at the forward and aft ends of the Mission Deck, to allow robust mooring and other miscellaneous usage such as tug service, etc.

2.0 REFERENCES/ENCLOSURES

2.1 REFERENCES:

2.1.1 NASSCO Dwg. No. 835-8468057, MISSION DECK MOORING FITTINGS FOUNDATION ANALYSIS

2.1.2 NASSCO Dwg. No. 101-8468312, DECK AND ELEVATIONS CONTROL FEATURES

2.1.3 MSC Dwg. No. 582-8613977 ESB Class Mission Deck Mooring Fittings - Structural INSTL

2.2 ENCLOSURES:

2.2.1 Vendor Information for Mooring Chocks and Bitts

2.2.2 Photograph of Similar Mooring Chocks onboard ESB 1

3.0 ITEM LOCATION/DESCRIPTION

3.1 LOCATION/QUANTITY:

3.1.1 Mission Deck, starboard side, frames 58 and 61; port side, frame 61p; and port and starboard sides, frames 92.

3.2 ITEM DESCRIPTION/MANUFACTURER’S DATA:

3.2.1 BILL OF MATERIALS: See Reference 2.1.3 Bill of Material

3.2.2 Quantities shown are considered estimates. The Contractor shall provide the exact quantities and additional materials which are not listed in the List of Materials, in order to install a fully functional system which meets the requirements of this specification.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES:

4.1 Government Furnished Equipment (GFE):

PC No.	QTY	DESCRIPTION
1	5	Panama Mooring Chock, Deck, 500x400mm Opening
2	2	Mooring Bitt, 457mm Diameter

4.2 Government Furnished Material (GFM): None

4.3 Government Furnished Services (GFS): None

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTR’s including but not limited to GTR’s 1 through 7, 25, and 29.

5.2 The Contractor and all Subcontractors, regardless of tier, are advised to review other work items under this contract, to determine their effect on the work required under this work item.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Arrangements/Outfitting:

7.1.1 Contractor shall be responsible for providing and maintaining gas free certificate for all hot work in all work areas and adjacent spaces in the completion of this work item.

7.1.2 The Contractor shall certify that voids in way of welding are gas freed and made safe for hot work. This shall include opening of voids and providing ventilation, and lighting, and reclosing of voids upon completion of work. Spaces include Void 3-65-0 and Void 3-97-0.

7.1.3 Contractor shall layout mooring fitting locations on the Mission Deck as shown in Reference 2.1.3. After initial mooring fitting placement; but, prior to start of welding, the Port Engineer will inspect and approve final locations.

7.1.3.1 Contractor shall conduct mooring fitting location layout as follows:

- One (1) Panama chock on Starboard Mission Deck at Frame 58 to support the Bitts already installed
- One (1) Panama chock on Starboard Mission Deck at Frame 61 to support the Bitts already installed
- One (1) Panama chock on Starboard Mission Deck at Frame 92
- One (1) set of mooring bitts on Starboard Mission Deck at Frame 92 to support chocks installed at Frame 92
- One (1) Panama chock on Port Mission Deck at Frame 61 to support the Bitts already installed
- One (1) Panama chock on Port Mission Deck at Frame 92
- One (1) set of mooring bitts on Port Mission Deck at Frame 92 to support chocks installed at Frame 92.

7.2 Structural:

7.2.1 Install new backup structure below deck at frames 92 port and starboard to support raised foundations for mooring chocks and mooring bitts, using Reference 2.1.3 as guidance. Foundations shall be sized for a safe working load of 100 tons, similar to existing mooring fitting foundations.

7.2.2 Fabricate and install five (5) new raised foundations for mooring chocks on the mission deck similar to existing raised foundation on the port side at frame 58, using Reference 2.1.3 and Enclosure 2.2.1 and 2.2.2 as guidance. Foundations shall align with new and existing below deck structure and shall be sized for a safe working load of 100 tons, similar to existing chock foundations. Deck edge coaming and liferails shall be modified as necessary.

7.2.3 Install five (5) new 500 x 400 mm deck mount Panama chocks (GFE Item No. 1) atop new raised chock foundations installed per paragraph 7.2.2, using Reference 2.1.3 and Enclosures 2.2.1 and 2.2.2 as guidance. Chocks shall have a safe working load of 100 tons. Deck edge coaming and liferails shall be modified as necessary.

7.2.4 Install two (2) new 475 mm double barrel mooring bits (GFE Item No. 2) atop new raised chock foundations installed per paragraph 7.2.2, using Reference 2.1.3 and Enclosure 2.2.1 guidance. Bitts shall have a safe working load of 100 tons. Deck edge coaming and liferails shall be modified as necessary.

7.2.5 Cut holes in bulwarks at frame 92 port and starboard to allow installation of Panama Chocks as described in Paragraphs 7.2.3 and 7.2.3, using Reference 2.1.3 as guidance. Horizontal stiffeners on the inboard sides of the bulwark shall be relocated to transition above the top of the opening.

7.3 Mechanical/Fluids: None additional

7.4 Electrical:

7.4.1 Temporarily reroute cable run in overhead of Void 3-97-0 in order to allow installation of backup structure. Cable shall be returned to original route upon completion of the installation.

7.5 Electronics: None additional

7.6 Preparation of Drawings/Documentation:

7.6.1 Contractor shall provide documentation on all tests and inspections required by the ABS attending surveyor to the MSC Port Engineer. This shall include visual inspections, NDT, and any other required tests.

7.7 Inspection/Tests:

7.7.1 The contractor shall be responsible for notifying all applicable Regulatory Bodies and MSC Port Engineer of scheduled inspections and testing. Failure to obtain Regulatory Body representatives to witness testing or inspections shall require rescheduling the test or inspection at the Contractor’s expense.

7.7.2 Fit-up and NDT of weldments shall be inspected as directed by the ABS attending surveyor.

7.8 Painting

7.8.1 All new and disturbed surface coatings, including but not limited to paint and deck covering, shall be restored to match the surrounding areas. This includes both side of steel where welding or burning has occurred including the interior of voids.

7.9 Marking:

7.9.1 New mooring chocks shall be permanently identified with weld bead as having a safe working load of 100 tons.

7.10 Manufacturer’s Representative: None

8.0 GENERAL REQUIREMENTS:

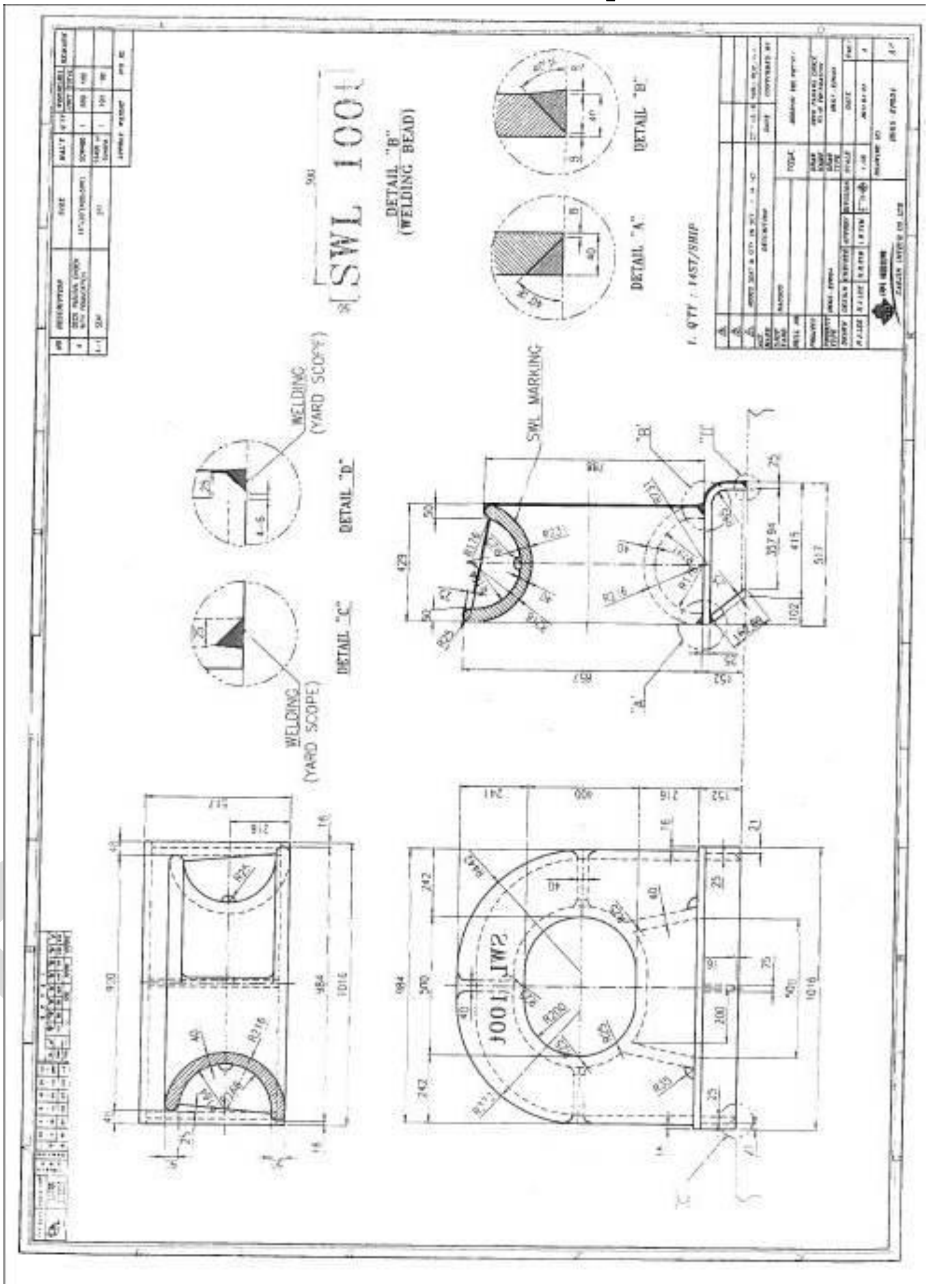
8.1 The Contractor shall furnish all labor and materials, which includes but is not limited to scaffolding, staging, ventilation, designated fire watch, tools, access to ballast tanks, and crane service necessary to accomplish this work item unless otherwise indicated. This also includes procuring and providing the services of a Marine Chemist.

8.2 Contractor shall be responsible for removal and reinstallation of all interferences including liferails, bulwarks, and insulation in all work areas for the completion of this work item.

8.3 The Contractor shall arrange for all U.S. Coast Guard and ABS inspections and surveys to obtain installation approval. The Contractor shall bear all costs for services of Regulatory Bodies and for obtaining final installation approval.

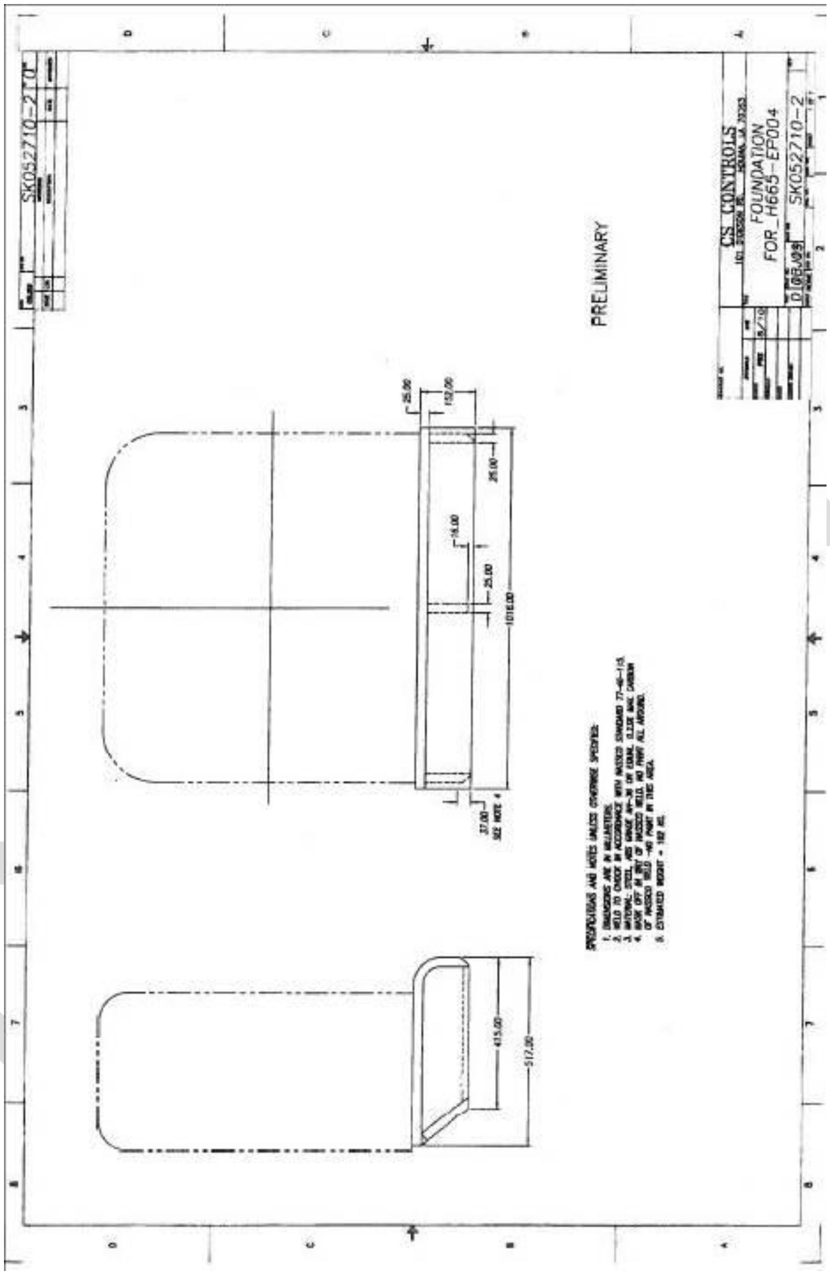
8.4 Inspection and acceptance of work shall be accomplished by the MSC Port Engineer.

For Guidance Only



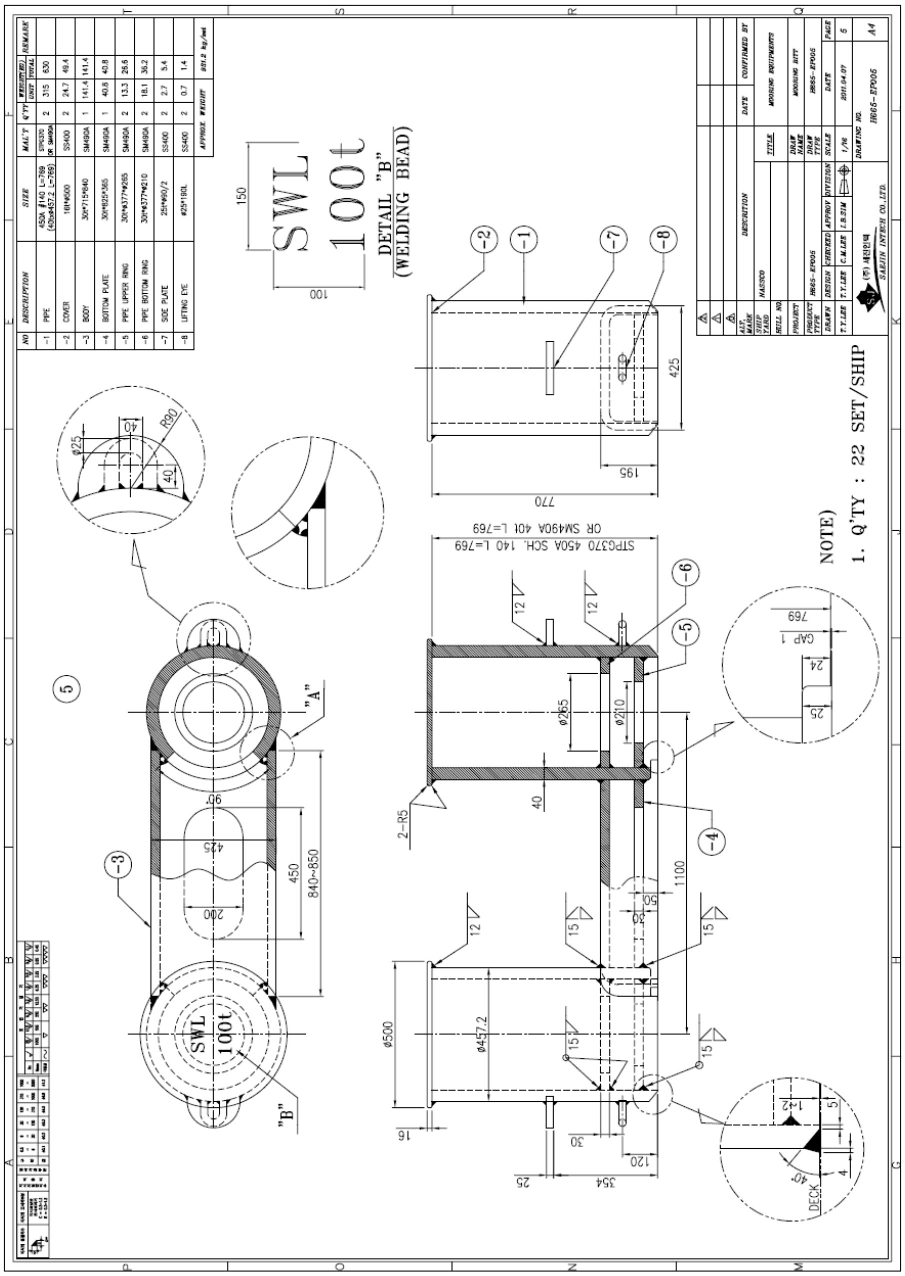
Enclosure 2.2.1
Vendor Information for Mooring Chocks and Bitts

For Guidance Only



Enclosure 2.2.1
(continued)

For Guidance Only



Enclosure 2.2.1
(continued)
For Guidance Only

ENCLOSURE 2.2.2
Photograph of Similar Mooring Chocks onboard ESB 1

USNS Hershel Williams
(T-ESB-3 CLASS)

STRUCTURAL/OUTFITTING **RFQ NO. TBD**
02 FEB 2022
ITEM NO. WI-623 **CATEGORY "A"** **CACI/RICKS/MK**

EXTERIOR STAIRWELL TRACTION MODIFICATIONS (T-ESB-21-018)

1.0 ABSTRACT:

1.1 This item describes the ship modifications necessary for the installation of positive traction stair covers on exterior stairwells. These modifications will improve the traction and safety of exterior stairwells.

2.0 REFERENCES/ENCLOSURES:

2.1 References

2.1.1 MSC Drawing No. 803-7081122, General Technical Requirements

2.1.2 MSC Drawing No. 803-7080703, Preparation of Computer Aided Design (CAD) Drawings

2.1.3 MSC Drawing No. 623-8615251, Positive Traction Stair Mods

2.1.4 NAVSEA Drawing 631-8393784, Paint Schedule

2.1.5 NAVSEA Drawing 631-8452132, Paint Code Matrix

2.1.6 Anti-Slip Hi-Traction Covers Installation Guideline

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations: Exterior stairwell locations throughout.

3.1.2 Quantity: Use Reference 2.1.3 as guidance.

3.2 Item Description/Manufacturer's Data:

3.2.1 Material/Components, including specific quantities and manufacturer make and model numbers, necessary to accomplish work required by this work item are included in the List of Materials of References 2.1.3.

3.2.2 Quantities specified in the List of Materials of References 2.1.3 are considered estimates. The contractor shall provide the exact quantities and additional material such as miscellaneous pipe fittings, elbows, caps, valves, fasteners, pipe and cable hangers, weld material (pads/tabs), cable tags, etc., which are not included in the List of Materials of References 2.1.3, to install a fully functional system which meets the requirements of this specification.

EXTERIOR STAIRWELL TRACTION MODIFICATIONS (T-ESB-21-018)

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES

4.1 Government Furnished Equipment (GFE): None additional.
4.2 Government Furnished Material (GFM): None additional.
4.3 Government Furnished Services (GFS): None additional.
4.4 Government Furnished Information (GFI): None additional.

5.0 NOTES

5.1 The ship's force will provide access to any required spaces.

5.2 The contractor and all subcontractors, regardless of tier, are advised to review other work items under this contract to determine their effect on the work required under this work item. Many of the definitions relating to performance of this Work Item are found in Work Item 0001.

5.3 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR), Reference 2.1.1, to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 27, 28, and 29.

6.0 QUALITY ASSURANCE REQUIREMENTS:

6.1 Inspection and acceptance of all work shall be by the MSCREP.

6.2 All work performed shall be in accordance with Regulatory Body rules and regulations.

6.3 Government will furnish ABS Surveyor attendance. Contractor is responsible for coordinating any or all onsite visits required for ABS representative to witness testing and inspections.

7.0 STATEMENT OF WORK REQUIRED

Provide all labor, materials, staging, and tools required to accomplish this specification.

7.1 Arrangements/Outfitting:

7.1.1 Removals: None additional.

7.1.2 Installation: None additional.

7.1.2.1 Fit each stair tread to the existing grating steps of each exterior stairwell using Reference 2.1.3 and 2.1.6 for guidance.

7.1.2.2 Drill (4) holes ea. to install (4) ea. grating clips and fasteners to secure each stair tread to existing grating using Reference 2.1.3 and 2.1.6 for guidance.

7.1.2.3 Note: The first and last stair tread installed in each stairwell shall be black and yellow colored; all other treads shall be all black.

7.2 Structural:

7.2.1 Repair deck coverings and insulation in way of removals/installation or otherwise damaged during the execution of this work item using References 2.1.1, 2.1.4, and 2.1.5 as guidance.

7.2.2 Removals: None additional.

7.2.3 Installation: None additional.

7.3 Mechanical/Fluids/Ventilation:

7.3.1 Removals: None additional.

7.3.2 Installation: None additional.

7.4 Electrical:

7.4.1 Removals: None additional.

7.4.2 Installation: None additional.

7.5 Electronics/Communications:

7.5.1 Removals: None additional.

7.5.2 Installation: None additional.

7.6 Preparation of Drawings/Documentation:

7.6.1 None additional.

7.7 Inspection/Test:

7.7.1 Inspections: None additional.

7.7.2 Tests: None additional.

EXTERIOR STAIRWELL TRACTION MODIFICATIONS (T-ESB-21-018)

7.8 Insulation and Painting:

7.8.1 Insulation: None additional.

7.8.2 Painting (General):

7.8.2.1 Prepare, prime and paint all disturbed areas to match the surrounding surfaces using contractor-supplied paint. Surface preparation and paint coating application requirements are provided in References 2.1.1, 2.1.4 and 2.1.5.

7.8.2.2 New installed stair traction treads shall not be coated or painted.

7.9 Markings:

7.9.1 None additional.

7.10 Manufacturer's Representative: None additional.

8.0 GENERAL REQUIREMENTS

8.1 None additional.

1.0 ABSTRACT

This item describes the expansion of the 1A UNREP FAS station and modifications of the FAS equipment.

2.0 REFERENCES

2.1 MSC Drawing No. 179-8615510, Fueling at Sea STA 1A Catwalk Mods & Deck Extension – Structural Modifications

2.2 MSC Drawing No. 390-8615843, Fueling at Sea STA 1A Catwalk Mods & Deck Extension – Power System Modifications

2.3 MSC Drawing No. 612-8615511, Fueling at Sea STA 1A Safety Nets and Handrail Modifications – Structural Modifications

2.4 Military Sealift Command (MSC) General Technical Requirements (GTRs), DWG 7081122

2.5 American Bureau of Shipping (ABS) Rules for Building and Classing Marine Vessels 2020 (Available Commercially)

2.6 Preparation of Computer Aided Designed (CAD) Drawings for USNS Ships, MSC CAD Drawing Standard, DWG 803-7080803

2.7 NAVSEA Drawing No. 804-6352179 Interface Requirements Gypsy Winch

2.8 NAVSEA Drawing No. 804-8436624 Safety Net, Deck Edge, CRES 316 Frame, CRES 316 Net

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Location: FAS Station Frame 63.25 STBD Side

3.1.2 Quantity: None

3.2 Item Description/Manufacturer’s Data

3.2.1 Bill of Materials

See References 2.1 through 2.2.

3.2.2 Quantities specified in the List of Materials of References 2.1 through 2.2 are considered estimates. The Contractor shall provide the exact quantities and additional material such as miscellaneous pipe fittings, elbows, caps, valves, pipe hangers, weld material, cable hangers, cable tags, buswork, etc., which are not included in the List of Materials of References 2.1 through 2.2, in order to install a fully functional system which meets the requirements of this specification.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICES: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs (Ref. 2.4) including but not limited to GTRs 1 through 7, 23, 25, 28, and 29.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract to determine their effect on the work required under this work item.

5.3 No deviations from Reference drawings shall be allowed unless authorized by the MSCREP.

5.4 Ship’s force shall make all tagouts and secure valves as required to support contractor in accomplishment of this work item.

6.0 QUALITY ASSURANCE REQUIREMENTS:

6.1 All welding certification, materials, and procedures shall be in accordance with Reference 2.5 requirements.

7.0 STATEMENT OF WORK REQUIRED

7.1 Arrangements/Outfitting: None additional

7.2 Structural

7.2.1 Removals

Using Reference 2.1 as guidance,

a. Remove the Aft Catwalk.

b. Remove the Aft Catwalk handrails.

c. Enlarge doorway in aft bulkhead.

d. Remove FAS Winch Control pedestal.

e. Remove the gypsy winch cleat.

7.2.2 Installations

Using Reference 2.1 and 2.3 as guidance,

a. Install new platform aft of FAS station.

b. Install new longitudinal bulkhead on inboard side of the new platform.

c. Install safety handrails and lifelines.

d. Reinstall Winch Control pedestal.

e. Install stbd deck edge life nets.

f. Install the aft messenger return chock.

7.2.3 Relocation: Relocate and reinstall the gypsy winch cleat using Reference 2.1 and 2.7 as guidance.

7.3 Mechanical/Fluids: None

7.4 Electrical

7.4.1 Prior to starting work ensure that all circuits are de-energized and that all affected circuits are tagged out and isolated. See NOTES 5.5 to 5.8 for electrical safety guidance.

7.4.2 Removals: None

7.4.3 Relocations

a. Relocate Light Dimmer Switch and Light Dimmer Sw. Disconnect located on Bulkhead 62 in FAS sta. Both shall be moved inboard using reference 2.2 as guidance.

b. Relocate one overhead HPS light fixture from above new platform extension to below new platform extension using reference 2.2 as guidance.

7.4.4 Installations

a. Install two new overhead fluorescent light fixtures above platform extension using reference 2.2 as guidance.

7.5 Electronics:

7.5.1 Relocate Telephone located on Bulkhead 62 outboard in FAS Station. Telephone shall be relocated on aft face of bulkhead 62 approx. 1ft from port edge of modified opening using reference 2.2 as guidance.

7.6 Preparation of Drawings/Documentation

7.6.1 Prepare working drawings to accomplish all work required by this item, using Reference 2.6 for guidance.

7.6.2 Prepare and submit red-lined markups of References 2.1 through 2.2 reflecting the as-built installation.

7.7 Inspection/Test

7.7.1 The contractor shall conduct preliminary tests on all completed work before calling for an official test using Reference 2.4 as guidance.

7.7.2 The contractor shall prove proper operation of all re-installed interferences and equipment removed and disturbed in the presence of the MSCREP.

7.7.3 The contractor shall conduct all testing using Reference 2.3 as guidance.

7.7.4 Testing shall be accomplished to the satisfaction of the MSCREP.

7.7.5 Operational Test: None

7.7.6 Insulation Resistance Test

7.7.7 Enclosure Ground Test

7.7.8 Perform Cleat Load Test using Reference 2.7 as guidance.

7.7.9 Load test new safety nets and frames using Reference 2.8 as guidance.

7.7.10 Perform Load Test on handrails utilizing Reference 2.4 as guidance.

7.8 Painting

Paint all new and disturbed surfaces to match surrounding surfaces.

7.9 Marking

7.9.1 Install name plates, notices, and markings for all new and modified systems (or areas) using Reference 2.4 as guidance.

7.9.2 Fabricate and install one label plate that documents testing of the winch cleat. Label plate shall read:

FAS MESSENGER WINCH CLEAT
STATIC TEST LOAD: 8000 LBS
TEST ACTIVITY: TBD
TEST DATE: XX/XX/202X

where the actual testing activity and date of test is specified.

The letters shall be at least 1/4” tall and engraved on a Type 316 stainless steel plate that is attached to the deck within 3 inches of the cleat, using Type 316 stainless steel, oval-head machine screws and thickened epoxy adhesive.

7.10 Manufacturer’s Representative: None

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
HULL AND STRUCTURAL		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0104	**CATEGORY "A"**	**2026-01-30**
ESB-CSI-FLIGHT DECK UNDERSIDE PRESERVATION (1 YR)		**Manovich, David**

1.0 ABSTRACT

The purpose of this item is to conduct a full underside wash and surface prepare and paint the underside of the flight deck, underside supporting structures, piping, and flight deck stanchions.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 801-8495724, Booklet of General Plans

2.2 NAVSEA Drawing No. 801-8499371, General Arrangement,(ESB 4)

2.3 NAVSEA Dwg 631-8393784, ESB 4 Paint Schedule

2.4 Surface Preparation Standard, SSPC-SP WJ-2/NACE WJ-2 Waterjet Cleaning of Metals – Very Thorough Cleaning (Available Commercially)

2.5 Surface Preparation Standard, SSPC-SP10/NACE 2, Near White Metal Blast Cleaning (Available Commercially)

2.6 Surface Preparation Standard, SSPC-SP-11, Power Tool Cleaning to Bare Metal (Available Commercially)

2.7 PPG Product Data Sheets, Amercoat 240 and Amershield (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: Flight Deck Underside

3.1.1 Zones

a. ZONE 1: FR 58 to 62

3.1.2 Quantity

a. 15,000 square feet in Zone 1

b. 10,000 sq feet on support stanchions.

c. 25,000 sq feet total area

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Red Oxide	210 gal
Amercoat 240	Epoxy (stripe coat)	Off-White	105 gal
Amercoat 240	Epoxy (2nd primer coat)	Off-White	210 gal
Amershield	Polyurethane (topcoat)	Haze Gray	210 gal
Amercoat T-10	Solvent	N/A	75 gal
Amercoat 65	Solvent	N/A	75 gal

4.3 Government Furnished Services (GFS)

4.3.1 PPG Coating Representative

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.3 THIS SHIP CONTAINS HIGH VOLTAGE (HV) ELECTRICAL SYSTEMS. OBEY ALL POSTED AND VERBAL INSTRUCTIONS REGARDING SAFETY AND EXCLUSION FROM HIGH VOLTAGE AREAS. AT NO TIME

APPROACH, WORK ON, OR ENTER A HIGH VOLTAGE AREA WITHOUT PROPER AUTHORIZATION.

5.4 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment as required to renew the paint coatings on the flight deck underside structures, cross beams, piping, piping hangers, cableways, cableway supports, and 4 flight deck support stanchions from Mission Deck to Underside of Flight Deck identified in section 3.0 using References 2.1 through 2.8 as guidance.

7.2 Prior to the start of any surface preparation:

7.2.1 Record all markings, label plates and placards documenting their positions, symbols, and text. Submit a copy of the record to the OMT REP.

7.2.2 Note the condition of any damaged equipment.

7.2.3 With assistance from the Ship’s Force, conduct an operational test in the presence of the OMT REP documenting “as found” conditions of adjacent lighting, speakers, and deck drains.

7.3 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.4 Coordinate with Ship's Force to tagout all lights and machinery in way of the areas being worked. Plug all overboard discharges in way of surfaces being prepared and painted.

7.5 Provide and erect staging required to access all applicable surfaces. Provide man lifts (cherry picker/JLG), any required safety gear, and operator for use by OMT REP and contractor's QA inspector when inspecting flight deck underside surface preparations and coatings.

7.6 Paint material will be stored within the Paint Manufacturer's recommended temperature range. When paint material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.7 Provide and maintain adequate lighting during all surface preparation, coating, and inspection activities.

7.8 Temporarily remove interferences, label plates and placards in way of surface preparation and paint application. Reinstall upon completion of all coatings work. Label plates that are riveted to the hull or otherwise permanently installed need not be removed.

7.9 Prior to the start of surface preparation and painting, take precautions to protect the vessel from contamination. Install drop cloths, masking, division shields, seals, blanks, and filtering materials to prevent abrasives and foreign substances from entering lights, machinery, piping, hatches, deck drains, ventilation systems, tank vents, valve stems, motor shafts, seals and temporary openings during blasting and painting operations. The following actions, at a minimum will be taken:

7.9.1 Mask all normally unpainted surfaces

7.9.2 Plug deck drains, deck sprinkler nozzles, and open pipes.

7.9.3 Wrap all valve stems and exposed portions of motor shafts, and hydraulic cylinders.

7.9.4 Install filter media on all tank vents and air intake vents.

7.9.5 Install protective coverings on all installed equipment, port lights, windows, permanently installed label plates, light globes, switches and pushbuttons, vent screens and appurtenances not intended to be painted.

7.9.6 Protect running rigging and mooring lines which cannot be stowed or removed for the duration of non-skid removal.

7.9.7 Protective covering will be inspected at regular intervals, but not less than the start of each work shift. Degraded covering will be repaired prior to restart of work.

7.10 Cleaning

7.10.1 Solvent clean the entire flight deck underside using SSPC-SP1, as guidance, using biodegradable detergent to remove all dirt, oil, grease, soluble salts or other organic matter from the specified surfaces. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Cleaning agents and the containment of runoff shall comply with all applicable local, state, and country laws and regulations. Conduct solvent cleaning at 2,500 to 3,000 PSI

7.10.2 Final wash-down will be made with clean, fresh water at 2,500 to 3,000 PSI. Containment of runoff shall comply with all applicable local, state, and country laws and regulations.

7.11 Upon completion of work, ship force will conduct final operational testing of all lights, speakers, flood and spot lights, sprinkler heads and drains in the vicinity of the flight deck underside work completed. Contractor and ship force will compare "post work" performance test results to the "as-found" conditions previously recorded in paragraphs 7.2.1 through

7.2.3. Correct any discrepancies noted between the “as found” and “post work” conditions. All repaired items will be satisfactorily re-demonstrated to ship force and OMT REP.

7.12 Surface Preparation

7.12.1 Solvent Cleaning the areas identified in section 3.0 using SSPC-SP1, as guidance, using biodegradable detergent to remove all dirt, oil, grease, soluble salts, or other organic matter from the specified surfaces. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible.

7.12.2 The final wash-down will be made with clean, fresh water at 2,500 to 3,000 PSI. Upon completion of all water washing, chloride testing will be performed at a rate of no less than one test per 500 Square Feet of surface.

7.12.3 The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.12.4 If contamination is found, additional water wash will be performed. Additional tests will be made as necessary to determine the extent of contamination and to prove the success of remediation.

7.12.5 Immediately following the completion of fresh water washing, conduct a survey of the areas to be blasted and coated with the following personnel in attendance:

a. Paint Manufacturer’s Representative

b. Contractor’s paint representative

c. OMT REP

d. Ship’s Master or his designated representative

7.12.6 Any pipe hanger and light bracket and other attaching hardware, handrails, stanchions, or other areas/items beyond ability to preserve shall be noted via Condition Report.

7.12.7 Method

a. Ultra-High Pressure water jet the designated areas of the flight deck underside using SSPC-SP WJ-2/NACE WJ-2 to a WJ-2 finish using Reference 2.2 as guidance. Ensure that the surface profile of the vacuum blasted decking is 2 to 3 mils.

b. Dry abrasive blast the designated flight deck underside areas using SSPC-SP10/NACE 2, Near White Metal Blast Cleaning; using Reference 2.3 as guidance. The surface profile achieved will be angular in nature, and within the range set by the coating manufacturer’s product data sheet for the coating system being applied. Prior to the start of abrasive blasting the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.12.8 Where spot blasting has been specified, edges of adjacent intact coating will be feathered-in.

7.12.9 No more than a light (L) grade flash rust will be allowed on the steel at the time of coating application. If heavier flash rust is present, the surface will be pressure washed at 2,500 to 3,000 PSI and allowed to air dry to restore the surface to a paint-able condition.

7.12.10 Accomplish the requirements of SSPC-SP-11 Power Tool Clean to Bare Metal to any areas inaccessible to the abrasive blast or water jetting including foundations and other appendages in way of the aforementioned areas to bare metal using Reference 2.4 as guidance.

7.12.11 After surface preparation, all surfaces will be blown down using clean dry air to remove all dust, dirt and debris.

7.12.12 Regardless of whether the surfaces are abrasive blasted or UHP Water Jetted, immediately prior to coating the surface will be re-tested for non-visual surface contaminants at a rate of no less than one test per 1,000 square feet of prepared area. The maximum allowable contamination concentrations for surfaces prepared by means of dry abrasive blasting will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment. For those surfaces prepared by means of UHP water jetting, the limits of NV-2 found in the appendix of SSPC-SP WJ-2/NACE WJ-2 will apply.

7.13 Coatings Application

7.13.1 Prior to application of any coating, the area to be painted will be inspected and approved by the OMT REP and the Paint Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include measuring and recording:

a. type of surface preparation used

b. surface cleanliness

c. surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape

d. environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity)

e. over coating interval (curing time) since previous application

f. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet. DFT gages will be calibrated, spot readings will be taken and DFT tolerances will be using SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages.

7.13.2 Ensure the following conditions are met prior to painting:

a. Surfaces will be clean, dry, and free of oil, grease or residue from abrasive blasting.

b. Surface appearance meets the definition of SSPC-SP WJ2/NACE WJ-2 Surface Preparation of Steel and Other Hard Materials by High and Ultra-High Pressure Water Jetting Prior to Recoating, WJ-2 Condition or SSPC-SP10/NACE 2, Near White Metal Blast Cleaning as applicable.

c. Air and substrate temperatures will be within the range published by the manufacturer, see Reference 2.5. Surface temp during application and curing is acceptable down to:

1. 23°F (-5°C) for the Amershield.

d. During application, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer, will not exceed 85%.

f. Condensation and/or rain is not to contact the uncured Amershield as it may change the color and gloss.

7.13.3 No coating will be applied below 35°F (-1.7°) without prior written approval of the OMT REP.

7.13.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.13.5 All coatings will be prepared and applied using the manufacturer’s requirements outlined in the Product Data Sheets, using Reference 2.5 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millage are being applied.

7.13.6 EPOXY

a. Apply the following 1st and 2nd Coats of Primer to all blasted and power tool cleaned surface.

b. One full coat of Amercoat 240, (red oxide) at 5-6 mils DFT.

c. One stripe coat of Amercoat 240, (off-white) at 2-3 mils DFT.

d. One full coat of Amercoat 240, (off-white) 5-6 mils DFT.

e. The stripe coat will be applied to all weldments, crevices, corners, edges, and other areas not conducive to proper coverage. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.13.7 POLYURETHANE

a. Apply One full top coat of Amershield, (haze gray) 3-4 mils DFT to the flight deck underside.

b. The environmental conditions, DFT of each coat, cure and over coating intervals are critical to the application. The application of all coatings will adhere to the Product Data Sheets and OMT Paint Rep. For weather conditions outside 35˚F (-1.7˚C) through 90˚F (32˚C) range consult the OMT Paint Representative for proper overcoat intervals.

7.14 During all surface preparation and coating operations, the area involved will be roped off or otherwise barricaded to prevent entry by persons other than those directly involved in the work underway or those inspecting same. Cordoning or barricading will remain in place until the topcoat is properly cured.

7.15 Upon completion the vessel will be cleaned of all residues resulting from the surface preparation and painting operations. Remove all protective coverings, debris and replace all interferences removed in the performance of this item.

7.16 Upon completion of work, ship's force will conduct final operational testing of all lights, speakers, flood and spot lights, sprinkler heads and drains in the vicinity of the flight deck underside work completed. Contractor and ship force will compare "post work" performance test results to the "as-found" conditions previously recorded in paragraphs 7.2.1 through 7.2.3. Correct any discrepancies noted between the “as found” and “post work” conditions. All repaired items will be satisfactorily re-demonstrated to ship force and OMT REP.

7.17 Reports

7.17.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.17.2 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.17.3 Prepare a Paint Report and submit same to the OMT REP within three days of completing the coating application. The report will include the following data:

a. The location, date, and time of each coating application.

b. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

c. Interval between coatings.

d. Dry film thickness readings at a rate of five spot readings per 1,000 sq. ft. of surface for each coat of paint. DFT gages will be calibrated, and spot readings will be taken using SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages as guidance.

e. Paint Manufacturer, Product Identification Number, color, and Batch Numbers for each coat of paint applied.

f. Surface profile measurements of the metal substrate.

g. Results of testing for non-visible surface contaminants (soluble salts).

7.18 Painting: None additional

7.19 Manufacturer's Representative

7.19.1 A Government Paint Technical Representative will be in attendance, on the government’s account, to observe the surface preparation and coating application on the government’s behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor’s efforts. He/She is not for supervision of the contractor’s workforce.

7.20 Preparation of Drawings

7.20.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
HULL AND STRUCTURAL		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0152	CATEGORY "A"	2026-01-30
TESB_CCSI_ABS ANNUAL SURVEY - TANK INSPECTION		Manovich, David

1.0 ABSTRACT

The purpose of this work item is to assist with the inspections of the vessel's ballast tanks, combined cargo/ballast tanks and peak tanks for an ABS Annual Tank Survey.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSEA Drawing No. 801-8499371, General Arrangement

2.1.2 NAVSEA Drawing No. 835-8495719, Capacity Plan

2.1.3 NAVSEA Drawing No. 631-8393784, Paint Schedule

2.2 Enclosure

2.2.1 Tank Survey Status Form

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations: Various

3.1.2 Quantity: See Enclosure 2.2.2

3.2 Tanks to be inspected:

3.2.1 Tank No: 11

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of

tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 THIS SHIP CONTAINS HIGH VOLTAGE (HV) ELECTRICAL SYSTEMS. OBEY ALL POSTED AND VERBAL INSTRUCTIONS REGARDING SAFETY AND EXCLUSION FROM HIGH VOLTAGE AREAS. AT NO TIME APPROACH, WORK ON, OR ENTER A HIGH VOLTAGE AREA WITHOUT PROPER AUTHORIZATION.

5.3 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

5.4 Fall protection must be provided to all personnel when the possibility of a fall of 4 feet or more is present on elevated surfaces, with unprotected sides, edges, or openings per SMS Procedure 2.1-014-ALL (latest revision). In addition, 29 CFR §1910.28 covers personal fall arrest system and ladder safety requirements for vertical ladders.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide labor and material to support tank inspections identified below by ABS and MSC using References 2.1.1 through 2.1.3 and Enclosures 2.2.1 as guidance.

7.1.1 Provide six personal fall arrest systems to the OMT REP for use during the entire performance period. The personal fall arrest systems will be used by the OMT REP, ABS

Surveyors, and Government furnished Technical Representatives while accessing tanks. The personal fall arrest systems are to consist of six sets of fall protection harnesses and six "Y" lanyards (six feet). The personal fall arrest systems are to be ANSI Z359 (current revision) and OSHA compliant and be maintained in good condition. The fall protection harnesses are to have four or more D-Rings, adjustable restraint lanyards, adjustable twin leg fall arrest lanyards (Y-type), and single shock absorber lanyards fitted with large “ladder” hooks and double locking carabineers.

7.1.2 Provide six personal floatation devices (PFD) to the OMT REP whenever working over the side or fall to water exists.

7.1.3 Provide a portable davit or tripod with block and tackle equal to a G4 Rescue System by Gravitec. The system is to be maintained onboard the vessel at the ships gangway for the entire period of performance for use by rescue personnel in the event of an emergency to aid in the extraction of personnel from tanks.

7.1.4 Provide a safety observer during all tank surveys to remain at the tank entrance to monitor the safety of all survey personnel inside the tank or space and raise the alarm if an emergency arises.

7.1.5 Provide a portable, explosion proof communication system between the survey party in the tank or space and the safety observer on deck. This communication system is to also include the personnel in charge of ballast pump operations if boats or rafts are to be used inside the tanks.

7.1.6 Provide the services of an ABS certified Thickness Measurement firm to perform thickness measurements of tank components in areas of substantial corrosion, where found or identified at previous surveys, or when considered necessary by the ABS Surveyor and as authorized by the OMT REP. Assume approximately 30 measurements.

7.2 Clean all tanks prior to ABS Surveyor and OMT REP inspection. Tanks are to be sufficiently clean and free from water, scale, dirt, and oil residues to reveal corrosion, deformation, fractures, damages, other structural deterioration, and condition of the coating. Remove from surfaces all loose accumulated corrosion scale.

7.3 Provide sufficient lighting to reveal corrosion, deformation, fractures, damages, other structural deterioration, and condition of the coating.

7.4 Provide a safe means of access to conduct the survey for all tanks being inspected using Reference 2.1.1 as guidance.

7.5 Provide a shipfitter to accompany the ABS Surveyor and OMT REP to record conditions and locations of all deficiencies as noted by the ABS Surveyor and OMT REP during tank inspections.

7.6 Submit a condition report for all tanks noting all the “as found” conditions and recommendations.

7.7 Reports

7.7.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.7.2 Complete and submit a weekly status report of the tank survey progress using Enclosure 2.2.2. The report is to identify completion of applicable survey milestones in each tank: Visual exam, UT exam, Close-Up survey, and Hydrostatic testing. The report shall be submitted to the OMT REP prior to the weekly Production and Progress meeting, refer to Work Item 013.

7.7.3 Submit a report that includes “as released” condition of the tanks effected by this work item. Submit one (1) electronic copy in Portable Document Format (PDF).

7.7.4 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.8 Painting: None additional

7.9 Manufacturer’s Representative: None

7.10 Preparation of Drawings:

7.10.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 – TANK SURVEY FORM

Tank (description)	Tank Type (*)	Overall Survey (**)	Close-Up Survey (**)	Thickness Measurement (**)	Tank Testing (**)	Remarks: (wastage, structural damage, fractures, suspect areas, tank vents, sounding tubes, striker plates, ladders, anodes, …)
SWB Tank 4-16-0	B					
SWB Tank 4-108-0	B					
SWB Tank 4-57-4	B					
SWB Tank 4-57-3	B					
SWB Tank 3-126-0	B					
SWB Tank 5-97-4	B					
SWB Tank 5-97-5	B					
Incinerator Sludge Tank A-29-1	Sludge					
Food Waste Tank A1-92.5-2	Sludge					
Grey Water Tank A1-92-6	Water					
Jp-5 Overflow Tank 5-97-3	FO					

ENCLOSURE 2.2.1 – TANK SURVEY FORM

Tank (description)	Tank Type (*)	Overall Survey (**)	Close-Up Survey (**)	Thickness Measurement (**)	Tank Testing (**)	Remarks: (wastage, structural damage, fractures, suspect areas, tank vents, sounding tubes, striker plates, ladders, anodes, …)

* - Ballast(B), Fuel Oil(FO), Lube Oil(LO), Freshwater(FW), Potable Water(PW), Void(V)
** - Not Started (Date), In Progress (Date), Complete (Date)

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect, and recoat the vessel's saltwater ballast tanks.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 835-8495719, Selected Record Drawing Capacity Plan

2.2 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.3 Surface Preparation Standard, SSPC-SP10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.5 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel (Available Commercially)

2.6 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.7 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: 4-16-0

3.2 Quantity:

3.2.1 Ballast Tank approx. surface area

a. 136,285 square feet.

3.2.2 Tank Zincs

a. Approx. amount of 12 lb zincs: 130

3.3 Description: Saltwater Ballast Tanks and Associated

Zincs

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1^{st} primer coat)	Buff	1136 gal
Amercoat 240	Epoxy (stripe coat)	Red Oxide	568 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	1136 gal
Amercoat T-10	Solvent	N/A	285 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government's account, to observe the surface preparation and coating application on the government's behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor's efforts. They are not for supervision of the contractor's workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated

with monitoring, containment, handling, collection, processing, clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. All labor and material costs associated with this level of effort, where required or found necessary, shall be included in the price of this work item.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS:

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections where required to ensure compliance with this Work Item including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.7 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor that shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval must be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces identified in Section 3.0.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient size to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must:

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C) differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to

personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation, submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report is to be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect the same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or coating spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coating work.

7.13 Prior to the start of surface preparation perform an inspection of all zincs in the ballast tank. Develop a report that details the location, size, and condition of the zincs to include percent wastage, quantity, replacement weight, and mounting method for each zinc in the tank.

7.13.1 Temporarily remove bolted zinc anodes in way of surface preparation and paint application. Reinstall bolt-on zincs upon completion of all coatings work.

a. For bidding purposes, base the estimate on removing and renewing 0% of bolted-on zincs. Replacement of welded zinc will be subject to issuance of change order.

7.13.2 If zincs are welded in, apply protective coverings over those zincs if wastage is less than 50%.

7.13.3 Submit report to OMT REP.

7.14 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.15 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the coating manufacturer’s representative. In no case will the paint be thinned more than 5% by volume.

7.16 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.17 Surface Preparation

7.17.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals, crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with initial water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

b. After completion of accomplishment of SSPC-SP1 and removal of all liquids, mud, dirt, and debris, area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative.

7.17.2 The entire interior of the tank, including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast Cleaning. The blast profile achieved must be angular in nature and within the range set by the manufacturer’s product data sheet for the coating system being applied.

7.17.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.17.4 Upon completion of abrasive blasting, all spent

grit, dust, and dirt generated by this work item will be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.17.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit a Condition Found Report documenting results of inspection and recommended action.

7.18 Coatings Application

7.18.1 Prior to application of any coating, the area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity).

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet.

7.18.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the coating manufacturer, use Reference 2.8 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.18.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.18.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.18.5 All coatings will be prepared and applied using the manufacturer’s requirements outlined in the Product Data Sheets, use Reference 2.8 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.18.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.18.7 EPOXY Apply the following Coats to all prepared surfaces:

a. One full coat of Amercoat 240, (buff) at 5-6 mils DFT.

b. One stripe coat of Amercoat 240, (red oxide) at 2-3 mils DFT.

c. One full coat of Amercoat 240, (off white) at 5-6 mils DFT.

7.19 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.19.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.20 During all surface preparation and coating operations, the area involved will only be accessed by people directly involved in the work underway or those inspecting the same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.21 Upon completion of coating activities, fit, weld, NDT, and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.21.1 Conduct tightness test.

a. Testing shall be accomplished prior to

preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.22 Upon completion of the air test, remove all blanks and plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in, and touched-up with the topcoat only.

7.23 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.24 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.25 Reports

7.25.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.25.2 Submit a report that includes “as released”

condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.25.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company's representative.

7.25.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Coating Manufacturer, Product Identification Number, quantity, color, and batch numbers for each coat applied.

7.26 Painting: None additional

7.27 Manufacturer's Representative: None additional

7.28 Preparation of Drawings

7.28.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect, and recoat the vessel's saltwater ballast tanks.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 835-8495719, Selected Record Drawing Capacity Plan

2.2 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.3 Surface Preparation Standard, SSPC-SP10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.5 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel (Available Commercially)

2.6 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.7 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: 4-108-0

3.2 Quantity:

3.2.1 Ballast Tank approx. surface area

a. 28,989 square feet.

3.2.2 Tank Zincs

a. Approx. amount of 12 lb zincs: 60

3.3 Description: Saltwater Ballast Tanks and Associated Zincs

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Buff	242 gal
Amercoat 240	Epoxy (stripe coat)	Red Oxide	121 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	242 gal
Amercoat T-10	Solvent	N/A	61 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government's account, to observe the surface preparation and coating application on the government's behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor's efforts. They are not for supervision of the contractor's workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated

with monitoring, containment, handling, collection, processing, clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. All labor and material costs associated with this level of effort, where required or found necessary, shall be included in the price of this work item.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS:

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections where required to ensure compliance with this Work Item including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.7 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor that shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval must be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces identified in Section 3.0.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient size to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must:

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C) differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to

personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation, submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report is to be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect the same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or coating spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coating work.

7.13 Prior to the start of surface preparation perform an inspection of all zincs in the ballast tank. Develop a report that details the location, size, and condition of the zincs to include percent wastage, quantity, replacement weight, and mounting method for each zinc in the tank.

7.13.1 Temporarily remove bolted zinc anodes in way of surface preparation and paint application. Reinstall bolt-on zincs upon completion of all coatings work.

a. For bidding purposes, base the estimate on removing and renewing 0% of bolted-on zincs. Replacement of welded zinc will be subject to issuance of change order.

7.13.2 If zincs are welded in, apply protective coverings over those zincs if wastage is less than 50%.

7.13.3 Submit report to OMT REP.

7.14 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.15 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the coating manufacturer’s representative. In no case will the paint be thinned more than 5% by volume.

7.16 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.17 Surface Preparation

7.17.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals, crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with initial water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

b. After completion of accomplishment of SSPC-SP1 and removal of all liquids, mud, dirt, and debris, area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative.

7.17.2 The entire interior of the tank, including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast Cleaning. The blast profile achieved must be angular in nature and within the range set by the manufacturer’s product data sheet for the coating system being applied.

7.17.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.17.4 Upon completion of abrasive blasting, all spent

grit, dust, and dirt generated by this work item will be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.17.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit a Condition Found Report documenting results of inspection and recommended action.

7.18 Coatings Application

7.18.1 Prior to application of any coating, the area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity).

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet.

7.18.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the coating manufacturer, use Reference 2.8 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.18.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.18.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.18.5 All coatings will be prepared and applied using the manufacturer's requirements outlined in the Product Data Sheets, use Reference 2.8 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.18.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.18.7 EPOXY Apply the following Coats to all prepared surfaces:

a. One full coat of Amercoat 240, (buff) at 5-6 mils DFT.

b. One stripe coat of Amercoat 240, (red oxide) at 2-3 mils DFT.

c. One full coat of Amercoat 240, (off white) at 5-6 mils DFT.

7.19 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.19.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.20 During all surface preparation and coating operations, the area involved will only be accessed by people directly involved in the work underway or those inspecting the same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.21 Upon completion of coating activities, fit, weld, NDT, and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.21.1 Conduct tightness test.

a. Testing shall be accomplished prior to

preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.22 Upon completion of the air test, remove all blanks and plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in, and touched-up with the topcoat only.

7.23 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.24 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.25 Reports

7.25.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.25.2 Submit a report that includes “as released”

condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.25.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company's representative.

7.25.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Coating Manufacturer, Product Identification Number, quantity, color, and batch numbers for each coat applied.

7.26 Painting: None additional

7.27 Manufacturer's Representative: None additional

7.28 Preparation of Drawings

7.28.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect, and recoat the vessel's saltwater ballast tanks.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 835-8495719, Selected Record Drawing Capacity Plan

2.2 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.3 Surface Preparation Standard, SSPC-SP10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.5 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel (Available Commercially)

2.6 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.7 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: 4-57-4

3.2 Quantity:

3.2.1 Ballast Tank approx. surface area

a. 48,778 square feet.

3.2.2 Tank Zincs

a. Approx. amount of 12 lb zincs: 35

3.3 Description: Saltwater Ballast Tanks and Associated

Zincs

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Buff	407 gal
Amercoat 240	Epoxy (stripe coat)	Red Oxide	203 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	407 gal
Amercoat T-10	Solvent	N/A	102 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government's account, to observe the surface preparation and coating application on the government's behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor's efforts. They are not for supervision of the contractor's workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated with monitoring, containment, handling, collection, processing,

clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. All labor and material costs associated with this level of effort, where required or found necessary, shall be included in the price of this work item.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS:

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections where required to ensure compliance with this Work Item including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.7 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor that shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval must be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces identified in Section 3.0.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient size to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must:

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C) differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to

personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation, submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report is to be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect the same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or coating spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coating work.

7.13 Prior to the start of surface preparation perform an inspection of all zincs in the ballast tank. Develop a report that details the location, size, and condition of the zincs to include percent wastage, quantity, replacement weight, and mounting method for each zinc in the tank.

7.13.1 Temporarily remove bolted zinc anodes in way of surface preparation and paint application. Reinstall bolt-on zincs upon completion of all coatings work.

a. For bidding purposes, base the estimate on removing and renewing 0% of bolted-on zincs. Replacement of welded zinc will be subject to issuance of change order.

7.13.2 If zincs are welded in, apply protective coverings over those zincs if wastage is less than 50%.

7.13.3 Submit report to OMT REP.

7.14 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.15 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the coating manufacturer’s representative. In no case will the paint be thinned more than 5% by volume.

7.16 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.17 Surface Preparation

7.17.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals, crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with initial water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

b. After completion of accomplishment of SSPC-SP1 and removal of all liquids, mud, dirt, and debris, area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative.

7.17.2 The entire interior of the tank, including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast Cleaning. The blast profile achieved must be angular in nature and within the range set by the manufacturer’s product data sheet for the coating system being applied.

7.17.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.17.4 Upon completion of abrasive blasting, all spent

grit, dust, and dirt generated by this work item will be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.17.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit a Condition Found Report documenting results of inspection and recommended action.

7.18 Coatings Application

7.18.1 Prior to application of any coating, the area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity).

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet.

7.18.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the coating manufacturer, use Reference 2.8 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.18.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.18.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.18.5 All coatings will be prepared and applied using the manufacturer's requirements outlined in the Product Data Sheets, use Reference 2.8 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.18.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.18.7 EPOXY Apply the following Coats to all prepared surfaces:

a. One full coat of Amercoat 240, (buff) at 5-6 mils DFT.

b. One stripe coat of Amercoat 240, (red oxide) at 2-3 mils DFT.

c. One full coat of Amercoat 240, (off white) at 5-6 mils DFT.

7.19 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.19.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.20 During all surface preparation and coating operations, the area involved will only be accessed by people directly involved in the work underway or those inspecting the same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.21 Upon completion of coating activities, fit, weld, NDT, and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.21.1 Conduct tightness test.

a. Testing shall be accomplished prior to

preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.22 Upon completion of the air test, remove all blanks and plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in, and touched-up with the topcoat only.

7.23 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.24 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.25 Reports

7.25.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.25.2 Submit a report that includes “as released”

condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.25.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company's representative.

7.25.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Coating Manufacturer, Product Identification Number, quantity, color, and batch numbers for each coat applied.

7.26 Painting: None additional

7.27 Manufacturer's Representative: None additional

7.28 Preparation of Drawings

7.28.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect, and recoat the vessel's saltwater ballast tanks.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 835-8495719, Selected Record Drawing Capacity Plan

2.2 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.3 Surface Preparation Standard, SSPC-SP10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.5 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel (Available Commercially)

2.6 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.7 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: 4-57-3

3.2 Quantity:

3.2.1 Ballast Tank approx. surface area

a. 47,778 square feet.

3.2.2 Tank Zincs

a. Approx. amount of 12 lb zincs: 35

3.3 Description: Saltwater Ballast Tanks and Associated

Zincs

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Buff	398 gal
Amercoat 240	Epoxy (stripe coat)	Red Oxide	100 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	398 gal
Amercoat T-10	Solvent	N/A	90 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government’s account, to observe the surface preparation and coating application on the government’s behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor’s efforts. They are not for supervision of the contractor’s workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated with monitoring, containment, handling, collection, processing,

clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. All labor and material costs associated with this level of effort, where required or found necessary, shall be included in the price of this work item.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS:

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections where required to ensure compliance with this Work Item including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.7 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor that shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval must be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces identified in Section 3.0.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient size to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must:

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C) differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to

personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation, submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report is to be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect the same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or coating spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coating work.

7.13 Prior to the start of surface preparation perform an inspection of all zincs in the ballast tank. Develop a report that details the location, size, and condition of the zincs to include percent wastage, quantity, replacement weight, and mounting method for each zinc in the tank.

7.13.1 Temporarily remove bolted zinc anodes in way of surface preparation and paint application. Reinstall bolt-on zincs upon completion of all coatings work.

a. For bidding purposes, base the estimate on removing and renewing 0% of bolted-on zincs. Replacement of welded zinc will be subject to issuance of change order.

7.13.2 If zincs are welded in, apply protective coverings over those zincs if wastage is less than 50%.

7.13.3 Submit report to OMT REP.

7.14 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.15 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the coating manufacturer’s representative. In no case will the paint be thinned more than 5% by volume.

7.16 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.17 Surface Preparation

7.17.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals, crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with initial water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

b. After completion of accomplishment of SSPC-SP1 and removal of all liquids, mud, dirt, and debris, area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative.

7.17.2 The entire interior of the tank, including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast Cleaning. The blast profile achieved must be angular in nature and within the range set by the manufacturer’s product data sheet for the coating system being applied.

7.17.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.17.4 Upon completion of abrasive blasting, all spent

grit, dust, and dirt generated by this work item will be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.17.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit a Condition Found Report documenting results of inspection and recommended action.

7.18 Coatings Application

7.18.1 Prior to application of any coating, the area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity).

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet.

7.18.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the coating manufacturer, use Reference 2.8 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.18.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.18.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.18.5 All coatings will be prepared and applied using the manufacturer's requirements outlined in the Product Data Sheets, use Reference 2.8 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.18.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.18.7 EPOXY Apply the following Coats to all prepared surfaces:

a. One full coat of Amercoat 240, (buff) at 5-6 mils DFT.

b. One stripe coat of Amercoat 240, (red oxide) at 2-3 mils DFT.

c. One full coat of Amercoat 240, (off white) at 5-6 mils DFT.

7.19 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.19.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.20 During all surface preparation and coating operations, the area involved will only be accessed by people directly involved in the work underway or those inspecting the same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.21 Upon completion of coating activities, fit, weld, NDT, and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.21.1 Conduct tightness test.

a. Testing shall be accomplished prior to

preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.22 Upon completion of the air test, remove all blanks and plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in, and touched-up with the topcoat only.

7.23 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.24 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.25 Reports

7.25.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.25.2 Submit a report that includes “as released”

condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.25.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company's representative.

7.25.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Coating Manufacturer, Product Identification Number, quantity, color, and batch numbers for each coat applied.

7.26 Painting: None additional

7.27 Manufacturer's Representative: None additional

7.28 Preparation of Drawings

7.28.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect, and recoat the vessel's saltwater ballast tanks.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 835-8495719, Selected Record Drawing Capacity Plan

2.2 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.3 Surface Preparation Standard, SSPC-SP10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.5 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel (Available Commercially)

2.6 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.7 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: 3-126-0

3.2 Quantity:

3.2.1 Ballast Tank approx. surface area

a. 66,760 square feet.

3.2.2 Tank Zincs

a. Approx. amount of 12 lb zincs: 52

3.3 Description: Saltwater Ballast Tanks and Associated

Zincs

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Buff	557 gal
Amercoat 240	Epoxy (stripe coat)	Red Oxide	180 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	557 gal
Amercoat T-10	Solvent	N/A	130 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government’s account, to observe the surface preparation and coating application on the government’s behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor’s efforts. They are not for supervision of the contractor’s workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated with monitoring, containment, handling, collection, processing,

clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. All labor and material costs associated with this level of effort, where required or found necessary, shall be included in the price of this work item.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS:

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections where required to ensure compliance with this Work Item including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.7 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor that shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval must be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces identified in Section 3.0.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient size to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must:

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C) differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to

personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation, submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report is to be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect the same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or coating spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coating work.

7.13 Prior to the start of surface preparation perform an inspection of all zincs in the ballast tank. Develop a report that details the location, size, and condition of the zincs to include percent wastage, quantity, replacement weight, and mounting method for each zinc in the tank.

7.13.1 Temporarily remove bolted zinc anodes in way of surface preparation and paint application. Reinstall bolt-on zincs upon completion of all coatings work.

a. For bidding purposes, base the estimate on removing and renewing 0% of bolted-on zincs. Replacement of welded zinc will be subject to issuance of change order.

7.13.2 If zincs are welded in, apply protective coverings over those zincs if wastage is less than 50%.

7.13.3 Submit report to OMT REP.

7.14 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.15 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the coating manufacturer’s representative. In no case will the paint be thinned more than 5% by volume.

7.16 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.17 Surface Preparation

7.17.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals, crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with initial water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

b. After completion of accomplishment of SSPC-SP1 and removal of all liquids, mud, dirt, and debris, area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative.

7.17.2 The entire interior of the tank, including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast Cleaning. The blast profile achieved must be angular in nature and within the range set by the manufacturer’s product data sheet for the coating system being applied.

7.17.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.17.4 Upon completion of abrasive blasting, all spent

grit, dust, and dirt generated by this work item will be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.17.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit a Condition Found Report documenting results of inspection and recommended action.

7.18 Coatings Application

7.18.1 Prior to application of any coating, the area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity).

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet.

7.18.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the coating manufacturer, use Reference 2.8 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.18.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.18.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.18.5 All coatings will be prepared and applied using the manufacturer's requirements outlined in the Product Data Sheets, use Reference 2.8 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.18.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.18.7 EPOXY Apply the following Coats to all prepared surfaces:

a. One full coat of Amercoat 240, (buff) at 5-6 mils DFT.

b. One stripe coat of Amercoat 240, (red oxide) at 2-3 mils DFT.

c. One full coat of Amercoat 240, (off white) at 5-6 mils DFT.

7.19 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.19.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.20 During all surface preparation and coating operations, the area involved will only be accessed by people directly involved in the work underway or those inspecting the same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.21 Upon completion of coating activities, fit, weld, NDT, and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.21.1 Conduct tightness test.

a. Testing shall be accomplished prior to

preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.22 Upon completion of the air test, remove all blanks and plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in, and touched-up with the topcoat only.

7.23 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.24 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.25 Reports

7.25.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.25.2 Submit a report that includes “as released”

condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.25.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company’s representative.

7.25.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Coating Manufacturer, Product Identification Number, quantity, color, and batch numbers for each coat applied.

7.26 Painting: None additional

7.27 Manufacturer's Representative: None additional

7.28 Preparation of Drawings

7.28.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect, and recoat the vessel's saltwater ballast tanks.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 835-8495719, Selected Record Drawing Capacity Plan

2.2 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.3 Surface Preparation Standard, SSPC-SP10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.5 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel (Available Commercially)

2.6 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.7 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: 5-97-4

3.2 Quantity:

3.2.1 Ballast Tank approx. surface area

a. 52,161 square feet.

3.2.2 Tank Zincs

a. Approx. amount of 12 lb zincs: 44

3.3 Description: Saltwater Ballast Tanks and Associated

Zincs

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Buff	435 gal
Amercoat 240	Epoxy (stripe coat)	Red Oxide	120 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	435 gal
Amercoat T-10	Solvent	N/A	100 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government's account, to observe the surface preparation and coating application on the government's behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor's efforts. They are not for supervision of the contractor's workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated with monitoring, containment, handling, collection, processing,

clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. All labor and material costs associated with this level of effort, where required or found necessary, shall be included in the price of this work item.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS:

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections where required to ensure compliance with this Work Item including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.7 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor that shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval must be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces identified in Section 3.0.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient size to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must:

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C) differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to

personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation, submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report is to be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect the same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or coating spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coating work.

7.13 Prior to the start of surface preparation perform an inspection of all zincs in the ballast tank. Develop a report that details the location, size, and condition of the zincs to include percent wastage, quantity, replacement weight, and mounting method for each zinc in the tank.

7.13.1 Temporarily remove bolted zinc anodes in way of surface preparation and paint application. Reinstall bolt-on zincs upon completion of all coatings work.

a. For bidding purposes, base the estimate on removing and renewing 0% of bolted-on zincs. Replacement of welded zinc will be subject to issuance of change order.

7.13.2 If zincs are welded in, apply protective coverings over those zincs if wastage is less than 50%.

7.13.3 Submit report to OMT REP.

7.14 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.15 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the coating manufacturer’s representative. In no case will the paint be thinned more than 5% by volume.

7.16 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.17 Surface Preparation

7.17.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals, crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with initial water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

b. After completion of accomplishment of SSPC-SP1 and removal of all liquids, mud, dirt, and debris, area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative.

7.17.2 The entire interior of the tank, including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast Cleaning. The blast profile achieved must be angular in nature and within the range set by the manufacturer’s product data sheet for the coating system being applied.

7.17.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.17.4 Upon completion of abrasive blasting, all spent

grit, dust, and dirt generated by this work item will be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.17.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit a Condition Found Report documenting results of inspection and recommended action.

7.18 Coatings Application

7.18.1 Prior to application of any coating, the area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity).

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet.

7.18.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the coating manufacturer, use Reference 2.8 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.18.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.18.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.18.5 All coatings will be prepared and applied using the manufacturer's requirements outlined in the Product Data Sheets, use Reference 2.8 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.18.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.18.7 EPOXY Apply the following Coats to all prepared surfaces:

a. One full coat of Amercoat 240, (buff) at 5-6 mils DFT.

b. One stripe coat of Amercoat 240, (red oxide) at 2-3 mils DFT.

c. One full coat of Amercoat 240, (off white) at 5-6 mils DFT.

7.19 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.19.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.20 During all surface preparation and coating operations, the area involved will only be accessed by people directly involved in the work underway or those inspecting the same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.21 Upon completion of coating activities, fit, weld, NDT, and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.21.1 Conduct tightness test.

a. Testing shall be accomplished prior to

preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.22 Upon completion of the air test, remove all blanks and plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in, and touched-up with the topcoat only.

7.23 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.24 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.25 Reports

7.25.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.25.2 Submit a report that includes “as released”

condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.25.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company's representative.

7.25.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Coating Manufacturer, Product Identification Number, quantity, color, and batch numbers for each coat applied.

7.26 Painting: None additional

7.27 Manufacturer's Representative: None additional

7.28 Preparation of Drawings

7.28.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect, and recoat the vessel's saltwater ballast tanks.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 835-8495719, Selected Record Drawing Capacity Plan

2.2 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.3 Surface Preparation Standard, SSPC-SP10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.5 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel (Available Commercially)

2.6 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.7 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: 5-97-5

3.2 Quantity:

3.2.1 Ballast Tank approx. surface area

a. 52,073 square feet.

3.2.2 Tank Zincs

a. Approx. amount of 12 lb zincs: 44

3.3 Description: Saltwater Ballast Tanks and Associated Zincs

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Buff	434 gal
Amercoat 240	Epoxy (stripe coat)	Red Oxide	120 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	434 gal
Amercoat T-10	Solvent	N/A	100 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government’s account, to observe the surface preparation and coating application on the government’s behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor’s efforts. They are not for supervision of the contractor’s workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated

with monitoring, containment, handling, collection, processing, clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. All labor and material costs associated with this level of effort, where required or found necessary, shall be included in the price of this work item.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS:

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections where required to ensure compliance with this Work Item including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.7 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor that shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval must be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces identified in Section 3.0.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient size to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must:

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C) differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to

personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation, submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report is to be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect the same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or coating spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coating work.

7.13 Prior to the start of surface preparation perform an inspection of all zincs in the ballast tank. Develop a report that details the location, size, and condition of the zincs to include percent wastage, quantity, replacement weight, and mounting method for each zinc in the tank.

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a. For bidding purposes, base the estimate on removing and renewing 0% of bolted-on zincs. Replacement of welded zinc will be subject to issuance of change order.

7.13.2 If zincs are welded in, apply protective coverings over those zincs if wastage is less than 50%.

7.13.3 Submit report to OMT REP.

7.14 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.15 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the coating manufacturer's representative. In no case will the paint be thinned more than 5% by volume.

7.16 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.17 Surface Preparation

7.17.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals, crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with initial water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

b. After completion of accomplishment of SSPC-SP1 and removal of all liquids, mud, dirt, and debris, area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative.

7.17.2 The entire interior of the tank, including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast Cleaning. The blast profile achieved must be angular in nature and within the range set by the manufacturer’s product data sheet for the coating system being applied.

7.17.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.17.4 Upon completion of abrasive blasting, all spent

grit, dust, and dirt generated by this work item will be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.17.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit a Condition Found Report documenting results of inspection and recommended action.

7.18 Coatings Application

7.18.1 Prior to application of any coating, the area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity).

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet.

7.18.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the coating manufacturer, use Reference 2.8 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.18.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.18.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.18.5 All coatings will be prepared and applied using the manufacturer's requirements outlined in the Product Data Sheets, use Reference 2.8 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.18.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.18.7 EPOXY Apply the following Coats to all prepared surfaces:

a. One full coat of Amercoat 240, (buff) at 5-6 mils DFT.

b. One stripe coat of Amercoat 240, (red oxide) at 2-3 mils DFT.

c. One full coat of Amercoat 240, (off white) at 5-6 mils DFT.

7.19 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.19.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.20 During all surface preparation and coating operations, the area involved will only be accessed by people directly involved in the work underway or those inspecting the same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.21 Upon completion of coating activities, fit, weld, NDT, and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.21.1 Conduct tightness test.

a. Testing shall be accomplished prior to

preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.22 Upon completion of the air test, remove all blanks and plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in, and touched-up with the topcoat only.

7.23 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.24 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.25 Reports

7.25.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.25.2 Submit a report that includes “as released”

condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.25.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company's representative.

7.25.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Coating Manufacturer, Product Identification Number, quantity, color, and batch numbers for each coat applied.

7.26 Painting: None additional

7.27 Manufacturer's Representative: None additional

7.28 Preparation of Drawings

7.28.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect and recoat the vessel's waste water/sludge tanks.

2.0 REFERENCES

2.1 MSC Drawing No. 835-8499355, Capacity Plan

2.2 MSC Drawing No. 801-8499371, General Arrangement

2.3 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP-10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.5 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.6 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel

2.7 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.8 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: A-29-1

3.2 Quantity:

3.2.1 Waste water/sludge Tank approx. surface area

a. 30 square feet.

3.2.2 Tank Zincs: None

3.3 Description: Incinerator Sludge

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Pastel Green	1 gal
Amercoat 240	Epoxy (stripe coat)	Buff	1 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	1 gal
Amercoat T-10	Solvent	N/A	1 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government’s account, to observe the surface preparation and coating application on the government’s behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor’s efforts. They are not for supervision of the contractor’s workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated with monitoring, containment, handling, collection, processing, clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. ALL LABOR AND MATERIAL COSTS ASSOCIATED WITH THIS LEVEL OF EFFORT, WHERE REQUIRED OR FOUND NECESSARY, SHALL BE INCLUDED IN THE PRICE OF THIS WORK ITEM.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections as required to ensure compliance with this Work Item, including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment as required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.9 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating

operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor that shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval will be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient size to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank, void, or cofferdam will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C). differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report is to be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or paint spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coatings work.

7.13 Prior to the start of surface preparation perform an inspection of all zincs in the ballast tank. Develop a report that details the location, size, and condition of the zincs to include percent wastage, quantity, replacement weight, and mounting method for each zinc in the tank.

7.13.1 Temporarily remove bolted zinc anodes in way of surface preparation and paint application. Reinstall bolt-on zincs upon completion of all coatings work.

a. For bidding purposes, base the estimate on removing and renewing 0% of bolted-on zincs. Replacement of welded zinc will be subject to issuance of change order.

7.13.2 If zincs are welded in, apply protective coverings over those zincs if wastage is less than 50%.

7.14 Submit report to OMT REP.

7.15 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.16 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the paint manufacturer's representative. In no case will the paint be thinned more than 5% by volume.

7.17 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.18 Surface Preparation

7.18.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5,000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals,

crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

7.18.2 The entire interior of the tank including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast Cleaning as guidance. The blast profile achieved must be angular in nature and within the range set by the manufacturer's product data sheet for the coating system being applied.

7.18.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.18.4 Upon completion of abrasive blasting, all spent grit, dust, and dirt generated by this work item is to be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.18.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit as found condition report documenting results of inspection and recommended action.

7.19 Coatings Application

7.19.1 Prior to application of any coating, the area to be painted will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity)

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet.

7.19.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the paint manufacturer, use Reference 2.9 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.19.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.19.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.19.5 All coatings will be prepared and applied using the manufacturer’s requirements outlined in the Product Data Sheets, use Reference 2.9 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.19.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.19.7 Apply coatings to all prepared surfaces as called for in Paragraph 4.2.1

a. One full coat of Epoxy, (pastel green) at 5-6 mils DFT.

b. One stripe coat of Epoxy, (buff) at 2-3 mils DFT.

c. One full coat of Epoxy, (off white) at 5-6 mils DFT.

7.20 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.20.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.21 During all surface preparation and coating operations, the area involved will only be accessed by persons directly involved in the work underway or those inspecting the same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.22 Upon completion of coating activities, fit, weld, NDT and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.22.1 Conduct tightness test.

a. Testing shall be accomplished prior to preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.23 Upon completion of the air test, remove all blanks and

plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in and touched-up with the topcoat only.

7.24 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.25 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.26 Reports

7.26.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.26.2 Submit a report that includes “as released” condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.26.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company’s representative.

7.26.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Paint Manufacturer, Product Identification Number, color, quantity, and batch numbers for each coat of paint applied.

7.27 Painting: None additional

7.28 Manufacturer's Representative: None additional

7.29 Preparation of Drawings

7.29.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect and recoat the vessel's waste water/sludge tanks.

2.0 REFERENCES

2.1 MSC Drawing No. 835-8499355, Capacity Plan

2.2 MSC Drawing No. 801-8499371, General Arrangement

2.3 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP-10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.5 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.6 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel

2.7 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.8 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: A1-92.5-2

3.2 Quantity:

3.2.1 Waste water/sludge Tank approx. surface area

a. 212 square feet.

3.2.2 Tank Zincs: None

3.3 Description: Food Waste

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Pastel Green	2 gal
Amercoat 240	Epoxy (stripe coat)	Buff	2 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	1 gal
Amercoat T-10	Solvent	N/A	1 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government’s account, to observe the surface preparation and coating application on the government’s behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor’s efforts. They are not for supervision of the contractor’s workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated with monitoring, containment, handling, collection, processing, clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. ALL LABOR AND MATERIAL COSTS ASSOCIATED WITH THIS LEVEL OF EFFORT, WHERE REQUIRED OR FOUND NECESSARY, SHALL BE INCLUDED IN THE PRICE OF THIS WORK ITEM.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections as required to ensure compliance with this Work Item, including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment as required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.9 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating

operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor that shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval will be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient size to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank, void, or cofferdam will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C). differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report is to be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or paint spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coatings work.

7.13 Prior to the start of surface preparation perform an inspection of all zincs in the ballast tank. Develop a report that details the location, size, and condition of the zincs to include percent wastage, quantity, replacement weight, and mounting method for each zinc in the tank.

7.13.1 Temporarily remove bolted zinc anodes in way of surface preparation and paint application. Reinstall bolt-on zincs upon completion of all coatings work.

a. For bidding purposes, base the estimate on removing and renewing 0% of bolted-on zincs. Replacement of welded zinc will be subject to issuance of change order.

7.13.2 If zincs are welded in, apply protective coverings over those zincs if wastage is less than 50%.

7.14 Submit report to OMT REP.

7.15 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.16 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the paint manufacturer's representative. In no case will the paint be thinned more than 5% by volume.

7.17 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.18 Surface Preparation

7.18.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5,000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals,

crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

7.18.2 The entire interior of the tank including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast Cleaning as guidance. The blast profile achieved must be angular in nature and within the range set by the manufacturer’s product data sheet for the coating system being applied.

7.18.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.18.4 Upon completion of abrasive blasting, all spent grit, dust, and dirt generated by this work item is to be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.18.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit as found condition report documenting results of inspection and recommended action.

7.19 Coatings Application

7.19.1 Prior to application of any coating, the area to be painted will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity)

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet.

7.19.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the paint manufacturer, use Reference 2.9 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.19.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.19.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.19.5 All coatings will be prepared and applied using the manufacturer's requirements outlined in the Product Data Sheets, use Reference 2.9 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.19.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.19.7 Apply coatings to all prepared surfaces as called for in Paragraph 4.2.1

a. One full coat of Epoxy, (pastel green) at 5-6 mils DFT.

b. One stripe coat of Epoxy, (buff) at 2-3 mils DFT.

c. One full coat of Epoxy, (off white) at 5-6 mils DFT.

7.20 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.20.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.21 During all surface preparation and coating operations, the area involved will only be accessed by persons directly involved in the work underway or those inspecting the same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.22 Upon completion of coating activities, fit, weld, NDT and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.22.1 Conduct tightness test.

a. Testing shall be accomplished prior to preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.23 Upon completion of the air test, remove all blanks and

plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in and touched-up with the topcoat only.

7.24 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.25 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.26 Reports

7.26.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.26.2 Submit a report that includes “as released” condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.26.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company’s representative.

7.26.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Paint Manufacturer, Product Identification Number, color, quantity, and batch numbers for each coat of paint applied.

7.27 Painting: None additional

7.28 Manufacturer's Representative: None additional

7.29 Preparation of Drawings

7.29.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect and recoat the vessel's waste water/sludge tanks.

2.0 REFERENCES

2.1 MSC Drawing No. 835-8499355, Capacity Plan

2.2 MSC Drawing No. 801-8499371, General Arrangement

2.3 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP-10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.5 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.6 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel

2.7 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.8 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: A1-92-6

3.2 Quantity:

3.2.1 Waste water/sludge Tank approx. surface area

a. 2,295 square feet.

3.2.2 Tank Zincs: None

3.3 Description: Grey Water Holding Tank

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Pastel Green	20 gal
Amercoat 240	Epoxy (stripe coat)	Buff	5 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	20 gal
Amercoat T-10	Solvent	N/A	5 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government's account, to observe the surface preparation and coating application on the government's behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor's efforts. They are not for supervision of the contractor's workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated with monitoring, containment, handling, collection, processing, clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. ALL LABOR AND MATERIAL COSTS ASSOCIATED WITH THIS LEVEL OF EFFORT, WHERE REQUIRED OR FOUND NECESSARY, SHALL BE INCLUDED IN THE PRICE OF THIS WORK ITEM.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections as required to ensure compliance with this Work Item, including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment as required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.9 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating

operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor that shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval will be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient size to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank, void, or cofferdam will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C). differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report is to be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or paint spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coatings work.

7.13 Prior to the start of surface preparation perform an inspection of all zincs in the ballast tank. Develop a report that details the location, size, and condition of the zincs to include percent wastage, quantity, replacement weight, and mounting method for each zinc in the tank.

7.13.1 Temporarily remove bolted zinc anodes in way of surface preparation and paint application. Reinstall bolt-on zincs upon completion of all coatings work.

a. For bidding purposes, base the estimate on removing and renewing 0% of bolted-on zincs. Replacement of welded zinc will be subject to issuance of change order.

7.13.2 If zincs are welded in, apply protective coverings over those zincs if wastage is less than 50%.

7.14 Submit report to OMT REP.

7.15 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.16 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the paint manufacturer's representative. In no case will the paint be thinned more than 5% by volume.

7.17 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.18 Surface Preparation

7.18.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5,000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals,

crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

7.18.2 The entire interior of the tank including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast Cleaning as guidance. The blast profile achieved must be angular in nature and within the range set by the manufacturer’s product data sheet for the coating system being applied.

7.18.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.18.4 Upon completion of abrasive blasting, all spent grit, dust, and dirt generated by this work item is to be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.18.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit as found condition report documenting results of inspection and recommended action.

7.19 Coatings Application

7.19.1 Prior to application of any coating, the area to be painted will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity)

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet.

7.19.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the paint manufacturer, use Reference 2.9 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.19.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.19.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.19.5 All coatings will be prepared and applied using the manufacturer’s requirements outlined in the Product Data Sheets, use Reference 2.9 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.19.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.19.7 Apply coatings to all prepared surfaces as called for in Paragraph 4.2.1

a. One full coat of Epoxy, (pastel green) at 5-6 mils DFT.

b. One stripe coat of Epoxy, (buff) at 2-3 mils DFT.

c. One full coat of Epoxy, (off white) at 5-6 mils DFT.

7.20 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.20.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.21 During all surface preparation and coating operations, the area involved will only be accessed by persons directly involved in the work underway or those inspecting the same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.22 Upon completion of coating activities, fit, weld, NDT and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.22.1 Conduct tightness test.

a. Testing shall be accomplished prior to preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.23 Upon completion of the air test, remove all blanks and

plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in and touched-up with the topcoat only.

7.24 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.25 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.26 Reports

7.26.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.26.2 Submit a report that includes “as released” condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.26.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company’s representative.

7.26.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Paint Manufacturer, Product Identification Number, color, quantity, and batch numbers for each coat of paint applied.

7.27 Painting: None additional

7.28 Manufacturer's Representative: None additional

7.29 Preparation of Drawings

7.29.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect, and coat approximately 2,000 square feet of the vessel's topside items.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 MSC Drawing No. 801-8495724, Booklet of General Plans

2.1.2 Surface Preparation Standard, SSPC-SP WJ-2/NACE WJ-2 Waterjet Cleaning of Metals – Very Thorough Cleaning (Available Commercially)

2.1.3 Surface Preparation Standard, SSPC-SP10/NACE 2, Near White Metal Blast Cleaning (Available Commercially)

2.1.4 Surface Preparation Standard, SSPC-SP11, Power Tool Cleaning to Bare Metal (Available Commercially)

2.1.5 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.1.6 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel (Available Commercially)

2.1.7 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.1.8 PPG-MSC Paint Handbook

2.1.9 PPG Product Data Sheets, Amercoat 240 and Amershield (Available Commercially)

2.2 Enclosure

2.2.1 Spot Blast Locations

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Various

3.1.2 Quantity

a. Topside: 2,000 sq. ft. of surface area

3.2 Description: All vertical surfaces and including overhangs and handrails, decks only calling for coating covering, masts including all attachments and platforms, all normally coated surfaces of davits and foundation, all normally coated surfaces of cranes and foundation, booms, and all normally coated surfaces of underway replenishment kingposts.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	**Type/Notes**	**Color**	**Qty**
Amercoat 240	Epoxy (1st primer coat)	Red Oxide	17 gal
Amercoat 240	Epoxy (stripe coat)	Off-White	5 gal
Amercoat 240	Epoxy (2nd primer coat)	Off-White	17 gal
Amershield	Polyurethane (topcoat)	Haze Gray	17 gal
Amercoat T-10	Solvent	N/A	6 gal
Amercoat 65	Solvent	N/A	2 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Coating Technical Representative will be in attendance, on the government’s account, to provide guidance, and observe the surface preparation and coating application on the government’s behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor’s efforts. They are not for supervision of the contractor’s workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated with monitoring, containment, handling, collection, processing, clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. ALL LABOR AND MATERIAL COSTS ASSOCIATED WITH THIS LEVEL OF EFFORT, WHERE REQUIRED OR FOUND NECESSARY, SHALL BE INCLUDED IN THE PRICE OF THIS WORK ITEM.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 The main mast is defined as all surfaces and appendages from the base up to and including the top of the highest point on the mast. The surfaces include, but are not limited to, the platforms, port and starboard yardarms, antenna foundations, ladders, handrails, and stanchions.

5.5 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections where required to ensure compliance with this Work Item, including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment as required to renew the paint coatings on the topside of the vessel as identified in section 3.0 using References 2.1.1 through 2.1.11 as guidance.

7.2 Drain and dispose of any liquid and materials generated as result of this work in accordance with all local, state, and federal laws.

7.3 Prior to the start of any surface preparation:

7.3.1 Record all markings, label plates and placards documenting their positions, symbols, and text. Submit a copy of the record to the OMT REP.

7.3.2 Note the condition of any damaged cable hanger, wireway supports, speakers, lights, switches, ladders, foundation, and climbing aids.

7.3.3 With assistance from the Ship’s Force, conduct an operational test in the presence of the OMT REP documenting “as found” conditions of all lighting, speakers, and deck drains.

7.3.4 In the first 5 Days of the availability and with assistance from the Ship’s Force, conduct an operational test of equipment with areas defined in 3.0 in the presence of the OMT REP documenting “as found” condition. This includes but is not limited to Cranes/Davits/Booms/UNREP gear/Radars.

7.4 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.5 Plug all overboard discharges in way of surfaces being prepared and coated.

7.6 Provide and erect staging required to access all applicable surfaces as identified in Section 3.0.

7.7 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When coating material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70°F.

7.8 Provide and maintain adequate lighting during all surface preparation, coating, and inspection activities.

7.9 Temporarily remove interferences, label plates and placards in way of surface preparation and coating application. Reinstall upon completion of all coatings work. Label plates that are riveted to the hull or otherwise permanently installed need not be removed.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. Install drop cloths, masking, division shields, seals, blanks, and filtering materials to prevent abrasives and foreign substances from entering lights, machinery, piping, hatches, deck drains, ventilation systems, tank vents, valve stems, motor shafts, seals, and temporary openings during surface preparation and coating operations. The following actions, at a minimum will be taken:

7.10.1 Mask all normally uncoated surfaces

7.10.2 Plug deck drains, deck sprinkler nozzles, and open pipes.

7.10.3 Wrap all valve stems and exposed portions of motor shafts, and hydraulic cylinders.

7.10.4 Install filter media on all tank vents and air intake vents.

7.10.5 Install protective coverings on all installed equipment, port lights, windows, permanently installed label plates, light globes, switches and pushbuttons, equipment, uncoated surfaces, vent screens and appurtenances not intended to be coated.

7.10.6 Protect mooring lines which cannot be stowed or removed for the duration of coating.

7.10.7 Protective covering will be inspected at regular intervals, but not less than the start of each work shift. Degraded covering will be repaired prior to restart of work.

7.11 Surface Preparation

7.11.1 Solvent Clean the vessel topside using Reference 2.1.7; SSPC-SP1 as guidance, using biodegradable detergent to remove all dirt, oil, grease, soluble salts, or other organic matter from the specified surfaces. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible.

7.11.2 Final wash-down will be made with clean, fresh water at 2,500 to 3,000 PSI. Upon completion of all water washing, chloride testing will be performed at a rate of no less than one test per 500 Square Feet of surface.

7.11.3 The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.11.4 If contamination is found, additional water washing will be performed. Additional tests will be made as necessary to determine the extent of contamination and to prove the success of remediation.

7.11.5 Method

a. Ultra-High-Pressure water jet the vessel topside area identified in section 3.0 using SSPC-SP WJ-2/NACE WJ-2 to a WJ-2 finish using Reference 2.1.4 as guidance. Ensure that the surface profile of the vacuum blasted decking is 2 to 3 mils. Report any areas under 2 mils to OMT Representative.

7.11.6 Where spot blasting has been specified in Enclosure 2.2.1, edges of adjacent intact coating will be feathered-in.

7.11.7 No more than a light (L) grade flash rust will be allowed on the steel at the time of coating application. If heavier flash rust is present, the surface will be pressure washed at 2,500 to 3,000 PSI and allowed to air dry to restore the surface to a coat-able condition.

7.11.8 Accomplish the requirements of SSPC-SP-11 Power Tool Clean to Bare Metal to any areas inaccessible to the blast or water jetting including foundations, fan housings, hose racks, pad eyes, life ring clips, scupper extensions, areas behind equipment mounted on stand-offs (small items like speakers, light globes, switches and pushbuttons are to be lifted free during power tooling), and other appendages in way of the aforementioned areas to bare metal using Reference 2.1.6 as guidance.

7.11.9 After surface preparation, all surfaces will be blown down using clean dry air to remove all dust, dirt, and debris. Perform a joint inspection to identify areas that need investigation or repair because of corrosion or damage.

7.11.10 Regardless of whether the surfaces are blasted or UHP Water Jetted, immediately prior to coating, the surface will be re-tested for non-visual surface contaminants at a rate of no less than one test per 1,000 square feet of prepared area. The maximum allowable contamination concentrations for surfaces prepared by means of blasting will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment. For those surfaces prepared by means of UHP water jetting, the limits of NV-2 found in the appendix of SSPC-SP WJ-2/NACE WJ-2 will apply.

7.12 Coatings Application

7.12.1 Prior to application of any coating, the area to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include measuring and recording:

a. Type of surface preparation used

b. Surface cleanliness

c. Upon completion of UHP Water jetting, measure the surface profile using a Keane-Tator (or equal) Surface Profile Comparator. Testex (or equal) Replica Tape is to be used at a rate of one reading per 1,000 SF for verification. Record and submit the measured surface profile readings to the OMT Rep within 24 hours. If the surface profile is below 2 mils, immediately notify the OMT Rep and await guidance prior to proceeding with application of the paint coating. Replica Tape is to be mounted, identified as to location, and included as part of the final paint report.

d. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity)

e. Overcoating interval (curing time) since previous application

f. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet. DFT gages will be calibrated, spot readings will be taken, and DFT tolerances will be using SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages.

7.12.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP WJ2/NACE WJ-2 Surface Preparation of Steel and Other Hard Materials by High and Ultra-High Pressure Water Jetting Prior to Recoating, WJ-2 Condition or SSPC-SP10/NACE 2, Near

White Metal Blast Cleaning as applicable.

c. Air and substrate temperatures will be within the range published by the coating manufacturer, see Reference 2.1.10.

d. During application, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer, will not exceed 85%.

f. Condensation and/or rain is not to contact the uncured Amershield as it may change the color and gloss.

7.12.3 No coating will be applied below 35°F (1.7°C) without prior written approval of the OMT REP.

7.12.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.12.5 All coatings will be prepared and applied using the manufacturer’s requirements outlined in the Product Data Sheets, using Reference 2.1.11 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millage are being applied.

7.12.6 Decks and vertical areas: Apply coating to areas defined in section 3.0 using References 2.1.1 through 2.1.11 as guidance.

a. EPOXY: Only the areas that were selected for preparation to bare metal and adjacent feathered area will receive Epoxy.

1. Apply the following 1st and 2nd Coats of Primer to all blasted and power tool cleaned surface.

2. One coat of Amercoat 240, (red oxide) at 5-6 mils DFT.

3. One stripe coat of Amercoat 240, (off-white) at 2-3 mils DFT.

4. One coat of Amercoat 240, (off-white) 5-6 mils DFT.

5. The stripe coat will be applied to all weldments, crevices, corners, edges, and other areas not conducive to proper coverage. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

b. POLYURETHANE

1. Prior to applying Amershield, all areas to be coated shall be prepared for overcoating by washing with Prep 88 in accordance with manufacturer's instruction, Reference 2.1.10.

2. Minimum concentration shall be two parts water to one part concentrate for general cleaning and to restore "overcoat ability."

3. Strictly adhere to dwell times, wetting and rinsing instructions.

4. Apply One topcoat of Amershield, (haze gray) 3-4 mils DFT to the vessel topside.

5. The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. For weather conditions outside 35°F (1.7°C) through 90°F (32°C) range consult the OMT Coating Representative for proper overcoat intervals.

7.12.7 Mast: Apply coating to areas defined in section 3.0 using References 2.1.1 through 2.1.11 as guidance.

a. EPOXY: Only the areas that were selected for preparation to bare metal and adjacent feathered area will receive Epoxy

1. Apply the following 1st and 2nd Coats of Primer to all blasted and power tool cleaned surface.

2. One coat of Amercoat 240, (red oxide) at 5-6

mils DFT.

3. One stripe coat of Amercoat 240, (off-white) at 2-3 mils DFT.

4. One coat of Amercoat 240, (off-white) 5-6 mils DFT.

5. The stripe coat will be applied to all weldments, crevices, corners, edges, and other areas not conducive to proper coverage. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

b. POLYURETHANE

1. Prior to applying Amershield, all areas to be coated shall be prepared for overcoating by washing with Prep 88 in accordance with manufacturer's instruction, Reference 2.1.10.

2. Minimum concentration shall be two parts water to one part concentrate for general cleaning and to restore "ovecoat ability."

3. Strictly adhere to dwell times, wetting and rinsing instructions.

4. Apply One topcoat of Amershield, (haze gray) 3-4 mils DFT to the vessel topside.

5. The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. For weather conditions outside 35°F (1.7°C) through 90°F (32°C) range consult the OMT Coating Representative for proper overcoat intervals.

7.12.8 Stores Crane: Apply coating to areas defined in section 3.0 using References 2.1.1 through 2.1.11 as guidance.

a. EPOXY: Only the areas that were selected for preparation to bare metal and adjacent feathered area will receive Epoxy

1. Apply the following 1st and 2nd Coats of Primer to all blasted and power tool cleaned surface.

2. One coat of Amercoat 240, (red oxide) at 5-6 mils DFT.

3. One stripe coat of Amercoat 240, (off-white) at 2-3 mils DFT.

4. One coat of Amercoat 240, (off-white) 5-6 mils DFT.

5. The stripe coat will be applied to all weldments, crevices, corners, edges, and other areas not conducive to proper coverage. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

b. POLYURETHANE

1. Prior to applying Amershield, all areas to be coated shall be prepared for overcoating by washing with Prep 88 in accordance with manufacturer’s instruction, Reference 2.1.10.

2. Minimum concentration shall be two parts water to one part concentrate for general cleaning and to restore “overcoat ability.”

3. Strictly adhere to dwell times, wetting and rinsing instructions.

4. Apply One topcoat of Amershield, (haze gray) 3-4 mils DFT to the vessel topside.

5. The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. For weather conditions outside 35°F (1.7°C) through 90°F (32°C) range consult the OMT Coating Representative for proper overcoat intervals.

7.12.9 Kingpost Apply coating to areas defined in section 3.0 using References 2.1.1 through 2.1.11 as guidance.

a. EPOXY: Only the areas that were selected for preparation to bare metal and adjacent feathered area will receive Epoxy

1. Apply the following 1st and 2nd Coats of Primer to all blasted and power tool cleaned surface.

2. One coat of Amercoat 240, (red oxide) at 5-6 mils DFT.

3. One stripe coat of Amercoat 240, (off-white) at 2-3 mils DFT.

4. One coat of Amercoat 240, (off-white) 5-6 mils DFT.

5. The stripe coat will be applied to all weldments, crevices, corners, edges, and other areas not conducive to proper coverage. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

b. POLYURETHANE

1. Prior to applying Amershield, all areas to be coated shall be prepared for overcoating by washing with Prep 88 in accordance with manufacturer’s instruction, Reference 2.1.10.

2. Minimum concentration shall be two parts water to one part concentrate for general cleaning and to restore “overcoat ability.”

3. Strictly adhere to dwell times, wetting and rinsing instructions.

4. Apply One topcoat of Amershield, (haze gray) 3-4 mils DFT to the vessel topside.

5. The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. For weather conditions outside 35°F (1.7°C) through 90°F (32°C) range consult the OMT Coating Representative for proper overcoat

intervals.

7.12.10 During all surface preparation and coating operations, the area involved will be roped off or otherwise barricaded to prevent entry by persons other than those directly involved in the work underway or those inspecting the same. Cordoning or barricading will remain in place until the topcoat is properly cured.

7.13 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations. Remove all protective coverings, debris and replace all interferences removed in the performance of this item.

7.14 Upon completion of work, ship's force will conduct final operational testing of ALL lights, speakers, floodlights, and spotlights, sprinkler heads, and drains in the vicinity of the vessel topside work completed. Contractor and ship's force will compare "post work" performance test results to the "as-found" conditions previously recorded in paragraphs 7.2.1 through 7.2.3. Correct any discrepancies noted between the "as found" and "post work" conditions. All repaired items will be satisfactorily re-demonstrated to ship's force and OMT REP.

7.15 Reports

7.15.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.15.2 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company's representative.

7.15.3 Prepare a Coating Report and submit same to the OMT REP within three days of completing the coating application. The report will include the following data:

a. The location, date, and time of each coating application.

b. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

c. Interval between coatings.

d. Dry film thickness readings at a rate of five spot readings per 1,000 sq. ft. of surface for each coat of paint. DFT gages will be calibrated, and spot readings will be taken using SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages as guidance.

e. Coating Manufacturer, Product Identification Number, quantity, color, and Batch Numbers for each coat of paint applied.

f. Surface profile measurements of the metal substrate.

g. Results of testing for non-visible surface contaminants (soluble salts).

7.16 Painting: None additional

7.17 Manufacturer's Representative: None additional

7.18 Preparation of Drawings

7.18.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - Spot Blast Locations

Location	Description	Square Footage
To Be determined	To Be determined	2,000

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect, and coat approximately 2,000 square feet of machinery space.

2.0 REFERENCES

2.1 MSC Drawing No. 801-8495724, Booklet of General Plans

2.2 Surface Preparation Standard, SSPC-SP1, Solvent Cleaning (Available Commercially)

2.3 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.4 Surface Preparation Standard, SSPC-SP11, Power Tool Cleaning to Bare Metal (Available Commercially)

2.5 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel (Available Commercially)

2.6 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.7 PPG Product Data Sheets, Amercoat 240 and 5450 (Available Commercially)

2.8 PPG Protective and Marine Coatings – MSC Paint Handbook

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Machinery Room #1, Upper Level

b. Machinery Room #2, Lower Level

c. Auxiliary Machinery Room

d. Fwd Pump Room

e. Bow Thruster Room

f. Anchor/Windlass Machinery Space

3.1.2 Quantity

a. Approx. 2000 sq feet in total to be surface prepared and coated

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1^{st} primer coat)	Red Oxide	17 gal
Amercoat 240	Epoxy (stripe coat)	Off-White	5 gal
Amercoat 240	Epoxy (2^{nd} primer coat – bulkheads and overheads)	White	17 gal
Amercoat 5450	Epoxy (Topcoat-bulkheads and overheads if insulation not used)	Soft White	17 gal
Amercoat T-10	Solvent	N/A	6 gal
Amercoat 15	Solvent	N/A	2 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Coating Technical Representative will be in attendance, on the government’s account, to provide guidance, and observe the surface preparation and coating application on the government’s behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor’s efforts. They are not for supervision of the contractor’s workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated with monitoring, containment, handling, collection, processing, clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. ALL LABOR AND MATERIAL COSTS ASSOCIATED WITH THIS LEVEL OF EFFORT, WHERE REQUIRED OR FOUND NECESSARY, SHALL BE INCLUDED IN THE PRICE OF THIS WORK ITEM.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections where required to ensure compliance with this Work Item, including, but not limited to, Blotter Test, Vial Test, Ph Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment as required to prepare and renew the paint coatings and insulation in the spaces identified in section 3.0 using References 2.1 through 2.8 as guidance.

7.2 Coordinate with ship’s force to tagout all valves, drains, level indicating, and alarm systems associated with the areas being worked. Install temporary blanks on drains to prevent entry of liquids during accomplishment of this item. Return to normal operating conditions on completion of work.

7.3 Cleaning and gas free certification requirements for the subject areas are covered under other work items 020 and 021.

7.4 In conjunction with initial water washing, pump out liquids, and remove all mud, dirt, and debris using contractor furnished pumps and hoses, as may be necessary to clean the machinery spaces.

7.4.1 Water removal needed to support this item shall be priced into the item. It is not considered Bilge water removal which is covered by and will be priced in work item 011.

7.5 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for surface preparation, inspection, and coating application requirements. Ventilation will be of sufficient size to dry all washed areas and maintain a clear atmosphere during surface prep and coating operations.

7.6 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination and overspray. The following actions, at a minimum will be taken:

7.6.1 Install protective coverings and take appropriate measures to ensure that the vessel's interior and its equipment are protected to prevent contamination from water, grit, dust, debris, or coating spray; to include all open vents, ducting, drains, piping, and suction lines, lighting, valve stems, reach rod universal joints, sight glasses, gauges, sensors, pump rotating elements, seals, rubber expansion joints, switches, motor casing vents, electrical equipment, and anything else that could be damaged by water, detergent, oil, debris, contamination and coating overspray.

7.6.2 In the areas to be scaled and coated, cover all labels plates, cable tags, and identification markings. Record the location, size, and color of any markings.

7.6.3 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.6.4 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.7 The contractor, ABS and OMT REP will inspect the areas to include foundations, structure, piping, and pipe hangers to determine if steel work will be required. Prepare and submit an as found condition report to the OMT REP identifying any discrepancies and recommended repair.

7.8 Surface Preparation

7.8.1 Exercise care during the cleaning and surface preparation processes to avoid damage to wireways, level sensors, and lighting.

7.8.2 Clean the overheads of each space using hot water and detergent.

7.8.3 Perform a hand wipe down of all bulkheads using hot water and detergents. Collect all cleaning materials and dispose of them in accordance with environmental rules and regulations.

7.8.4 Perform a walkdown of the spaces to be worked on with the OMT REP to identify all areas that shall be scaled to bare metal.

7.8.5 Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, Reference 2.2, degreasing by performing a low-pressure water wash with degreaser and warmed freshwater to 120°F of the entire machinery space removing all oil, grease, slime, dirt, mud, soluble salts, and other foreign matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Particular attention will be given to piping, valves, the undersides of stiffeners, ledges, snipes, limber holes in longitudinals, crevices, and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Heavily soiled areas will be scrubbed, and hand wiped with rags to remove excess oil and grease prior to pressure washing. Accomplish a final rinse with hot, clean, freshwater upon completion of the degreasing and solvent cleaning process.

7.8.6 Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.8.7 2,000 square feet of the machinery space, including structural members, valves, manifolds, piping, pipe brackets, foundations, and manhole covers will be power tool cleaned removing all loose mill scale, loose rust, loose coating, and other loose detrimental foreign matter using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, Reference 2.3 as guidance.

7.8.8 For bidding purposes, assume 10% of the machinery space area including structural members, valves, manifolds, piping, pipe brackets, foundations, and manhole covers will be required to be power tool cleaned to bare metal removing mill scale, rust, coating, oxides, corrosion products, and other foreign matter using SSPC-SP11, Power Tool Cleaning to Bare Metal, Reference 2.4 as guidance. These areas will be identified by the OMT REP and OMT Coating Representative.

7.8.9 Upon completion of surface preparation, all surfaces will be vacuumed clean and blown down with dry, oil free air to remove all dust, dirt, and debris.

7.9 Coatings Application

7.9.1 Prior to application of any coating, the areas to be coated will be inspected and approved by the OMT REP and the OMT Coating Representative. Conditions will follow Product Data sheets using Reference 2.7 as guidance. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Type of surface preparation used.

b. Surfaces are clean, dry, and free of all contaminants including salt deposits.

c. Verify surface appearance meets the definition of SSPC-SP3, Power Tool Cleaning or SSPC-SP-11, Power Tool Cleaning to Bare Metal as appropriate.

d. Test for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

e. Check surface profile.

f. Measure and record environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity). Confirm they are within the coating manufacturers' allowable range using References 2.7 and 2.8 as guidance.

1. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

2. Relative Humidity will not exceed 85%.

g. Record overcoating intervals (curing time) since previous application.

h. Measure and record Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet. DFT gages will be calibrated, spot readings will be taken, and DFT tolerances will be using SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages

7.9.2 No coating will be applied at temperatures below 45°F without prior written approval of the OMT REP.

7.9.3 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.9.4 All coatings will be prepared and applied in accordance with the manufacturer's requirements outlined in the Product Data Sheets using Reference 2.7 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.9.5 Stripe coating will be performed by brush, roller or other mechanical means that will ensure that the coating is applied to the areas where needed. Stripe all limber holes, snipes, corners, weldments, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application. In no case will spray application of a stripe coat be allowed.

7.9.6 EPOXY

a. Apply the following Coats to prepared surfaces:

b. For all surfaces - One full coat of Amercoat 240, (Pastel Green) at 5-6 mils DFT.

c. For all surfaces - One stripe coat of Amercoat 240, (Buff) at 2-3 mils DFT.

d. For Bulkheads and Overheads – One full coat of Amercoat 240, (Red Oxide) at 5-6 mils DFT.

e. For Foundations – One full coat of Amercoat 240, (Red) at 5-6 mils DFT.

f. For Bulkheads and Overheads – One full coat of Amercoat 5450, (White) at 2-3 mils DFT.

g. The stripe coat will be applied to all weldments, crevices, corners, edges, and other areas not conducive to proper coverage. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.10 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the area coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.10.1 Upon completion, the areas are to be ventilated to promote curing. Drying times are greatly dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Cure time prior to immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.11 During all surface preparation and coating operations, the area involved will only be accessed by people directly involved in the work underway or those inspecting the same. All people entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.12 A final inspection of the spaces will be conducted by the OMT REP and Chief Engineer. Removal of all blanks, filters, and protective covers from all open vents, ducting, drains, piping, and suctions, lighting, valve stems, reach rod universal joints, sight glasses, gauges, sensors, pump rotating elements, seals, rubber expansion joints, switches, motor casing vents, and electrical equipment. With the assistance of the Chief Engineer, verify proper operation of lighting, level indicating and alarm systems.

7.13 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.14 Reports

7.14.1 Prepare a Coating Report and submit same to the OMT REP within three days of completing the coating application. The report will include the following data:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings at a rate of five spot readings per 1,000 sq. ft. of surface for each coat of paint. DFT gages will be calibrated, and spot readings will be taken using SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages as guidance.

g. Coating Manufacturer, Product Identification Number, quantity, color, and Batch Numbers for each coat of paint applied.

7.15 Painting/Lagging

7.15.1 Prepare, prime, coat, and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.15.2 Renew lagging where damaged or disturbed.

7.16 Manufacturer’s Representative: None additional

7.17 Preparation of Drawings

7.17.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to surface prepare, inspect, and recoat the vessel's cargo petroleum tanks.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 835-8499355, Capacity Plan

2.2 Surface Preparation Standard, SSPC-SP10/NACE No. 2, Near White Blast Cleaning (Available Commercially)

2.3 Surface Preparation Standard, SSPC-SP3, Power Tool Cleaning (Available Commercially)

2.4 ASTM D4417, Method C, Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel (Available Commercially)

2.5 Surface Preparation Standard, SSPC-PA2, Measurement of Dry Coating Thickness with Magnetic Gages (Available Commercially)

2.6 PPG Product Data Sheets, Amercoat 240 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: 5-97-3

3.2 Quantity:

3.2.1 Cargo Petroleum Tank approx. surface area

a. 1,821 square feet.

3.3 Description: Mission JP-5 OVRFL-RCLM

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

4.2.1 Coatings List

Product	Type/Notes	Color	Qty
Amercoat 240	Epoxy (1st primer coat)	Buff	15 gal
Amercoat 240	Epoxy (stripe coat)	Red Oxide	5 gal
Amercoat 240	Epoxy (2nd primer coat)	Off White	15 gal
Amercoat T-10	Solvent	N/A	4 gal

4.3 Government Furnished Services (GFS)

4.3.1 OMT Coating Representative

a. A Government Coating Technical Representative will be in attendance, on the government's account, to observe the surface preparation and coating application on the government's behalf and to advise the OMT REP as to the quality, and appropriateness of the contractor's efforts. They are not for supervision of the contractor's workforce.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier, must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor, and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor is responsible for all costs associated with monitoring, containment, handling, collection, processing, clean up, recycling, disposing of wash water, rinse water, dissolved solvents, used and generated during work accomplished by this work item, whether on the ship, on the pier adjacent to the ship, or in the dry dock basin. ALL LABOR AND MATERIAL COSTS ASSOCIATED WITH THIS LEVEL OF EFFORT, WHERE REQUIRED OR FOUND NECESSARY, SHALL BE INCLUDED IN THE PRICE OF THIS WORK ITEM.

5.3 Solvents supplied as GFM are for viscosity control only. Solvents required for equipment clean-up are the responsibility of the contractor.

5.4 Contractors must conform to all Local, State, and Federal environmental requirements and that the extent and cost of those additional requirements must be included in the work item bid.

5.5 This ship contains high voltage (HV) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 MSC reserves the right to require at no additional cost that the contractor perform additional tests or inspections where required to ensure compliance with this Work Item, including, but not limited to, Blotter Test, Vial Test, pH Test, Non-Visual Surface Contaminant Tests, Adhesion Tests and Sieve Tests.

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools, and equipment as required to renew the paint coatings in the Tanks identified in Section 3.0 using References 2.1 through 2.6 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Open manholes for tank(s) identified in Section 3.0. Cleaning and gas free certification requirements for the subject tanks are covered under other work items 020 and 021.

7.5 For ventilation and access during blasting and coating operations, the contractor may elect to make a maximum of two temporary access cuts. Prior to making any cut(s), submit a sketch to the OMT REP and the local ABS Surveyor which shows the location and size of each proposed cut and the relationship of the proposed cut(s) to adjacent structural members and weld seams. OMT REP and ABS approval will be obtained prior to making the cut(s).

7.6 Provide and erect all necessary staging required to access all applicable surfaces.

7.7 Provide and maintain adequate lighting and ventilation. Lighting will be sufficient for preparation, inspection, and application requirements. Ventilation will be of sufficient volume and strength to maintain a clear atmosphere during blasting and coating operations.

7.8 During abrasive blasting activities, the use of dehumidification (DH) equipment is discretionary on the part of the contractor. However, it is expected that the entire tank, will be presented for inspection after abrasive blasting at one time. Piece-meal presentation will not be accepted. During coating and curing activities, DH equipment is mandatory. When used, the DH equipment must:

7.8.1 Effect a complete air change in the space at least three times per hour.

7.8.2 Maintain the relative humidity in the space to at least 50%.

7.8.3 Maintain a minimum of 5° F (3°C). differential between the substrate temperature of the space and the dew point, with the dew point being the lower temperature.

7.9 Ducting will be run so as not to create hazards to personnel transiting the areas through which the ducting is run. Ducting will be maintained airtight and in good repair such that it does not contribute to contamination of the vessel's interior or equipment, with abrasive blast grit, dust, or coating.

7.10 Prior to the start of surface preparation and coating, take precautions to protect the vessel from contamination. The following actions, at a minimum will be taken:

7.10.1 All lines servicing the tank will be broken at the first joint off the tank and blanked. Lines which are of all welded construction, or which do not have a mechanical joint within one foot of the tank will be plugged on the inside of the tank. Prior to the start of surface preparation, submit report to the OMT REP documenting the tank, and the quantity and location of all blanks, caps, and plugs used for the isolation of the tank; this includes but is not limited to suction and fill lines, vents, sounding tubes, etc. The report must be used as a checklist when the tank is presented and inspected prior to closure.

7.10.2 Valve stems and reach rod universal joints within the tank will be greased and wrapped to protect same from blasting and coating.

7.10.3 As necessary, based upon the location of ventilation and dehumidification exhausts, grease and wrap valve stems and exposed portions of hydraulic cylinders.

7.10.4 Install filters on air intake vents.

7.10.5 Install covers on fuel tank vents.

7.10.6 Appropriate measures will be taken to ensure that the vessel's interior and its equipment is not subject to contamination from blasting grit, dust, or paint spray.

7.10.7 In the area to be blasted, record the location, size, and color of all ship's markings.

7.10.8 Protective covering will be inspected at regular intervals, but not less than at the start of each work shift. Degraded protective covering will be repaired prior to the restart of work.

7.11 Contamination of the vessel and its equipment will be reported to the OMT REP verbally, immediately upon its discovery, followed by a written report within four hours of the verbal notification. Clean the contaminated equipment and show that the contamination has not caused damage to the equipment or vessel. The cost to repair equipment damaged by such contamination will be borne by the Contractor.

7.12 Temporarily remove all tank level indicators. Reinstall upon completion of all coatings work.

7.13 After surface preparation, all surfaces will be blown down using clean dry air and/or vacuumed to remove all dust, dirt, and debris.

7.14 The thinning of the paint will be allowed for viscosity control, only if determined to be necessary by the coating manufacturer's representative. In no case will the paint be thinned more than 5% by volume.

7.15 Coating material will be stored within the Coating Manufacturer's recommended temperature range. When paint material is being applied, ensure that the material's temperature is within the Manufacturer's recommended range, but in any case, not less than 70° F.

7.16 Surface Preparation

7.16.1 Perform a high-pressure water wash of the entire tank interior using fresh water at a minimum of 5000 psi to remove slime, dirt, mud, soluble salts, and other foreign matter. Particular attention will be given to the undersides of stiffeners, ledges, snipes, limber holes in longitudinals, crevices and areas of rust, rust scale, blistered, cracked, peeling, or flaking coatings. Accomplish the requirements of Surface Preparation Standard SSPC-SP1, Solvent Cleaning, to remove all dirt, oil, grease, soluble salts, or other organic matter. Biodegradable cleaners or other eco-friendly methods/agents will be used to the greatest extent possible. Upon completion of all water washing, chloride testing will be performed at random locations. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants as determined by field or laboratory analysis using reliable, reproducible test equipment.

a. In conjunction with initial water washing, pump out liquids, and remove all mud, dirt and debris using contractor furnished pumps and hoses, as may be necessary to clean the tanks.

7.16.2 The entire interior of the tank including all structural members, internal piping, reach rods and reach rod support brackets, interior of the manhole covers, and interior of removed access cuts, will be abrasive blasted using Surface Preparation Standard SSPC-SP10/NACE 2, Near White Metal Blast

Cleaning, as guidance. The blast profile achieved must be angular in nature and within the range set by the manufacturer's product data sheet for the coating system being applied.

7.16.3 Prior to the start of abrasive blasting, the blast media will be tested to ensure that it is not contaminated with chlorides, as determined by field or laboratory analysis using reliable, reproducible test equipment.

7.16.4 Upon completion of abrasive blasting, all spent grit, dust, and dirt generated by this work item will be removed. All internal surfaces will be blown-down with dry, oil free air at a maximum pressure of ten psi and/or vacuumed clean.

7.16.5 After completion of blasting and initial cleaning, a joint inspection of the tank will be conducted by OMT Representative, ABS Surveyor, and contractor to determine if there are structural defects which require repair. Submit Condition Found Report documenting results of inspection and recommended action.

7.17 Coatings Application

7.17.1 Prior to application of any coating, the area to be painted will be inspected and approved by the OMT REP and the OMT Coating Representative. This includes not only the initial coat of paint, but all subsequent coats as well. Inspection will include observation, measuring and recording:

a. Surface cleanliness to include tests for non-visual surface contaminants at random locations to ensure that surfaces have not been re-contaminated. The maximum allowable contamination concentrations will be less than 10 ug/cm2 of chloride contaminants.

b. Surface profile as determined by Keane-Tator Comparator (or equal) examination and/or replica tape, using a minimum profile of 2 mils per Reference 2.4.

c. Environmental conditions (surface temperature, ambient air temperature, dew point temperature and relative humidity).

d. The overcoating interval (curing time) since previous application.

e. Dry Film Thickness (DFT) of previous coating taken at a rate of five per 1,000 square feet per Reference 2.5.

7.17.2 Ensure the following conditions are met prior to coating:

a. Surfaces will be clean, dry, and free of oil, grease, or residue.

b. Surface appearance meets the definition of SSPC-SP10, Near White Metal Blast Cleaning.

c. Air and substrate temperatures will be within the range published by the coating manufacturer, use Reference 2.6 as guidance.

d. During application and curing, the substrate temperature will be at least 5°F (3°C) above Dew Point.

e. The Relative Humidity is within the range set by the manufacturer but will not exceed 85%.

7.17.3 No coating will be applied at temperatures below 35°F (1.7°C) without prior written approval of the OMT REP.

7.17.4 No coating will be applied between the hours of sunset and 0800 without prior written approval of the OMT REP.

7.17.5 All coatings will be prepared and applied using the manufacturer’s requirements outlined in the Product Data Sheets, use Reference 2.6 as guidance. Random Wet Film Thickness (WFT) readings are to be taken during the application of coatings to verify correct millages are being applied.

7.17.6 Stripe coat all weldments, crevices, corners, edges, limber holes, snipes, pitted areas, threaded surfaces, or other areas that are not conducive to proper coverage by spray application.

a. Stripe coating will be performed by brush, roller, paint mitt, or other mechanical means that will ensure that the coating is applied to the areas where needed. In no case will spray application of a stripe coat be allowed.

b. Stripe coats will extend a minimum of 2" from each edge of the area being stripe coated.

7.17.7 Apply coatings to all prepared surfaces as called for in Paragraph 4.2.1:

a. One full coat of Epoxy, (buff) at 5-6 mils DFT.

b. One stripe coat of Epoxy, (red oxide) at 2-3 mils DFT.

c. One full coat of Epoxy, (off white) at 5-6 mils DFT.

7.18 The environmental conditions, DFT of each coat, cure and overcoating intervals are critical to the application of the tank coatings. The application of all coatings will adhere to the Product Data Sheets and OMT Coating Representative. Consult the OMT Coating Representative for proper overcoat intervals.

7.18.1 Upon completion, the tanks are to be left open and ventilated to promote curing before being immersed in water and cargo. Drying times are dependent on air and surface temperatures, film thickness, ventilation, and relative humidity. The production schedule must consider this cure time. Forced ventilation and heat are permitted under the guidance of the OMT Coating Representative. For planning purposes:

Substrate Temp	Dry to touch	Dry to handle	Service - water immersion
23°F (-5°C)	36 hours	60 hours	21 days
32°F (0°C)	24 hours	36 hours	14 days
50°F (10°C)	10 hours	16 hours	10 days
68°F (20°C)	5 hours	10 hours	6 days
86°F (30°C)	3 hours	8 hours	3 days

7.19 During all surface preparation and coating operations, the area involved will only be accessed by persons directly involved in the work underway or those inspecting same. All persons entering the work area will wear disposable booties over their shoes to prevent contamination of the surface. New booties will be made available at the tank entry and be put on each time a person enters the area.

7.20 Upon completion of coating activities, fit, weld, NDT, and tightness test the removed access plates to the satisfaction of the ABS Surveyor and OMT REP.

7.20.1 Conduct tightness test.

a. Testing shall be accomplished prior to preservation of any weld seams.

b. Unless otherwise approved by the OMT REP, the tightness test will be accomplished by closing tank, and applying a 1 1/2 psi air test to tank and checking welds with soap solution to prove the installed plates are tight in the presence of ABS and OMT REP.

c. Upon completion of the air test, re-open the tank and allow it to ventilate.

7.21 Upon completion of the air test, remove all blanks and plugs and make-up all lines as original. Re-install all interferences. All replacements will be proven operational. Clean welds and disturbed areas using Surface Preparation Standard SSPC-SP3, Power Tool Cleaning, and coat the access plate(s) weld seams and all other areas disturbed by re-installations with two coats of epoxy. The edges of disturbed areas will be feathered-in. Touch-up coats will be applied in the same order as the original coatings. Where only the topcoat of the system has been damaged, affected areas will be sanded to abrade the surface, feathered-in, and touched-up with the topcoat only.

7.22 Prior to closure, a final inspection of the tank interior will be conducted by the OMT REP. Using the checklist, removal of all blanks and plugs from piping, suction and fill lines, vents, and sounding tubes is to be verified as well as proper operation of tank level indicating and alarm systems. Upon acceptance by the OMT REP, re-install the manhole cover(s) with new gaskets.

7.23 Upon completion, the vessel will be cleaned of all residues resulting from the surface preparation and coating operations.

7.24 Reports

7.24.1 When inspection, service, and testing reveal any deficiency, a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.24.2 Submit a report that includes “as released” condition of the tanks effected by this work item. Submit one electronic copy in Portable Document Format (PDF).

7.24.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the company’s representative.

7.24.4 Prepare a Coating Report and submit to the OMT REP within three days of completing the coating application. Include following information:

a. The location, date, and time of each coating application.

b. Surface profile measurements of the metal substrate.

c. Results of testing for non-visible surface contaminants (soluble salts).

d. The air and substrate temperatures, relative humidity, and dewpoint temperature at the time of each coating application.

e. Interval between coatings.

f. Dry film thickness readings.

g. Coating Manufacturer, Product Identification Number, color, quantity, and batch numbers for each coat applied.

7.25 Painting: None additional

7.26 Manufacturer's Representative: None additional

7.27 Preparation of Drawings

7.28 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
ELECTRICAL		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0303	**CATEGORY "A"**	**2026-02-01**
ESB-CSI-HIGH VOLTAGE SHIP SERVICE AND EMERGENCY SWITCHBOARD INSPECTION AND CLEANING (2.5 YR)		**Port Engineer's Name**

1.0 ABSTRACT

The purpose of this item is to provide qualified service technicians to perform cleaning and maintenance on the High and Medium Voltage Switchboards, Ship's Service Switchboards, Emergency and Bow Thruster Switchboards, Load Centers, Control Centers, Distribution Panels and Transformers to include breaker maintenance.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 Safety Management System 2.1-016 AKE/ESB, High Voltage System

2.1.2 NAVSEA Drawing No. 320-8468257, Electrical System One Line Diagram

2.1.3 Tech Manual S6265-A9-MMC-010, Delta Synchronous Generators, 7650 KVA Including Generator Excitation Control Panel

2.1.4 Tech Manual S9202-BX-MMC-010, Propulsion Control System Volume I (used for Harmonic Filters, Excitation and Control Cubicles, SD7000 F3 Synchro-Converter, Converter Cooling, Local and Remote Control Panels, Input/Output Racks, Maintenance Tables, Factory Acceptance Test Report)

2.1.5 Tech Manual S9314-HL-MMC-010, High Voltage Transformers, (6.6KV Primary)(used for High Voltage SWBDS (THT 8FN 4-5M/N), Propulsion Motor (10BS 6M4/G), and Bow Thruster (TGT 8FN 2M7/N)

2.1.6 Tech Manual TE670-AJ-MMC-010, High Voltage

Switchboards, 6.6KV 2000A Nos. 1 and 2 Main and 6.6KV 1000A No. 1A and No. 2A Sub

2.1.7 Tech Manual SE670-AS-MMC-010, 480VAC Ship Service Distribution and Emergency Switchboards, Shore and Mission Power Stations, and Forward Load Center

2.1.8 Cutler Hammer, I.B. 32-255-1G, Instructions for Installation, Operation and Maintenance of Type VCP-W Vacuum Circuit Breakers

2.1.9 Tech Manual S9314-HK-MMC-010, Bow Thruster Motor Variable Speed Drive Controller Type: ALSPA GD DELTA

2.1.10 Tech Manual S6263-CM-MMC-010, Motor Control Center, Individual Controllers and Remote Control Stations

2.2 Enclosures

2.2.1 ANSI-National Electrical Testing Association 2011, Table 100.12.1 thru 4

2.2.2 Switchboard, Converter, Transformer, Load Center, and Control Center List

3.0 ITEM LOCATION/DESCRIPTION

3.1 Locations

a. HV Switchboard 1, 3-55-1, High Voltage Room 2S

b. HV Switchboard 2, 3-55-2, High Voltage Room 2P

c. HV Switchboard 1A, 2-55-1, High Voltage Room 1S

d. HV Switchboard 2A, 2-55-2, High Voltage Room 1P

e. MV Switchboard 1, 2-56-1, ECR 1

f. MV Switchboard 2, 2-56-2, ECR 2

g. MV Switchboard 1A, 1-34-1, Casing (S)

h. MV Switchboard 2A, 1-34-1, Casing (S)

i. Emergency Switchboard, A-34-1, EDG Room

j. Converter, Transformer including HV Transformer, Load Center, and Control Center locations are shown in Reference 2.1.1

3.1.2 Quantity

a. Switchboards/Converters: 13

b. Transformers: 45

c. Load Centers: 3

d. Control Centers: 18

e. High Voltage Transformers: 14

f. Distribution Panels: 8

3.2 Item Description/Manufacturer’s Data

3.2.1 HV Switchboard #1 and #2: 6.6KV, 2000A, 3Ø, 3W 60Hz, 31.5 KA RMS Systems, Point Eight Power, Dwg 1117 and 1118, consisting of two busses with eight sections per bus and a bus transition section.

3.2.2 HV Switchboard #1A and #2A: 6.6KV, 2000A, 3Ø, 3W 60Hz, 31.5 KA RMS Systems, Point Eight Power, Dwg 1117 and

1118, consisting of two busses with two sections per bus.

3.2.3 MV Switchboard #1 and #2: 480V, 5000A, 3Ø, 3W 60Hz, 100 KA RMS Systems, Point Eight Power, Dwg 1061 and 1062, consisting of two busses with seven sections per bus and a bus transition section.

3.2.4 MV Switchboard #1A and #2A: 480V, 6300A, 3Ø, 3W 60Hz, 100 KA RMS Systems, Point Eight Power, Dwg 1065 and 1065, consisting of two busses with five sections per bus.

3.2.5 Emergency Switchboard: 480V, 1000A, 3Ø, 3W 60Hz, 100 KA RMS Systems, Point Eight Power, Dwg 1063 consisting of one bus with seven sections.

3.2.6 HV Switchboard Breakers

a. 1200 Amp Eaton 72VCPW-ND32 breakers, Qty: 16

b. 1200 Amp Eaton 72VCP-W32 breakers, Qty: Four

c. 2000 Amp Eaton 72VCP-W32 breakers, Qty: Two

3.2.7 HV Switchboard Contactor

a. Cutler Hammer Model ZN1N0AK 360 A Rollout NEMA E2 Vacuum Contactor Assembly with test plug and receptacle, CLF 1.0E primary and 7A secondary CPT 5.1KV fuses, Qty: One

3.2.8 MV Switchboard Breakers

a. Cutler Hammer Model Magnum DS IEC Breaker 6300 AF, 100 KA, 3 Pole ABCABC, Draw out, 6300A Sensor/Plug, 520 LSI Trip Unit with 110/127 VAC Undervoltage Release, 6A/6B Auxiliary Contacts, 2 Form C OTS/Bell Alarm Contacts, Qty: Two.

b. Cutler Hammer Model Magnum DS breaker 5000AF, 100KA, 3 Pole, Draw out, 5000A Sensor/Plug 520 LSI trip unit, 110/127 VAC Shunt trip, 110/125VAC 5 sec motor operator, 110/127 VAC spring release, Qty: Two.

c. Cutler Hammer Model Magnum DS breaker 4000AF, 100KA, 3 Pole, Draw out, 4000A Sensor/Plug 520 LSI trip unit, 110/127 VAC Shunt trip, 110/125VAC 5 sec motor operator, 110/127 VAC spring release, 110/127 VAC UVR, 6A/6B Aux. Contacts, 2 Form C OTS Manual Reset, Charge After Close, Qty: Two.

d. Cutler Hammer Model Magnum DS breaker 2500AF, 100/85KA, 3 Pole, Draw out, 2500A Sensor/Plug 520 LSI Trip Unit, 120 VAC Shunt Trip, 110-125VAC Motor Operator, 110/127 VAC/DC Spring Release, 110 VAC UV, 6A/6B Aux. Contacts, 2A/2B OTS Switch, 110-127VAC UV, Qty: Two.

e. Cutler Hammer Model Magnum DS breaker 2000AF, 100KA, 3 Pole, Draw Out, 2000A Sensor/Plug 520 LSI trip unit, 110/127 VAC Shunt trip, 110/125VAC 5 sec motor operator, 110/127 VAC spring release, 6A/6B Aux. Contacts, 2 Form C OTS Manual Reset, Charge After Close, Quantity: Two.

3.2.9 Emergency Switchboard Breakers

a. Cutler Hammer Model Magnum DS breaker 2000AF, 65KA, 3 Pole, Draw Out, 2000A Sensor/Plug, 520 LSI Trip Unit, 120 VAC Shunt Trip, 110/125VAC 5 Sec. Motor Operator, 110/127 VAC spring release, 110/127VAC UVR INST, 6A/6B Aux. Contacts, 2 Form C OTS Manual Reset, Quantity: Three.

3.3 Bill of Materials (Contractor Furnished): None

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of

this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 THIS SHIP CONTAINS HIGH VOLTAGE (HV) ELECTRICAL SYSTEMS. OBEY ALL POSTED AND VERBAL INSTRUCTIONS REGARDING SAFETY AND EXCLUSION FROM HIGH VOLTAGE AREAS. AT NO TIME APPROACH, WORK ON, OR ENTER A HIGH VOLTAGE AREA WITHOUT PROPER AUTHORIZATION.

5.3 THE ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

5.4 Before starting work, all equipment shall be locked out and tagged out using the ship’s HV lock out procedures, and the ship's LV lock out procedures. Switchboards shall be proven de-energized prior to beginning any work. Keeping half the switchboard energized is not authorized.

5.5 Under torquing or over torquing may produce overheating and/or cause damage to the conductor.

5.6 Do not apply insulation test voltage to electronic devices or equipment with solid state components.

5.7 Compressed air should be avoided because it may force dirt and dust into coil insulation.

5.8 Do not work on breaker/contactor element with primary power applied.

5.9 Do not work on a breaker/contactor element with secondary contacts connected.

5.10 Do not defeat any safety interlocks.

5.11 Do not leave maintenance tool in the socket after charging the closing springs.

5.12 Always remove the breakers/contactor from the enclosure before performing any maintenance. Failure to do so could result in electrical shock leading to death, severe personal injury or property damage.

5.13 The maintenance for the four propulsion motor transformers is covered in Work Items 0301A/0301B.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 Final close-out inspection of each panel to be performed by the OMT REP or the specified designee.

7.0 STATEMENT OF WORK REQUIRED

7.1 With the assistance of ship’s force, secure all power to switchboards and equipment identified in section 3.0 prior to work being performed. Institute safety procedures that will ensure that switchboards and equipment will not be energized while servicing and or repairs are in progress. The emergency switchboard shall be cleaned and restored to service prior to commencing work on the main switchboards. Secure power only after coordinating the exact time with the Chief Engineer and the OMT REP.

7.2 Coordinate all work contained in this item with the Chief Engineer and Chief Electrician. Switchboard work shall be done after normal hours between 1800 and 0400 to reduce interference and impact with the day shift work. Both HV Switchboard busses are to be de-energized, isolated, and proven dead for this work. Keeping half the switchboard energized is not authorized.

7.3 Provide industrial support and services of qualified Marine Electrical Specialists, labor personnel, and tools to inspect, service, and test equipment listed in section 3.0 using References 2.1.1 through 2.1.10 and Enclosures 2.1.1 and 2.1.2 as guidance. The Reps shall be experienced with the T-AKE, ESB, or T-ESD Classes and their HV switchboards and load centers.

7.4 Marine Electrical Specialists are required to hold a valid certification of approved high voltage qualification as specified in Reference 2.1.1.

7.5 Erect temporary lighting as required in areas where electrical power is secured for electrical equipment cleaning.

7.6 Present Calibrated torque wrenches to be used in execution of this work item to the OMT REP for inspection.

7.7 Paint marker color used to mark all torqued bolts must be approved by the OMT REP.

7.8 Authorized Technical Representatives are to perform the following work:

7.8.1 Clean, test, and inspect the following Main Switchboards, Ship’s Service Switchboards, Emergency Switchboards, Load Centers, Motor Control Centers, and HV Vacuum Circuit Breakers and perform gauge calibration:

a. MV Switchboard No. 1

b. MV Switchboard No. 2

c. MV Switchboard No. 1A

d. MV Switchboard No. 2A

e. HV Switchboard No. 1

f. HV Switchboard No. 2

g. HV Switchboard No. 1A

h. HV Switchboard No. 2A

i. Emergency Switchboard

j. H402-EP014 Power Substation LC01

k. Motor Control Center MCC05

l. Motor Control Center MCC06

m. Motor Control Center MCC07

n. Motor Control Center MCC08

o. Motor Control Center MCC09

p. Motor Control Center MCC10

q. Motor Control Center MCC11

r. Motor Control Center MCC12

s. Motor Control Center MCC13

t. Motor Control Center MCC14

u. Motor Control Center MCC15

v. Motor Control Center MCC16

w. Motor Control Center MCC20

x. MV Switchboard No. 1 120V Distribution Panel

y. MV Switchboard No. 2 120V Distribution Panel

z. Emergency Switchboard 120V Distribution Panel

aa. P400A Distribution Panel

bb. P400B Distribution Panel

cc. P100A Distribution Panel

dd. P100B Distribution Panel

ee. EP100 Distribution Panel

7.8.2 Tighten all bus bars, lug connections, terminals, relays, trips and contact points.

7.8.3 Identify corroded fasteners and hardware requiring replacement.

7.8.4 Submit an “as found” condition report with test measurements.

7.8.5 Clean/Inspect four Propulsion Frequency Converters

NOTE: Refer to Reference 2.1.4 – Tech Manual S9202-BX-MMC-010 for detailed procedures.

a. Visually inspect converter components for general/overall clean condition of the interior and exterior cabinets.

b. Inspect frame paint for signs of corrosion.

c. Inspect bars for possible signs overheating.

d. Inspect cabinet for signs of water leakage.

e. Inspect semi-conductor-heat sink sets.

f. Inspect capacitors for signs of overheating and swelling.

g. Inspect resistors for signs of overheating.

h. Inspect terminal blocks for overheating or corrosion.

i. Inspect wiring for damage, overheating, or corrosion.

j. Inspect associated electronic boards for overheating or corrosion.

k. Check the water pressure. Pressure should be set at 1.5 bar (21.8 PSI)

l. Bleed de-ionized water system via designated vent valve. Close tap as soon as water appears.

m. Check the condition of air filters. Clean air filters if it can be done or replace if necessary. (Have enough air filters on hand to replace if needed. Any air filters not used will be turned over to Ship's force.)

7.8.6 HV Breaker Maintenance

a. All HV breaker maintenance shall be performed by an Authorized OEM Rep.

b. Refer to References 2.1.6 and 2.1.8 for detailed maintenance procedures.

c. Separately tag out, lock out, rack out, and temporarily remove each HV Circuit Breaker approved for inspection, maintenance and testing.

d. Visually inspect the drive insulator, barriers and stand-off insulators for damage. Clean all of the accessible insulating surfaces with a dry lint free cloth or same moistened with Denatured Alcohol (or similar mild solvent) when needed for hardened dirt.

e. Manually charge the closing springs and close the circuit breaker. Trip then repeat several times. Leave breaker in closed position.

f. Using a Micro-Ohmmeter, check the primary resistance of each pole and record the values for reference. The resistance should not exceed 60 microohms.

g. Check the insulation integrity of the Main Circuit to Ground, between Main circuit Terminals and Control circuit to ground with a hip potential (HI pot) tester as follows:

NOTE: The test voltage depends upon the maximum rated voltage of the breaker. For breaker elements rated at 8.25 KV, the test voltage is 27 KVAC.

h. After removing the HV circuit breaker from the switchboard using References 2.1.6 and 2.1.8 as guidance and accomplishing preliminary cleaning and inspections perform the following tests.

1. Close the breaker. Connect the High Potential lead of the test machine to one of the poles of the breaker. Connect the remaining poles and breaker frame to ground.

2. Start the machine with output potential at zero and increase to the test voltage. Maintain the test voltage for one minute. Repeat for the remaining poles. Successful withstand indicates satisfactory insulation strength of the primary circuit.

3. Visually check the vacuum interrupters by closing the circuit breakers and observing if all the green marks on the moving stems are visible. If a mark is not visible, perform a contact wipe check.

4. The adequacy of the contact wipe can be determined by observing the vacuum interrupter side of the operating rod assembly on a closed circuit breaker. Refer to References 2.1.7 and 2.1.8 for the procedure in determining the contact wipe of blue or red contact springs. If the wipe is not adequate, the vacuum interrupter assembly (pole unit) must be replaced. A field adjustment is not possible.

NOTE: Failure to replace a pole unit assembly when contact erosion mark is not visible or wipe is unsatisfactory will cause the breaker to fail to interrupt and thereby cause property damage or personal injury.

NOTE: Vacuum Interrupters used in Type VCP-W Vacuum Circuit Breaker Elements are highly reliable interrupting elements. Satisfactory performance of these devices are dependent upon the integrity of the vacuum in the interrupter and the internal dielectric strength. Both of these parameters can be easily checked by a one minute AC High Potential test.

WARNING: Applying abnormally high voltage across a pair of contacts in vacuum may produce x-radiation. The radiation increases with the increase in voltage and or decreases in contact spacing. X radiation produced during this test with recommended voltage and normal contact spacing is extremely low and well below maximum permitted by standards. However, as a precautionary measure against possibility of application of higher than recommended voltage and or below normal contact spacing, it is recommended that all operating personnel stand at least four meters away in front of the breaker element.

WARNING: After the high potential is removed, an electrical charge may be retained by the vacuum interrupters. Failure to discharge the residual electrostatic charge could result in an electrical shock. All six primary terminals and the center ring of each vacuum interrupter of the circuit breaker should be grounded to reduce this electrical charge before coming in contact with the primary circuit.

i. Perform vacuum interrupter integrity test as follows:

1. With the breaker element open, connect all top primary studs (bars) together and to the high potential machine lead. Connect all bottom studs together and ground them along with the breaker frame. Start the machine at zero potential, increase to appropriate test voltage and maintain for one minute.

2. Test voltage is 27 KV AC 60hz, per Reference 2.1.8, section 6-4.

3. A successful withstand indicates that all interrupters have a satisfactory vacuum level. If there is a breakdown, the defective interrupter or interrupters should be identified by an individual test and replaced before placing the breaker in service.

NOTE: To avoid any ambiguity in the AC high potential test due to leakage or displacement (capacitive) current, the test unit should have sufficient volt ampere capacity. It is recommended that the equipment be capable of delivering 25 milliamperes for one minute.

4. Inspect the primary disconnects for evidence of burning or damage. Replace if burned, damaged or eroded,

5. Test the closing and tripping device including disconnects of the circuit breaker twice. Replace any defective device. Identify per troubleshooting chart in

Reference 2.1.8.

6. Visually check that wiring is securely tied in proper place. Repair or tie as necessary.

7. Visually check that the terminals are tight. Tighten or replace if necessary.

8. Conduct a functional test of the charging motor. Replace brushes if necessary or motor.

9. Make a careful visual inspection of the operating mechanism for any loose or missing parts. Tighten or reinstate if necessary with appropriate tools. Visually check for dust or foreign matter. Clean as necessary.

10. Check for the smooth operation of the operating mechanism. Check for deformation or excessive wear. Remove cause and replace parts if necessary.

11. Lubricate operating mechanisms very sparingly with light mineral oil; refer to Reference 2.1.8, Figure 6-1, for lubrications points. After lubrication, operate the circuit breaker several times manually and electrically.

12. Check for smooth operation during manual charging, closing and tripping.

13. Conduct a Closure Test in accordance with Reference 2.1.8, Section 6-9.1

14. Notify Ships force that the HV Circuit Breaker is ready for final inspection and testing.

15. After the Chief Engineer or designee has accomplished and approved the CB inspection for cleanliness and completeness, connect the test jig and demonstrate electrical and mechanical cycling, charging, closing and tripping of the

circuit breaker.

16. Replace all remaining covers and insulators and reinstall breakers into the switchboard cassette cradle.

17. Sign the HV Permit to Work to acknowledge and confirm that the work has been completed and return to the Authorizing Engineer. Notify ALL workers that it is no longer safe to work on that equipment.

7.8.7 HV Contactor Maintenance

a. All HV Contactor maintenance shall be performed by and authorized OEM Rep.

b. Separately tag out and lockout each HV contactor approved for inspection, maintenance and testing.

c. Temporarily remove contactor from the switchboard.

d. Visually inspect contactor for dirt and/or grease. Clean with dry lint free cloth or same moistened with denatured alcohol when needed.

e. To verify the integrity of the vacuum interrupters, apply a voltage of 16KV-AC across the open contacts of the interrupters. The voltage should be applied for 60 seconds without breakdown. Breakdown is defined as a current of 5mA or more flowing across the open contacts. Note that approximately 1.3mA of current will flow through each interrupter during the AC test due to the capacitance of the vacuum interrupter. If a DC high potential test unit is used, make certain that the peak voltage does not exceed 23KV, the peak of the corresponding AC RMS test voltage. A megger cannot be used to verify vacuum integrity due to its limited output voltage.

WARNING: When a vacuum bottle is tested with voltage over 5000 volts across its open gap, there is some possibility of generating X-rays. Test should be minimized and personnel should be no closer than 10 feet. This is a precaution until such time as the possible hazard is better understood and standards are published.

NOTE: Periodic dielectric tests across open contacts should not be omitted on the basis of satisfactory contactor performance since under certain operating conditions, the contactor may perform satisfactorily even though one vacuum interrupter has become defective.

f. Contact material vaporizes from the contact faces during every interruption and condenses elsewhere inside the bottle. This is normal and is provided for by over travel or wear allowance. When the contactor is fully closed, there is a gap between the lower bottle nut and a pivot plate. As the contacts wear, this gap decreases. When the gap goes below 0.020 inches on any pole, all the bottle subassemblies should be replaced. Use the 0.020 inch thick fork shaped over travel feeler gauge supplied for this measurement, part no 5259C11. DO NOT RE-ADJUST THE BOTTLE NUTS TO RESET OVER TRAVEL AS THE CONTACTS WEAR. Once the contactor is place in service, over travel should be checked but not adjusted.

g. Inspect the primary disconnects for evidence of burning or damage. Replace if burned, damaged or eroded.

h. Test the closing and tripping device including disconnects of the contactor twice. Replace any defective devices.

i. Visually check that wiring is securely tied in proper place. Repair or tie as necessary.

j. Visually check that the terminals are tight.

Tighten or replace if necessary.

k. Make a careful visual inspection of the operating mechanism for any loose or missing parts. Tighten or reinstate if necessary with appropriate tools. Visually check for dust or foreign matter.

l. Check for the smooth operation of the operating mechanism. Check for deformation or excessive wear. Remove cause and replace parts if necessary.

m. Notify Ships force that the HV Contactor is ready for final inspection.

n. After Chief Engineer or designee inspection and acceptance reinstall the contactor in the cubicle.

o. Reconnect all mechanical linkages, hardware and electrical connectors. Prior to restoring line voltage, use the Earthing Key to check the contactor for proper operational cycling by opening and closing the contactor several times. If no electrical or mechanical issues are observed, leave contactor in the ‘Open’ position and remove ‘Earthing Key’. The contactor is now ready to place back into normal service.

7.8.8 Post Switchboard cleaning, prior to closing all panels conduct a visual inspection of each Switchboard in the presence of the OMT REP, Chief Engineer and Chief Electrician. Check each for cleanliness ensuring they are free of dirt, dust, debris and other foreign matter.

7.8.9 Upon completion of all work, replace equipment covers and close the Switchboards leaving them in a ready for service condition.

7.8.10 Sign the HV Permit to Work to acknowledge and confirm that the work has been completed and return to the Authorizing Engineer. Notify ALL workers that it is no longer safe to work on that equipment.

7.8.11 In coordination with Chief Engineer, follow proper startup procedures to re-energize the system.

7.8.12 Perform an operational test of all functions of the switchboards.

7.8.13 Submit a report to the Port Engineer describing all work and all repairs accomplished.

7.8.14 In conjunction with the Dock Trial:

a. With assistance of ships force perform an operational test of all the switchboard functions. Testing shall include the following:

1. Opening and closing of Generator and Shore Power breakers

2. Manual and automatic startup of generators, paralleling and bus transfer functions.

3. Reverse power, reverse current and under voltage function for each generator.

7.8.15 Reports

a. When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

b. Submit a final as released condition report detailing at a minimum, conditions found, work accomplished, parts replaced, and any new recommendations to Chief Engineer and Port Engineer. Report shall be in electronic Portable Document Format (PDF) format and delivered within 10 days of completion of this task.

c. All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.9 Painting

7.9.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.10 Manufacturer’s Representative

7.10.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the equipment identified in section 3.0 ensuring compliance with the manufacturer’s standards and performance specifications.

7.10.2 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only original equipment manufacturer (OEM) authorized technical field service providers or MSC qualified non-OEM technical field service providers and OEM authorized or MSC qualified non-OEM parts shall be used to accomplish the requirements of this work item for this ship critical safety item including oversight and guidance on all aspects of equipment as-found condition inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable. The technical service provider is to attend dock and sea trials.

7.10.3 Contractors interested in gaining MSC qualification as a non-OEM parts and/or service provider should visit this website for qualification and submittal requirements: https://www.msc.usff.navy.mil/Business-Opportunities/Contracts/Qualification-for-Items-Critical-to-Shipboard-Safety-on-MSC-Vessels/

7.10.4 Possible Sources

a. Certified High Voltage support contractors with T-AKE, ESB, or T-ESD High Voltage System experience:

PowerGen Controls
8600 Telephone Road
Houston, TX 77061
Office Phone: +1(832)730-5220
Email: Info@powergencontrols.com

7.10.5 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.11 Preparation of Drawings

7.11.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

<u>ENCLOSURE 2.2.1 - ANSI-National Electrical Testing Association 2011, Table 100.12.1 thru</u>

TABLE 100.12.1

Bolt-Torque Values for Electrical Connections

US Standard Fasteners [a]
Heat-Treated Steel – Cadmium or Zinc Plated [b]

Grade	SAE 1&2	SAE 5	SAE 7	SAE 8
Head Marking				
Minimum Tensile (Strength) (lbf/in^2)	64K	105K	133K	150K
Bolt Diameter (Inches)	**Torque (Pound-Feet)**			
1/4	4	6	8	8
5/16	7	11	15	18
3/8	12	20	27	30
7/16	19	32	44	48
1/2	30	48	68	74
9/16	42	70	96	105
5/8	59	96	135	145
3/4	96	160	225	235
7/8	150	240	350	380
1.0	225	370	530	570

a. Consult manufacturer for equipment supplied with metric fasteners.
b. Table is based on national coarse thread pitch.

Table 100.12.2

US Standard Fasteners [a]
Silicon Bronze Fasteners [b c]
Torque (Pound-Feet)

Bolt Diameter (Inches)	Nonlubricated	Lubricated
5/16	15	10
3/8	20	15
1/2	40	25
5/8	55	40
3/4	70	60

a. Consult manufacturer for equipment supplied with metric fasteners.
b. Table is based on national coarse thread pitch.
c. This table is based on bronze alloy bolts having a minimum tensile strength of 70,000 pounds per square inch.

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ENCLOSURE 2.2.1 - ANSI-National Electrical Testing Association 2011, Table 100.12.1 thru 4

TABLE 100.12.3

US Standard Fasteners [a]
Aluminum Alloy Fasteners [b c]
Torque (Pound-Feet)

Bolt Diameter (Inches)	Lubricated
5/16	10
3/8	14
1/2	25
5/8	40
3/4	60

a. Consult manufacturer for equipment supplied with metric fasteners.
b. Table is based on national coarse thread pitch.
c. This table is based on aluminum alloy bolts having a minimum tensile strength of 55,000 pounds per square inch.

TABLE 100.12.4

US Standard Fasteners [a]
Stainless Steel Fasteners [b c]
Torque (Pound-Feet)

Bolt Diameter (Inches)	Uncoated
5/16	15
3/8	20
1/2	40
5/8	55
3/4	70

a. Consult manufacturer for equipment supplied with metric fasteners.
b. Table is based on national coarse thread pitch.
c. This table is to be used for the following hardware types:
Bolts, cap screws, nuts, flat washers, locknuts (18-8 alloy)
Belleville washers (302 alloy).

Tables in 100.12 are compiled from Penn-Union Catalogue and Square D Company, Anderson Products Division, *General Catalog*: Class 3910 *Distribution Technical Data, Class 3930 Reference Data Substation Connector Products*.

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ENCLOSURE 2.2.2 – SWITCHBOARD, CONVERTER, TRANSFORMER, LOAD CENTER, AND CONTROL CENTER LIST	
	SWITCHBOARDS/CONVERTERS
1	Ship Service Switchboard No. 1
2	Ship Service Switchboard No. 2
3	High Voltage Switchboard No. 1
4	High Voltage Switchboard No. 2
5	High Voltage Switchboard No. 1A
6	High Voltage Switchboard No. 2A
7	MV Switchboard No. 1
8	MV Switchboard No. 2
9	MV Switchboard No. 1A
10	MV Switchboard No. 2A
11	Emergency Switchboard (EDG)
12	Bow Thruster Converter
13	Shore Power Station No. 1
14	Mission Module Power Station 1
15	Mission Module Power Station 2
	TRANSFORMERS
1	Ship Service High Voltage Transformer DT1
2	Ship Service High Voltage Transformer DT2
3	Ship Service High Voltage Transformer DT3
4	Ship Service High Voltage Transformer DT4
5	Bow Thruster SWBD Transformer
6	L402-EP009 Load Center Transformer
7	L402-EP010 Load Center Transformer
8	L402-EP011 Load Center Transformer
9	L402-EP012 Load Center Transformer
10	L402-EP013 Load Center Transformer
11	L402-EP014 Load Center Transformer
12	L402-EP015 Load Center Transformer
13	L402-EP016 Load Center Transformer
14	L402-EP017 Load Center Transformer
15	L402-EP018 Load Center Transformer
16	L402-EP019 Load Center Transformer

ENCLOSURE 2.2.2 - SWITCHBOARD, CONVERTER, TRANSFORMER, LOAD CENTER, AND CONTROL CENTER LIST	
17	L402-EP020 Load Center Transformer
18	P400A-22 Transformer
19	T-L116 Transformer
20	T-P100B Transformer
21	T-P100A Transformer
22	T-P205 Transformer
23	T-P206 Transformer
24	T-P123 Transformer
25	T-P431 Transformer
26	T-TRH02 Transformer
27	T-TRH01 Transformer
28	T-L117 Transformer
29	T-P118 Transformer
30	TE-4 Transformer
31	T-TRH04 Transformer
32	T-TRH03 Transformer
33	T-P115 Transformer
34	T-P426 Transformer
35	T-L118 Transformer
36	T-P119 Transformer
37	TE-3 Transformer
38	T-L105 Transformer
39	T-L105 Transformer
40	T-L121 Transformer
41	HT SYS HEAT TR Transformer
42	P400A-20 Transformer
43	T-P121 Transformer
44	T-P122 Transformer
45	HEAT TRACE Transformer
	LOAD CENTERS
1	Load Center LC-01 FWD AMR
2	Emergency Load Center 1, EDG Room (A-34-2)
3	Emergency Load Center 2, EDG Room (A-34-2)

ENCLOSURE 2.2.2 – SWITCHBOARD, CONVERTER, TRANSFORMER, LOAD CENTER, AND CONTROL CENTER LIST	
	CONTROL CENTERS
1	Motor Control Center 1A
2	Motor Control Center 1B
3	Motor Control Center 2A
4	Motor Control Center 2B
5	Motor Control Center 05
6	Motor Control Center 06
7	Motor Control Center 07
8	Motor Control Center 08
9	Motor Control Center 09
10	Motor Control Center 10
11	Motor Control Center 11
12	Motor Control Center 12
13	Motor Control Center 13
14	Motor Control Center 14
15	Motor Control Center 15
16	Motor Control Center 16
17	Motor Control Center 20
18	Emergency MCC Unit 1 through 6
19	Motor Control Center “A01”
20	Motor Control Center “A02”
21	Motor Control Center “MV01”
22	Motor Control Center “MV02”
23	Motor Control Center “P02”
24	Motor Control Center “P02”
	DISTRIBUTION PANELS
1	MV Switchboard No. 1 120V Distribution Panel
2	MV Switchboard No. 2 120V Distribution Panel
3	Emergency Switchboard 120V Distribution Panel
4	480V Distribution Panel
5	480V Distribution Panel
6	120V Distribution Panel
7	120V Distribution Panel

ENCLOSURE 2.2.2 - SWITCHBOARD, CONVERTER, TRANSFORMER, LOAD CENTER, AND CONTROL CENTER LIST	
8	Emergency Distribution Panel

USS Hershel Williams		
(ESB 4)		
ELECTRICAL	CONTRACT NO. N68171-26-RN-003	
ITEM NO. 0306	CATEGORY "A"	2026-01-30
TESB_CSI_EMERGENCY GENERATOR - INSPECT AND SERVICE (2.5 YR) (SCSI)	Manovich, David	

1.0 ABSTRACT

The purpose of this item is to perform the 2.5-year Service on the Generator side of the Emergency Diesel Generator.

2.0 REFERENCES

2.1 Tech Manual T9233-CM-MMC-010 and T9233-CM-MMC-020 Emergency Diesel Engine and Generator (Model 1000 HS50V16RC)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Emergency Diesel Generator Room (A-34-2)

3.1.2 Quantity: One

3.2 Item Description/Manufacturer’s Data

Manufacturer: G&M Tex LTD/Wartsila
Model Number: 1000HS50V16RC

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor and all subcontractors regardless of tier are advised to review other work items under this contract.

Many of the definitions relating to performance of this work item are found in Work Item 001.

5.3 THIS SHIP CONTAINS HIGH VOLTAGE (HV) ELECTRICAL SYSTEMS. OBEY ALL POSTED AND VERBAL INSTRUCTIONS REGARDING SAFETY AND EXCLUSION FROM HIGH VOLTAGE AREAS. AT NO TIME APPROACH, WORK ON, OR ENTER A HIGH VOLTAGE AREA WITHOUT PROPER AUTHORIZATION.

5.4 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Perform 2.5 year service and inspection on the generator identified section 3.0 using Reference 2.1 as guidance.

7.2 With the assistance of ship's force, tag-out all required systems and equipment. Drain and dispose of any liquid in accordance with all federal, state, and local regulations.

7.3 Temporarily remove the generator outlet box cover.

7.4 Check all exposed electrical connections for tightness.

7.5 Check transformers, fuses, capacitors and lightning arrestors for loose mountings or physical damage.

7.6 Check insulation resistance to ground on all generator windings (Minimum acceptance criteria of 1 Mohm):

7.6.1 Main rotating assembly.

7.6.2 Main stator assembly.

7.6.3 Exciter and PMA stationary field.

7.6.4 Exciter armature assembly.

7.7 Check space heaters for proper operation.

7.8 Check exciter armature for proper rotating rectifier connection tightness.

7.9 Visually inspect generator windings for oil or dirt contamination. Excessive contamination may necessitate surface cleaning with compressed air and electrical solvent.

7.10 Report findings and condition to OMT REP. Note any deficiencies and correct as required.

7.11 With the assistance of ship’s force, clear tags and prepare system for operational test.

7.12 Perform an operational test of the generator in the presence of the OMT REP.

7.13 Reports

7.13.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.13.2 Submit a report that includes “as released” condition of the Emergency Generator. Submit one electronic copy in Portable Document Format (PDF).

7.13.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.14 Painting/Lagging

7.14.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.14.2 Renew lagging where damaged or disturbed.

7.15 Manufacturer’s Representative

7.15.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the Emergency Generator identified in 0 ensuring compliance with the manufacturer’s standards and performance specifications.

7.15.2 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only original equipment manufacturer (OEM) authorized technical field service providers or MSC qualified non-OEM technical field service providers and OEM authorized or MSC qualified non-OEM parts shall be used to accomplish the requirements of this work item for this ship critical safety item including oversight and guidance on all aspects of equipment as-found condition inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable. The technical service provider is to attend dock and sea trials.

7.15.3 Contractors interested in gaining MSC qualification as a non-OEM parts and/or service provider should visit this website for qualification and submittal requirements: https://procurement.msc.navy.mil/.

7.15.4 Possible Sources

Wärtsilä Defense, Inc. Headquarters
3617 Koppens Way
Chesapeake, VA 23323

Phone: 757-558-3625
Fax: 757-558-3627

7.15.5 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.16 Preparation of Drawings: None additional

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
ELECTRICAL		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0353A	**CATEGORY "A"**	**2026-01-30**
ESB-CCSI-THERMOGRAPHIC SURVEY (1 YR)		**Manovich, David**

1.0 ABSTRACT

The purpose of this work item is to conduct a thermographic survey of the ship's service switchboards, emergency switchboards, load centers, motor control centers, and miscellaneous electrical equipment.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSEA Drawing No. 320-8468257, Electrical System One Line Diagram

2.1.2 Tech Manual TE670-AJ-MMC-010, High Voltage Switchboards, 6.6KV 2000A Nos. 1 and 2 Main and 6.6KV 1000A No. 1A and No. 2A Sub

2.1.3 Tech Manual SE670-AS-MMC-010, 480 VAC Ship Service Distribution and Emergency Switchboards, Shore and Mission Power Stations, and Forward Load Center

2.1.4 Tech Manual S9202-BX-MMC-010, Propulsion Control System Volume I

2.1.5 ASTM E 1934-99a Examining Electrical and Mechanical Equipment with Infrared Thermography (Available Commercially)

2.2 Enclosures

2.2.1 Equipment List

2.2.2 Quantitative Infrared Inspection Report (Sample)

2.2.3 Thermographic Survey, Suggested Action Based on Temperature Rise

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations: Locations of switchboards, load centers, motor control centers and additional miscellaneous equipment including transformers, motor starters, fan control unit controllers, power and lighting distribution panels, and circuit breakers are identified in Enclosure 2.2.1.

3.1.2 Quantity: Various

3.2 Item Description/Manufacturer’s Data

3.2.1 Electrical Equipment identified in Enclosure 2.2.1.

3.2.2 Manufacturer: Various

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM): None

4.3 Government Furnished Services (GFS)

4.3.1 A ship’s force representative will escort and assist the thermographic survey technician to identify the location of equipment during the inspection.

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Provide all approved Personnel Protection Equipment (PPE) required for shipboard work and High Voltage Space access (refer to Reference 2.1.5).

5.3 Do not apply insulation test voltage to electronic devices or equipment with solid state components.

5.4 Do not defeat any safety interlocks.

5.5 Always remove the breakers/contactor from the enclosure before performing any maintenance. Failure to do so could result in electrical shock leading to death, sever personal injury or property damage.

5.6 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide the services of a qualified contractor to perform a quantitative infrared inspection of electrical equipment identified in section 3.0 and Enclosure 2.2.1. Use References 2.1.1 through 2.1.5 and Enclosures 2.2.2 and 2.2.3 as guidance.

7.1.1 Thermographic survey representatives will have to ride the vessel prior to arrival of the ship to the shipyard to inspect propulsion related items listed in enclosure 2.2.1. Propulsions equipment must be under a 40% load to obtain proper readings.

7.1.2 For bidding purposes, estimate the thermographic survey representative will be onboard an additional 16 hours at the end of the availability to recheck areas where significant discrepancies were discovered and repaired and to inspect other areas as directed by OMT REP.

7.2 The infrared thermographer is to perform calibration tests, before each quantitative infrared examination, to ensure that all temperature-measuring equipment is within the manufacturers’ standard specifications for accuracy.

7.3 In coordination with the Chief Engineer and OMT REP, perform a quantitative infrared inspection (thermographic survey) of all equipment identified in section 3.0 and Enclosure 2.2.1 as soon as possible after ship arrival to view as many live circuits as possible. Inspections should take place during periods of maximum possible loading but not less than 40 percent of rated load of the equipment being inspected.

7.4 Accomplish the survey, identifying any hotspots found and their severity. Ship’s force electrician will accompany the survey team to assist in locating and safely exposing live circuits. Any condition categorized as Immediate or Mandatory as defined in Enclosure 2.2.2 is to be brought to the attention of the OMT REP immediately.

7.5 All work is only to be accomplished by experienced and qualified personnel. They are also to be trained in electrical safety to avoid personnel injury. Infrared inspection requires the inspector to take pictures of equipment while energized and operating under normal conditions. This may expose the inspector to risks from rotating machines, pressurized pipes and energized conductors. These risks are compounded by potential loss of depth perception that may occur while operating a camera. As such the worker must be fully aware of the risks associated with his surroundings and be fully knowledgeable of health and safety standards. When necessary, an assistant is to be assigned to ensure the safety of the camera operator.

7.6 Coordinate all work contained in this item with the OMT REP to reduce interference and impact with system operation and shift work.

7.7 Any repairs deemed necessary by the OMT REP will be addressed via a change order. The infrared thermographer is to re-examine each exception after repair to assure that its operating temperature is normal, and the potential problem corrected.

7.8 Upon completion of all work, replace equipment covers and close the Switchboards, Motor Control Centers, UPS and other miscellaneous electrical equipment leaving them in a ready for service condition.

7.9 Prior to departure, the Infrared Thermographic Survey contractor is to provide a verbal report to the OMT REP disclosing any significant and/or emergent recommendations classified as Immediate or Mandatory.

7.10 Reports

7.10.1 Provide a quantitative infrared thermographic survey report in accordance with Section 7 of Reference 2.1.5 using Enclosure 2.2.2 as guidance. Submit the report to the OMT REP within three days after completion of the survey. The report is to at a minimum identify the:

a. Name, affiliation, address, and telephone number of the infrared thermographer

b. Thermographer certification level and number

c. The manufacturer, model and serial number of the infrared imaging system used

d. Description of equipment to be tested

e. Discrepancies and their severity

f. Temperature difference between the area of concern and the reference area

g. Probable cause of temperature difference

h. Identify load conditions at time of inspection

i. Provide both digital color photographs and thermograms of the deficient area

j. Recommended actions to repair each hot spot found

7.10.2 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.10.3 Submit a report that includes “as released” condition of the Switchboards, Motor Control Centers, UPS and other miscellaneous electrical equipment. Submit one electronic copy in Portable Document Format (PDF).

7.10.4 Submit all thermographic images and readings identified by equipment item to the OMT Rep.

7.10.5 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.11 Painting/Lagging

7.11.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.11.2 Renew lagging where damaged or disturbed.

7.12 Manufacturer’s Representative

7.12.1 The thermographer is to be at least Level 2 certified using either NETA (International Electrical Testing Association) or ANSI (American National Standards Institute) training standards. Submit proof of qualifications, certificates and experience to the OMT REP for review and approval prior to the start of any inspections.

7.12.2 Thermographer Training and Qualification Levels

a. Level 1 - Level 1 training and qualification is ideal for those who are new to thermography. It will enable you to gather high-quality data and sort the data based on written pass/fail test criteria.

b. Level 2 - Level 2 is for thermographers who are experienced in thermography and troubleshooting. The level 2 training along with your prior experience qualifies you to set up and calibrate equipment, interpret data, create reports, and supervise Level 1-qualified personnel.

c. Level 3 - Level 3 is the most advanced level thermographer. This training qualifies you to develop inspection procedures and severity criteria, interpret relevant codes, and manage a thermography program including overseeing and providing training and testing, and calculating the return on investment for the program.

7.13 Preparation of Drawings

7.13.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER	LOCATION	REPAIR PRIORITY (OBSERVED)
150 HP FOAM PUMP	BALLAST CONTROL ROOM A DECK (A-56-0)	
50 HP FWD FOAM PROPORTIONING UNIT POWER PACK	AUXILIARY MACHINERY ROOM (6-108-0)	
50 HP AFT FOAM PROPORTIONING UNIT	BALLAST CONTROL ROOM A DECK (A-56-0)	
10 TON KNUCKLE BOOM	WEATHER DECK (B DECK FR 57)	
AC UNIT 1S (ECR1) SELF CONTAINED	AC ROOM (2-56-3)	
AC UNIT 1P (ECR2) SELF CONTAINED	AC ROOM (2-56-3)	
ANCHOR WINDLASS/MOORING WINCH W1 POWER PACK	BOSUNS STOREROOM (1-126-0)	
ANCHOR WINDLASS/MOORING WINCH W2 POWER PACK	BOSUNS STOREROOM (1-126-0)	
ANCHOR WINDLASS/MOORING WINCH M3 POWER PACK	BOSUNS STOREROOM (1-126-0)	
ANCHOR WINDLASS/MOORING WINCH M4 POWER PACK	BOSUNS STOREROOM (1-126-0)	
AUXILIARY FUEL TRANSFER PUMP	MACHINERY ROOM 2 (5-56-2)	
BALLAST PUMP/MOTOR NO. 1	PUMP ROOM (5-57-0)	
BALLAST PUMP/MOTOR NO. 2	PUMP ROOM (5-57-0)	
BALLAST PUMP/MOTOR NO. 3	PUMP ROOM (5-57-0)	
BALLAST PUMP ROOM EXHAUST FAN M7	PUMP ROOM (5-57-0)	
BALLAST PUMP ROOM EXHAUST FAN M8	PUMP ROOM (5-57-0)	
NO. 1 BALLAST WATER TREATMENT SYSTEM	BALLAST CONTROL ROOM A DECK (A-56-0)	
NO. 1 BALLAST PUMP CONTROLLER	BALLAST CONTROL ROOM A DECK (A-56-0)	
NO. 2 BALLAST WATER TREATMENT SYSTEM	BALLAST CONTROL ROOM A DECK (A-56-0)	
NO. 2 BALLAST PUMP CONTROLLER	BALLAST CONTROL ROOM A DECK (A-56-0)	
NO. 3 BALLAST WATER TREATMENT SYSTEM	BALLAST CONTROL ROOM A DECK (A-56-0)	
NO. 3 BALLAST PUMP CONTROLLER	BALLAST CONTROL ROOM A DECK (A-56-0)	
BOAT HANDLING CRANE	MISSION DECK FR 70-79P (MD-79-2)	
BOW THRUSTER CONVERTER	AUXILIARY MACHINERY ROOM (6-108-0)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER	LOCATION	REPAIR PRIORITY (OBSERVED)
BOW THRUSTER HYDRAULIC PUMP	AUXILIARY MACHINERY ROOM (6-108-0)	
BOW THRUSTER LIFTING PUMP	AUXILIARY MACHINERY ROOM (6-108-0)	
CATHODIC PROTECTION POWER SUPPLY		
CENTRAL LT FW COOLING PUMP 1P	MACHINERY ROOM 2 (5-56-2)	
CENTRAL LT FW COOLING PUMP 1S	MACHINERY ROOM 1 (5-56-1)	
CENTRAL LT FW COOLING PUMP 2P	MACHINERY ROOM 2 (5-56-2)	
CENTRAL LT FW COOLING PUMP 2S	MACHINERY ROOM 1 (5-56-1)	
CENTRAL SW COOLING PUMP 1P	MACHINERY ROOM 2 (5-56-2)	
CENTRAL SW COOLING PUMP 1S	MACHINERY ROOM 1 (5-56-1)	
CENTRAL SW COOLING PUMP 2P	MACHINERY ROOM 2 (5-56-2)	
COMMUNICATION MODULE 460V PP	COMMUNICATION MODULE (E-48-0)	
DISTILLING PLANT (PORT)	MACHINERY ROOM 2 (5-56-2)	
DISTILLING PLANT (STBD)	MACHINERY ROOM 1 (5-56-1)	
DISTILLER NO. 1 SW ANTI-FOULING SYS	MACHINERY ROOM 1 (5-56-1)	
DISTILLER NO. 2 SW ANTI-FOULING SYS	MACHINERY ROOM 2 (5-56-2)	
EMERGENCY DIESEL GENERATOR (EDG)	EDG ROOM (A-34-2)	
EDG DAMPER	EDG ROOM (A-34-2)	
EDG HYDRAULIC 1C START RECHARGE PUMP	EDG ROOM (A-34-2)	
EDG SUPPLY FAN M29	EDG ROOM (A-34-2)	
EDG PRE-LUBE PUMP BATTERY CHARGER	EDG ROOM (A-34-2)	
EMCC	EDG ROOM (A-34-2)	
ENGINEERING WORKSHOP EXHAUST FAN M10	ENGINEERING WORKSHOP (3-55-1)	
FIRE & GENERAL SERVICE PUMP PORT	MACHINERY ROOM 2 (5-56-2)	
FIRE & GENERAL SERVICE PUMP STBD	MACHINERY ROOM 1 (5-56-1)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER	LOCATION	REPAIR PRIORITY (OBSERVED)
FIRE PUMP NO. 3	PUMP ROOM (5-57-0)	
FORKLIFT BATTERY CHARGER NO. 1	MHE STOWAGE/MAINTENANCE/REPAIR AREA (A5-89-4)	
FORKLIFT BATTERY CHARGER NO. 2	MHE STOWAGE/MAINTENANCE/REPAIR AREA(A5-89-4)	
FOAM PROPORTIONING UNIT (FWD)		
FUEL OIL SUPPLY PUMP 1S		
FUEL OIL SUPPLY PUMP 2S		
FUEL OIL SUPPLY PUMP 1P		
FUEL OIL SUPPLY PUMP 2P		
FUEL SUPPLY ROOM/FUEL VALVE TEST ROOM EXHAUST FAN M17	FUEL SUPPLY ROOM (3-56-4)	
GENERATOR 1P DE LO PUMP	MACHINERY ROOM 2 (5-56-2)	
GENERATOR 1P NDE LO PUMP	MACHINERY ROOM 2 (5-56-2)	
GENERATOR 2P DE LO PUMP	MACHINERY ROOM 2 (5-56-2)	
GENERATOR 2P NDE LO PUMP	MACHINERY ROOM 2 (5-56-2)	
GENERATOR 1S DE LO PUMP	MACHINERY ROOM 1 (5-56-1)	
GENERATOR 1S NDE LO PUMP	MACHINERY ROOM 1 (5-56-1)	
GENERATOR 2S DE LO PUMP	MACHINERY ROOM 1 (5-56-1)	
GENERATOR 2S NDE LO PUMP	MACHINERY ROOM 1 (5-56-1)	
HAZARDOUS MATERIAL CONTAINER RECEPTACLE	UPPER DECK FR 16-56 PORT (1-WD-12)	
HARMONIC FILTER 1 FAN 1	HV ROOM 1S (3-55-3)	
HARMONIC FILTER 1 FAN 2	HV ROOM 1S (3-55-3)	
HARMONIC FILTER 2 FAN 1	HV ROOM 1P (2-55-2)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER	LOCATION	REPAIR PRIORITY (OBSERVED)
HARMONIC FILTER 2 FAN 2	HV ROOM 1P (2-55-2)	
HOT WATER HEATER NO. 1	SUPPORT SERVICE ROOM (A1-96-0)	
HOT WATER HEATER NO. 2	SUPPORT SERVICE ROOM (A1-96-0)	
HT FW COOLING PUMP 1P	MACHINERY ROOM 2 (5-56-2)	
HT FW COOLING PUMP 2P	MACHINERY ROOM 2 (5-56-2)	
HT FW COOLING PUMP 1S	MACHINERY ROOM 1 (5-56-1)	
HT FW COOLING PUMP 2S	MACHINERY ROOM 1 (5-56-1)	
INCINERATOR	INCINERATOR ROOM (A-29-1)	
INCINERATOR SLUDGE TANK HEATER (STBD)	INCINERATOR ROOM (A-29-1)	
LO TRANSFER PUMP PORT	MACHINERY ROOM 2 (5-56-2)	
LO TRANSFER PUMP STBD	MACHINERY ROOM 1 (5-56-1)	
MACHINERY ROOM 1 SUPPLY FAN M1	MACHINERY ROOM 1 (5-56-1)	
MACHINERY ROOM 2 SUPPLY FAN M2	MACHINERY ROOM 2 (5-56-2)	
MACHINERY ROOM 1 SUPPLY FAN M3	MACHINERY ROOM 1 (5-56-1)	
MACHINERY ROOM 2 SUPPLY FAN M4	MACHINERY ROOM 2 (5-56-2)	
MACHINERY ROOM 1 SUPPLY FAN M5	MACHINERY ROOM 1 (5-56-1)	
MACHINERY ROOM 2 SUPPLY FAN M6	MACHINERY ROOM 2 (5-56-2)	
MACHINERY VENT MISSION DECK	SUPPORT SERVICE ROOM (A1-96-0)	
MAIN GENERATOR 1P	MACHINERY ROOM 2 (5-56-2)	
MAIN GENERATOR 2P	MACHINERY ROOM 2 (5-56-2)	
MAIN GENERATOR 1S	MACHINERY ROOM 1 (5-56-1)	
MAIN GENERATOR 2S	MACHINERY ROOM 1 (5-56-1)	
MAIN FUEL TRANSFER PUMP	MACHINERY ROOM 2 (5-56-2)	
MDE F.O. SUPPLY PUMP 1S	FUEL SUPPLY ROOM 1 (3-56-3)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST		
EQUIPMENT/CIRCUIT NUMBER	**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
MDE F.O. SUPPLY PUMP 2S	FUEL SUPPLY ROOM 1 (3-56-3)	
MDE F.O. SUPPLY PUMP 1P	FUEL SUPPLY ROOM 2 (3-56-4)	
MDE F.O. SUPPLY PUMP 2P	FUEL SUPPLY ROOM 2 (3-56-4)	
MDG PRE-LUBE PUMP 1S	MACHINERY ROOM 1 (5-56-1)	
MDG PRE-LUBE PUMP 2S	MACHINERY ROOM 1 (5-56-1)	
MDG-PRE-LUBE PUMP 1P	MACHINERY ROOM 2 (5-56-2)	
MDG PRE-LUBE PUMP 2P	MACHINERY ROOM 2 (5-56-2)	
MISSION MODULE SEAWATER PUMP	MACHINERY ROOM 1 (5-56-1)	
MISSION MODULE SERVICE POWER STATION NO. 1	OPEN DECK FR 59-61 STBD A DECK (A-WD-7A)	
MISSION MODULE SW PUMP NO. 2	MACHINERY ROOM 1 (5-56-1)	
MOORING WINCH UPPER DECK AFT (PORT)	UPPER DECK	
MOORING WINCH UPPER DECK AFT (STBD)	UPPER DECK	
MOORING WINCH CENTERLINE (FWD)	UPPER DECK	
MOORING WINCH M6	SHT 21 BOTTOM RIGHT CORNER	
MOORING WINCH FWD M2 (PORT)	FWD MOORING DECK FOC'SLE	
MOORING WINCH FWD M1 (STBD)	FWD MOORING DECK FOC'SLE	
OILY WATER SEPARATOR		
PROPULSION MOTOR (PORT)	MACHINERY ROOM 2 (5-56-2)	
PROPULSION MOTOR (STBD)	MACHINERY ROOM 1 (5-56-1)	
PROPULSION MOTOR 1 FAN 1	MACHINERY ROOM 1 (5-56-1)	
PROPULSION MOTOR 1 FAN 2	MACHINERY ROOM 1 (5-56-1)	
PROPULSION MOTOR 1 FAN 3	MACHINERY ROOM 1 (5-56-1)	
PROPULSION MOTOR 1 FAN 4	MACHINERY ROOM 1 (5-56-1)	
PROPULSION MOTOR 1 NDE BEARING LO PUMP 1	MACHINERY ROOM 1 (5-56-1)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER	LOCATION	REPAIR PRIORITY (OBSERVED)
PROPULSION MOTOR 1 NDE BEARING LO PUMP 2	MACHINERY ROOM 1 (5-56-1)	
PROPULSION MOTOR 1 DE BEARING LO PUMP 1	MACHINERY ROOM 1 (5-56-1)	
PROPULSION MOTOR 1 DE BEARING LO PUMP 2	MACHINERY ROOM 2 (5-56-1)	
PROPULSION MOTOR 1 TURNING GEAR	MACHINERY ROOM 2 (5-56-1)	
PROPULSION SHAFT THRUST BEARING LO SYSTEM(STBD)	MACHINERY ROOM 1 (5-56-1)	
PROPULSION MOTOR CONTROL/EXCITATION UNIT (STBD)	HV ROOM 1S (2-55-1)	
PROPULSION MOTOR 2 FAN 1	MACHINERY ROOM 2 (5-56-2)	
PROPULSION MOTOR 2 FAN 2	MACHINERY ROOM 2 (5-56-2)	
PROPULSION MOTOR 2 FAN 3	MACHINERY ROOM 2 (5-56-2)	
PROPULSION MOTOR 2 FAN 4	MACHINERY ROOM 2 (5-56-2)	
PROPULSION MOTOR 2 NDE BEARING L.O. PUMP 1	MACHINERY ROOM 2 (5-56-2)	
PROPULSION MOTOR 2 NDE BEARING L.O. PUMP 2	MACHINERY ROOM 2 (5-56-2)	
PROPULSION MOTOR 2 DE BEARING L.O. PUMP 1	MACHINERY ROOM 2 (5-56-2)	
PROPULSION MOTOR 2 DE BEARING L.O. PUMP 2	MACHINERY ROOM 2 (5-56-2)	
PROPULSION MOTOR 2 TURNING GEAR	MACHINERY ROOM 2 (5-56-2)	
PROPULSION MOTOR CONTROL/EXCITATION UNIT (PORT)	HV ROOM 1P (2-55-2)	
PROPULSION SHAFT THRUST BEARING LO SYSTEM (PORT)	MACHINERY ROOM 2 (5-56-2)	
PROPULSION TRANSFORMER 1A FAN 1	HV ROOM 1S (2-55-1)	
PROPULSION TRANSFORMER 1A FAN 2	HV ROOM 1S (2-55-1)	
PROPULSION TRANSFORMER 1B FAN 1	HV ROOM 1S (2-55-1)	
PROPULSION TRANSFORMER 1B FAN 2	HV ROOM 1S (2-55-1)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER	LOCATION	REPAIR PRIORITY (OBSERVED)
PROPULSION TRANSFORMER 2A FAN 1	HV ROOM 12 (2-55-2)	
PROPULSION TRANSFORMER 2A FAN 2	HV ROOM 1P (2-55-2)	
PROPULSION TRANSFORMER 2B FAN 1	HV ROOM 1P (2-55-2)	
PROPULSION TRANSFORMER 2B FAN 2	HV ROOM 1P (2-55-2)	
PURIFIER ROOM EXHAUST FAN M9	PURIFIER ROOM (3-56-2)	
SALTWATER ANTI-FOULING SYSTEM		
SERVICE TRUNK SUPPLY FAN M23		
SHORE POWER STATION		
SPRINKLING PUMP	BALLAST CONTROL ROOM A DECK (A-56-0)	
START AIR COMPRESSOR & RECEIVER 1P	HV ROOM 1P (2-55-2)	
START AIR COMPRESSOR & RECEIVER 2P	HV ROOM 1P (2-55-2)	
START AIR COMPRESSOR & RECEIVER 1S	HV ROOM 1S (2-55-1)	
START AIR COMPRESSOR & RECEIVER 2S	HV ROOM 1S (2-55-1)	
STERN TUBE SALTWATER SUPPLY PUMP PORT	MACHINERY ROOM 2 (5-56-2)	
STERN TUBE SALTWATER SUPPLY PUMP STBD	MACHINERY ROOM 1 (5-56-1)	
STERN TUBE SUPPLY PUMP (STBY)		
STEERING GEAR 1P	STEERING GEAR ROOM 2 (2-16-2)	
STEERING GEAR 2P	STEERING GEAR ROOM 1 (2-16-2)	
STEERING GEAR 1S	STEERING GEAR ROOM 1 (2-16-1)	
STEERING GEAR 2S	STEERING GEAR ROOM 1 (2-16-1)	
SW ANTI-FOULING SYSTEM	AUXILIARY MACHINERY ROOM	
SYNCHROCONVERTER (PC1) DEIONIZER WATER PUMP 1	HV ROOM 1P (2-55-2)	
SYNCHROCONVERTER (PC1) DEIONIZER WATER PUMP 2	HV ROOM 1P (2-55-2)	
SYNCHROCONVERTER (PC2) DEIONIZER WATER PUMP 1	HV ROOM 1P (2-55-2)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST		
EQUIPMENT/CIRCUIT NUMBER	**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
SYNCHROCONVERTER (PC2) DEIONIZER WATER PUMP 2	HV ROOM 1P (2-55-2)	
SYNCHROCONVERTER (SC1) DEIONIZER WATER PUMP 1	HV ROOM 2S (3-55-3)	
SYNCHROCONVERTER (SC1) DEIONIZER WATER PUMP 2	HV ROOM 2S (3-55-3)	
SYNCHROCONVERTER (SC2) DEIONIZER WATER PUMP 1	HV ROOM 2S (3-55-3)	
SYNCHROCONVERTER (SC2) DEIONIZER WATER PUMP 2	HV ROOM 2S (3-55-3)	
SYNCHROCONVERTER 1A	HV ROOM 2S (3-55-3)	
SYNCHROCONVERTER 1B	HV ROOM 2S (3-55-3)	
SYNCHROCONVERTER 2A	HV ROOM 1P (3-55-2)	
SYNCHROCONVERTER 2B	HV ROOM 1P (3-55-2)	
LIST OF ELECTRICAL SWITCHBOARDS		
HV1, 6.6 KV	HV ROOM 1S (3-55-3)	
HV1A, 6.6 KV	HV ROOM 1S (2-55-1)	
HV2, 6.6 KV	HV ROOM 1P (3-55-2)	
HV2A, 6.6 KV	HV ROOM 1P (2-55-2)	
MV1	ENGINE CONTROL ROOM 1 (2-56-1)	
MV2	ENGINE CONTROL ROOM 2 (2-56-2)	
MV1A	CASING STBD UPPER DECK (1-34-1)	
MV2A	CASING STBD UPPER DECK (1-34-1)	
EMERGENCY SWITCHBOARD	EMERGENCY DIESEL GENERATOR ROOM (A-34-2)	
LIST OF HARMONIC FILTERS		
HARMONIC FILTER 1	HV ROOM 1S (2-55-1)	
HARMONIC FILTER 2	HV ROOM 1P (2-55-2)	
LIST OF PROPULSION TRANSFORMERS		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
PT1	6.6/1.60KV 6400KVA PROPULSION TRANSFORMER 2A	HV ROOM 2P (3-55-2)	
PT2	6.6/1.60KV 6400KVA PROPULSION TRANSFORMER 2B	HV ROOM 2P (3-55-2)	
ST1	6.6/1.60KV 6400KVA PROPULSION TRANSFORMER 1A	HV ROOM 2S (3-55-3)	
ST2	6.6/1.60KV 6400KVA PROPULSION TRANSFORMER 1B	HV ROOM 2S (3-55-3)	
BOW THRUSTER TRANSFORMER			
BT1	6.6/742V 2700KVA	HV ROOM 1P (2-55-2)	
ABT 1A	MDG & HV PWR UPS NO. 1	ENGINE CONTROL ROOM (ECR) 1 (2-56-1)	
ABT 1B	IACMS UPS NO. 1	ECR 1 (2-56-1)	
ABT 1C	PROPULSION UPS NO. 1	ECR 1 (2-56-1)	
ABT 2A	MDG & HV PWR UPS NO. 2	ECR 2 (2-56-2)	
ABT 2B	IACMS UPS NO. 2	ECR 2 (2-56-2)	
ABT 2C	PROPULSION UPS NO. 2	ECR 2 (2-56-2)	
UPS	EDG & EMERGENCY SWITCHBOARD UPS		
UPS	MDG HV PWR SYS UPS NO. 1	ECR 1 (2-56-1)	
UPS	MDG HV PWR SYS UPS NO. 2	ECR 2 (2-56-2)	
UPS	IACMS UPS NO. 1	ECR 1 (2-56-1)	
UPS	IACMS UPS NO. 2	ECR 2 (2-56-2)	
UPS	PROPULSION UPS NO. 1	ECR 1 (2-56-1)	
UPS	PROPULSION UPS NO. 2	ECR 2 (2-56-2)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST			
EQUIPMENT/CIRCUIT NUMBER		**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
UPS	TANK LEVEL INDICATOR (TLI) UPS (SERVICE TRUNK)		
UPS	TLI UPS		
LIST OF TRANSFORMERS			
DT1	DISTRIBUTION TRANSFORMER #1 6.6/0.48KV 4000A	HV ROOM 1S (2-55-1)	
DT2	DISTRIBUTION TRANSFORMER #2 6.6/0.48KV 4000A	HV ROOM 1P (2-55-2)	
DT3	DISTRIBUTION TRANSFORMER #3 6.6/0.48KV 4000A	UPPER DECK CASING STBD (1-34-1)	
SHIP'S SERVICE TRANSFORMERS			
DT4	DISTRIBUTION TRANSFORMER #4 6.6/0.48KV 4000A	UPPER DECK CASING STBD (1-34-1)	
LC01	POWER SUBSTATION LC01 6.6/0.48KV 2500KVA	AUXILIARY MACHINERY ROOM (AMR) (6-108-0)	
LOAD CENTER TRANSFORMERS			
T-P417	480/480V 200KVA	ELECT EQUIPMENT ROOM (1-56-2)	
T-P426	480/480V 200KVA	FAN ROOM/ELEC EQUIPMENT (A4-96-1)	
T-P431	480/480V 200KVA	EMBARKED PERSONNEL LAUNDRY (A2-96-0)	
T-P432	480/480V 150KVA	SUPPORT SERVICES ROOM (A2-93-2)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
480/240 VOLT TRANSFORMERS			
T-P201	480/240V 37.5KVA	ELECT EQUIPMENT ROOM (D-50-0)	
T-P202	480/240V 75KVA	ENLISTED LAUNDRY (A2-96-2)	
T-TRH01	480/240V 45KVA	FAN ROOM/ELECT EQUIPMENT ROOM (A2-96-1)	
T-TRH02	480/240V 45KVA	FAN ROOM/ELECT EQUIPMENT ROOM (A2-96-1)	
T-TRH03	480/240V 45KVA	FAN ROOM/ELECT EQUIPMENT ROOM (A3-96-3)	
T-TRH04	480/240V 45KVA	FAN ROOM/ELECT EQUIPMENT ROOM (A3-96-3)	
120/120 VOLT TRANSFORMERS			
T-L101	120/120V 15KVA	MACHINERY ROOM 1 (5-56-1)	
T-L102	120/120V 15KVA	ENGINE CONTROL ROOM 2 (2-56-2)	
T-L106	120/120V 15KVA	LOWER DECK HOUSE	
T-L108	120/120V 15KVA	UPPER DECK HOUSE	
T-L116	120/120V 15KVA	SUPPORT SERVICES ROOM (A1-96-0)	
T-L117	120/120V 30KVA	FAN ROOM/ELECT EQUIPMENT ROOM (A2-96-1)	
T-L118	120/120V 30KVA	SUPPORT SERVICES ROOM (A1-96-0)	
480/120 VOLT TRANSFORMERS			
T-P100A	480/120V 75KVA	SUPPORT SERVICES ROOM (A1-96-0)	
T-P100B	480/120V 75KVA	SUPPORT SERVICES ROOM (A1-96-0)	
T-P103	480/120V 15KVA	ELECT EQUIPMENT ROOM (1-56-2)	
T-P104	480/120V 30KVA	ELECT EQUIPMENT ROOM (1-56-2)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
T-P105	480/120V 15KVA	ELECT EQUIPMENT ROOM (1-56-2)	
T-P115	480/120V 30KVA	FAN ROOM /ELECT EQUIPMENT ROOM (A4-96-1)	
T-P118	120/120V 30KVA	FAN ROOM/ELECT EQUIPMENT ROOM (A2-96-1)	
T-P119	480/120V 30KVA	FAN ROOM /ELECT EQUIPMENT ROOM (A4-96-1)	
T-P120	480/120V 30KVA	SUPPORT SERVICES ROOM (A4-96-0)	
T-P121	480/120V 100KVA	FAN ROOM /ELECT EQUIPMENT ROOM (A7-96-0)	
T-P122	480/120V 100KVA	FAN ROOM /ELECT EQUIPMENT ROOM (A7-96-0)	
TRANSFORMER	112.6 KVA, 480/480 (A7-96-0) PCS ISOLATION	FAN ROOM /ELECT EQUIPMENT ROOM (A7-96-0)	
TRANSFORMER	112.5 KVA, 480/480 VAC (A7-96-0)	FAN ROOM /ELECT EQUIPMENT ROOM (A7-96-0)	
T-P123	480/120V 75KVA	FAN ROOM (A1-89-1)	
T-P124	480/120V 75KVA	UNASSIGNED (1-56-1)	
TMV1	480/120V 150KVA	ENGINE CONTROL ROOM 1 (2-56-1)	
TMV2	480/120V 150KVA	ENGINE CONTROL ROOM 2 (2-56-2)	
TE1	480/120V 150KVA	EMERGENCY DIESEL GENERATOR ROOM (A-34-2)	
TE2	480/120V 75KVA	EMERGENCY DIESEL GENERATOR ROOM (A-34-2)	
TE-4	480/120V 150KVA	FAN ROOM/ELEC EQUIPMENT (A2-96-1)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST			
EQUIPMENT/CIRCUIT NUMBER		**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
480/208/120 VOLT TRANSFORMERS			
T-P205	480/208/120V 150KVA	SUPPORT SERVICES ROOM (A1-96-0)	
T-P206	480/208/120V 150KVA	SUPPORT SERVICES ROOM (A1-96-0)	
LIST OF MOTOR CONTROL CENTERS			
MCC05	AUXILIARY MACHINERY	MACHINERY ROOM NO. 1 (5-56-1)	
MCC06	AUXILIARY MACHINERY	MACHINERY ROOM NO. 2 (5-56-2)	
MCC07	AUXILIARY MACHINERY	STEERING GEAR ROOM (2-16-1)	
MCC08	ACCOMMODATION VENTILATION	ELEC EQUIPMENT ROOM (1-56-2)	
MCC09	MISC VENTILATION	CO2 RM (1-34-2)	
MCC10	MISC VENTILATION	AMR (6-108-0)	
MCC11	MACHINERY SERVICES	AMR GRADING LEVEL (6-108-0)	
MCC12	AUXILIARY MACHINERY	JP-5 PUMP ROOM (5-108-1)	
MCC13	BALLAST PUMP	BALLAST PUMP ROOM (5-57-0)	
MCC14	AUXILIARY MACHINERY	HOTEL SERVICES (2-48-1)	
LIST OF LOADS BY MCCS			
MCC05, AUXILIARY MACHINERY, MACHINERY ROOM (5-56-1)			
MCC05-01	DEEPWELL BILGE PUMP STBD		
MCC05-02	STERN TUBE SALTWATER PUMP STBD		
MCC05-04	OILY WATER SEPARATOR		
MCC05-05	HOT WATER CIRC PUMP		
MCC06, AUXILIARY MACHINERY, MACHINERY ROOM NO. 2 (5-56-2)			
MCC06-01	DEEPWELL BILGE PUMP PORT		
MCC06-02	STERN TUBE SALTWATER		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
	SUPPLY PUMP PORT		
MCC06-03	MISSION DISTILLED WATER SERVICE PUMP		
MCC07, AUXILIARY MACHINERY, STEERING GEAR ROOM 1			
MCC07-01	STEERING GEAR ROOM 1 SUPPLY FAN M14		
MCC07-02	STEERING GEAR ROOM 2 SUPPLY FAN M15		
MCC07-03	MOORING WINCH M5 POWER PACK	AFT MOORING DECK, FR 8	
MCC07-04	MOORING WINCH M7 POWER PACK	AFT MOORING DECK, FR 2 STBD	
MCC07-06	MOORING WINCH M3 POWER PACK	UPPER DECK, AFT OF LIFEBOAT, FR 38.5 STBD	
MCC07-07	MOORING WINCH M4 POWER PACK	UPPER DECK, AFT OF LIFEBOAT, FR 38.5 PORT	
MCC07-08	CO2 RM EXHAUST FAN M21		
MCC07-09	MSD ROOM EXHAUST FAN M22		
MCC08 ACCOMMODATION VENTILATION, ELECTRICAL EQUIPMENT ROOM (1-56-2)			
MCC08-01	“E” DECK HEATERS AND FANS		
MCC08-02	GALLEY SPACE HEATERS AND FANS		
MCC08-03	MISC SANITATION EXHAUST FAN E1		
MCC08-05	ACCOMMODATION (AC1 & AC2) RETURN FAN R1		

ENCLOSURE 2.2.1 – EQUIPMENT LIST			
EQUIPMENT/CIRCUIT NUMBER		**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
MCC08-06	DRY PROVISIONS EXHAUST FAN E3		
MCC08-07	HOSPITAL EXHAUST FAN E4		
MCC08-08	HYDRAULIC EQUIPMENT ROOM EXHAUST FAN E5		
MCC09, MISC VENTILATION, CO2 LOCKER ROOM (1-34-2)			
MCC09-01	CHEMICAL STORE EXHAUST FAN M12		
MCC09-02	PAINT & LAMP STORE EXHAUST FAN M11		
MCC09-03	ACETYLENE & OXYGEN STOWAGE ROOM EXHAUST FAN M16		
MCC10, MISCELLANEOUS VENTILATION, AMR (6-108-0)			
MCC10-01	JP-5 PUMP ROOM SUPPLY FAN M25		
MCC10-02	JP-5 PUMP ROOM EXHAUST FAN M26		
MCC10-03	AMR SUPPLY FAN M27		
MCC10-04	AMR EXHAUST FAN M28		
MCC10-05	SERVICE TRUNK EXHAUST FAN M24		
MCC10-06	BOSUNS STOREROOM EXHAUST FAN M13		
MCC11, MACHINERY SERVICES, AMR GRATING LEVEL (6-108-0)			
MCC11-01	BOW THRUSTER L.T. F.W.		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
	COOLING PUMP NO. 1		
MCC11-02	BOW THRUSTER L.T. F.W. COOLING PUMP NO. 2		
MCC11-03	BOW THRUSTER S.W. COOLING/DEWATERING PUMP NO. 1		
MCC11-04	BOW THRUSTER S.W. COOLING/DEWATERING PUMP NO. 2		
MCC11-05	MISSION SERVICE POTABLE WATER PUMP		
MCC11-06	REVERSE OSMOSIS UNIT		
MCC11-07	FWD GENERAL SERVICE PUMP		
MCC11-08	RO RECIRC BROMINATOR		
MCC11-09	BOW THRUSTER ANTI-FOULING SYSTEM		
MCC12, AUXILIARY MACHINERY, JP-5 PUMP ROOM (5-108-1)			
MCC12-01	JP-5 MISSION TRANSFER PUMP		
MCC12-02	JP-5 MISSION SERVICE PUMP		
MCC13, BALLAST PUMP, BALLAST PUMP ROOM (5-57-0)			
MCC13-01	NO. 1 BALLAST PUMP		
MCC13-02	NO. 2 BALLAST PUMP		
MCC13-03	NO. 3 BALLAST PUMP		
MCC14, AUXILIARY MACHINERY, HOTEL SERVICES (2-48-1)			
MCC14-01	NO. 1 AC CHILL WATER PUMP		

ENCLOSURE 2.2.1 – EQUIPMENT LIST			
EQUIPMENT/CIRCUIT NUMBER		**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
MCC14-02	NO. 2 AC CHILL WATER PUMP		
LIST OF ELECTRICAL LOAD CENTERS			
LC01	POWER SUBSTATION	AMR (6-108-0)	
ELC1	EMERGENCY LOAD CENTER 1	EDG ROOM (A-34-2)	
ELC2	EMERGENCY LOAD CENTER 2	EDG ROOM (A-34-2)	
LIST OF ELECTRICAL PANELS			
LIST OF 480 POWER PANELS			
P400A	DISTRIBUTION PANEL (LIGHTING)	SUPPORT SERVICE ROOM (A1-96-0)	
P400B	MISCELLANEOUS EQUIPMENT	SUPPORT SERVICE ROOM (A1-96-0)	
P401	HOTEL	HOTEL SERVICE ROOM (2-48-1)	
P402	AUXILIARY MACHINERY	MACHINERY ROOM 2 (5-56-2)	
P403	AUXILIARY MACHINERY	MACHINERY ROOM 1 (5-56-1)	
P404	AUXILIARY MACHINERY	STORES (2-48-2)	
P405	ENGINEERING WORKSHOP	ENGINEERING WORKSHOP (3-55-1)	
P406	AUXILIARY MACHINERY	MACHINERY ROOM 2 (5-56-2)	
P407	AUXILIARY MACHINERY	HOTEL SERVICE ROOM (2-48-1)	
P408	DECK MACHINERY	ELECT EQUIPMENT ROOM 2 (1-56-2)	
P409	AUXILIARY MACHINERY	MACHINERY ROOM 1 (5-56-1)	
P411	MISCELLANEOUS POWER	ELEC EQUIPMENT ROOM (1-56-2)	
P412	MOV	MACHINERY ROOM 2 (5-56-2)	
P414	MOV	MACHINERY ROOM 2 (5-56-2)	
P417	COMMISSARY	GALLEY (A-47-0)	
P419	MOV	MACHINERY ROOM 1 (5-56-1)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
P420	BALLAST PUMP ROOM HEATERS	BALLAST PUMP ROOM (5-57-0)	
P421	AMR HEATERS & MOV	AMR (6-108-0)	
P423	HEATERS	SUPPORT SERVICE ROOM (A1-96-0)	
P424	HEATERS	SUPPORT SERVICE ROOM (A1-96-0)	
P425	MISSION SERVICES	MHE STOWAGE/MAINTENANCE/REPAIR AREA (A5-89-4)	
P426	GFI POWER PANEL	GALLEY (A4-92-1)	
P427	AVIATION SERVICES	FAN ROOM/ELEC EQUIPMENT ROOM (A6-96-0)	
P428	AVIATION SERVICES	FAN ROOM/ELEC EQUIPMENT ROOM (A6-96-0)	
P429	DECK MACHINERY	SUPPORT SERVICE ROOM (A1-96-0)	
P430	AUXILIARY MACHINERY	SUPPORT SERVICE ROOM (A1-96-0)	
P431		EMBARKED PERSONNEL LAUNDRY (A2-96-0)	
P432	GFI POWER PANEL	SERVICE TRUNK (A2-93-2)	
P433	TEU SERVICES FLIGHT DECK	ELEC EQUIPMENT ROOM (1-56-2)	
P434	TEU SERVICES MISSION DECK FWD	FAN ROOM (A1-89-0)	
P435	TEU SERVICES MISSION DECK FWD	FAN ROOM (A1-89-0)	
P436	TEU SERVICES A6 LVL	STAIRTOWER (A1-91,5-2)	
PPHV01	VENTILATION POWER PANEL	FAN RM/ELEC EQUIPMENT (A4-96-1)	
LIST OF 240V POWER PANELS			
P201	GALLEY/LAUNDRY MISC 240V	ELEC EQUIPMENT ROOM (D-50-0)	
P202	ENLISTED LAUNDRY MISC 240V	ENLISTED LAUNDRY (A2-96-2)	
P203	GFI POWER PANEL	OFFICERS LAUNDRY (A3-96-0)	
P205	MILVAN SERVICE	SUPPORT SERVICE ROOM (A1-96-0)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST			
EQUIPMENT/CIRCUIT NUMBER		**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
P206	GFI POWER PANEL	SUPPORT SERVICE ROOM (A1-96-0)	
LIST OF 120V POWER PANELS			
PDB1	POWER DISTRIBUTION BOX 1 ELECTRONIC NAC PANEL	WHEELHOUSE (E-57-0)	
PDB2	POWER DISTRIBUTION BOX 2 ELECTRONIC NAC PANEL	WHEELHOUSE (E-57-0)	
P100A	LIGHTING	SUPPORT SERVICES ROOM (A6-96-0)	
P100B	LIGHTING	SUPPORT SERVICES ROOM (A1-96-0)	
P101	AUXILIARY MACHINERY	MACHINERY ROOM 1 (5-56-1)	
P102	AUXILIARY	MACHINERY ROOM 2 (5-56-2)	
P103	COMMISSARY	GALLEY A-47-0	
P104	COMMISSARY & LAUNDRY	CREW LAUNDRY (B-49-1)	
P105	COMMISSARY	OFFICERS MESS A-45-2	
P106	HVAC REHEATERS B DECK	ELEC EQUIPMENT ROOM (1-56-2)	
P107	HVAC REHEATER A DECK	ELEC EQUIPMENT ROOM (1-56-2)	
P108		ELEC EQUIPMENT ROOM (D-50-0)	
P109	HVAC REHEATER C DECK	ELEC EQUIPMENT ROOM (D-50-0)	
P110	HVAC REHEATER UPPER DECK	ELECT EQUIPMENT ROOM (1-56-2)	
P111	FORECASTLE	BOSUN’S STOREROOM (1-126-0)	
P113	AUXILIARY MACHINERY	MACHINERY ROOM 1 (5-56-1)HG	
P115	GFI POWER PANEL	GALLEY (A4-92-1)	
P118	GFI POWER PANEL	FAN ROOM/ELEC EQUIPMENT ROOM (A2-96-1)	
P119	GFI POWER PANEL	FAN ROOM/ELEC EQUIPMENT ROOM (A4-96-1)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
P120	GFI POWER PANEL	ENLISTED MESSROOM (A4-90-0)	
P121	VARIOUS ELECTRONICS	FAN ROOM/ELEC EQUIPMENT ROOM (A7-96-0)	
P121A	COMMUNICATION CENTER	COMMUNICATION CENTER (A8-94-0)	
P121B	TACTICAL OP CENTER	TACTICAL OPERATIONS CENTER (A8-91-0)	
P121C	COMMUNICATION CENTER	COMMUNICATION CENTER (A8-94-0)	
P121D	BRIEFING ROOM	BRIEFING ROOM (A4-94-4)	
P122	VARIOUS ELECTRONICS	FAN ROOM/ELEC EQUIPMENT ROOM (A7-96-0)	
P122A	TRAINING CENTER	TRAINING ROOM (A8-89-0)	
P122B	PLANNING & INTEL CENTER	PLANNING CENTER (A8-93-1)	
P122C	HELICOPTER CONTROL	HELICOPTER CONTROL (A8-87-0)	
P123	120V POWER PANEL	FAN ROOM (A1-89-1)	
P124	120V POWER PANEL	UNASSIGNED (1-56-1)	
LIST OF EMERGENCY POWER PANELS			
EP401	MOV	MACHINERY ROOM 1 (5-56-1)	
EP402	MOV	MACHINERY ROOM 2 (5-56-2)	
EP403	LIGHTING PNL WEATHER DECK	ELEC EQUIPMENT ROOM (1-56-2)	
EP404	HEATED WINDOW WHEELHOUSE	ELEC EQUIPMENT ROOM (D-50-0)	
EP405	MOV	BALLAST PUMP ROOM (5-57-0)	
EP406	MOV	SERVICE TRUNK (5-97-2)	
EP407	MOV	SERVICE TRUNK (5-97-2)	
EP4O8	MOV	SERVICE TRUNK (5-97-2)	
EP409	MISC EQUIPMENT/LIGHTING	SUPPORT SERVICE ROOM (A1-96-0)	
EP410	HEATED WINDOWS (HCS)	FAN ROOM/ELEC EQUIPMENT ROOM (A7-96-0)	
EP100	MISSION SERVICES/LIGHTING	FAN ROOM/ELEC EQUIPMENT ROOM (A2-96-1)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST			
EQUIPMENT/CIRCUIT NUMBER		**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
EP101	AUXILIARY MACHINERY	MACHINERY ROOM 1 (5-56-1)	
EP102	AUXILIARY MACHINERY	MACHINERY ROOM 2 (5-56-2)	
EP103	MISC DECKHOUSE	ELEC EQUIP ROOM (D-50-0)	
EP104	WINDOW WIPERS	ELEC EQUIP ROOM (D-50-0)	
EP105	EMERGENCY SERVICES	FAN ROOM/ELEC EQUIPMENT ROOM (A6-96-0)	
EP106	MISSION SPACE	FAN ROOM/ELEC EQUIPMENT ROOM (A8-96-0)	
EP107	MISSION SPACE	FAN ROOM/ELEC EQUIPMENT ROOM (A8-96-0)	
LIST OF LIGHTING PANELS			
L401	WEATHER DECK AFT	ELEC EQUIPMENT ROOM (1-56-2)	
L403	FLOODLIGHTS	BOSUN'S STOREROOM (1-26-0)	
L101	MACHINERY ROOM 1	MACHINERY ROOM 1 (5-56-1)	
L102	GFI RECEPTACLE	MACHINERY ROOM 2 (5-56-2)	
L103	MACHINERY ROOM 1	MACHINERY ROOM 1 (5-56-1)	
L104	AUXILIARY LIGHTS	MACHINERY ROOM 2 (5-56-2)	
L105	UPPER DECK	ELEC EQUIPMENT ROOM (1-56-2)	
L106	GFI RECEPTACLE LOWER DECKHOUSE	ACCESS A DECK (1-52-2)	
L107	LOWER DECK HOUSE	ACCESS B DECK (1-52-2)	
L108	LIGHTING UPPER DECK HOUSE	ELEC EQUIPMENT ROOM (D-50-0)	
L109	UPPER DECK HOUSE	ELEC EQUIPMENT ROOM (D-50-0)	
L110	LIGHTING	MACHINERY ROOM 2 (5-56-2)	
L111	MACHINERY ROOM 1	MACHINERY ROOM 1 (5-56-1)	
L112	LIGHTING WHEELHOUSE & WEATHER DECK	ELEC EQUIPMENT ROOM (D-50-0)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
L115	LIGHTING	BOSUN'S STOREROOM (1-26-0)	
L116	LIGHTING	SUPPORT SERVICE ROOM (A1-96-0)	
L117	LIGHTING	FAN ROOM/ELEC EQUIPMENT ROOM (A2-96-1)	
L118	LIGHTING	FAN ROOM/ELEC EQUIPMENT ROOM (A4-96-1)	
L119	LIGHTING	SUPPORT SERVICE ROOM (A1-96-0)	
L120	LIGHTING	FAN ROOM/ELEC EQUIPMENT ROOM (A2-96-1)	
L121	LIGHTING	SUPPORT SERVICE ROOM (A1-96-0)	
L122	LIGHTING	FAN ROOM/ELEC EQUIPMENT ROOM (A3-96-3)	
L123	LIGHTING	SUPPORT SERVICE ROOM (A1-96-0)	
L124	GFI RECEPTACLE	A5-A7 LVL	
L125	GFI RECEPTACLE	A2-A4 LVL	
L126	LIGHTING	SUPPORT SERVICE ROOM (A1-96-0)	
L127	LIGHTING	SUPPORT SERVICE ROOM (A1-96-0)	
EL101	MACHINERY ROOM 1	MACHINERY ROOM 1 (5-56-1)	
EL102	MACHINERY ROOM 2	MACHINERY ROOM 2 (5-56-2)	
EP104	ACCOMMODATION “A,” “B” & UPPER DECK	ELEC EQUIPMENT ROOM (1-56-2)	
EL105	C, D & WHEELHOUSE DECK	SERVICE TRUNK D DECK (1-52-3)	
EL106	LIGHTING, FORECASTLE	AUXILIARY MACHINERY ROOM (6-108-0)	
EL107	LIGHTING	SERVICE SUPPORT ROOM (A1-96-0)	
EL108	LIGHTING	FAN ROOM/ELEC EQUIPMENT ROOM (A3-96-3)	
EL109	LIGHTING	FAN ROOM/ELEC EQUIPMENT ROOM (A8-96-0)	
EL110	LIGHTING	FAN ROOM/ELEC EQUIPMENT ROOM (A8-96-0)	
LIST OF LOADS BY PANELS			

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
P401, HOTEL LOADS, 20.7M FLAT (2-48-1)			
P401-01	A/C PLANT 2S		
P401-02	A/C PLANT 1S		
P401-03	SHIP SERVICE REFRIGERATION UNIT 1S		
P401-04	SHIP SERVICE REFRIGERATION UNIT 2S		
P401-05	MSD		
P401-06	REFRIGERATION CONDENSING UNIT		
P401-07	POTABLE WATER HEATER		
P401-08	MCC14		
P401-09	SEWAGE HANDLING TANK AERATION BLOWER		
P402-01	MDO PURIFIER NO. 1		
P402-02	MDO PURIFIER NO. 2		
P402-03	LO PURIFIER 1P		
P402-04	LO PURIFIER 2P		
P402-05	LO PURIFIER 3P		
P402-06	LO PURIFIER 4P		
P402-07	BRIDGE CRANE (PORT)		
P402-09	VARIOUS PROPULSION HEATERS		
P402-10	FUEL OIL AUTOMATIC FILTER (PORT)		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
P402-11	DG1P POWER PANEL		
P402-12	DG2P POWER PANEL		
P402-13	DG1P ENGINE CONTROL CABINET		
P402-14	DG2P ENGINE CONTROL CABINET		
P403, AUXILIARY MACHINERY, 13.6M FLAT (5-56-1)			
P403-01	BRIDGE CRANE (STBD)		
P403-02	VARIOUS PROPULSION HEATERS		
P403-03	DG1S POWER PANEL		
P403-04	DG2S POWER PANEL		
P403-06	PORT RECIRC BROMINATOR		
P403-07	DG1S ENGINE CONTROL CABINET		
P403-08	DG2S ENGINE CONTROL CABINET		
P404, AUXILIARY MACHINERY, STORES (2-48-2)			
P404-01	GENERAL SERVICE AIR COMPRESSOR (PORT)		
P404-02	GREY WATER DISCHARGE PUMP		
P404-03	MDG 1P LUBE OIL AUTO CLEANING FILTER		
P404-04	MDG 2P LUBE OIL AUTO CLEANING FILTER		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
P404-05	GREY WATER HOLDING TANK AERATION BLOWER		
P404-07	SYNCHROCONVERTER 2A(SC2A) GROUNDING SW MOTOR		
P404-08	SYNCHROCONVERTER 2B(SC2B) GROUNDING SW MOTOR		
P405, ENGINEERS WORKSHOP (3-55-1)			
P405-01	SELF CONTAINED AC UNIT 2S		
P405-02	DRILL PRESS PEDESTAL		
P405-03	SYNCHROCONVERTER 1A(SC1A) GROUNDING SW MOTOR		
P405-04	LATHE, PRECISION		
P405-05	PEDESTAL GRINDER		
P405-06	PEDESTAL GRINDER		
P405-07	SYNCHROCONVERTER 1B(SC1B) GROUNDING SW MOTOR		
P405-08	WELDING MACHINE		
P405-09	PROPULSION TRANSFORMER 1B PREMAG TRANSFORMER		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
P405-10	TEST PANEL ELECTRICAL		
P405-12	FUEL OIL AUTOMATIC FILTER (STBD)		
TRANSFORMER	3 KVA, 480/240 VAC	ENGINEERS WORKSHOP (3-55-1)	
P406, AMR, (5-56-2)			
P406-01	DISTILLING PLANT (PORT)		
P406-02	LCL FIRE EXTINGUISHING SYSTEM		
P406-03	PROPULSION TRANSFORMER 2B PREMAG TRANSFORMER		
P406-04	ME1P SHAFT TURNING GEAR		
P406-05	ME2P SHAFT TURNING GEAR		
P406-07	DG1P OIL MIST ELIMINATOR		
P406-08	DG2P OIL MIST ELIMINATOR		
P407, AUXILIARY MACHINERY HOTEL SERVICES, (2-48-1)			
P407-01	GENERAL SERVICE AIR COMPRESSOR (STBD)		
P407-02	MDG 1S LUBE OIL AUTO CLEANING FILTER		
P407-03	MDG 2S LUBE OIL AUTO CLEANING FILTER		
P408, DECK MACHINERY, ELECTRICAL EQUIPMENT ROOM (1-56-2)			
P408-02	MACHINERY REMOVAL/STORES CRANE (PORT)		

USSHerhel Williams		
(ESB 4)		
ELECTRICAL		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0353A	CATEGORY "A"	2026-01-30
ESB-CCSI-THERMOGRAPHIC SURVEY (1 YR)		Manovich, David

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
P408-03	LIFEBOAT WINCH (PORT)		
P408-04	RESCUE BOAT WINCH		
P409, AMR, (5-56-1)			
P409-01	DISTILLING PLANT (STBD)		
P409-02	CATHODIC PROTECTION (AFT)		
P409-03	FRESH POTABLE WATER HYDROPHORE UNIT		
P409-04	ME1S SHAFT TURNING GEAR		
P409-05	ME2S SHAFT TURNING GEAR		
P409-06	DG1S OIL MIST ELIMINATOR		
P409-07	DG2S OIL MIST ELIMINATOR		
P411, MISCELLANEOUS POWER, ELECTRICAL EQUIPMENT ROOM (1-56-2)			
P411-02	MACHINERY REMOVAL/STORES CRANE (STBD)		
P411-04	LIFEBOAT WINCH (STBD)		
P411-06	MISSION DECK JIB HOIST		
P411-07	AFT MISSION DECK RECEPTACLE SYSTEM		
P412, MOTOR OPERATED VALVES, MACHINERY ROOM 2 (5-56-2)			
P412-01	MOV FP-V-13		
P412-02	MOV FP-V-14		
P412-03	SPARE		
P412-04	MOV GF-V-88		
P412-05	MOV GF-V-72		

ENCLOSURE 2.2.1 – EQUIPMENT LIST			
EQUIPMENT/CIRCUIT NUMBER		**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
P413, MOTOR OPERATED VALVES, MACHINERY ROOM 2 (5-56-2)			
P413-01	MOV MD-V-29		
P413-02	MOV MD-V-32		
P413-03	MOV FT-V-140		
P413-04	MOV MD-V-34		
P413-05	MOV MD-V-36		
P417, GFI POWER PANEL, COMMISSARY, GALLEY (A-47-0)			
P417-01	PRESSURELESS STEAMER		
P417-02	RANGE/OVEN		
P417-03	CONVECTION OVEN		
P417-04	DEEP FAT FRYER		
P417-05	SANITIZATION SINK HEAT BOOSTER		
P417-06	DISHWASHER		
P417-07	HOOD CLEANING BOOSTER HEATER		
P417-08	STEAM JACKET KETTLE		
P417-09	DEEP FAT FRYER		
P417-10	RANGE/OVEN		
P417-11	GARBAGE DISPOSAL		
P417-12	GARBAGE DISPOSAL		
P419, MOTOR OPERATED VALVES, MACHINERY ROOM 1 (5-56-1)			
P419-02	MOV FP-V-11		
P419-03	MOV FP-V-12		
P419-04	MOV GF-V-38		

ENCLOSURE 2.2.1 – EQUIPMENT LIST			
EQUIPMENT/CIRCUIT NUMBER		**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
P419-05	MOV GF-V-22		
P419-07	MOV MD-V-37		
P419-12	MOV FT-V-136		
P420, BALLAST PUMP ROOM HEATERS, BALLAST PUMP ROOM (5-57-0)			
P420-01 THROUGH P420-06	VARIOUS BALLAST PUMP ROOM HEATERS		
P420-07	SALTWATER ANTI-FOULING SYSTEM		
P421, AMR HEATERS & MOV, AMR (6-108-0)			
P420-01 THROUGH P420-06	VARIOUS AMR MOVS & HEATERS		
P201, MISCELLANEOUS 240V GALLEY & LAUNDRY, ELECTRICAL EQUIPMENT ROOM (D-50-0)			
P201-01	DRYER 1 (CREW LAUNDRY)		
P201-02	DRYER 2 (CREW LAUNDRY)		
P201-03	DRYER 3 (CREW LAUNDRY)		
P201-04	DRYER 1 (OFFICERS LAUNDRY)		
P201-05	DRYER 2 (OFFICERS LAUNDRY)		
P201-06	HOT WATER UNIT		
PDB1, POWER DISTRIBUTION BOX 1 ELECTRONIC NAC PANEL, WHEELHOUSE (E-57-0)			
PDB1-01	DOPPLER SPEED LOG TRANSCEIVER		
PDB1-02	X-BAND RADAR TRANSFORMER BOX		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
PDB1-03	X-BAND RADAR SCANNER		
PDB1-04	STBD RUDDER ANGLE INDICATOR		
PDB1-05	BNWAS & CONNING DISPLAY		
PDB1-06	ECDIS, VMS 1		
PDB1-08	COURSE RECORDER PRINTER		
PDB1-09	SAT/COM FLEET BROADBAND 500		
PDB1-10	GMDSS RADIO POWER UNIT NO. 1		
PDB1-11	24VDC POWER SUPPLY NO. 1		
PDB1-12	DOPPLER SPEED LOG DISPLAY		
PDB1-13	MASTER CLOCK		
PDB1-14	MASTER CLOCK DIMMER		
PDB1-15	BNWAS, ETHERNET MODULE		
PDB1-16	AIS PILOT LAPTOP CONNECTION		
PDB1-17	X-BAND RADAR DISPLAY		
PDB2, POWER DISTRIBUTION BOX 2 ELECTRONIC NAC PANEL, WHEELHOUSE (E-57-0)			
PDB2-01	MAGNETIC COMPASS		
PDB2-02	BNWAS		
PDB2-03	ECHO DEPTH SOUNDER		
PDB2-05	ECHO DEPTH SOUNDER PRINTER		
PDB2-06	S-BAND RADAR SCANNER		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
PDB2-08	RADAR INTERSWITCH		
CIRCUIT NO.	**MODEL/NOMENCLATURE/TYPE**	**LOCATION**	
PDB2-10	S-BAND RADAR DISPLAY		
PDB2-11	VDR		
PDB2-13	24VDC POWER SUPPLY NO. 2		
PDB2-14	ROUTE PLANNING VMS 2, IBS UPS		
PDB2-17	S-BAND RADAR TRANSFORMER BOX		
PDB2-20	PORT RUDDER ANGLE INDICATOR		
P101, AUXILIARY MACHINERY ROOM 1 (5-56-1)			
P101-01	MISSION MODULE POWER STATION 1 HEATER		
P101-02	BAND SAW		
P101-03	ELECTRICAL TEST PANEL		
P101-04	GENERAL SERVICE AIR DEHYDRATOR (STBD)		
P101-05	INSTRUMENT AIR DEHYDRATOR (STBD)		
P101-06	SUBMERSIBLE SUMP PUMP (MSD SINK)		
P101-08	GENERATOR 1S BEARING OIL HEATER		
P101-09	GENERATOR 2S BEARING OIL		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
	HEATER		
P101-10	STBD DISTILLER BYPASS VALVE WC-V-097		
P101-11	22W CHEMICAL TREATMENT PUMP		
P101-12	PROPULSION MOTOR CONTROLLER/EXCITATION UNIT (STBD)		
P101-13	MOV GF-V-31 MDG EMERGENCY HEAD TANK (STBD)		
P101-14	SHORE POWER STATION HEATER		
P101-15	HV SWBD 1 INTERNAL 120V BUS		
P101-16	SALTWATER PUMP 1S BFA CONTROL & ANALYZER		
P101-17	PROPULSION MOTOR CONTROLLER/EXCITATION UNIT (STBD)		
P101-18	DISTILLING PLANT 1 BFA CONTROL & ANALYZER		
P101-19	SYNCHROCONVERTER 1B SPACE HEATER		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
P102, AUXILIARY MACHINERY ROOM 2 (5-56-2)			
P102-01	GENERAL SERVICE AIR DEHYDRATOR (PORT)		
P102-02	INSTRUMENT AIR DEHYDRATOR (PORT)		
P102-03	PORT DISTILLER BYPASS VALVE WC-V-098		
P102-04	GENERATOR 1P BEARING OIL HEATER		
P102-05	GENERATOR 2P BEARING OIL HEATER		
P102-06	22W CHEMICAL TREATMENT PUMP		
P102-07	SYNCHROCONVERTER 2A SPACE HEATER		
P102-08	PROPULSION MOTOR CONTROLLER/EXCITATION UNIT (PORT)		
P102-09	MOV-GF-V-81 MDG EMERGENCY HEAD TANK (PORT)		
P102-10	PUMP ROOM SALTWATER PUMP BFA CONTROL & ANALYZER		
P102-11	HV SWITCHBOARD INTERNAL 120V BUS		

ENCLOSURE 2.2.1 – EQUIPMENT LIST			
EQUIPMENT/CIRCUIT NUMBER		**LOCATION**	**REPAIR PRIORITY (OBSERVED)**
P102-12	SPARE		
P102-13	SALTWATER PUMP (1P/2P) BFA CONTROL & ANALYZER		
P102-14	DISTILLING PLANT 2 BFA CONTROL & ANALYZER		
P102-15	MISSION MODULE POWER STATION 2 HEATER		
P102-16	PROPULSION MOTOR CONTROLLER/EXCITATION UNIT (PORT)		
P102-17	SYNCHROCONVERTER 2B SPACE HEATER		
P102-18	CYCLONE SEPARATOR (PORT)		
P103, GFI POWER PANEL COMMISSARY, GALLEY (A-47-0)			
P103-01	HOT PLATE DISPENSER		
P103-02	REFRIGERATOR		
P103-03	FREEZER		
P103-04	BENCH MOUNTED MIXER		
P103-05	CAN OPENER		
P103-06	MICROWAVE		
P103-07	COLD SERVING UNIT		
P103-09	MEAT SLICER		
P103-10	ICEMAKER		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
P103-11	SPARE		
P103-12	PORTABLE COOKING OIL FILTER		
P103-13	CHEST FREEZER		
P103-14	UNDERCOUNTER REFRIGERATOR		
P103-15	ANSUL R-102 DISCHARGE HORN/STROBE		
P103-16	GAYLORD VENT HOOD BOOSTER PUMP		
P104, GFI POWER PANEL COMMISSARY & LAUNDRY, CREWS LAUNDRY (B-49-1)			
P104-01	DRINKING FOUNTAIN	(CREW MESS)	
P104-02	4-SLICE TOASTER	(CREW MESS)	
P104-03	REFRIGERATOR/FREEZER WITH ICE MAKER	(CREW MESS)	
P104-04	COFFEE MAKER	(CREW MESS)	
P104-05	MICROWAVE	(CREW MESS)	
P104-06	WASHING MACHINE 1	(CREW LAUNDRY)	
P104-07	WASHING MACHINE 2	(CREW LAUNDRY)	
P104-08	WASHING MACHINE 3	(CREW LAUNDRY)	
P104-09	WASHING MACHINE 1	(OFFICER LAUNDRY)	
P104-10	WASHING MACHINE 2	(OFFICER LAUNDRY)	
P104-11	COFFEE MAKER	(CREW LOUNGE/TV ROOM)	
P104-12	TRASH COMPACTOR	(WASTE MANAGEMENT ROOM)	
P104-13	SINK HEATER	(WASTE MANAGEMENT ROOM)	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
P104-14	JUICE DISPENSER	(CREW MESS)	
P104-15	REFRIGERATOR	(CREW LOUNGE/TV ROOM)	
P105, GFI POWER PANEL COMMISSARY, OFFICERS MESS (A-45-2)			
P105-01	DRINK FOUNTAIN	(OFFICER MESS)	
P105-02	4-SLICE TOASTER		
P105-03	REFRIGERATOR	(OFFICER LOUNGE/TV ROOM)	
P105-04	COFFEE MAKER		
P105-05	REFRIGERATOR/FREEZER WITH ICEMAKER		
P105-06	COFFEE MAKER	(OFFICER LOUNGE/TV ROOM)	
P105-07	MICROWAVE		
P105-08	DRINK FOUNTAIN	(BALLAST CONTROL ROOM & PASSAGEWAY A-54-0)	
P105-09	DRINKING FOUNTAINS	(PASSAGEWAYS 1-54-0 & B-54-0)	
P105-10	DRINK FOUNTAIN	(PASSAGEWAYS C-54-0 & D-54-0)	
P105-11	DRINK FOUNTAIN	(WHEELHOUSE E-57-0)	
P105-12	JUICE DISPENSER	(OFFICER MESS)	
P106, HVAC REHEATERS “B” DECK, ELECTRICAL EQUIPMENT ROOM (1-56-2)			
P106-01 - P106-06	VARIOUS HEATER ON “B” DECK FRAME 46 - 56		
P107, HVAC REHEATERS “A” DECK, ELECTRICAL EQUIPMENT ROOM (1-56-2)			
P107-01 - P107-07	VARIOUS HEATER ON “A” DECK FRAME 45 - 56		
P108, HVAC REHEATERS “D” & NAV BRIDGE DECK, ELECTRICAL EQUIPMENT ROOM (D-50-0)			
P108-01 - P108-06	VARIOUS HEATER ON “D”		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
	DECK FRAME 50 - 56		
P109, HVAC REHEATERS "C" DECK, ELECTRICAL EQUIPMENT ROOM (D-50-0)			
P109-01 - P109-08	VARIOUS HEATER ON "C" DECK FRAME 49 - 56		
P110, HVAC REHEATERS "UPPER" DECK, ELECTRICAL EQUIPMENT ROOM (1-56-2)			
P110-01 - P110-09	VARIOUS HEATER ON "MAIN" DECK FRAME 42 - 56		
P111, FORECASTLE, BOSUNS STOREROOM (1-126-0)			
P111-01	AFFF TRANSFER PUMP P/P	6-108-0	
P111-02	TRANSFORMER HEATER/SPACE HEATER/HOOD LIGHT		
P111-03	SOUND POWER TELEPHONE	1-126-0	
P111-04	SOLENOID VALVE CABINET	6-108-0	
P111-05	BT BFA CONTROLLER FOR SALTWATER PUMP	6-108-0	
P111-06	UHF RADIO		
P111-07	CCTV		
P111-14	RESERVED FOR GOVERNMENT USE		
P111-15	BOW THRUSTER CONVERTER	6-108-0	
P113, AUXILIARY MACHINERY POWER PANEL, MACHINERY ROOM 1 (5-56-1)			
P113-01	SOLENOID VALVE PANEL	MACHINERY ROOM 1	
P113-02			

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
EP401, EMERGENCY POWER PANEL MOV, MACHINERY ROOM 1 (5-56-1)			
EP401-01 - EP401-13	VARIOUS MOVS IN MACHINERY ROOM 1		
EP402, EMERGENCY POWER PANEL MOV, MACHINERY ROOM 2 (5-56-2)			
EP402-01 - EP402-14	VARIOUS MOVS IN MACHINERY ROOM 2		
EP403, EMERGENCY POWER PANEL MOV, ELECTRICAL EQUIPMENT ROOM (1-56-2)			
EP403-01 - EP403-03	VARIOUS WTC MOVS		
EP404, EMERGENCY POWER PANEL ELECTRICAL WINDOW HEATERS WHEELHOUSE, ELECTRICALEQUIPMENT ROOM (D-50-0)			
EP404-01 - EP404-11	VARIOUS WHEELHOUSE WINDOW HEATERS		
EP405, MOV, BALLAST PUMP ROOM (5-57-0)			
EP405-01 - EP405-14	VARIOUS MOVS IN THE BALLAST PUMP ROOM		
EP406, MOV, SERVICE TRUNK AFT (5-97-2)			
EP406-01 - EP406-10	VARIOUS MOVS		
EP407, MOV, SERVICE TRUNK MID (5-97-2)			
EP407-01 - EP407-12	VARIOUS MOVS		
EP408, MOV, SERVICE TRUNK FWD (5-97-2)			
EP408-01 - EP408-11	VARIOUS MOVS		
EP408-12	TE3, EL106 LIGHTING PANEL FORECASTLE	AUXILIARY MACHINERY ROOM (6-108-0)	
EP101, AUXILIARY MACHINERY EMERGENCY POWER PANEL, MACHINERY ROOM 1 (5-56-1)			
EP101-01	SOLENOID VALVE RELAY		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
	PANEL (STBD)		
EP101-02	SOUND POWER TELEPHONE		
EP101-03	RUDDER WATER INDICATOR SENSOR (STBD)		
EP101-04	INCINERATOR ROOM FIRE DAMPER		
EP101-05	ALARM COLUMN RELAY PANEL (STBD)		
EP101-06	SPARE		
EP101-07	HV SWITCHBOARD 1 CIRCUIT BREAKER RECHARGE		
EP101-08	HV SWITCHBOARD 1 CIRCUIT BREAKER RECHARGE		
EP101-09	RELAY FOR PUBLIC ADDRESS SYSTEM 1-23-1		
EP101-12	MV SWITCHBOARD 1A CIRCUIT BREAKER RECHARGE		
EP102, AUXILIARY MACHINERY EMERGENCY POWER PANEL, MACHINERY ROOM 2 (5-56-2)			
EP102-01	SOLENOID VALVE RELAY PANEL (PORT)		
EP102-02	SOUND POWER TELEPHONE		
EP102-03	ALARM COLUMN RELAY PANEL (PORT)		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
EP102-04	HV SWITCHBOARD 2 CIRCUIT BREAKER RECHARGE		
EP102-05	HV SWITCHBOARD 2 CIRCUIT BREAKER RECHARGE		
EP102-06	RUDDER WATER INDICATOR SENSOR (PORT)		
EP102-10	MV SWITCHBOARD 2A CIRCUIT BREAKER RECHARGE		
EP103, MISCELLANEOUS DECKHOUSE EMERGENCY POWER PANEL, ELECTRICAL EQUIPMENT ROOM(D-50-0)			
EP103-03	WT SLIDING DOOR SYSTEM		
EP103-04	AUTO TELEPHONE PBX		
EP103-08	ENTERTAINMENT SYSTEM AMPLIFIER		
EP103-10	REFRIGERATION LOCK-IN ALARM PANEL		
EP103-13	TANK LEVEL INDICATOR (TLI) UPS (SERVICE TRUNK)		
EP103-14	TLI UPS		
EP103-18	PA/GEA		

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
MISCELLANEOUS ELECTRICAL EQUIPMENT			
TRANSFORMER	75 KVA, 460/120 VAC	(1-102-0)	
TRANSFORMER	15 KVA, 460/120 VAC	(1-102-0)	
TRANSFORMER	15 KVA, 120/120 VAC (5-56-2)	(3-36-2)	
TRANSFORMER	15 KVA, 120/120 VAC (D-50-0)	(D-49-0)	
TRANSFORMER	15 KVA, 120/120 VAC (1-53-2)	(A-52-3)	
TRANSFORMER	15 KVA, 120/120 VAC (A1-96-0)	SUPPORT SERVICE ROOM (A1-96-0)	
AC171	AIR HANDLING UNIT CONTROLLER	Fan Room A2 Level	
AC270/ER270	AIR HANDLING UNIT CONTROLLER	Fan Room A2 Level	
AC260/ER260	AIR HANDLING UNIT CONTROLLER	Fan Room A3 Level	
AC160/ER160	AIR HANDLING UNIT CONTROLLER	Fan Room A3 Level	
AC240/ER240	AIR HANDLING UNIT CONTROLLER	Fan Room A4 Level	
AC250/ER250	AIR HANDLING UNIT CONTROLLER	Fan Room A4 Level	
AC140/ER140	AIR HANDLING UNIT CONTROLLER	Fan Room A4 Level	
AC150	AIR HANDLING UNIT CONTROLLER	Fan Room A4 Level	
AC116	AIR HANDLING UNIT CONTROLLER	Fan Room A6 Level	
AC117	AIR HANDLING UNIT CONTROLLER	Fan Room A6 Level	
AC120	AIR HANDLING UNIT CONTROLLER	Fan Room A6 Level	
AC121	AIR HANDLING UNIT CONTROLLER	Fan Room A6 Level	
AC134	AIR HANDLING UNIT CONTROLLER	Fan Room A6 Level	
AC110/ER110	AIR HANDLING UNIT CONTROLLER	Fan Room A7 Level	
FCU110 (FCU111-Stacked)	FAN COIL UNIT, CONTROLLER	Fan Room A8 Level	

ENCLOSURE 2.2.1 – EQUIPMENT LIST

EQUIPMENT/CIRCUIT NUMBER		LOCATION	REPAIR PRIORITY (OBSERVED)
AC136	AIR HANDLING UNIT CONTROLLER	HANGAR STWG	
NO.1 ACCOMMODATION	AIR HANDLING UNIT (AHU) CONTROLLER	1-52-5	
NO.2 ACCOMMODATION	AIR HANDLING UNIT (AHU) CONTROLLER	1-52-5	
NO.3 ACCOMMODATION	AIR HANDLING UNIT (AHU – 1- 45 - 1) CONTROLLER	1-52-5	

USS Hershel Williams		
(ESB 4)		
ELECTRICAL	CONTRACT NO. N68171-26-RN-003	
ITEM NO. 0353A	CATEGORY "A"	2026-01-30
ESB-CCSI-THERMOGRAPHIC SURVEY (1 YR)	Manovich, David	

ENCLOSURE 2.2.2 - QUANTITATIVE INFRARED INSPECTION REPORT

[redacted]

QUANTITATIVE INFRARED THERMOGRAPHIC INSPECTION REPORT (SAMPLE)

ITEM	REPAIR PRIORTY	TEMP O/A °C	ABSOLUTE TEMP
26	IMMEDIATE	*71.29	*99.49

EQUIPMENT

#1 WATER HEATER CONTROLLER 2nd BREAKER FROM LEFT

LOCATION

02 LEVEL WATER HEATER ROOM

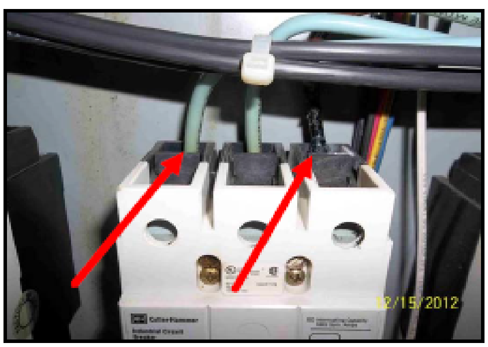
STATEMENT OF ANALYSIS

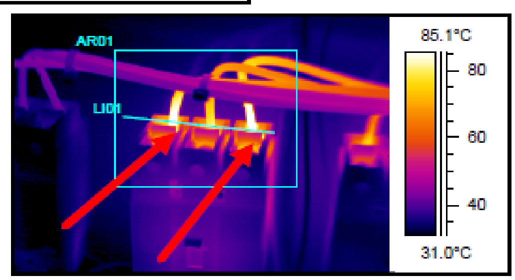
"A" & "C" PHASE TERMINAL CONNECTIONS ON THE TOP OF THE BREAKER ARE EXTREMELY OVERHEATED. MILLIVOLT DROP TEST RESULTS ARE 650MV, 110MV, 650MV

RECOMMENDATION

Replace the breaker. Pull the affected leads,(inspect leads for discoloration, trim back and recrimp if necessary) clean all metal contact points on the affected leads and fasteners until shiny (use a clean green scotchbrite pad if available). Replace the affected leads (securing firmly) and return the equipment to its normal operating condition.

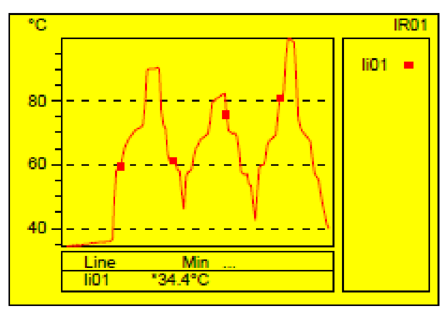
IR information	Value
Date of creation	12/16/2012
Time of creation	1:38:04 AM
Camera type	MIKRON
Object parameter	**Value**
Emissivity	1.00
Object distance	0.0 m
Ambient temperature	20.0°C
Label	**Value**
IR : max	>99.6°C
IR : min	<28.2°C
LI01 : cursor	-
LI01 : max	*99.5°C
LI01 : min	*34.4°C
LI01 : max-min	*65.1°C
AR01 : max	*99.5°C
AR01 : min	*30.9°C
AR01 : max-min	*68.5°C





MAINTENANCE ACTION

TEMPERATURE PROFILE



(SAMPLE)

ENCLOSURE 2.2.3 – THERMOGRAPHIC SURVEY, SUGGESTED ACTIONS BASED ON TEMPERATURE RISE

Temperature difference (ΔT) based upon comparisons between component and ambient air temperatures (O/A)	Recommended Action	Repair Priority (Established by MSC)
0° C – 24° C	Electrical **failure improbable** but corrective action required at next maintenance period	**Desirable (green)**
25° C – 39° C	Component **failure probable** unless corrective action taken	**Important (yellow)**
40° C – 69° C	Component **failure almost certain** unless corrective action taken	**Mandatory (orange)**
>70° C	Component **failure imminent**, repair immediately	**Immediate (red)**

O/A = Over ambient

Source ΔT criteria: Systems Energy Audit Company per direction of N7X.
Note: Temperature specifications vary depending on the exact type of equipment. Even in the same class of equipment (i.e., cables) there are various temperature ratings. Heating is generally related to the square of the current; therefore, the load current will have a major impact on T. In the absence of consensus standards for T, the values in this table will provide reasonable guidelines.

USS Hershel Williams		
(ESB 4)		
COMMUNICATION AND NAVIGATION		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0451	CATEGORY "A"	2026-02-23
ESB-CCSI-ANNUAL RADAR SERVICE		Port Engineer's Name

1.0 ABSTRACT

The purpose of this work item is to inspect and service the ships radar systems.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 Tech Manual TE211-AP-MMC-010, Synapsis Radar with Nautoscan Pedestal

2.1.2 IMO MSC.192 (79), Adoption of the Revised Performance Standards for RADAR Equipment (Available commercially)

2.2 Enclosure

2.2.1 Table 2 of Reference 2.1.2, Minimum Detection Ranges in Clutter-free Conditions

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location

3.1.1 Bridge, Fr. E-57-0

3.1.2 Aft House Top – Deck Level

3.1.3 Forecastle Deck – FR 108 on Forward House

3.2 Description/Quantity

Equipment	Manufacturer	Model Number	Quantity
10CM “S” Band Radar, 12 FT	Raytheon	LPR-A3	1
3CM “X” Band Radar, 8 FT	Raytheon	LPR-A25	1
3CM “X” Band Radar, 8 FT	Raytheon	LPR-A25	1
10CM RADAR UPS	Unknown	Unknown	1
3CM RADAR UPS	Unknown	Unknown	1

3CM RADAR UPS	Unknown	Unknown	1

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

5.3 THE ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

5.4 The radar antenna emits electromagnetic radio frequency (RF) energy which can be harmful, particularly to your eyes. All personnel are to be trained on the risks and safety procedures.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the inspection and servicing of the RADARs using References 2.1.1 through 2.1.3 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship's force to have associated equipment unlocked and/or tagged in per ship's lock out/tag out policy.

7.2.1 Prevent the potential risk of being struck by the rotating antenna while men are aloft. Turn off the radar power switch before servicing the antenna unit. Post a warning sign near the switch indicating men are aloft and it should not be turned on while the antenna is being serviced.

7.3 All work will only be accomplished by experienced and qualified personnel. They are also to be trained in radar and electrical safety to avoid personnel injury. As such the worker must be fully aware of the risks associated with his surroundings and be fully knowledgeable of health and safety standards.

7.4 Temporarily remove access panels for cleaning and inspection. Retain and mark all fasteners for reinstallation.

7.5 Record the running hours of the Antenna motors, Monitor displays and Magnetrons.

7.6 Record the version of software loaded on the systems. Submit CFR to OMT Rep identifying current software version installed and if a newer software version is available.

7.7 Visually inspect the physical condition of the antennas, processors, monitors, control stations, wave guides and wiring for any damage, corrosion, contamination, deterioration or signs of overheating.

7.7.1 Open antenna covers to check terminal strips and plug connections inside. Check for loose connections and contacts and plugs for proper seating. Inspect the rubber gaskets on these covers for deterioration.

7.7.2 Inspect all system components exposed to the elements for verification of watertight integrity.

7.7.3 Inspect the Antenna drive mechanism.

7.7.4 Check processors, monitors and control stations for loose connections and contacts and plugs for proper seating. Verify all grounds straps and cable shielding are secure and in good condition.

7.7.5 Check all foundations, bracing, fittings, wireways, bolts and clamps for condition and tightness.

7.7.6 Check all stuffing tubes and cable transits verifying all penetrations are watertight.

7.8 Clean the antenna radiators of all dirt, salt and debris using a soft cloth, fresh water and mild soap detergent. Inspect for cracks on the radiator surface.

7.9 Thoroughly clean the processors, monitors and control stations with vacuum, brush and lint free rags. Use a soft cloth to clean the monitor screens. If available, use an anti-static spray. Water, isopropyl, alcohol and similar nonabrasive cleaning fluids may also be used to clean the screen. Do not use solvents or chemical-based cleaners. Check for loose connections and contacts and plugs for proper seating. Verify all grounds straps and cable shielding are secure and in good condition.

7.10 Replace the following components:

7.10.1 All fan filters (annual)

7.10.2 All fuses (annual)

7.11 Check the last change out date of UPS batteries, if more than five years or OEM recommended time period, submit a condition report to replace UPS batteries.

7.12 Check and adjust alignment with ship's gyro systems. The accuracy of azimuth alignment of the radar presentation should be within 0.5°.

7.13 Test, inspect and verify proper operation of all features, secondary inputs, warnings and alarms of the radar systems using Manufacturer and IMO requirements as guidance. To include:

7.13.1 Display Mode: True Motion, North Up and Course Up

7.13.2 Independent, Interswitch and Repeater modes

7.13.3 Range Scales and Range Rings

7.13.4 Auto and manual target acquisition

7.13.5 Range and Bearings

7.13.6 Range detection, see Table 2 below

7.13.7 Target Tracking

7.13.8 ARPA

7.13.9 AIS

7.13.10 Closest Point of Approach (CPA) and predicted Time to CPA (TCPA)

7.13.11 Target Data: actual range of target, actual bearing of target, predicted target range at the CPA, predicted time to CPA (TCPA), true course of target, true speed of target.

7.13.12 AIS Target Data: ship’s identification, navigational status, position, range, bearing, COG, SOG, CPA and TCPA.

7.13.13 Operation with SARTs and RADAR beacons

7.13.14 Watch Zones (infringe limits)

7.13.15 Target Trails

7.13.16 Heading Lines, Electronic Bearing Lines (EBLs) and Parallel Index lines (PI)

7.13.17 Routes and Waypoints

7.13.18 Collision avoidance

7.13.19 Operational Alarms

7.13.20 Display of Charts (if applicable)

7.14 Provide a condition report to the OMT REP by individual radar summarizing all as found conditions to include running hours and revision of software and any recommended parts and repairs.

7.15 Mechanically clean, prime and paint all new and disturbed surfaces to match surrounding areas.

7.16 Upon approval, reinstall all access panels and covers using existing fasteners leaving the radars in a ready for service condition. The OMT REP and Chief Mate will inspect the radar antennas prior to reinstalling any covers.

7.17 Upon completion of repairs, adjustments and tests optimize and tune the systems. Demonstrate satisfactory operation of the ship's radar systems to the OMT REP, Captain and Navigator. Complete all final test and adjustments with radar when vessel is on ships power.

7.18 Provide four hours of onboard training to the ship’s Crew.

7.19 Reports

7.19.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.19.2 Submit a report that includes “as released” condition of the radars. Submit one electronic copy in Portable Document Format (PDF). The report should include any documented radar blocked zones and obstructions, total running hours on the system, and the current version of software installed.

7.19.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.20 Painting: None additional

7.21 Manufacturer's Representative

7.21.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the radars identified in section 3.0 ensuring compliance with the manufacturer's standards and performance specifications.

7.21.2 Possible Source

Raytheon Technologies
870 Winter Street
Waltham, MA 02451
(781) 522-3000

Raytheon Missiles and Defense
1151 East Hermans Rd
Tucson, AZ 85756
(520)794-3000

Jeff Farlow
J & L Marine
P.O. Box 9248
Chesapeake, VA 23320
Office Phone: 757-472-6555
Email: jeff_farlow@netzero.net

Mackay World Service
Phone: 888-361-8532 (U.S. Toll Free)
281-478-6245 (International)
Fax: 212-500-0013
service@mackaymarine.com

Raytheon-Anscheutz GmbH
Mitch A. Pawlowski or Melissa Vannocker
Email: Mitchell_A_Pawlowski@raytheon.com mvannocker@raytheon.com
Phone: 858.522.2741 / 858-522-2295

7.21.3 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.22 Preparation of Drawings

7.22.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1

TABLE 2 OF REFERENCE 2.1.2

MINIMUM DETECTION RANGES IN CLUTTER-FREE CONDITIONS

Target Description	Target Feature	Detection Range in NM	
	Height above sea level (Meters)	X-Band (NM)	S-Band (NM)
Shorelines	Rising to 60	20	20
Shorelines	Rising to 6	8	8
Shorelines	Rising to 3	6	6
SOLAS ships (>5,000 gross tonnage)	10	11	11
SOLAS ships (>500 gross tonnage)	5	8	8
Small vessel with radar reflector meeting IMO Performance Standards	4	5	3.7
Navigation buoy with corner reflector	3.5	4.9	3.6
Typical Navigation buoy	3.5	4.6	3
Small vessel of length 10 m with no radar reflector	2	3.4	3

1.0 ABSTRACT

The purpose of this item is to inspect and service the ships ECDIS systems.

2.0 REFERENCES

2.1 Tech Manual TE211-AQ-MMC-010, Synapsis ECDIS Version, E01.00 or Higher (ESB 3-5)

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location

3.1.1 Bridge, E-57-0 (Primary)

3.1.2 CO’s Stateroom, D-56-0 (Secondary)

3.1.3 Master’s Day Room/Office, D-56-1 (Secondary)

3.1.4 CO’s Office, A3-89.5-12 (Secondary)

3.2 Description/Quantity

Equipment	Manufacturer	Model/Type	Quantity
ECDIS, primary	**Raytheon**	**950-012**	**1**
ECDIS, secondary	**Raytheon**	**950-012**	**1**

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

5.3 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the inspection and servicing of the ECDIS system using Reference 2.1 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 All work will only be accomplished by experienced and qualified personnel. They are also to be trained in electrical safety to avoid personnel injury. As such the worker must be fully aware of the risks associated with his surroundings and be fully knowledgeable of health and safety standards. The equipment may contain Electrostatic Sensitive Devices (ESSDs). Take care not to damage these devices by discharge of electrostatic voltages.

7.4 Temporarily remove access panels for cleaning and inspection. Retain and mark all fasteners for reinstallation.

7.5 Record the latest version of software, patches and cartography loaded on the systems. Submit CFR to OMT Rep identifying current software version installed and if a newer software version is available.

7.6 Visually inspect the physical condition of the processors, monitors, control stations and wiring for any damage, corrosion, contamination, deterioration or signs of overheating.

7.6.1 Check processors, monitors and control stations for loose connections and contacts and plugs for proper seating. Verify all grounds straps and cable shielding are secure and in good condition.

7.6.2 Check all foundations, bracing, fittings, wireways, bolts and clamps for condition and tightness.

7.6.3 Check all stuffing tubes and cable transits verifying all penetrations are watertight.

7.7 Clean the GPS Antennas of all dirt, salt and debris using a soft cloth, fresh water and mild soap detergent. Inspect for cracks on the surface.

7.8 Thoroughly clean the processors, monitors and control stations with vacuum, brush and lint free rags. Use a soft cloth to clean the monitor screens. If available, use an anti-static spray. Water, isopropyl, alcohol and similar nonabrasive cleaning fluids may also be used to clean the screen. Do not use solvents or chemical-based cleaners.

7.9 Replace the following components:

7.9.1 All fan filters (annual)

7.9.2 All fuses (annual)

7.9.3 The hard disk in Processor Unit (as recommended by OEM or 10 years Sperry Vision Master)

7.10 Check the last change out date of UPS batteries, if more than 5 years or OEM recommended time period, submit a condition report to replace UPS batteries.

7.11 Verify Gyro input matches Gyrocompass system heading. Adjust ECDIS to match Gyrocompass.

7.12 Test, inspect and verify proper operation of all features, secondary inputs, warnings and alarms of the ECDIS systems using Manufacturer and IMO requirements as guidance. To include:

7.12.1 Display Mode: True Motion, North Up and Course

Up (if applicable):

7.12.2 Multi-Nodes,

7.12.3 Scales and Range,

7.12.4 Range and Bearings,

7.12.5 ARPA and AIS Targets,

7.12.6 Target Tracking,

7.12.7 Closest Point of Approach (CPA) and predicted Time to CPA (TCPA),

7.12.8 Target Data: range, bearing, COG, SOG, CPA and TCPA,

7.12.9 AIS Target Data: ship's identification, heading, range, bearing, COG, SOG, CPA and TCPA,

7.12.10 GPS / DGPS input,

7.12.11 Gyrocompass input,

7.12.12 Sounder input,

7.12.13 Wind Speed and Direction input (if applicable),

7.12.14 Tides,

7.12.15 Watch Zones (infringe limits),

7.12.16 Target Trails,

7.12.17 Heading Lines, Electronic Bearing Lines (EBLs) and Parallel Index lines (PI),

7.12.18 Route Planning,

7.12.19 Route Monitoring,

7.12.20 Waypoints,

7.12.21 Voyage Recording (Logging),

7.12.22 Backup Arrangement,

7.12.23 Operational Alarms,

7.12.24 Emergency source of electrical power, and

7.12.25 Radar overlay (if applicable).

7.13 Provide a condition report to the OMT REP by individual ECDIS summarizing all as found conditions to include revision of software and cartography and any recommended parts and repairs.

7.14 Upon approval, reinstall all access panels and covers using existing fasteners leaving the ECDIS systems in a ready for service condition.

7.15 Upon completion of repairs, adjustments and tests optimize and tune the systems. Demonstrate satisfactory operation of the ship's ECDIS systems to the OMT REP, Captain and Navigator.

7.16 Provide four hours of onboard training to the ship’s Crew.

7.17 Reports

7.17.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.17.2 Submit a report that includes “as released” condition of the ECDIS. Submit one electronic copy in Portable Document Format (PDF).

7.17.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.18 Painting: None additional

7.19 Manufacturer’s Representative

7.19.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the ECDIS identified in section 3.0 ensuring compliance with the manufacturer’s standards and performance specifications.

7.19.2 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only original equipment manufacturer (OEM) authorized technical field service providers or MSC qualified non-OEM technical field service providers and OEM authorized or MSC qualified non-OEM parts will be used to accomplish the requirements of this work item for this ship critical safety item including oversight and guidance on all aspects of equipment as-found condition inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable. The technical service provider is to attend dock and sea trials.

7.19.3 Contractors interested in gaining MSC qualification as a non-OEM parts and/or service provider should visit this website for qualification and submittal requirements: https://www.msc.usff.navy.mil/Business-Opportunities/Contracts/Qualification-for-Items-Critical-to-Shipboard-Safety-on-MSC-Vessels/.

7.19.4 Possible Sources

Jeff Farlow
J & L Marine
P.O. Box 9248
Chesapeake, VA 23320
Office Phone: 757-472-6555
Email: jeff_farlow@netzero.net

Mackay World Service
Phone: 888-361-8532 (U.S. Toll Free)
281-478-6245 (International)
Fax: 212-500-0013
service@mackaymarine.com

Raytheon-Anscheutz GmbH
Mitch A. Pawlowski or Melissa Vannocker
Email: Mitchell_A_Pawlowski@raytheon.com,

mvannocker@raytheon.com
Phone: 858.522.2741/858-522-2295

7.19.5 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.20 Preparation of Drawings

7.20.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to inspect and service the ships Gyrocompass systems.

2.0 REFERENCES

2.1 Tech Manual S9426-AT-MMC-010, Gyro Compass, STD 22, DUAL SYSTEMS, TYPE 110-233 NG002 (MLP-1 Class)

2.2 NAVSEA Drawing No. 438-8468428, BWD, Ship Control Integrated Bridge and Navigation System

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location

3.1.1 Chart Room/Bridge/Bridge Wings (E-57-0)

3.1.2 Electronics Equipment Room (D-50-0)

3.1.3 Steering Gear Room No.1 & No.2 (2-16-1 & 2-16-2)

3.1.4 Engine Control Room No.1 & No.2 (2-56-1 & 2-56-2)

3.2 Description/Quantity

Equipment	Manufacturer	Model Number	Quantity
Gyro, Master	Raytheon	STD 22	2
Gyro, Repeater	Raytheon	STD 22	6
UPS Battery	**Wartsila**	**JOVYSTAR Ocean**	**2**

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to

determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 After a cold power up the Gyro requires a minimum of four hours to settle out before reliable heading data is available. After power down it make take up to one hour for the gyroscopes to stop rotating.

5.3 The permitted ambient temperature for operating the gyrocompass system is -10° to 55°C. Gyros with fluid in the gyrosphere container will start to freeze below 0°C. When the ambient temperature of the gyrocompass location falls below -10°C when in operation or below 0°C when not in operation the gyrosphere container must be removed from the compass housing and stored in a location which will not fall below 0°C. This removal is only to be conducted by authorized service personnel.

5.4 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the inspection and servicing of the Gyrocompass system using References 2.1 as guidance; SOLAS V/12 and 19.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 All work will only be accomplished by trained, experienced and authorized service personnel for the specific system. They are also to be trained in electrical safety to avoid personnel injury. As such the worker must be fully aware of the risks associated with his surroundings and be fully knowledgeable of health and safety standards. The equipment does contain Electrostatic Sensitive Devices (ESSDs). Take care not to damage these devices by discharge of electrostatic voltages.

7.4 Temporarily remove access panels for cleaning and inspection. Retain and mark all fasteners for reinstallation.

7.5 Record the Manufacturer name, Model and Serial number of the systems.

7.6 Visually inspect the physical condition of the equipment/systems listed in section 3.0 and associated wiring for any damage, corrosion, contamination, deterioration or signs of overheating.

7.6.1 Check gyrocompass system for loose connections and contacts and plugs for proper seating. Verify all grounds straps and cable shielding are secure and in good condition. Verify with the ships Chief Engineer that there are no known issues with gyro signal data being received by Repeaters, RADARs, Autopilot, Course Recorder, ODM, AIS, SVDR, NGW, TACAN, INMARSAT, ECDIS, and TVDTS.

7.6.2 Check all foundations, bracing, gimbals, fittings, wireways, bolts and clamps for condition and tightness.

7.6.3 Check all stuffing tubes and cable transits verifying all penetrations are watertight.

7.7 Clean the compass housing and repeaters of all dirt, salt and debris using a soft cloth, fresh water and mild soap detergent. Do not use acetone, solvents or chemical-based cleaners.

7.8 Perform the manufacturers recommended 12-18 month maintenance using Reference 2.1 as guidance. Clean the gyrospheres, inner surface of the container and replace all fluids. Inspect the Centering Pin for wear. Replace the following components:

7.8.1 All fan filters (annual)

7.8.2 All fuses (annual)

7.9 Unless Gyro Compass UPS is listed as NA in Paragraph 3.2:

7.9.1 Verify Make, Model of unit.

7.9.2 Check UPS for proper operation.

7.9.3 Check the last change out date of UPS batteries, note whether batteries have been changed within five years or within OEM recommended time period, whichever is shorter.

7.9.4 Submit report of Gyro Compass UPS finding along with recommendations for repair or battery replacement to OMT REP.

7.10 Check and adjust alignment of ships Gyrocompass systems. The compass is to display the ships true heading to an accuracy of at least 0.5°.

7.10.1 Check for any variable error.

7.10.2 Verify speed and latitude automatic correction inputs are valid.

7.11 Test, inspect and verify proper operation of all features, warnings and alarms of the Gyrocompass systems using Manufacturer and IMO requirements as guidance. To include:

7.11.1 Gyro Failure

7.11.2 Loss of Magnetic Compass data

7.11.3 Variation of Magnetic Compass

7.11.4 Loss of Speed data

7.11.5 Variation of Speed data

7.11.6 Loss of Position data

7.11.7 Heading differential

7.11.8 Failure of Main Power

7.11.9 Failure of Backup Power

7.11.10 Operational Alarms

7.11.11 Emergency source of electrical power

7.11.12 All illuminations function properly

7.11.13 Auto-shift functionality

7.12 Select Gyroscope No.1 and verify all repeaters follow.

7.13 Select Gyroscope No.2 and verify all repeaters follow.

7.14 Check Gyrocompass system repeater outputs for correct output.

7.15 Ensure Binnacle is free in mounts.

7.15.1 Ensure cable(s) from Binnacle is/are free, clean, and flexible.

7.16 Verify synchro outputs from synchro distribution panel.

7.17 Provide a condition report to the OMT REP by individual Gyro summarizing all as found conditions to include revision of software and any recommended parts and repairs.

7.18 Upon approval, reinstall all access panels and covers using existing fasteners leaving the Gyrocompass systems in a ready for service condition.

7.19 Clean all dust and oil deposits from the Master Compasses and the Electronics Control Unit.

7.20 Upon completion of repairs, adjustments and tests optimize and tune the systems. Demonstrate satisfactory operation of the ship's Gyrocompass systems to the OMT REP, Captain and Navigator. Ensure Gyro Speed Correction is turned ON at the conclusion of testing and demonstrations.

7.21 Provide four hours of onboard training to the ship's Crew.

7.22 Reports

7.22.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.22.2 Submit a report that includes "as released" condition of the Gyro. Submit one electronic copy in Portable Document Format (PDF).

7.22.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.23 Painting: None additional

7.24 Manufacturer's Representative

7.24.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the Gyrocompass identified in section 3.0 ensuring compliance with the manufacturer's standards and performance specifications. The Manufacturer's Representative is to ride the ship for sea trials and demonstrate satisfactory operation of the Gyro Systems at sea.

7.24.2 Possible Source

Raytheon Technologies
870 Winter Street
Waltham, MA 02451
(781) 522-3000

J&L Marine
P.O. Box 9248
Chesapeake VA 23320
POC: Jeff Farlow
Phone: 757-472-6555
jeff_farlow@netzero.net

Mackay World Service
Phone: 888-361-8532 (U.S. Toll Free)
281-478-6245 (International)
Fax: 212-500-0013
service@mackaymarine.com

7.24.3 Provide the OMT REP with a signed letter of qualification from the original equipment manufacturer and certificate of training for the Gyro systems technician working on the Gyro systems identified in section 3.0 prior to the start of any work.

7.24.4 During dock trials, verify that all equipment that requires and utilizes a gyro input is operating correctly with the input sent from the gyro. This is to be verified with the ship’s Chief Engineer, Navigator, and Ship’s Communications Officer within the first two hours of dock trials.

7.25 Preparation of Drawings

7.25.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
COMMUNICATION AND NAVIGATION		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0454	CATEGORY "A"	2026-02-03
ESB-CCSI-ANNUAL VDR RECERTIFICATION		Port Engineer's Name

1.0 ABSTRACT

The purpose of this item is to inspect and service the ships Voyage Data Recorder (VDR).

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 Tech Manual TE030-AF-MMC-010, Voyage Data Recorder Danelec Model: DM100-2014 VDR

2.1.2 USCG NVIC 8-01, Approval of Navigation Equipment for Ships

2.2 Enclosures

2.2.1 ABS CHECK SHEET FOR SOLAS SURVEY FOR “VDR” AND “S-VDR” RADIO TECHNICIANS SURVEY; Latest Revision Available at https://ww2.eagle.org/en/rules-and-resources/form.html

2.2.2 Mandatory Bridge Alarms

2.2.3 USCG approved Service Provider; available at https://cgmix.uscg.mil/EQLabs/EQLabsSearch.aspx Number/Series: 165.152 for VDR

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Location: Pilot House

3.1.2 Quantity: One VDR

3.2 Description/Quantity

Equipment	Manufacturer	Model Number	Quantity
VDR	Danelec	DM100-2014	One

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

5.3 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the inspection and servicing of the VDR system using Reference 2.1.1 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 All work will only be accomplished by trained, experienced and authorized service personnel for the specific system. They are also to be trained in electrical safety to avoid personnel injury. As such the worker must be fully aware of the risks associated with his surroundings and be fully knowledgeable of health and safety standards. The equipment does contain Electrostatic Sensitive Devices (ESSDs). Take care not to damage these devices by discharge of electrostatic voltages.

7.4 Temporarily remove access panels for cleaning and inspection. Retain and mark all fasteners for reinstallation.

7.5 Record the Manufacturers name, Model and Serial number of the system.

7.6 Visually inspect the physical condition of the VDR system including Data Acquisition Unit, Protected Data Capsule, Bridge Alarm Unit, Bridge Microphones, Sensor Interface Unit and wiring for any damage, corrosion, contamination, deterioration or signs of overheating.

7.6.1 Check VDR system for loose connections and contacts and plugs for proper seating. Verify all grounds straps and cable shielding are secure and in good condition.

7.6.2 Check all foundations, bracing, fittings, wireways, bolts and clamps for condition and tightness.

7.6.3 Check all stuffing tubes and cable transits verifying all penetrations are watertight.

7.7 Clean the VDR components of all dirt, salt and debris using a soft cloth, fresh water and mild soap detergent. Do not use acetone, solvents or chemical-based cleaners.

7.8 Using SOLAS V, Regulation 18.8 as guidance, conduct an annual performance check and certification of the VDR system. Complete and submit the ABS survey form in Enclosure 2.2.1 to the OMT REP. The test must be conducted by an approved testing or servicing facility to verify the accuracy, duration and recoverability of the recorded data. In addition, tests and inspections are conducted to determine the serviceability of all protective enclosures and devices fitted to aid location. A copy of the certificate of compliance issued by the testing facility, stating the date of compliance and the applicable performance standards, must be retained on board the ship.

7.8.1 Test, inspect and verify proper operation of all features, warnings and alarms of the VDR system in accordance with Manufacturer and IMO requirements. IEC 61996 provides detailed information on the required data sets to be recorded for VDR and S-VDR.

A.861(20) REF	DATA ITEM	SOURCE
5.4.1	Date and time	Preferably external to ship (e.g. GNSS)
5.4.2	Ship’s position	Electronic Positioning system
5.4.3	Speed (through water or over ground)	Ship’s SDME
5.4.4	Heading	Ship’s compass
5.4.5	Bridge Audio	1 or more bridge microphones
5.4.6	Comms. Audio	VHF
5.4.7	Radar / AIS data	Master radar display
5.4.8	Water depth	Echo Sounder
5.4.9	Main alarms	All mandatory alarms on bridge. See Enclosure 2.2.2
5.4.10	Rudder order and response	Steering gear and autopilot
5.4.11	Engine order and response	Telegraphs, controls and thrusters
5.4.12	Hull openings status	All mandatory status information displayed on bridge
5.4.13	Watertight and fire door status	All mandatory status information displayed on bridge
5.4.14	Acceleration and hull stresses	Hull stress and response monitoring equipment where fitted

A.861(20) REF	DATA ITEM	SOURCE
5.4.15	Wind speed and direction	Anemometer when fitted

7.8.2 The inspection is also to ensure batteries, enclosures and location aids are in good condition and operational. This should be undertaken in accordance with the manufacturer's instructions by qualified, trained and experienced personnel. Successful completion of the maintenance routine should be recorded in the ship’s onboard planned maintenance log.

7.8.3 Evaluate the performance of the VDR by analyzing a 12-hour period of data. The 12-hour recording should cover an operational period when most sensors will be exercised. Download the recorded data or exchange of recording medium. Conduct offline analysis of the recorded data by the manufacturer’s certified representative to verify the accuracy, duration and recoverability of the recorded data. A certificate confirming the satisfactory results of this test should be retained onboard.

7.9 Provide a condition report to the OMT REP summarizing all as found conditions to include revision of software and any recommended parts and repairs.

7.10 Upon approval, reinstall all access panels and covers using existing fasteners leaving the VDR system in a ready for service condition.

7.11 Upon completion of repairs, adjustments and tests optimize and tune the system. Demonstrate satisfactory operation of the ship's VDR systems to the OMT REP, Captain and Navigator.

7.12 Provide one hour of onboard training to the ship’s crew.

7.13 Reports

7.13.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.13.2 Submit a report that includes "as released" condition of the VDR. Submit one electronic copy in Portable Document Format (PDF).

7.13.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.14 Painting: None additional

7.15 Manufacturer's Representative

7.15.1 Using reference 2.1.2 as guidance, provide the services of an OEM and USCG approved representative to accomplish all work and testing on the VDR identified in section 3.0 ensuring compliance with the manufacturer's standards and performance specifications. A list of accepted USCG laboratories is available in enclosure 2.2.3 or online at the link below.

https://cgmix.uscg.mil/EQLabs/EQLabsSearch.aspx Number/Series: 165.152 for VDR

7.15.2 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 and ABS Certificate # prior to the start of any work.

7.16 Preparation of Drawings

7.16.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.2 MANDATORY BRIDGE ALARMS

MANDATORY ALARMS ON THE BRIDGE			
IMO Clause	**Function**	**Notes**	**Signal type and quantity**
SOLAS Ch II-1, 29.5.2,30.3	Main and auxiliary steering gears power units.	Failure of power to any steering gear power unit. Operation of devices for above circuit protection overload, loss of phase in three phase system.	
SOLAS Ch II-1, 29.8.4	Main and auxiliary steering gear control system.	Failure of power to supply system.	
SOLAS Ch II-1, 29.1.2.2	Steering gear, low hydraulic fluid level.	Low level of fluid in hydraulic fluid reservoir.	
SOLAS Ch II-1, 31.2.7, 49.5	Propulsion machinery remote control system failure.	For vessels with bridge control of propulsion machinery and manned engine room failure of remote control of propulsion machinery.	
SOLAS Ch II-1, 31.2.9, 49.7	Propulsion machinery low starting air pressure.	For vessels with bridge control of propulsion machinery and manned engine room, low start air pressure but further starting of propulsion machinery possible.	
SOLAS Ch II-1, 52	Automatic shutdown of propulsion machinery.	Vessels with periodically unattended machinery spaces. Shut down of propulsion and other machinery due to serious malfunction.	
SOLAS Ch II-1, 51.3	Fault requiring action by or attention of the OOW.	Vessels with periodically unattended machinery spaces.	
SOLAS Ch II-1, 51.2.2	Alarm system normal power supply failure.	Vessels with periodically unattended machinery spaces. Failure of normal power	

ENCLOSURE 2.2.2 MANDATORY BRIDGE ALARMS

MANDATORY ALARMS ON THE BRIDGE			
IMO Clause	**Function**	**Notes**	**Signal type and quantity**
		supply to alarm plant.	
SOLAS Ch II-1, 15.7.3.1	Watertight door low hydraulic fluid level.	Passenger ships constructed on or after 1st February 1992 with hydraulic power operated sliding watertight doors. Low level of fluid in hydraulic reservoir.	
SOLAS Ch II-1, 15.7.3.1	Watertight door low gas pressure, loss of stored energy	Passenger ships constructed on or after 1st February 1992 with hydraulic power operated sliding watertight doors. Low gas pressure or loss of stored energy in operating hydraulic accumulator	

1.0 ABSTRACT

This work item describes the requirements to inspect and service the ships radio communication equipment.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 Tech Manual SE061-AB-MMC-010 (Rev 1), Global Maritime Distress and Safety System (GMDSS), Sailor 6110

2.1.2 Tech Manual L-T7226-COT-010 (Version 1.0), Satellite Communication System IMARSAT, NGW Network

2.1.3 Tech Manual SE230-AC-MMC-010, W-AIS R4SW Secure Transponder, Model 7000 System

2.1.4 Tech Manual EE174-AD-OMI-010, Defense Advanced GPS Receiver (DAGR) Satellite Signals Navigation Set AN/PSN-13 (V)

2.1.5 Operation Manual EE150-AW-1MC-010, AN/PRC-150(V)(C) Manpack Radio

2.1.6 Tech Manual SE109-AY-MMC-010 (Rev 1), Radiotelex, MF/HF Sailor 6300

2.1.7 Tech Manual SE061-AA-MMC-010 (Rev 1), Survival Radio Sailor SP3520 VHF GMDSS

2.1.8 Tech Manual SE109-AX-MMC-010 (Rev 2), Technical Manual, Radiotelephone, VHF DSC Sailor 6222

2.1.9 Tech Manual TE061-AJ-MMC-010 (Rev 1), NAVTEX Receiver, Model NCR-333

2.1.10 Tech Manual SE175-AF-MMC-010 (Rev 2), Emergency Position Indicating Radio Beacon (EPIRB), (Sailor SE406-II)

2.1.11 Tech Manual SE175-AG-MMC-010 (Rev 1), Search and Rescue Transponder (SART), Type II, 9GHZ

2.1.12 Tech Manual TE130-A5-MMC-010 (Rev 1), SSA, LRIT, Maritime, and Non-SOLAS Sailor 6120/30/40/50

2.2 Enclosures

2.2.1 ABS CHECK SHEET FOR SOLAS SURVEY FOR SLR/GMDSS; Latest Revision Available at https://ww2.eagle.org/en/rules-and-resources/forms.html

2.2.2 ABS CHECK SHEET FOR SOLAS SURVEY FOR “AIS” RADIO TECHNICIANS SURVEY; Latest Revision Available at https://ww2.eagle.org/en/rules-and-resources/forms.html

2.2.3 ABS CHECK SHEET FOR SOLAS SURVEY FOR SSAS; Latest Revision Available at https://ww2.eagle.org/en/rules-and-resources/forms.html

2.2.4 FCC GMDSS Inspection Check Sheet; Latest Revision Available at https://www.fcc.gov/sites/default/files/gmdss-checklist.pdf

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location

3.1.1 Pilot House

3.2 Description/Quantity

3.2.1 One GMDS Suite containing the following equipment:

Equipment	Manufacturer	Model Number	Quantity
GMDSS including two Sailor 6110 Mini-C GMDSS Systems one SS Series 500 GMDSS Console	Thrane & Thrane	Sailor 6000 A3 150W	One

Equipment	Manufacturer	Model Number	Quantity
INMARSAT	Thrane & Thrane	Sailor 61X0	As Installed
AIS	SAAB SCANIA AB	7000 100-575C (ESB3) /R5 Supreme (ESB4 & 5)	One
GPS	SAAB	7000 109-140	Two
VHF	Harris Corp	14304	One
MF/HF Radiotelephone	Thrane & Thrane A/S	Sailor 6301	As Installed
MF/HF Radiotelex	Thrane & Thrane A/S	Sailor 6300	As Installed
NAVTEX	Japan Radio Co. LTD.	NCR-333	One
EPIRB	Sailor	SE406-II	1
SART	Orolia Limited	SAILOR SART II	2
MF/HF DSC Class A One Channel Watch Receiver	Thrane & Thrane	Sailor 6311	1
SSAS	Thrane & Thrane	Sailor 61X0	X
Two-Way VHF Radiotelephone	Thrane & Thrane	Sailor 6222	2
ECDIS	Raytheon	950-012	1
LRIT	Thrane & Thrane	Sailor 61X0	As Installed
Accessory Connection Box	Thrane & Thrane A/S	Sailor 6208 Model TT-6208A Control; Unit Connection Box P/N 406208A	1
Parallel printer, 12/24V	Thrane & Thrane A/S	SAILOR H1252B/TT-3608A, P/N 801250006	2
EDS-205A (CV-C) Moxa Switch 5 Port	Thrane & Thrane A/S	SAILOR 6197; Ethernet Switch; Model TT-6197; P/N	3

Equipment	Manufacturer	Model Number	Quantity
		406197A	
Power Supply and Charger Inclusive Wall Tray 300W/28V DC	Thrane & Thrane A/S	Model TT-6080A; P/N 406080A-THR	3
AC-DC Power Supply 300W/28VDC	Thrane & Thrane A/S	Model TT-6070A; P/N 406080A-THR	1

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Perform this work item in conjunction with work item 451 Annual RADAR Service.

5.3 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

5.4 THE ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the inspection and servicing of the systems using Enclosures 2.2.1 through 2.2.4 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 All work shall only be accomplished by trained, experienced and authorized service personnel for the specific system. They are also to be trained in electrical safety to avoid personnel injury. As such the worker must be fully aware of the risks associated with his surroundings and be fully knowledgeable of health and safety standards. The equipment may contain Electrostatic Sensitive Devices (ESSDs). Take care not to damage these devices by discharge of electrostatic voltages.

7.4 Temporarily remove access panels for cleaning and inspection, where applicable. Retain and mark all fasteners for reinstallation.

7.5 Record the Manufacturer’s Name, Model and Serial number including revision of software of the equipment identified in section 3.0.

7.6 Visually inspect the physical condition of the equipment and systems identified in section 3.0 including associated antennas and wiring for any damage, corrosion, contamination, deterioration or signs of overheating.

7.6.1 Check the systems for loose connections and contacts and plugs for proper seating. Verify all grounds straps and cable shielding are secure and in good condition.

7.6.2 Check all foundations, bracing, fittings, wireways, bolts and clamps for condition and tightness.

7.6.3 Check all stuffing tubes and cable transits verifying all penetrations are watertight.

7.7 Clean the system components of all dirt, salt and debris using a soft cloth, fresh water and mild soap detergent. Do not use acetone, solvents or chemical-based cleaners.

7.8 Using SOLAS IV as guidance, conduct an annual performance check and certification of the Radio and GMDSS systems identified in section 3.0 using Enclosures 2.2.1 through 2.2.4 as guidance. Complete and submit the ABS survey forms in Enclosures 2.2.1 through 2.2.4 to the OMT REP. The tests must be conducted by approved testing or servicing facilities as outlined in paragraph 7.17. In addition, tests and inspections are to be conducted to determine the serviceability of all protective enclosures, hydrostatic releases, antennas and devices fitted to aid location. A copy of the certificate of compliance issued by the testing facilities, stating the date of compliance and the applicable performance standards, must be retained on board the ship.

7.8.1 Test, inspect and verify proper operation of all features, warnings and alarms of the systems using Manufacturer and IMO requirements as guidance.

7.8.2 The inspection is also to ensure emergency sources of power, antennas, batteries, hydrostatic releases, enclosures and location aids are in periodicity, good condition and operational. This should be undertaken using the manufacturer's instructions as guidance by qualified, trained and experienced personnel. Successful completion of the maintenance should be recorded in the ship’s onboard planned maintenance log.

7.8.3 Check that emergency sources of power and batteries have capacity to operate the basic equipment for a minimum of 1 hour per SOLAS IV/3. Check that the chargers are the automatic type and can recharge the reserve batteries within 10 hrs.

7.8.4 Verify the ships identification information has been programmed correctly into the AIS.

7.8.5 Inspect and test the EPIRBs using MSC/Circ.1040 as guidance. Check hydrostatic release and batteries for expiration date.

7.8.6 Inspect and test the LRIT system for proper operation.

7.8.7 For SSAS survey, ensure all email addresses for SSAS are current prior to testing. All required passwords shall be recorded and submitted to the ship’s Master to maintain.

7.9 Provide a condition report to the OMT REP, by individual Communication equipment, summarizing all as found conditions to include running hours and revision of software and any recommended parts and repairs.

7.10 Upon approval, reinstall all access panels and covers using existing fasteners leaving the systems in a ready for service condition.

7.11 Upon completion of repairs, adjustments and tests optimize and tune the system, as applicable. Demonstrate satisfactory operation of the ship's systems to the OMT REP, Captain and Navigator.

7.12 Verify all shipboard and equipment manuals required to be on board with latest revisions.

7.13 Provide four hours of onboard training to the ship’s Crew.

7.14 Provide the Master with new annual Shore Based Maintenance Agreement.

7.15 Reports

7.15.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.15.2 Submit a report that includes “as released” condition of the radio communication equipment recertification. Submit one electronic copy in Portable Document Format (PDF).

7.15.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.16 Painting: None

7.17 Manufacturer’s Representative

7.17.1 Provide the services of a Manufacturer’s

Authorized Technical Representative to perform any and all system tests, adjustments and maintenance to ensure full and complete operation of the systems using IMO and manufacturer's performance specifications as guidance. In addition, the Technical Rep is to be an ABS recognized External Specialist for the specific system.

https://ww2.eagle.org/en/rules-and-resources/recognized-external-specialists.html.

https://www.eagle.org/ABSEaglePrograms/es/es-search.jsp

7.17.2 Provide the OMT REP with a signed letter of qualification from the original equipment manufacturer, certificate of training for the system identified in section 3.0 and ABS Certificate # prior to the start of any work.

7.18 Preparation of Drawings

7.18.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
COMMUNICATION AND NAVIGATION		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0460	**CATEGORY "A"**	**2026-02-03**
ESB-CCSI-ANNUAL INSPECTION OF MANUALLY OPERATED ALARMS		**Port Engineer's Name**

1.0 ABSTRACT

The purpose of this item is to inspect, service and test the ships Manually Operated Alarms.

2.0 REFERENCES

2.1 Drawing No. 433-8468423 BWD, Public Address and General Emergency Alarm

2.2 Drawing No. 432-8499353 Selected record Drawing Interior Communication Block Diagram

2.3 Drawing No. 436-8467973 BWD, Alarm Extension Panel/Alarm Column

2.4 Drawing No. 541-344-8662, GENERAL ALARM & ANNOUNCING SYSTEM TESTS

2.5 Tech Manual T9ESB-AD-MAN-010, Engineers’ Operating Manual, Vol. 1, Mechanical

2.6 Tech Manual TE101-BD-MMC-010, PUBLIC ADDRESS/TALKBACK/GENERAL ALARM INTEGRATED COMMUNICATION SYSTEM

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location: Various throughout the vessel

3.2 Description

3.2.1 Public Address / General Alarm System

a. General Alarm

1. Speakers

2. Contact makers

3. Rotating beacons

b. Public Address speakers

c. Flight Crash Alarm

d. Ship’s Whistle

e. Main Equipment Rack

f. 5MC Equipment Rack

g. UPS

3.2.2 Engineer’s Alarms

3.2.3 Alarm Columns

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE:

4.1 Government Furnished Equipment: None

4.2 Government Furnished Material: None

4.3 Government Furnished Services: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

5.3 THE ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT

CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the inspection and servicing of the Manually Operated Alarm systems identified in section 3.0 using References 2.1 through 2.6 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Coordinate any public address announcements notifying all those onboard of imminent tests.

7.4 All work will only be accomplished by trained, experienced and authorized service personnel for the specific system. They are also to be trained in electrical safety to avoid personnel injury.

7.5 Temporarily remove access panels for cleaning and inspection. Retain and mark all fasteners for reinstallation.

7.6 Using ABS (7-6-2/1.1.11) as guidance and with assistance from ships force, conduct an annual performance check of the Manually Operated Alarm systems to include the Manually Operated Alarm systems for the Public Address-General Alarm Systems (1MC) and the Public Address-General Alarm Systems (5MC) in accordance with Reference 2.4 using References 2.1 through 2.3 as guidance and the Alarm Column System, Engineer Report System, Manned ECR Condition, Unmanned ECR Condition, Engineer Calling Alarm, Selective Call and Extension Alarm System using Reference 2.5 as guidance.

7.6.1 Visually inspect the physical condition of the alarm systems including all contact makers, power sources, klaxons, visual alarms (red-flashing or rotating beacon), speakers, push buttons and wiring for any damage, corrosion, contamination, missing components, deterioration or signs of overheating.

7.6.2 Test, inspect and verify proper operation of all features, warnings and alarms of the alarm systems identified in section 3.0.

7.6.3 Verify the general emergency alarm system is capable of sounding the alarm signal consisting of seven or more short blasts followed by one long blast on the ship's whistle and additionally on an electrically operated klaxon warning system. Inspect and verify proper operation of both the main and emergency sources of electrical power.

7.6.4 The inspection is also to ensure batteries, chargers and enclosures are in good condition and operational. This should be undertaken using the manufacturer's instructions as guidance by qualified, trained and experienced personnel.

7.6.5 Verify the Public Address system can be heard from the navigation bridge throughout spaces where crew members or passengers are normally present, the muster stations, the embarkation stations and the machinery spaces.

7.6.6 Troubleshoot ten loudspeakers and ten general alarm bells. Contractor to replace with GFM where items found to be defective.

7.6.7 For estimating purposes assume 40 hours for troubleshooting are required and time for replacing up to six speakers.

7.7 Provide a condition report to the OMT REP summarizing all as found conditions to include any recommended parts and repairs.

7.8 Upon approval, reinstall all access panels and covers using existing fasteners leaving the systems in a ready for service condition.

7.9 Upon completion of repairs, adjustments and tests demonstrate satisfactory operation of the ship's Manually Operated Alarm systems to the ABS Surveyor, OMT REP and Captain.

7.10 Reports

7.10.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.10.2 Submit a report that includes “as released” condition of the Manually Operated Alarms. Submit one electronic copy in Portable Document Format (PDF).

7.10.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.11 Painting: None additional

7.12 Manufacturer’s Representative: None

7.13 Preparation of Drawings

7.13.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USSHershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0503	**CATEGORY "A"**	**2026-01-30**
ESB-CSI-CARGO FUEL HOSE TESTING (1 YR)		**Manovich, David**

1.0 ABSTRACT

The purpose of this item is to conduct the annual maintenance and testing of cargo fuel hoses.

2.0 REFERENCES

2.1 33 CFR §156.170 (c)(1) and (f)(3), Equipment Tests and Inspections

2.2 33 CFR §154.500, Hose Assemblies

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. FAS Station 1A (two hoses)

b. Two spare hoses, coordinate receipt from Chief Mate or Cargo Mate

3.1.2 Quantity: Total of four hoses, each hose is ten linear feet long and six inches in diameter

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor and all subcontractors regardless of tier are advised to review other work items under this contract. Many of the definitions relating to performance of this work item are found in Work Item 001.

5.3 Ensure hose couplings do not touch deck or metal surface. For MIL-PRF-370 hoses, resistance must not exceed 1.5 ohms per 1 foot of hose length and +10 ohms total.

5.4 Hydrostatic test pressure is 250 psi unless otherwise specified by the hose manufacturer.

5.5 Numerous lengths of hose, coupled together, may be tested simultaneously provided hose can be examined.

5.6 The hose assembly should be supported throughout its length so as to slope upward to eliminate air pockets when vented.

5.7 Replace any hoses, fitting, coupling, nozzle, gasket, strainer or other equipment not meeting the test or inspection requirements with CFM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect and test each hose, hose fitting, hose nozzle, and hose coupling identified in Section 3.0 using References 2.1 and 2.2 as guidance.

NOTE: Replacement of material, including hoses, that fall inspection or testing and subsequent retesting, as required, will be the subject of a Request for Proposal (RFP).

7.2 Contractor is responsible for temporarily removing two hoses installed at fueling at sea rig and temporarily installing a blank flange with gasket while hoses are removed for testing. The two remaining hoses are spares. Coordinate receiving spare hoses with Chief Mate or Cargo Mate.

7.2.1 Inspect each hose section for cuts, or abrasions that expose hose reinforcement threads, bulges, blisters, general swelling, fraying of fabric, and loosening of interior tube.

7.2.2 Clean hose fittings and inspect all fittings and/or coupling for nicks and burrs.

7.2.3 Inspect all hose fitting gaskets.

7.3 During any test or inspection, the entire external surface of the hose must be accessible. Each nonmetallic transfer hose must:

7.3.1 Have no unrepaired loose covers, kinks, bulges, soft spots or any other defect which would permit the discharge of oil or hazardous material through the hose material, and no gouges, cuts or slashes that penetrate the first layer of hose reinforcement.

7.3.2 Have no external deterioration and to the extent internal inspection is possible with both ends of the hose open, no internal deterioration.

7.3.3 Not burst, bulge, leak, or abnormally distort under static liquid pressure at least 1- 1/2 times the maximum allowable working pressure (MAWP).

7.3.4 Hoses not meeting the requirements may be found to be acceptable after a static liquid pressure test is successfully completed in the presence of the OMT REP. The test medium is not required to be water.

7.4 Hydrostatically test each hose listed in section 3.0 to at least 1-1/2 times the maximum allowable working pressure or the pressure required by paragraph 5.4, whichever is greater.

7.4.1 During any test or inspection, the entire external surface of the hose must be accessible.

7.4.2 Each hose must:

a. Have no unrepaired loose covers, kinks, bulges, soft spots or any other defect which would permit the discharge of oil or hazardous material through the hose material, and no gouges, cuts or slashes that penetrate the first layer of hose reinforcement.

b. Have no external deterioration and, to the extent internal inspection is possible with both ends of the hose open, no internal deterioration;

c. Not burst, bulge, leak, or distort.

7.5 Measure Resistance of Fuel Hose

7.5.1 Measure ground circuit resistance from nozzle to deck and record resistance reading in Ohms. Include resistance reading for each hose tested in condition report to OMT REP.

7.6 Drain and remove two hoses from the FAS Station 1A. Temporarily install blanks with gaskets in way of removals. After fabrication of hose assembly with 4 other spare hoses, layout in manner so it is not twisted or kinked. Run chalk line around hose parallel to edge of each hose coupling. Measure distance from chalk line to edge of coupling.

7.7 Tie down the hose assembly securely at least every 10 feet.

7.8 Install the test flange fitted with vent valve at upper end of hose.

7.9 Connect the lower end of hose assembly to suitable pump, capable of producing enough pressure to complete the test.

WARNING: Pressurized hoses can fail violently. Maintain a safe distance from pressurized hoses except when inspecting.

7.10 Fill the hose assembly with water; vent all air from hose assembly at upper end.

7.11 Start the pump and raise the pressure in 25 lbs. increments to 1.5 x MAWP.

7.12 Stop the pump and maintain the test pressure for 5 minutes.

7.13 During the test, inspect hose sections, couplings, and flanges for leaks and other signs of weakness.

7.14 When pressure test and inspection is completed, slowly reduce the pressure to zero by opening vent valve at upper end of hose.

7.15 Remove the test flange.

7.16 Drain hydraulic test fluid from the hose assembly.

7.17 Replace all hoses that bulge or burst. Hydrostatically test all new replacement hose sections.

7.18 Check the distance from chalk line to edge of coupling in 4 places; verify couplings did not slip.

7.19 Replace all delivery hose couplings that leak or slip with new couplings. Hydrostatically test hose with new couplings a second time. If hose fails second test, replace with a new hose and hydrostatically test new section.

7.20 Using hot freshwater (130 degrees to 180 degrees F), flush the hose assembly for 5 to 10 minutes. Do not recirculate flushing fluid.

7.21 Disconnect pump from hose assembly.

7.22 Stencil date of test and psig (1-1/2 times MAWP) on hose.

7.23 Let the hose drain and allow hose to completely dry inside. If hose will be installed immediately on an alongside fuel receiving rig; install hose on rig, leaving lower couplings disconnected until hose is fully drained and hose is dry inside.

7.24 Once dry, rig or place the hose in storage. For hoses that dried in place on an alongside fuel receiving rig, connect lower couplings and ensure rig is connected to riser.

7.25 Reports

7.25.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.25.2 Submit a report that includes “as released” condition of the hoses listed in section 3.0. Submit one electronic copy in Portable Document Format (PDF).

7.25.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.26 Painting

7.26.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.27 Manufacturer’s Representative: None

7.28 Preparation of Drawings: None additional

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0517A	**CATEGORY "A"**	**2026-01-30**
ESB-CSI-LTFW COOLER CLEANING AND INSPECTION (2.5 YR)		**Manovich, David**

1.0 ABSTRACT

This item describes the requirement to clean and inspect the ship's Central Low Temperature Freshwater (LTFW) coolers.

2.0 REFERENCES

2.1 Tech Manual S9532-B3-MMC-010, Heat Exchanger, Plate Type, Installation, Operation, and Maintenance Manual

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Machinery Room 1 (5-56-1)

b. Machinery Room 2 (5-56-2)

3.1.2 Quantity: Two Central Low Temp Fresh Water Coolers (plate type)

3.2 Item Description/Manufacturer's Data

MFR: KPHE Heating and Cooling
MFR DWG: 10M02-MLP-CFW-002
MODEL: GHP3003-FRC7/4
TYPE: PLATE TYPE
DESIGN PRESSURE: 6.9 BAR @ 94 DEG C
NO OF PLATES: 303
PLATE MATERIAL: TITANIUM

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of

this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTR's 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract, including but not limited to Work Items 023 to determine their effect on the work required under this work item. Many of the definitions relating to performance of this work item are found in Work Item 001.

5.3 THIS SHIP CONTAINS HIGH VOLTAGE (HV) ELECTRICAL SYSTEMS. OBEY ALL POSTED AND VERBAL INSTRUCTIONS REGARDING SAFETY AND EXCLUSION FROM HIGH VOLTAGE AREAS. AT NO TIME APPROACH, WORK ON, OR ENTER A HIGH VOLTAGE AREA WITHOUT PROPER AUTHORIZATION.

5.4 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Coordinate with ship's force to have associated equipment locked and/or tagged out per ship's lock out/tag out policy. When the maintenance is complete, coordinate with ship's force to have associated equipment unlocked and/or tagged in per ship's lock out/tag out policy.

7.2 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.3 Open each heat exchanger using Section 2, Paragraph 5.1.2 of Reference 2.1 as guidance.

7.3.1 Only one heat exchanger may be taken apart at a time.

7.4 Identify and record the clamping dimension of each heat exchanger using Section 2, Paragraph 5.3.2 of Reference 2.1 as guidance.

7.5 Provide containment to prevent contamination and/or damage to equipment near or below each heat exchanger. Particular attention shall be made to water spray and drainage caused by pressure cleaning.

7.6 Manually clean each heat exchanger identified in section 3.0 using Section 2, Paragraph 5.2 of Reference 2.1 as guidance.

7.6.1 All loose material shall be removed.

7.6.2 No more than 5% of the area of each plate may have contamination, such as coatings on the plate surface, when cleaning process is complete.

7.6.3 Hard cleaning tools (e.g. brushes with metal bristles) shall not be used as they can damage the metallic surfaces of the heat exchanger plates and the surfaces of the gaskets.

7.6.4 High-pressure water of not more than 3000 psi may be used to clean plates so long as water/steam is not directed at or under a plate gasket.

7.7 Inspect each heat exchanger plate.

7.7.1 Ensure the gaskets are free of foreign matter and contamination.

7.7.2 Ensure the gaskets fit correctly in the grooves of the heat exchanger plates.

7.7.3 Inspect heat exchanger plates for corrosion, damage, foreign matter, and contamination.

7.8 Close each heat exchanger using Section 2, Paragraph 5.3 of Reference 2.1 as guidance.

7.9 Inspection/Test

7.9.1 Complete leak detection and elimination process identified in Section 2, Paragraph 4.4 and Section 3 of Reference 2.1.

7.9.2 Complete an operational test for a minimum of 12-hours using Reference 2.1 start-up procedures identified in section 2. No leakage is allowed after plates and gaskets have reached design working temperatures and pressures.

7.10 Reports

7.10.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.10.2 Submit a report that includes “as released” condition of the Central Low Temperature Freshwater Coolers Submit one electronic copy in Portable Document Format (PDF).

7.10.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.11 Painting/Lagging

7.11.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.11.2 Renew lagging where damaged or disturbed.

7.12 Manufacturer's Representative: None

7.13 Preparation of Drawings: None additional

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0518A	**CATEGORY "A"**	**2026-01-30**
ESB-CSI-HTFW COOLER CLEANING AND INSPECTION (2. 5YR)		**Manovich, David**

1.0 ABSTRACT

This item describes the requirement to clean and inspect the ship's Central High Temperature Freshwater (HTFW) coolers.

2.0 REFERENCES

2.1 Tech Manual S9532-B3-MMC-010, Heat Exchanger, Plate Type, Installation, Operation, and Maintenance Manual

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Machinery Room 1 (5-56-1)

b. Machinery Room 2 (5-56-2)

3.1.2 Quantity: Two High Temperature Fresh Water Coolers (plate type)

3.2 Item Description/Manufacturer's Data

MFR: KPHE Heating & Cooling
MFR DWG: 10M02-MLP-HT-001(002)
MODEL: GHP1102S-FRC10/2
TYPE: PLATE TYPE
DESIGN PRESSURE: 6.9 BAR @ 98 DEG C
NO OF PLATES: 67
PLATE MATERIAL: TITANIUM

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of

this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTR's 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor and all subcontractors, regardless of tier are advised to review other work items under this contract, including but not limited to Work Items 023 to determine their effect on the work required under this work item. Many of the definitions relating to performance of this work item are found in Work Item 001.

5.3 THIS SHIP CONTAINS HIGH VOLTAGE (HV) ELECTRICAL SYSTEMS. OBEY ALL POSTED AND VERBAL INSTRUCTIONS REGARDING SAFETY AND EXCLUSION FROM HIGH VOLTAGE AREAS. AT NO TIME APPROACH, WORK ON, OR ENTER A HIGH VOLTAGE AREA WITHOUT PROPER AUTHORIZATION.

5.4 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Coordinate with ship's force to have associated equipment locked and/or tagged out per ship's lock out/tag out policy. When the maintenance is complete, coordinate with ship's force to have associated equipment unlocked and/or tagged in per ship's lock out/tag out policy.

7.2 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.3 Open each heat exchanger using Section 2, Paragraph 5.1.2 of Reference 2.1 as guidance.

7.3.1 Only one heat exchanger may be taken apart at a time.

7.4 Identify and record the clamping dimension of each heat exchanger using Section 2, Paragraph 5.3.2 of Reference 2.1 as guidance.

7.5 Provide containment to prevent contamination and/or damage to equipment near or below each heat exchanger. Particular attention shall be made to water spray and drainage caused by pressure cleaning.

7.6 Manually clean each heat exchanger identified in Paragraph 3.1.2 using Section 2, Paragraph 5.2 of Reference 2.1 as guidance.

7.6.1 All loose material shall be removed.

7.6.2 No more than 5% of the area of each plate may have contamination, such as coatings on the plate surface, when cleaning process is complete.

7.6.3 Hard cleaning tools (e.g. brushes with metal bristles) shall not be used as they can damage the metallic surfaces of the heat exchanger plates and the surfaces of the gaskets.

7.6.4 High-pressure water of not more than 3000 psi may be used to clean plates so long as water/steam is not directed at or under a plate gasket.

7.7 Inspect each heat exchanger plate.

7.7.1 Ensure the gaskets are free of foreign matter and contamination.

7.7.2 Ensure the gaskets fit correctly in the grooves of the heat exchanger plates.

7.7.3 Inspect heat exchanger plates for corrosion, damage, foreign matter, and contamination.

7.8 Close each heat exchanger using Section 2, Paragraph 5.3 of Reference 2.1 as guidance.

7.9 Inspection/Test

7.9.1 Complete leak detection and elimination process identified in Section 2, Paragraph 4.4 and Section 3 of Reference 2.1.

7.9.2 Complete an operational test for a minimum of 12-hours using Reference 2.1 start-up procedures identified in section 2. No leakage is allowed after plates and gaskets have reached design working temperatures and pressures.

7.10 Reports

7.10.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.10.2 Submit a report that includes “as released” condition of the High Temperature Freshwater Coolers Submit one electronic copy in Portable Document Format (PDF).

7.10.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.11 Painting/Lagging

7.11.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.11.2 Renew lagging where damaged or disturbed.

7.12 Manufacturer's Representative: None

7.13 Preparation of Drawings: None additional

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0527	CATEGORY "A"	2026-01-30
ESB-CSI-EDG COUPLING INSPECTION (1 YR)		Manovich, David

1.0 ABSTRACT

The purpose of this item is to perform the annual inspection of the Emergency Diesel Generator (EDG) Flexible Coupling.

2.0 REFERENCES

2.1 T9233-CM-MMC-010/020, Generator Set, Emergency Diesel Installation, Operation and Maintenance

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Emergency Diesel Generator Room (A-34-2)

3.1.2 Quantity

a. One Emergency Diesel Generator

3.2 Item Description/Manufacturer's Data

3.2.1 Stromag Inc.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 THIS SHIP CONTAINS HIGH VOLTAGE (HV) ELECTRICAL SYSTEMS. OBEY ALL POSTED AND VERBAL INSTRUCTIONS REGARDING SAFETY AND EXCLUSION FROM HIGH VOLTAGE AREAS. AT NO TIME APPROACH, WORK ON, OR ENTER A HIGH VOLTAGE AREA WITHOUT PROPER AUTHORIZATION.

5.3 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.2 Provide the services of a certified Stromag Field Service Technical Representative to inspect the EDG Flexible coupling using Reference 2.1 as guidance.

7.3 Ensure that all fasteners are bagged, tagged, and labeled for reuse. Inspect all fasteners prior to reuse.

7.4 Visual Inspections

7.4.1 Clean all surfaces of the Flexible Coupling prior to performing visual inspections.

7.4.2 Perform Visual Inspection of the rubber elements for any defects such as cracks.

7.4.3 Perform Visual Inspection of all accessible membrane and ring parts.

7.5 Torque Checks

7.5.1 All instruments/tools that require calibration shall have a current calibration label affixed to it.

7.5.2 Check coupling bolt torques to OEM specification and verify no loose bolts.

7.5.3 Perform hydraulic torque checks on slewing ring bolts, indicating 20% of the total bolts.

7.5.4 Ensure segments are installed in the proper orientation and fasteners are installed in the correct location and torque.

7.5.5 Connecting Link Fasteners

a. Validate that the Flexible Coupling Connection link torque meets the OEM specification.

b. Torque shall be witnessed by OMT REP and documented. Provide data to the OMT REP via Condition Report.

7.6 Alignment Check

7.6.1 Perform generator to engine alignment checks with laser alignment tool using Reference 2.1 as guidance.

7.7 Reports

7.7.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.7.2 Submit a report that includes “as released” condition of the Stromag Coupling. Submit one electronic copy in

Portable Document Format (PDF).

7.7.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.8 Painting: None additional

7.9 Manufacturer's Representative

7.9.1 Provide the services of an OEM authorized field service representative to accomplish all work and inspections on the Stromag Couplings identified in section 3.0 ensuring compliance with the manufacturer's standards and performance specifications.

7.9.2 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only original equipment manufacturer (OEM) authorized technical field service providers or MSC qualified non-OEM technical field service providers and OEM authorized or MSC qualified non-OEM parts shall be used to accomplish the requirements of this work item for this ship critical safety item including oversight and guidance on all aspects of equipment as-found condition inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable. The technical service provider is to attend dock trials.

7.9.3 Contractors interested in gaining MSC qualification as a non-OEM parts and/or service provider should visit this website for qualification and submittal requirements: https://www.msc.usff.navy.mil/Business-Opportunities/Contracts/Qualification-for-Items-Critical-to-Shipboard-Safety-on-MSC-Vessels/.

7.9.4 Possible Source

STROMAG REP (SINGAPORE)
TEL: 65 6861 5100
sgvoith@pacific.net.sg
info@stromag.com

7.9.5 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.10 Preparation of Drawings

7.10.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0547	CATEGORY "A"	2026-01-30
ESB-CSI-REPLACE STEERING GEAR LIMIT SWITCHES (2.5 YR)		Manovich, David

1.0 ABSTRACT

The purpose of this item is to have OEM replace the steering gear limit switches for the rudder feedback unit.

2.0 REFERENCES

2.1 NAVSEA Technical Manual T9561-BV-MMC-010, Steering Gear System Mechanical and Hydraulic Volume 1

2.2 NAVSEA Technical Manual T9561-BV-MMC-020, Steering Gear System Mechanical and Hydraulic Volume 2

2.3 NAVSEA Technical Manual T9561-BV-MMC-030, Steering Gear System Mechanical and Hydraulic Volume 3

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Steering Gear Room (3-26-1)

b. Steering Gear Room (3-26-2)

3.1.2 Quantity

a. Two Steering Gears

b. Eight Limit Switches per steering gear

3.2 Item Description/Manufacturer’s Data

3.2.1 Steering Gear

a. Manufacturer: Rolls-Royce

b. Model Number: RV 2700-2

3.2.2 Rubber Feedback Unit AS

a. Manufacturer: Raytheon Anschutz

b. Model Number: Type 101-532

3.3 Contractor Furnished Material (CFM)

3.3.1 Limit Switches B1-B4

a. Quantity: Sixteen

b. Part Number: 101-532.02 E01

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 THIS SHIP CONTAINS HIGH VOLTAGE (HV) ELECTRICAL SYSTEMS. OBEY ALL POSTED AND VERBAL INSTRUCTIONS REGARDING SAFETY AND EXCLUSION FROM HIGH VOLTAGE AREAS. AT NO TIME APPROACH, WORK ON, OR ENTER A HIGH VOLTAGE AREA WITHOUT PROPER AUTHORIZATION.

5.3 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Perform replacement of steering gear limit switches identified in section 3.0 with assistance from the OEM representative using References 2.1 through 2.3 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Remove limit switches and replace with CFM.

7.5 Perform operational test of steering system with OMT REP present.

7.6 Reports

7.6.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.6.2 Submit a report that includes “as released” condition of the steering gear limit switches. Submit one electronic copy in Portable Document Format (PDF).

7.6.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.7 Painting/Lagging

7.7.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.7.2 Renew lagging where damaged or disturbed.

7.8 Manufacturer's Representative

7.8.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the steering gear limit switches identified in section 3.0 ensuring compliance with the manufacturer's standards and performance specifications.

7.8.2 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only original equipment manufacturer (OEM) authorized technical field service providers or MSC qualified non-OEM technical field service providers and OEM authorized or MSC qualified non-OEM parts shall be used to accomplish the requirements of this work item for this ship critical safety item including oversight and guidance on all aspects of equipment as-found condition inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable. The technical service provider is to attend dock and sea trials.

7.8.3 Contractors interested in gaining MSC qualification as a non-OEM parts and/or service provider should visit this website for qualification and submittal requirements: https://www.msc.usff.navy.mil/Business-Opportunities/Contracts/Qualification-for-Items-Critical-to-Shipboard-Safety-on-MSC-Vessels/.

7.8.4 Possible Sources

CS Controls, Inc.
101 Dickson Rd.
Houma, LA 70363
Ph: 985-879-1577

7.8.5 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.9 Preparation of Drawings

7.9.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied.

8.0 GENERAL REQUIREMENTS: None additional

<table>
<tr><td colspan="6">USS Hershel Williams</td></tr>
<tr><td colspan="6">(ESB 4)</td></tr>
<tr><td colspan="3">AUXILIARY MACHINERY</td><td colspan="3">CONTRACT NO. N68171-26-RN-003</td></tr>
<tr><td colspan="2">ITEM NO. 0554</td><td colspan="2">CATEGORY "A"</td><td colspan="2">2026-01-30</td></tr>
<tr><td colspan="3">ESB-CCSI-INSPECT TEST AND CERTIFY MACHINERY SPACE CRANES AND HOISTS (1 YR)</td><td colspan="3">Manovich, David</td></tr>
</table>

1.0 ABSTRACT

The purpose of this item is to inspect, test, and certify machinery space cranes and hoists.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 Tech Manual TG812-AL-MMC-010, Engine Room Crane (Model No. 3912)

2.1.2 Tech Manual T00000921-MMC-010 CRANE, JIB, 5MT PUMP ROOM ACCESS SUPPORT SYSTEMS (With DEMAG DC-PRO 25 CHAIN HOIST)

2.1.3 ASME B30.2: Overhead and Gantry Cranes (Available Commercially)

2.1.4 ASME B30.11: Monorails and Underhung Cranes (Available Commercially)

2.1.5 ASME B30.16: Overhead Underhung and Stationary Hoists (Available Commercially)

2.2 Enclosures

2.2.1 List of Machinery Space Cranes and Hoists

2.2.2 Inspection, Test, and Certification Form

2.2.3 Equipment Definitions

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Machinery Room 2 (5-56-2), Port Side, 20.874 M Flat, Frame 35 Port

b. Machinery Room 1 (5-56-1), Stbd Side, 20.874 M Flat, Frame 35 Stbd

c. Weather Deck (A-56)

3.1.2 Quantity

a. Three Cranes

3.2 Item Description/Manufacturer’s Data

3.2.1 3.5 Metric Ton Crane, Bridge Type, Engine Room CS-Controls, Inc. Model #3912

3.2.2 Manufacturer
HA EAN Machinery Ind. Co. LTD Korea

3.2.3 CRANE, JIB, 5MT PUMP ROOM ACCESS SUPPORT SYSTEMS (With DEMAG DC-PRO 25 CHAIN HOIST)

3.2.4 Manufacturer: DEMAG CRANES & COMPONENTS GmbH, Germany

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Monorail hoists/engine room cranes not listed in the vessel's ABS Register of Lifting Appliances will be maintained and tested to ASME standards.

5.3 The replacement of load chain or wire rope does not necessitate a load test. However, an operational test of the hoist should be made in accordance with ASME prior to putting the hoist back in service.

5.4 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, test, and certify machinery space cranes and hoists identified in section 3.0 using References 2.1.1 through 2.1.5 as guidance. Accomplish all tasks in accordance with IMO, SOLAS, and the Manufacturer's requirements.

7.2 Coordinate with ship's force to have associated equipment locked and/or tagged out per ship's lock out/tag out policy. When the maintenance is complete, coordinate with ship's force to have associated equipment unlocked and/or tagged in per ship's lock out/tag out policy.

7.3 Conduct an annual inspection of the machinery space cranes and hoists, to meet the manufacturer's design, installation, maintenance instructions and service bulletin requirements. Coordinate the inspection with the Chief Engineer to ensure the vessel is always aware of system status.

7.4 The inspection will determine whether conditions found constitute a hazard, and whether repairs or disassembly are required for additional inspection. Inspect the following items:

7.4.1 Operating mechanisms for proper functionality, proper adjustment, and unusual sounds

7.4.2 Crane structure including bridge, runway rails, trolley, hoist, load-carrying flange of all track systems, hangers and sway braces for deformation, excessive wear, corrosion, damage or fractures

7.4.3 Loose or missing fasteners, bolts, nuts, pins or rivets

7.4.4 Cracked or worn sheaves, drums, drive/load/idler chain sprockets, and excessive drive chain stretch

7.4.5 Cracked, worn or distorted parts such as load blocks, suspension housing, chain attachments, clevises, yokes, suspension bolts, pins, bearings, wheels, shafts, gears, rollers, locking and clamping devices, bumpers, end stops, switch baffles, and interlock bolts

7.4.6 Hooks and hook latches for deformation, excessive wear, or fractures. Accomplish nondestructive testing of hooks and inspect for fractures.

7.4.7 Hook attachment and securing means

7.4.8 Excessive wear of drive tires, trolley guide, and drive wheels

7.4.9 Leakage in pneumatic or hydraulic systems (if applicable)

7.4.10 Wire rope for proper reeving through sheaves and spooling onto drums

7.4.11 Brake system parts for evidence of excessive wear, glazed or oil-contaminated friction disks; worn pawls, cams, or ratchets; and corroded, stretched, or broken pawl springs

7.4.12 Excessive wear of drive chain sprockets and excessive drive chain stretch

7.4.13 Deterioration of controllers, master switches, contacts, limit switches, and push button stations

7.4.14 Travel limit devices for proper operation, including the motion limiting devices that interrupt power or cause audible or visual warnings/alarms to be activated. Each motion must be executed at low speed into the limit device with no load on the hoist

7.4.15 Missing guards, covers, stops or bumpers

7.4.16 Wire rope including end attachments. Identify any gross damage that may become a hazard, such as:

a. Distortion of the wire rope, such as: kinking, crushing, un-stranding, bird-caging, main strand displacement or core protrusion

b. General corrosion

c. Broken or cut strands

d. Quantity, distribution, and type of visible broken or damaged wires

7.4.17 Welded link chain including end attachments. Identify any gross damage that may become a hazard, such as:

a. Gouges, nicks, weld spatter, corrosion, and distorted links

b. Wear or stretch

7.4.18 Roller chain including end attachments.

Identify any gross damage that may become a hazard, such as:

a. Elongation

b. Twist

c. Sidebow

7.4.19 If chain or wire rope damage is discovered, the chain/rope must either be removed from service or inspected as detailed in Reference 2.1.1 and 2.1.3. This additional inspection or replacement will be covered by a change order if deemed necessary by the OMT REP.

7.4.20 Inspect all function, instruction, caution, and warning labels, markings, or placards for legibility.

7.4.21 Immediately report any adverse conditions to the OMT REP. The report is to include photographs for record-keeping purposes.

7.5 Conduct maintenance on all machinery space cranes and hoists, to meet the manufacturer's design, installation, maintenance instructions, and service bulletin requirements. The maintenance includes/verifies:

7.5.1 Clean and lubricate all moving parts, where specified in Reference 2.1.5.

7.6 Conduct testing of the machinery space cranes and hoists meeting the manufacturer's design, installation, maintenance instructions and service bulletin requirements. Coordinate testing with OMT REP. Install barricades and warning signs to prevent access to areas of active testing. Erect and maintain control lines, warning lines, railings or similar barriers to mark the boundaries of the hazard areas. The testing includes/verifies:

7.6.1 Conduct a functional test, under no load, evaluating the performance of individual machinery space cranes and hoists. The test includes the following functions:

a. Lifting and Lowering

b. Trolley Travel

c. Bridge Travel

d. Hoist Limit Devices

e. Travel Limiting Devices

f. Locking and Indicating Devices

7.7 Testing is to be coordinate with the OMT REP and ABS to allow observation if deemed necessary.

7.8 Maintain workspaces in a clean condition during and upon completion of the work requirements. Prevent contamination of any adjacent equipment and spaces.

7.9 Upon completion of all inspections, tests and repairs, return the machinery space cranes and hoists to a ready for service condition.

7.10 With the assistance of ship’s force, tag-in all required systems and equipment.

7.11 Reports

7.11.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.11.2 Submit a report that includes “as released” condition of the machinery space cranes and hoists. Submit one electronic copy in Portable Document Format (PDF).

7.11.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.11.4 Prepare and submit an Inspection, Test and Certification Form (Enclosure 2.2.2) detailing all cranes and hoists examined, tested and certified. All certificates are to be signed by the attending manufacturer's representative (or authorized technical representative) that performed the inspections and testing.

7.12 Painting

7.12.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.13 Manufacturer's Representative

7.13.1 Provide the services of a Contractor recognized as a crane certifying authority to inspect, test and certify the machinery space cranes and hoists using Reference 2.1.1 through 2.1.5 as guidance. Qualifications and certificates for the company and field service rep are to be submitted to the OMT REP for review and approval prior to starting any work. Suggested source:

CS Controls Inc.
101 Dickson Rd, Houma LA 70363
Phone:(958)876-6040
Fax:(985)876-0751
Email: Info@cscontrols.com

7.14 Preparation of Drawings

7.14.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the

OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0554	CATEGORY "A"	2026-01-30
ESB-CCSI-INSPECT TEST AND CERTIFY MACHINERY SPACE CRANES AND HOISTS (1 YR)		Manovich, David

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ENCLOSURE 2.2.1 LIST OF MACHINERY SPACE CRANES AND HOISTS (1 YR)

No.	Description	Location	Mfg.	Model No.	Serial No.	Capacity SFWL (Metric Tons)
1	Port Engine Room Bridge Crane	5-56-2	HA EAN Machinery / CS Controls Inc.	3912		3.5
2	Starboard Engine Room Bridge Crane	2-56-1	HA EAN Machinery / CS Controls Inc.	3912		3.5
3	JIB Crane Pump Room Access	A-56-WEA	DEMAG	DC PRO 25-5000	94817890	5

ENCLOSURE 2.2.2 INSPECTION, TEST AND CERTIFICATION FORM

Equipment Identification:	
Location:	
Mfg.:	
Serial No:	
Safe Working Load (SFWL):	
Hoist Type (chain/wire):	

	Periodic Inspection (Annual – Normal Service)	
Inspect	Acceptable (✓)	Discrepancy (✗)
Operating mechanisms for proper operation, adjustment and unusual sounds		
Structural members for deformation, cracks, or excessive wear on any part of the crane		
For loose or missing fasteners, bolts, nuts, pins or rivets		
For cracked or worn sheaves, drums, sprockets or chain stretch		
For cracked, worn or distorted parts		
Load hooks, latches, support clevises and pins for deformation, cracks, or excessive wear		
Load hook attachment and securing means		
For excessive wear of drive tires, trolley guide and drive wheels		
For leakage in pneumatic or hydraulic systems (if applicable)		
wire rope for proper reeving and spooling		
Condition of Brake system including friction disks/pads, pawls, cams or ratchets		
Wear of drive chain sprockets and drive chain stretch		
Controllers, switches, contacts, limit switches, pushbutton stations, lights, beacons, gongs, sirens, horns for damage, excessive wear, maladjustment or adverse condition on any part		
Travel limit devices for proper performance.		
For missing guards, covers, stops or bumpers		
Wire rope for damage that may be a hazard		

Load chain for damage that may be a hazard		
Missing or illegible function, instruction, caution and warnings, markings or placards		

ENCLOSURE 2.2.2 INSPECTION, TEST AND CERTIFICATION FORM

Test	Functional Test (Annual – Normal Service)	
	Acceptable (✓)	Discrepancy (✗)
Lifting and lowering		
Trolley travel		
Bridge travel		
Hoist limit devices		
Travel limiting devices		
Locking and indicating devices		

Deficiencies Noted (✗): ______________________________

____
____
____
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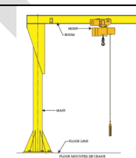
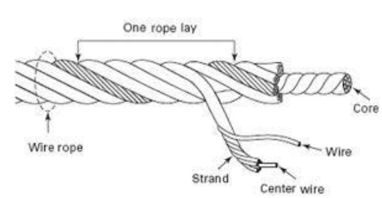
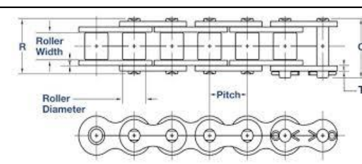
Name of Certifying Company ______________________________

Name of Person Certifying ______________________________

Date ______________________________

ENCLOSURE 2.2.3 EQUIPMENT DEFINITIONS

Overhead Crane - An overhead crane, commonly called a Bridge Crane or Traveling Bridge Crane, is a type of crane found in industrial environments. An overhead crane runs on an elevated runway system consisting of parallel rails with a traveling beam (bridge) spanning the gap. A hoist travels along the bridge and provides three axes of hook motion.	
Gantry Crane - Is a type of overhead crane that is similar to a bridge crane, but instead of moving on suspended runways, the crane uses legs to support the bridge, trolley, and hoist. These legs travel on tires or on rails that are embedded in the floor or ground structure.	
Jib Crane - A type of crane, which has a horizontal member (known as jib or boom) that supports a moveable hoist fixed to a wall or to a floor-mounted pillar is known as jib crane. The jib may swing through an arc, giving additional lateral movement.	
Monorail Crane - This type of overhead lifting system is specialized where the hoist and trolley run along a single stationary beam. They are very effective where materials are to be moved from one fixed point to another fixed location. However, only two directions of hook travel are possible: up/down and along the axis of the monorail beam.	
Hoists - A machinery unit that is used for lifting or lowering a freely suspended (unguided) load. There are two main types; **Chain hoists** and **Wire rope hoists**.	 
Wire Lay - a rope lay is the length along the rope in which one strand makes a complete revolution around the rope.	
Chain - Roller chain is a power chain consisting of parallel pairs of flat links joined by pins covered with rollers and engaging with the teeth of sprockets.	

Chain - Welded Link chain is a series of links or rings connected to or fitted into one another.	

1.0 ABSTRACT

This work item describes the requirements to conduct the Annual inspection, test and certification of the ships mission equipment deck cranes.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 ABS Guide for Certification of Lifting Appliances, (Latest Edition) (Available Commercially)

2.1.2 International Standard (ISO) 4309, Cranes - Wire Rope - Care and maintenance, inspection and discard (Latest Edition) (Available Commercially)

2.1.3 Tech Manual TG811-BY-MMC-010, Crane, Boom, Knuckle, Boat Handling: Model KEB450-52.5-42.5

2.1.4 Tech Manual TG811-BU-MMC-010, Crane, Boom, Extend, Knuckle, Electro-Hydraulic: Model KEB450-89-49

2.2 Enclosure

2.2.1 Deck Cranes

2.2.2 Wire Rope Inspection Record

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location

3.1.1 Mission Deck Boat Handling Crane (A1-73-0)

3.1.2 Aft Knuckle Boom Crane (1-55-2)

3.2 Description

3.2.1 See Enclosure 2.2.1

3.2.2 Crane Age: 09 Years

3.3 Quantity: See Enclosure 2.2.1

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE/INFORMATION

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

Item	Part Number	NSN/TNICN	Description	Qty[1]	UOM[2]
1.	Mobilgear 600 XP 150	XXXX	Mission Deck Boat Handling Crane - Swing Drive (x2)	11.4	liter
2.	Mobilgear 600 XP 150	XXXX	Mission Deck Boat Handling Crane - Winch Gearbox	4	liter
3.	Mobil DTE 10 Excel 46	XXXX	Mission Deck Boat Handling Crane - Hydraulic Tank	2080	liter
4.	Mobilgear 600 XP 150	XXXX	Aft Knuckle Boom Crane - Swing Drive (x2)	11.4	liter
5.	Mobilgear 600 XP 150	XXXX	Aft Knuckle Boom Crane - Winch Gearbox	4	liter
6.	Mobil DTE 10 Excel 46	XXXX	Aft Knuckle Boom Crane - Hydraulic Tank	2080	liter

1 Qty = Quantity
2 UOM = Unit of measure

4.3 Government Furnished Service (GFS): None

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Definitions

5.2.1 Offshore Cranes: a lifting appliance mounted on a vessel, used for lifting and moving cargo, equipment, supplies and other loads under the environmental conditions while the vessel is at open sea and/or when there may be motion relative to the other vessel during crane operations.

5.2.2 Shipboard Cranes: lifting appliances mounted on surface-type vessels, used for lifting and moving loads of less than 160 tf (1570 kN, 352800 lbf) such as cargo, containers, equipment and other loads; or for handling hoses, while the vessel is within a harbor or sheltered area under mild environmental conditions.

5.3 Per References 2.1.1, 2-7/9 and the bearing manufacturer’s instructions, Rocking Tests are to be taken every six months. The results of these tests are to be recorded in the Register of Lifting Appliances for review by the attending ABS surveyor at each annual survey.

5.4 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the thorough examination, maintenance and testing of the ships deck cranes using ABS and USCG requirements as guidance. Include staging and/or man-lift for the mission deck and aft knuckle boom crane(s).

7.2 With the assistance of ships force, conduct an operational test prior to the start of work to confirm operability of ship’s equipment. Submit condition found report to OMT REP.

7.3 With assistance from the Chief Engineer tag out the systems for inspection to avoid damage or injury during this work item.

7.4 Coordinate the inspection and testing of equipment with the Chief Engineer to ensure the vessel is always aware of system status until the work is completed.

7.5 Conduct the Annual inspections, maintenance and testing of the ships deck cranes shown Enclosures 2.2.1 using References 2.1.1 through 2.1.5 as guidance.

7.5.1 Conduct an inspection of the equipment listed in Enclosure 2.2.1. The examination shall include:

a. Visual inspection of the entire crane structure and support structures to include boom rest for deformation, excessive wear, corrosion, damage or fractures, as necessary. The boom is to be lowered for this examination.

b. Visual inspection of crane hooks for deformation, excessive wear or fractures.

c. The slewing ring is to be examined for slack bolts, damaged bearings and deformation or fractured weldments (where applicable).

d. Visual external examination and operational test of crane machinery including hydraulics, prime mover, clutches, brakes, clutches, sheaves; hoisting, slewing and luffing machinery. Check all oil levels and for leaks.

e. Visual inspection of sheaves, rope guides and wire rope including end attachments. Conduct a 'Periodic Inspection' of the wire rope in accordance with Reference 2.1.2 along its entire length. Particular attention shall be given to:

1. The sheaves by checking for wear and corrugation (the rope imprint in the groove surface) with a groove gauge, broken or chipped flanges, cracks in the hub, freedom of rotation without drag and bearing wear,

2. The wire in way of drum anchorage, rope terminations, sections that travel thru sheaves/safe load indicators/hook block/spooling device, sections that spool on the drum, sections subjected to abrasion from external features

(ie; hatch coamings) and any part of the rope exposed to heat,

f. The wire rope may be examined by visual means and/or measurement or by the Magnetic Rope Test (MRT) method. Wire rope inspection and discard requirements are to be in accordance with Section 6 of Reference 2.1.2 (Discard Criteria). The wire rope is not to be used if it meets the Wire Rope Rejection Criteria for any failure modes outlined in Reference 2.1.2:

1. Visible broken wires
2. Decrease in rope diameter
3. Fracture of strands
4. Corrosion
5. Deformation
6. Damage

g. Visual inspection of the Operators station, controls, windows, and communications.

h. Where the crane is approved for varying capacities, verify that a crane capacity rating chart indicating the maximum SWL are conspicuously posted near the controls and visible to the operator.

i. Immediately report any adverse conditions noted to the OMT REP. The report is to include photographs for record purposes.

7.5.2 Conduct maintenance on the cranes using the manufacturers design, installation, maintenance instructions and service bulletins as guidance. The maintenance includes/verifies:

a. Adjust, set and test all safety shutdown mechanisms and limit switches.

b. Grease all lubrication points as described in Reference 2.1.3 through 2.1.6. Open and Gas free the pedestals and voids of the mission deck and aft knuckle boom crane(s).

7.5.3 Conduct various tests and examinations of the cranes. Prior to operating any equipment or commencing any work, the Contractor, along with Ship’s Force, shall inspect all equipment and/or rigging to ensure safe operation. The testing shall include/verify:

CAUTION: Install barricades and warning signs to prevent access to areas of active testing. Erect and maintain control lines, warning lines, railings or similar barriers to mark the boundaries of the hazard areas.

a. Conduct a general examination of machinery, piping and electrical equipment

b. Conduct a Functional Test including main and auxiliary load hoisting and lowering, boom raising and lowering, slewing (swinging), safety protective (fail-safe) and limiting devices and load and boom angle or radius indicators. Test all limits and safeties.

c. Conduct a Rocking Test, using the bearing manufacturer’s instructions as guidance. Provide a written report to the OMT REP and ABS Surveyor documenting the results.

7.6 Testing is to be coordinated with the OMT REP, ABS and USCG Inspectors to allow for observation if deemed necessary.

7.7 Record the software versions installed for each crane and include this information in the final service report.

7.8 Clean, prime and paint all new and disturbed surfaces to match the surrounding areas.

7.9 Reports

7.9.1 When examination, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repair and parts required. Submit one electronic copy in Portable Document Format (PDF) to the OMT REP.

7.9.2 Provide a final service report detailing all cranes and hoists examined, tested and certified. Report to include listing the software versions installed for each crane. Submit one electronic copy in Portable Document Format (PDF). All certificates are to be signed by the attending Technical Representative who performed the exam and testing.

[NOTE for ESB Class PPE: the Aft Knuckle Boom Crane final service report with the software version data must be forwarded to the NAVSEA 05D Ship Design Manager.]

7.9.3 The competent person shall prepare and submit a wire rope inspection record, Enclosure 2.2.2, in accordance with Reference 2.1.2 to the OMT REP.

7.10 Painting: None additional

7.11 Manufacturer's Representative

CARGO CRANES ARE CONSIDERED CRITICAL EQUIPMENT UNDER COMSC INSTRUCTION 4490.1C AND QUALIFIED CONTRACTORS ARE REQUIRED.

7.11.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the mission equipment deck cranes identified in section 3.0 ensuring compliance with the manufacturer's standards and performance specifications.

7.11.2 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only original equipment manufacturer (OEM) authorized technical field service providers or MSC qualified non-OEM technical field service providers and OEM authorized or MSC qualified non-OEM parts will be used to accomplish the requirements of this work item for this ship critical safety item including oversight and guidance on all aspects of equipment as-found condition inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable.

7.11.3 Contractors interested in gaining MSC qualification as a non-OEM parts and/or service provider should visit this website for qualification and submittal requirements: https://www.msc.usff.navy.mil/Business-Opportunities/Contracts/Qualification-for-Items-Critical-to-

Shipboard-Safety-on-MSC-Vessels/.

7.11.4 Possible Sources

Appleton Marine, Inc.
3030 E. Pershing Street
Appleton, WI 54911
Phone: 940-738-5432

7.11.5 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.12 Preparation of Drawings

7.12.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

No.	Description	Location	Mfg.	Model No.	Serial No.	Mode (if applicable)	Capacity SFWL (LTons)	Radius (ft)
1	Mission Deck Boat Handling Crane	A1-73-0	Appleton Marine	KEB450-52.5-42.5	20494	Cargo Personnel	11	52.5
2	Aft Knuckle Boom Crane	1-55-2	Appleton Marine	KEB450-89-49	20426	Cargo / OP4	10	88.6

ENCLOSURE 2.2.2 – WIRE ROPE INSPECTION RECORD

VISUAL INSPECTION			
Crane Reference:		**Rope Application:**	

Rope Category Number [3] *(01-31)*			
Brand Name *(if known)*			
Nominal Diameter *(mm)*			
Construction			
	☐ IWRC	☐ FC	☐ WSC

ENCLOSURE 2.2.2 - WIRE ROPE INSPECTION RECORD

<table>
<tr><td colspan="2">Core [1]</td><td colspan="7"></td></tr>
<tr><td colspan="2">Wire Finish [1]</td><td colspan="7">☐ Uncoated ☐ Zinc/Galv</td></tr>
<tr><td rowspan="2">Direction & Type of Lay [1]</td><td>(Right)</td><td colspan="7">☐ sZ ☐ zZ ☐ Z</td></tr>
<tr><td>(Left)</td><td colspan="7">☐ zS ☐ sS ☐ S</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
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<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="2">Permissible number of visible broken outer wires in:</td><td>6d</td><td colspan="2"></td><td>30d</td><td colspan="3"></td></tr>
<tr><td colspan="2"></td><td colspan="7"></td></tr>
</table>

ENCLOSURE 2.2.2 – WIRE ROPE INSPECTION RECORD

Reference diameter *(mm)*:			
Permissible decrease in diameter from reference diameter *(mm)*:			
Date Installed *(mm/dd/yy)*		**Date Discarded** *(mm/dd/yy)*	

Visible Broken Wires				**Diameter**			**Corrosion**	**Damage and/or Deformation**		**Position in rope**	**Overall Assessment**
Number in length of		Severity Rating ²		Measured Diameter (mm)	Actual decrease from reference (mm)	Severity Rating ²	Severity Rating ²	Severity Rating ²	Nature		Combined Severity Rating ²
6d	30d	6d	30d								

ENCLOSURE 2.2.2 - WIRE ROPE INSPECTION RECORD

Other Observations/Comments											
Performance to date											

ENCLOSURE 2.2.2 - WIRE ROPE INSPECTION RECORD

(cycles/hrs/days/months/etc)		
Date of Inspection *(mm/dd/yy)*		
Name of Competent Person		
	(Print)	*(Signature)*
1 Check box ✓		
2 Describe severity rating as: slight or 20%; medium or 40%; high or 60%; very high or 80%; or discard or 100%		
3 The RCN is used to indicate how many broken wires over a length of 30 times the rope's nominal diameter or six times the rope's nominal diameter, respectively, are acceptable.		

1.0 ABSTRACT

This work item describes the requirements to conduct the Annual inspection, test and certification of the ships non-mission equipment deck cranes.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 ABS Guide for Certification of Lifting Appliances, (Latest Edition) (Available Commercially)

2.1.2 International Standard (ISO) 4309, Cranes - Wire Rope - Care and maintenance, inspection and discard (Latest Edition) (Available Commercially)

2.1.3 Tech Manual TG811-BR-MMC-010, Crane, Provisioning, Electric/Hydraulic (Type: HPC 140-0517)

2.2 Enclosure

2.2.1 Deck Cranes

2.2.2 Wire Rope Inspection Record

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location

3.1.1 Port Provision Crane (B-45-2)

3.1.2 Starboard Provision Crane (B-45-1)

3.2 Description

3.2.1 See Enclosure 2.2.1

3.2.2 Crane Age: 09 Years

3.3 Quantity: See Enclosure 2.2.1

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE/INFORMATION

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

Item	Part Number	NSN/TNICN	Description	Qty[1]	UOM[2]
1.	Mobilgear 600 XP 150	N/A	Port Provision Crane – Slewing Gearbox	7	liter
2.	Mobilgear 600 XP 150	N/A	Port Provision Crane – Winch Gearbox	3.3	liter
3.	Mobil DTE 10 Excel 32	N/A	Port Provision Crane – Hydraulic Tank	480	liter
4.	Mobilgear 600 XP 150	N/A	Stbd Provision Crane – Slewing Gearbox	7	liter
5.	Mobilgear 600 XP 150	N/A	Stbd Provision Crane – Winch Gearbox	3.3	liter
6.	Mobil DTE 10 Excel 32	N/A	Stbd Provision Crane – Hydraulic Tank	480	liter

1 Qty = Quantity
2 UOM = Unit of measure

4.3 Government Furnished Service (GFS): None

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Definitions

5.2.1 Offshore Cranes: a lifting appliance mounted on a vessel, used for lifting and moving cargo, equipment, supplies and other loads under the environmental conditions while the vessel is at open sea and/or when there may be motion relative to the other vessel during crane operations.

5.2.2 Shipboard Cranes: lifting appliances mounted on surface-type vessels, used for lifting and moving loads of less than 160 tf (1570 kN, 352800 lbf) such as cargo, containers, equipment and other loads; or for handling hoses, while the vessel is within a harbor or sheltered area under mild environmental conditions.

5.3 Per References 2.1.1, 2-7/9 and the bearing manufacturer’s instructions, Rocking Tests are to be taken every six months. The results of these tests are to be recorded in the Register of Lifting Appliances for review by the attending ABS surveyor at each annual survey.

5.4 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the thorough examination, maintenance and testing of the ships deck cranes using ABS and USCG requirements as guidance. Include staging and/or man-lift for the Port and STBD Provision crane(s).

7.2 With the assistance of ships force, conduct an operational test prior to the start of work to confirm operability of ship’s equipment. Submit condition found report to OMT REP.

7.3 With assistance from the Chief Engineer tag out the systems for inspection to avoid damage or injury during this work item.

7.4 Coordinate the inspection and testing of equipment with the Chief Engineer to ensure the vessel is always aware of system status until the work is completed.

7.5 Conduct the Annual inspections, maintenance and testing of the ships deck cranes shown Enclosures 2.2.1 using References 2.1.1 through 2.1.3 as guidance.

7.5.1 Conduct an inspection of the equipment listed in Enclosure 2.2.1. The examination shall include:

a. Visual inspection of the entire crane structure and support structures to include boom rest for deformation, excessive wear, corrosion, damage or fractures, as necessary. The boom is to be lowered for this examination.

b. Visual inspection of crane hooks for deformation, excessive wear or fractures.

c. The slewing ring is to be examined for slack bolts, damaged bearings and deformation or fractured weldments (where applicable).

d. Visual external examination and operational test of crane machinery including hydraulics, prime mover, clutches, brakes, clutches, sheaves; hoisting, slewing and luffing machinery. Check all oil levels and for leaks.

e. Visual inspection of sheaves, rope guides and wire rope including end attachments. Conduct a ‘Periodic Inspection’ of the wire rope in accordance with Reference 2.1.2 along its entire length. Particular attention shall be given to:

1. The sheaves by checking for wear and corrugation (the rope imprint in the groove surface) with a groove gauge, broken or chipped flanges, cracks in the hub, freedom of rotation without drag and bearing wear,

2. The wire in way of drum anchorage, rope terminations, sections that travel thru sheaves/safe load indicators/hook block/spooling device, sections that spool on the drum, sections subjected to abrasion from external features (ie; hatch coamings) and any part of the rope exposed to heat,

f. The wire rope may be examined by visual means and/or measurement or by the Magnetic Rope Test (MRT) method. Wire rope inspection and discard requirements are to be in accordance with Section 6 of Reference 2.1.2 (Discard Criteria).

The wire rope is not to be used if it meets the Wire Rope Rejection Criteria for any failure modes outlined in Reference 2.1.2:

1. Visible broken wires
2. Decrease in rope diameter
3. Fracture of strands
4. Corrosion
5. Deformation
6. Damage

g. Visual inspection of the Operators station, controls, windows, and communications.

h. Where the crane is approved for varying capacities, verify that a crane capacity rating chart indicating the maximum SWL are conspicuously posted near the controls and visible to the operator.

i. Immediately report any adverse conditions noted to the OMT REP. The report is to include photographs for record purposes.

7.5.2 Conduct maintenance on the cranes using the manufacturers design, installation, maintenance instructions and service bulletins as guidance. The maintenance includes/verifies:

a. Adjust, set and test all safety shutdown mechanisms and limit switches.

7.5.3 Grease all lubrication points as described in Reference 2.1.3. Open and Gas free the pedestals and voids of the Port and STBD Provision crane(s).

7.5.4 Conduct various tests and examinations of the cranes. Prior to operating any equipment or commencing any work, the Contractor, along with Ship’s Force, shall inspect all equipment and/or rigging to ensure safe operation. The testing shall include/verify:

CAUTION: Install barricades and warning signs to prevent access to areas of active testing. Erect and maintain control lines, warning lines, railings or similar barriers to mark the boundaries of the hazard areas.

a. Conduct a general examination of machinery, piping and electrical equipment

b. Conduct a Functional Test including main and auxiliary load hoisting and lowering, boom raising and lowering, slewing (swinging), safety protective (fail-safe) and limiting devices. Test all limits and safeties.

c. Conduct a Rocking Test, using the bearing manufacturer’s instructions as guidance. Provide a written report to the OMT REP and ABS Surveyor documenting the results.

7.6 Testing is to be coordinated with the OMT REP, ABS and USCG Inspectors to allow for observation if deemed necessary.

7.7 Record the software versions installed for each crane and include this information in the final service report.

7.8 Clean, prime and paint all new and disturbed surfaces to match the surrounding areas.

7.9 Reports

7.9.1 When examination, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repair and parts required. Submit one electronic copy in Portable Document Format (PDF) to the OMT REP.

7.9.2 Provide a final service report detailing all cranes and hoists examined, tested and certified. Report to include listing the software versions installed for each crane. Submit one electronic copy in Portable Document Format (PDF). All certificates are to be signed by the attending Technical Representative who performed the exam and testing.

7.9.3 The competent person shall prepare and submit a wire rope inspection record, Enclosure 2.2.2, in accordance with Reference 2.1.2 to the OMT REP.

7.10 Painting: None additional

7.11 Manufacturer's Representative

7.11.1 Provide the services of a Contractor recognized as a crane certifying authority to inspect, repair, weight test and recertify the cranes using References 2.1.1 through 2.1.3 as guidance. Qualifications and certificates for the company and field service rep are to be submitted to the OMT REP for review and approval prior to starting any work.

7.12 Preparation of Drawings

7.12.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 – DECK CRANES

No.	Description	Location	Mfg.	Model No.	Serial No.	Mode (if applicable)	Capacity SFWL (LTons)	Radius (ft)
1	Port Provision Crane	B-45-2	Oriental Precision & Engineering Co.	HPC 140-0517	33590P	N/A	5	55.7
2	Starboard Provision Crane	B-45-1	Oriental Precision & Engineering Co.	HPC 140-0517	33590S	N/A	5	55.7

ENCLOSURE 2.2.2 - WIRE ROPE INSPECTION RECORD

VISUAL INSPECTION			
Crane Reference:		**Rope Application:**	

Rope Category Number [3] *(01-31)*	
Brand Name *(if known)*	
Nominal Diameter *(mm)*	
Construction	
	☐ IWRC ☐ FC ☐ WSC

ENCLOSURE 2.2.2 - WIRE ROPE INSPECTION RECORD

Core [1]							
Wire Finish [1]		☐ Uncoated		☐ Zinc/Galv			
Direction & Type of Lay [1]	*(Right)*	☐ sZ		☐ zZ		☐ Z	
	(Left)	☐ zS		☐ sS		☐ S	
Permissible number of visible broken outer wires in:		6d		30d			

ENCLOSURE 2.2.2 – WIRE ROPE INSPECTION RECORD

Reference diameter *(mm)*:			
Permissible decrease in diameter from reference diameter *(mm)*:			
Date Installed *(mm/dd/yy)*		**Date Discarded** *(mm/dd/yy)*	

Visible Broken Wires				**Diameter**			**Corrosion**	**Damage and/or Deformation**		**Position in rope**	**Overall Assessment**
Number in length of		Severity Rating ²		Measured Diameter	Actual decrease from reference						
6d	30d	6d	30d	(mm)	(mm)	Severity Rating ²	Severity Rating ²	Severity Rating ²	Nature		Combined Severity Rating ²

ENCLOSURE 2.2.2 - WIRE ROPE INSPECTION RECORD

Other Observations/Comments	
Performance to date	

ENCLOSURE 2.2.2 - WIRE ROPE INSPECTION RECORD

(*cycles/hrs/days/months/etc*)		
Date of Inspection (*mm/dd/yy*)		
Name of Competent Person		
	(Print)	*(Signature)*
1 Check box ✓		
2 Describe severity rating as: slight or 20%; medium or 40%; high or 60%; very high or 80%; or discard or 100%		
3 The RCN is used to indicate how many broken wires over a length of 30 times the rope's nominal diameter or six times the rope's nominal diameter, respectively, are acceptable.		

1.0 ABSTRACT

The purpose of this item describes the requirements to inspect, test and calibrate various gauges, thermometers, meters and special tools.

2.0 REFERENCES

2.1 ISO/IEC 17025:2017, General Requirements for the Competence of Testing and Calibration Laboratories (Available Commercially)

2.2 MSC QMS N7.71.4734.101-Q Shipboard Instrumentation Calibration Requirements

2.3 USNS Hershel Williams (ESB 4) Annual SMS Calibration List Mechanical

2.4 USNS Hershel Williams (ESB 4) Annual SMS Calibration List Electrical

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location: Throughout the vessel

3.2 Description: See References 2.3 and 2.4.

3.3 Quantity

3.3.1 1500 Gauges

3.3.2 200 Thermometers

3.3.3 400 Meters

3.3.4 250 Special tools

NOTE: Gauges, thermometers, meters, and special tools to be calibrated are highlighted in References 2.3 and 2.4.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of

tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the thorough examination, testing and calibration of the ship’s gauges, thermometers, meters and special tools using ANSI and ISO requirements as guidance.

7.1.1 Gauges, thermometers, meters and special tools to be examined, tested, and calibrated are highlighted in References 2.3 and 2.4.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.2.1 Ensure that all power to switchboards, power panels, controllers or other equipment that is to be worked on are secured and tagged out.

7.3 Isolate gauges, meters and thermometers designated for calibration. Clear gauge lines of obstructions by blowing clean dry air through the lines.

7.4 Coordinate the inspection, testing and calibration of equipment with the Chief Engineer to ensure the vessel is always aware of system status until the work is completed.

7.5 Examine, test, adjust and calibrate gauges, meters, thermometers and special tools highlighted in References 2.3 and 2.4 to a transfer measurement standard, traceable to the National Institute of Standards and Technology (NIST), using manufacturer's specifications, ANSI and Reference 2.1 as guidance. Utilize standards which have an accuracy level at least four times that of the instrument being calibrated. Demonstrate current calibration of all test equipment prior to any testing.

7.6 All gauges, meters, thermometers and accessories are to be calibrated in place with calibrated test equipment and standards. All gauges, thermometers and meters located out on the weather decks will have their calibration stickers installed on the inside of the glass cover where applicable.

7.7 Affix a tamper proof calibration label/sticker denoting the name of the calibrating facility, the date of calibration and the due date of the next calibration to the face of the unit. Use References 2.2, 2.3, 2.4 and the Table below as guidance for identifying the time period between calibration. Labels are to be placed over calibration adjusting screws where applicable. The sticker shall be affixed in a visible location that does not interfere with reading the instrument, the name of the process the instrument measures, or the units the instrument measures.

Periodicity (year)	**Instrument**
Annual	Test equipment including any device used for periodic assessment of machinery, components, or the environment such as tachometers, torque wrenches, dynameters, flow meters, test gages and electrical test equipment.
Annual	Magazine sprinkler system pressure gages [open loop, fire main pressure, and pneumatically released pilot valve (PRPV)] are to be calibrated every 12 months.

7.8 Calibration Exceptions: The following are to be replaced on an as needed basis and are not to be calibrated using Reference 2.2 as guidance:

7.8.1 Direct reading fluid thermometers,

7.8.2 Control gages on low pressure (0-60 psi) pneumatic regulators,

7.8.3 Pressure gages on pneumatic stations designed for general use,

7.8.4 Manometers, and

7.8.5 FM-200-cylinder liquid level gages.

7.9 Bi-metallic thermometers and fire main gages will have tamper-resistant seals (Calibration Void if Seal Broken) affixed to controls or adjustments that, if moved, will invalidate the calibration. Seals should not be used to cover adjustments or controls, which are part of the normal use and operation of the instrument. This label may also be used to prevent removal and/or interchange of plug-ins, modules, and subassemblies when such removal or interchange will affect the calibration.

7.10 Any instrument failing calibration will have a label affixed indicating the calibration activity, date of calibration attempt, and either the word “Failed” or “Rejected.” This label will differ (size, color, or shape) from labels used for successful calibration.

7.11 Replace with new all seals, packing, gaskets, mounts, defective terminal lugs and fasteners.

7.12 All gauges, thermometers, meters and their disturbed connections will be tested under operational conditions during Dock and Sea Trials. Correct any deficiencies noted.

7.13 Reports

7.13.1 When examination, testing and calibration reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repair and parts required. Submit one copy in Portable Document Format (PDF).

7.13.2 Provide a certificate of calibration for each instrument which will include the ID number, location of the instrument, description of the measuring point of the gage, or functional description of the test instrument, manufacturer, model and serial number, range, units, date of calibration, full name of the individual performing the calibration, and the traceability number of the standard used in the calibration.

7.13.3 Provide a final service report and Calibration Certificates detailing all examines, tests and calibrations. All certificates are to be signed by the attending Technical Representative who performed the exam and calibration. Submit one copy in Portable Document Format (PDF).

7.13.4 Update references 2.3 and 2.4 based on the results of calibrations performed and provide an updated calibration list in Microsoft Excel format to the OMT REP.

7.14 Painting

7.14.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.15 Manufacturer’s Representative

7.15.1 Provide the services of an accredited gauge calibration company meeting ANSI Z540 or ISO/IEC 17025:2017. Submit proof of accreditation, qualifications and any certificates for the company and field service rep to the OMT REP for review and approval prior to starting any work.

7.15.2 Provide skilled technicians, portable instruments, standards and all necessary equipment to calibrate all designated gauges, meters and thermometers.

7.16 Preparation of Drawings

7.16.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification, and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to inspect, service, test, and certify the Scott Safety Self-Contained Breathing Apparatus (SCBA).

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 Tech Manual, S6220-EN-MMO-010, Self-Contained Breathing Apparatus Scott Air Pak Operation and Maintenance Manual

2.1.2 49 CFR §180, Subpart C - Qualification, Maintenance and Use of Cylinders

2.1.3 NFPA 1852 Standard on Selection, Care, and Maintenance of Open-Circuit Self-Contained Breathing Apparatus (SCBA) - (Available Commercially)

2.1.4 NFPA 1989 Standard on Breathing Air Quality for Emergency Services Respiratory Protection - (Available Commercially)

2.1.5 NAVSEA Drawing No. 830-8496378, Fire Control Plan

2.1.6 USNS Hershel Williams (ESB 4) SCBA Service Report

2.2 Enclosures

2.2.1 SCBA Facepiece and Pack Inventory

2.2.2 SCBA Cylinder Inventory

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location/Description/Quantity

3.1.1 Locations

a. Damage Control Locker, 1-34-3

b. Damage Control Locker, A1-90-1

c. SCBA Room, 1-29-2

d. Helo Crash Locker

e. Fire Truck

f. ECR 1

g. ECR 2

3.1.2 Description

SCBA Mfg:	Scott Safety (3M)
SCBA Model:	Model 4.5, Series 50 or 75
Cylinder Material:	Carbon Fiber
Cylinder Duration:	45 minute
Cylinder Pressure:	4500 psig
Cylinder Lifespan:	30 years
Cylinder Hydro Date:	See Enclosure 2.2.2
Special Permit (15yr)	SP-10915
Special Permit (30yr)	SP-14232

3.1.3

3.1.4 Quantity

a. 40 SCBA outfits, complete.

b. 120 cylinders

3.2 Item Description/Manufacturer's Data

3.2.1 See Enclosure 2.2.1 for the USNS Hershel Williams (ESB 4) SCBA Outfits (facepiece and pack assemblies).

3.2.2 See Enclosure 2.2.2 for the USNS Hershel Williams (ESB 4) Air Cylinders.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Each self-contained breathing apparatus must be of the pressure-demand, open-circuit type, approved by the Mine Safety and Health Administration (MSHA) and the National Institute for Occupational Safety and Health (NIOSH) per 46 CFR §35.20(Tanker), 46 CFR §77.35-5(Passenger Vessels), 46 CFR §96.35(General Cargo). Additionally, each SCBA must be accredited to NFPA 1981.

5.3 SCBA that are built to NFPA 1981 Edition 2007 or earlier will alarm at 25% remaining air, whereas NFPA 1981 Edition 2013 or later will alarm at 33% remaining air.

5.4 When using the term "SCBA", "SCBA" refers to the SCBA pack, mask and cylinder.

5.5 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Contractor must only provide the service of an Original Equipment Manufacturers (OEM) NFPA Scott SCBA Authorized Service Center (ASC) and NFPA OEM Scott technician(s) to perform service on the Scott SCBAs identified in section 3.0.

7.2 Prior to the removal of the Scott SCBAs from the vessel, provide the OMT REP with documents in a condition report that verify the Scott SCBA ASC and Scott SCBA technician are eligible to provide service on the SCBAs identified in section 3.0. All documents must be current at the time of serving.

7.2.1 Scott ASC must have letter(s) or certificate(s) issued by Scott stating the service facility is permitted to provide service and repair for Scott SCBAS, to the NFPA standard. Other societies standards such as EN(European) are not permitted to be substituted. ASC must have access to the following OEM Scott items; service bulletins, repair manuals, repair parts, specialized repair tools and a SCBA functional testing machine meeting the standards of Reference 2.1.3. A Posicheck3 Test Machine by Honeywell Analytics, or equal, is to be used, and must be running the most recent version of software from Scott, version 4.1.1.166S or later, additionally it must have been calibrated with the last 12 months.

7.2.2 Technician providing service on the Scott SCBA must be factory trained by Scott and possess a minimum Scott Repair Technician Certificate Level II for the model/type of SCBA identified in section 3.0.

7.2.3 Cylinder hydro testing facility must be certified as U.S. Department of Transportation (DOT) cylinder re-qualifier facility. Facility must have a U.S. DOT Requalification Identification Number (RIN) Letter that states that facility is permitted to requalify Scott cylinders identified in section 3.0. List of approved cylinder re-qualifier facilities and their U.S. DOT RIN can be found at the following URLs:

https://www.phmsa.dot.gov/hazmat/pressure-vessels-approvals/cylinders-requalifiers

https://www.phmsa.dot.gov/approvals-and-permits/hazmats/approvals-search

7.2.4 The above Scott ASC, Scott SCBA Technician and U.S. DOT RIN cylinder re-qualification facility is not permitted to be changed without prior approval from the OMT REP.

7.3 Inspect, service, test, and certify the Self-Contained Breathing Apparatus (SCBA) using References 2.1.1 through 2.1.6 and Enclosures 2.2.1 and 2.2.2 as guidance.

7.3.1 With assistance from ship's force, take custody of the SCBA outfits and spare cylinders, and temporarily remove from the ship and transport to the OEM Scott Safety certified authorized service facility. Only half (50%) of the SCBA outfits and cylinders are to be removed from the vessel at a time to maintain ship's force firefighting capabilities.

7.3.2 Custody for removal of the Scott Safety SCBAs must be arranged through the OMT REP and the vessel's Safety Officer.

7.4 SCBA technician will conduct a biannual visual inspection of the SCBA outfits and cylinders to meet the manufacturer's design, installation, maintenance instructions and service bulletin requirements. The inspection must encompass the entire SCBA outfit and cylinder with attention to the following:

7.4.1 Inspection of the face mask assembly includes the following:

a. Inspect all sealing surface and edges for damage, cracks, tears, holes, pliability and tackiness.

b. Inspect the head harness buckles, straps, and webbing for breakage, loss of elasticity and excessive wear.

c. Inspect the lens for holes, cracks, scratches, distortion, heat-damaged areas, and a proper seal with the face piece.

d. Inspect the regulator seating surface.

e. Inspect the nose cup for cuts or damage.

f. Inspect the regulator connections for damage and proper functionality.

g. Inspect voicemitters for dents or damage and properly installed in the ducts.

h. Inspect the EPIC/EPIC 3 voice amplification unit to ensure it operates.

7.4.2 Inspection of the backframe and harness assembly includes the following:

a. Inspect harness straps and backframe for broken welds, cuts, tears, abrasions, and indications of heat damage or chemical- related damage.

b. Inspect all buckles, fasteners, and adjustments to ensure proper functionality.

c. Inspect the cylinder retention system for damage and proper functionality.

d. Inspect the cylinder for secure attachment to the backframe.

e. Inspect harness straps for full extension.

f. Ensure waist belt has a regulator holder.

7.4.3 Inspection of the SCBA breathing air cylinder assembly includes the following:

a. Verify the hydrostatic test date on the cylinder is valid.

b. Record the serial number, age, and date of the latest hydrostatic test.

c. Inspect gauges for damage and pressure accuracy.

d. Inspect cylinder body for cracks, dents, weakened areas, and indications of heat damage or chemical damage.

e. Inspect the composite portions of the cylinder for cuts, gouges, loose composite materials, and the absence of resin.

f. Inspect the cylinder valve outlet sealing surface and threads for damage.

g. Inspect the cylinder valve handwheel for damage, alignment, serviceability, and secure attachment.

h. Inspect the burst disc outlet area for debris.

i. Verify full charge of each cylinder.

7.4.4 Inspection of the HP and LP hoses includes the following:

a. Inspect hoses for cuts, splitting, brittleness abrasions, bubbling, cracks, heat damage, and chemical damage.

b. Inspect external fittings for visual signs of damage.

c. Verify hoses have proper tight connections.

7.4.5 Inspection of the regulator includes the following:

a. Inspect the regulator for damage or missing components.

b. Inspect the regulator gasket is in good condition.

c. Inspect purge valve for damage and turns smoothly one-half turn by hand.

d. Inspect housing and components for damage.

e. Inspect the regulator and bypass (bleeder valve) for proper functionality.

f. Inspect the Heads Up Display (HUD) for damage and ensure unit operates. Verify that the rubber guard is in place and is not torn or damaged.

g. Inspect regulator for any unusual sounds such as whistling, chattering or rattling during operation.

7.4.6 Inspection of the remote gauge console includes the following:

a. Inspect the remote pressure gauge for damage.

b. Verify the cylinder pressure gauge and remote pressure gauge read within 10% of each other.

7.4.7 Inspection of any optional accessories and components must meet the manufacturer’s design, installation, maintenance instructions, and service bulletin requirements. Include the following:

a. Verify complete assembly of components.

b. Conduct assessment of wear and damage.

c. Verify secure attachment.

d. Verify adequate power source.

e. Verify all operating modes are inspected for proper functionality.

7.4.8 Identify any cylinders that have reached their maximum service life of 15/30 years, maximum for carbon fiber.

30 year service life cylinders have a blue band around the cylinder with Scott SEL in the band. Vessels may have both, the 15 and 30 year cylinders onboard. Ensure Enclosure 2.2.2 has the end of service life column filled out by the contractor.

7.5 During the SCBA Inspection the following must be accomplished:

7.5.1 Each SCBA must have a fully functional HUD and voice amplification unit. No exceptions.

7.5.2 SBCA technician must clean and disinfect each regulator and face piece identified in section 3.0 only using OEM recommendations using Reference 2.1.1 as guidance. OEM recommends using Wescodyne-G disinfectant-detergent. CAUTION: Bleach, alcohol and products containing alcohol should not be used on the facemask or other rubber based components.

7.5.3 SCBA technician must replace HUD batteries in the remote gauge console and EPIC Voice Amp using Reference 2.1.1 as guidance.

Remote Gauge Console (Two each AA batteries)
-Energizer Max E91
-Energizer Industrial EN91
-Duracell Copper Top MN1500
-Duracell Pro-Cell PC1500

EPIC 3 Voice Amp (Three each AAA batteries)
-Energizer Max E92
-Energizer Industrial EN92
Duracell Copper Top MN2400
Duracell Pro-Cell PC2400
Rayovac 824

7.5.4 Service Notice: All Scott Safety SCBAs labeled as CBRN must be rated for CBRN use. CBRN breathing valves are the color black, whereas non-CBRN breathing valves are the color orange. Breathing valves that are the color orange are not permitted on the Scott Safety SCBA detailed in section 3.0 of this work item. Orange, Non-CBRN breathing valves must be replaced with black CBRN breathing valves. All SCBAs must be CBRN rated.

7.5.5 Service Notice: CBRN rated SCBA masks have grey nose cones. Non-CBRN SCBA masks have a black nose cone. Black, non-CBRN rated nose cones must be replace with grey CBRN nose cones. All SCBA masks must be CBRN rated.

a. For estimating purposes assume that 10 each non-CBRN nose cones require replacement.

7.5.6 Conduct an air exchange by fully depleting the air and refilling each cylinder to 4500 psi with minimum grade D air using Reference 2.1.4 as guidance.

7.6 SCBA technician must conduct a functional performance test on each SCBA. The serial number of the Honeywell Analytics PosiChek3 machine, or equal, must be printed on the report for each SCBA. Coordinate testing with the OMT REP and the ship’s Safety Officer to witness testing. The functional performance test must include but not limited to the following per OEM requirements:

7.6.1 Visual inspection

7.6.2 Exhalation pressure

7.6.3 Facepiece leakage

7.6.4 Positive Pressure

7.6.5 Primary Lockup

7.6.6 Primary Creep

7.6.7 Air Saver Switch Activation

7.6.8 Transfer

7.6.9 Secondary Lockup

7.6.10 Secondary Creep

7.6.11 High Pressure Leakage

7.6.12 Secondary Pressure at High Cylinder

7.6.13 Purge

7.6.14 Pressure gauge accuracy at 3000 PSI, 2000 PSI and 1000 PSI rated cylinder pressure

7.6.15 First stage pressure during breathing resistance test at standard (40 LPM) work rate

7.6.16 First stage pressure during breathing resistance test at maximum (100 LPM) work rate

7.6.17 Heads Up Display (HUD) display and Vibralert

7.7 SCBA technician must collect and record the SCBA details required to complete Enclosures 2.2.1 and 2.2.2.

7.8 Conduct a visual examination on each cylinder to validate if the cylinder has reached its maximum service life or requires hydrostatic testing. SCBA cylinder manufacture month and year will be displayed on the cylinder label.

7.8.1 Scott cylinders have a service life of 15/30 years maximum. SCBA cylinders that have reached its maximum service life or fail hydrostatic testing are to be removed from service and disposed. Additionally, each cylinder must have a label affixed to it with the words “CONDEMNED” overcoated with epoxy, but not obscuring the original manufactures label using Reference 2.1.2 as guidance.

7.8.2 Certified cylinder re-qualifier facility must conduct a hydrostatic test on each Scott Safety SCBA cylinder during this service period for Scott Safety SCBA cylinders that have less than 10 months remaining before the next overhaul date.

a. For estimating purposes assume that 50 each SCBA Cylinders require hydrostatic testing.

b. Apply hydro testing label to the cylinder. The label must bear the company name, month and year of hydro testing using Reference 2.1.2 as guidance.

7.9 Compressed, gaseous, breathing air must meet the requirements specified in Reference 2.1.4.

7.10 Maintain workspaces in a clean condition during and upon completion of the work requirements. Prevent contamination of any adjacent equipment and spaces.

7.11 Upon completion of all inspections, tests and repairs, return the SCBA and cylinders to a ready for service condition.

7.12 Return the SCBA outfits and cylinders to the vessel, place them in their original Damage Control Locker, and turn over to the OMT REP upon completion. Return the SCBA outfits and cylinders to their original location onboard the vessel. Chain of custody and equipment return to be verified by OMT REP and vessel’s Safety Officer.

7.13 Reports

7.13.1 Submit an initial report with all documents detailed in paragraph 7.2 through 7.2.3.

7.13.2 Complete and submit one electronic copy in Portable Document Format (PDF) of Enclosures 2.2.1 and 2.2.2 documenting all inspections, tests and certifications. All certificates are to be signed by the attending manufacturer’s representative (or authorized technical representative) who performed the inspections and calibration.

7.13.3 When inspections, maintenance, servicing and testing reveal any deficiency, a condition report is to be submitted with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

NOTE TO PPE: When reviewing the condition reports, you can send the CR and failed flow test to N732 for assistance at MSC_N73_DamageControl@us.navy.mil.

7.13.4 Submit a final report that includes “as released” condition of the SCBAs affirming they are safe for use. Each report and checklist must be completed and signed by the person who carried out the inspections, maintenance servicing and testing then by countersigned by the ASC manger. Submit one electronic copy in Portable Document Format (PDF) containing the following:

a. Service facility’s Breathing Air Compressor (BAC) air sample test certification. Certification report to be dated within the past four months and is to include documentation of air quality test results.

b. PosiChek3, or equal, functional test results for each SCBA, all pages required.

c. Hydrostatic testing results for each cylinder.

d. Completed copies of Enclosure 2.2.1 and 2.2.2.

e. A detailed list of all parts that were replaced for each SCBA.

f. Letter and certificates for the ASC and SCBA technician certificate.

7.13.5 All reports and checklists are to be completed and signed by the person who carried out the inspections, maintenance, servicing, and testing and are to be countersigned by the Company's representative.

7.14 Painting: None additional

7.15 Manufacturer’s Representative

7.15.1 Provide the services of an Original Equipment Manufacturers (OEM) Scott service facility and factory trained SCBA technician to perform all examinations, servicing, testing and certification of the SCBAs and cylinders in accordance with manufacturer’s performance specifications.

7.15.2 Provide the OMT REP with a signed letter of qualification from the OEM, certificate of training for the SCBA identified in section 3.0 prior to the start of any work. Functional tests are to be performed by a manufacturer certified technician.

7.15.3 Provide skilled Level II technicians, service bulletins, standards, spare parts and all necessary equipment and tools to service and certify the SCBAs and cylinders.

7.15.4 The SCBA technician is to conduct a two hour group instruction with ship’s force personnel on the proper maintenance, care and use of the SCBA outfits.

7.16 Preparation of Drawings

7.16.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 – FACEPIECE AND PACK INVENTORY

No.	Location	Manufacturer	SCBA Model NFPA 1981 - Edition (1981-XXXX)	Facepiece Model Number	HP Regulator Serial No.	Breathing Valve Assembly Serial No.	LP Hose Change (last date)	Annual Functional Test (last date)	Remarks
1	DCC	SCOTT	2007						
2	DCC	SCOTT	2007						
3	ECR	SCOTT	2007						
4	ECR	SCOTT	2007						
5	HELO CRASH LK	SCOTT	2007						
6	HELO CRASH LK	SCOTT	2007						
7	HELO CRASH LK	SCOTT	2007						
8	HELO CRASH LK	SCOTT	2007						
9	HELO CRASH LK	SCOTT	2007						
10	HELO CRASH LK	SCOTT	2007						
11	HELO CRASH LK	SCOTT	2007						
12	REPAIR 1	SCOTT	2007						
13	REPAIR 1	SCOTT	2007						
14	REPAIR 1	SCOTT	2007						
15	REPAIR 1	SCOTT	2007						
16	REPAIR 1	SCOTT	2007						
17	REPAIR 1	SCOTT	2007						
18	REPAIR 1	SCOTT	2007						

ENCLOSURE 2.2.1 - FACEPIECE AND PACK INVENTORY

No.	Location	Manufacturer	SCBA Model NFPA 1981 - Edition (1981-XXXX)	Facepiece Model Number	HP Regulator Serial No.	Breathing Valve Assembly Serial No.	LP Hose Change (last date)	Annual Functional Test (last date)	Remarks
19	REPAIR 1	SCOTT	2007						
20	REPAIR 1	SCOTT	2007						
21	REPAIR 1	SCOTT	2007						
22	REPAIR 1	SCOTT	2007						
23	REPAIR 1	SCOTT	2007						
24	REPAIR 1	SCOTT	2007						
25	REPAIR 1	SCOTT	2007						
26	REPAIR 1	SCOTT	2007						
27	REPAIR 1	SCOTT	2007						
28	REPAIR 1	SCOTT	2007						
29	REPAIR 1	SCOTT	2007						
30	REPAIR 2	SCOTT	2007						
31	REPAIR 2	SCOTT	2007						
32	REPAIR 2	SCOTT	2007						
33	REPAIR 2	SCOTT	2007						
34	REPAIR 2	SCOTT	2007						
35	REPAIR 2	SCOTT	2007						
36	REPAIR 2	SCOTT	2007						
37	REPAIR 2	SCOTT	2007						
38	REPAIR 2	SCOTT	2007						
39	SAFETY EQUIP RM	SCOTT	2007						
40	SAFETY EQUIP RM	SCOTT	2007						

Service Facility (name) ______________________________

Service Technician (name) ______________________________ Date ____________________

ENCLOSURE 2.2.2 – SCBA CYLINDER INVENTORY

No.	Location	Manufacturer	Serial No.	DOT No. (DOT-SP XXXX)	Cyl. Rating (45/60 min.)	Cyl. Rating (pressure)	Date of Mfr. (DOM)	1st Hydro Test (date)	2nd Hydro Test (date)	End of Service Life (**)	Remarks
1		LUXFER GAS CYLINDERS	ABW30790	14232	45min	4500	7/2013	7/2018	7/2023	7/2028	Replace in 2027
2		LUXFER GAS CYLINDERS	ABW30108	14232	45min	4500	7/2013	7/2018	4/2021	7/2028	Hydro in 2025
3		LUXFER GAS CYLINDERS	ABW30154	14232	45min	4500	7/2013	7/2018	4/2021	7/2028	Hydro in 2025
4		LUXFER GAS CYLINDERS	ABW30161	14232	45min	4500	7/2013	7/2018	7/2023	7/2028	Replace in 2027
5		LUXFER GAS CYLINDERS	ABW30217	14232	45min	4500	7/2013	7/2018	7/2023	7/2028	Replace in 2027
6		LUXFER GAS CYLINDERS	ABW30771	14232	45min	4500	8/2013	7/2018	4/2021	8/2028	Hydro in 2025
7		LUXFER GAS CYLINDERS	ABW30779	14232	45min	4500	8/2013	7/2018	4/2021	8/2028	Hydro in 2025
8		LUXFER GAS CYLINDERS	ABW30783	14232	45min	4500	8/2013	7/2018	7/2023	8/2028	Replace in 2027

9		LUXFER GAS CYLINDERS	ABW30790	14232	45min	4500	8/2013	7/2018	7/2023	8/2028	Replace in 2027
10		LUXFER GAS CYLINDERS	ABW30793	14232	45min	4500	8/2013	7/2018	7/2023	8/2028	Replace in 2027
11		LUXFER GAS CYLINDERS	ABW43137	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
12		LUXFER GAS CYLINDERS	ABW43168	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
13		LUXFER GAS CYLINDERS	ABW43169	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
14		LUXFER GAS CYLINDERS	ABW43185	14232	45min	4500	2/2016	1/2022		2/2031	Hydro in 2026
15		LUXFER GAS CYLINDERS	ABW43213	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
16		LUXFER GAS CYLINDERS	ABW43318	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
17		LUXFER GAS CYLINDERS	ABW91612	14232	45min	4500	2/2016	6/2021		2/2031	Hydro in 2025
18		LUXFER GAS	ABW43398	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025

		CYLINDERS									
19		LUXFER GAS CYLINDERS	ABW91599	14232	45min	4500	2/2016	6/2021		2/2031	Hydro in 2025
20		LUXFER GAS CYLINDERS	AWB43430	14232	45min	4500	2/2016	4/2021		2/3031	Hydro in 2025
21		LUXFER GAS CYLINDERS	ABW43467	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
22		LUXFER GAS CYLINDERS	ABW43471	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
23		LUXFER GAS CYLINDERS	ABW43472	14232	45min	4500	2/2016	1/2022		2/2031	Hydro in 2026
24		LUXFER GAS CYLINDERS	ABW43481	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
25		LUXFER GAS CYLINDERS	ABW43482	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
26		LUXFER GAS CYLINDERS	ABW43185	14232	45min	4500	2/2016	1/2022		2/2031	Hydro in 2026
27		LUXFER GAS CYLINDERS	ABW43488	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
28		LUXFER	ABW43530	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in

		GAS CYLINDERS									2025
29		LUXFER GAS CYLINDERS	ABW43563	14232	45min	4500	2/2016	1/2022		2/2031	Hydro in 2026
30		LUXFER GAS CYLINDERS	ABW43578	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
31		LUXFER GAS CYLINDERS	ABW43601	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
32		LUXFER GAS CYLINDERS	ABW43640	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
33		LUXFER GAS CYLINDERS	ABW43650	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
34		LUXFER GAS CYLINDERS	ABW43674	14232	45min	4500	2/2016	1/2022		2/2031	Hydro in 2026
35		LUXFER GAS CYLINDERS	ABW43679	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
36		LUXFER GAS CYLINDERS	ABW43711	14232	45min	4500	2/2016	1/2022		2/2031	Hydro in 2026
37		LUXFER GAS CYLINDERS	ABW43213	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025

38		LUXFER GAS CYLINDERS	ABW43739	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
39		LUXFER GAS CYLINDERS	ABW43743	14232	45min	4500	2/2016	1/2022		2/2031	Hydro in 2026
40		LUXFER GAS CYLINDERS	ABW43744	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
41		LUXFER GAS CYLINDERS	ABW43761	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
42		LUXFER GAS CYLINDERS	ABW43779	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
43		LUXFER GAS CYLINDERS	ABW43808	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
44		LUXFER GAS CYLINDERS	ABW43812	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
45		LUXFER GAS CYLINDERS	ABW43813	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
46		LUXFER GAS CYLINDERS	ABW43818	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
47		LUXFER GAS	ABW43825	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025

		CYLINDERS									
48		LUXFER GAS CYLINDERS	ABW43833	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
49		LUXFER GAS CYLINDERS	ABW43834	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
50		LUXFER GAS CYLINDERS	ABW43835	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
51		LUXFER GAS CYLINDERS	ABW43839	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
52		LUXFER GAS CYLINDERS	ABW43840	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
53		LUXFER GAS CYLINDERS	ABW43843	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
54		LUXFER GAS CYLINDERS	ABW43847	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
55		LUXFER GAS CYLINDERS	ABW43851	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
56		LUXFER GAS CYLINDERS	ABW43852	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
57		LUXFER	ABW43862	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in

		GAS CYLINDERS									2025
58		LUXFER GAS CYLINDERS	ABW43872	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
59		LUXFER GAS CYLINDERS	ABW43876	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
60		LUXFER GAS CYLINDERS	ABW43886	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
61		LUXFER GAS CYLINDERS	ABW43890	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
62		LUXFER GAS CYLINDERS	ABW43898	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
63		LUXFER GAS CYLINDERS	ABW43904	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
64		LUXFER GAS CYLINDERS	ABW43909	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
65		LUXFER GAS CYLINDERS	ABW43911	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
66		LUXFER GAS CYLINDERS	ABW43912	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025

67		LUXFER GAS CYLINDERS	ABW43915	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
68		LUXFER GAS CYLINDERS	ABW43918	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
69		LUXFER GAS CYLINDERS	ABW43921	14232	45min	4500	2/2016	1/2022		2/2031	Hydro in 2026
70		LUXFER GAS CYLINDERS	ABW43925	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
71		LUXFER GAS CYLINDERS	ABW43927	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
72		LUXFER GAS CYLINDERS	ABW43929	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
73		LUXFER GAS CYLINDERS	ABW43931	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
74		LUXFER GAS CYLINDERS	ABW43934	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
75		LUXFER GAS CYLINDERS	ABW43937	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
76		LUXFER GAS	ABW43946	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025

		CYLINDERS									
77		LUXFER GAS CYLINDERS	ABW43949	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
78		LUXFER GAS CYLINDERS	ABW43958	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
79		LUXFER GAS CYLINDERS	ABW43995	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
80		LUXFER GAS CYLINDERS	ABW43999	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
81		LUXFER GAS CYLINDERS	ABW44030	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
82		LUXFER GAS CYLINDERS	ABW44080	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
83		LUXFER GAS CYLINDERS	ABW44095	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
84		LUXFER GAS CYLINDERS	ABW44120	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
85		LUXFER GAS CYLINDERS	ABW44134	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
86		LUXFER	ABW44137	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in

		GAS CYLINDERS									2025
87		LUXFER GAS CYLINDERS	ABW44146	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
88		LUXFER GAS CYLINDERS	ABW44166	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
89		LUXFER GAS CYLINDERS	ABW44224	14232	45min	4500	2/2016	1/2022		2/2031	Hydro in 2026
90		LUXFER GAS CYLINDERS	ABW44242	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
91		LUXFER GAS CYLINDERS	ABW46822	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
92		LUXFER GAS CYLINDERS	ABW47620	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
93		LUXFER GAS CYLINDERS	ABW48349	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
94		LUXFER GAS CYLINDERS	ABW48351	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
95		LUXFER GAS CYLINDERS	ABW48352	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025

96		LUXFER GAS CYLINDERS	ABW48353	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
97		LUXFER GAS CYLINDERS	ABW48358	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
98		LUXFER GAS CYLINDERS	ABW48369	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
99		LUXFER GAS CYLINDERS	ABW48374	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
100		LUXFER GAS CYLINDERS	ABW48377	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
101		LUXFER GAS CYLINDERS	ABW48378	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
102		LUXFER GAS CYLINDERS	ABW48379	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
103		LUXFER GAS CYLINDERS	ABW48381	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
104		LUXFER GAS CYLINDERS	ABW48393	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
105		LUXFER GAS	ABW48394	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025

		CYLINDERS									
106		LUXFER GAS CYLINDERS	ABW48396	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
107		LUXFER GAS CYLINDERS	ABW48409	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
108		LUXFER GAS CYLINDERS	ABW48411	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
109		LUXFER GAS CYLINDERS	ABW48420	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
110		LUXFER GAS CYLINDERS	ABW48421	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
111		LUXFER GAS CYLINDERS	ABW48426	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
112		LUXFER GAS CYLINDERS	AWB48437	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
113		LUXFER GAS CYLINDERS	ABW48439	14232	45min	4500	2/2016	4/2021		2/2031	Hydro in 2025
114		LUXFER GAS CYLINDERS	ABW48440	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
115		LUXFER	ABW48443	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in

		GAS CYLINDERS									2025
116		LUXFER GAS CYLINDERS	ABW48445	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
117		LUXFER GAS CYLINDERS	ABW48446	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
118		LUXFER GAS CYLINDERS	ABW48447	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
119		LUXFER GAS CYLINDERS	ABW48456	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025
120		LUXFER GAS CYLINDERS	ABW48462	14232	45min	4500	7/2016	4/2021		7/2031	Hydro in 2025

****NOTE**: ESL is typically 5 years from the 2nd Hydrostatic Test unless the cylinder is an extended service life cylinder.

Service Facility (name) ______________________________

Service Technician (name) ______________________________ Date ______________________

1.0 ABSTRACT

The purpose of this item is to inspect, service, test, and certify the Self-Contained Breathing Apparatus (SCBA) Compressor and Refilling Station.

2.0 REFERENCES

2.1 Tech Manual MNL-0172, Bauer SCBA Refilling Station, Model No. CFS55-3S45DF

2.2 Tech Manual MNL-0145, MAXI Verticus 08 High Pressure Air Compressor Unit Instruction and Replacement Parts List

2.3 NFPA 1989, Standard on Breathing Air Quality for Emergency Services Respiratory Protection (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations: SCBA Compressor Room (1-29-1)

3.1.2 Quantity

a. One SCBA Compressor and Refilling Station, Current Operating Hours on Compressor:TBD

3.2 Item Description/Manufacturer’s Data

3.2.1 SCBA Compressor, Bauer Model No. MNVT8-E3-460-P25

3.2.2 SCBA Refilling Station, Bauer Model No. CFS 5.5

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service, test, and certify the Self-Contained Breathing Apparatus (SCBA) Compressor and Refilling Station identified in section 3.0, using References 2.1 through 2.3 as guidance. Accomplish all tasks in accordance with IMO, SOLAS, and the Manufacturer’s requirements.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

CAUTION: Verify that all pressure has been released from the system before carrying out any work.

7.3 Conduct an inspection of the SCBA Compressor and Refilling Station to meet the manufacturer’s design, installation, maintenance instructions and service bulletin requirements. The inspection must encompass the entire system with particular attention to the following:

7.3.1 Conduct a visual inspection of the system including: SCBA cylinder enclosure including door and tray, operator’s panel, compressor, purification system, motor, hour meter, breakers and fuses, gauges, controls, all labels and plates such as caution, warning and instruction placards, alarms, cooling systems, condensate traps and drains, pneumatic piping, hoses, pressure maintaining valve, pressure regulator, relief valves, valves and fittings.

7.3.2 Verify no unauthorized alterations have been made to the system and components.

7.3.3 Verify a sign or placard is posted near the air intake identifying it as an intake source for breathing air.

7.3.4 Verify a WARNING label is installed at the purification chamber, instructing: Prior to changing purifier cartridges, performing service or maintenance, to release all air pressure in the compressor system.

7.3.5 Inspect purification cartridges and filters. Record their expiration dates and verify they meet OEM requirements.

7.3.6 Inspect the audible and visual alarms and any controls that shutdown the compressor:

a. Carbon Monoxide (CO) Alarm

b. Discharge Air High Temperature Alarm

c. Low Oil Level and Low Oil Pressure Alarm

d. Moisture in the compressed air at the Purification System Outlet

e. Compressor Compartment High Temperature Alarm

f. Inclination alarm

Inspect all the pressure relief valves. Verify and document the calibration date.

7.3.7 Inspection of any optional accessories and components must meet the manufacturer’s design, installation, maintenance instructions, and service bulletin requirements. Include the following:

a. Verify complete assembly of components.

b. Conduct assessment of wear and damage.

c. Verify all operating modes are inspected for proper functionality.

7.4 Conduct annual and two year maintenance on the SCBA Compressor and Refilling Station to meet the manufacturer's design, installation, maintenance instructions, and service bulletin requirements. The maintenance includes/verifies:

7.4.1 Verify all operating fluids, including lubricant and coolant, are at the manufacturer's recommended levels.

7.4.2 Drain the crankcase and replace the oil and oil filters using Reference 2.2 as guidance.

7.4.3 Inspect the condition of the drive belts including the alignment and tension.

7.4.4 Verify the expiration dates of the purification cartridges and filters.

7.4.5 Lubricate all parts.

7.5 Conduct testing of the SCBA Compressor and Refilling Station to meet the manufacturer's design, installation, maintenance instructions and service bulletin requirements. Coordinate testing with OMT REP. The testing includes the following:

7.5.1 Conduct a one hour functional performance test of the SCBA Compressor and Refilling System. The test is to include the following:

a. Verify proper operation using Reference 2.1 as guidance.

b. Verify inter-stage pressures are within the manufacturer's specifications.

c. Verify proper operation of all audible and visual alarms.

d. Verify proper operation of all shutdowns.

e. Verify set point of the Pressure Relief Valve.

f. Verify set point of the Pressure Maintaining and Regulator Valves.

g. Verify proper operation of the Emergency Stop.

7.5.2 Draw an air sample and submit to an accredited laboratory for air quality testing using Reference 2.3 as guidance. The breathing air must be tested for:

a. Oxygen Content

b. Carbon Monoxide Content

c. Carbon Dioxide Content

d. Condensed Oil and Particulate Content

e. Water Content

f. Non-methane Volatile Organic Compounds (VOC) Content

g. Nitrogen Content

h. Odor

7.6 Compressed, gaseous breathing air must meet the requirements for breathing air specified in Reference 2.3.

7.7 Maintain workspaces in a clean condition during and upon completion of the work requirements. Prevent contamination of any adjacent equipment and spaces.

7.8 Upon completion of all inspections, tests and repairs, return the SCBA Compressor and Refilling Station to a ready for service condition.

7.9 Reports

7.9.1 Submit one electronic copy in Portable Document Format (PDF) of the examination, servicing, and certification report for the SCBA Compressor and Refilling Station to the OMT REP and ship’s Master upon completion of testing.

7.9.2 Submit one electronic copy in Portable Document Format (PDF) of the Air Quality Sample results to the OMT REP and ship’s Master. All certificates are to be signed by the manufacturer’s representative (or authorized technical representative) performing the test at the accredited laboratory.

7.9.3 When inspections, maintenance, servicing and testing reveal any deficiency, a condition report is to be submitted with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.9.4 Submit a report that includes “as released” condition of the SCBAs. Submit one electronic copy in Portable Document Format (PDF).

7.9.5 All reports and checklists are to be completed and signed by the person who carried out the inspections, maintenance, servicing, and testing and are to be countersigned by the Company's representative.

7.10 Painting: None additional

7.11 Manufacturer’s Representative

7.11.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the air compressor and filling stations identified in section 3.0 ensuring compliance with the manufacturer’s standards and performance specifications.

7.11.2 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.11.3 Provide the services of a Laboratory accredited to test compressed breathing air quality in accordance with Reference 2.2. Provide the OMT REP with written proof of qualification and accreditation from the Authority Having Jurisdiction (AHJ).

7.12 Preparation of Drawings

7.12.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 3 4)		
AUXILIARY MACHINERY		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0565A	**CATEGORY "A"**	**2026-02-02**
ESB-CCSI-FIXED GAS FIREFIGHTING SYSTEM INSPECTION AND TEST (1 YR) (SCSI)		**Port Engineer's Name**

1.0 ABSTRACT

The purpose of this item is to inspect, service and certify the ship’s fixed CO2 fire extinguishing systems.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSEA Drawing No. 830-8496378, Fire Control and Safety Plan

2.1.2 Tech Manual S9543-AA-MMC-010, High Pressure Carbon Dioxide Marine Fire Suppression System

2.1.3 IMO MSC.1/Circ.1318, Rev 1, 25 May 2021, Guidelines for the Maintenance and Inspection of Fixed Carbon Dioxide Fire-Extinguishing Systems

2.2 Enclosures

2.2.1 Fixed Gas Firefighting System – 1 YR Check Sheet

2.2.2 Fixed Gas Firefighting System – Cylinder Service Record

2.2.3 Placards and Markings

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Spaces Protected by Fixed CO2 Firefighting Systems

1. Incinerator Room (A-29-1)

2. EDG Room (A-34-2)
3. Casing (P) (1-23-2)
4. Casing (S) (1-34-1)
5. Paint and Lamp Store (1-34-4)
6. Lubt/Chemical Room (1-34-5)
7. MSC Cosal Storeroom (2-48-2)
8. High Voltage Room No. 1P (2-55-2)
9. High Voltage Room No. 1S (2-55-1)
10. AC Room (2-56-4)
11. AC Room (2-56-3)
12. Engine Control Room 1 (2-56-1)
13. Engine Control Room 2 (2-56-2)
14. Hotel Services Room (2-48-1)
15. Fuel Storage Room (3-56-3)
16. Elec Workshop (3-56-1)
17. Purifier Room (3-56-2)
18. Engineers Workshop (3-55-1)
19. High Voltage Room No. 2S (3-55-3)
20. High Voltage Room No. 2P (3-55-2)
21. Fuel Supply Room (3-56-4)
22. Fuel Supply Room (3-56-5)

23. Machinery Room 1 (5-56-1)

24. Machinery Room 2 (5-56-2)

25. FWD Incinerator Room (A4-96-4)

26. FWD Paint Locker (A1-89-4)

27. JP-5 Pump Room (5-108-1)

b. Locations/Quantity of Fixed CO2 Bottles

1. CO2 Room (1-34-2): 243 100# CO2 cylinders

2. CO2 Room NO.1 (A4-94.5-2): Three 100# CO2 cylinders

3. JP-5 Pump Room (5-108-1) Three 100# CO2 cylinders

c. Nitrogen Pilot and Time Delay Cylinders

1. CO2 Room (1-34-2) Quantity: 9, 108 cu. In. pilot cylinders including (2 each contained inside 2 IMO release cabinets-MR 1 and MR 2) and 5 108 cu. in. time delay cylinders.

2. CO2 Room NO.1 (A4-94.5-2), Quantity: 2, 108 cu. In. pilot cylinders and 2 108 cu. in. time delay cylinders.

3. Fire Control Room (1-56-6), Quantity: 4, 108 cu. In. pilot cylinders (2 each contained inside 2 IMO release cabinets-MR 1 and MR 2)

4. A1 Level, Port Side Fr. 88, Quantity 2, 108 cu. In. pilot cylinders (2 each contained inside 1 IMO release cabinet-Paint Locker FWD)

5. Main Deck, Port Side Fr. 33, Quantity 2, 108 cu. In. pilot cylinders (2 each contained inside 1 IMO release cabinet-Paint/Lamp Store)

6. Passage (A4-94-0), Quantity 2, 108 cu. In. pilot cylinders (2 each contained inside 1 IMO release cabinet-Incinerator Room FWD)

7. A Deck, Fr. 34 (outside EDG Room) Quantity 2, 108 cu. In. pilot cylinders (2 each contained inside 1 IMO release cabinet-EDG Room)

8. Machinery Room 2 (5-56-2)(3rd Deck, outside Purifier Room) 2, 108 cu. In. pilot cylinders (2 each contained inside 1 IMO release cabinet-Purifier Room)

3.2 Item Description: Fixed Gas Firefighting systems

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 46 CFR §95.16-1(a) “Clean agent” means a halocarbon or inert gas used as a fire extinguishing agent. Inert gasses are defined as using one or more of the gasses nitrogen, argon, or helium.

5.3 MSC QMS N7.722.10440.5-WP, FLEXIBLE HOSE ASSEMBLY

INSPECTION AND REPLACEMENT established policy for the test and replacement periodicities for all MSC vessels.

5.4 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service, test and certify the fixed gas fire extinguishing systems identified in section 3.0 using References 2.1.1 through 2.1.3 and Enclosures 2.2.1 through 2.2.3 as guidance. Accomplish inspection, service, and testing in accordance with IMO, SOLAS, and the Manufacturer’s requirements.

7.2 Coordinate with ship’s force to have the associated equipment and system locked and/or tagged out per ship’s lock out/tag out policy to prevent accidental discharge. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.2.1 Tag-out the fixed gas firefighting systems and controls to prevent accidental discharge.

7.3 Prior to accomplishing any work, conduct a meeting with the vessels Chief Engineer and OMT Rep to determine what ship’s equipment will be shut down by the testing. It may be necessary to temporarily disconnect interlock circuits before beginning the work. Any ship’s circuits that are disconnected during the servicing must be re-connected and functionally tested when the servicing is completed to ensure the extinguishing system is operable.

7.4 Conduct an annual inspection of all fixed gas firefighting systems identified in paragraphs 3.1.1 through 3.1.3 and record the results on Enclosures 2.2.1 and 2.2.2.

7.4.1 Inspect all piping, controls, actuators, remote pulls, valves, time delays, pressure switches, sirens and alarms, extinguishing agent, expelling means and physical condition to verify the system is in good order.

7.4.2 Verify all markings, operating instructions and warning placards are in place, unobstructed and marked legible using Enclosure 2.2.3, 46 CFR §97.37 and §95.15 as guidance. Ensure operating instructions are located in a conspicuous place at or near pull boxes, stop valve controls and in the cylinder storage room. On systems where the cylinders are within the protected space, ensure the instructions include a schematic diagram of the system and instructions detailing alternate methods of discharging the system should the manual release or stop valve fail to operate.

7.4.3 Conduct an external visual examination of all cylinders identified in paragraphs 3.1.1 through 3.1.3 and complete Enclosures 2.2.1 through 2.2.2 for each system by location. Identify any with leaks or show obvious physical damage, dents, bulging, and corrosion or have been involved in a fire. Verify each cylinder is clearly marked and record all data including name of manufacturer, year of manufacturer, serial number and size and determine if a hydrostatic test is due.

7.4.4 Weigh the extinguishing agent cylinders and pilot cylinders for gross weight, subtract the cylinders tare weight to determine the net weight of the charge in each. Record the net weight results on Enclosure 2.2.2. Identify the Design Required Charge Weight and Supplied Charge Weight using References 2.1.1 through 2.1.3 and record the results on Enclosure 2.2.2. Compare the measured charge weight or liquid level to the systems required values. If the weight loss of a cylinders charge exceeds the thresholds below immediately notify the OMT and submit a condition report. A cylinder recharge is required if the loss exceeds:

a. CO_2 - 10% of charge weight

b. Inert gas - 5% of cylinder gauge pressure (adjusted for temperature)

c. Halon, Halocarbons, FM-200, NOVEC 1230 - 5% of charge weight

OR

d. Halon, Halocarbons, FM-200, NOVEC 1230 - if the cylinder is fitted with a pressure gauge 10% of the cylinder pressure (adjusted for temperature)

OR

e. Halon, Halocarbons, FM-200, NOVEC 1230 - if the cylinder is fitted with an integral floating dipstick liquid level indicator 5% of the liquid level (adjusted for temperature)

1. Any recharging will be the subject of a Contract Change Order (CCO) only as approved by the OMT Rep. When recharging is required it is to be performed using the manufacturer's service manual as guidance.

2. After recharging, a leak test is to be performed proving the cylinder is tight.

3. In no case is a cylinder/container to be recharged if it is beyond its specified hydrostatic test date.

7.4.5 Verify containers/cylinders fitted with a pressure gauge are in the proper range and the installation is free from leakage.

7.4.6 Check connections of all manifolds and distribution piping for tightness.

7.4.7 Inspect all hoses for damage or decay.

7.4.8 Visually examine all flexible hoses (hoses, loops,..) and their associated components connecting the high pressure cylinders and pilot cylinders to the distribution manifold/piping using the manufacturer's recommendations as guidance. Submit a condition report identifying any found damaged, decayed, or unserviceable.

7.4.9 Verify the discharge piping and nozzles are clear of any external obstructions. Nozzles should be inspected to ensure they have not been blocked or obstructed by the storage of spare parts or a new installation of structure or machinery.

7.4.10 Verify all control/section valves are in the correct position.

7.4.11 Visually inspect the boundaries of the protected space to confirm that no modifications have been made to the enclosure that have created un-closeable openings that would render the system ineffective. If cylinders are installed inside the protected space, verify the integrity of the double release lines inside the protected space, and check low pressure or circuit integrity monitors on release cabinet, as applicable.

7.4.12 Submit a condition report to OMT REP with recommended repairs and parts required when inspections reveal any deficiency.

7.5 For estimating purposes, assume 10% of the cylinders identified in 3.1.1 through 3.1.3 will be due for a hydrostatic test. Perform a hydrostatic test and internal visual examination of those cylinders using References 2.1.1 through 2.1.3 for guidance in accordance with 46 CFR §147.60 through 67.

7.5.1 Temporarily remove the fire extinguishing agent and store during the hydrostatic testing and internal exams. Refill the cylinders upon successful completion.

7.5.2 Upon successful completion, re-mark the cylinders in accordance with 46 CFR §147.60 through 67.

7.6 Secure a tag or label to all cylinders, indicating inspection and maintenance was performed. The tag at a minimum is to include the type of extinguishing agent, test date (month/year) and name of test facility.

7.7 With the assistance of ship’s force, functionally test all remote pulls, fuel shut-off controls, actuators, main stop and distribution valves, time delays, audible and visual alarms, and ventilation shutdowns interconnected with fire protection systems using References 2.1.1 through 2.1.2 and manufacturer’s instruction manual as guidance.

7.7.1 Establish and implement a safety plan to prevent personnel entry into the protected spaces until testing is completed and the spaces have been ventilated and determined to be safe for human occupancy. In cases where the agent storage tank or high pressure cylinders are physically disconnected from the manifold before testing, the need for evacuation should be determined on a case-by-case basis.

7.7.2 The time delay devices must have an accuracy of -0/+20 percent of the rated time delay period through the operating temperature range and range of delay settings.

a. Test cylinders are to be sized for a minimum of 50lbs for the delay plus an additional 15lbs/min for each siren in the system.

b. b. Test cylinders must be equipped with a syphon tube.

c. c. Test cylinders must be allowed to stabilize a minimum of 72 hours after filling prior to use in testing.

d. d. Install a new nitrogen cartridge on Time Delays (if fitted) prior to testing.

e. e. Upon completion of testing, reset the Time Delay and replace nitrogen cartridge (if fitted) returning it to service.

f. f. Test cylinder is to be used only once per time delay device test. New test cylinder shall be used for each test.

7.7.3 Measure the force on remote pulls to ensure they require less than 40 lbs to operate and do not require an excessive amount of travel to activate the system.

7.8 Affix cylinder and system tags and labels documenting inspection, maintenance, recharging or hydrostatic testing so as not to obstruct system use, classification or manufacturer’s labels.

7.9 Coordinate and conduct a final functional test demonstrating satisfactory operation of the fixed gas firefighting systems in the presence of OMT REP and ABS.

7.10 Upon completion of all inspections, tests and repairs return the Firefighting Systems to a ready for service condition. Conduct a final walk around survey with the ships Master and OMT REP to verify status.

7.11 Reports

7.11.1 Upon completion of all inspections, tests, maintenance and recharging, prepare and submit the System Check sheet and Service Record forms (Enclosures 2.2.1 through 2.2.2). The reports are to be typewritten and detail the examination of the systems and final “as released” conditions. Submit one electronic copy in Portable Document Format (PDF) to the OMT.

7.11.2 For cylinders being requalified, Pressure test and visual inspection records are to adhere to 49 CFR §180.215.

7.11.3 Upon completion, submit a final report certifying that the fixed gas firefighting systems have been inspected and returned to active service. Submit one electronic copy in Portable Document Format (PDF) to the OMT.

7.11.4 All reports and checklists are to be completed and signed by the person who carried out the inspections, maintenance, servicing, and testing and are to be countersigned by the Company's representative.

7.12 Painting

7.12.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.13 Manufacturer’s Representative

7.13.1 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only fire extinguishing system service companies certified and recognized as an ABS Recognized Service Supplier shall be used. To include oversight of all aspects of equipment inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable. In addition, only OEM authorized parts will be used.

https://www.msc.usff.navy.mil/Business-Opportunities/Contracts/Qualification-for-Items-Critical-to-Shipboard-Safety-on-MSC-Vessels/

7.13.2 It is critical that all persons working on the system must be certified and fully knowledgeable in its operation and repair. Persons in training to be certified are permitted to perform maintenance and recharging of extinguishing systems under the direct supervision and in the immediate presence of a certified person.

7.13.3 Companies and persons performing maintenance and recharging of extinguishing systems are to have available the appropriate servicing manuals, correct tools, recharging materials, lubricants, and manufacturer’s replacement parts.

7.14 Preparation of Drawings

7.14.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - FIXED GAS FIREFIGHTING SYSTEM – 1YR CHECKSHEET

Vessel: ____________ **IMO #:** ____________
System Location: ____________
Space Protected: ____________ **FX Agent:** ____________

Shipyard: ____________
Service Company: ____________ **Inspector:** ____________
RIN #: ____________ **Date:** ____________

To be completed by the system inspector at the time of inspection or test. The inspection of the firefighting system shall at a minimum include checks and tests of the following:

ENCLOSURE 2.2.1 - FIXED GAS FIREFIGHTING SYSTEM – 1YR CHECKSHEET

#	COMPONENT	SAT	UNSAT	NA	COMMENTs
1	Fire extinguishing system and controls secured to prevent accidental discharge	☐	☐	☐	
2	Placards and Instructions in place and legible	☐	☐	☐	
3	Cylinder hydrostatic test due dates verified and recorded	☐	☐	☐	
4	Hose replacement due dates verified and recorded	☐	☐	☐	
5	Cylinder pressure gauge reading or indicator is in the operable range	☐	☐	☐	
6	Cylinder charge determined by weighing or other authorized means	☐	☐	☐	
7	Cylinders and cylinder valves visually examined externally	☐	☐	☐	
8	Flexible hoses visually examined	☐	☐	☐	
9	All cylinder clamps, brackets and connections checked for tightness	☐	☐	☐	
10	Manifold visually inspected	☐	☐	☐	
11	Control valves externally examined (main and distribution)	☐	☐	☐	
12	Control valves internally examined (5YRs) (main and distribution)	☐	☐	☑	Not due
13	Control valves tested (main and distribution)	☐	☐	☐	
14	Activating heads on cylinder valves removed and tested, where possible(5YR)	☐	☐	☑	Not due

ENCLOSURE 2.2.1 - FIXED GAS FIREFIGHTING SYSTEM – 1YR CHECKSHEET

#	COMPONENT	SAT	UNSAT	NA	COMMENTs
15	Cylinders and pilot cylinders visually examined internally (10YRs)	☐	☐	☑	Not due
16	Cylinders and pilot cylinders hydrostatically pressure tested (10YRs)	☐	☐	☑	Not due
17	Flexible hoses were tested (5YR) (ONLY:FM-200/ Novec 1230/N2)	☐	☐	☑	Not due
18	Flexible hoses were renewed (10YR)	☐	☐	☑	Not due
19	Check the connections of all pilot release piping and tubing for tightness	☐	☐	☐	
20	Servo tubing/pilot lines pressure tested at working pressure and checked for leakage and blockage (5YR)	☐	☐	☑	Not due
21	Automatic Timers and Time Delays tested	☐	☐	☐	
22	Remote release stations and system inspected	☐	☐	☐	
23	Remote release system tested	☐	☐	☐	
24	Manual pull cables, pulleys, gang releases tested, serviced and tightened/adjusted as necessary	☐	☐	☐	
25	Inspect discharge piping and nozzles verifying they are clear of external obstruction	☐	☐	☐	
26	Discharge piping and nozzles proved clear by blowing dry air or nitrogen thru system (2YR and 5YR)	☐	☐	☑	Not due
27	Boundaries of the protected space visually examined to confirm no mods have been made creating un-closeable openings	☐	☐	☐	
28	Alarms tested (Audible and Visual)	☐	☐	☐	
29	Test shutdown of ventilation fans	☐	☐	☐	
30	Test closure of vent dampers	☐	☐	☐	
31	Auto shutdown of engines tested	☐	☐	☐	
32	Remote fuel shutoffs tested	☐	☐	☐	

ENCLOSURE 2.2.1 - FIXED GAS FIREFIGHTING SYSTEM – 1YR CHECKSHEET

#	COMPONENT	SAT	UNSAT	NA	COMMENTs
33	System returned to active service	☐	☐	☐	
34	Test and inspection tags attached	☐	☐	☐	

SAT = Satisfactory **UNSAT** = Unsatisfactory **N/A** = Not applicable

Inspector Certification:

This system, as specified, has been inspected and tested according to the standards cited in the work item.

Signed: ______________________ Date: ______________________

Printed Name: ______________________ Phone: ______________________

Title: ______________________

Organization: ______________________

ENCLOSURE 2.2.2 - FIXED GAS FIREFIGHTING SYSTEM – CYLINDER SERVICE RECORD

VESSEL ______________________ SHIPYARD ______________________ DATE ____________

SYSTEM LOCATION ______________________ SERVICE COMPANY ______________________

SPACE PROTECTED ______________________ INSPECTOR ______________________

ENCLOSURE 2.2.2 - FIXED GAS FIREFIGHTING SYSTEM – CYLINDER SERVICE RECORD

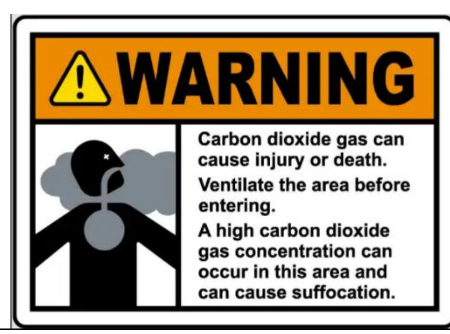
CYL No.	BANK (Main/ Reserve/ Pilot)	FX AGENT	CYLINDER MARKINGS				SYSTEM DESIGN		WEIGHED CHARGE (Net WT) (lbs)	EXTERNAL VISUAL EXAM (date)	INTERNAL VISUAL EXAM (date)	CYLINDER HYDRO TEST (date)	CYLINDER HYDRO TEST (psi)	HOSE VISUAL EXAM (last date)	HOSE REPLACED (last date)
			SERIAL No.	MFG DATE	LAST TEST DATE	WORKING PRESSURE (psi)	CHARGE REQUIRED (Net WT) (lbs)	CHARGE SUPPLIED (Net WT) (lbs)							
1															
2															
3															
4															
5															
6															
7															
8															
9															
10															
11															
12															
13															

USS Hershel Williams		
(ESB 3 4)		
AUXILIARY MACHINERY		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0565A	CATEGORY "A"	2026-02-02
ESB-CCSI-FIXED GAS FIREFIGHTING SYSTEM INSPECTION AND TEST (1 YR) (SCSI)		Port Engineer's Name

ENCLOSURE 2.2.2 - FIXED GAS FIREFIGHTING SYSTEM – CYLINDER SERVICE RECORD

CYL No.	BANK (Main/	FX AGENT	CYLINDER MARKINGS				SYSTEM DESIGN		WEIGHED CHARGE	EXTERNAL VISUAL	INTERNAL VISUAL	CYLINDER	CYLINDER HYDRO	HOSE VISUAL	HOSE REPLACED
14															
15															
16															
17															
18															
19															
20															
21															
22															
23															
24															
25															
26															
27															
28															
29															
30															
31															

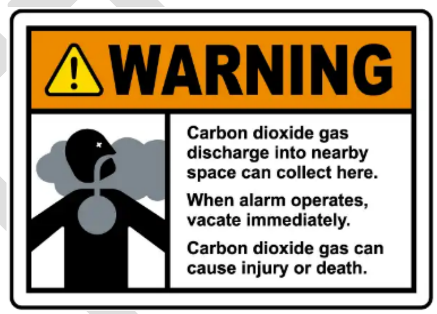
ENCLOSURE 2.2.3 - PLACARDS AND MARKINGS

Gas	Requirement	Placard verbiage
All	46 CFR §97.37-9	Each fire extinguishing alarm must be conspicuously marked: "WHEN ALARM SOUNDS VACATE AT ONCE. CARBON DIOXIDE/HALON/FM-200/NOVEC 1230 BEING RELEASED."
All	46 CFR §97.37-10 and §95.15-10	Each control valve to branch lines of all fire extinguishing systems will be plainly and permanently marked indicating the spaces served.
All	46 CFR §97.37-13	Control cabinets or spaces containing valves or manifolds for the various fire extinguishing systems must be distinctly marked in conspicuous red letters at least 2 inches high: "CARBON DIOXIDE/HALON/FM-200/NOVEC 1230 FIRE APPARATUS."
CO_2	46 CFR §97.37-11(a)	Each entrance to spaces storing carbon dioxide must be conspicuously marked — "CARBON DIOXIDE GAS CAN CAUSE INJURY OR DEATH. VENTILATE THE AREA BEFORE ENTERING. A HIGH CONCENTRATION CAN OCCUR IN THIS AREA AND CAN CAUSE SUFFOCATION." 

ENCLOSURE 2.2.3 - PLACARDS AND MARKINGS

Gas	Requirement	Placard verbiage
		https://www.safetysign.com/
CO_2	46 CFR §97.37-11(b)	Each entrance to spaces protected by carbon dioxide must be conspicuously marked — "CARBON DIOXIDE GAS CAN CAUSE INJURY OR DEATH. WHEN ALARM OPERATES OR WINTERGREEN SCENT IS DETECTED, DO NOT ENTER UNTIL VENTILATED. LOCK OUT SYSTEM WHEN SERVICING." The reference to wintergreen scent may be omitted for carbon dioxide systems not required to have odorizing units and not equipped with such units.  https://www.mysafetysign.com/
CO_2	46 CFR §97.37-11(c)	Each entrance to spaces into which carbon dioxide might migrate must be conspicuously marked — "CARBON DIOXIDE GAS CAN CAUSE INJURY OR DEATH. DISCHARGE INTO NEARBY SPACE CAN COLLECT HERE. WHEN ALARM OPERATES OR WINTERGREEN SCENT IS DETECTED VACATE IMMEDIATELY." The

ENCLOSURE 2.2.3 - PLACARDS AND MARKINGS

Gas	Requirement	Placard verbiage
		reference to wintergreen scent may be omitted for carbon dioxide systems not required to have odorizing units and not equipped with such units  https://www.safetysign.com/
CO2	46 CFR §95.15-60 Odorizing units	Each carbon dioxide extinguishing system installed or altered after July 9, 2013, must have an approved odorizing unit to produce the scent of wintergreen, the detection of which will serve as an indication that carbon dioxide gas is present in a protected area and any other area into which the carbon dioxide may migrate.

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY	CONTRACT NO. N68171-26-RN-003	
ITEM NO. 0567A	CATEGORY "A"	2026-02-02
ESB-CCSI-FIXED FOAM FIREFIGHTING SYSTEM INSPECTION AND TEST (1 YR) (SCSI)	Port Engineer's Name	

1.0 ABSTRACT

The purpose of this item is to inspect and test the ship's fixed foam firefighting systems.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NASSCO Drawing No. 830-8496378, Fire Control and Safety Plan

2.1.2 NASSCO Drawing No. 555-8499369, Fixed Foam Fire Extinguishing System Diagram

2.1.3 Tech Manual S9555-FL-MMC-010, Aqueous Film Forming Foam (AFFF) Fire Fighting Equipment

2.1.4 IMO MSC.1/Circ.1432, 31 May 2012, Revised Guidelines for the Maintenance and Inspection of Fire Protection Systems and Appliances (Available Commercially)

2.1.5 Foam Concentrate Safety Data Sheet, National Foam, Universal CGC6 (Available Commercially)

2.1.6 MSC SMS Memo#: 198; Memo to Masters, Chief Engineers, and Shore Staff – Aqueous Film Forming Foam (AFFF) Conservation

2.2 Enclosures

2.2.1 Fixed Foam Firefighting System – Check Sheet

2.2.2 Fixed Foam Firefighting System – Service Record

2.2.3 Placards and Markings

2.2.4 AFFF Drain Lines

2.2.5 Testing Laboratories by Foam Type

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations Protected by Fixed Foam Firefighting Systems

a. Flight Deck and Hangar (A5-86-0) and (A5-96-0)

b. Mission Deck (A1-87-0)

c. Portable Foam Applicators - Six total identified on Reference 2.1.1: One located at (5-108-1), one located at (A-34-2), one located at (A-33-3), one located at (5-56-1), one located at (5-56-2), and one located at (A4-96-4)

3.2 Description

3.2.1 See References 2.1.1 through 2.1.2.

3.3 Location/Quantity of Foam Proportioning Systems

3.3.1 Aft Foam Station (1-56-4): 1,740 gals of MILSPEC 3% Ansul Ansulite

3.3.2 Forward Foam Station (A5-90-4): 1,740 gals of MILSPEC 3% Ansul Ansulite

3.4 Location/Quantity of Foam Equipment

3.4.1 Foam Hose Reels – 17 locations as identified on Reference 2.1.1: (A5-58-1), (A5-58-2), (A5-65-1), (A5-65-2), (A5-73-1), (A5-73-2), (A5-78-1), (A5-78-2), (A5-84-1), (A5-84-2), (A5-89-1), (A5-89-2), (A5-94-2), (A5-64-2), (A5-64-0), (A5-76-2), (A5-76-0)

3.4.2 Portable Foam Applicators - Six total identified on Reference 2.1.1: One located at

Fr. 97 Partial 4524 Stringer), Two located at (A-34-1), (A-34-2), Two located at (3rd Deck Fr. 35), and One located at (A4-94-2)

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The Oily Water Separator (OWS) cannot process foam. Secure the OWS before draining foam to the bilges. Foam should be stored for discharge to a shore facility if the ship is in restricted waters. Foam may be discharged overboard if not in restricted waters.

5.3 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all labor, materials, staging, tools and equipment as required to inspect, maintain and test the ship’s fixed foam firefighting systems identified in section 3.0 using References 2.1.1 through 2.1.6 and Enclosure 2.2.1 through 2.2.3 as guidance. Accomplish inspections and testing in accordance with USCG, IMO/SOLAS, and the Manufacturer’s requirements.

7.2 Coordinate with the ship’s Chief Engineer (CHENG) to have associated equipment locked out and tagged out per ship’s policy. Secure and tag out the fixed foam extinguishing system and controls to prevent accidental discharge. Ensure manual valves at the tanks are closed to prevent the loss of foam or possible dilution or contamination by water in the lines.

7.3 When the maintenance is complete, coordinate with the CHENG to have the locks and tags removed from the associated equipment.

7.4 Conduct an **INSPECTION** of all fixed foam firefighting systems using References 2.1.1 through 2.1.5 and Enclosures 2.2.1 through 2.2.3 as guidance.

7.4.1 Inspect the fixed foam fire extinguishing systems and fill out Enclosures 2.2.1 and 2.2.2. It is to be a thorough examination of the system including all piping, pipe supports, controls, actuators, release stations, monitors, nozzles, valves, pressure-vacuum vent, proportioner system, gauges, pressure switches, flow switches, hoses, hose reels, portable applicators, audible and visual alarms, foam, expelling means and physical condition to verify that the system is in good operating condition. Immediately report any deficient conditions to the OMT REP.

7.4.2 Verify all markings, operating instructions and warning placards are in place, unobstructed and clearly legible. Use Enclosure 2.2.3 as guidance. Operating instructions are to be located in a conspicuous place at or near the controls.

7.4.3 Conduct an external visual examination of all foam concentrate storage tanks and complete Enclosure 2.2.1 and 2.2.2. Identify any obvious physical damage, dents, bulging, corrosion, and leaks. Verify each tank is clearly marked and record the foam data including name of manufacturer, type, concentration, and tank capacity.

7.4.4 Visually inspect the pressure and temperature gauges and tank level indicators. Identify any damage, incorrect indication and range, or issues inhibiting their proper function.

7.4.5 Verify and record the level of the foam concentrate in each tank. Compare the measured liquid levels with the recommended fill levels and previous readings to verify the proper quantity of foam concentrate is in the tank.

7.4.6 Confirm with the ships Master the foam concentrate has been maintained within the manufacturer’s limits for temperature, 35°F and 120°F (2°C and 49°C). Verify the foam concentrate is being stored in a cool, dry, well ventilated area under cover and out of direct sunlight.

7.4.7 Visually inspect all valves, fittings, piping, expansion joints and connections for valve alignment, system tightness, condition and evidence of leakage.

7.4.8 Externally examine the foam concentrate pumps, couplings and motors. Check alignment of the pumps and motors. Rotate the pumps by hand to confirm freedom of movement. With assistance from ships force run the concentrate pumps and check for proper rotation, output pressure, leakage, excessive noise or vibration or overheating. Check electrical voltage to motors.

7.4.9 Inspect and verify all pump and system relief valves are properly set and operating correctly.

7.4.10 Visually inspect all nozzles focusing particular attention in areas where nozzles are subject to weather or physical damage (Cargo Tank deck, RO-RO decks, Flight decks, Hangars, etc…).

7.4.11 Visually verify all control and distribution valves are in the correct position.

7.4.12 Cycle all control valves manually. Verify proper operation and identify any damage or degradation.

7.4.13 Cycle all automated control valves from the control panel. Verify their proper operation and identify any damage, degradation or deficient conditions.

7.4.14 Inspect all remote release stations for physical condition and proper operation. Identify any damage, degradation or deficient conditions.

7.4.15 Inspect all foam monitors (if applicable) for damage, blockage, and freedom of movement.

7.4.16 Inspect all foam hose reels and hoses identified in paragraph 3.4.1.

a. Inspection should include, but not be limited to, the hose reel, hose, hose connections and fittings and vari-nozzle assembly.

b. Unwind hose from reel to its full length. Verify the hose moves freely. Grease if there are any grease points. Inspect the entire length of hose.

c. Inspect the condition and tightness of hose connection to reel.

d. Operate vari-nozzle to ensure it does not stick.

7.4.17 Vessels fitted with fixed deck firefighting systems, shall inspect any Fiber Reinforced Plastic (FRP) gratings in platforms and access ways for firefighting equipment. FRP gratings must be USCG class L1 (NVIC 9-97 Ch1) which provides the highest degree of safety and protection after fire exposure. All FRP gratings should be permanently stamped with the USCG Type Approval Number and Approval Level (L1, L2 or L3) to facilitate inspections.

7.4.18 Visually examine all Portable Foam Applicators as shown on the ships Fire Control Plan, Reference 2.1.1 and listed in paragraph 3.4. Identify any that are missing or damaged. Verify all portable foam applicators are set to the correct proportioning ratio for the foam concentrate supplied.

7.4.19 Verify all portable foam concentrate containers listed in paragraph 3.4 are factory sealed and the manufacturer’s service life interval has not been exceeded.

7.4.20 When inspections reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required.

7.5 Accomplish the following **MAINTENANCE** on the fixed foam firefighting system:

7.5.1 Temporarily remove and inspect the pressure-vacuum vent valve on the concentrate storage tanks. While vent is removed, make certain that the opening is not blocked and that dirt or other foreign objects do not enter the tank. Flush, clean, inspect and lubricate the valve and all components ensuring proper operation. Reassemble and install with new gaskets upon completion.

7.5.2 Open all filters/strainers, saltwater supply sensing lines and foam sensing lines. Examine, clean and flush all debris and contamination and reassemble with new gaskets.

7.6 Conduct the following **TESTS** on the fixed foam firefighting system using References 2.1.1 through 2.1.3 and Enclosures 2.2.1 and 2.2.2 as guidance. Testing is to be coordinated and conducted in the presence of the OMT REP and ABS Surveyor. Fill out and submit Enclosure 2.2.1 and 2.2.2 upon completion.

7.6.1 Upon vessel arrival at the contractor’s facility, blow test each fixed foam firefighting system confirming the discharge piping and nozzles are clear of any obstructions, debris and contamination. The test shall be performed by isolating the discharge piping from the system and flowing dry air or nitrogen, or other suitable means as recommended by manufacturer, through the piping to confirm the pipework and nozzles are clear of any debris and obstructions. This may require the removal of nozzles, if applicable. Identify, record and report the location and quantity of any non-functioning nozzles.

NOTE: The following 3 sections (7.6.2, 7.6.3, 7.6.4) are for ESB Class only

7.6.2 Before conducting blow down test, drain all Zones using the Mission Deck Low Point Drains using Enclosure 2.2.5 for locations.

NOTE: Use Caution when draining from the low point drains. AFFF and sea water sitting for long period of time can produce H2S Gas which is harmful for personnel.

7.6.3 Use of Air to test nozzles is limited to the Flight Deck Nozzles. Ensure that initially all Flight Deck Nozzle Covers are installed and then only a few at a time are removed to test for air flow. Compressed air should be tied into the FWD and AFT skids for testing.

7.6.4 Testing of the Hanger Bay overhead nozzles will require the installation of plastic ducting to be installed. Manlift will be required for the installation and removal of the ducting.

7.6.5 Within the first two weeks of an availability, draw a one pint (500 ml) representative sample of the foam concentrate from each concentrate storage tank and submit to the applicable testing laboratory based on the table in Enclosure 2.2.5.

Sample Collection: While a thief sample obtained through the manhole access is ideal, a bottom sample drawn from the drain valve is acceptable for practicality. If obtaining a bottom sample, utilize a clean five-gallon bucket to collect foam concentrate from the bottom drain valve. Continue drawing concentrate until the color appears consistent, then obtain the one pint (500 ml) sample. Return any remaining foam concentrate from the bucket to the tank, or store it in a designated container for proper disposal with the tank contents when they are disposed.

NOTE: Do not take the sample from the surrounding pipework.

7.6.6 The foam manufacturer is to issue a lab certificate indicating specific gravity, sedimentation, pH value, expansion ratio, drainage time, volumetric mass and percentage of water dilution using MSC.1/Circ.1312 for low expansion foam or MSC/Circ.670 for high expansion foam as guidance. Submit a copy in Portable Document Format (PDF) format to the OMT REP upon receipt.

7.6.7 Test the diaphragm valve for holes or tears in accordance with the OEMs procedures. With assistance of ships force, pressurize <u>either</u> the foam concentrate side <u>or</u> the water side of the proportioning system. Open the diaphragm valve Flush-Out Petcock for the side of the proportioning system <u>not</u> pressurized. Observe the liquid being discharged from the appropriate flush-out petcock. If liquid from pressurized side is found in discharge from flush-out petcock, the valve diaphragm has a hole or tear.

7.6.8 Functionally test all audible and visual alarms warning of the release of the system. Test for their satisfactory operation in each machinery space, cargo pump-room, vehicle space, RO-RO space and special category space being protected by the fixed foam systems. Test all alarms ensuring they operate for the length of time needed to evacuate the space, but in no case less than 20 seconds.

7.6.9 Functionally test all remote release stations for proper operation. This is to include demonstration of system automation such as auto startup of fire pumps, etc... upon activation.

7.6.10 Functionally test all ventilation and fuel shut-off controls connected to fire-protection systems for proper operation.

7.6.11 Test all system cross connections to other sources of water supply for proper operation.

7.6.12 Hydrostatically test the hose reel hoses to working pressure.

7.7 Tags and Labels

7.7.1 Tags and labels documenting inspection, maintenance, recharging or testing shall be affixed so as not to obstruct the equipment use, classification or manufacturers labels.

7.7.2 Each storage tank shall have a tag or label securely attached that indicates inspection and maintenance was performed. The tag at a minimum shall identify the following:

a. Date (month/day/year) of Inspection

b. Gallons of AFFF

c. Name of the Agency/Company performing the Inspection

7.8 Upon completion of all inspections, tests and repairs return the Foam Firefighting Systems to a ready for service condition. Conduct a final walk around survey with the vessels Master and OMT REP to verify system alignment and status.

7.9 Reports

7.9.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.9.2 Upon completion of all inspections, maintenance and tests, prepare and submit the results of the foam concentrate tests, foam mix ratio tests, System Check sheet and System Service form shown in Enclosure 2.2.1 and 2.2.2. The report is to document the examination of the system and final “as released” conditions. Submit one electronic copy in Portable Document Format (PDF) to the OMT REP.

7.9.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.10 Painting/Lagging

7.10.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.10.2 Renew lagging where damaged or disturbed.

7.11 Manufacturer's Representative

7.11.1 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only fire extinguishing system service companies certified and recognized as an ABS Recognized Service Supplier shall be used. To include oversight of all aspects of equipment inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable. In addition, only OEM authorized parts will be used.

https://www.msc.usff.navy.mil/Business-Opportunities/Contracts/Qualification-for-Items-Critical-to-Shipboard-Safety-on-MSC-Vessels/

7.11.2 It is critical that all persons working on the system must be certified and fully knowledgeable in its operation and repair. Persons in training to be certified are permitted to perform maintenance and recharging of extinguishing systems under the direct supervision and in the immediate presence of a certified person.

7.11.3 Companies and persons performing maintenance and recharging of extinguishing systems are to have available the appropriate servicing manuals, correct tools, recharging materials, lubricants, and manufacturer's replacement parts.

7.12 Preparation of Drawings

7.12.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - FIXED FOAM FIREFIGHTING SYSTEM - CHECKSHEET

Vessel: ____________________ **IMO #:** ____________________

System Location: ____________________

Area Protected: ____________________ **Foam Mfg:** ____________________

Tank Capacity: ____________________ **Foam Type:** ____________________

Shipyard: ____________________ **Inspector:** ____________________

Service Company: ____________________ **Date:** ____________________

To be completed by the system inspector at the time of inspection or test. The inspection of the fire fighting system will at a minimum include checks and tests of the following:

#	COMPONENT	SAT	UNSAT	NA	COMMENTS
1	Fire extinguishing system and controls secured to prevent accidental discharge	☐	☐	☐	
2	Placards and Instructions in place and legible	☐	☐	☐	
3	Visually examine foam concentrate storage tank externally	☐	☐	☐	
4	Verify tank level indicator is functioning properly	☐	☐	☐	
5	Foam concentrate level is in the operable range	☐	☐	☐	
6	PV valve on the concentrate storage tank inspected.	☐	☐	☐	
7	Foam concentrate sampled from storage tanks and sent to Mfg for analysis	☐	☐	☐	
8	Foam concentrate being stored within Mfg’s environmental parameters	☐	☐	☐	
9	System piping, hoses, valves and nozzles visually inspected	☐	☐	☐	

SAT = Satisfactory **UNSAT** = Unsatisfactory **N/A** = Not applicable

ENCLOSURE 2.2.1 - FIXED FOAM FIREFIGHTING SYSTEM – CHECKSHEET

#	COMPONENT	SAT	UNSAT	NA	COMMENTS
10	Relief valve setting verified and tested	☐	☐	☐	
11	Control valves cycled manually	☐	☐	☐	
12	Control valves cycled from control panel	☐	☐	☐	
13	Internally examine control valves (5YR)	☐	☐	☑	Not due
14	Remote release stations examined and tested	☐	☐	☐	
15	Monitors visually examined (if applicable)	☐	☐	☐	
16	Hose reels and applicators visually examined	☐	☐	☐	
17	Hydrostatically test hoses to working pressure	☐	☐	☐	
18	Examine foam concentrate pump and motor alignment, freedom of movement, correct rotation	☐	☐	☐	
19	Test foam concentrate pump operation; output pressure, vibration and motor voltage	☐	☐	☐	
20	Discharge piping and nozzles on foam systems proved clear by blowing air thru system (5YR)	☐	☐	☑	Not due
21	Discharge piping and nozzles on **weather deck and flight deck** foam systems proved clear by flowing water thru system (2.5YR)	☐	☐	☑	Not due
22	Perform functional test of system at sea	☐	☐	☐	

SAT = Satisfactory **UNSAT** = Unsatisfactory **N/A** = Not applicable

ENCLOSURE 2.2.1 FIXED FOAM FIREFIGHTING SYSTEM - CHECKSHEET

#	COMPONENT	SAT	UNSAT	NA	COMMENTS
23	Alarms tested (Audible and Visual)	☐	☐	☐	
24	Test ventilation and fuel shut-off controls interconnected to the foam system (if applicable)	☐	☐	☐	
25	Sample and test the mix ratio from each proportioner (2.5YR)	☐	☐	☑	Not due
26	Flush RO-RO deck deluge system piping w/water, drain and purge w/air (if applicable) (5YR)	☐	☐	☑	Not due
27	System returned to active service	☐	☐	☐	
28	Inspection tags attached	☐	☐	☐	

SAT = Satisfactory **UNSAT** = Unsatisfactory **N/A** = Not applicable

Inspector Certification:

This system, as specified, has been inspected and tested according to the standards cited in the work item.

Signed: ______________________ Date: ______________________

Printed Name: ______________________ Phone: ______________________

Title: ______________________

Organization: ______________________

ENCLOSURE 2.2.2 - FIXED FOAM FIREFIGHTING SYSTEM – SERVICE RECORD

Vessel: ____________ **IMO #:** ____________
System Location: ____________
Area Protected: ____________ **Foam Mfg:** ____________
Tank Capacity: ____________ **Foam Type:** ____________

Shipyard: ____________ **Inspector:** ____________
Service Company: ____________ **Date:** ____________

To be completed by the system inspector at the time of inspection or test. The inspection of the fire fighting system will at a minimum include checks and tests of the following:

ENCLOSURE 2.2.2 – FIXED FOAM FIREFIGHTING SYSTEM – SERVICE RECORD

COMPONENT	EXAM	RECORDED DATA	SET POINT
Proportioner Skid	MFG		
	Model No.		
	Serial No.		
AFFF Storage Tank	Capacity (gals)		1740 gals
	External Visual Inspection (date)		1YR
	Internal Visual Inspection (date)		1YR
	Measured AFFF Charge (date)		1YR
	Measured AFFF Charge (gals)		gal
	AFFF Liquid Level (% Full)		95 %
	PV Valve examined		1YR
	Low level alarm setting (%)		%
	Samples of foam concentrate tested by MFG (date)		1YR
AFFF System	Visual inspection of system made (date)		1YR
	Flow test all water supply and foam pumps for proper pressure and capacity		1YR

ENCLOSURE 2.2.2 – FIXED FOAM FIREFIGHTING SYSTEM – SERVICE RECORD

COMPONENT	EXAM	RECORDED DATA	SET POINT
	Test all system cross connections to other sources of water supply		1YR
	Blow test discharge piping and nozzles proving clear		5YR
	Test all ventilation and fuel shut-off controls connected to the fire extinguishing system		1YR
	Functional test of weather deck and flight deck foam systems at sea. Discharge foam for 15 sec from nozzles chosen by ABS Surveyor. Discharge water from all lines and nozzles proving they are clear of obstruction		2.5YR
	Test all proportioners to confirm mix ratio tolerance within +30 to – 10% of design approval		5YR
Alarms	Functional Test (audible and visual alarms)		1YR
Relief valve	Pump relief valve pop test (date)		1YR
	Pump relief valve pop test (psi)		psi
Control valves	Verify all Control and Distribution valves are in correct position		1YR
	Main Stop valve internally examined (date)		5YR
	Distribution valves internally examined (date)		5YR
	Balanced Proportioner valves internally examined (date)		5YR
Filters & Strainers	Examined and verified free of debris and contamination		1YR
Release Stations	Functional test		1YR
Portable Foam Applicators	Verify eqpt is in place and in proper order		1YR
	Verify portable foam		1YR

ENCLOSURE 2.2.2 - FIXED FOAM FIREFIGHTING SYSTEM - SERVICE RECORD

COMPONENT	EXAM	RECORDED DATA	SET POINT
	concentrate containers are factory sealed and within service life		
Foam Hose Reels	Inspect hose reel, hose and vari-nozzle		1YR
	Hydro test hose to working pressure		1YR
	Functional test		2.5YR
Foam Monitors (if applicable)	Inspect		1YR
	Functional test		2.5YR
Firemain	Pressure		psi
RO-RO Deck Deluge system	Flush w/water, drain and purge w/air (if applicable)		5YR

Inspector Certification:

This system, as specified, has been inspected and tested according to the standards cited in the work item.

Signed: ______________________ Date: ______________________

Printed Name: ______________________ Phone: ______________________

Title: ______________________

Organization: ______________________

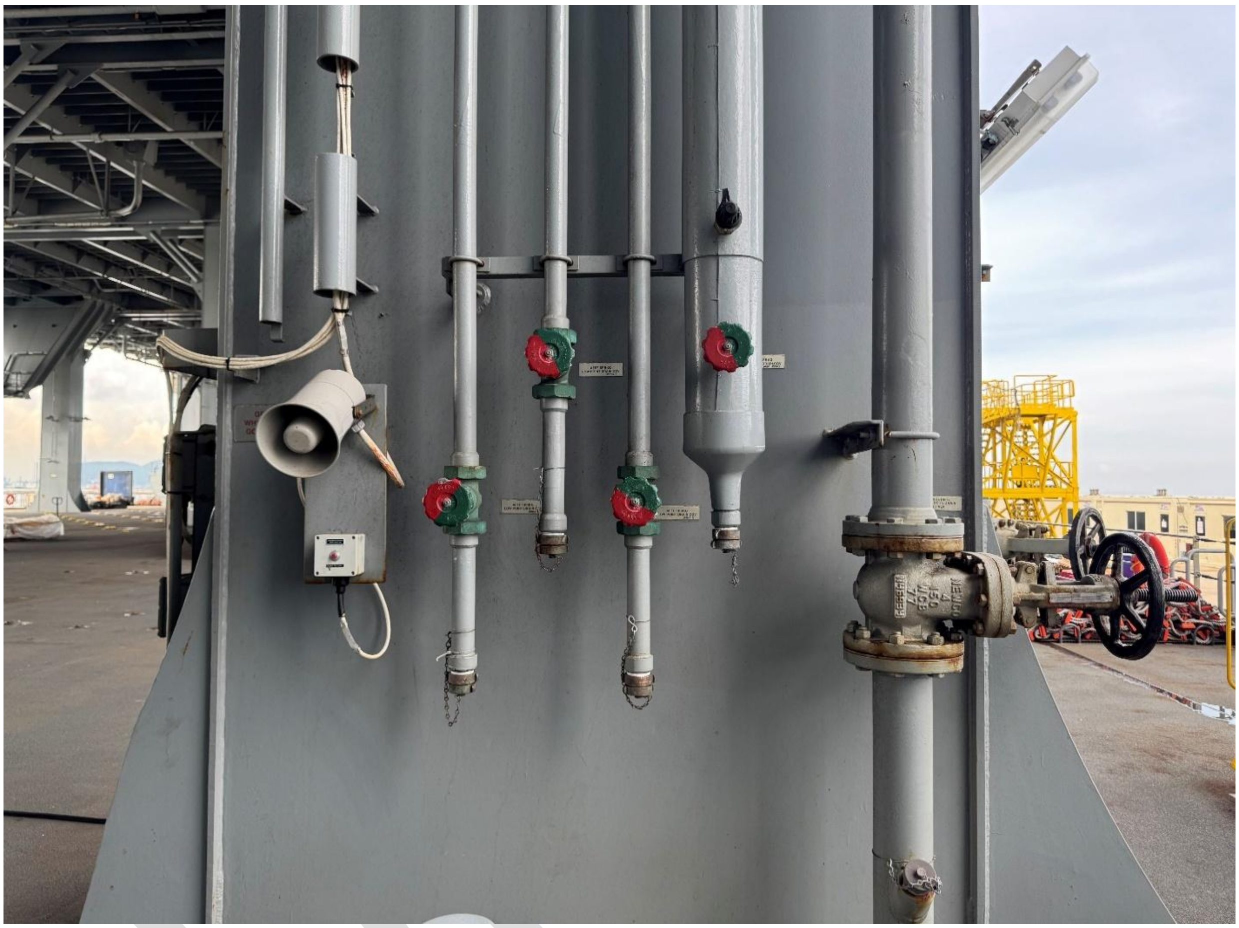
ENCLOSURE 2.2.3 - PLACARD AND MARKINGS

Foam	Requirement	Placard verbiage
Cargo and Miscellaneous Vessels	46 CFR §97.37-10 and §95.15-10	Each branch line control valve of all fire extinguishing systems will be plainly and permanently marked indicating the spaces served.
Cargo and Miscellaneous Vessels	46 CFR §97.37-13 and §35.40-10	Control cabinets or spaces containing valves or manifolds for the various fire extinguishing systems must be distinctly marked in conspicuous red letters at least 2 inches high: "FOAM FIRE APPARATUS."
Tank vessels	46 §35.40-10	Each foam firefighting apparatus must be marked "FOAM FIRE APPARATUS" in red letters at least 2 inches high. Branch pipe valves leading to the several compartments must be distinctly marked to indicate the compartments or parts of the vessel to which they lead.
Tank vessels	46 §35.40-17	At each required foam hose/monitor valve there will be marked in not less than 2-inch red letters and figures: "FOAM STATION 1," 2, 3, etc.

ENCLOSURE 2.2.4 – AFFF DRAIN LOCATIONS

Mission Deck Fr 57 Centerline

ENCLOSURE 2.2.4 – AFFF DRAIN LOCATIONS



Mission Deck Fr86 PORT Stanchion

ENCLOSURE 2.2.5 - TESTING LABORATORIES BY FOAM TYPE

Laboratory by Foam Type

Please use the table below to determine which laboratory is correct for your vessel and/or system. If there are any questions, please contact Drew Marine at MSC-Technical@drew-marine.com. The laboratory addresses are below:

TYCO FIRE PROTECTION PRODUCTS
2700 Industrial PKWY South
BLDG 130, Receiving #7
Marinette, WI 54143-2542, USA

NATIONAL FOAM
Attn: TECH SERV. LAB
141 Junny Road
Angier, NC 27501 USA

VIKING LIFESAVING EQUIPMENT A/S
Hestehaven 19
5260 Odense, Denmark

Laboratory	Class/Vessel	System Name	Foam Concentrate
TYCO	T-EPF Class	PORT HEF Tank	JET-X 2-3/4%
		STBD HEF Tank	JET-X 2-3/4%
National Foam	T-AKE	FWD Tank	UNIVERSAL GOLD 1/3%
	T-AKE	AFT Tank	UNIVERSAL GOLD 1/3%
	T-AO 188, 194, 196, 199, 201, 202, 203	FWD Tank	UNIVERSAL CG6%
	T-AO 188, 194, 196, 199, 201, 202, 203	AFT Tank	UNIVERSAL CG6%
	T-AH	HELO Deck	UNIVERSAL CG6%
	T-AH	AFT Tank	UNIVERSAL GOLD 1/3%
	T-AKR USNS PILILAAU	HELO DECK	UNIVERSAL CG6%
	T-AKR USNS SEAY	HELO DECK	UNIVERSAL CG6%
	T-AK (Except USNS STOCKHAM)	FOAM Tank	AER-O-FOAM XL 3-3%
	HST	Foam Tank	UNIVERSAL GOLD 1/3%
Viking	T-EPF	PORT AFFF TANK	MILPSEC 3%
	T-EPF	STBD AFFF TANK	MILPSEC 3%
	T-AO 187, 189, 195, 200, 204	FWD Tank	ANSULITE 3X3%
	T-AO 187, 189, 195, 200, 204	AFT Tank	ANSULITE 3X3%
	T-AO 205 CLASS	FWD Tank	SOLBERG/HILLER 3X3%
	T-AO 205 CLASS	AFT Tank	SOLBERG/HILLER 3X3%
	T-AKR (Except SEAY and PILILAAU)	FWD Tank	ANSULITE 3X3%
	T-AKR (Except SEAY and PILILAAU)	AFT Tank	ANSULITE 3X3%
	ESB	FWD Tank	MILSPEC 3%
	ESB	AFT Tank	MILSPEC 3%
	T-ESD	FWD Tank	ANSULITE 3X3%
	T-ESD	AFT Tank	ANSULITE 3X3%
	T-AOE	1-490-3-Q Tank	MILSPEC 6%
	T-AOE	1-488-1-Q Tank	MILSPEC 6%
	T-AOE	1-289-1-Q Tank	MILSPEC 6%
	T-ARS	FWD Foam Tank	MILSPEC 6%
	T-ARS	AFT Foam Tank	MILSPEC 6%
	T-AGS	AFT Foam tank	ANSULITE 3X3%
	AS (USS CABLE AND USS LAND)	FOAM Tank	MILSPEC 6%
	USNS SEAY & USNS PILILAAU	FWD Tank	AER-O-FOAM XL 3-3%
	USNS SEAY & USNS PILILAAU	AFT Tank	AER-O-FOAM XL 3-3%
	USNS STOCKHAM	FWD Tank	ANSULITE 3X3%
	USNS STOCKHAM	AFT Tank	ANSULITE 3X3%
	USNS STOCKHAM	HELO Deck Tank	ANSULITE 3X3%
	USS MOUNT WHITNEY	E/R Foam Tank	MILSPEC 6%
	USS MOUNT WHITNEY	HELO DECK Tank	MILSPEC 6%
	SBX-1	PORT Foam Tank	ANSULITE 3X3%
	SBX-1	STBD Foam Tank	ANSULITE 3X3%
	USNS WHEELER	FOAM Tank	ANSULITE 3X3%
	USNS IMPECCABLE	AFFF STATION	ANSULITE 3X3%
	USNS CATAWBA	FOAM Tank	ANSULITE 3X3%

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0568A	**CATEGORY "A"**	**2026-02-03**
ESB-CCSI-WATER MIST-SPRAY-SPRINKLER INSPECTION AND TEST (1 YR) (SCSI)		**Port Engineer's Name**

1.0 ABSTRACT

The purpose of this item is to describe the requirements to inspect, service and test the fixed water mist, water spray and sprinkler firefighting systems (non-magazine).

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSEA Drawing No. 830-8496378, Fire Control and Safety Plan

2.1.2 Test Procedure No. 543-344-8737, Local Water Based Fire Extinguishing System

2.1.3 Tech Manual No. S9555-FJ-MMC-010, Fire Extinguishing System X-Flow Water Mist (CS-4000)

2.1.4 IMO MSC.1/Circ.1432, 31 May 2012, Revised Guidelines for the Maintenance and Inspection of Fire Protection Systems and Appliances

2.1.5 MSC Drawing No. 118501, XFLOW WATER MIST SYSTEM

2.1.6 MSC Drawing No. 118501-002, XFLOW WATER MIST SYSTEM DETECTOR SYSTEM

2.1.7 MSC Drawing No. 118501-004, XFLOW WATER MIST SYSTEM LOOP 1 DIAGRAM

2.1.8 MSC Drawing No. 118501-005, XFLOW WATER MIST SYSTEM LOOP 2 DIAGRAM

2.1.9 MSC Drawing No. 118501-006, XFLOW WATER MIST SYSTEM PIPE DIAGRAM

2.2 Enclosures

2.2.1 Mist/Spray/Sprinkler Firefighting System - Check sheet

2.2.2 Mist/Spray/Sprinkler Firefighting System - Service Record

2.2.3 Placards and Markings

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location/Quantity

3.1.1 Locations:

a. Machinery Room 1 (5-56-1)

b. Machinery Room 2 (5-56-2)

c. Purifier Room (3-56-2)

d. Emergency Diesel Generator Room (A-34-2)

3.1.2 Quantity

a. Local Water Mist Skid, Novenco, Model X_Flow, One

3.2 Description: See References 2.1.2 and 2.1.3

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Definitions

5.2.1 Dry Standpipe: A standpipe system designed to have piping contain water only when the system is being utilized.

5.2.2 Wet Standpipe: A standpipe system designed to have piping contain water at all times.

5.2.3 Automatic System: A fire suppression or control device that operates immediately and automatically when its heat activated element is opened by heat from a fire, allowing water to discharge over a specified area. Automatic sprinkler systems are of the wet pipe type.

5.2.4 Deluge System: A sprinkler system employing open sprinklers or nozzles attached to a piping system and water supply. When the water supply valve opens, water flows into the piping system and discharges from all sprinklers or nozzles attached.

5.2.5 Nozzle: A device for use in applications requiring special water discharge patterns, directional spray, or other unusual discharge characteristics.

5.2.6 Sprinkler: The component of a fire sprinkler system that discharges water when the effects of a fire have been detected. They can be open or thermally activated.

5.3 This work item is not intended for systems protecting Ammunition Magazines.

5.4 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the thorough examination, testing and repair of the fixed Water Mist/Water Spray/Sprinkler firefighting systems using IMO/SOLAS and USCG requirements as guidance.

7.2 Coordinate with ship's Chief Engineer (CHENG) to have the associated equipment and system locked and/or tagged out during the performance of this work item. Secure and tag out the fixed Water Mist/Water Spray/Sprinkler firefighting systems and controls to prevent accidental discharge.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Conduct an annual **INSPECTION** of the fixed Water Mist/Water Spray/Sprinkler firefighting systems using References 2.1.1 through 2.1.9 and Enclosures 2.2.1 through 2.2.3 as guidance.

7.4.1 Inspect the fixed Water Mist/Water Spray/Sprinkler firefighting systems and fill out Enclosures 2.2.1 and 2.2.2. It is to be a thorough examination of the entire system including all sprinklers, nozzles, piping, pipe supports, controls, sensors, actuators, release stations, pressure tank, control/section valves, system valves, gauges, switches, signal devices, sirens and alarms and physical condition to verify that the system is in good operating condition.

7.4.2 Verify all markings, operating instructions and warning placards are in place, unobstructed and clearly legible. See Enclosure 2.2.3 for guidance. Operating instructions are to be in a conspicuous place at or near the controls.

7.4.3 Conduct an external visual examination of the associated tanks (pressure and accumulator) completing Enclosures 2.2.1 and 2.2.2. Identify any obvious physical damage, dents, bulging, corrosion, and leaks. Verify each tank is clearly marked and record the water treatment data including name of manufacturer, type, concentration, and tank capacity.

7.4.4 Inspect the condition of the tank level and pressure indicating system.

7.4.5 Verify and record the air pressure and water level in each tank. The measured data must be compared with the manufacturers recommended parameters and previous readings.

7.4.6 Draw a representative sample of the freshwater and/or antifreeze solution from each tank and test its specific gravity. Inspect the water quality for evidence of sedimentation, corrosion, or scale-forming minerals. Test for adequate freeze protection. Provide a copy of the test results to the OMT Rep.

7.4.7 Verify all control/section valves are in the correct position.

7.4.8 Check all valves, fittings, piping and connections for valve alignment, damage, corrosion, and evidence of leakage.

7.4.9 Inspect sprinkler pipe hangers, braces, and supports verifying they are not damaged, loose or unattached.

7.4.10 Inspect all alarms (control valve monitoring, fire pump status, sprinkler pump status, water tank low level, water tank pressure, zone waterflow, etc.…), pressure switches, flame detectors, smoke detectors and supervisory signal initiating devices for proper operation. An audible alarm shall be given at the central safety station and the navigation bridge within 30 seconds of waterflow.

7.4.11 Inspect heat tracing (if applicable).

7.4.12 Externally examine the sprinkler pumps, couplings, and motors. Check alignment of the pumps and motors. Rotate the pumps by hand to confirm freedom of movement. With assistance from ships force run the sprinkler pumps and check for proper rotation, output pressure, leakage, excessive noise or vibration or overheating. Check electrical voltage to motors.

7.4.13 Verify all pump relief valves are properly set and operating correctly.

7.4.14 Cycle all control valves manually.

7.4.15 Cycle all automated control valves from the control panel.

7.4.16 Inspect and test all remote release stations.

7.4.17 Visually inspect all sprinklers, nozzles and sprinkler guards. Any that show signs of leakage, corrosion that would impair performance, physical damage, loss of fluid in the glass bulb heat-responsive element, loading detrimental to performance or paint other than applied by the Manufacturer require replacement.

7.4.18 Inspect and verify the discharge piping, sprinklers and nozzles are clear of any obstructions. Sprinklers and nozzles should be visually examined to ensure they have not been obstructed by the storage of spare parts or a new installation of structure or machinery.

7.4.19 Inspect and verify the necessary quantity of spare sprinkler heads are carried onboard for all types and ratings installed on the ship as follows:

Total number of heads in the system	*Required number of spares*
< 300	6
300 to 1000	12
> 1000	24

7.4.20 Inspect and verify that a sprinkler wrench is onboard.

7.4.21 Inspect the International Shore Connection fittings, both port and starboard side. Verify the connections are colored and marked so they are easily located and identified.

7.4.22 If the system is pre-primed with water containing a fire suppression enhancing additive and/or an antifreeze agent, it is to be inspected and tested as specified by the manufacturer to verify its effectiveness.

7.4.23 When inspections reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required.

7.5 Conduct annual **MAINTENANCE** of the fixed Water Mist/Water Spray/Sprinkler firefighting systems using References 2.1.1 through 2.1.9 and Enclosures 2.2.1 through 2.2.3 as guidance.

7.5.1 Open all filters/strainers (including saltwater supply as applicable), examine, clean of all debris and contamination and reassemble.

7.6 Conduct annual **TESTING** of the fixed Water Mist/Water Spray/Sprinkler firefighting systems using References 2.1.1 through 2.1.9 and Enclosures 2.2.1 through 2.2.3 as guidance.

7.6.1 Temporarily install plastic socks using transparent four mil poly tubing roll over all water mist nozzles (inclusive of nozzles in the overhead) directing water to bilge or containment area away from equipment. Remove socks after all testing is completed.

NOTE: Contractor shall be responsible for removal of all liquids generated and cleaning of bilges, to ensure affected bilges are left in a clean and dry condition.

7.6.2 Test all system cross connections to other sources of water supply for proper operation.

7.6.3 Test emergency power supply switchover (where applicable).

7.6.4 Flow test all pumps for proper pressure and capacity.

7.6.5 For Dry Pipe systems, confirm the sprinklers, nozzles and pipes are free of obstructions, by blowing through the lines with compressed air or nitrogen as applicable or other suitable means recommended by the manufacturer. Nozzles are to be removed for this test. To be coordinated and conducted in the presence of the OMT REP and ABS Surveyor.

7.6.6 For automatic systems, functionally test two randomly selected sprinkler head/nozzles per section in ten sections; twenty 20 sprinkler head/nozzles in total. Sprinkler heads/nozzles are to be chosen by the ABS Surveyor but are to be different from any previously tested within the last five years. The functional test is defined as a test that demonstrates operation and flow of the water from sprinkler head/nozzle. To be coordinated and conducted in the presence of the OMT REP and ABS Surveyor.

[FOR ESB, ESD and TAKE CLASS ONLY] Note: Nozzles in the Engine Rooms are located approximately 20 Meters (60 Feet) above the deck in the Machinery Rooms 1 & 2.

a. Test the water quality of each corresponding section for evidence of sedimentation, corrosion, or scale-forming minerals. Test for adequate freeze protection.

b. Document any sprinkler head/nozzles that fail and note where any two have failed within the same test section.

c. Submit a test report summary to the OMT Rep.

d. If three or more of each type sprinkler head/nozzle fails then ‘Part 2 - Extended Testing’ is required per Reference 2.1.4. Extended Testing will be the subject of a CCO if deemed necessary by the OMT rep.

7.6.7 Coordinate and conduct a functional test of the system, with assistance from ships force, demonstrating satisfactory operation of the fixed Water Mist/Water Spray/Sprinkler firefighting systems in the presence of the OMT REP and ABS Surveyors (without releasing water into the space). Verify proper operation using the test valves for each section. To include satisfactory operation of all sources of water supply, release stations, power sources, controls, automation, sensors, actuators, valves, pumps, audible and visual alarms as stated in References 2.1.2 through 2.1.9.

7.6.8 Each section drained or flushed shall be refilled upon completion in accordance with the manufacturer’s guidelines. Test and record the water quality as a baseline for future monitoring. Submit results to the OMT Rep.

7.7 Tags and Labels

7.7.1 Tags and labels documenting inspection, maintenance, recharging or testing shall be affixed so as not to obstruct the equipment use, classification or manufacturers labels.

7.7.2 Each storage tank shall have a tag or label securely attached that indicates inspection and maintenance was performed. The tag at a minimum shall identify the following:

a. Date (month/day/year) of Inspection

b. Name of the Agency/Company performing the Inspection

7.8 Upon completion of all inspections, tests and repairs return the Water Mist/Water Spray/Sprinkler Firefighting Systems to a ready for service condition. Conduct a final walk around survey with the OMT REP and ships Master to verify system alignment and status.

7.9 Reports

7.9.1 Upon completion of all inspections, maintenance and tests the contractor will prepare and submit the results of the water quality tests, System Check & Service Record forms shown in Enclosures 2.2.1 and 2.2.2. They will document the examination of the system and final “as released” conditions. Submit one electronic copy in Portable Document Format (PDF) to the OMT REP.

7.9.2 All reports, records and checklists will be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.10 Painting/Lagging

7.10.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.10.2 Renew lagging where damaged or disturbed.

7.11 Manufacturer’s Representative

7.11.1 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only fire extinguishing system service companies certified and recognized as an ABS Recognized Service Supplier shall be used. To include oversight of all aspects of equipment inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable. In addition, only OEM authorized parts will be used.

https://www.msc.usff.navy.mil/Business-Opportunities/Contracts/Qualification-for-Items-Critical-to-Shipboard-Safety-on-MSC-Vessels/

7.11.2 It is critical that all persons working on the system must be certified and fully knowledgeable in its operation and repair. Persons in training to be certified are permitted to perform maintenance and recharging of extinguishing systems under the direct supervision and in the immediate presence of a certified person.

7.11.3 Companies and persons performing maintenance and recharging of extinguishing systems are to have available the appropriate servicing manuals, correct tools, recharging materials, lubricants, and manufacturer’s replacement parts.

7.12 Preparation of Drawings

7.12.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied.

8.0 GENERAL REQUIREMENTS: None additional

MIST/SPRAY/SPRINKLER FIREFIGHTING SYSTEM – CHECKSHEET

Vessel: ____________ **IMO #:** ____________
System Location: ____________
Area Protected: ____________ **System Operation:** automatic/manual
Tank Capacity: ____________ **Water Treatment:** ____________

Shipyard: ____________ **Inspector:** ____________
Service Company: ____________ **Date:** ____________

To be completed by the system inspector at the time of inspection or test. The inspection of the firefighting system will at a minimum include checks and tests of the following:

#	COMPONENT	SAT	UNSAT	NA	COMMENTs
1	Fire extinguishing system and controls secured to prevent accidental discharge	☐	☐	☐	
2	Placards and Instructions in place and legible	☐	☐	☐	
3	Externally examine storage/pressure tanks	☐	☐	☐	
4	Internally examine storage/pressure tanks (5YR)	☐	☐	☑	Not due
5	Verify tank level and pressure indicators are functioning properly	☐	☐	☐	
6	Water level and pressure are in the operable range	☐	☐	☐	
7	Fresh water sampled from tanks and analyzed	☐	☐	☐	
8	System piping, valves and sprinklers/nozzles visually inspected	☐	☐	☐	

ENCLOSURE 2.2.1

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0568A	**CATEGORY "A"**	**2026-02-03**
ESB-CCSI-WATER MIST-SPRAY-SPRINKLER INSPECTION AND TEST (1 YR) (SCSI)		**Port Engineer's Name**

#	COMPONENT	SAT	UNSAT	NA	COMMENTs
9	Sprinkler pipe hangers, braces and supports inspected	☐	☐	☐	
10	Inspect waterflow device, pressure switches and signal devices	☐	☐	☐	
11	Inspect heat tracing (if applicable)	☐	☐	☐	
12	Replace system batteries (5YR)	☐	☐	☑	Not due
13	Examine sprinkler pump and motor alignment, freedom of movement, correct rotation	☐	☐	☐	
14	Test sprinkler pump operation; output pressure, vibration and motor voltage	☐	☐	☐	
15	Relief valve setting verified and tested	☐	☐	☐	
16	Verify all Control/Section valves are in correct position	☐	☐	☐	
17	Control valves cycled manually	☐	☐	☐	
18	Control valves cycled from control panel	☐	☐	☐	
19	Internally examine control/section valves (5YR)	☐	☐	☑	Not due
20	Internally examine the non-return valve connecting the sprinkler system to firemain (5YR)	☐	☐	☑	Not due
21	Remote release stations examined and tested	☐	☐	☐	

ENCLOSURE 2.2.1

#	COMPONENT	SAT	UNSAT	NA	COMMENTs
22	Inspect all sprinklers and nozzles	☐	☐	☐	
23	Verify qty of spare nozzles and sprinkler wrench carried onboard	☐	☐	☐	
24	Inspect international shore connections	☐	☐	☐	
25	Test quality of system water and additives	☐	☐	☐	
26	Open and clean SW strainers	☐	☐	☐	
27	Test all system cross connections to other sources of water supply	☐	☐	☐	
31	Test emergency power supply switchover	☐	☐	☐	
32	Flow test all sprinkler pumps for proper pressure and capacity	☐	☐	☐	
33	Dry System - Blow test discharge piping and nozzles proving clear (if applicable)	☐	☐	☐	
34	Automatic System – functionally test randomly selected sprinkler/nozzles (if applicable)	☐	☐	☐	
	Flush all RO-RO deck deluge system piping. (if applicable) (5YR)	☐	☐	☑	Not due
	Gas and water cylinders hydrostatically pressure tested and internal exam (10YRs)	☐	☐	☑	Not due

ENCLOSURE 2.2.1

#	COMPONENT	SAT	UNSAT	NA	COMMENTs
35	Perform a final functional test of system	☐	☐	☐	
36	Alarms tested (Audible and Visual)	☐	☐	☐	
37	Detectors and supervisory signal initiating devices tested	☐	☐	☐	
38	System refilled and treated	☐	☐	☐	
39	System returned to active service	☐	☐	☐	
40	Inspection tags attached	☐	☐	☐	

SAT = Satisfactory **UNSAT** = Unsatisfactory **N/A** = Not applicable

This system, as specified, has been inspected and tested according to the standards cited in the work item.

Signed: ____________________ Date: ____________________

Printed Name: ____________________ Phone: ____________________

Title: ____________________

Organization: ____________________

ENCLOSURE 2.2.1

MIST/SPRAY/SPRINKLER FIREFIGHTING SYSTEM – 1YR SERVICE RECORD

Vessel: ____________ **IMO #:** ____________

System Location: ____________

Area Protected: ____________ **System Operation:** automatic/manual

Tank Capacity: ____________ **Water Treatment:** ____________

Shipyard: ____________ **Inspector:** ____________

Service Company: ____________ **Date:** ____________

To be completed by the system inspector at the time of inspection or test. The inspection of the fire fighting system will at a minimum include checks and tests of the following:

COMPONENT	EXAM	RECORDED DATA	SET POINT
Detector-Alarm System	MFG		
	Model No.		
	Serial No.		
Sprinkling Pump(s)	MFG		
	Model No.		
	Serial No.		
	Serial No.		
	Serial No.		
Fixed Sprinklers and Nozzles	Mfg		
	Model No.		qty
	Mfg		
	Model No.		qty
	Mfg		
	Model No.		qty

ENCLOSURE 2.2.2

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0568A	CATEGORY "A"	2026-02-03
ESB-CCSI-WATER MIST-SPRAY-SPRINKLER INSPECTION AND TEST (1 YR) (SCSI)		Port Engineer's Name

COMPONENT	EXAM	RECORDED DATA	SET POINT
Fresh Water Storage Tank	Capacity (gals)		gals
	External Visual Inspection (date)		1YR
	Internal Visual Inspection (date)		5YR
	Measured FW Charge (date)		1YR
	Measured FW Charge (gals)		gal
	FW Liquid Level (% Full)		%
	FW Tank Pressure (psi)		psi
	Low level alarm setting (%)		%
	Samples of water tested (date)		1YR
	Hydro test and internal exam on gas and water cylinders (date)		10YR
Water Mist/Water Spray/Sprinkler System	Visual inspection of system made including sprinklers/nozzles(date)		1YR
	Test all system cross connections to other sources of water supply		1YR
	Obstruction test of Dry Pipe systems. Blow dry compressed air or N2 through discharge piping and nozzles proving clear		1YR
	Functional test of Automatic systems. Flow test 2 randomly selected sprinklers/nozzles per section in 10 sections as chosen by ABS surveyor. Discharge water proving they are clear of obstruction		1YR
	Examine all sprinkler pumps and motors		1YR

	Flow test all water supply and sprinkler pumps for proper pressure and capacity		1YR

ENCLOSURE 2.2.2

COMPONENT	EXAM	RECORDED DATA	SET POINT
	Emergency power switchover tested		1YR
	Test all ventilation and fuel shut-off controls connected to the fire extinguishing system		1YR
	Functional test of system demonstrating SAT operation		1YR
	International Shore Connection examined		1YR
	System batteries replaced (date)		5YR
	Flush RO-RO deck deluge system (date)		5YR
Alarms	Inspect all alarms including status indication, flame/heat/smoke detectors, switches and signal initiating devices		1YR
	Functional Test (audible and visual alarms)		1YR
Relief valve	Pump relief valve pop test (date)		1YR
	Pump relief valve pop test (psi)		psi
Control valves	Verify all Control/Section valves are in correct position (date)		1YR
	Control valves cycled by local and remote (date)		1YR
	Control/Section valves internally examined (date)		5YR
	Sprinkler System-Firemain cross connection valve internally examined (date)		5YR
Filters & Strainers	Examined and verified free of debris and contamination		1YR

Release Stations	Functional test		1YR
Firemain	Pressure		XXX psi

ENCLOSURE 2.2.2

Inspector Certification:

This system, as specified, has been inspected and tested according to the standards cited in the work item.

Signed: ____________________ Date: ____________________

Printed Name: ____________________ Phone: ____________________

Title: ____________________

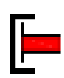
Organization: ____________________

ENCLOSURE 2.2.2

PLACARD AND MARKINGS

Vessel Type	Requirement	Placard verbiage
Cargo and Miscellaneous Vessels	46 CFR §97.37-10	Each branch line control valve of all fire extinguishing systems will be plainly and permanently marked indicating the spaces served.
Cargo and Miscellaneous Vessels	46 CFR §97.37-13	The control cabinets or spaces containing valves or manifolds for the various fire extinguishing systems must be distinctly marked in conspicuous red letters at least 2 inches high: “[STEAM/CARBON DIOXIDE/CLEAN AGENT/FOAM/WATER SPRAY—as appropriate] FIRE APPARATUS.”.
Tank Vessels	§35.40-18	(a) Water spray system apparatus will be marked: “WATER SPRAY SYSTEM,” as appropriate, in not less than 2-inch red letters. (b) The control valve, and its control if located remotely, will be distinctly marked to indicate the compartment protected.
Passenger Vessels	46 CFR §78.47-13	(a) The fire detection, alarm, and automatic sprinkler indicators in the engine room must be identified by at least 1-inch red lettering as “FIRE ALARM” or “SPRINKLER ALARM” as appropriate. Where such alarm indicators on the bridge or in the fire control station do not form a cabinet, the indicators must be suitably identified as above.
Passenger Vessels	46 CFR §78.47-17	Each control cabinet or space containing valves or manifolds for a fire extinguishing system must be distinctly marked in conspicuous red letters at least 2

PLACARD AND MARKINGS

Vessel Type	Requirement	Placard verbiage
		inches high: "[CARBON DIOXIDE/STEAM/FOAM/WATER SPRAY/MANUAL SPRINKLING/AUTOMATIC SPRINKLING/CLEAN AGENT—as appropriate] FIRE SYSTEM.".
All	MSC.1/Circ.1387	Where automatically operated systems are installed for localized fire protection without the necessity of engine shut-down, personnel evacuation, shutting down of forced ventilation fans, or sealing of the space a warning notice must be posted outside each entry point stating the type of medium being used and the possibility of automatic release.
Ships above 500 tons gross ton	SOLAS II-2, 10 46 §76.10-10(C)	International Shore Connection shall be colored and marked so they are easily access from the shore access point (ie; gangway location) 

ENCLOSURE 2.2.3

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0569	**CATEGORY "A"**	**2026-02-03**
ESB-CCSI-FIRE AND SMOKE DETECTION AND ALARM SYSTEM (1 YR) (SCSI)		**Port Engineer's Name**

1.0 ABSTRACT

The purpose of this item is to inspect, service, and test the ship’s fire and smoke detection system and associated alarm systems.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSEA Drawing No. 830-8496378, Fire Control and Safety Plan

2.1.2 NAVSEA Drawing No. 436-8468424 BWD Fire & Smoke Detect System

2.1.3 NASSCO Test Procedure 541-344-8665, Test Procedure Detection System Tests

2.1.4 Tech Manual TG511-AD-MMC-010, Addressable Fire Detection System, Installation, Operation and Maintenance Manual

2.1.5 IMO MSC.1/Circ.1432, 31 May 2012, Revised Guidelines for the Maintenance and Inspection of Fire Protection Systems and Appliances

2.1.6 NFPA Code 72 National Fire Alarm and Signaling Code (Available Commercially)

2.2 Enclosures

2.2.1 Fire and Smoke Detection and Alarm System – Inspection and Test Form

2.2.2 Fire and Smoke Detection and Alarm System – Initiating Device Test Results Form

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations: See References 2.1.1 and 2.1.2

3.1.2 Quantity: See References 2.1.1 and 2.1.2

a. 195 Smoke Detectors, Dry Location (I)

b. 240 Smoke detectors, Wet Location (Iw)

c. 11 Heat Detectors, Wet Location (T2w)

d. Six Heat Detectors, Wet Location (T1w)

e. Six Heat Detectors, Cold Space (T)

f. 14 Combined Smoke and Heat Detectors, (Tx)

g. Four Flame Detectors, Wet Location (F)

h. Ten Flame Detectors, Wet Space (Fir)

i. 17 Smoke and Heat Detectors, (S/H)

j. One Main Fire Control Cabinet, Fwd

k. One Repeater Fire Control Panel, M 4.3 Aft

3.2 Item Description/Manufacturer’s Data: Fire and smoke detection and alarm system
Manufacturer: Consillium Marine and Safety AB
Salwico Cargo Addressable Fire Detection System.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service, and test the fire and smoke detection system identified in section 3.0 using References 2.1.1 through 2.1.5 as guidance. Accomplish inspection and testing in accordance with IMO, SOLAS, and manufacturer’s requirements.

7.2 Fire detection and alarm systems may output signals to other fire safety systems including, but not limited to, paging systems, fire alarm or public address systems, fan stops, fire doors, fire dampers, sprinkler systems, smoke extraction systems, low-location lighting systems, fixed local application fire extinguishing systems, and closed-circuit television systems.

7.3 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship's lock out/tag out policy.

7.4 Conduct an annual inspection of all fire and smoke detection and alarm systems and complete Enclosures 2.2.1 and 2.2.2. It is to be a thorough examination of the equipment and system ensuring there are no changes that affect performance and verifying that the system and components are in good operating condition. It is to include inspection for ship modifications, changes in environmental conditions, device locations, physical obstruction, physical damage and degree of cleanliness.

7.4.1 Examine the control equipment to verify normal system condition to include fuses, interfaced equipment, lamps and LEDs, power supply and trouble signals.

7.4.2 Verify location and condition of emergency communication equipment including public address system, and general alarm.

7.4.3 Examine batteries for corrosion or leakage. Verify month and year of manufacture, tightness of connections and level of electrolyte.

7.4.4 Examine UPS and verify meters, lamps and LEDs indicate normal operating status. Verify proper fuse ratings.

7.4.5 Examine location and condition of the remote annunciators.

7.4.6 Verify location and condition of fiber optic cable connections.

7.4.7 Conduct an external visual examination of all detectors and pull stations. Identify any obvious physical damage, dirt, corrosion, and missing device. Verify each is in the correct location, rigidly mounted, clearly marked and record all data including device/sensor type, name of manufacturer, sensitivity/range, explosion proof, and if weatherproof. For air sampling devices, verify that any in-line filters are clean. For detectors, verify the area requiring detection is not obstructed or outside the detectors field of view.

7.4.8 Clean detectors whenever there is evidence of dirt or other contamination.

7.4.9 Verify condition of any fire alarm control interface with other safety systems to include such as fire dampers, fire zone doors, sliding watertight doors, and automatic releases.

7.4.10 Verify location and condition of all notification appliances (audible alarms, visual alarms, combination a/v alarms, and exit markings).

7.4.11 Verify audible alarms are identified by at least 1-inch red lettering as “FIRE ALARM” as required by 46 CFR §78.47-13.

7.4.12 Verify location of the mass notification system and inspect to ensure the system is functioning properly. Inspection is to include control equipment, fuses, interfaces, lamps and LEDs, main power supply, UPS, initiating devices, notification devices, antennas and transceivers.

7.4.13 Verify system devices comply with the following standards, per 46 CFR §161.002-1:

a. Control units

b. Detectors

c. Smoke detectors

d. Flame detectors

e. Audible alarms

f. Visual alarms

g. Manual Signaling Boxes

7.4.14 Verify location and condition of supervising station – receivers.

7.4.15 Verify that clear information about the operation of the fire detection and alarm system is displayed on or adjacent to its control panels.

7.4.16 Verify the detection and alarm system has had no modifications made since last inspection.

7.4.17 When inspections reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required.

7.5 Functionally test the fire and smoke detection and alarm system. It is to include all devices, fault alarms and signals, and visual and audible alarms using References 2.1.1 through 2.1.3 as guidance. Complete Enclosures 2.2.1 and 2.2.2 recording the results of all inspections and tests.

7.5.1 Coordinate all testing prior to accomplishment. Advance notification is to be given to the Shipyard Superintendent and OMT REP. With assistance from ship crew, make announcements over the public address system prior to initiating any alarms.

7.5.2 Test and verify the control equipment functions by receiving correct alarm, supervisory and trouble signals (input); operation of evacuation signals and auxiliary functions (outputs); circuit supervision including detection of open circuits and ground faults; and power supply supervision for detection of loss of AC power and disconnection of secondary batteries. Verify rating and supervision of fuses. Illuminate all lamps and LEDs. Test the primary power under maximum load conditions, including all alarm appliances requiring simultaneous operation. Test the secondary power supply separately.

7.5.3 Develop a testing schedule to ensure that every initiating device is tested. Actuate every initiating device, including detectors and pull stations and verify proper operation and receipt of correct signal at the supervising station within 90 seconds. Upon completion of the test, restore the system to its functional operating condition. Activation of any detector or manual pull station must cause an audible and visual alarm signal at the control panel. Complete Enclosures 2.2.1 and 2.2.2 recording the results.

7.5.4 Verify that the detector/alarm initiating device is alarming the proper zone on the Master Panel and all Remote Annunciators.

7.5.5 Arrange the testing schedule so that all zones will be alarmed during testing. Conduct a cross-zone detection test. The last alarm given on a particular zone should be silenced only (not reset), then, to verify that the subsequent alarm is functioning properly, place another zone into alarm, then the system can be reset.

7.5.6 Test the detectors using the manufacturer’s guidance. Care and caution are to be used to prevent damage to the detectors and pull stations while testing.

7.5.7 In addition, test all smoke and flame type detectors to ensure that each is within its listed and marked sensitivity range using one of the following methods:

a. Calibrated test method

b. Manufacturers calibrated sensitivity test instrument

c. Listed control unit arranged for the purpose

d. Other approved calibrated flame sensitivity test method that is directly proportional to the input signal from a fire.

e. Other calibrated smoke sensitivity test method that is approved by the authority having jurisdiction (AHJ).

7.5.8 If detectors are designed to be adjustable, adjust to bring them into the approved range.

NOTE: DO NOT DETERMINE FLAME DETECTOR SENSITIVITY BY USING A LIGHT SOURCE THAT ADMINISTERS AN UNMEASURED QUANTITY OF RADIATION AT AN UNDEFINED DISTANCE FROM THE DETECTOR

7.5.9 Verify that the detectors are self-restoring.

7.5.10 Verify that if the alarm signal has not been acknowledged within two minutes, an audible fire alarm is automatically sounded throughout the crew accommodations and service spaces, control stations, and manned machinery spaces. Verify that each Bridge, Quarter-Deck Office and the Engine Room Control Station is receiving the proper delay before any alarms are sent.

7.5.11 Demonstrate that all detection and alarm system monitors have functional power supplies necessary for the operation of the system during loss of power and fault conditions. Disconnect the primary (main) power supplies and verify a trouble or fault indication is given. Measure the systems standby and alarm current demand and verify the batteries ability to meet these requirements. Verify communication between sending and receiving units under both primary and secondary power. Reconnect the primary power upon completion.

7.5.12 Test the UPS for proper operation.

7.5.13 Prior to conducting any battery test, verify that all system software stored in volatile memory is protected from loss. Test the performance of the battery charger by measuring the voltage across the battery when they are fully charged and connected to the charger. Test the performance of the batteries by disconnecting the charger and conducting a load test (full fire alarm load) on the batteries verifying the voltage does not fall below the manufacturer’s specified levels.

7.5.14 Verify the local and remote annunciators operate and identify correctly under all conditions.

7.5.15 Test every notification device (audible and visual alarm) verifying each operates properly.

7.5.16 Check the supervisory circuits and indicators on the control panel and each annunciator panel by operating the Lamp Test Switch. All visual indicators should be lit, and the trouble horn should sound.

7.5.17 Test for automatic supervision (monitoring) of any fault in all external wiring such as an open circuit, short circuit, or ground. Any of these conditions should result in a visual fault/trouble signal.

7.5.18 Verify that every fault (such as an open circuit, short circuit, or ground fault) does not prevent the continued individual identification of the remaining detectors and manual pull stations.

7.5.19 Test and verify that the system automatically resets to a normal operating condition after alarm and fault situations are cleared.

7.5.20 Detection and alarm systems may accept signals from other safety systems. Test proper functioning of all signals received from other safety systems (For example, a signal initiated from actuation of an automatic sprinkler valve or fusible link may be sent to a fire detection and alarm system).

7.5.21 Test all interconnected auxiliary equipment or systems such as remotely controlled fire doors, sliding watertight doors, fire dampers, fan shutdowns, and equipment shutdowns for proper activation.

7.6 Accomplish maintenance on the fire and smoke detection system:

7.6.1 Replace those smoke or flame detectors found outside the approved range of sensitivity.

7.6.2 For estimating purposes, assume ten smoke, five heat, five smoke/heat, five flame detectors and ten pull stations will require replacement.

7.6.3 Replace any system batteries when the recharged battery voltage or current falls below the manufacturer’s recommendation.

7.6.4 Install new detector labels where found missing. Replace detector labels where found to be incorrectly labeled. Estimate for 3% of all labels on systems listed in 3.1.1 to be replaced.

7.7 Coordinate and conduct a final functional test demonstrating satisfactory operation of the fire and smoke detection and alarm systems in the presence of the OMT REP.

7.8 Upon completion of all inspections, tests and repairs return the Fire and Smoke Detection and Alarm Systems to a ready for service condition. Conduct a final walk around survey with the OMT REP to verify status.

7.9 Company certified for this work item per 7.12 to provide operation and maintenance training to crew, approximately four hours, and provide certificate to each individual in attendance.

7.10 Reports

7.10.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.10.2 Upon completion of all inspections, tests and maintenance, prepare and submit the System Check sheet and Test Result forms shown in Enclosure 2.2.1 and 2.2.2. The reports are to be typewritten reports documenting the examination of the system and final “as released” conditions. Submit one electronic copy in Portable Document Format (PDF).

7.10.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.11 Painting: None additional

7.12 Manufacturer’s Representative

7.12.1 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only fire extinguishing system service companies certified and recognized as an ABS Recognized Service Supplier shall be used. To include oversight of all aspects of equipment inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable. In addition, only OEM authorized parts will be used.

https://www.msc.usff.navy.mil/Business-

Opportunities/Contracts/Qualification-for-Items-Critical-to-Shipboard-Safety-on-MSC-Vessels/

7.12.2 It is critical that all persons working on the system must be certified and fully knowledgeable in its operation and repair. Persons in training to be certified are permitted to perform maintenance under the direct supervision and in the immediate presence of a certified person.

7.12.3 Companies and persons performing maintenance are to have available the appropriate servicing manuals, correct tools, materials, lubricants, and manufacturer’s replacement parts.

7.13 Preparation of Drawings

7.13.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 Fire and Smoke Detection and Alarm System – Inspection and Test Form

Vessel: ____________ **IMO #:** ____________

Shipyard: ____________ **Date of Inspection:** ____________

Service Company: ____________ **Inspector:** ____________

Phone: ____________

Email address: ____________

To be completed by the system inspector at the time of inspection or test. The inspection of the fire and smoke detection system will at a minimum include checks and tests of the following:

SYSTEM VERIFICATION

1. TYPE

- ☐ Fire Alarm System
- ☐ Smoke Alarm System
- ☐ Security System
- ☐ Fire/Smoke Combination
- ☐ Other (specify): ____________

2. CONTROL UNIT

Manufacturer: ____________ Model: ____________

Serial #: ____________

☐ An owner's manual, manufacturer's instructions, a written sequence of operation and a copy of the system drawing is stored onboard.

3. SYSTEM SOFTWARE

Software revision ____________ Last ____________

ENCLOSURE 2.2.1 Fire and Smoke Detection and Alarm System – Inspection and Test Form

☐ A copy of the ship specific software is stored onboard.

4. SYSTEM POWER

Primary Power

Control Panel Input voltage: ____________ Control panel amps: ____________

Secondary System

Nominal voltage: ____________ Amp/hour rating: ____________

In standby mode: ____________ In alarm mode: ____________

Batteries (date of mfg): ____________

5. ANNUNCIATORS

Manufacturer - No.1: ______________ Model: ______________

Manufacturer - No.2: ______________ Model: ______________

Manufacturer - No.3: ______________ Model: ______________

Manufacturer - No.4: ______________ Model: ______________

6. DETECTORS

Thermal (qty): __________

Smoke (qty): __________

Flame (qty): __________

Other (qty): __________

7. PULL STATIONS

Manual stations (qty): __________

8. ALARMS

Audible (qty): __________ A/V Combination ______________

_____________ (qty): _________________

Visual (qty): _____________

9. NOTIFICATIONS MADE PRIOR TO TESTING

ENCLOSURE 2.2.1 Fire and Smoke Detection and Alarm System – Inspection and Test Form

Shipyard Superintendent: ________________________________ (time/date)

Vessel Master: ________________________________ (time/date)

Vessel Chief Engineer: ________________________________ (time/date)

Port Engineer: ________________________________ (time/date)

ENCLOSURE 2.2.2 FIRE AND SMOKE DETECTION AND ALARM SYSTEM INSPECTION AND TEST FORM

1. SUMMARY of RESULTs

#	DESCRIPTION	VISUAL INSPECTION	FUNCTIONAL TEST	SAT	UNSAT	NA	COMMENTs
1	Secure interconnected fire extinguishing systems and controls to prevent accidental discharge	☐	☐	☐	☐	☐	
2	Placards and Instructions in place and legible	☐	☐	☐	☐	☐	
3	Verify no modifications have been made since last inspection.	☐	☐	☐	☐	☐	
4	Control Unit and Related Equipment	☐	☐	☐	☐	☐	
	Lamps/LEDs/LCDs	☐	☐	☐	☐	☐	
	Fuses	☐	☐	☐	☐	☐	
	Input Signals (alarm/supervisory/trouble)	☐	☐	☐	☐	☐	

	Output Signals	☐	☐	☐	☐	☐	
	Supervision (wire faults/power loss/fuses)	☐	☐	☐	☐	☐	

ENCLOSURE 2.2.2 FIRE AND SMOKE DETECTION AND ALARM SYSTEM INSPECTION AND TEST FORM

#	DESCRIPTION	VISUAL INSPECTION	FUNCTIONAL TEST	SAT	UNSAT	NA	COMMENTs
	Local Annunciators	☐	☐	☐	☐	☐	
	Remote Annunciators	☐	☐	☐	☐	☐	
	Input from other safety systems	☐	☐	☐	☐	☐	
	Fiber Optic Cable Connections	☐	☐	☐	☐	☐	
	Other (specify)	☐	☐	☐	☐	☐	
5	Power Supplies	☐	☐	☐	☐	☐	
	Primary AC power	☐	☐	☐	☐	☐	
	Secondary Power	☐	☐	☐	☐	☐	
	UPS	☐	☐	☐	☐	☐	

	Battery condition	☐	☐	☐	☐	☐	
	Load voltage	☐	☐	☐	☐	☐	
	Discharge test	☐	☐	☐	☐	☐	
	Charger test	☐	☐	☐	☐	☐	
6	Notification Appliance	☐	☐	☐	☐	☐	

ENCLOSURE 2.2.2 FIRE AND SMOKE DETECTION AND ALARM SYSTEM INSPECTION AND TEST FORM

#	DESCRIPTION	VISUAL INSPECTION	FUNCTIONAL TEST	SAT	UNSAT	NA	COMMENTs
	Audible alarm	☐	☐	☐	☐	☐	
	Visual alarm	☐	☐	☐	☐	☐	
	Combination A/V alarm	☐	☐	☐	☐	☐	
	General alarm	☐	☐	☐	☐	☐	
7	Initiating Device – Detectors	☐	☐	☐	☐	☐	
	Heat	☐	☐	☐	☐	☐	
	Flame	☐	☐	☐	☐	☐	
	Security	☐	☐	☐	☐	☐	

	Flooding	☐	☐	☐	☐	☐	
	Other (specify)	☐	☐	☐	☐	☐	
8	Initiating Device - Pull Stations	☐	☐	☐	☐	☐	
9	Emergency Communications	☐	☐	☐	☐	☐	
	Public Address system	☐	☐	☐	☐	☐	
	Phone handsets	☐	☐	☐	☐	☐	
	Emergency Elevator comm	☐	☐	☐	☐	☐	

ENCLOSURE 2.2.2 FIRE AND SMOKE DETECTION AND ALARM SYSTEM INSPECTION AND TEST FORM

#	DESCRIPTION	VISUAL INSPECTION	FUNCTIONAL TEST	SAT	UNSAT	NA	COMMENTs
	Other	☐	☐	☐	☐	☐	
10	Other Monitored Systems	☐	☐	☐	☐	☐	
	Fire Pumps	☐	☐	☐	☐	☐	
	Special Fire Suppression system	☐	☐	☐	☐	☐	
	Other (specify)	☐	☐	☐	☐	☐	
11	Auxiliary Functions	☐	☐	☐	☐	☐	

USS Hershel Williams			
(ESB 4)			
AUXILIARY MACHINERY		CONTRACT NO. N68171-26-RN-003	
ITEM NO. 0569	CATEGORY "A"	2026-02-03	
ESB-CCSI-FIRE AND SMOKE DETECTION AND ALARM SYSTEM (1 YR) (SCSI)		Port Engineer's Name	

	Door releasing device	☐	☐	☐	☐	☐	
	Door unlocking	☐	☐	☐	☐	☐	
	Fan shutdown	☐	☐	☐	☐	☐	
	Smoke damper operation	☐	☐	☐	☐	☐	
	Elevator recall	☐	☐	☐	☐	☐	
	Elevator shunt trip	☐	☐	☐	☐	☐	
	Paging system	☐	☐	☐	☐	☐	
	Mass notification system	☐	☐	☐	☐	☐	
	Closed-circuit television systems	☐	☐	☐	☐	☐	

ENCLOSURE 2.2.2 FIRE AND SMOKE DETECTION AND ALARM SYSTEM INSPECTION AND TEST FORM

#	DESCRIPTION	VISUAL INSPECTION	FUNCTIONAL TEST	SAT	UNSAT	NA	COMMENTs
	Other (specify)	☐	☐	☐	☐	☐	
12	Alarm Initiating Device	☐	☐	☐	☐	☐	
	Device test result sheets attached listing all devices and test results	☐	☐	☐	☐	☐	

13	Supervisory Alarm Initiating Device	☐	☐	☐	☐	☐	
	Device test result sheets attached listing all devices and test results	☐	☐	☐	☐	☐	
14	Alarm Notification Appliances	☐	☐	☐	☐	☐	
	Appliance test result sheets attached listing all appliances and test results	☐	☐	☐	☐	☐	
15	Supervisory Station Monitoring	☐	☐	☐	☐	☐	
	Alarm signal	☐	☐	☐	☐	☐	
	Alarm restoration	☐	☐	☐	☐	☐	
	Fault/Trouble restoration	☐	☐	☐	☐	☐	

ENCLOSURE 2.2.2 FIRE AND SMOKE DETECTION AND ALARM SYSTEM INSPECTION AND TEST FORM

#	DESCRIPTION	VISUAL INSPECTION	FUNCTIONAL TEST	SAT	UNSAT	NA	COMMENTs
	Supervisory signal	☐	☐	☐	☐	☐	
	Supervisory restoration	☐	☐	☐	☐	☐	

16	Final functional system test with MSC and ABS in attendance	☐	☐	☐	☐	☐	
17	System returned to active service	☐	☐	☐	☐	☐	
18	Inspection tags attached	☐	☐	☐	☐	☐	
SAT = Satisfactory **UNSAT** = Unsatisfactory **N/A** = Not applicable							

ENCLOSURE 2.2.2 FIRE AND SMOKE DETECTION AND

2. CERTIFICATION

Inspector Certification:

This system, as specified, has been inspected and tested according to the standards cited in the work item.

Signed:	________________	Printed Name:	________________	Date:	________________
Organization:	________________	Title:	________________	Phone:	________________

ENCLOSURE 2.2.2 FIRE AND SMOKE DETECTION AND FIRE AND SMOKE DETECTION AND ALARM SYSTEM INITIATING DEVICE TEST RESULTS FORM (attach additional sheets as required)				
Device Type	Address	Location	Test Results Function	Test Results Sensitivity
Smoke detector	Zone 1	X-XX-XX	SAT	
Heat detector	Zone 1	X-XX-XX	SAT	XX
Smoke/Heat detector	Zone 1	X-XX-XX	SAT	XX
Flame detector	Zone 1	X-XX-XX	SAT	XX
Manual pull station	Zone 1	X-XX-XX	SAT	
Smoke detector	Zone 2	X-XX-XX	SAT	
Heat detector	Zone 2	X-XX-XX	SAT	XX
Smoke/Heat detector	Zone 1	X-XX-XX	SAT	XX
Flame detector	Zone 1	X-XX-XX	SAT	XX
Manual pull station	Zone 2	X-XX-XX	SAT	

ENCLOSURE 2.2.2 FIRE AND SMOKE DETECTION AND FIRE AND SMOKE DETECTION AND ALARM SYSTEM INITIATING DEVICE TEST RESULTS FORM (attach additional sheets as required)

Device Type	Address	Location	Test Results Function	Test Results Sensitivity
Ground fault detector	Control	X-XX-XX	SAT	
Short circuit detector	Control	X-XX-XX	SAT	
Open circuit detector	Control	X-XX-XX	SAT	
Loss of primary power	Control	X-XX-XX	SAT	

<table>
<tr><td colspan="4">USS Hershel Williams</td></tr>
<tr><td colspan="4">(ESB4)</td></tr>
<tr><td colspan="2">AUXILIARY MACHINERY</td><td colspan="2">CONTRACT NO. N68171-26-RN-003</td></tr>
<tr><td>ITEM NO. 0570</td><td>CATEGORY "A"</td><td colspan="2">2026-02-03</td></tr>
<tr><td colspan="2">ESB-CCSI-PORTABLE FIRE EXTINGUISHER SERVICE (1 YR)</td><td colspan="2">Port Engineer's Name</td></tr>
</table>

1.0 ABSTRACT

The purpose of this item is to inspect, service and test the ship’s portable and semi-portable fire extinguishers.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSEA Drawing No. 830-8496378, Fire Control and Safety Plan

2.1.2 USNS Hershel Williams, Last Portable Fire Extinguisher Service Record

2.1.3 Resolution A.951(23), Adopted on 5 December 2003, Improved Guidelines for Marine Portable Extinguishers (Available Commercially)

2.1.4 National Fire Protection Association, NFPA 10, Standard for Portable Fire Extinguishers (Available Commercially)

2.2 Enclosures

2.2.1 Portable Fire Extinguisher - Service Record

2.2.2 Portable Fire Extinguisher – Periodic Inspection

2.2.3 NFPA 10, Annex I, Maintenance Procedures

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location/Quantity

3.1.1 Locations: See Reference 2.1.1

3.1.2 Quantity

a. 123 CO2 extinguishers

b. Six Halogenated extinguishers

c. 153 BC extinguishers

d. Four class K extinguishers

e. 15 PKP extinguishers

f. 26 ABC extinguishers

g. 12 Foam extinguishers

3.2 Description: See Reference 2.1.2

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 46 CFR §71.25-20 requires all portable fire extinguishers and semi-portable fire extinguishing systems be maintained in accordance with NFPA 10, chapter 7.

5.3 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service, recharge, and test the ship’s portable and semi-portable fire extinguishers identified in section 3.0 in accordance with IMO, SOLAS, and CFR requirements using References 2.1.1 through 2.1.4 and Enclosures 2.2.1 through 2.2.3 as guidance.

7.2 Dispose of any liquids and powders removed in accordance with all federal, state, and local regulations.

7.3 Conduct inspection, maintenance, and recharging of all portable and semi-portable fire extinguishers using References 2.1.1 through 2.1.4 and Enclosures 2.2.1 through 2.2.3 as guidance.

7.3.1 Inspection

a. Conduct a periodic inspection of all extinguishers within the first week of the vessel’s arrival in the contractor’s facility.

b. Submit completed Enclosure 2.2.2 to the OMT REP along with any recommended service and parts required.

7.3.2 Maintenance

a. Temporarily remove the extinguishers off the ship to the authorized service facility. No more than 25% are to be removed from the vessel at any point. Care is to be used not to remove all fire protection from anyone space at a time.

b. Conduct annual inspection and maintenance on all extinguishers and complete Enclosure 2.2.1. It is to be a thorough examination of the extinguisher including mechanical parts, extinguishing agent, expelling means and physical condition. Maintenance is to include those tasks detailed in the manufacturers service manual.

c. Verify each fire extinguisher is clearly marked with the: name of the manufacturer, class fire rating, type and quantity of extinguishing medium, approval details, instructions for use and recharge, year of manufacture, temperature range for SAT operation and test pressure.

d. Conduct an annual external examination of all extinguishers using Enclosures 2.2.1 through 2.2.3 as guidance. All removable extinguisher boots, foot rings and attachments are to be temporarily removed to accommodate the cylinder exams. Identify obvious physical damage, corrosion, hose or nozzle blockage, operating instructions are present and legible and facing forward and determine if internal exam or hydrostatic test is due.

e. Conduct an internal examination of the extinguishers at intervals not to exceed those specified in NFPA 10.

Extinguisher Type	Internal Examination Interval (years)
Dry Chemical, cartridge and cylinder operated, with mild steel shells	1
Dry Powder, cartridge and cylinder operated, with mild steel shells	1
AFFF (aqueous film-forming foam)	3*
FFFP (film-forming fluoroprotein foam)	3*
Dry chemical, stored pressure,	5

Extinguisher Type	Internal Examination Interval (years)
with stainless steel shells	
Carbon Dioxide	5
Wet Chemical	5
Dry chemical, stored pressure, with mild steel shells, brazed brass shells and aluminum shells	6
Halogenated agents	6
Dry powder, stored pressure, with mild steel shells	6

NOTE: The extinguishing agent in liquid charge AFFF and FFFP extinguishers is to be replaced every 3 years and an internal exam is conducted at that time.

f. Perform an annual conductivity test on all Carbon Dioxide hose assemblies. Any hose that fails the test is to be replaced. Carbon dioxide hose assemblies have a continuous metal braid that connects to both couplings to minimize the static shock hazard. The reason for the conductivity test is to determine that the hose is conductive from the inlet coupling to the outlet orifice.

g. Discharge hoses on wheeled type fire extinguishers are to be completely uncoiled and examined annually. They are to be recoiled preventing kinks and to allow rapid deployment.

h. Pressure regulators on wheeled type fire extinguishers are to be tested annually for outlet static pressure and flow rate using the manufacturer's instructions as guidance.

i. When the inspection reveals any deficiency a condition report is to be submitted to the OMT REP with recommended repair and parts required using Enclosure 2.2.3 as guidance.

7.3.3 Recharging

a. When recharging is required, perform in accordance with the manufacturers service manual.

b. The amount of the recharge is to be verified by weighing.

c. After recharging, perform a leak test on stored-pressure and self-expelling types of extinguishers.

d. In no case is an extinguisher to be recharged if it is beyond its specified hydrostatic test date.

e. Recharge CO2 extinguishers if weight loss exceeds 10 percent of weight of charge.

f. AFFF and FFFP agent is to be replaced at least once every three (3) years.

7.4 Conduct a hydrostatic test on all portable and semi-portable fire extinguishers that are due using References 2.1.1 through 2.1.4 as guidance. See Reference 2.1.2 for last hydrostatic test dates.

7.4.1 Include in estimate the internal examination, hydrostatic testing, and recharging of:

a. 30 CO2 extinguishers

b. 10 AFFF extinguishers

c. 5 ABC extinguishers

d. 5 PKP extinguishers

7.4.2 Cylinders and cartridges with US Department of Transportation (DOT) or Transport Canada (TC) markings are to be retested in accordance with applicable DOT or TC regulations.

7.4.3 Per 29 CFR §1910.157 (Table L-1) and NFPA 10, test intervals are not to exceed those shown below:

Extinguisher Type	**Test Interval (years)**
Stored-pressure water, water mist, loaded stream, and/or antifreeze	5
Wetting Agent	5
AFFF (aqueous film-forming foam)	5
FFFP (film-forming fluoroprotein foam)	5
Dry chemical, stored pressure, with stainless steel shells	5
Carbon Dioxide	5
Wet Chemical	5
Dry chemical, stored pressure, with mild steel	10**

Extinguisher Type	Test Interval (years)
shells, brazed brass shells and aluminum shells	
Dry Chemical, cartridge or cylinder-operated, with mild steel shells	10**
Halon 1211 or 1301	10**
Dry powder, stored pressure, cartridge or cylinder-operated,with mild steel shells	10**

7.4.4 Conduct a hydrostatic test on all hose assemblies that have a shut-off nozzle at the discharge end of the hose and are due using References 2.1.1, 2.1.3, and 2.1.4 as guidance. In addition, all high-pressure and low-pressure accessory hoses (other than agent discharge hose) used on wheeled extinguishers are to be hydrostatically tested. Test intervals are the same as those for the fire extinguisher or cylinder. Per para 8.6.3 of NFPA 10, Hose assemblies that require a hydrostatic test will be tested at:

a. CO_2 - 1250 psi (8619 kPa)***

b. Dry Chem, Dry Powder, Water, Foam or Halogenated - 300 psi (2068 kPa) or service pressure if higher

c. Wheeled extinguisher - Low pressure accessory hose - 300 psi (2068 kPa)

d. Wheeled extinguisher - High pressure accessory hose - 3000 psi (20.68 MPa)

7.4.5 Hydrostatic testing is to be performed by persons certified and trained in pressure testing procedures and safeguards. All testing is to be conducted using clean fresh water as the test medium.

a. Hydrostatic testing is to always include an internal and external visual examination of the cylinder prior.

b. Tags and labels documenting inspection documenting inspection, maintenance, recharging or hydrostatic testing are to be installed so as not to obstruct the extinguishers use, classification or manufacturers labels.

7.4.6 Securely attach a tag or label to each extinguisher that indicates inspection and maintenance was performed. The tag at a minimum is to identify the following:

a. Month and Year of maintenance

b. Name of person performing the work

c. Name of the company performing the work

7.4.7 Each cylinder that has undergone maintenance that included an internal examination or has been recharged requiring the removal of the valve assembly is also to have a verification-of-service collar installed around the neck of the container using Reference 2.1.4 as guidance. The collar is not to interfere with the operation of the extinguisher and be a single circular piece unable to move over the neck of the container unless the valve is completely removed. The collar as a minimum is to identify the following:

a. Month and Year of the recharge or internal exam

b. Name of the company performing the work

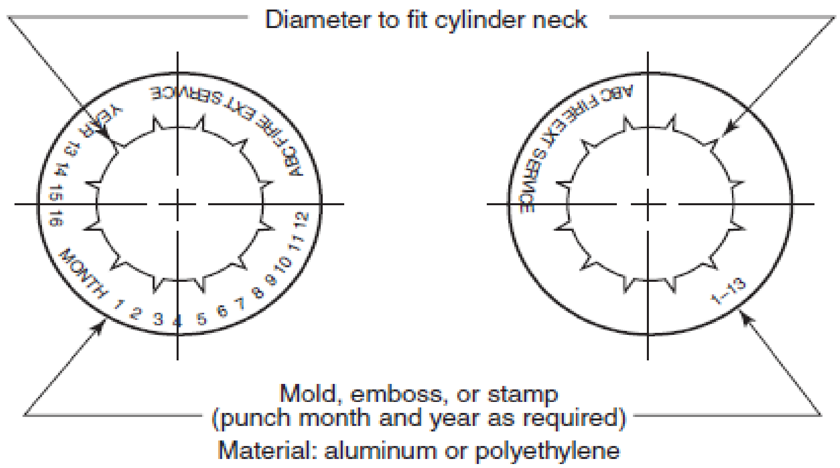


FIGURE A.7.10 Design of a Verification-of-Service Collar.

NOTE: Service Collar Exemptions: New AFFF, FFFP or Wet Chemical extinguishers that required an initial charge in the field, liquefied gas, halogenated agent and CO2 extinguishers that have been recharged without valve removal. Cartridge and cylinder-operated extinguishers.

7.4.8 Fire extinguishers identified in 7.3.2.d that require the Six-Year Internal Examination and pass are to have the maintenance info recorded on a durable weatherproof label that is a minimum of 2” x 3.5” (51mm x 89mm) affixed to the shell. Any previous six-year exam labels are to be removed. The labels are to be of the self-destructive type when their removal is attempted. The label is to at a minimum identify the following:

a. Month and Year of six-year internal examination

b. Name of person performing the work

c. Name of the company performing the work

7.4.9 Fire extinguishers that are low pressure non-DOT type that pass a hydrostatic test are to have the information recorded on a durable weatherproof label that is a minimum of 2" x 3.5" (51mm x 89mm) affixed by to the shell. Any previous test labels are to be removed. The labels are to be of the self-destructive type when their removal is attempted. The label is to at a minimum identify the following:

a. Month and Year of test

b. Test pressure used

c. Name or initials of person performing the test

d. Name of the company performing the test

7.4.10 High pressure cylinders or cartridges that pass a hydrostatic test are to be stamped with the retesters' ID number and the month and year of the retest per DOT/TC requirements. Stamping will be placed on the shoulder, top, head, neck or foot ring of the cylinder or in accordance with 49 CFR 180.213 (c)(1).

7.4.11 Carbon Dioxide hose assemblies that pass a conductivity test are to have the test information recorded on a durable weatherproof label that is a minimum of ½" x 3" (13mm x 76mm) affixed to the hose by a heatless process. Any previous labels are to be removed. The label is to include the following:

a. Month and Year of test, indicated by perforation, such as done by hand punch

b. Name or initials of person performing the work

c. Name of the company performing the work

7.5 Testing is to be coordinated with the OMT REP and the ABS Surveyor.

7.6 Upon completion of all inspections, tests and repairs return the Fire Extinguishers to the vessel and leave them in their original location in a ready for service condition. Conduct a final inspection aboard with the OMT and the ship's Master.

7.7 Reports

7.7.1 Complete the Periodic Inspection form shown in Enclosure 2.2.2, documenting the initial condition of every Fire Extinguisher. Submit one electronic copy in Portable Document Format (pdf) as soon as the data becomes available.

7.7.2 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.7.3 Upon completion of all inspections, tests, maintenance and recharging, prepare and submit the Service Record form shown in Enclosure 2.2.1. Submit one electronic copy in Portable Document Format (pdf).

7.7.4 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.8 Painting: None additional

7.9 Manufacturer's Representative

7.9.1 Persons performing annual and periodic maintenance and recharging of fire detection and extinguishing equipment are to be certified and recognized as an authorized specialist by ABS.

7.9.2 Persons training to be certified are permitted to perform maintenance and recharging of extinguishers under the direct supervision and in the immediate presence of a certified person.

7.9.3 Companies and persons performing maintenance and recharging of extinguishers are to have available the appropriate servicing manuals, correct tools, recharging materials, lubricants, and manufacturer’s replacement parts.

7.9.4 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only fire detection and extinguishing service companies certified and recognized as an authorized specialist by ABS and OEM authorized parts will be used to accomplish the requirements of this work item for this ship critical safety item including oversight and guidance on all aspects of equipment as-found condition inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable.

7.10 Preparation of Drawings

7.10.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 PORTABLE FIRE EXTINGUISHER - SERVICE RECORD

Vessel: ____________________ **Date:** ____________________

Location: ____________________ **Company:** ____________________

Inspector: ____________________

FX No.	Location	Size (lbs)	Class A/B/C	FX Medium	Serial No.	Year of Mfg	Visual Inspection		Hydrostatic Test		Recharge	Hose Inspection
							Last External	Last Internal	Last Test	Pressure (psi)		

LEGEND

Fire Extinguishing Medium

- Carbon Dioxide (CO2)
- Halogenated (H)
- Dry Powder (DP)

USS Hershel Williams		
(ESB4)		
AUXILIARY MACHINERY	CONTRACT NO. N68171-26-RN-003	
ITEM NO. 0570	CATEGORY "A"	2026-02-03
ESB-CCSI-PORTABLE FIRE EXTINGUISHER SERVICE (1 YR)	Port Engineer's Name	

FX No.	Location	Size (lbs	Class	FX Medium	Serial No.	Year of	Visual Inspection	Hydrostatic Test	Recharge	Hose Inspectio

Dry Chemical (DC)
Wet Chemical (WC)
Aqueous film-forming foam (AFFF)
Film-forming fluoroprotein foam (FFFP)

USS Hershel Williams		
(ESB4)		
AUXILIARY MACHINERY	CONTRACT NO. N68171-26-RN-003	
ITEM NO. 0570	CATEGORY "A"	2026-02-03
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		)	A/B/C			**Mfg**	**Last External**	**Last Internal**	**Last Test**	**Pressure** (psi)		**n**

ENCLOSURE 2.2.1 PORTABLE FIRE EXTINGUISHER - SERVICE RECORD

USS Hershel Williams		
(ESB4)		
AUXILIARY MACHINERY	CONTRACT NO. N68171-26-RN-003	
ITEM NO. 0570	CATEGORY "A"	2026-02-03
ESB-CCSI-PORTABLE FIRE EXTINGUISHER SERVICE (1 YR)	Port Engineer's Name	

ENCLOSURE 2.2.2 Portable Fire Extinguisher – Periodic Inspection

Vessel:	______________________	**IMO #:**	______________________
Shipyard:	______________________	**Date:**	______________________
Company:	______________________	**Inspector:**	______________________

To be completed by the extinguisher inspector at the time of inspection or test. The inspection of the extinguishers will at a minimum include checks and tests of the following:

1) Extinguisher is in designated place
2) No obstruction to access or visibility
3) Pressure gauge reading or indicator is in the operable range or position
4) Fullness determined by weighing
5) Condition of tires, wheels, carriage, hose and nozzle for wheeled extinguishers
6) Indicator for non-rechargeable extinguishers using push-to-test indicators
7) Verify the operating instructions are legible and face forward
8) Check for missing or broken safety seals and tamper indicators
9) Examine for obvious physical damage, corrosion, leakage or clogged nozzle.
10) Note deficiencies.

Y = Satisfactory **N** = Unsatisfactory (explain below)
N/A = Not applicable

ENCLOSURE 2.2.2 Portable Fire Extinguisher – Periodic Inspection

No.	Location	1	2	3	4	5	6	7	8	9	10

Description of any unsatisfactory conditions noted:

ENCLOSURE 2.2.3 NFPA 10, Annex I, Maintenance

Annex I Maintenance Procedures

This annex is not a part of the requirements of this NFPA document but is included for informational purposes only.

I.1 Maintenance Checklists. For convenience, the following checklists are organized into two parts. The first, Table I.1(a), is arranged by mechanical parts (components and containers) common to most fire extinguishers. The second, Table I.1(b), is arranged by extinguishing material and expelling means and involves a description of the problems peculiar to each agent.

Table I.1(a) Mechanical Parts Maintenance Checklist

Cylinder/Shell	Corrective Action
1. Hydrostatic test date or date of manufacture	1. Retest, if needed
2. Corrosion	2. Conduct hydrostatic test and refinish, or condemn
3. Mechanical damage (denting or abrasion)	3. Conduct hydrostatic test and refinish, or condemn
4. Paint condition	4. Refinish
5. Presence of repairs (welding, soldering, brazing, etc.)	5. Condemn
6. Damaged threads (corroded, crossthreaded, or worn)	6. Condemn
7. Broken hanger attachment, carrying handle lug	7. Condemn
8. Sealing surface damage (nicks or corrosion)	8. Condemn
Nameplate	**Corrective Action**
1. Illegible wording	1. Clean or replace (Note: Only labels without a listing mark can be replaced.)
2. Corrosion or loose plate	2. Inspect shell under plate (see cylinder/shell check points) and reattach plate
Nozzle or Horn	**Corrective Action**
1. Deformed, damaged, or cracked	1. Replace
2. Blocked openings	2. Clean
3. Damaged threads (corroded, crossthreaded, or worn)	3. Replace
4. Aged (brittle)	4. Replace
Hose Assembly	**Corrective Action**
1. Damaged (cut, cracked, or worn)	1. Replace
2. Damaged couplings or swivel joint (cracked or corroded)	2. Replace
3. Damaged threads (corroded, crossthreaded, or worn)	3. Replace
4. Inner tube cut at couplings	4. Replace or consult manufacturer
5. Electrically nonconductive between couplings (CO_2 hose only)	5. Replace
6. Hose obstruction	6. Remove obstruction or replace
7. Hydrostatic test date	7. Retest if needed
Pull/Ring Pin	**Corrective Action**
1. Damaged (bent, corroded, or binding)	1. Replace
2. Missing	2. Replace
Gauge or Pressure-Indicating Device	**Corrective Action**
1. Immovable, jammed, or missing pointer (pressure test)	1. Depressurize and replace gauge
2. Missing, deformed, or broken crystal	2. Depressurize and replace gauge
3. Illegible or faded dial	3. Depressurize and replace gauge
4. Corrosion	4. Depressurize and check calibration, clean and refinish, or replace gauge
5. Dented case or crystal retainer	5. Depressurize and check calibration, or replace gauge
6. Immovable or corroded pressure-indicating stem (nongauge type)	6. Depressurize and discard shell
7. Verify gauge compatibility	7. Depressurize and replace

Procedures

ENCLOSURE 2.2.3 NFPA 10, Annex I, Maintenance Procedures

Table I.1(a) *Continued*

Shell or Cylinder Valve	Corrective Action
1. Corroded, damaged, or jammed lever, handle, spring, stem, or fastener joint	1. Depressurize, check freedom of movement, and repair or replace
2. Damaged outlet threads (corroded, crossthreaded, or worn)	2. Depressurize and replace
Nozzle Shutoff Valve	**Corrective Action**
1. Corroded, damaged, jammed, or binding lever, spring, stem, or fastener joint	1. Repair and lubricate, or replace
2. Plugged, deformed, or corroded nozzle tip or discharge passage	2. Clean or replace
Puncture Mechanism	**Corrective Action**
1. Damaged, jammed, or binding puncture lever, stem, or fastener joint	1. Replace
2. Dull or damaged cutting or puncture pin	2. Replace
3. Damaged threads (corroded, crossthreaded, or worn)	3. Replace
Expellant/Gas Cartridge	**Corrective Action**
1. Corrosion	1. Replace with correct expellant gas cartridge
2. Damaged seal disc (injured, cut, or corroded)	2. Replace with correct expellant gas cartridge
3. Damaged threads (corroded, crossthreaded, or worn)	3. Replace with correct expellant gas cartridge
4. Illegible weight markings	4. Replace with correct expellant gas cartridge
5. Improper gas cartridge	5. Replace with correct expellant gas cartridge
6. Improper cartridge seal	6. Replace with correct expellant gas cartridge
Gas Cylinders	**Corrective Action**
1. Hydrostatic test date or date of manufacture	1. Retest if needed
2. Corrosion	2. Conduct hydrostatic test and refinish, or discard
3. Paint condition	3. Refinish
4. Presence of repairs (welding, soldering, brazing, etc.)	4. Condemn
5. Damaged threads (corroded, crossthreaded, or worn)	5. Condemn
Fill Cap	**Corrective Action**
1. Corroded, cracked, or broken	1. Replace
2. Damaged threads (corroded, crossthreaded, or worn)	2. Replace
3. Sealing surface damage (nicked, deformed, or corroded)	3. Clean, repair, and leak test, or replace
4. Obstructed vent hole or slot	4. Clean
Nonrechargeable Shell/Cylinder	**Corrective Action**
1. Corrosion	1. Depressurize and discard
2. Damaged seal disc (injured, cut, or corroded)	2. Depressurize and discard
3. Damaged threads (corroded, crossthreaded, or worn)	3. Depressurize and discard
4. Illegible weight or date markings	4. Depressurize and discard
Carriage and Wheels	**Corrective Action**
1. Corroded, bent, or broken carriage	1. Repair or replace
2. Damaged wheel (buckled or broken spoke, bent rim or axle, loose tire, low pressure, jammed bearing)	2. Clean, repair, and lubricate, or replace
Carrying Handle	**Corrective Action**
1. Broken handle lug	1. Condemn cylinder or consult manufacturer regarding repair
2. Broken handle	2. Replace
3. Corroded, jammed, or worn fastener	3. Clean or replace

(continues)

ENCLOSURE 2.2.3 NFPA 10, Annex I, Maintenance Procedures

Table I.1(a) *Continued*

Tamper Seals or Indicators	Corrective Action
1. Broken or missing	1. Check Table I.1(b) for specific action
2. Fill cap indicator corroded or inoperative	2. Repair, clean, or replace
3. Fill cap indicator operated	3. Depressurize unit, check content, refill
Hand Pump	**Corrective Action**
1. Corroded, jammed, or damaged pump	1. Repair and lubricate, or replace
2. Improper adjustment of packing nut	2. Adjust
Pressurizing Valve	**Corrective Action**
1. Leaking seals	1. Depressurize and replace valve or core
Gasket and "O" Ring Seals	**Corrective Action**
1. Damaged (cut, cracked, or worn)	1. Replace and lubricate
2. Missing	2. Replace and lubricate
3. Aged or weathered (compression set, brittle, cracked)	3. Replace and lubricate
Brackets and Hangers	**Corrective Action**
1. Corroded, worn, or bent	1. Repair and refinish, or replace
2. Loose or binding fit	2. Adjust fit or replace
3. Worn, loose, corroded, or missing screw or bolt	3. Tighten or replace
4. Worn bumper, webbing, or grommet	4. Replace
5. Improper type	5. Replace
Gas Tube and Siphon or Pickup Tube	**Corrective Action**
1. Corroded, dented, cracked, or broken	1. Replace
2. Blocked tube or openings in tube	2. Clean or replace
Safety Relief Device	**Corrective Action**
1. Corroded or damaged	1. Depressurize and replace
2. Broken, operated, or plugged	2. Depressurize and replace
Pressure Regulators	**Corrective Action**
1. External condition: damaged or corroded	1. If damaged, replace regulator; if corroded, clean regulator or replace
2. Pressure relief (corroded, plugged, dented, leaking, broken, or missing)	2. Disconnect regulator from pressure source, replace pressure relief, or replace regulator
3. Protective bonnet relief hole (tape missing or seal wire broken or missing)	3. Replace regulator
4. Adjusting screw (lock pin missing)	4. Replace regulator

Table I.1(b) Agent and Expelling Means Maintenance Checklist

AFFF and FFP	Corrective Action
1. Recharging date due	1. Empty, clean, and recharge
2. Improper fill levels	2. Empty, clean, and recharge
3. Agent condition (check for sediment)	3. Empty, clean, and recharge
4. Improper fill level (by weight or observation)	4. Empty and recharge with new solution
5. Agent condition (presence of precipitate or other foreign matter)	5. Empty and recharge with new solution
6. Improper gauge pressure	6. Repressurize and leak test
7. Broken or missing tamper indicator	7. Leak test, replace indicator
Self-Expelling	
Carbon Dioxide	**Corrective Action**
1. Improper weight	1. Recharge to proper weight
2. Broken or missing tamper indicator	2. Leak test and weigh, also recharge or replace seal

ENCLOSURE 2.2.3 NFPA 10, Annex I, Maintenance Procedures

Table I.1(b) *Continued*

Halon 1301 Bromotrifluoromethane	Corrective Action
1. Punctured cylinder seal disc	1. Replace shell
2. Improper weight	2. Replace shell or return to manufacturer for refilling
3. Broken or missing tamper seal	3. Examine cylinder seal disc, replace seal
Combination Halon 1211/1301	**Corrective Action**
1. Improper weight	1. Return to manufacturer *(See 7.2.3.3.)*
2. Broken or missing tamper seal	2. Return to manufacturer *(See 7.2.3.3.)*
Manually Operated	
Mechanical Pump Water and Loaded Stream	**Corrective Action**
1. Improper fill level	1. Refill to proper level
2. Defective pump	2. Clean, repair, and lubricate, or replace
Dry Powder Pail	**Corrective Action**
1. Improper fill level	1. Refill
2. Agent condition (contamination or caking)	2. Discard and replace
3. Missing scoop	3. Replace
Gas Cartridge or Cylinder	
Dry Chemical and Dry Powder Types	**Corrective Action**
1. Improper weight or charge level	1. Refill to correct weight or charge level
2. Agent condition (contamination, caking, or wrong agent)	2. Empty and recharge with new agent
3. Cartridge	3.
(a) Punctured seal disc	(a) Replace cartridge
(b) Improper weight	(b) Replace cartridge
(c) Broken or missing tamper indicator	(c) Examine seal disc, replace
(d) Improper cartridge seal	(d) Replace cartridge seal
4. Gas cylinder with gauge	4.
(a) Low pressure	(a) Replace or recharge cylinder
(b) Broken or missing tamper seal	(b) Leak test, replace
5. Gas cylinder without gauge	5.
(a) Low pressure (attach gauge and measure pressure)	(a) Leak test (if low, replace or recharge cylinder)
(b) Broken or missing tamper seal	(b) Measure pressure, leak test, replace seal
Stored-Pressure	
Combination Halon 1211/1301	**Corrective Action**
1. Refillable	1.
(a) Improper extinguisher agent	(a) Return to manufacturer *(See 7.2.3.3.)*
(b) Improper gauge pressure	(b) Return to manufacturer *(See 7.2.3.3.)*
(c) Broken or missing tamper seal	(c) Examine extinguisher, leak test, replace tamper seal
2. Nonrechargeable extinguisher with pressure indicator	2.
(a) Low pressure	(a) Return to manufacturer *(See 7.2.3.3.)*
(b) Broken or missing tamper seal	(b) Return to manufacturer *(See 7.2.3.3.)*
Dry Chemical and Dry Powder Types	**Corrective Action**
1. Rechargeable	1.
(a) Improper extinguisher weight	(a) Leak test and refill to correct weight
(b) Improper gauge pressure	(b) Repressurize and leak test
(c) Broken or missing tamper seal	(c) Leak test, check weight, and replace seal
2. Disposable shell with pressure indicator	2.
(a) Punctured seal disc	(a) Depressurize and discard
(b) Low pressure	(b) Depressurize and discard
(c) Broken or missing tamper indicator	(c) Depressurize and discard

(continues)

ENCLOSURE 2.2.3 NFPA 10, Annex I, Maintenance Procedures

Table I.1(b) *Continued*

3. Disposable shell without pressure indicator	3.
(a) Punctured seal disc	(a) Depressurize and discard
(b) Low weight	(b) Depressurize and discard
(c) Broken or missing tamper seal	(c) Depressurize and discard
4. Nonrechargeable extinguisher with pressure indicator	4.
(a) Low pressure	(a) Depressurize and discard
(b) Broken or missing tamper indicator	(b) Depressurize and discard
Wet Chemical Type	**Corrective Action**
1. Improper fill level (by weight or observation)	1. Empty and recharge with new agent to correct weight fill line
2. Improper gauge pressure	2. Repressurize and leak test or consult manufacturer
3. Broken or missing tamper seal	3. Verify fill level, recharge if required, replace tamper seal
Halogenated-Type Agents	**Corrective Action**
1. Broken or missing tamper seal	1. Verify level and pressure, recharge if required, replace tamper seal
2. Improper gauge pressure	2. Weigh, repressurize, and leak test or consult manufacturer
3. Improper weight	3. Leak test and recharge to correct weight
Water and Loaded Stream	**Corrective Action**
1. Improper fill level (by weight or observation)	1. Recharge to correct level in accordance with the manufacturer's manual
2. Agent condition if antifreeze or loaded stream	2. Empty and recharge with new agent
3. Improper gauge pressure	3. Repressurize and leak test or consult manufacturer
4. Broken or missing tamper seal	4. Leak test, replace seal

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0571	**CATEGORY "A"**	**2026-01-30**
ESB-CCSI-FIRE AND SMOKE DAMPER SERVICE (1 YR)		**Manovich, David**

1.0 ABSTRACT

The purpose of this item is to inspect, service and test the fire and smoke dampers.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSEA Drawing No. 830-8496378, Fire Control and Safety Plan

2.1.2 Drawing No. 512-8499376 HVAC FWD Accommodation Diagram

2.1.3 Drawing No. 512-8468337 HVAC Diagram Accommodations

2.2 Enclosures

2.2.1 USS Hershel Williams (ESB 4) Fire Damper List

2.2.2 NFPA Form 17-C

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations: Various; See Enclosure 2.2.1

3.1.2 Quantity

a. 22 Fire Damper Serving Accommodation and Service Spaces

b. 35 Shutoff/Smoke Damper Serving Accommodation and Service Spaces

c. 25 Fire Damper Serving Machinery Spaces

d. 22 Shutoff/Smoke Damper Serving Machinery Spaces

e. 22 Remote Control for Fire Damper Serving Accommodation and Service Spaces

f. Six Remote Control for Shut-Off/Smoke Damper Serving Accommodation and Service Spaces

g. 27 Remote Control for Fire Damper Serving Machinery Spaces

h. Four Remote Control for Shut-off/Smoke Damper Serving Machinery Spaces

3.2 Item Description/Manufacturer's Data: None

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Definitions

5.2.1 An automatic fire damper is a fire damper that closes independently in response to exposure to fire products.

5.2.2 A manual fire damper is a fire damper that is intended to be opened or closed by the crew by hand at the damper itself.

5.2.3 A remotely operated fire damper is a fire damper that is closed by the crew through a control located at a distance away from the controlled damper.

5.3 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

5.4 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service and test the Fire and Smoke Dampers identified in Enclosure 2.2.1 in accordance with SOLAS and USCG requirements using References 2.1.1 through 2.1.3 as guidance. Fill in any missing information in Enclosure 2.2.1 and submit to OMT REP.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Temporarily remove all overhead access doors and panels in staterooms, passageways, and compartments, as well as any other interferences to gain access to all fire dampers, fire damper reach rods, and reach rod universals.

7.4 Inspect, service, and test each fire and smoke damper identified in Enclosure 2.2.1. Ensure that all fire and smoke dampers are:

7.4.1 Accessible and unobstructed.

7.4.2 Fitted with a marked inspection door that clearly identifies the damper. Ensure remote controls for applicable dampers are also marked as such.

7.4.3 Marked with red letters at least one-half inch (½") or twelve and seven-tenths millimeters (12.7 mm) in height stating, "VENTILATION FIRE DAMPER".

7.4.4 Fitted with an indicator showing whether the damper is "OPEN" or "CLOSED".

7.4.5 Manually cycled between "OPEN" and "CLOSED". Visually examine the damper blades. When "CLOSED", the fire dampers must make contact with the steel strip that runs along the edge of their internal cross-section.

7.4.6 Free of rusting, bending, misaligned components, damaged blades, or defective hinges that would interfere with or hinder proper operation of the damper.

7.4.7 Not penetrated by foreign objects.

7.4.8 Maintaining fire integrity of the divisions at openings and penetrations.

7.4.9 Where damper is fitted with a fusible link, temporarily disconnect link and verify damper functions as designed.

7.4.10 Closing completely and latching "CLOSED" (where latches are applicable).

7.4.11 Not damaged or painted. Special attention is required for fusible links. If a fusible link is damaged or painted, it must be replaced.

7.4.12 Dry-lubricated, including all exposed moving parts, using the manufacturer's instructions.

7.4.13 Re-tested for proper operation following any repairs.

7.5 Fire dampers which are operated automatically must be in the “CLOSED” position within two minutes after the commencement of the test [(2010 FTP CODE) ANNEX 1 - FIRE TEST PROCEDURES; 2.2.5].

7.6 Fire dampers which are operated manually must be “CLOSED” within one minute after the commencement of the test. [(2010 FTP CODE) ANNEX 1 - FIRE TEST PROCEDURES; 2.2.6].

7.7 Each automatic fire damper is subject to additional testing aside from the tasks outlined in paragraph 7.4. Each automatic fire damper must:

7.7.1 Be operated both automatically and manually from both sides of the division.

7.7.2 Demonstrate proper operation of the fail-safe mechanism in the event of a fire, by closing the damper upon loss of electrical power, hydraulic or pneumatic pressure.

7.7.3 Demonstrate remote-control capability for de-energizing the Galley Range Hood, Laundry, and Drying Room Exhaust and Supply Fans from within the space, as well as for providing operation of the fire damper.

7.7.4 Complete Enclosures 2.2.1 and 2.2.2 for each fire and smoke damper and submit, along with a report, to the OMT REP.

7.8 Testing is to be coordinated with the OMT REP.

7.9 Maintain workspaces in a clean condition during and upon completion of the work requirements. Prevent contamination of any adjacent equipment and spaces.

7.10 Upon completion of all inspections, tests and repairs, return the fire and smoke dampers to a ready for service condition.

7.11 Reports

7.11.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.11.2 Upon completion of all inspections, maintenance, servicing, and testing, prepare and submit a typewritten Service Report documenting the final "as released" condition of all fire and smoke dampers affirming they are safe for use. Submit one electronic copy in Portable Document Format (PDF) of Enclosures 2.2.1 and 2.2.2.

7.11.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.12 Painting: None additional

7.13 Manufacturer's Representative: None

7.14 Preparation of Drawings

7.14.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 USS Hershel Williams (ESB 4) Fire Damper List

PART NO.	LOCATION NO.	NASSCO PART NO.	DECK	TYPE/MODEL	ACTUATOR
FCU110-FD01	FAN ROOM A8 LEVEL	HS09-EL333	A8 LEVEL	FDL-EL	BF 120-2
FCU110-FD02	FAN ROOM A8 LEVEL	HS09-EL334	A8 LEVEL	FDL-EL	BF 120-2
FCU110-FD03	FAN ROOM A8 LEVEL	HS09-EL335	A8 LEVEL	FDL-EL	BF 120-2
AC120-FD01	FAN ROOM A6 LEVEL	HS09-EL191	A6 LEVEL	FDL-EL	BF 120-2
AC121-FD01	FAN ROOM A6 LEVEL	HS09-EL192	A6 LEVEL	FDL-EL	BF 120-2
AC134-FD01	FAN ROOM A6 LEVEL	HS09-EL348	A5 LEVEL	FDL-EL	BF 120-2
AC136-FD01	MISSION SUPPORT NO.4	HS09-EL305	A5 LEVEL	FDL-EL	BF 120-2
E136-FD01	MISSION SUPPORT NO.4	HS09-EL306	A5 LEVEL	FDL-EL	BF 120-2
E136-FD02	PLENUM A6 LEVEL PS	HS09-EL378	A6 LEVEL	FDL-EL	BF 120-2
AC140-FD01	FAN ROOM A4 LEVEL	HS09-EL213	A4 LEVEL	FDL-EL	BF 120-2
AC140-FD02	FAN ROOM A4 LEVEL	HS09-EL214	A4 LEVEL	FDL-EL	BF 120-2
ER140-FD01	FAN ROOM A4 LEVEL	HS09-EL215	A4 LEVEL	FDL-EL	BF 120-2
E141-FD01	TRASH ROOM	HS09-EL216	A4 LEVEL	FDL-EL	BF 120-2
AC150-FD01	FAN ROOM A4 LEVEL	HS09-EL217	A4 LEVEL	FDL-EL	BF 120-2
AC150-FD02	FAN ROOM A4 LEVEL	HS09-EL218	A4 LEVEL	FDL-EL	BF 120-2
E150-FD01	FAN ROOM A6 LEVEL	HS09-EL219	A6 LEVEL	FDL-EL	BF 120-2
E151-FD01	FAN ROOM A4 LEVEL	HS09-EL220	A4 LEVEL	FDL-EL	BF 120-2
AC171-FD01	FAN ROOM A2 LEVEL	HS09-EL274	A2 LEVEL	FDL_EL	BF 120-2
E171-FD01	FAN ROOM A2 LEVEL	HS09-EL275	A2 LEVEL	FDL EL	BF 120-2
AC110-MD01	FAN ROOM A8 LEVEL	HS09-EL221	A8 LEVEL	UTG	BF 120-2
ER110-MD01	FAN ROOM A8 LEVEL	HS09-EL222	A8 LEVEL	UTG	BF 120-2
AC116-MD01	FAN ROOM AS LEVEL	HS09-EL325	AS LEVEL	UTG	BF 120-2
AC117-MD01	FAN ROOM A6 LEVEL	HS09-EL326	A6 LEVEL	UTG	BF 120-2
AC120-MD01	FAN ROOM A6 LEVEL	HS09-EL223	A6 LEVEL	UTG	BF 120-2
AC121-MD01	FAN ROOM A6 LEVEL	HS09-EL224	A6 LEVEL	UTG	BF 120-2
E120-MD01	HANGAR A6 LEVEL	HS09-EL327	A6 LEVEL	UTG	EXMAX-15-F
E121-MD01	HANGAR A6 LEVEL	HS09-EL328	A6 LEVEL	UTG	EXMAX-15-F
S130-MD01	FAN ROOM A6 LEVEL	HS09-EL329	AS LEVEL	UTG	BF 120-2

S131-MD01	FAN ROOM A6 LEVEL	HS09-EL226	AS LEVEL	UTG	BF 120-2
S132-MD01	FAN ROOM A6 LEVEL	HS09-EL330	AS LEVEL	UTG	BF 120-2
PART NO.	**LOCATION NO.**	**NASSCO PART NO.**	**DECK**	**TYPE/MODEL**	**ACTUATOR**
S133-MD01	FAN ROOM A6 LEVEL	HS09-EL331	AS LEVEL	UTG	BF 120-2
AC134-MD01	FAN ROOM A6 LEVEL	HS09-EL332	AS LEVEL	UTG	BF 120-2
AC140-MD01	FAN ROOM A4 LEVEL	HS09-EL227	A4 LEVEL	UTG	BF 120-2
AC150-MD01	FAN ROOM A4 LEVEL	HS09-EL228	A4 LEVEL	UTG	BF 120-2
AC160-MD01	FAN ROOM A3 LEVEL	HS09-EL229	A3 LEVEL	UTG	BF 120-2
AC170-MD01	FAN ROOM A2 LEVEL	HS09-EL230	A2 LEVEL	UTG	BF 120-2
AC171-MD01	FAN ROOM A2 LEVEL	HS09-EL278	A2 LEVEL	UTG	BF 120-2
E141-MD01	TRASH ROOM	HS09-EL231	A4 LEVEL	UTG	BF 120-2
E141-MD02	PROCESSED WASTE STOR ROOM	HS09-EL307	A4 LEVEL	UTG	BF 120-2
E141-MD03	SOLID WASTE MGT ROOM	HS09-EL308	A4 LEVEL	UTG	BF 120-2
E142-MD01	INCINERATOR ROOM	HS09-EL232	A4 LEVEL	UTG	BF 120-2
E142-MD02	INCINERATOR ROOM	HS09-EL233	A4 LEVEL	UTG	BF 120-2
S180-MD01	STAIRTOWER. PS. A5 LEVEL	HS09-EL279	A5 LEVEL	UTG	BF 120-2
S181-MD01	FAN ROOM A4 LEVEL	HS09-EL312	A4 LEVEL	UTG	BF 120-2
S182-MD01	STAIRTOWER. SB, A5 LEVEL	HS09-EL313	A5 LEVEL	UTG	BF 120-2
AC3-FD01	FAN ROOM 1-52-5	H509-EL080	UPPER DECK	FDL-EL550-450	BF 120-2
E2-FD01	GALLEY A-47-0	H509-EL081	A-DECK	FDL-EL600-300	BF 120-2
E2-FD02	GALLEY A-47-0	H509-EL082	A-DECK	FDL-EL800-350	BF 120-2
AC3-FD01	Fan Room 1-52-5	H509-EL080	Upper Deck	FDL-EL550-450	BF 120-2
E2-FD01	Galley A-47-0	H509-EL081	A-39-1	FDL-EL600-300	BF 120-2
E2-FD02	Galley A-47-0	H509-EL082	A-39-1	FDL-EL800-350	BF 120-2

ITEM NO.	LOCATION NO.	NASSCO PART NO.	DECK	NOTES
FD01	FIRE DAMPER	28-637074	3-39-2	
FD02	FIRE DAMPER	28-637074	3-55-2	Interlocked with Fan M4
FD03	FIRE DAMPER	28-637078	1-29-2	
FD04	FIRE DAMPER	28-637078	1-25-2	

ITEM NO.	LOCATION NO.	NASSCO PART NO.	DECK	NOTES
FD05	FIRE DAMPER	28-637078	1-29-4	Interlocked with Fan M11
FD06	FIRE DAMPER	28-637082	A-28-1	
FD07	FIRE DAMPER	28-638064	A-26-2	
FD08	FIRE DAMPER	28-637081	A-30-1	
FD09	FIRE DAMPER	Not Used	1-103-1	
FD10	FIRE DAMPER	28-637082	5-104-1	Interlocked with Fan M26
FD11	FIRE DAMPER	28-637082	5-103-1	Interlocked with Fan M25
FD12	FIRE DAMPER	28-637075	1-100-2	Interlocked with Fan M28
FD13	FIRE DAMPER	28-638069	1-100-1	Interlocked with Fan M27
FD14	FIRE DAMPER	28-637082	1-101-1	Interlocked with Fan M25
FD15	FIRE DAMPER	28-637082	1-108-1	Interlocked with Fan M26
FD16	FIRE DAMPER	28-638056	6-108-0	Interlocked with Fan M24
FD17	FIRE DAMPER	28-638056	6-108-0	Interlocked with Fan M24
FD18	FIRE DAMPER	HV-28-1004	5-57-2	Interlocked with Fan M23
FD19	FIRE DAMPER	28-637083	3-54.5-2	Interlocked with Fan M9
FD20	FIRE DAMPER	28-637083	3-55-3	Interlocked with Fan M17
FD21	FIRE DAMPER	28-638079	2-55.5-2	Interlocked with Fan M6
FD22	FIRE DAMPER	28-637086	2-55.5-1	Interlocked with Fan M5
FD23	FIRE DAMPER	28-637078	1-24-1	
FD24	FIRE DAMPER		A1-90-0	Interlocked with Fan M31
FD25	FIRE DAMPER		A1-90-0	Interlocked with Fan M32
FD26	FIRE DAMPER		A1-90-1	Interlocked with Fan M32
FD27	FIRE DAMPER		1-29-1	Interlocked with Fan M43
MD01	MOTORIZED DAMPER M1	28-626072	1-97-1	Interlocked with Fan M13
MD02	MOTORIZED DAMPER M1	HV-28-1088	1-57-0	
MD03	MOTORIZED DAMPER M1	HV-28-1087	1-57-0	
MD04	MOTORIZED DAMPER M1	Not Used		
MD05	MOTORIZED DAMPER M1	Not Used		
MD06	MOTORIZED DAMPER M1	HV-28-1283	1-106-1	Interlocked with Fan M13
MD07	MOTORIZED DAMPER M1	HV-28-0456	2-37-1	Interlocked with Fan M10
PD01	PNEUMATIC DAMPER	HV-28-0177	D-16-2	Pneumatic Operated

PD02	PNEUMATIC DAMPER	HV-28-0177	D-16-4	Pneumatic Operated	
PD03	PNEUMATIC DAMPER	HV-28-0177	D-16-6	Pneumatic Operated	
ITEM NO.	**LOCATION NO.**	**NASSCO PART NO.**	**DECK**	**NOTES**	
PD04	PNEUMATIC DAMPER	HV-28-0177	D-16-1	Pneumatic Operated	
PD05	PNEUMATIC DAMPER	HV-28-0177	D-16-3	Pneumatic Operated	
PD06	PNEUMATIC DAMPER	HV-28-0177	D-16-5	Pneumatic Operated	
MD08	MOTORIZED DAMPER M1	HV-28-1289	2-40-1	Interlocked with Fan M22	
MD09	MOTORIZED DAMPER M1	28-627054	A-34-2	Interlocked with Fan M29	
MD10	MOTORIZED DAMPER M1	HV-28-1891	1-103-1	Intlckd w Exh Hood Fan	
FCU110-FD01	FAN ROOM A8 LEVEL	HS09-EL333	A8 LEVEL	FDL-EL	BF 120-2
FCU110-FD02	FAN ROOM A8 LEVEL	HS09-EL334	A8 LEVEL	FDL-EL	BF 120-2
FCU110-FD03	FAN ROOM A8 LEVEL	HS09-EL335	A8 LEVEL	FDL-EL	BF 120-2
AC120-FD01	FAN ROOM A6 LEVEL	HS09-EL191	A6 LEVEL	FDL-EL	BF 120-2
AC121-FD01	FAN ROOM A6 LEVEL	HS09-EL192	A6 LEVEL	FDL-EL	BF 120-2
AC134-FD01	FAN ROOM A6 LEVEL	HS09-EL348	A5 LEVEL	FDL-EL	BF 120-2
AC136-FD01	MISSION SUPPORT NO.4	HS09-EL305	A5 LEVEL	FDL-EL	BF 120-2
E136-FD01	MISSION SUPPORT NO.4	HS09-EL306	A5 LEVEL	FDL-EL	BF 120-2
E136-FD02	PLENUM A6 LEVEL PS	HS09-EL378	A6 LEVEL	FDL-EL	BF 120-2
AC140-FD01	FAN ROOM A4 LEVEL	HS09-EL213	A4 LEVEL	FDL-EL	BF 120-2
AC140-FD02	FAN ROOM A4 LEVEL	HS09-EL214	A4 LEVEL	FDL-EL	BF 120-2
ER140-FD01	FAN ROOM A4 LEVEL	HS09-EL215	A4 LEVEL	FDL-EL	BF 120-2
E141-FD01	TRASH ROOM	HS09-EL216	A4 LEVEL	FDL-EL	BF 120-2
AC150-FD01	FAN ROOM A4 LEVEL	HS09-EL217	A4 LEVEL	FDL-EL	BF 120-2
AC150-FD02	FAN ROOM A4 LEVEL	HS09-EL218	A4 LEVEL	FDL-EL	BF 120-2
E150-FD01	FAN ROOM A6 LEVEL	HS09-EL219	A6 LEVEL	FDL-EL	BF 120-2
E151-FD01	FAN ROOM A4 LEVEL	HS09-EL220	A4 LEVEL	FDL-EL	BF 120-2
PART NO.	**LOCATION NO.**	**NASSCO PART NO.**	**DECK**	**TYPE/MODEL**	**ACTUATOR**
AC171-FD01	FAN ROOM A2 LEVEL	HS09-EL274	A2 LEVEL	FDL_EL	BF 120-2
E171-FD01	FAN ROOM A2 LEVEL	HS09-EL275	A2 LEVEL	FDL EL	BF 120-2
AC110-MD01	FAN ROOM A8 LEVEL	HS09-EL221	A8 LEVEL	UTG	BF 120-2
ER110-MD01	FAN ROOM A8 LEVEL	HS09-EL222	A8 LEVEL	UTG	BF 120-2
AC116-MD01	FAN ROOM AS LEVEL	HS09-EL325	AS LEVEL	UTG	BF 120-2
AC117-MD01	FAN ROOM A6 LEVEL	HS09-EL326	A6 LEVEL	UTG	BF 120-2

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0571	CATEGORY "A"	2026-01-30
ESB-CCSI-FIRE AND SMOKE DAMPER SERVICE (1 YR)		Manovich, David

AC120-MD01	FAN ROOM A6 LEVEL	HS09-EL223	A6 LEVEL	UTG	BF 120-2
AC121-MD01	FAN ROOM A6 LEVEL	HS09-EL224	A6 LEVEL	UTG	BF 120-2
PART NO.	**LOCATION NO.**	**NASSCO PART NO.**	**DECK**	**TYPE/MODEL**	**ACTUATOR**
E120-MD01	HANGAR A6 LEVEL	HS09-EL327	A6 LEVEL	UTG	EXMAX-15-F
E121-MD01	HANGAR A6 LEVEL	HS09-EL328	A6 LEVEL	UTG	EXMAX-15-F
S130-MD01	FAN ROOM A6 LEVEL	HS09-EL329	AS LEVEL	UTG	BF 120-2
S131-MD01	FAN ROOM A6 LEVEL	HS09-EL226	AS LEVEL	UTG	BF 120-2
S132-MD01	FAN ROOM A6 LEVEL	HS09-EL330	A6-96-9	UTG	BF 120-2
S133-MD01	FAN ROOM A6 LEVEL	HS09-EL331	A6-96-4	UTG	BF 120-2
AC134-MD01	FAN ROOM A6 LEVEL	HS09-EL332	AS LEVEL	UTG	BF 120-2
AC140-MD01	FAN ROOM A4 LEVEL	HS09-EL227	A4 LEVEL	UTG	BF 120-2
AC150-MD01	FAN ROOM A4 LEVEL	HS09-EL228	A4 LEVEL	UTG	BF 120-2
AC160-MD01	FAN ROOM A3 LEVEL	HS09-EL229	A3 LEVEL	UTG	BF 120-2
AC170-MD01	FAN ROOM A2 LEVEL	HS09-EL230	A2 LEVEL	UTG	BF 120-2
AC171-MD01	FAN ROOM A2 LEVEL	HS09-EL278	A2 LEVEL	UTG	BF 120-2
E141-MD01	TRASH ROOM	HS09-EL231	A4 LEVEL	UTG	BF 120-2
E141-MD02	PROCESSED WASTE STOR ROOM	HS09-EL307	A4 LEVEL	UTG	BF 120-2
E141-MD03	SOLID WASTE MGT ROOM	HS09-EL308	A4 LEVEL	UTG	BF 120-2
E142-MD01	INCINERATOR ROOM	HS09-EL232	A4 LEVEL	UTG	BF 120-2
E142-MD02	INCINERATOR ROOM	HS09-EL233	A4 LEVEL	UTG	BF 120-2
S180-MD01	STAIRTOWER. PS. A5 LEVEL	HS09-EL279	A5 LEVEL	UTG	BF 120-2
S181-MD01	FAN ROOM A4 LEVEL	HS09-EL312	A1-915-2	UTG	BF 120-2
S182-MD01	STAIRTOWER. SB, A5 LEVEL	HS09-EL313	A5-90-1	UTG	BF 120-2
AC3-FD01	FAN ROOM 1-52-5	H509-EL080	1-52-5	FDL-EL550-450	BF 120-2
E2-FD01	GALLEY A-47-0	H509-EL081	A-39-1	FDL-EL600-300	BF 120-2
E2-FD02	GALLEY A-47-0	H509-EL082	A-39-1	FDL-EL800-350	BF 120-2
AC3-FD01	Fan Room 1-52-5	H509-EL080	Upper Deck	FDL-EL550-450	BF 120-2
E2-FD01	Galley A-47-0	H509-EL081	A-Deck	FDL-EL600-300	BF 120-2
E2-FD02	Galley A-47-0	H509-EL082	A-Deck	FDL-EL800-350	BF 120-2

ENCLOSURE 2.2.2 NFPA FORM 17-C

FIRE AND SMOKE DAMPERS
NFPA FORM 17-C

Inspection and Maintenance

Vessel: ________________ **IMO #:** ________________

Shipyard: ________________ **Date:** ________________

Service Company: ________________ **Inspector:** ________________

1) Full unobstructed access to dampers is provided.
2) Smoke damper needs to be cycled through its closure and opening in accordance with NFPA 72.
3) Fusible link, if provide, is removed and damper closes completely and latches closed (where latch provided).
4) No interference with damper operation by rusted, bent, or misaligned parts, damaged blades, or defective hinges.
5) Damper frame not penetrated by foreign objects to prevent proper operation.
6) Fusible link is not damaged or painted. If damaged, it should be replaced.
7) All exposed moving parts to be dry lubricated per manufacturer's instructions.
8) Note deficiencies and how they were corrected. After any repairs, the damper should be retested for proper operation.

ENCLOSURE 2.2.2 NFPA FORM 17-C

Y = Satisfactory **N** = Unsatisfactory (explain below) **N/A** = Not applicable

No.	Location	1	2	3	4	5	6	7	8

ENCLOSURE 2.2.2 NFPA FORM 17-C

Y = Satisfactory **N** = Unsatisfactory (explain below)
N/A = Not applicable

No.	Location	1	2	3	4	5	6	7	8

1.0 ABSTRACT

The purpose of this item is to inspect, service, recharge, and test the Deep Fat Fryer, Grill and Range Hood Fire Extinguishing Systems.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NFPA 17 Standard for Dry Chemical Extinguishing Systems (Available Commercially)

2.1.2 NFPA 17A Standard for Wet Chemical Extinguishing Systems (Available Commercially)

2.1.3 ISO 15371, Ships and Marine Technology — Fire-Extinguishing Systems for Protection of Galley Cooking Equipment (Available Commercially)

2.1.4 SOLAS Chapter II-2 Construction - Fire Protection, Fire Detection and Fire Extinction, Part C Suppression of Fire, Regulation 10 Firefighting, Para 6 (Available Commercially)

2.1.5 Tech Manual, T6161-LG-FSE-010, Galley Hood Fire Suppression System

2.1.6 Tech Manual T6161-LP-MMC-010, Vapor Hood, Exhaust Ventilator and Command Control Center; Installation, Operation And Maintenance Manual

2.2 Enclosures

2.2.1 Design of a Verification of Service Collar

2.2.2 Deep Fat Fryer, Grill and Range Hood Fire Extinguishing Systems – Annual Inspection

2.2.3 Deep Fat Fryer, Grill and Range Hood Fire Extinguishing Systems - Service Record

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Aft Galley (A-47-0)

b. Forward Galley (A4-92-1)

3.1.2 Quantity

a. Deep Fat Fryer: **Two Systems in Aft Galley**

b. Grill: **One System in Forward Galley**

c. Three Range Hoods

3.1.3 Item Description/Manufacturer’s Data

a. Range Hood Extinguishing System Manufacturer: Ansul by Tyco Fire Suppression and Building Products, Model R-102 Restaurant Fire Suppression System.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service, recharge, and test all Deep Fat Fryer, Grill and Range Hood Fire Extinguishing Systems using References 2.1.1 through 2.1.6 and Enclosures 2.2.1 through 2.2.3 as guidance.

7.2 Coordinate with ship's force to have associated equipment locked and/or tagged out per ship's lock out/tag out policy. When the maintenance is complete, coordinate with ship's force to have associated equipment unlocked and/or tagged in per ship's lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Temporarily remove bulkhead and overhead panels only as necessary to gain access to the ducting and thermostatic switches. Temporarily remove access covers and plates to the ventilation systems to allow inspection. Existing openings will be used to the maximum extent possible. New access openings will be approved in advance by the OMT REP only. New openings in watertight ducting sections are not permitted. Upon completion and acceptance of all work, reinstall access covers and panels using new gaskets and sheet metal fasteners.

7.5 Conduct an annual inspection of Deep Fat Fryer, Grill and Range Hood Fire Extinguishing Systems meeting the manufacturer's design, installation, maintenance instructions and service bulletin requirements using References 2.1.1 through 2.1.6 and Enclosures 2.2.1 through 2.2.3 as guidance. The inspection includes/verifies:

7.5.1 Neither the extinguishing system nor the protected equipment has been modified or relocated.

7.5.2 The exhaust system and filters have been cleaned.

7.5.3 The Deep Fat Fryer is fitted with an automatic or manual extinguishing system tested to an international standard using Reference 2.1.4 as guidance.

7.5.4 The grease extraction hood is equipped with a dry or wet chemical fire extinguishing system meeting 46 CFR§118.425.

7.5.5 Using Reference 2.1.2 as guidance, ensure that upon activation of the fire extinguishing system, electrical power is automatically shut off to the cooking appliances.

7.5.6 Using Reference 2.1.4 as guidance, ensure there is an alarm in the galley for indicating activation of the fire extinguishing system

7.5.7 The detectors, the expellant gas containers, the agent containers, releasing devices, dampers, piping, hose assemblies, nozzles, signals, fire extinguishing agent levels and all auxiliary equipment are in proper operating condition.

7.5.8 The systems show no physical damage, corrosion, leakage or condition that might prevent operation.

7.5.9 The pressure gauges are indicating within their operable range.

7.5.10 Using Reference 2.1.3 as guidance, ensure there is at least one manual actuator located in the path of egress.

7.5.11 The manual actuators are marked, visible and unobstructed.

7.5.12 The operating instructions are visible and legible.

7.5.13 Electrical breakers are properly marked.

7.5.14 The tamper indicators and seals are intact.

7.5.15 The maintenance tag or certificate is in place.

7.5.16 Using Reference 2.1.3 as guidance, ensure a placard is conspicuously displayed near each Class K portable fire extinguisher that states that the fixed fire protection system is to be activated prior to using the portable extinguisher.

7.5.17 The nozzle blow off caps or foil sealing discs installed in the nozzles are intact and undamaged.

7.5.18 The agent distribution piping is not obstructed. Use dry air or nitrogen and blow through the agent distribution piping with the nozzle blow-off caps removed, verifying that dry air or nitrogen is discharging at each nozzle location.

7.5.19 Verify upon activation of the ventilation hood fire protection system, the fire dampers close, and the ventilation fans stop. If equipped, fire smothering water spray is released into the interior of the ventilator.

7.5.20 Verify and record the date the temperature sensing elements of the fusible metal alloy were last replaced.

7.5.21 Using Reference 2.1.3 as guidance, in locations where automatic fire extinguishing systems provide protection for hoods and ducts containing a water wash system, the water wash system must be delayed for a minimum of 60 seconds upon operation of the automatic fire extinguishing system using Reference 2.1.3 as guidance.

7.6 Conduct annual maintenance of Deep Fat Fryer, Grill and Range Hood Fire Detection, Alarm and Fire Extinguishing Systems meeting the manufacturer’s design, installation, maintenance instructions and service bulletin requirements using References 2.1.1 through 2.1.6 and Enclosures 2.2.1 through 2.2.3 as guidance. The maintenance includes/verifies:

7.6.1 Verify each fire extinguisher is clearly marked with the name of the manufacturer, class fire rating, type and quantity of extinguishing medium, approval details and year of manufacture.

7.6.2 Examination of all extinguishers. All removable extinguisher boots, foot rings and attachments are to be removed to accommodate the cylinder exams. Identify physical damage, corrosion, or any condition that may lead to improper operation, and determine if internal exam or hydrostatic test is due.

7.6.3 Using Reference 2.1.1 as guidance, verify dry chemical within a stored pressure system is examined at least every 6 years.

7.6.4 Where inspection or maintenance of any dry or wet chemical containers reveal pitting or corrosion in excess of the manufacturer's limits, or show symptoms of structural damage, fire damage, or repair by welding or brazing, the container is to be hydrostatically tested or replaced.

7.6.5 Using Reference 2.1.1 as guidance, perform the following minimum maintenance of the restorable type heat detectors:

a. Clean and visually exam all fixed temperature sensing heat detectors for damage or buildup of debris.

b. Perform an operational/functional test of the detectors using the manufacturer's instructions to verify their settings.

c. Conduct a calibration verification test of the detector using the manufacturer's instructions for guidance.

7.6.6 Fixed temperature sensing elements of the fusible metal alloy type are to be replaced at least annually from the date of installation. The old sensing elements are to be destroyed when removed. The year of manufacture and date of installation of the fixed temperature sensing element are to be marked on the system inspection tag.

7.6.7 Calibrate all Detroit switches associated with each range hood.

7.7 If hydrostatic test is due or insufficient charge is noted during inspection or maintenance:

7.7.1 The system is to be recharged using the manufacturers design, installation, and service manual requirements for guidance.

7.7.2 Using References 2.1.1 through 2.1.6 as guidance, discard dry and wet chemical agents removed from the containers prior to hydrostatic testing.

7.7.3 After recharging, a leak test must be performed on stored-pressure and self-expelling types of extinguishers.

7.7.4 In no case is an extinguisher to be recharged if it is beyond its specified hydrostatic test date.

7.8 Conduct hydrostatic testing of Deep Fat Fryer, Grill and Range Fire Extinguishing Systems meeting the manufacturer's design, installation, maintenance instructions and service bulletin requirements using References 2.1.1 through 2.1.6 and Enclosures 2.2.1 through 2.2.3 as guidance. The hydrostatic testing includes/verifies:

7.8.1 Wet and dry chemical containers, auxiliary pressure containers, and hose assemblies are hydrostatically tested every 10 years per SOLAS/II-2/14.

NOTE: 29 CFR AND NFPA 17 AND A7A (7.5.1) PERMIT TEST INTERVALS OF 12 YEARS HOWEVER SOLAS/II-2/14 MANDATES INTERVALS NOT TO EXCEED TEN YEARS

7.8.2 Hydrostatic testing is to be performed by persons certified and trained in pressure testing procedures and safeguards. All testing is to be conducted using clean fresh water as the test medium.

7.8.3 Hydrostatic testing always includes an internal and external visual examination prior to beginning the test.

7.9 Conduct annual system test of Deep Fat Fryer, Grill and Range Hood Fire Detection, Alarm and Fire Extinguishing Systems meeting the manufacturer's design, installation, maintenance instructions and service bulletin requirements using References 2.1.1 through 2.1.6 and Enclosures 2.2.1 through 2.2.3 as guidance. The maintenance includes/verifies:

7.9.1 Simulate a functional test of the fire detection, control, alarm and extinguishing systems demonstrating proper operation both in manual and automatic mode. Using Reference 2.1.3 as guidance, manual activation using cable operated pull stations is not to require more than 40 lbs. of force.

7.9.2 Demonstrate proper operation of the ventilator hood as related to fire protection (damper closure, vent fan shutdown, and water mist).

7.9.3 Using References 2.1.1 through 2.1.3 and 2.1.5 through 2.1.6 as guidance, demonstrate proper operation of the deep fat fryer and cooking appliances as related to fire protection (power is shut off upon activation of the fire extinguishing system).

7.10 Tags and Labels

7.10.1 Tags and labels documenting inspection, maintenance, recharging or hydrostatic testing are to be affixed so as not to obstruct the extinguishers use, classification or manufacturers labels.

7.10.2 Each extinguishing system is to have a tag or label securely attached that indicates inspection and maintenance performed. The tag at a minimum identifies the following:

a. Month and year of maintenance.

b. Name of person performing the work.

c. Name of the company performing the work.

d. The year of manufacture and date of installation of the fixed temperature sensing element.

7.10.3 Using Reference 2.1.3 as guidance, ensure each cylinder that has undergone maintenance that included an internal examination or has been recharged requiring the removal of the valve assembly must also have a verification-of-service collar installed around the neck of the container as shown in Enclosure 2.2.1. The collar must not interfere with the operation of the extinguisher and must be a single circular piece unable to move over the neck of the container unless the valve is completely removed. The collar at a minimum identifies the following:

a. Month and year of the recharge or internal exam.

b. Name of the company performing the work.

7.10.4 Using Reference 2.1.1 as guidance, determine the Dry Chemical extinguishers that passed the 6-year internal examination. The maintenance info for each of the identified cylinders are to be recorded on a durable weatherproof label that is a minimum of 2” x 3.5” (51mm x 89mm) affixed to the shell. Any previous exam labels are to be removed. The labels are to be of the self-destructive type when their removal is attempted. The label, at a minimum, identifies the following:

a. Month and year of six-year internal examination.

b. Name of person performing the work.

c. Name of the company performing the work.

7.10.5 Fire extinguishers that are low pressure non-DOT type that pass a hydrostatic test are to have the info recorded on a durable weatherproof label that is a minimum of 2” x 3.5” (51mm x 89mm) affixed by to the shell. Any previous test labels are to be removed. The labels must be of the self-destructive type when their removal is attempted. The label, at a minimum, identifies the following:

a. Month and year of test.

b. Test pressure used.

c. Name or initials of person performing the test.

d. Name of the company performing the test.

7.10.6 Fire extinguishers that are high pressure cylinders or cartridges that pass a hydrostatic test are to be stamped with the retesters’ ID number and the month and year of the retest as per DOT/TC requirements. Stamping is to be placed on the shoulder, top, head, neck or foot ring of the cylinder or in other acceptable locations as identified by 49 CFR 180.213.

7.10.7 Testing is to be coordinated with the OMT REP and ABS and USCG to allow for observation if deemed necessary.

7.10.8 Upon completion of all inspections, tests and repairs, return the Fire Extinguishing Systems to a ready for service condition. Demonstrate and verify system status with the OMT REP and ship’s Master.

7.11 Maintain work spaces in a clean condition during and upon completion of the work requirements. Prevent contamination of any adjacent equipment and spaces.

7.12 Reports

7.12.1 When inspections, service and testing reveal any deficiency, a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.12.2 Complete the Annual Inspection form included as Enclosure 2.2.2 of this work item. The initial condition of every Fire Extinguishing system must be documented. Submit one electronic copy in Portable Document Format (PDF).

7.12.3 Upon completion of all inspections, maintenance, servicing, recharging, and testing, prepare and submit a Service Record, included in Enclosure 2.2.3 of this work item, documenting the final “as released” condition of all Fire Extinguishing Systems affirming they are safe for use. Submit one electronic copy in Portable Document Format (PDF).

7.12.4 All reports and checklists are to be completed and signed by the person who carried out the inspections, maintenance, servicing, recharging, and testing and are to be countersigned by the Company's representative.

7.13 Painting: None additional

7.14 Manufacturer’s Representative

7.14.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the deep fat fryer and range hood identified in section 3.0 ensuring compliance with the manufacturer’s standards and performance specifications.

7.14.2 This work item requires maintenance and repair actions to be performed on a MSC ship critical safety item. Only original equipment manufacturer (OEM) authorized technical field service providers or MSC qualified non-OEM technical field service providers and OEM authorized or MSC qualified non-OEM parts will be used to accomplish the requirements of this work item for this ship critical safety item including oversight and guidance on all aspects of equipment as-found condition inspection, removal, disassembly, reassembly, repairs, modifications, reinstallation, and testing as applicable. All tests shall be completed prior to Galley/Habitability turn over.

7.14.3 Contractors interested in gaining MSC qualification as a non-OEM parts and/or service provider should visit this website for qualification and submittal requirements: https://www.msc.usff.navy.mil/Business-Opportunities/Contracts/Qualification-for-Items-Critical-to-Shipboard-Safety-on-MSC-Vessels/

7.14.4 Possible Sources

Lang World, Inc.
265 Hobson Street
Smithville, TN 37166
Phone: (800) 264-7827
info@langworld.com

Hiller Systems
POC: Dan Clift
Email:dclift@hillercompanies.com

7.14.5 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

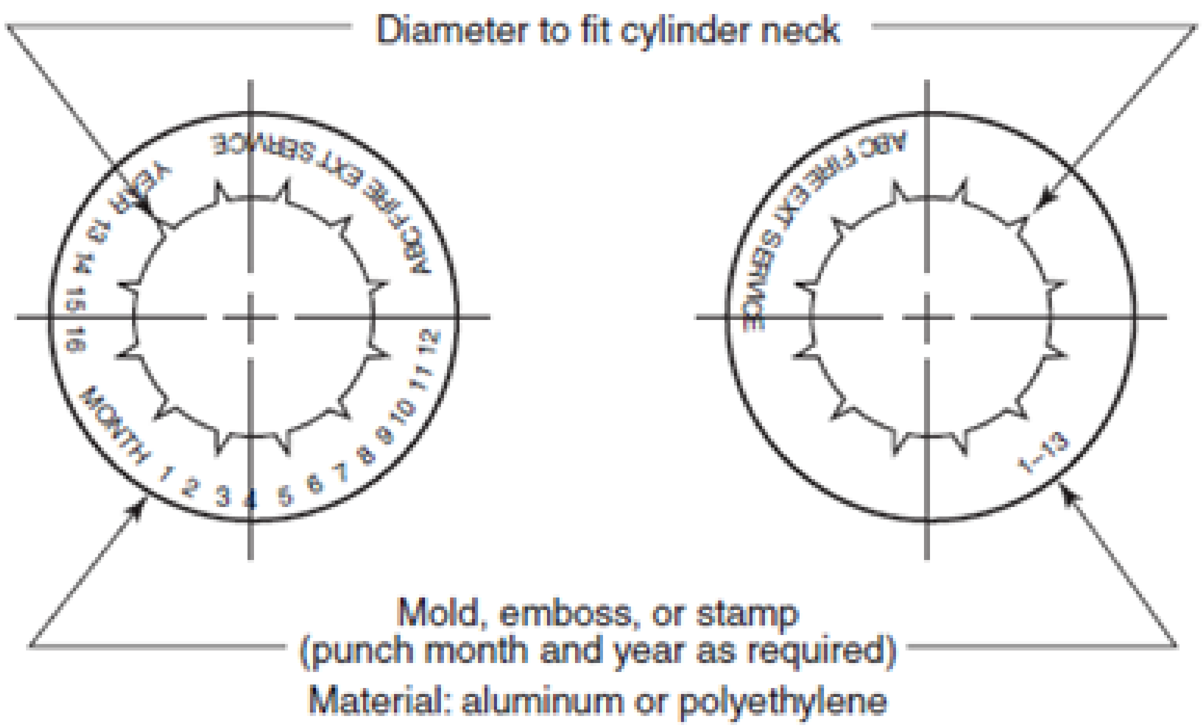
7.15 Preparation of Drawings

7.15.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - DESIGN OF A VERIFICATION OF SERVICE COLLAR

FOR GUIDANCE ONLY



ENCLOSURE 2.2.2 - DEEP FAT FRYER, GRILL AND RANGE HOOD FIRE EXTINGUISHING SYSTEMS - ANNUAL INSPECTION

Vessel: ______________________ **IMO #:** ______________________

Shipyard: ______________________ **Date:** ______________________

FX Company: ______________________ **Inspector:** ______________________

To be completed by the extinguishing system inspector at the time of inspection or test. The annual inspection shall at a minimum include checks and tests of the following:

S = Satisfactory U = Unsatisfactory (explain below) **N/A** = Not applicable

ENCLOSURE 2.2.2 – DEEP FAT FRYER, GRILLL AND RANGE HOOD FIRE EXTINGUISHING SYSTEMS – ANNUAL INSPECTION

No.	Description	Condition (S/U/NA)	Initials
1	Neither the extinguishing system nor the protected equipment has been modified or relocated.		
2	The exhaust system and filters have been cleaned.		
3	Record the type of deep fat fryer oil used and its flashpoint. Type: Flashpoint:		
4	The flashpoint of the deep fat fryer oil is in accordance with the User Manual requirements.		
5	The deep fat fryer is fitted with an automatic or manual extinguishing system tested to an international standard per ref 2.1.4.		
6	The grease extraction hood is equipped with a dry or wet chemical fire extinguishing system meeting NFPA 17 or 17A per 46 CFR§118.425.		
7	That upon activation of the fire		

ENCLOSURE 2.2.2 – DEEP FAT FRYER, GRILLL AND RANGE HOOD FIRE EXTINGUISHING SYSTEMS – ANNUAL INSPECTION

No.	Description	Condition (S/U/NA)	Initials
	extinguishing system electrical power is automatically shut off to the cooking appliances per ref 2.1.2 (9.3.5) and 2.1.6.		
8	There is an alarm in the galley for indicating activation of the fire extinguishing system per ref 2.1.4.		
9	The systems show no physical damage, corrosion, leakage or condition that might prevent operation.		
10	The pressure gauge(s), if provided, are indicating within their operable range.		
11	There is at least one manual actuator located in the path of egress, ref 2.1.3 (10.5.1.1).		
12	The manual actuators are marked, visible and unobstructed.		
13	The operating instructions are visible and legible.		
14	Electrical breakers are properly marked.		
15	The tamper indicators and seals are intact.		
16	The maintenance tag or certificate is in place.		
17	A placard is conspicuously displayed near each Class K portable extinguisher that states that the fixed fire protection system shall be activated prior to using the portable extinguisher per ref 2.1.3.		
18	The nozzle blowoff caps or foil sealing discs installed in the nozzles, preventing grease buildup, are intact and undamaged.		
19	Verify the agent distribution piping is not obstructed. Use dry air or nitrogen and blow through the agent distribution piping with the nozzle blow-off caps removed, verifying that dry air or nitrogen is discharging at each nozzle		

ENCLOSURE 2.2.2 – DEEP FAT FRYER, GRILLL AND RANGE HOOD FIRE EXTINGUISHING SYSTEMS – ANNUAL INSPECTION

No.	Description	Condition (S/U/NA)	Initials
	location.		
20	Verify internal condition of ducting and thermostatic switches.		
21	Verify upon activation of the ventilation hood fire protection system the fire dampers close and the ventilation fans stop. And if so equipped, fire smothering water spray is released into the interior of the ventilator.		
22	Verify & record the date the temperature sensing elements of the fusible metal alloy type were last replaced. Date:		
23	Verify where automatic fire extinguishing systems, in accordance with NFPA 17A, are providing protection for hoods & ducts containing a water wash system that the water wash system is delayed for a minimum of 60 seconds upon activation of the automatic fire extinguishing system.		
24	Verify the detectors, the expellant gas container(s), the agent container(s), releasing devices, dampers, piping, hose assemblies, nozzles, signals, fire extinguishing agent levels and all auxiliary equipment are in good operating condition.		

Description of any unsatisfactory conditions noted:

ENCLOSURE 2.2.3 DEEP FAT FRYER, GRILL AND RANGE HOOD FIRE EXTINGUISHING SYSTEMS - SERVICE RECORD (TO BE FILLED OUT/PROVIDED BY PPE)

Vessel: ____________________ **IMO #:** ____________________

Shipyard: ____________________ **Date:** ____________________

FX Company: ____________________ **Inspector:** ____________________

Location	Size (lbs.)	Class A/B/K	FX Medium	Serial No.	Year of Mfg.	Visual Inspection		Hydrostatic Test		Recharge (date)	Hose Inspection	Fusible Link Replaced (date)
						Last External (date)	Last Internal (date)	Last Test (date)	Pressure (psi)			

LEGEND -Fire Extinguishing Medium

Dry Chemical (DC)

Wet Chemical (WC)

1.0 ABSTRACT

The purpose of this item is to inspect, service and test the ship's fire hoses and stations.

2.0 REFERENCES

2.1 MSC Drawing No. 830-8496378, Fire Control and Safety Plan

2.2 46 CFR §95.10-10, Fire Hydrants and Hose

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. 120 fire station locations identified in Reference 2.1.

b. Fire Locker 1 (1-126-0)

c. Fire Locker 2 (1-43-1)

3.1.2 Quantity

a. 229, 50 and 75-foot lengths of 1 ½ inch fire hose

b. 50, hoses 50-foot lengths at fire stations

c. 30, spare hoses, total, located in Damage Control Repair Lockers

d. Seven, hoses 50-foot length of 2 ½ inch fire hose located in Damage Control Repair Lockers

e. 20, hoses length of 4 inch fire hoses located in Damage Control Repair Lockers

3.2 Item Description: Fire Hoses

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Per MSC-USCG MOA dtd 2018, approval is granted for the use of MILSPEC MIL-H-24606B 1.75 inch diameter fire hose with 1.50 inch brass couplings with NH thread in lieu of UL 19. In addition, the use of wye-gate valves with 2-1/2” inlet and 1-1/2” outlets for fire hose connections in which the closure is composed of resilient nonmetallic material is permissible provided the station can be isolated from the firemain by a cut-out valve (COV) with metal to metal seats. The couplings for all fire hydrants, nozzles, hose connections, and other fittings are to have uniform thread configuration, for each diameter hose, to accommodate interchangeability.

5.3 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 All items will be serviced and maintained in accordance with OEMs specifications and certification requirements as set forth by the USCG, ABS, and National Fire Protection Agency (NFPA).

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service, and test the ship’s fire hoses and fire stations identified in Reference 2.1 and section 3.0 using Reference 2.2 as guidance. Accomplish inspection, servicing, and testing in accordance with IMO, SOLAS, USCG and the Manufacturer’s requirements.

7.2 Coordinate with ship’s force to tagout all required systems and equipment. Drain and dispose of any liquid, to vessel tanks.

7.3 Provide temporary fire hoses and equipment to adequately protect the ship from fire during the accomplishment of this item. No more than 25% of the ship’s hoses should be removed at any one time. Coordinate removals with OMT REP to ensure the vessel always has adequate coverage until the hoses are returned and reinstalled.

7.4 Conduct an annual inspection of the fire hoses and stations using Reference 2.2 as guidance. The inspection is to verify:

7.4.1 Each fire hydrant has at least one length of fire hose, a spanner wrench, and a hose rack or other device for stowing the hose. Spanners are suitable for use on the hose at that station.

7.4.2 Each fire station hydrant or “y” branch is equipped with a valve so that the hose may be removed while there is pressure on the fire main.

a. Inspect all flanged connections to fire main piping and hydrants for gasket material that has been identified by APB to be "readily rendered ineffective by heat". An example of gasket material that has been accepted by ABS for us in fire main and hydrant flanged connections is reddish brown Garlock Series 9900 or equal ABS Type Approved material per ABS Rules 4-6-1/3.5, 4-6-2/9.5 vi), 4-7-3/1.11.1. Include a listing of connection that are unacceptable or questionable to the OMT REP.

7.4.3 Firehoses are connected to the outlets. On open decks where no protection is afforded to the hose in heavy weather, or where the hose may be liable to damage from the handling of cargo, the hose may be temporarily removed from the hydrant and stowed in an accessible nearby location.

7.4.4 Each firehose on each hydrant has a combination solid stream and water spray firehose nozzle approved under subpart 46 CFR §162.027.

7.4.5 Visually examine each hose for broken, damaged, or missing parts.

7.4.6 In each propulsion machinery space containing an oil-fired boiler or internal combustion machinery, each fire hose having combination nozzle must have a low-velocity water spray applicator that is approved under subpart §162.027. The length of the applicator must be less than 1.8 meters (6 feet). For Tank Vessel requirements see 46 CFR Table 34.10-10(E).

7.4.7 Each low-velocity water spray applicator has fixed brackets, hooks, or other means for stowing next to the hydrant.

7.4.8 Firehose is not being used for any other purpose than fire extinguishing, drills, and testing.

7.4.9 Inspect the physical condition of all hoses checking for damage, cuts, abrasions, and mildew.

7.4.10 Visually examine the hose lining at each end for signs of delamination.

7.4.11 Fire station hydrant connections are brass, bronze, or other equivalent metal.

7.4.12 Fire hydrants, nozzles, hose couplings, and other fittings have threads to accommodate the hose connections noted below. Couplings are to:

a. Use National Standard firehose coupling threads for the 1-1⁄2 inch (38 millimeter) and 2-1⁄2 inch (64 millimeter) hose sizes (9 threads per inch for 1-1⁄2 inch hose, and 7-1⁄2 threads per inch for 2-1⁄2 inch hose); or

b. To be a uniform design, for each diameter hose, throughout the vessel to allow for interchangeability.

c. Each section of firehose is lined commercial firehose that conforms to Underwriter’s Laboratories, Inc. Standard 19 or Federal Specification ZZ-H-451E.

d. Ensure there is a shore connection to the fire main on each side of the vessel in an accessible location. Ensure there is an international voyage complying with ASTM F 1121. The international connection is to be capable of being used on either side of the vessel.

7.5 Conduct annual maintenance on all fire hoses and stations in accordance with the manufacturer’s design, installation, maintenance instructions and service bulletins. The maintenance will include/verify:

7.5.1 Thoroughly wash out and clean the interior surfaces of all exterior fire station boxes, removing all dirt, debris, mildew, and water prior to re-stowing fire hoses.

7.6 Conduct annual testing of the fire hoses using Reference 2.2 as guidance and per 46 CFR §91.25-20(4). The testing is to include:

7.6.1 Hydrostatically test all firehoses to a test pressure equivalent to the maximum pressure to which they may be subjected in service, but not less than 100 psi (690 kPa). Before testing, the firehoses should be removed from any space containing cargo.

7.6.2 The total length of any hose line in the hose test layout to be service tested is not to exceed 300 feet per Reference 2.2.

7.6.3 The hose test layout will be straight, without kinks or twists.

7.6.4 The test medium is to be clean fresh water.

7.6.5 Ensure all air has been bled from the hoses before pressure testing.

7.6.6 Test pressure will be held for a minimum of three minutes.

7.6.7 Ensure the test gauge has been calibrated within the last 12 months.

7.6.8 Upon successful completion, stencil the fire hoses with date of each service test and the service test pressure. Any new hoses will also be stenciled with the Vessel’s Name. Only a paint marker, spray paint, or fast drying stenciling ink that meets MILSPEC A-A-00508D is permitted to stencil or cover each fire hose. Apply product as recommended by the stenciling ink manufacturer. Apply the minimal amount of coverage over the previous hydrostatic testing date so that it no longer shows. Stencil and covering should be white or black, whichever color provides the most contrast. Recommend a non-toxic/non-corrosive metal barrel tube type permanent paint marker; spray paint that is exclusively used for stenciling, or a fast drying stenciling ink applied by a roller.

7.7 Testing is to be coordinated with OMT REP and ABS and USCG. Any firehose found to be defective and incapable of repair is to be destroyed in the presence of OMT REP. Once the failed firehoses are destroyed the area must be cleaned up and all debris removed from the vessel. Defective hoses are to be replaced with new ship’s spares.

7.8 Care is to be used to protect the hoses and coupling threads from damage during the accomplishment of this work item. To prevent mechanical damage, firefighting appliances are not to be dropped or dragged.

7.9 Upon completion of all inspections, tests and repairs, return the fire hoses to the vessel and their respective station, apply a light coat of silicon-based lubricant to coupling threads, and leave fire hoses in a ready for service condition. All hoses are to be drained and thoroughly dried before being placed in storage.

7.10 Reports

7.10.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.10.2 Submit a report that includes “as released” condition of the firehoses. Submit one electronic copy in Portable Document Format (PDF).

7.10.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.11 Painting: None additional

7.12 Manufacturer’s Representative: None

7.13 Preparation of Drawings

7.13.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

<table>
<tr><td colspan="3">USS Hershel Williams</td></tr>
<tr><td colspan="3">(ESB 4)</td></tr>
<tr><td colspan="2">AUXILIARY MACHINERY</td><td>CONTRACT NO. N68171-26-RN-003</td></tr>
<tr><td>ITEM NO. 0574A</td><td>CATEGORY "A"</td><td>2026-02-02</td></tr>
<tr><td colspan="2">ESB-CCSI-FIRE DOORS AND SHUTTERS - SLIDING WATERTIGHT DOORS (1 YR)</td><td>Port Engineer's Name</td></tr>
</table>

1.0 ABSTRACT

The purpose of this item is to describe the annual requirements to inspect, service, and test ships critical fire doors and shutters.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 830-8496378, Fire Control and Safety Plan

2.2 Tech Manual T9167-BX-MMC-010, Doors, Sliding, Watertight-Models USA-1046-7L, USA-1046-7R and USA-1046-8L

2.3 Tech Manual T9167-BY-MMC-010, Door, Rollup, Curtain Type HH8000, HD8000, HH9000

2.4 NAVSEA Drawing No. 624-8468294, Joiner Door and Passing Window Schedule

2.5 NAVSEA Drawing No. 624-8452231, Joiner Door and Passing Window Details

2.6 NAVSEA Drawing No. 624-8468286, Joiner Door and Passing Window Schedule

2.7 Subpart 46 CFR 111.99, Fire Door Holding and Release Systems

2.8 SOLAS II-2, Construction, Fire Protection, Fire Detection and Fire Extinction, Part C Suppression of Fire, Regulation 9 Containment of Fire (Available commercially)

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Door Locations

3.1.1 Sliding Watertight Doors

a. Door No. 1 - Left Hand Opening 31-1/2” X 71”, 73 ft. Head Pressure, Located on Tank Top, Service Trunk, Frame 97

b. Door No. 2 - Left Hand Opening 31-1/2” X 71”, 73 ft. Head Pressure, Located on Tank Top, Service Trunk, Frame 57

c. Door No. 3 - Right Hand Opening 31-1/2” X 71”, 73 ft. Head Pressure, Located on Fourth Deck, Machinery Room 1 and 2, Frame 50

d. Door No. 4 - Right Hand Opening 31-1/2” X 71”, 73 ft. Head Pressure, Located on Second Deck, Engine Control Room 1 and 2, Frame 45

e. Door No. 5 - Left Hand Opening 35-1/2” X 71”, 53 ft. Head Pressure, Located on Third Deck, Machinery Room 1 and 2, Frame 20

f. Door No. 6 - Left Hand Opening 31-1/2” X 71”, 73 ft. Head Pressure, Located on Second Deck, Steering Gear Room 1 and 2, Frame 4

g. Door No. 7 - Right Hand Opening 31-1/2” X 71”, 73 ft. Head Pressure, Located on Tank Top, Machinery Room 1 and 2, Frame 20

3.1.2 Roller Doors

a. Port Roller Door (A5-86-4) to MHE Stowage/Maintenance Repair Area (A5-89-4)

b. Starboard Roller Door (A5-86-3) serving Passage (A5-96-3)

c. Hangar Roller Door (A5-86-0) to (A5-96-0)

3.1.3 Passing Windows

a. Aft Galley Passing Window with rolling shutter (A-47-0)

b. Fwd Galley Passing Window with rolling shutter (A4-92-1)

3.1.4 Joiner Doors

a. Quantity: 400, Various Locations, see Reference 2.4 through 2.6

3.2 Description/Quantity

a. Seven Sliding Firetight/Watertight Doors

Manufacturer: USA Sliding Door

Model: USA-1046

b. Three Rolling Curtain Fire Doors

Manufacturer: Federal Equipment Company

Model: HH9000, Quantity: Two

Model: HH8000, Quantity: One

c. Four Rolling Shutter/Passing Windows

Model: US Joiner Roller Shutter

d. 400 Fire Rated Joiner Doors

Model: US Joiner Doors, see References 2.4 through 2.6

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Fire door; means a door that is in a fire boundary, such as a stairway enclosure or main vertical zone bulkhead, that is not usually kept closed.

5.3 Fire door holding magnet; means an electromagnet for holding a fire door open.

5.4 Local control panel; means a manually-operated device next to a fire door for releasing the door so that the fire door self-closing mechanism may close the door.

5.5 Central control panel; means a manually-operated device on the navigating bridge or in the fire control room for releasing one or more fire doors.

5.6 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

5.7 THE ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Examine, test, and repair fire doors and shutters identified in section 3.0 in accordance with NFPA, SOLAS, and ABS and USCG requirements using References 2.1 through 2.8 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.2.1 Temporarily remove access doors/panels, overhead panels in passageways and compartments, paneling and any other interferences to access all fire doors and shutters and their operators listed in section 3.0.

7.3 Conduct inspections, maintenance, and testing of the fire doors, shutters and controls using References 2.1 through 2.8 as guidance.

7.4 Conduct an annual inspection of the fire doors, shutters and controls listed in section 3.0 using References 2.1 through 2.8 as guidance. The examination will include/verify:

7.4.1 A visual inspection of both sides of the door or shutter.

7.4.2 No field modifications have been made.

7.4.3 Full unobstructed access to doors and shutters is provided.

7.4.4 There is no rusted, bent, misaligned, defective or missing parts.

7.4.5 The door or shutter frame and guides are not warped, damaged or penetrated and the fire boundary integrity is maintained at openings and penetrations.

7.4.6 With assistance from ships force, cycle each door through its closure and opening. Check for proper operation and full closure as well as travel and proper operation of limit switches. Verify it is possible for each door to be opened and closed from each side of the bulkhead by one person only.

7.4.7 Doors in stairtowers, stairways, main vertical zone bulkheads and galley boundaries are self-closing and hold-back hooks are not fitted, unless subject to central control station release.

7.4.8 Chains and cables are in good operating condition. Inspect chains and cables for excessive wear, stretching and binding and verify they are not kinked, pinched, twisted, inflexible, painted or coated in grease.

7.4.9 The fusible link, if fitted, is temporarily removed and shutter closes completely and latches closed (where latch provided).

7.4.10 The fusible link is not damaged or painted. If damaged, it is to be replaced.

7.4.11 Each remotely operated fire door or shutter indicator displays whether the door or shutter is OPEN or CLOSED.

7.5 Conduct maintenance of all fire doors, shutters and controls listed in section 3.0 using References 2.1 through 2.8 and manufacturers maintenance instructions and service bulletins as guidance. The maintenance is to include/verify:

7.5.1 All exposed moving parts are lubricated per manufacturer’s instructions.

7.5.2 Grease roller doors internal gearbox components for proper function of hand cranks using

Reference 2.3 for guidance.

7.5.3 Adjust spring tensioning of roller doors using Reference 2.3 for guidance.

7.5.4 Realign hard stops of roller doors for unobstructed operation using Reference 2.3 for guidance.

7.6 With assistance from ships force, conduct an operational test of all fire doors, shutters and controls listed in section 3.0 to verify the assemblies operate correctly and will close under fire conditions.

7.6.1 Check the clearance between the hydraulic sliding door and its frame with a .003” (0.08mm) feeler gage using Reference 2.2 for guidance. Degree of contact should be such to reject the gage at any sealing point around the perimeter.

7.6.2 Sliding or rolling fire doors and shutters are to have an average closing speed of not less than 6”/sec or not more than 24”/sec in accordance with NFPA 80, Standard for Fire Doors and Other Openings, 5.2.3.6.2 (15).

7.6.3 The approximate time of closure for hinged fire doors is to be no more than 40 seconds and no less than ten seconds from the beginning of their movement with the ship in upright position using Reference 2.8 as guidance.

7.6.4 Automatic and Centrally Controlled doors using Reference 2.1.6 as guidance are to:

a. Be able to be operated both automatically and manually from both sides of the division.

b. Demonstrate the failsafe mechanism closing the door or shutter in a fire upon loss of electrical power or hydraulic or pneumatic pressure loss.

c. Demonstrate remote-control arrangements for shutting of the fire doors and shutters from the central control station. Verify indication at the fire door indicator panel in the continuously manned central control station displays door status; open and closed.

d. Remote-released sliding or power-operated doors are to be equipped with an alarm that sounds at least five seconds but no more than ten seconds after the door being released from the central control station and before the door begins to move and continues sounding until the door is completely closed.

e. A door closed remotely from the central control station is to be capable of being re-opened from both sides of the door by local control. After such local opening, the door is to automatically close again.

7.6.5 After any repairs, the door and shutters are to be retested for proper operation.

7.7 Reset all automatic closing devices using the manufacturer’s instructions as guidance. Verify that the automatic closing device has been reset correctly.

7.8 Testing is to be coordinated with the OMT REP and ABS to allow for random observation if deemed necessary.

7.9 Upon completion of all inspections, tests and repairs leave the fire doors and shutters in a ready for service condition. Conduct a final inspection with the OMT REP.

7.10 Reports

7.10.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.10.2 Submit a report that includes "as released" condition of the fire doors and shutters. Submit one electronic copy in Portable Document Format (PDF).

7.10.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.11 Painting

7.11.1 Prepare, prime, and paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.12 Manufacturer's Representative

7.12.1 Possible Source

7.13 Preparation of Drawings

7.13.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to describe the annual requirements to inspect, service, and test ships critical fire doors and shutters.

2.0 REFERENCES

2.1 MSC Drawing No. 830-8496378, Fire Control Plan

2.2 NAVSEA Drawing No. 624-8468294, Joiner Door and Passing Window Schedule

2.3 NAVSEA Drawing No. 624-8452231, Joiner Door and Passing Window Details

2.4 NAVSEA Drawing No. 624-8468286, Joiner Door and Passing Window Schedule

2.5 Subpart 46 CFR 111.99, Fire Door Holding and Release Systems

2.6 SOLAS II-2, Construction, Fire Protection, Fire Detection and Fire Extinction, Part C Suppression of Fire, Regulation 9 Containment of Fire (Available commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location

3.1.1 Passing Windows

a. Aft Galley Passing Window with rolling shutter (A-47-0)

b. Fwd Galley Passing Window with rolling shutter (A4-92-1)

3.1.2 Joiner Doors

a. Quantity: 400, Various Locations, see References 2.2 through 2.4

3.2 Description/Quantity

3.2.1 Four Rolling Shutter/Passing Windows

Model: US Joiner Roller Shutter

3.2.2 400 Fire Rated Joiner Doors

Model: US Joiner Doors, see References 2.2 through 2.4

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Fire door; means a door that is in a fire boundary, such as a stairway enclosure or main vertical zone bulkhead, that is not usually kept closed.

5.3 Fire door holding magnet; means an electromagnet for holding a fire door open.

5.4 Local control panel; means a manually-operated device next to a fire door for releasing the door so that the fire door self-closing mechanism may close the door.

5.5 Central control panel; means a manually-operated device on the navigating bridge or in the fire control room for releasing one or more fire doors.

5.6 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Examine, test, and refurbish fire doors and shutters identified in section 3.0 in accordance with NFPA, SOLAS, and ABS and USCG requirements using References 2.1.1 through 2.1.5 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.2.1 Temporarily remove access doors/panels, overhead panels in passageways and compartments, paneling and any other interferences to access all fire doors and shutters and their operators listed in section 3.0.

7.3 Conduct inspections, maintenance, and testing of the fire doors, shutters and controls using References 2.1.1 through 2.1.5 as guidance.

7.4 Conduct an annual inspection of the fire doors, shutters and controls listed in section 3.0 using References 2.1.1 through 2.1.5 as guidance. The examination will include/verify:

7.4.1 A visual inspection of both sides of the door or shutter.

7.4.2 No field modifications have been made.

7.4.3 Full unobstructed access to doors and shutters is provided.

7.4.4 There is no rusted, bent, misaligned, defective or missing parts.

7.4.5 The door or shutter frame and guides are not warped, damaged or penetrated and the fire boundary integrity is maintained at openings and penetrations.

7.4.6 With assistance from ships force, cycle each door through its closure and opening. Check for proper operation and full closure as well as travel and proper operation of limit switches. Verify it is possible for each door to be opened and closed from each side of the bulkhead by one person only.

7.4.7 Doors in stairtowers, stairways, main vertical zone bulkheads and galley boundaries are self-closing and hold-back hooks are not fitted, unless subject to central control station release.

7.4.8 Chains and cables are in good operating condition. Inspect chains and cables for excessive wear, stretching and binding and verify they are not kinked, pinched, twisted, inflexible, painted or coated in grease.

7.4.9 The fusible link, if fitted, is temporarily removed and shutter closes completely and latches closed (where latch provided).

7.4.10 Each remotely operated fire door or shutter indicator displays whether the door or shutter is OPEN or CLOSED.

7.5 Conduct maintenance of all fire doors, shutters and controls listed in section 3.0 using References 2.1.1 through 2.1.5 and manufacturers maintenance instructions and service bulletins as guidance. The maintenance is to include/verify:

7.5.1 All exposed moving parts are lubricated per manufacturer’s instructions.

7.5.2 Grease roller doors internal gearbox components for proper function of hand cranks using Reference 2.1.3 for guidance.

7.5.3 Replace all fusible links.

7.5.4 Adjust spring tensioning of roller doors using Reference 2.1.3 for guidance.

7.5.5 Realign hard stops of roller doors for unobstructed operation using Reference 2.1.3 for guidance.

7.6 With assistance from ships force, conduct an operational test of all fire doors, shutters and controls listed in section 3.0 to verify the assemblies operate correctly and will close under fire conditions.

7.6.1 Check the clearance between the hydraulic sliding door and its frame with a .003” (0.08mm) feeler gage using Reference 2.1.2 for guidance. Degree of contact should be such to reject the gage at any sealing point around the perimeter.

7.6.2 Sliding or rolling fire doors and shutters are to have an average closing speed of not less than 6”/sec or not more than 24”/sec in accordance with NFPA 80, Standard for Fire Doors and Other Openings, 5.2.3.6.2 (15).

7.6.3 The approximate time of closure for hinged fire doors is to be no more than 40 seconds and no less than 10 seconds from the beginning of their movement with the ship in upright position using Reference 2.1.5 as guidance.

7.6.4 Automatic and Centrally Controlled doors using Reference 2.1.5 as guidance are to:

a. Be able to be operated both automatically and manually from both sides of the division.

b. Demonstrate the failsafe mechanism closing the door or shutter in a fire upon loss of electrical power or hydraulic or pneumatic pressure loss.

c. Demonstrate remote-control arrangements for shutting of the fire doors and shutters from the central control station. Verify indication at the fire door indicator panel in the continuously manned central control station displays door status; open and closed.

d. Remote-released sliding or power-operated doors are to be equipped with an alarm that sounds at least 5 seconds but no more than 10 seconds after the door being released from the central control station and before the door begins to move and continues sounding until the door is completely closed.

e. A door closed remotely from the central control station is to be capable of being re-opened from both sides of the door by local control. After such local opening, the door is to automatically close again.

7.6.5 After any repairs, the door and shutters are to be retested for proper operation.

7.7 Reset all automatic closing devices using the manufacturer’s instructions as guidance. Verify that the automatic closing device has been reset correctly.

7.8 Testing is to be coordinated with the OMT REP and ABS and USCG to allow for random observation if deemed necessary.

7.9 Upon completion of all inspections, tests and repairs leave the fire doors and shutters in a ready for service condition. Conduct a final inspection with the OMT REP.

7.10 Reports

7.10.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.10.2 Submit a report that includes “as released” condition of the fire doors and shutters. Submit one electronic copy in Portable Document Format (PDF).

7.10.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.11 Painting

7.11.1 Prepare, prime, and paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.12 Manufacturer’s Representative

7.12.1 Possible Source

7.13 Preparation of Drawings

7.13.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the

OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY	CONTRACT NO. N68171-26-RN-003	
ITEM NO. 0574C	CATEGORY "A"	2025-09-12
ESB-CCSI-FIRE DOORS AND SHUTTERS - ROLLER CURTAINS (1 YR)	Manovich, David	

1.0 ABSTRACT

The purpose of this item is to describe the annual requirements to inspect, service, and test ships critical fire doors and shutters.

2.0 REFERENCES

2.1 NAVSEA Drawing No. 830-8496378 Fire Control and Safety Plan

2.2 Tech Manual T9167-BY-MMC-010, Door, Rollup, Curtain Type HH8000, HD8000, HH9000

2.3 NAVSEA Drawing No. 624-8468294, Joiner Door and Passing Window Schedule

2.4 NAVSEA Drawing No. 624-8452231, Joiner Door and Passing Window Details

2.5 NAVSEA Drawing No. 624-8468286, Joiner Door and Passing Window Schedule

2.6 Subpart 46 CFR 111.99, Fire Door Holding and Release Systems

2.7 SOLAS II-2, Construction, Fire Protection, Fire Detection and Fire Extinction, Part C Suppression of Fire, Regulation 9 Containment of Fire (Available commercially)

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Door Locations

3.1.1 Roller Doors

a. Port Roller Door (A5-86-4) to MHE Stowage/Maintenance Repair Area (A5-89-4)

b. Starboard Roller Door (A5-86-3) serving Passage (A5-96-3)

c. Hangar Roller Door (A5-86-0) to (A5-96-0)

3.1.2 Passing Windows

a. Aft Galley Passing Window with rolling shutter (A-47-0)

b. Fwd Galley Passing Window with rolling shutter (A4-92-1)

3.2 Description/Quantity

a. Three Rolling Curtain Fire Doors

Manufacturer: Federal Equipment Company

Model: HH9000, Quantity: Two

Model: HH8000, Quantity: One

b. Four Rolling Shutter/Passing Windows

Model: US Joiner Roller Shutter

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Fire door; means a door that is in a fire boundary, such as a stairway enclosure or main vertical zone bulkhead, that is not usually kept closed.

5.3 Fire door holding magnet; means an electromagnet for holding a fire door open.

5.4 Local control panel; means a manually-operated device next to a fire door for releasing the door so that the fire door self-closing mechanism may close the door.

5.5 Central control panel; means a manually-operated device on the navigating bridge or in the fire control room for releasing one or more fire doors.

5.6 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

5.7 THE ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Examine, test, and repair fire doors and shutters identified in section 3.0 in accordance with NFPA, SOLAS, and ABS and USCG requirements using References 2.1 through 2.9 as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship's force to have associated equipment unlocked and/or tagged in per ship's lock out/tag out policy.

7.2.1 Temporarily remove access doors/panels, overhead panels in passageways and compartments, paneling and any other interferences to access all fire doors and shutters and their operators listed in section 3.0.

7.3 Conduct inspections, maintenance, and testing of the fire doors, shutters and controls using References 2.1 through 2.7 as guidance.

7.4 Conduct an annual inspection of the fire doors, shutters and controls listed in section 3.0 using References 2.1 through 2.9 as guidance. The examination will include/verify:

7.4.1 A visual inspection of both sides of the door or shutter.

7.4.2 No field modifications have been made.

7.4.3 Full unobstructed access to doors and shutters is provided.

7.4.4 There is no rusted, bent, misaligned, defective or missing parts.

7.4.5 The door or shutter frame and guides are not warped, damaged or penetrated and the fire boundary integrity is maintained at openings and penetrations.

7.4.6 With assistance from ships force, cycle each door through its closure and opening. Check for proper operation and full closure as well as travel and proper operation of limit switches. Verify it is possible for each door to be opened and closed from each side of the bulkhead by one person only.

7.4.7 Doors in stairtowers, stairways, main vertical zone bulkheads and galley boundaries are self-closing and hold-back hooks are not fitted, unless subject to central control station release.

7.4.8 Chains and cables are in good operating condition. Inspect chains and cables for excessive wear, stretching and binding and verify they are not kinked, pinched, twisted, inflexible, painted or coated in grease.

7.4.9 The fusible link, if fitted, is temporarily removed and shutter closes completely and latches closed (where latch provided).

7.4.10 The fusible link is not damaged or painted. If damaged, it is to be replaced.

7.4.11 Each remotely operated fire door or shutter indicator displays whether the door or shutter is OPEN or CLOSED.

7.5 Conduct maintenance of all fire doors, shutters and controls listed in section 3.0 using References 2.1 through 2.9 and manufacturers maintenance instructions and service bulletins as guidance. The maintenance is to include/verify:

7.5.1 All exposed moving parts are lubricated per manufacturer’s instructions.

7.5.2 Grease roller doors internal gearbox components for proper function of hand cranks using Reference 2.3 for guidance.

7.5.3 Adjust spring tensioning of roller doors using Reference 2.3 for guidance.

7.5.4 Realign hard stops of roller doors for unobstructed operation using Reference 2.3 for guidance.

7.6 With assistance from ships force, conduct an operational test of all fire doors, shutters and controls listed in section 3.0 to verify the assemblies operate correctly and will close under fire conditions.

7.6.1 Check the clearance between the hydraulic sliding door and its frame with a .003” (0.08mm) feeler gage using Reference 2.2 for guidance. Degree of contact should be such to reject the gage at any sealing point around the perimeter.

7.6.2 Sliding or rolling fire doors and shutters are to have an average closing speed of not less than 6”/sec or not more than 24”/sec in accordance with NFPA 80, Standard for Fire Doors and Other Openings, 5.2.3.6.2 (15).

7.6.3 The approximate time of closure for hinged fire doors is to be no more than 40 seconds and no less than ten seconds from the beginning of their movement with the ship in upright position using Reference 2.8 as guidance.

7.6.4 Automatic and Centrally Controlled doors using Reference 2.1.6 as guidance are to:

a. Be able to be operated both automatically and manually from both sides of the division.

b. Demonstrate the failsafe mechanism closing the door or shutter in a fire upon loss of electrical power or hydraulic or pneumatic pressure loss.

c. Demonstrate remote-control arrangements for shutting of the fire doors and shutters from the central control station. Verify indication at the fire door indicator panel in the continuously manned central control station displays door status; open and closed.

d. Remote-released sliding or power-operated doors are to be equipped with an alarm that sounds at least five seconds but no more than ten seconds after the door being released from the central control station and before the door begins to move and continues sounding until the door is completely closed.

e. A door closed remotely from the central control station is to be capable of being re-opened from both sides of the door by local control. After such local opening, the door is to automatically close again.

7.6.5 After any repairs, the door and shutters are to be retested for proper operation.

7.7 Reset all automatic closing devices using the manufacturer's instructions as guidance. Verify that the automatic closing device has been reset correctly.

7.8 Testing is to be coordinated with the OMT REP and ABS to allow for random observation if deemed necessary.

7.9 Upon completion of all inspections, tests and repairs leave the fire doors and shutters in a ready for service condition. Conduct a final inspection with the OMT REP.

7.10 Reports

7.10.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.10.2 Submit a report that includes "as released" condition of the fire doors and shutters. Submit one electronic copy in Portable Document Format (PDF).

7.10.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.11 Painting

7.11.1 Prepare, prime, and paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.12 Manufacturer's Representative

7.12.1 Possible Source

E.STech Marine Co. OEM (Original Equipment Manufacturer) Rep or an OEM Authorized Technical Representative

E.STech Marine Co.,LTD
287-19, Noksansaneopjung-ro,
Gangseo-gu, Busan, Korea, Republic Of

E.STech Marine Co., LTD
1761-12, Songjeong-Dong,
Gangseo-Gu, Busan, Korea, Republic Of

Tel: +82-51-831-6644
Fax: +82-51-831-6642
Email: est@estmarine.co.kr
Website: www.ESTMARINE.CO.KR

7.13 Preparation of Drawings

7.13.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

This work item describes the requirements to clean, inspect and service the Boll and Kirch Auto flush lube oil filters.

2.0 REFERENCES

2.1 T6437-A3-MMC-010, Boll and Kirch Automatic Filter

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location:

3.1.1 Main Engine Room No 1, 5-56-1

3.1.2 Main Engine Room No 2, 5-56-2

3.2 Description/Quantity: Main Engine LO Filter

3.2.1 Mfg: Boll & Kirch Filter

3.2.2 Model/Type: 6.61 DN 200

3.2.3 Serial/Fabr No:

a. LO Filter No.1 - 3872198/2

b. LO Filter No.2 - 3872199/2

c. LO Filter No.3 - 3872200/2

d. LO Filter No.4 - 3872201/2

3.2.4 Filtration (micron): 34 µm

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTR’s.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Remove and replace all interferences, rig and unrig, stage and unstage, make all disassemblies and subsequent reassemblies. Provide and operate equipment and supply all services and assistance to accomplish the thorough examination and testing of the ships Boll and Kirch auto backflushing filters using Reference 2.1 as guidance.

7.2 With assistance from the Chief Engineer coordinate the inspection and testing of the Auto Filter to avoid damage or injury during the course of this work item. Use the Boll Special Tools in the performance of this work where required using Reference 2.1 as guidance. The Manufacturer's Authorized Service Provider shall provide Boll Special Tools to accomplish work required. The ships crew will operate all onboard equipment.

7.2.1 Secure the filter heating and allow the filter to cool down.

7.2.2 Secure the compressed air supply to prevent it from being switched on unintentionally.

7.2.3 Depressurize and drain the filter chambers.

7.2.4 Tagout the Auto Filter system.

7.3 Coordinate the inspection and testing of equipment with the OMT Rep and Chief Engineer to ensure the vessel is aware of equipment status at all times until the work is completed.

7.4 Identify, and record the following data for each Auto-Filter assembly:

7.4.1 Filter No:

7.4.2 Mfg Name:

7.4.3 Model/Type:

7.4.4 Serial/Fabr No:

7.4.5 Filtration (micron):

7.4.6 Year of mfg:

7.5 Inspect the auto-filter system. Report any leaks, wear, corrosion, damage or adverse conditions noted to the OMT REP.

7.6 Remove and clean the bypass filter elements.

7.7 Inspect the differential pressure indicator.

7.8 Screw the guide pins into the change-over housing to facilitate the removal of the filter housing. Lift the filter housings off the upper section of housing using care. Remove the candle filter elements and inspect for damage. Record the candle part number and mesh size found. The filter elements should be fine mesh 34 micron (µm) using Reference 2.1 as guidance.

NOTE: Filters must be Boll OEM only. After market filters do not fit properly.

7.9 Remove and record any particles collected on the magnetic candle (if fitted).

7.10 Clean all sealing surfaces.

7.11 Clean the candle filter elements using BOLL-CLEAN 2000 and the BOLL High-Pressure Cleaning Unit, TYPE 5.04 using Reference 2.1 as guidance. Proceed with extreme caution when performing all cleaning work to prevent damage to the candle filter elements.

7.12 Replace all O-rings, seals and gaskets.

7.13 Reinstall the candle filter elements and housing. Adhere to the OEM recommended tightening torque values for filter candles and components. Make sure that the seals are properly in place.

7.14 Check the condition of the vent (automatic) and check that it is functioning properly. Inspect the condition of the float ball and make sure it moves freely and easily.

7.15 Unscrew and remove the guide pins from the change-over housing.

7.16 Reassemble the auto-filter leaving it in a ready for service condition.

7.17 With assistance of ships force and the OEM rep, follow the Commissioning procedure for the Auto-Filter. Demonstrate proper operation of the unit and all its alarms, features and modes to include backflushing to the OMT Rep.

7.18 Conduct an Operational Test, during Dock and Sea Trials, evaluating the performance of each unit. Testing is to be coordinated with the OMT Rep to allow for observation if deemed necessary.

7.19 With assistance of the ships crew inventory and examine the vessels onboard spares for the Auto-Filter. Ensure the candle filter elements are OEM products, undamaged and the correct filtration (micron). Each vessel should have a complete set of candle filter elements (114 pieces), gaskets and sealing rings onboard.

7.20 Reports

7.20.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.20.2 Submit a report that includes “as released” condition of the Auto-Flush Lube Oil Filters. Submit one electronic copy in Portable Document Format (PDF).

7.20.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.21 Painting/Lagging

7.21.1 Clean, prime and paint all new and disturbed surfaces to match the surrounding areas.

7.22 Manufacturer's Representative

7.22.1 Provide the services of a Manufacturer's Authorized Service Provider to inspect, service and test the Auto-Filter listed in paragraph 3.2.

7.22.2 Possible Boll and Kirch sources:

Boll Filter Corporation
22635 Venture Drive
Novi, MI 48375 - USA
Tel.: +1/(0)248/773-8200
www.bollfilterusa.com

Motor-Services Hugo Stamp, Inc.
3005 SW 3rd Ave.
Ft. Lauderdale, Florida 33315 - USA
Tel.: +1/(0)954/763 3660
https://www.mshs.com/

Boll & Kirch, Filterbau GmbH
Siemensstr. 10-14,
D-50170 Kerpen
Germany
Tel.: +49/(0)2273/562-0
https://www.bollfilter.com/en.html

FHG Marine Engineering
2004 NW 25th Avenue
Pompano Beach, FL 33069 – USA
Tel.: +1 (954) 504-2853
https://www.fhgmarineengineering.com

MAN Energy Solutions

Stadtbachstrae 1
86153 Augsburg, Germany
Tel.: +49 821 322-0
https://www.man-es.com/

7.23 Preparation of Drawings

7.23.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0581	CATEGORY "A"	2026-01-30
ESB-CCSI-OXYGEN BOTTLE INSPECTION AND HYDRO (5 YR)		Manovich, David

1.0 ABSTRACT

This item describes the requirements to inspect, hydrostatically test and refill the ship’s medical grade oxygen portable bottles.

2.0 REFERENCES/ENCLOSURES

2.1 49 CFR 180.209(b)

2.2 MIL-STD-1411C Department of Defense Inspection and Maintenance of Compressed Gas Cylinders

2.3 S9086-SX-STM-010 Naval Ships' Technical Manual for Industrial Gases: Generating, Handling, and Storage

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Sick Bay (A3-93-6)

3.1.2 Quantity

a. 7 Size H medical grade oxygen cylinders

b. 12 Pony bottle sized medical grade oxygen cylinders

3.2 Item Description/Manufacturer’s Data: None

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to

determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide the services of an authorized technical representative and DOT and ABS authorized service facility to service the shipboard oxygen cylinders. Provide all materials and special equipment and supply all services and assistance to accomplish inspection and hydrostatic testing of all shipboard oxygen cylinders identified in section 3.0.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 With assistance from ship’s force, take custody of the oxygen cylinders and temporarily remove from the ship to an authorized service facility (DOT and ABS certified facility).

7.5 Perform inspection of each oxygen cylinder for damage.

7.6 Hydrostatically test each oxygen cylinder in accordance with DOT and ABS requirements.

7.6.1 Provide verification that all oxygen cylinders meet 49CFR180.205 and MIL-STD-1411A.

7.7 Refill each oxygen cylinder with medical grade oxygen.

7.7.1 Provide certification to verify the gas is medical grade oxygen of 99.8% purity.

7.8 Upon completion of all inspection and hydrostatic testing, return each cylinder to the ship and re-stow each bottle in their original location.

7.9 Reports

7.9.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF). Report shall include at minimum:

a. Condition found report documenting “as found” condition of each cylinder.

b. Hydrostatic test reports for each cylinder.

c. Material certification for the medical grade oxygen used to refill the cylinders.

7.9.2 Submit a report that includes “as released” condition of the oxygen bottle inspection and hydro. Submit one electronic copy in Portable Document Format (PDF).

7.9.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.10 Painting/Lagging

7.10.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.10.2 Renew lagging where damaged or disturbed.

7.11 Manufacturer’s Representative

7.11.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the oxygen bottles identified in section 3.0 ensuring compliance with the manufacturer’s standards and performance specifications. In addition, the service facility is to be an ABS recognized External Specialist for the system.

https://ww2.eagle.org/en/rules-and-resources/recognized-external-specialists.html

https://www.eagle.org/ABSEaglePrograms/es/es-search.jsp

7.11.2 Provide the OMT REP with a signed letter of qualification, ABS Certificate # and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.12 Preparation of Drawings

7.12.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
AUXILIARY MACHINERY		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0585A	CATEGORY "A"	2026-01-30
ESB-CSI-REVERSE OSMOSIS UNIT SERVICE (1 YR)		Manovich, David

1.0 ABSTRACT

The purpose of this item is to perform annual reverse osmosis unit maintenance.

2.0 REFERENCES

2.1 Tech Manual T9531-AX-MMC-010, Seawater Reverse Osmosis Unit, 25,000 USGPD

2.2 Tech Manual T6225-QT-MMC-010, Mission RO Seawater Feed Pump Assembly

2.3 Tech Manual T6226-AP-MMC-010, Mission RO Recirculating Brominator Pump Motor Assembly

2.4 Tech Manual T9531-BK-MMC-010, Mission RO High Pressure Pump and High Pressure Pump Motor

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Location: Auxiliary Machinery Room (3-97-0)

3.1.2 Quantity: **one**

3.2 Item Description/Manufacturer's Data

3.2.1 Description: Model MLP 25,000 GPD

3.2.2 Manufacturer: Aqua-Chem, Inc.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to

determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 The contractor and all subcontractors regardless of tier are advised to review other work items under this contract. Many of the definitions relating to performance of this work item are found in Work Item 001.

5.3 THIS SHIP CONTAINS HIGH VOLTAGE (HV) ELECTRICAL SYSTEMS. OBEY ALL POSTED AND VERBAL INSTRUCTIONS REGARDING SAFETY AND EXCLUSION FROM HIGH VOLTAGE AREAS. AT NO TIME APPROACH, WORK ON, OR ENTER A HIGH VOLTAGE AREA WITHOUT PROPER AUTHORIZATION.

5.4 THE ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.2 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.3 Provide the services of an Aqua Chem field service representative to complete the following listed maintenance items using References 2.1 through 2.4 as guidance:

7.3.1 Replace mechanical seal kit components for the reverse osmosis unit centrifugal seawater feed pump using Reference 2.2, Section 1 as guidance.

7.3.2 Replace mechanical seal kit components for the reverse osmosis unit bromine feeder recirculating pump using Reference 2.3, Section 1 as guidance.

7.3.3 Lubricate the reverse osmosis HP pump motor, using Reference 2.4, Part 2 as guidance.

7.3.4 Replenish calcium carbonate in pH adjuster tank using Reference 2.1, Part 1, and Section 3.11 as guidance.

NOTE: AC-4710 Calcium Carbonate pH adjuster shown in Reference 2.1, Appendix B is to be used for pH adjuster tank replenishment.

7.3.5 Provide an “as found” condition report to the OMT REP detailing any required repairs.

7.4 Inspection/Test

7.4.1 Perform operational testing upon reinstallation of all equipment in the presence of the OMT REP.

7.5 Reports

7.5.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.5.2 Submit a report that includes “as released” condition of the reverse osmosis unit. Submit one electronic copy in Portable Document Format (PDF).

7.5.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.6 Painting/Lagging

7.6.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.6.2 Renew lagging where damaged or disturbed.

7.7 Manufacturer's Representative

7.7.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the reverse osmosis unit identified in section 3.0 ensuring compliance with the manufacturer's standards and performance specifications.

7.7.2 Possible Source

Aqua-Chem, Inc.
3001 East Gov. John Sevier Highway
Knoxville, Tennessee 37914
Phone:(865)544-2065
Fax:(865)546-4330

7.7.3 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.8 Preparation of Drawings: None additional

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
HABITABILITY OUTFITTING AND FURNISHINGS		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0604	**CATEGORY "A"**	**2026-01-30**
ESB-CSI-ANNUAL LIFERAFT DAVIT CERTIFICATION		**Manovich, David**

1.0 ABSTRACT

The purpose of this work item is to inspect, service and test the ships liferaft davit.

2.0 REFERENCES

2.1 Tech Manual T9583-BF-MMC-010, DAVIT, CRANE TYPE, LIFE RAFT TYPE SCM L AND TYPE SCM 21-4.0L (LH) AUTOMATIC RELEASE HOOK TYPE CAR 35

3.0 ITEM LOCATION/DESCRIPTION

3.1 Liferaft Davit Location/Quantity/Description

3.1.1 One Liferaft Davit, Mission Deck, Port, Frame 85

3.1.2 Manufacturer: Palfinger NED-DECK Launch and Recovery Systems

3.1.3 Model: SCM L

3.2 Contractor Furnished Material

3.2.1 The following items will be procured by the contractor. 90001 Service set A(page 1-41 or 2-46 of Reference 2.1) applies:

a. 10268 R Lip Seal, Quantity: One

b. 10262 R Lip Seal, Quantity: Two

c. 3-63-21-00-22 Gasket (brake), Quantity: Three

d. 3-63-21-00-55 Gasket (cover), Quantity: One

e. 00471.000.090 Circlip, Quantity: Two

f. 3-63-21-00-30 Brake Shoe, Quantity: One Set

g. Brake shoe springs, set of two, Quantity: One Set

1. 3-63-21-00-36TA Wire diameter Ø1,5mm, Type A
2. 3-63-21-00-36TB Wire diameter Ø2,0mm, Type B
3. 3-63-21-00-36TC Wire diameter Ø2,5mm, Type C

h. 00094.400.40.40 Split Pin Ø4x40, Quantity: Two

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

Or

4.1 Government Furnished Equipment: None

4.2 Government Furnished Material

Item	Part Number	NSN/TNICN	Description	Qty[1]	UOM[2]
1.	AS Reqd	AS Reqd	Replacement Winch Gear Oil(provided by ships force)	AS Reqd	AS Reqd

1 Qty = Quantity
2 UOM = Unit of measure

4.3 Government Furnished Service: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 SOLAS regulation III/20 – Operational readiness, maintenance and inspections

SOLAS regulation III/36 – Instructions for onboard maintenance.

5.3 THIS SHIP CONTAINS HIGH VOLTAGE (HV) ELECTRICAL SYSTEMS. OBEY ALL POSTED AND VERBAL INSTRUCTIONS REGARDING SAFETY AND EXCLUSION FROM HIGH VOLTAGE AREAS. AT NO TIME APPROACH, WORK ON, OR ENTER A HIGH VOLTAGE AREA WITHOUT PROPER AUTHORIZATION.

5.4 THE ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the thorough examination, testing and repair of the Liferaft Davit using SOLAS and USCG requirements by authorized service providers. Use Reference 2.1 as guidance.

7.2 Liferaft Davit Inspection/Maintenance/Testing

7.2.1 All shipboard lock-out tag out procedures shall be followed and coordinated with the Chief Engineer.

7.2.2 Provide an authorized Manufacturer's Technical

Representative to perform all inspections and testing.

7.2.3 Perform a thorough inspection of the entire davit structure for corrosion, misalignments, deformations, excessive free play and loose or missing parts, such as pins, screws, and nuts.

7.2.4 Ensure all components are adequately lubricated. Tighten or replace fasteners, as required. Apply a manufacturer recommended lubricant to wires, sheaves and moving parts as necessary. Wire rope shall be slushed from bitter end to bitter end utilizing a mechanical wire rope lubricator to ensure adequate grease is applied.

7.2.5 Purchase and install new CFM identified in paragraph 3.2.1. Turn all unused material over to Chief Engineer.

7.2.6 Inspect the gripes for damage to the vinyl covering, loose seizing at the end of the fittings, cracks or distortions in the fittings.

7.2.7 Annually inspect the floating blocks, for proper engagement, cracks or distortions that would preclude further use.

7.2.8 Annually inspect the trackways to ensure they are free from debris and damage.

7.2.9 Annually inspect the falls for any signs of damage, kinks or deterioration with special regard for areas passing through sheaves.

7.2.10 Inspect the releasing hooks for proper operation and any visual signs of wear.

7.2.11 Inspect stressed structural parts including foundations and load bearing members for cracks and visible damage.

7.2.12 Protective covers for the raft and other davit components must be replaced and properly secured upon completion of testing.

7.2.13 Accomplish the following tests in the presence of the OMT REP, ABS and USCG:

a. Perform annual operational testing of the raft davit by lowering the empty craft or boat or equivalent load. When the craft has reached its maximum lowering speed and before the craft enters the water, the brake shall be abruptly applied. Following these tests, the stressed structural parts shall be reinspected where the structure permits the reinspection.

b. Perform operational test of the davit launched automatic release hook function as follows:

1. Manually release the hook with a load of 150 kg on the hook.

2. Automatically release the hook with a dummy weight of 200 kg on the hook when it is lowered to the ground. If a raft is used for the test instead of a dummy weight, the automatic release function shall release the raft when waterborne.

3. Examine the release hook and hook fastening to ensure that the hook is completely reset and no damage has occurred.

c. Test limit switches (ESB 3-5 only)

1. Test the winch brake limit switch, energize the winch circuit. With the manual hand brake set, attempt to payout wire rope by the power. The brake limit switch should prevent power operation in the power payout mode.

2. Test the hand crank limit switch, rotate the hand crank shaft shift lever to the engaged position. Attempt to operate the winch in both the power payout and hoist directions. The limit switch should prevent operation. Return the hand crank shaft shift lever to the disengaged position.

3. Test the davit arm latch limit switches, gravity lower davit arm about four feet from the stowed position. Turn the master switch to hoist and hoist the davit arms and manually throw the limit switch on one side of the davit. This should secure the davit's power. Repeat this action for the other davit arm.

7.2.14 Upon completion of davit testing, provide a condition found report, citing any noted deficiencies resulting from testing evolutions and provide corrective action recommendations to the OMT REP. Any repair actions required will be the subject of contract modification.

7.2.15 Return Davit and winch to readiness condition and record the results of the inspection in the ship's log.

7.3 All work shall only be accomplished by trained, experienced and authorized service personnel for the specific liferafts.

7.4 Reports

7.4.1 All reports and checklists shall be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.4.2 Provide all documentation required by ABS and/or the USCG Representative for certification of Life Raft Davit Annual Inspection. Provide a Service Report documenting all maintenance, repairs, testing, and inspections performed. Provide Service Report Portable Document Files (pdfs) to the ABS

Surveyor, Chief Engineer, and Port Engineer. Ensure any/all ABS Surveyor reports and final certificates are received and turned over to the OMT REP and Ship’s force.

7.5 Painting: None additional

7.6 Manufacturer’s Representative

7.6.1 Provide the services of a Manufacturer’s Authorized Service Facility and personnel to perform any and all inspections, maintenance and tests to ensure proper operation and repair of the Liferaft Davit using IMO and USCG requirements and manufacturer’s performance specifications as guidance.

7.6.2 Provide the OMT REP with a signed letter of qualification from the original liferaft davit manufacturer, certificate of training for the equipment identified in section 3.0 and ABS Certificate # prior to the start of any work.

7.6.3 Possible Source for Liferaft Davit Support

Palfinger Ned-Deck
Launch & Recovery Systems
Ambachtsweg 10
3771 MG Barneveld
The Netherlands
Phone: +31 (0) 342 - 42 21 05
www.palfingerneddeck.com/service
service-neddeck@palfinger.com
spareparts-neddeck@palfinger.com

7.7 Preparation of Drawings

7.7.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to accomplish the annual requirements to inspect, service and test the ships Lifeboats.

2.0 REFERENCE/ENCLOSURE

2.1 Reference

2.1.1 IMO Resolution MSC.402(96) + Corr.1 (adopted on 19 May 2016) Requirements for maintenance, thorough examination, operational testing, overhaul and repair of Lifeboats, Rescue Boats, Launching Appliances and Release Gear

2.1.2 USCG NVIC 03-19, Maintenance, thorough examination, operational testing, overhaul and repair of Lifeboats and Rescue Boats, Launching Appliances and Release Gear

2.1.3 Tech Manual S9583-C5-MMC-010, Lifeboat, Fassmer (Model: CLR-T 5.9)

2.2 Enclosure

2.2.1 Lifeboat Required Equipment (LSA Code, Chapter IV Survival Craft)

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location: Upper deck, Frame 55, Port and Starboard

3.2 Description

3.2.1 Lifeboat

a. Mfg: Fassmer Services GmbH & Co. KG

b. Model: CLR-T Series 5.9

c. Capacity: 49 men

3.3 Quantity: Two systems

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Off-load release mechanism means a release mechanism which releases the survival craft/rescue boat/fast rescue boat when it is waterborne or when there is no load on the hooks. On-load release mechanism means a release mechanism which releases the survival craft/rescue boat/fast rescue boat with load on the hooks. This release mechanism will be provided with a hydrostatic interlock unless other means are provided to ensure that the boat is waterborne before the release mechanism can be activated. In case of failure or when the boat is not waterborne, there will be a means to override the hydrostatic interlock or similar device to allow emergency release. This interlock override capability will be adequately protected against accidental or premature use. Adequate protection will include special mechanical protection not normally required for off-load release, in addition to a danger sign.

5.3 The maintenance and adjustment of release gear are critical operations with regard to the safe operation of lifeboats, rescue boats, fast rescue boats and davit launched liferafts. Utmost care will be taken when carrying out all inspection and maintenance operations on the equipment.

5.4 SOLAS regulation III/20 – Operational readiness, maintenance and inspections

5.5 SOLAS regulation III/36 – Instructions for onboard maintenance

5.6 A “Fall Preventer Device” (FPD) can be used to minimize the risk of injury or death by providing a secondary alternate load path in the event of failure of the on-load hook or its release mechanism or of accidental release of the on-load hook.

5.7 The Coast Guard considers the “make” to refer to the equipment manufacturer.

5.8 The Coast Guard considers the “type” to refer to the Coast Guard approval series for the equipment.

5.9 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None Additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service and test the ships Lifeboats using References 2.1.1 through 2.1.3 as guidance.

7.2 Adhere to the requirements of 29 CFR §1915.86 for occupational safety and health standards for shipyard personnel working on Lifeboats.

7.2.1 Before any employee works in or on a stowed or suspended lifeboat, the employer will secure the lifeboat independently from the releasing gear to prevent it from falling or capsizing.

7.2.2 The employer will not permit any employee to be in a lifeboat while it is being hoisted or lowered, except when the employer demonstrates that it is necessary to conduct operational tests or drills over water, or in the event of an emergency.

7.2.3 The employer will not permit any employee to work on the outboard side of a lifeboat that is stowed on chocks unless the lifeboat is secured by gripes or another device that prevents it from swinging.

7.3 Conduct the weekly and monthly inspections one time within two weeks of vessel’s arrival, and routine maintenance as specified in the equipment maintenance manual(s), by an authorized service provider. Perform all items listed in checklists for the weekly/monthly inspections required by SOLAS regulations III/20.6 and III/20.7.

7.3.1 Weekly inspection (III/20.6)

a. All survival craft, rescue boats and launching appliances will be visually inspected to ensure that they are ready for use. The inspection will include, but is not limited to, the condition of hooks, their attachment to the lifeboat and the on-load release gear being properly and completely reset.

b. With the assistance of ship’s force, all engines in lifeboats and rescue boats will be run for a total period of not less than three minutes, provided the ambient temperature is above the minimum temperature required for starting and running the engine. During this period of time, it should be demonstrated that the gear box and gear box train are engaging satisfactorily.

c. Lifeboats, except free-fall lifeboats, on cargo ships will be moved from their stowed position, without any persons on board, to the extent necessary to demonstrate satisfactory operation of launching appliances.

d. The general emergency alarm will be tested.

7.3.2 Monthly inspections (III/20.7)

a. All lifeboats, except free-fall lifeboats, will be turned out from their stowed position, without any persons on board.

b. Inspection of the life-saving appliances, including lifeboat equipment, will be carried out monthly using the SOLAS required checklist (SOLAS III, Regulation 36.1) to ensure that they are complete and in good order. A report of the inspection will be submitted.

7.4 With assistance of the ship’s Captain the OEM Authorized Rep is to review the vessels records of inspections and routine on-board maintenance carried out by the ship's crew and the applicable certificates for the equipment.

7.5 Lifeboats

7.5.1 Record the following information on each Lifeboat and submit a report with the data to the OMT REP:

a. Manufacturer' s name and address

b. Lifeboat model and serial number

c. Month and year of manufacture

d. Make and model of the on-load release gear

e. Number of persons the lifeboat is approved to carry

f. The approval information

g. Breathing air bottle serial #s and hydrostatic dates, if fitted

7.5.2 Thoroughly examine and test each Lifeboat for satisfactory condition and operation, to include:

a. Condition of the boat structure including fixed and loose equipment (including a visual examination of the external boundaries of the void spaces, as far as practicable)

b. Condition of the gel coat

c. Hatches and doors for watertightness

d. Engine and propulsion system, one hour in-water operational test with Mate and OEM Authorized Rep onboard. Demo to include full speed, forward and reverse operations.

1. Lifeboats will be capable of maneuvering at a speed of at least Six knots in calm water.

e. Sprinkler system, where fitted

f. Breathing air supply system, where fitted

g. Maneuvering system

h. Power supply system

i. Recharging of all engine starting, radio and searchlight batteries

j. Fixed two-way VHF radiotelephone with an antenna, where fitted

k. Fixed compass, where fitted

l. Bailing system

m. Markings and reflective tape

n. Fender/skate arrangements

o. Inventory all required equipment, see Enclosure 2.2.1.

7.5.3 Thoroughly check and service each Lifeboat, to include:

a. Replace bottom valve floater.

b. Replace window wiper blade.

c. Replace stuffing box packing.

d. Drain and replace hydraulic steering system oil. (If applicable)

7.5.4 Temporarily remove the boats from the ship to a protected location. Provide protection for lifeboats to include covering and protection from the elements at all times. Block the boat in the upright position, with clearance under the hull to permit inspection and access. In all cases, spreader bars shall be used in lieu of straps for lifting the boat using reference 2.1.3 as guidance.

7.5.5 All removed equipment, items, and rations will be temporarily stowed in a secure area, out of the weather, for inventory and reinstallation. The inventory report will be submitted within seven days of vessel arrival and include conditions of removed gear, expiration date of provisions and any shortages.

7.5.6 Temporarily open access covers for any installed tanks, empty contents and thoroughly clean of all dirt debris and residues. Clean, free up, lubricate and test all moving parts including interlocking devices, hinges, hand propelling gear, releasing gear and bilge pumps. Each fuel tank must be emptied, cleaned, and refilled with fresh fuel. Replace all fuel filters.

7.5.7 Replace the engine oil and filters.

7.5.8 Fresh water wash the boats with a mild detergent removing all salt, oils, and contaminants, cleaning the entire exterior of the boat. Clean and wipe down the interior of the boats. Apply a coat of wax to the exterior of the entire boat, hull and canopy.

7.5.9 Remove nozzles and clean out any debris. Prove unobstructed flow at each nozzle by connecting a fresh water source to the lifeboat sprinkler system to show free flow of water to the nozzles. After testing is completed and witnessed, reinstall nozzles. OMT REP to witness unobstructed flow.

7.6 Submit a condition report to the OMT REP summarizing all as found conditions, testing and inspections accomplished, and any discrepancies noted on each lifeboat and any recommended repairs and parts that are outside the original scope of work. Include a copy of the manufacturer’s authorized servicing company’s service report.

7.7 All work will only be accomplished by trained, experienced and authorized service personnel for the specific system.

7.8 Reports

7.8.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.8.2 Submit a report that includes “as released” condition of the lifeboat. Submit one electronic copy in Portable Document Format (PDF).

7.8.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.9 Painting: None additional

7.10 Manufacturer's Representative

7.10.1 Provide the services of certified personnel of either the Manufacturer or an Authorized Service Provider in accordance with References 2.1.1 and 2.1.2 to perform any and all system inspections, maintenance and tests to ensure proper operation of the Lifeboats using IMO and manufacturer's performance specifications as guidance.

a. *Manufacturer* means the original equipment manufacturer.

b. *Authorized Service Provider* (ASP) means an entity authorized by the Administration (USCG). The Coast Guard considers a service provider to be authorized if an Authorized Classification Society (ie; ABS/DNV/LLOYDS.) has verified the service provider is capable of servicing life-saving equipment in accordance with 46 CFR §199 and Reference 2.1.1.

7.10.2 Requirements for authorization as an ASP are defined in Reference 2.1.1. As a minimum:

a. Personnel will be certified by the manufacturer or authorized service provider for each make and type of the equipment to be worked on

b. Have sufficient tools, and in particular any specialized tools specified in the manufacturer's instructions, including portable tools as needed for the work

c. Have access to appropriate parts and accessories as specified for maintenance and repair

d. A documented and certified quality system

7.10.3 Provide the OMT REP with a copy of the ASP authorization document issued by an Authorized Classification Society defining the scope of services provided for the makes and types of equipment identified in section 3.0. The expiry date will be clearly written on the document. The authorization documentation will be submitted prior to the start of any work.

7.11 Preparation of Drawings

7.11.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - LIFEBOAT REQUIRED EQUIPMENT

	Every Lifeboat will carry:
1	Except for free-fall lifeboats, sufficient buoyant oars to make headway in calm seas. Thole pins, crutches or equivalent arrangements will be provided for each oar provided. Thole pins or crutches will be attached to the boat by lanyards or chains;
2	Two boat-hooks;
3	One buoyant bailer and two buckets;
4	One survival manual
5	One operational compass which is luminous or provided with suitable means of illumination. In a totally enclosed lifeboat, the compass will be permanently fitted at the steering position; in any other lifeboat, it will be provided with a binnacle if necessary to protect it from the weather, and suitable mounting arrangements;
6	One sea-anchor of adequate size fitted with a shock-resistant hawser which provides a firm hand grip when wet. The strength of the sea-anchor, hawser and tripping line if fitted will be adequate for all sea conditions;
7	Two efficient painters of a length equal to not less than twice the distance from the stowage position of the lifeboat to the waterline in the lightest seagoing condition or 15 m, whichever is the greater. On lifeboats to be launched by free-fall launching, both painters will be stowed near the bow ready for use;
8	Two hatchets, one at each end of the lifeboat;
9	Watertight receptacles containing a total of 3 liters of fresh water for each person the lifeboat is permitted to accommodate, of which either 1 liter per person may be replaced by a desalting apparatus capable of producing an equal amount of fresh water in 2 days, or 2 liters per

ENCLOSURE 2.2.1 - LIFEBOAT REQUIRED EQUIPMENT

	Every Lifeboat will carry:
	person may be replaced by a manually powered reverse osmosis desalinator capable of producing an equal amount of fresh water in 2 days;
10	One rustproof dipper with lanyard;
11	One rustproof graduated drinking vessel;
12	Food rations totaling not less than 10,000 kJ for each person the lifeboat is permitted to accommodate; these rations will be kept in airtight packaging and be stowed in a watertight container;
13	Four rocket parachute flares;
14	Six hand flares;
15	Two buoyant smoke signals;
16	One waterproof electric torch suitable for Morse signaling together with one spare set of batteries and one spare bulb in a waterproof container;
17	One daylight signaling mirror with instructions for its use for signaling to ships and aircraft;
18	One copy of the life-saving signals prescribed by regulation V/16 on a waterproof card or in a waterproof container;
19	One whistle or equivalent sound signal;
20	First-aid outfit in a waterproof case capable of being closed tightly after use;
21	Anti-seasickness medicine sufficient for at least 48 hrs and one seasickness bag for each person;

ENCLOSURE 2.2.1 - LIFEBOAT REQUIRED EQUIPMENT

	Every Lifeboat will carry:
22	One jack-knife to be kept attached to the boat by a lanyard;
23	Three tin openers;
24	Two buoyant rescue floats, attached to not less than 30 m of buoyant line;
25	If the lifeboat is not automatically self-bailing, a manual pump suitable for effective bailing;
26	One set of fishing tackle;
27	Sufficient tools for minor adjustments to the engine and its accessories;
28	Portable fire-extinguishing equipment of an approved type suitable for extinguishing oil fires;
29	One searchlight with a horizontal and vertical sector of at least 6°and a measured luminous intensity of 2500 candela (**cd**), which can work continuously for not less than three hrs;
30	An efficient radar reflector, unless a survival craft radar transponder is stowed in the lifeboat;
31	Thermal protective aids sufficient for 10% of the number of persons the lifeboat is permitted to accommodate or two, whichever is the greater;

1.0 ABSTRACT

The purpose of this item is to accomplish the annual requirements to inspect, service and test the ships Lifeboat Launching Appliances and Releasing Gear.

2.0 REFERENCES

2.1 Reference

2.1.1 IMO Resolution MSC.402(96) + Corr.1 (adopted on 19 May 2016) Requirements for maintenance, thorough examination, operational testing, overhaul and repair of Lifeboats, Rescue Boats, Launching Appliances and Release Gear

2.1.2 USCG NVIC 03-19, Maintenance, thorough examination, operational testing, overhaul and repair of Lifeboats and Rescue Boats, Launching Appliances and Release Gear

2.1.3 Tech Manual S9583-C6-MMC-010, Davit, Lifeboat Model GSP.FP.90/6,25 Winch, Lifeboat Model W20L.20

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location: Upper Deck, Frame 55, Port and Starboard

3.2 Description

3.2.1 Lifeboat Davit

a. Mfg: Global Davit GmbH, Survival & Deck Equipment

b. Model: Gsp.FP.90/6,25

3.3 Quantity: Two systems

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

Item	Part Number	NSN/TNICN	Description	Qty[1]	UOM[2]
1.	As Reqd	As Reqd	Replacement winch gearbox oil (Ship's Force to provide)	As Reqd	As Reqd

1 Qty = Quantity
2 UOM = Unit of measure

4.3 Government Furnished Services (GFS): None

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Off-load release mechanism means a release mechanism which releases the survival craft/rescue boat/fast rescue boat when it is waterborne or when there is no load on the hooks. On-load release mechanism means a release mechanism which releases the survival craft/rescue boat/fast rescue boat with load on the hooks. This release mechanism will be provided with a hydrostatic interlock unless other means are provided to ensure that the boat is waterborne before the release mechanism can be activated. In case of failure or when the boat is not waterborne, there will be a means to override the hydrostatic interlock or similar device to allow emergency release. This interlock override capability will be adequately protected against accidental or premature use. Adequate protection will include special mechanical protection not normally required for off-load release, in addition to a danger sign.

5.3 The maintenance and adjustment of release gear are critical operations with regard to the safe operation of lifeboats, rescue boats, fast rescue boats and davit launched liferafts. Utmost care will be taken when carrying out all inspection and maintenance operations on the equipment.

5.4 SOLAS regulation III/20 – Operational readiness, maintenance and inspections

5.5 SOLAS regulation III/36 – Instructions for onboard maintenance

5.6 A “Fall Preventer Device” (FPD) can be used to minimize the risk of injury or death by providing a secondary alternate load path in the event of failure of the on-load hook or its release mechanism or of accidental release of the on-load hook.

5.7 The Coast Guard considers the “make” to refer to the equipment manufacturer.

5.8 The Coast Guard considers the “type” to refer to the Coast Guard approval series for the equipment.

5.9 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None Additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service and test the ships Lifeboat Launching Appliances and Releasing Gear using References 2.1.1 through 2.1.3 as guidance.

7.2 Adhere to the requirements of 29 CFR §1915.86 for occupational safety and health standards for shipyard personnel working on Lifeboats.

7.2.1 Before any employee works in or on a stowed or suspended lifeboat, the employer will secure the lifeboat independently from the releasing gear to prevent it from falling or capsizing.

7.2.2 The employer will not permit any employee to be in a lifeboat while it is being hoisted or lowered, except when the employer demonstrates that it is necessary to conduct operational tests or drills over water, or in the event of an emergency.

7.2.3 The employer will not permit any employee to work on the outboard side of a lifeboat that is stowed on chocks unless the lifeboat is secured by gripes or another device that prevents it from swinging.

7.3 Conduct the weekly and monthly inspections and routine maintenance as specified in the equipment maintenance manual(s), by an authorized service provider. Perform all items listed in checklists for the weekly/monthly inspections required by SOLAS regulations III/20.6 and III/20.7.

7.3.1 Weekly inspection (III/20.6)

a. All survival craft, rescue boats and launching appliances will be visually inspected to ensure that they are ready for use. The inspection will include, but is not limited to, the condition of hooks, their attachment to the lifeboat and the on-load release gear being properly and completely reset.

b. Lifeboats, except free-fall lifeboats, on cargo ships will be moved from their stowed position, without any persons on board, to the extent necessary to demonstrate satisfactory operation of launching appliances;

c. The general emergency alarm will be tested.

7.3.2 Monthly inspections (III/20.7)

a. All lifeboats, except free-fall lifeboats, will be turned out from their stowed position, without any persons on board.

b. Inspection of the life-saving appliances, including lifeboat equipment, will be carried out monthly using the SOLAS required checklist (SOLAS III, Regulation 36.1) to ensure that they are complete and in good order. A report of the inspection will be submitted.

7.4 With assistance of the ship’s Captain the OEM Authorized Rep is to review the vessels records of inspections and routine on-board maintenance carried out by the ship's crew and the applicable certificates for the equipment.

7.5 Launching Appliances

7.5.1 Thoroughly examine and test the Launching Appliances for satisfactory condition and operation, to include:

a. Davit or other launching structures, in particular with regard to corrosion, misalignments, deformation and excessive free play

b. Wires and sheaves, possible damage such as kinks and corrosion, with special regard for areas passing through sheaves

c. Suspension chains

d. Lubrication of wires, sheaves and moving parts

e. Functioning of limit switches

f. Stored power systems, if applicable

g. Hydraulic systems, if applicable

h. Winch braking system using winch manual as guidance

i. Replace winch brake pads, when necessary

j. Winch foundation

k. Remote control system, if applicable. Verify the proper spooling of the remote-control wire, verify the proper position of the remote-control wire weight, and verify material condition of the shackle that connects the pull cable to the remote-control wire within the lifeboat per US Coast Guard Safety Alert Number 07-22.

l. Power supply system

m. Operating instruction placards, will be on or in the vicinity of survival craft and their launching controls

7.5.2 Drain and replace the winch gearbox oil with GFM oil identified in paragraphs 4.2.1. Before filling the gearbox with new oil, the gearbox should be thoroughly cleaned and inspected by the OMT REP. Lubricate all grease points and fittings.

7.6 Release Gear

7.6.1 Thoroughly examine the Release Gear for satisfactory condition. No maintenance or adjustment of the release gear will be undertaken while the hooks are under load. The release gear is to be examined prior to its operational test.

a. Operation of devices for activation of release gear

b. Excessive free play (tolerances)

c. Where fitted, examine the hydrostatic interlock system to include floats, membranes, diaphragms, pins, rods, linkages, for any signs of damage, and ensure same is intact and operates as designed

d. Emergency hydrostatic interlock override and danger sign

e. Cables for control and release

f. Hook fastening

7.7 Submit a condition report to the OMT REP summarizing all as found conditions, testing and inspections accomplished, and any discrepancies noted on each davit and any recommended repairs and parts that are outside the original scope of work. Include a copy of the manufacturer’s authorized servicing company’s service report.

7.8 Testing

7.8.1 Testing is to be conducted in the presence of the OMT REP, ABS Surveyors and USCG Inspectors.

7.8.2 Conduct an annual operational test of the launching appliance. Perform a dynamic test of the winch brake at maximum lowering speed. The load to be applied will be the weight of the survival craft without persons on board. When the craft has reached its maximum lowering speed and before it enters the water, the brake will be abruptly applied. Following these tests, the stressed structural parts will be reinspected where the structure permits the reinspection.

7.8.3 Conduct an operational test of the davit-launched lifeboats' on-load release function as follows:

a. Position the boat partially in the water such that the weight of the boat is substantially supported by the falls and the hydrostatic interlock system, where fitted, is not triggered;

b. Operate the on-load release gear;

c. Reset the on-load release gear; and

d. Examine the release gear and hook fastening to ensure that the hook is completely reset, and no damage has occurred.

7.8.4 Conduct an operational test of the davit-launched lifeboats off-load release function as follows:

a. Position the boat so that it is fully waterborne

b. Operate the off-load release gear

c. Reset the off-load release gear upon completion

7.8.5 The release gear is to be re-examined after its operational test and the operational test of the winch brake. Special consideration will be given to ensure that no damage has occurred during the winch brake test, especially to the hook fastening.

a. Operation of devices for activation of release gear

b. Excessive free play (tolerances)

c. Hydrostatic interlock system, where fitted

d. Emergency hydrostatic interlock override and danger sign

e. Cables for control and release

f. Hook fastening.

7.9 Prior to hoisting, the release gear will be checked by the OEM Authorized Tech Rep that is completely and properly reset. Recover the boat to the stowed position and leave it in a ready for service condition. The final turning-in of the boat will be done without any persons on board.

7.10 All work will only be accomplished by trained, experienced and authorized service personnel for the specific system.

7.11 Reports

7.11.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.11.2 Submit a report that includes “as released” condition of the davit. Submit one electronic copy in Portable Document Format (PDF).

7.11.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.12 Painting: None additional

7.13 Manufacturer’s Representative

7.13.1 Provide the services of certified personnel of either the Manufacturer or an Authorized Service Provider in accordance with References 2.1.1 and 2.1.2 to perform any and all system inspections, maintenance and tests to ensure proper operation of the Launching Appliances and Release Gear systems using IMO and manufacturer’s performance specifications as guidance.

a. *Manufacturer* means the original equipment manufacturer.

b. *Authorized Service Provider* (ASP) means an entity authorized by the Administration (USCG). The Coast Guard considers a service provider to be authorized if an Authorized Classification Society (ie; ABS/DNV/LLOYDS.) has verified the service provider is capable of servicing life-saving equipment in accordance with 46 CFR §199 and Reference 2.1.1.

7.13.2 Requirements for authorization as an ASP are defined in Reference 2.1.1. As a minimum:

a. Personnel will be certified by the manufacturer or authorized service provider for each make and type of the equipment to be worked on

b. Have sufficient tools, and in particular any specialized tools specified in the manufacturer's instructions, including portable tools as needed for the work

c. Have access to appropriate parts and accessories as specified for maintenance and repair

d. Have available the manufacturer's instructions for repair work involving disassembly or adjustment of on-load release mechanisms and davit winches

e. A documented and certified quality system

7.13.3 Provide the OMT REP with a copy of the ASP authorization document issued by an Authorized Classification Society defining the scope of services provided for the makes and types of equipment identified in section 3.0. The expiry date will be clearly written on the document. The authorization documentation will be submitted prior to the start of any work.

7.14 Preparation of Drawings

7.14.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to accomplish the annual requirements to inspect, service and test the ships Rescue Boats.

2.0 REFERENCE/ENCLOSURE

2.1 References

2.1.1 IMO Resolution MSC.402(96) + Corr.1 (adopted on 19 May 2016) Requirements for maintenance, thorough examination, operational testing, overhaul and repair of Lifeboats, Rescue Boats, Launching Appliances and Release Gear.

2.1.2 USCG NVIC 03-19, Maintenance, thorough examination, operational testing, overhaul and repair of Lifeboats and Rescue Boats, Launching Appliances and Release Gear

2.1.3 Tech Manual T9008-DB-MMC-010, RESCUE BOAT FASSMER FFR 6.1 ID

2.1.4 Rescue Boat Required Equipment (LSA Code, Chapter V Rescue Boats) 46CFR§199.175 Survival craft and rescue boat equipment. Table 199.175—Survival Craft Equipment

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location

3.1.1 Upper Deck, Starboard, Frame 35

3.2 Description

3.2.1 Rescue Boat

a. Mfg: Fr. Fassmer GmbH & Co. KG

b. Model: FASSMER FFR 6.1 ID, 6.1m x 2.23m x 2.55m motor propelled

c. Capacity: Six men

3.2.2 Quantity: One systems

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Off-load release mechanism means a release mechanism which releases the survival craft/rescue boat/fast rescue boat when it is waterborne or when there is no load on the hooks. On-load release mechanism means a release mechanism which releases the survival craft/rescue boat/fast rescue boat with load on the hooks. This release mechanism will be provided with a hydrostatic interlock unless other means are provided to ensure that the boat is waterborne before the release mechanism can be activated. In case of failure or when the boat is not waterborne, there will be a means to override the hydrostatic interlock or similar device to allow emergency release. This interlock override capability will be adequately protected against accidental or premature use. Adequate protection will include special mechanical protection not normally required for off-load release, in addition to a danger sign.

5.3 The maintenance and adjustment of release gear are critical operations with regard to the safe operation of lifeboats, rescue boats, fast rescue boats and davit launched liferafts. Utmost care will be taken when carrying out all inspection and maintenance operations on the equipment.

5.4 SOLAS regulation III/20 – Operational readiness, maintenance and inspections

5.5 SOLAS regulation III/36 – Instructions for onboard maintenance.

5.6 The Coast Guard considers the “make” to refer to the equipment manufacturer.

5.7 The Coast Guard considers the “type” to refer to the Coast Guard approval series for the equipment.

5.8 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service and test the ships Rescue Boats, using SOLAS and USCG requirements, using References 2.1.1 through 2.1.3 and Enclosure 2.2.1 as guidance by authorized service providers.

7.2 Adhere to the requirements of 29 CFR §1915.86 for occupational safety and health standards for shipyard personnel working on rescue boats.

7.2.1 Before any employee works in or on a stowed or suspended rescue boat, the employer will secure the rescue boat independently from the releasing gear to prevent it from falling or capsizing.

7.2.2 The employer will not permit any employee to be in a rescue boat while it is being hoisted or lowered, except when the employer demonstrates that it is necessary to conduct operational tests or drills over water, or in the event of an emergency.

7.2.3 The employer will not permit any employee to work on the outboard side of a rescue boat that is stowed on chocks unless the rescue boat is secured by gripes or another device that prevents it from swinging.

7.3 Conduct the weekly and monthly inspections one time within two weeks of vessel’s arrival, and routine maintenance as specified in the equipment maintenance manual(s), by an authorized service provider. Perform all items listed in checklists for the weekly/monthly inspections required by SOLAS regulations III/20.6 and III/20.7.

7.3.1 Weekly inspection (III/20.6)

a. All survival craft, rescue boats and launching appliances will be visually inspected to ensure that they are ready for use. The inspection will include, but is not limited to, the condition of hooks, their attachment to the lifeboat and the on-load release gear being properly and completely reset.

b. With the assistance of ship’s force, all engines in lifeboats and rescue boats will be run for a total period of not less than three minutes, provided the ambient temperature is above the minimum temperature required for starting and running the engine. During this period of time, it should be demonstrated that the gear box and gear box train are engaging satisfactorily;

c. Rescue boats on cargo ships will be moved from their stowed position, without any persons on board, to the extent necessary to demonstrate satisfactory operation of launching appliances;

d. The general emergency alarm will be tested.

7.3.2 Monthly inspections (III/20.7)

a. All rescue boats will be turned out from their stowed position, without any persons on board.

b. Inspection of the life-saving appliances, including rescue boat equipment, will be carried out monthly using the SOLAS required checklist (SOLAS III, Regulation 36.1) to ensure that they are complete and in good order. A report of the inspection will be submitted.

7.4 With assistance of the ship’s Captain the OEM Authorized Rep is to review the vessels records of inspections and routine on-board maintenance carried out by the ship's crew and the applicable certificates for the equipment.

7.5 RIB/Rescue Boats/Fast Rescue Boats

7.5.1 Rescue boats may be either of rigid or inflated construction or a combination of both. Record the following information on each Rescue Boat and submit a report with the data to the OMT REP:

a. Manufacturer' s name and address

b. Rescue Boat model and serial number

c. Month and year of manufacture

d. Make and model of the on-load release gear

e. Number of persons the Rescue Boat is approved to carry

f. The approval information

g. Self-righting bottle serial #s and hydrostatic dates, if fitted

7.5.2 Thoroughly examine and test each Rescue Boat for satisfactory condition and operation, to include:

a. Condition of the boat structure including fixed and loose equipment (including a visual examination of the external boundaries of the void spaces, as far as practicable)

b. Condition of the gel coat

c. Hatches and doors for watertightness

d. Engine and propulsion system, one hour in-water operational test with Mate and OEM Authorized Rep onboard. Demo to include full speed, forward and reverse operations.

1. Rescue boats will be capable of maneuvering at a speed of at least six knots and maintaining that speed for a period of at least four hours, when loaded with its full complement of persons

2. Fast rescue boats will be capable of maneuvering, for a period of at least four hours, at a speed of at least 20 kn in calm water with a crew of three persons and at least eight knots when loaded with its full complement of persons

e. Maneuvering system

f. Power supply system

g. Recharging of all engine starting, radio and searchlight batteries

h. Fixed two-way VHF radiotelephone with an antenna, where fitted

i. Fixed compass, where fitted

j. Bailing system

k. Markings and reflective tape

l. Fender/skate arrangements

m. Rescue boat righting system, where fitted. This is to include manual and automatic features, the pressurized cylinder, discharge valve and inflatable bag.

n. authorized service center.

o. Fast rescue boats will stop automatically or be stopped by the helmsman's emergency release switch

7.5.3 Temporarily remove the boats from the ship to a protected location. Provide protection for rescue boats to include covering and protection from the elements at all times. Block the boat in the upright position, with clearance under the hull to permit inspection and access.

7.5.4 Clean, free up, lubricate and test all moving parts including interlocking devices, hinges, releasing gear and bilge pumps. Each fuel tank must be emptied, cleaned, and refilled with fresh fuel. Replace all fuel filters.

7.5.5 Replace the engine oil and filters.

7.5.6 Fresh water wash the boats with a mild detergent removing all salt, oils, and contaminants cleaning the entire interior and exterior of the boat. Apply a coat of wax to the exterior of the entire boat.

7.5.7 All repairs and maintenance of inflated rescue boats will be carried out using the manufacturer's instructions as guidance. Repairs will be effected at an approved servicing station.

7.5.8 Inspect the condition of the buoyancy tubes forming the boundary of the inflated rescue boat. Verify proper operation of all nonreturn valves for manual inflation, means for deflation and safety relief valves on each buoyancy compartment.

7.5.9 Rescue boats will be stowed in a state of continuous readiness for launching in not more than five minutes, and if the inflated type, in a fully inflated condition at all times (SOLAS III, Regulation 14 Stowage of rescue boats). Recovery time of the rescue boat will be not more than five minutes in moderate sea conditions when loaded with its full complement of persons and equipment (SOLAS III, Regulation 17 Rescue boat embarkation, launching and recovery arrangements).

7.6 Submit a condition report to the OMT REP summarizing all as found conditions, testing and inspections accomplished, and any discrepancies noted on each Rescue Boat and any recommended repairs and parts that are outside the original scope of work. The contractor’s report will include a copy of the manufacturer’s authorized servicing company’s service report.

7.7 All work will only be accomplished by trained, experienced and authorized service personnel for the specific system.

7.8 Reports

7.8.1 All reports and checklists will be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.8.2 When thorough examination, operational testing, overhaul and repair are completed, a statement confirming that the Rescue Boat arrangements remain fit for service will be issued by the manufacturer or authorized service provider that conducted the work to the OMT REP. Submit one electronic copy in Portable Document Format (PDF). A copy of valid documents of certification and authorization as appropriate will also be included with the statement.

7.9 Painting: None additional

7.10 Manufacturer’s Representative

7.10.1 Provide the services of certified personnel of either the Manufacturer or an Authorized Service Provider in accordance with References 2.1.1 and 2.1.2 to perform any and all system inspections, maintenance and tests to ensure proper operation of the Rescue Boat using IMO and manufacturer’s performance specifications as guidance.

a. *Manufacturer* means the original equipment manufacturer.

b. *Authorized Service Provider* (ASP) means an entity authorized by the Administration (USCG). The Coast Guard considers a service provider to be authorized if an Authorized Classification Society (ie; ABS/DNV/LLOYDS.) has verified the service provider is capable of servicing life-saving equipment in accordance with 46 CFR §199 and Reference 2.1.1.

7.10.2 Requirements for authorization as an ASP are defined in Reference 2.1.1. As a minimum:

a. Personnel will be certified by the manufacturer or authorized service provider for each make and type of the equipment to be worked on

b. Have sufficient tools, and in particular any specialized tools specified in the manufacturer's instructions, including portable tools as needed for the work

c. Have access to appropriate parts and accessories as specified for maintenance and repair

d. Have available the manufacturer's instructions for repair work involving disassembly or adjustment of on-load release mechanisms and davit winches

e. A documented and certified quality system

7.10.3 Provide the OMT REP with a copy of the ASP authorization document issued by an Authorized Classification Society defining the scope of services provided for the makes and types of equipment identified in section 3.0. The expiry date will be clearly written on the document. The authorization documentation will be submitted prior to the start of any work.

7.11 Preparation of Drawings

7.11.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - RESCUE BOAT REQUIRED EQUIPMENT

	Every Rescue Boat will carry:
1	Sufficient buoyant oars or paddles to make headway in calm seas. Thole pins, crutches or equivalent arrangements will be provided for each oar. Thole pins or crutches will be attached to the boat by lanyards or chains;
2	One buoyant bailer;
3	One binnacle containing an efficient compass which is luminous or provided with suitable means of illumination;
4	One sea-anchor and tripping line if fitted with a hawser of adequate strength not less than 10 m in length;
5	One painter of sufficient length and strength, attached to the release device complying with the requirements of LSA Code, Chapter IV Survival Craft, paragraph 4.4.7.7 and placed at the forward end of the rescue boat;
6	One buoyant line, not less than 50 m in length, of sufficient strength to tow a liferaft as required by paragraph 5.1.1.7;
7	One waterproof electric torch suitable for Morse signaling, together with one spare set of batteries and one spare bulb in a waterproof container;
8	One whistle or equivalent sound signal;
9	One first-aid outfit in a waterproof case capable of being closed tightly after use;
10	Two buoyant rescue floats, attached to not less than 30 m of buoyant line;
11	One searchlight with a horizontal and vertical sector of at least 6° and a measured luminous intensity of 2 500 cd which can work continuously for not less than 3 h;

ENCLOSURE 2.2.1 - RESCUE BOAT REQUIRED EQUIPMENT

12	An efficient radar reflector;
13	Thermal protective aids complying with the requirements of LSA Code, Chapter II, Section 2.5 sufficient for 10% of the number of persons the rescue boat is permitted to accommodate or two, whichever is the greater; and
14	Portable fire-extinguishing equipment of an approved type suitable for extinguishing oil fires.

ENCLOSURE 2.2.1 - RESCUE BOAT REQUIRED EQUIPMENT

	In addition to the equipment required by every Rescue Boat, the normal equipment of every rigid rescue boat will include:
1	One boat-hook;
2	One bucket; and
3	One knife or hatchet.
	In addition to the equipment required by every Rescue Boat, the normal equipment of every inflated rescue boat will consist of:
1	One buoyant safety knife;
2	Two sponges;
3	An efficient manually operated bellows or pump;
4	One repair kit in a suitable container for repairing punctures; and
5	One safety boat-hook.
	In addition to the equipment required by every Rescue Boat, the normal equipment of every fast rescue boat will consist of:
1	VHF radiocommunication set which is hands-free and watertight

1.0 ABSTRACT

The purpose of this item is to accomplish the annual requirements to inspect, service and test the ships Rescue Boat Launching Appliances and Releasing Gear.

2.0 REFERENCE

2.1 References

2.1.1 IMO Resolution MSC.402(96) + Corr.1 (adopted on 19 May 2016) Requirements for maintenance, thorough examination, operational testing, overhaul and repair of Lifeboats, Rescue Boats, Launching Appliances and Release Gear.

2.1.2 USCG NVIC 03-19, Maintenance, thorough examination, operational testing, overhaul and repair of Lifeboats and Rescue Boats, Launching Appliances and Release Gear

2.1.3 Tech Manual S9583-C7-MMC-010, Rescue Boat Winch and Davit (Winch Model: W08E.40.01, Davit Model: Rhs 26/5.0)

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location

3.1.1 Upper Deck, Starboard, Frame 35

3.2 Description

3.2.1 Rescue Boat Davit

a. Mfg: Global Davit GmbH

b. Model: Rhs.26/5.0

3.2.2 Quantity: One systems

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None

4.2 Government Furnished Material (GFM)

Item	Part Number	NSN/TNICN	Description	Qty[1]	UOM[2]
1.	As Reqd	As Reqd	Replacement winch gearbox oil (provided by ship’s force)	As Reqd	As Reqd
2.	As Reqd	As Reqd	Replacement constant tension winch gearbox oil (provided by ship’s force)	As Reqd	As Reqd

1 Qty = Quantity
2 UOM = Unit of measure

4.3 Government Furnished Services (GFS): None

4.4 Government Furnished Information (GFI): None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 Off-load release mechanism means a release mechanism which releases the survival craft/rescue boat/fast rescue boat when it is waterborne or when there is no load on the hooks. On-load release mechanism means a release mechanism which releases the survival craft/rescue boat/fast rescue boat with load on the hooks. This release mechanism will be provided with a hydrostatic interlock unless other means are provided to ensure that the boat is waterborne before the release mechanism can be activated. In case of failure or when the boat is not waterborne, there will be a means to override the hydrostatic interlock or similar device to allow emergency release. This interlock override capability will be adequately protected against accidental or premature use. Adequate protection will include special mechanical protection not normally required for off-load release, in addition to a danger sign.

5.3 The maintenance and adjustment of release gear are critical operations with regard to the safe operation of lifeboats, rescue boats, fast rescue boats and davit launched liferafts. Utmost care will be taken when carrying out all inspection and maintenance operations on the equipment.

5.4 SOLAS regulation III/20 – Operational readiness, maintenance and inspections

5.5 SOLAS regulation III/36 – Instructions for onboard maintenance.

5.6 The Coast Guard considers the “make” to refer to the equipment manufacturer.

5.7 The Coast Guard considers the “type” to refer to the Coast Guard approval series for the equipment.

5.8 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service and test the ships Rescue Boat Launching Appliances and Releasing Gear using SOLAS and USCG requirements, using References 2.1.1 through 2.1.3 as guidance by authorized service providers.

7.2 Adhere to the requirements of 29 CFR §1915.86 for occupational safety and health standards for shipyard personnel working on Lifeboats.

7.2.1 Before any employee works in or on a stowed or suspended rescue boat, the employer will secure the rescue boat independently from the releasing gear to prevent it from falling or capsizing.

7.2.2 The employer will not permit any employee to be in a lifeboat while it is being hoisted or lowered, except when the employer demonstrates that it is necessary to conduct operational tests or drills over water, or in the event of an emergency.

7.2.3 The employer will not permit any employee to work on the outboard side of a rescue boat that is stowed on chocks unless the rescue boat is secured by gripes or another device that prevents it from swinging.

7.3 Conduct the weekly and monthly inspections and routine maintenance as specified in the equipment maintenance manual(s), by an authorized service provider. Perform all items listed in checklists for the weekly/monthly inspections required by SOLAS regulations III/20.6 and III/20.7.

7.3.1 Weekly inspection (III/20.6)

a. All survival craft, rescue boats and launching appliances will be visually inspected to ensure that they are ready for use. The inspection will include, but is not limited to, the condition of hooks, their attachment to the lifeboat and the on-load release gear being properly and completely reset.

b. Rescue boat on cargo ships will be moved from their stowed position, without any persons on board, to the extent necessary to demonstrate satisfactory operation of launching appliances;

c. The general emergency alarm will be tested.

7.3.2 Monthly inspections (III/20.7)

a. All rescue boat will be turned out from their stowed position, without any persons on board.

b. Inspection of the life-saving appliances, including lifeboat equipment, will be carried out monthly using the SOLAS required checklist (SOLAS III, Regulation 36.1) to ensure that they are complete and in good order. A report of the inspection will be submitted.

7.4 With assistance of the ship’s Captain the OEM Authorized Rep is to review the vessels records of inspections and routine on-board maintenance carried out by the ship's crew and the applicable certificates for the equipment.

7.5 Launching Appliances

7.5.1 Thoroughly examine and test the Launching Appliances for satisfactory condition and operation, to include:

a. Davit or other launching structures, in particular with regard to corrosion, misalignments, deformation and excessive free play

b. Wires and sheaves, possible damage such as kinks and corrosion, with special regard for areas passing through sheaves

c. Lubrication of wires, sheaves and moving parts

d. Functioning of limit switches

e. Stored power systems, if applicable

f. Hydraulic systems, if applicable

g. Winch braking system using winch manual as guidance

h. Replace winch brake pads, when necessary

i. Winch foundation

j. Remote control system, if applicable. Verify the proper spooling of the remote-control wire, verify the proper position of the remote-control wire weight, and verify material condition of the shackle that connects the pull cable to the remote-control wire within the lifeboat per US Coast Guard Safety Alert Number 07-22.

k. Power supply system

l. Operating instruction placards, will be on or in the vicinity of survival craft and their launching controls

7.5.2 Perform a walk around inspection looking for loose fasteners, leaking connections or seals, worn or damaged hoses and tubing, or any other defects or dangerous conditions. Promptly report any defects to the OMT REP.

7.5.3 Lubrication

a. Check all oil levels; gear reducers and hydraulics

b. Replace hydraulic filters.

c. Lubricate all points and fittings; Swing Bearing, Pinions and Ring Gear, Swing Drive, Constant Tension Winch, Sheaves and Bushings.

d. Drain and replace the Swing Drive and Constant Tension Winch gearbox oil with GFM oil identified in paragraphs 4.2.1 and 4.2.2. Before filling the gearbox with new oil, the gearbox should be inspected by the OMT REP.

e. Check Accumulator precharge

7.5.4 Launching Appliances fitted with slewing rings are to undergo Rocking Tests, using the bearing manufacturer’s instructions as guidance. The slewing ring assembly is to be examined for slack bolts, damaged bearings and deformation or fractured weldments. The OMT REP and ABS Surveyor are to witness.

7.6 Release Gear

7.6.1 Thoroughly examine the Release Gear for satisfactory condition. No maintenance or adjustment of the release gear will be undertaken while the hooks are under load. The release gear is to be examined prior to its operational test.

a. Operation of devices for activation of release gear

b. Excessive free play (tolerances)

c. Cables for control and release

d. Hook fastening

e. Where fitted, examine the hydrostatic interlock system to include floats, membranes, diaphragms, pins, rods, linkages, for any signs of damage, and ensure same is intact and operates as designed

7.7 Submit a condition report to the OMT REP summarizing all as found conditions, testing and inspections accomplished, and any discrepancies noted on each Rescue Boat and davit and any recommended repairs and parts that are outside the original scope of work. The contractor’s report will include a copy of the manufacturer’s authorized servicing company’s service report.

7.8 Testing

7.8.1 Testing is to be conducted in the presence of the OMT REP, ABS Surveyors and USCG Inspectors. Perform OP test including main and auxiliary load hoisting and lowering, slewing (swinging), safety protective (fail-safe) and limiting devices.

7.8.2 Conduct an annual operational test of the launching appliance. Perform a dynamic test of the winch brake at maximum lowering speed. The load to be applied will be the weight of the survival craft without persons on board. When the craft has reached its maximum lowering speed and before it enters the water, the brake will be abruptly applied. Following these tests, the stressed structural parts will be reinspected where the structure permits the reinspection.

7.8.3 Conduct an operational test of the davit-launched Rescue Boats' on-load release function (if fitted) as follows:

a. Position the boat partially in the water such that the weight of the boat is substantially supported by the falls and the hydrostatic interlock system, where fitted, is not triggered;

b. Operate the on-load release gear

c. Reset the on-load release gear

d. Examine the release gear and hook fastening to ensure that the hook is completely reset, and no damage has occurred

7.8.4 Conduct an operational test of the davit-launched Rescue Boats' off-load release function as follows:

a. Position the boat so that it is fully waterborne

b. Operate the off-load release gear

c. Reset the off-load release gear upon completion

7.8.5 The release gear is to be re-examined after its operational test and the operational test of the winch brake. Special consideration will be given to ensure that no damage has occurred during the winch brake test, especially to the hook fastening.

a. Operation of devices for activation of release gear

b. Excessive free play (tolerances)

c. Hydrostatic interlock system, where fitted

d. Cables for control and release

e. Hook fastening

7.9 Prior to hoisting, the release gear will be checked by the OEM Authorized Tech Rep that is completely and properly reset. Recover the boat to the stowed position and leave it in a ready for service condition. The final turning-in of the boat will be done without any persons on board.

7.10 All work will only be accomplished by trained, experienced and authorized service personnel for the specific system.

7.11 Reports

7.11.1 All reports and checklists will be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.11.2 When thorough examination, operational testing, overhaul and repair are completed, a statement confirming that the Rescue Boat arrangements remain fit for service will be issued by the manufacturer or authorized service provider that conducted the work to the OMT REP. Submit one electronic copy in Portable Document Format (PDF). A copy of valid documents of certification and authorization as appropriate will also be included with the statement.

7.12 Painting: None additional

7.13 Manufacturer’s Representative

7.13.1 Provide the services of certified personnel of either the Manufacturer or an Authorized Service Provider in accordance with References 2.1.1 and 2.1.2 to perform any and all system inspections, maintenance and tests to ensure proper operation of the Launching Appliances and Release Gear systems using IMO and manufacturer’s performance specifications as guidance.

a. *Manufacturer* means the original equipment manufacturer.

b. *Authorized Service Provider* (ASP) means an entity authorized by the Administration (USCG). The Coast Guard considers a service provider to be authorized if an Authorized Classification Society (ie; ABS/DNV/LLOYDS.) has verified the service provider is capable of servicing life-saving equipment in accordance with 46 CFR §199 and Reference 2.1.1.

7.13.2 Requirements for authorization as an ASP are defined in Reference 2.1.1. As a minimum:

a. Personnel will be certified by the manufacturer or authorized service provider for each make and type of the equipment to be worked on

b. Have sufficient tools, and in particular any specialized tools specified in the manufacturer's instructions, including portable tools as needed for the work

c. Have access to appropriate parts and accessories as specified for maintenance and repair

d. Have available the manufacturer's instructions for repair work involving disassembly or adjustment of on-load release mechanisms and davit winches

e. A documented and certified quality system

7.13.3 Provide the OMT REP with a copy of the ASP authorization document issued by an Authorized Classification Society defining the scope of services provided for the makes and types of equipment identified in section 3.0. The expiry date will be clearly written on the document. The authorization documentation will be submitted prior to the start of any work.

7.14 Preparation of Drawings

7.14.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this item is to inspect, service and test the ships Immersion Suits.

2.0 REFERENCES

2.1 IMO MSC/Circ.1114, Dated 25 May 2004, Guidelines for Periodic Testing of Immersion Suits and Anti-Exposure Suit Seams and Closures

2.2 MSC/Circ.1047, Dated 28 May 2002, Guidelines for Monthly Shipboard Inspection of Immersion Suits and Anti-Exposure Suits by Ships Crew

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location: Various throughout the ship

3.2 Description

3.2.1 Mfg: Stearns

3.2.2 Model: 1590 US Coast Guard Approved

3.3 Quantity: Three-hundred and eighty (380) suits

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 SOLAS regulation III/20 – Operational readiness, maintenance and inspection

5.3 SOLAS regulation III/36 – Instructions for onboard maintenance.

5.4 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide all services and assistance to accomplish the thorough examination, testing and repair of Immersion Suits using SOLAS and USCG requirements and References 2.1 and 2.2 as guidance by authorized service providers.

7.2 Convey the Immersion Suits identified in section 3.0 from the ship to an ABS, USCG, and SOLAS approved Immersion suit inspection, testing, and repair facility.

7.3 Conduct the monthly inspections and routine maintenance as specified in the equipment maintenance manuals, by an authorized service provider. Perform all items listed in checklists for the monthly inspections required by SOLAS regulations III and Reference 2.2.

7.3.1 Monthly inspections

a. Check closures on storage bag as well as general condition of bag for ease of removal of suit. Ensure donning instructions are legible. Confirm that suit is the type and size identified on the bag.

b. Lay the suit on a clean, flat surface. Make sure the suit is dry inside and out. Visually check for damage. Rips, tears or punctures should be repaired using manufacturer's instructions by a suitable repair station. A "suitable repair station" is one authorized by the suit manufacturer and/or acceptable to the Administration.

c. Check the zipper by sliding it up and down to check for ease of operation. Using lubricant recommended by the manufacturer, lubricate the front and back of the zipper and the slide fastener. If the zipper is not functional, the suit should be removed from service and discarded or returned to the manufacturer or a suitable repair station.

d. If fitted, check inflatable head support and/or buoyancy ring for damage and ensure that it is properly attached. Check inflation hoses for deterioration. At least quarterly, the head support/buoyancy ring should be inflated and tested for leaks (this test does not apply to integral inflatable lifejackets). Leaks should be repaired using manufacturers' instructions by a suitable repair station.

e. Check retro reflective tape for condition and adhesion. Replace if necessary.

f. If fitted, check whistle and expiration date of light and battery.

g. Replace suits in the bag with zippers fully opened.

7.4 Conduct the 3 Year inspection, service and routine maintenance on the Immersion Suits as required by SOLAS and Reference 2.1.1. Deterioration of seams and closures may not be readily apparent by visual inspection. Such deterioration can be detected by pressurization of the suit with air and testing of the seams and closures for leaks with a soapy water solution. Each suit is to be subjected to an air pressure test, at intervals not exceeding three years, or more frequently for suits over ten years of age:

a. A suitable head piece, fitted with a means to inject air into the suit, should be inserted into the face orifice of the suit and secured to minimize leakage around the face seal. A low-pressure monitoring device, either integral to the fitting for air injection or as a separate device, should also be inserted. If the suit is fitted with detachable gloves and/or boots, the wrists and/or cuffs should be sealed by inserting a short length of suitable diameter plastic pipe and securing the gloves and/or boots with suitable wire ties or hose clamps. The zipper should be fully zipped, and any face flap closed. The suit should then be inflated to a pressure of 0.7 to 1.4 kPa (0.1 to 0.2 psi). If an auxiliary inflatable means of buoyancy is provided, it should be inflated through the oral valve to a pressure of 0.7 kPa (0.1 psi) or until firm to the touch.

b. Each seam and closure of the suit - and each seam, oral tube and attachment points and joint or valve of any auxiliary inflatable means of buoyancy - should then be covered with a soapy water solution containing enough soap to produce bubbles (if leakage is noted at a foot valve to the extent that air pressure cannot be maintained, the valves should be sealed for the test).

c. If leaks are revealed by the propagation of bubbles at seams or closures, the leaking areas should be marked and, after cleaning the suit thoroughly with fresh water and drying it, repaired using the suit manufacturer's recommendations as guidance.

7.5 Testing is to be conducted in the presence of the OMT REP and ABS Surveyors. The air pressure test is to be performed at a suitable shore-based facility equipped to make any necessary repairs using the manufacturer's recommendations as guidance. Due to the wide variety of materials and adhesives used in immersion suits and anti-exposure suits, any repairs to a suit must be carried out by a facility which has access to the original manufacturer's recommended servicing instructions, parts and adhesives, and suitably trained personnel.

7.6 Return all Immersion Suits to the ship and turn over to the Chief Mate. Provide a signed receipt to the OMT REP.

7.7 All work will only be accomplished by trained, experienced and authorized service personnel for the specific suits.

7.8 Reports

7.8.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.8.2 When thorough examination, operational testing and repairs are completed, a statement confirming that the immersion suits remain fit for service shall be issued by the manufacturer or authorized service provider that conducted the work to the OMT REP. A copy of valid documents of certification and authorization as appropriate shall also be included with the statement. Submit a report that includes “as released” condition of the immersion suits. Submit one electronic copy in Portable Document Format (PDF).

7.8.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.9 Painting: None additional

7.10 Manufacturer’s Representative

7.10.1 Provide the services of a Manufacturer’s Authorized Technical Representative to perform any and all inspections, maintenance, and tests to ensure proper operation and repair of the Immersion Suits using IMO and manufacturer’s performance specifications as guidance.

https://www.eagle.org/ABSEaglePrograms/es/es-search.jsp

If repairs are required to a suit it should be accomplished by a facility authorized by the OEM or acceptable to the local USCG OCMI.

7.10.2 Provide the OMT REP with a signed letter of qualification from the original equipment manufacturer, certificate of training for the equipment identified in 3.0 prior to the start of any work.

7.11 Preparation of Drawings: None additional

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

The purpose of this work item is to describe the requirements to inspect, service, test and certify the ships Inflatable Liferafts and preserve all life raft cradles.

2.0 REFERENCES/ENCLOSURE

2.1 References

2.1.1 46 CFR §160.151-57 - Servicing Procedure

2.1.2 46 CFR §160.151-37 - Servicing Manual

2.1.3 46 CFR §160.151-21 Equipment required for SOLAS A and SOLAS B inflatable liferafts

2.1.4 USCG NVIC 4-86, 28 MAR 1986, Hydraulic Release Units for Liferafts, Life Floats and Buoyant Apparatus, and Alternate Float-Free Arrangements

2.2 Enclosure

2.2.1 Matrix of Liferafts Onboard

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location: See Enclosure 2.2.2

3.2 Description

3.2.1 Manufacturer: Elliot (RFD Beaufort)

a. Model: Mk IV TO

b. Capacity: 25 men

3.2.2 Manufacturer: Elliot (RFD Beaufort)

a. Model: Mk IV TO

b. Capacity: Ten men

3.2.3 Manufacturer: Survitec (RFD Beaufort)

a. Model: N20

b. Capacity: 50 men

3.2.4 Manufacturer: Survitec (RFD Beaufort)

a. Model: DL

b. Capacity: 25 men

3.3 Quantity

3.3.1 Ten Survitec 50-person liferaft

3.3.2 Ten Survitec 25-person liferaft

3.3.3 Two Elliot 10-person liferaft

3.3.4 Four Elliot 25-person liferaft

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the thorough examination, testing and repair of Inflatable Liferafts using SOLAS and USCG requirements by authorized service providers. Use References

2.1.1 through 2.1.4 and Enclosure 2.2.1 as guidance.

7.2 Record the following information on each Inflatable Liferaft and submit a report with the data to the OMT REP:

7.2.1 Manufacturer' s name and address

7.2.2 Liferaft model and serial number

7.2.3 Month and year of manufacture

7.2.4 SOLAS Pack (A or B)

7.2.5 Number of persons the Liferaft is approved to carry

7.2.6 The approval information

7.2.7 Date last serviced

7.2.8 Make and model of the hydrostatic release

7.2.9 Length of painter

7.3 Provide rigging and crane service to temporarily remove all liferafts from cradle locations identified in section 3.0 and enclosure 2.2.1, to the pier. Securely package and convey the Liferafts identified in section 3.0 and Enclosure 2.2.1 from the ship to an ABS and USCG approved Liferaft servicing facility for inspection, testing and repair.

7.4 Perform all required service, inspection, testing and certification on all Inflatable Liferafts, in accordance with Reference 2.1.1.

7.4.1 Each inflatable liferaft serviced by a servicing facility approved by the Coast Guard and ABS must be inspected and tested using Reference 2.1.1 and the manufacturer's servicing manual approved using Reference 2.1.2 as guidance.

7.4.2 Replace all batteries, power cells and survival equipment as required by References 2.1.1 through 2.1.3.

7.5 Replace the Painter system, Weak link and Hydrostatic

Release Unit on each Liferaft with USCG and SOLAS approved components rated for the intended service. See Reference 2.1.4 as guidance.

7.6 Coordinate testing with the OMT REP, ABS and USCG Inspectors to allow for random observation as required.

7.7 Submit a Condition Report to the OMT REP describing all as found conditions and any recommended repairs or replacements by liferaft, outside the scope of this work item.

7.7.1 Include in recommended replacement items, items that will expire prior seventeen months past the service sticker.

7.8 Mechanical clean all life raft cradles and foundations of all scale, loose and flaking paint and corrosion, using SSPC-SP3, Power Tool Cleaning as guidance. Apply two coats of Amercoat 240 followed by one full topcoat of Haze Gray Amershield to match the surrounding areas. Grease life raft ramps and grease pinned joints. Renew rubber matting to match existing.

7.9 Upon completion of all inspections, tests and repairs, provide rigging and crane service required, convey and return the Liferafts to the ship and reinstall in their original locations. Reconnect sea painter, hydrostatic releases, and related securing hardware; leaving each liferaft in a ready for service condition. Conduct a final inspection with the OMT REP.

7.10 Reports

7.10.1 All reports and checklists shall be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.10.2 The servicing facility shall issue a certificate to the OMT REP, in accordance with Reference 2.1.1, (p) for each liferaft it services.

7.11 Painting: None additional

7.12 Manufacturer’s Representative

7.12.1 Provide the services of a Manufacturer’s Authorized Service Facility and personnel to perform any and all inspections, maintenance and tests to ensure proper operation and repair of the Liferafts using IMO and USCG requirements and manufacturer’s performance specifications as guidance. In addition, the Servicing Facility is to be a USCG Liferaft Servicing Facility and ABS recognized service supplier for service type “Inflatable Liferaft Servicing Firms”; both listed for the specific raft model(s) being serviced.

https://www.eagle.org/ABSEaglePrograms/es/es-search.jsp

https://cgmix.uscg.mil/LifeRaftSearch/LiferaftSearch.aspx

7.12.2 Provide the OMT REP with a signed letter of qualification from the original equipment manufacturer, certificate of training for the equipment identified in section 3.0 and ABS Certificate # prior to the start of any work.

7.13 Preparation of Drawings

7.13.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

Mfg Name	Model #	Serial #	Year of mfg	Capacity (men)	Date last serviced (M/YR)
SurvitecZodiac	DL	XDC0GD27H717	AUG-2017	25	
SurvitecZodiac	DL	XDC0GD29H717	AUG-2017	25	
SurvitecZodiac	DL	XDC0GD31H717	AUG-2017	25	
SurvitecZodiac	DL	XDC0GD32H717	AUG-2017	25	
SurvitecZodiac	DL	XDC0GD34H717	AUG-2017	25	
SurvitecZodiac	EPS	XDC0GD85H717	AUG-2017	25	
SurvitecZodiac	TO	XDC2GK22J919	OCT-2019	50	
SurvitecZodiac	TO	XDC3GK00J919	OCT-2019	50	
SurvitecZodiac	TO	XDC3GK01J919	OCT-2019	50	
SurvitecZodiac	TO	XDC3GK02J919	OCT - 2019	50	
SurvitecZodiac	TO	XDC1GK45J919	OCT-2019	50	
SurvitecZodiac	TO	XDC2GK46J919	OCT-2019	50	

Mfg Name	Model #	Serial #	Year of mfg	Capacity (men)	Date last serviced (M/YR)
SurvitecZodiac	TO	XDC1GK47J919	OCT-2019	50	
SurvitecZodiac	TO	XDC1GK49J919	OCT-2019	50	
Elliot	Mk IV TO	5085900200176 LOT 1661	AUG-2014	10	
Elliot	Mk IV TO	5085900200377 LOT 1661	APR-2017	10	
Elliot	Mk IV TO	5086310213270 LOT 1674	AUG-2014	25	
Elliot	Mk IV TO	5086310213275 LOT 1674	AUG-2014	25	
Elliot	Mk IV TO	5086300200390 LOT 1674	Sept - 2014	25	
Elliot	Mk IV TO	5086300201062 LOT 1674	Jun - 2016	25	
Elliot	Mk IV TO	5086300201117 LOT 1674	JUL-2016	25	
Elliot	MK IV TO	5086300201057 LOT 1606	Apr-2016	25	
Elliot	MK IV TO	5086300201055 LOT 1606	Apr-2016	25	
Elliot	MK IV TO	50863002201210 LOT 1674	JUL-2016	25	

Mfg Name	Model #	Serial #	Year of mfg	Capacity (men)	Date last serviced (M/YR)
Elliot	MK IV TO	5086310213227 LOT 1674	AUG-2014	25	
Elliot	MK IV TO	5086310213216 LOT 1674	AUG-2016	25	

USS Hershel Williams		
(ESB 4)		
HABITABILITY OUTFITTING AND FURNISHINGS		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0657	**CATEGORY "A"**	**2026-01-30**
ESB-CCSI-ACCOMMODATION LADDER INSPECTION (1 YR)		**Manovich, David**

1.0 ABSTRACT

The purpose of this work item is to inspect, service and test the ships accommodation ladder and gangway.

2.0 REFERENCES

2.1 Schoellhorn-Albrecht Drawing No. BAL-30FP-25UL-G, Accommodation Ladder

2.2 Tech Manual T9623-AR-MMC-010, Accommodation Ladder and Winch

2.3 Guidelines for Construction, Installation, Maintenance and Inspection/Survey of Means of Embarkation and Disembarkation, MSC.1/Circ.1331, 11 June 2009

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location

3.1.1 Accommodation Ladder, A-69-2

3.2 Quantity: One accommodation ladder and winch system(s)

3.3 Description

3.3.1 Ladder

a. Mfg: Schoellhorn-Albrecht

b. Model: BAL-30FP-40

3.3.2 Wire Rope

a. Length: 76 meters (250 ft)

b. Diameter: 13 mm (1/2”)

c. Type: 5 x 19 IWRC

d. Lay: R.H.O.

e. Material: Galvanized

f. Min Breaking Strength: 10,113 Kg (22,250 lbs)

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE

4.1 Government Furnished Equipment (GFE): None.

4.2 Government Furnished Material (GFM)

Item	Part Number	NSN/TNICN	Description	Qty[1]	UOM[2]
1.	AS Reqd	AS Reqd	Grease (To be supplied by ship)	AS Reqd	As Reqd

1 Qty = Quantity

2 UOM = Unit of measure

4.3 Government Furnished Service (GFS): None.

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

5.3 THE T-ESB CLASS ELECTRICAL COLOR CODING DOES NOT FOLLOW CONSISTENT COLOR SCHEME. THE SHIP IS KNOWN TO HAVE

CURRENT CARRYING CONDUCTORS COLORED GREEN, BLUE, BLACK, AND WHITE. ASSUME ALL ELECTRICAL CABLES ARE ENERGIZED UNTIL TESTED AND PROVEN OTHERWISE. TEST ALL ELECTRICAL CABLES TO ENSURE THEY ARE DE-ENERGIZED PRIOR TO WORKING ON THE ITEM.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Provide and operate equipment and supply all services and assistance to accomplish the thorough examination, testing, service and certification of the accommodation ladders and gangways using IMO, SOLAS, USCG and the Manufacturer's requirements as guidance.

7.2 Provide temporary means of access to the vessel whenever the accommodation ladders and gangways are unavailable due to inspection, service and testing.

7.3 Conduct inspections, maintenance and testing of the accommodation ladders and gangways using References 2.1 through 2.3 as guidance.

7.3.1 Inspection: Conduct an annual inspection of the accommodation ladders and gangways verifying condition and conformance using References 2.1 through 2.3 as guidance. The examination will include/verify:

a. Each accommodation ladder or gangway is clearly marked at each end with a plate showing the restrictions on the safe operation and loading, including the maximum and minimum permitted design angles of inclination, design load, and maximum load on bottom end plate. Where the maximum operational load is less than the design load, it should also be shown on the marking plate.

b. Confirm satisfactory condition of all steps, treads, platforms, non-slip, side stringers, cross-members, decking, deck plates all support points such as pivots, wheels, and rollers, all suspension points such as lugs, brackets, stanchions, rigid handrails, hand ropes and turntables, davit structure, wire and sheaves. Pay careful attention to the condition of the hoist wire giving it a thorough inspection; SOLAS II-1, A-1, 3-9 and SOLAS III, B, I, 20.

c. Note any signs of damage, distortion, wear, cracks and corrosion.

d. Confirm satisfactory condition of the winch including the brake mechanism, remote control system and power supply system (motor). Open, inspect, and adjust brake mechanism (if required). OMT REP to witness adjustment prior to reassembly.

e. Confirm satisfactory condition of all fittings and davits associated with accommodation ladders and gangways

f. Confirm satisfactory condition of fittings or structures used for means of access to decks such as handholds in a gateway or bulwark ladder and stanchions.

7.3.2 Maintenance: Conduct annual maintenance on all accommodation ladders and gangways using the manufacturer's maintenance instructions and service bulletins as guidance. The maintenance will include/verify:

a. Lubricate all rollers, wheels, pivot tables, wire ropes, sheaves and associated parts with GFM provided by ship.

7.3.3 Annual System Test: Conduct annual testing of the accommodation ladders and gangways using Reference 2.1 through 2.3 as guidance. The testing will include:

a. Upon vessel arrival and with assistance from ships force conduct an operational test (no load) of each winch and accommodation ladder and gangway to confirm their proper operation and condition.

7.4 Testing is to be coordinated with the OMT REP and ABS

Surveyors to allow for observation if deemed necessary.

7.5 Upon completion of all inspections, tests and repairs return the accommodation ladders and gangways to a ready for service condition.

7.6 Reports

7.6.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.6.2 Submit a report that includes “as released” condition of the accommodation ladder. Submit one electronic copy in Portable Document Format (PDF).

7.6.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.7 Painting: None additional

7.8 Manufacturer’s Representative: None

7.9 Preparation of Drawings

7.9.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

USS Hershel Williams		
(ESB 4)		
HABITABILITY OUTFITTING AND FURNISHINGS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0662	CATEGORY "A"	2026-01-30
ESB-CCSI-DEEP FAT FRYER INSPECTION (1 YR)		Manovich, David

1.0 ABSTRACT

The purpose of this item is to inspect, service and test the Deep Fat Fryers.

2.0 REFERENCES

2.1 Tech Manual T6161-K6-FSE-010, Electric Marine Fryer (Model: 130F-480VM)

2.2 NAVSEA Drawing No. 651-8468281, Galley/Messroom Arrangement and Details

2.3 SOLAS Chapter II-2 Construction - Fire Protection, Fire Detection and Fire Extinction, Part C Suppression of Fire, Regulation 10 Firefighting, Para 6.4 (Available Commercially)

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

a. Aft Galley (A-47-0)

3.1.2 Quantity: Two Deep Fat Fryers for space identified in paragraph 3.1.1.

3.2 Item Description/Manufacturer’s Data

3.2.1 Lang (Star Manufacturer)
Model: 130F-440VM
Serial #: Unknown

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, service, and test the Deep Fat Fryers identified in section 3.0 using References 2.1 through 2.3 as guidance. Accomplish inspection, servicing, and testing in accordance with IMO, SOLAS, USCG, and the manufacturer’s requirements.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Temporarily remove access covers and panels to the Deep Fat Fryers to allow for inspection. Upon completion and acceptance of all work, reinstall access covers and panels.

7.5 Conduct an annual inspection of the Deep Fat Fryers to meet the manufacturer’s design, installation, maintenance instructions and service bulletin requirements. The inspection includes/verifies:

7.5.1 The extinguishing system and the protected equipment have not been modified or relocated.

7.5.2 Record the type of Deep Fat Fryer oil being used and its flashpoint.

7.5.3 The flashpoint of the Deep Fat Fryer oil meets the manufacturer’s requirements.

7.5.4 The Deep Fat Fryer is fitted with a primary and backup thermostat with an alarm to alert the operator in the event of failure of either thermostat.

7.5.5 The Deep Fat Fryer is fitted with a “shunt trip” bypass to secure the power in the event of over temperature in the cooking oil.

7.5.6 The Deep Fat Fryer shows no physical damage, leakage or condition that might prevent safe operation.

7.5.7 The controls and indicating lights are functioning properly and properly marked.

7.5.8 Electrical breakers are properly marked.

7.5.9 When the inspection reveals any deficiency a condition report is to be submitted to the OMT REP with the recommended repair and parts required.

7.6 Conduct annual maintenance of the Deep Fat Fryers to meet the manufacturer’s design, installation, maintenance instructions and service bulletin requirements using References 2.1 through 2.3 as guidance. The maintenance includes/verifies:

7.6.1 Inspect and service the systems including mechanical parts, electric parts and physical condition.

7.6.2 Remove the fryer oil/shortening. Clean the fryer completely. Provide and install new cooking oil using the manufacturer’s requirements as guidance. Provide details of the replacement cooking oil to the OMT REP.

7.7 Conduct annual testing of the Deep Fat Fryers to meet the manufacturer’s design, installation, maintenance instructions and service bulletin requirements. The testing includes/verifies:

7.7.1 Proper operation of the Deep Fat Fryer.

7.7.2 Proper operation of the fire protection and extinguishing system including control shutoff, high oil temperature cutout, high oil temperature alarm, and shunt trip features using References 2.1 through 2.3 as guidance. Ensure there is an alarm in the galley for indicating activation of the fire extinguishing system. Testing is to be coordinated with the OMT REP. No other work is to take place in the galley during this testing. Note that the flash point for cooking oil decreases with age. The cooking oil is to be changed prior to testing and the fire extinguishing system must be operational during testing.

WARNING: The flash point for cooking oil decreases with age. The cooking oil is to be changed prior to testing and the fire extinguishing system is to be operational during testing.

7.7.3 Accomplish testing in the presence of the OMT REP. Accomplish testing in the following sequence:

a. Use a calibrated pyrometer when performing tests. Test pyrometer (measure boiling water at 212°F) prior to the test.

b. Test the fire control shutoff (total loss of power to the Deep Fat Fryer upon release of the extinguishing system - to be simulated).

c. Test the operating thermostat for correct calibration using References 2.1 through 2.3 as guidance. The temperature should be within the manufacturer’s allowable range of the set temperature. The average temperature should only be off by 340-360˚F (171-182°C). If temperature is off, then a 1/16” flat blade screwdriver with a 2” shaft is required to make adjustments on the thermostat.

d. Test the over-temperature thermostat (power secured to heating elements, visual and audible alarms activate at the manufacturer’s setpoint).

e. Test the “shunt trip” bypass (total loss of power at the manufacturer’s setpoint). Do not test the “shunt trip” bypass until the operating thermostat, and over-temperature thermostat have first been successfully tested.

NOTE: Do not allow the oil temperature to exceed 475˚F (NFPA 96, Chapter 12.2).

f. Continuously monitor oil temperature throughout the test. Upon completion, allow the fryer to cool down. Restore power to the unit by resetting the “shunt trip” circuit breaker.

7.8 Maintain workspaces in a clean condition during and upon completion of the work requirements. Prevent contamination of any adjacent equipment and spaces.

7.9 Upon completion of all inspections, maintenance, servicing, and testing, return the Deep Fat Fryers to a ready for service condition.

7.10 Reports

7.10.1 When inspections, maintenance, servicing and testing reveal any deficiency, a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.10.2 Upon completion of all inspections, maintenance, servicing, and testing, prepare and submit a typewritten Service Report documenting the final “as released” condition of all Deep Fat Fryers affirming they are safe for use. Submit one electronic copy in Portable Document Format (PDF).

7.10.3 All reports and checklists are to be completed and signed by the person who carried out the inspections, maintenance, servicing, and testing and are to be countersigned by the Company's representative.

7.11 Painting: None additional

7.12 Manufacturer’s Representative

7.12.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the deep fat fryer identified in section 3.0 ensuring compliance with the manufacturer’s standards and performance specifications.

7.12.2 Possible Source

The Source
502 Rotary Street
Hampton, VA 23661
Phone: 757-825-1400 Toll Free: 1-800-497-2144
Email: sales@thesource2000.com

7.12.3 Provide the OMT REP with a signed letter of qualification from the Original Equipment Manufacturer and certificate of training for the field service representative for the equipment listed in section 3.0 prior to the start of any work.

7.13 Preparation of Drawings: None additional

8.0 GENERAL REQUIREMENTS: None additional

1.0 ABSTRACT

This work item describes the requirements to conduct an annual inspection, test and certification of the ships, material handling equipment (MHE) - articulated boom lifts.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSUP Tech Manual 0530-LP-642-6711, NAVSUP Publication 538; Management of Material Handling Equipment (MHE) and Shipboard Mobile Support Equipment (SMSE)

2.1.2 Tech Manual TG810-AJ-MMC-010, Boom Lift (Model: JLG 600S)

2.2 Enclosure

2.2.1 NAVSUP Publication 538, Fig. 8-12, MHE Initial Receipt Inspection Form

2.2.2 JLG, Annual Machine Inspection Report

2.2.3 MHE Safety Certification Form

3.0 ITEM LOCATION/DESCRIPTION/QUANTITY

3.1 Location

3.1.1 MHE Stowage, A5-89-4

3.2 Description/Quantity

Mfg	Model	Type	Platform Height (ft)	Power	Truck Wt No Load (lbs)	Qty
JLG	600S	Articulated Boom lift	60	Diesel	21,425 (2WS)	1

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 A minimum of two people (one on the ground/deck and the other in the basket) are required at all times when operating an aerial work platform.

5.3 Charging, testing and maintenance shall not be performed in magazines or other areas/spaces where ammunition and explosives are present.

5.4 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Remove and replace all interferences, rig and unrig, stage and unstage, make all disassemblies and subsequent reassemblies. Provide and operate equipment, and supply all services and assistance to accomplish the thorough examination and testing of the ships Material Handling Equipment (MHE) using References 2.1.1 through 2.1.2 as guidance.

7.2 With assistance from the Chief Engineer coordinate the movements, inspection and testing of MHE to avoid damage or injury during the course of this work item. The ship’s crew will operate all MHE while aboard ship.

7.3 Coordinate the inspection and testing of equipment with the OMT Rep and Chief Engineer to ensure the vessel is aware of equipment status at all times until the work is completed.

7.4 Only competent, licensed and trained Operators in compliance with 29 CFR § 1910.178(l)(6) shall operate the MHE while ashore. Use good operating practices to prevent accidents. It should be anticipated that some features of the MHE may not operate correctly upon initial receipt. Careful attention should be practiced by the Operators to maintain control of the vehicle at all times.

7.5 Conduct an inventory of the ships MHE listed in paragraph 3.2. Identify and record the following data for each onto a MHE Initial Receipt Inspection Form, Enclosure 2.1.1.

7.5.1 Vessel Name:

7.5.2 MSC Vehicle No:

7.5.3 Mfg:

7.5.4 Serial No:

7.5.5 Safe Working Load (lbs):

7.5.6 Designation:

7.5.7 Load Test Expiration (date):

7.5.8 Hourmeter reading:

NOTE: Check fuel system for leaks prior to any service or maintenance activity.

7.6 Conduct periodic inspection and testing of the MHE, using References 2.1.1 through 2.1.2 as guidance, to verify proper operation and adjustment. The inspection shall determine whether conditions found constitute a hazard and whether repairs or disassembly is required for additional inspection. Record the results for each onto a MHE Initial Receipt Inspection Form, Enclosures 2.2.1, and submit to the OMT REP. The examination shall include the following items:

7.6.1 Confirm the MHE is free of unauthorized modifications or additions.

7.6.2 Tires and Rims. Inspect tires for excessive wear, damage and proper pressure. Remove foreign material from tire treads. Reject tires for illegible or missing markings or labels. Reject pneumatic tires when the tire tread has worn down to the tread wear mark or if fabric is exposed through the sidewall. Inspect the rims for dents, bends, and cracks. Confirm wheel rim nuts are torqued properly.

7.6.3 Engine Oil and Fluid Levels. Always inspect floor/deck under MHE for any fluid puddles. Check:

a. Engine oil,

b. Hydraulic oil,

c. Transmission oil,

d. Brake fluid,

e. Radiator coolant. CAUTION: Do not check radiator coolant level when engine is hot.

f. Record if any levels are low.

7.6.4 Belts. Inspect engine belts for cracks, wear, damage, nicks or cuts, and proper tension.

7.6.5 Battery. Inspect battery cables for damage, cuts and abrasions. Verify cables are securely fastened to connector lugs and are free of corrosion, verdigris, arcing, pitting, exposed conductor material, and loose connections.

7.6.6 Electric trucks have color coded battery indicator power band indicating remaining charge level. Charge battery when indicator drops into yellow zone.

WARNING: For internal combustion start batteries, do not jump start any battery with low fluid level.

7.6.7 Access Covers. Inspect all access covers (e.g., battery or engine) for loose, missing, broken, or corroded covers. Ensure latches snugly secure covers when fastened.

7.6.8 Fuel System (Fuel-Powered MHE only). Visually inspect the entire fuel system assembly for any leaks while the MHE is operating. The MHE shall be rejected if any fuel leaks are discovered. Where accessible, inspect the fuel tank or gas cylinder for leakage, denting, bulging, corrosion, pitting, gouges not exposed to fire, or evidence of rough usage. Valves are protected from physical damage.

7.6.9 Neutral Start Switch (Fuel-Powered MHE only). Test the neutral start switch by attempting to start the engine with the directional control lever in either the forward or reverse position. If engine starts, the MHE shall be rejected.

7.6.10 Machine Controls. Check that all controls function properly at the platform and ground control station. Demonstrate speed(s), smoothness and limits of motion for the lift, swing, drive, telescope, platform rotation and leveling, emergency lowering, etc. controls using References 2.1.1 through 2.1.2 as guidance.

7.6.11 Indicator Lights, Fuses, and Relays. Turn key to ON position. All warning light and indicator lights will light up for two seconds (start check) when system power is turned ON. Check all indicator lights for correct operation. If any indicator lights do not operate correctly, check the fuses.

7.6.12 Engine. Start engine and check:

a. Unusual Engine Noises. Should any unusual noises be noted with the engine running, turn off MHE, reject and discontinue this check.

b. Idle Speed

c. Governed Speed

7.6.13 Lights. Check that the headlights, brake lights, and any other installed lights are working. All lights must operate properly for night work.

7.6.14 Horn. Depress the horn push button to verify that the horn is operating properly.

7.6.15 Verify back-up alarm operates.

7.6.16 Turntable Assembly Check. Confirm the turntable bearing, swing drive and gear are free of damage, or excess wear and properly lubricated.

7.6.17 Boom Assembly Check. Confirm the boom assembly is free of damage, distortion or excess wear. Examine all fasteners, bearings, wear pads, sheaves, pins, chains, cables, hoses and hydraulic cylinders.

7.6.18 Platform Assembly Check. Examine the platform, mounting hardware, gate, latches, guardrails, floor and lanyard anchorage points ensuring all are proper, secure and undamaged.

7.6.19 Hoses and cables. Inspect all hoses for cracked coverings, wear, bulges or leaks. Verify all fittings are free of cracks or leaks. Inspect cables for breaks, damage or chafing.

7.6.20 Transmission/Clutch. Verify that the transmission/clutch operates smoothly with no unusual noises.

7.6.21 Directional Controls. Shift directional controls into forward, neutral and reverse directions to verify the MHE operates properly and smoothly. Ensure steering operation functions smoothly.

7.6.22 Brake System Check.

a. Check the operation of the Parking Brake.

b. Visually inspect that no fluid is leaking from the brake system.

c. Check the Service Brakes. Check brake lining and parts of the brake assembly for wear or damage. Verify they stop the MHE smoothly and evenly without pulling or binding.

d. Where applicable, check the dead-man brake or travel control disconnect device for proper operation.

7.6.23 Gauges/Meters. Where applicable, inspect the following:

a. Warning Indicators. Verify that all warning and alarm devices, such as horns, bells, lights, etc., fitted on MHE are operational. With the engine running at normal operating temperature, check the oil pressure gauge for normal operating pressure. If any warning indicator is inoperable or signals a malfunction, the MHE shall be rejected until repaired.

b. Coolant Temperature Gauge. With the engine running at normal operating temperature, check that the gauge is indicating within the proper indicating range.

c. Fuel Gauge. Check the fuel gauge for proper reading. Electric powered types should be in "green" power range.

d. Voltmeter/Ammeter. With the engine running, check the voltmeter/ammeter to verify that it’s in the green range when the engine is running at governed speed.

e. Hourmeter. Verify that the hourmeter is registering while the engine is running.

f. Load Sensors. Verify proper operation. Adjust accordingly.

7.6.24 Fire Extinguisher. When equipped, visually inspect the extinguisher cylinder for dents. Check that the gauge is registering in the green (if so equipped) and check that the wire seal has not been broken. Verify periodic checks are current. Check nozzle and hose for defects. Reject extinguisher if not serviceable.

7.6.25 Operator Restraint System. If MHE is equipped with an operator restraint system (e.g., harness, seat belt, hip restraint) it shall be inspected to verify that it operates correctly and does not exhibit any evidence of wear or damage.

7.6.26 Visually inspect for spark emission in a dark location. The MHE shall be rejected if there is any spark emission.

a. For fuel-powered MHE, inspect with the engine on and in neutral. Special attention should be given to the entire exhaust systems (pipes), the ignition wiring, the alternator/generator housing, and the area under the MHE where the engine is located.

b. For electric-powered MHE, inspect while operating each electrical motor (e.g., drive motor, hydraulic, etc.). The drive wheels may be raised off the deck/floor to test the drive motor. Special attention should be given to the area around the covers and under the MHE where the wiring, contactors, and motors are located.

7.6.27 Structural Cracks/Broken Weldments. Inspect all external weldments for structural cracks or defects. Reject MHE until repaired or replaced.

7.6.28 Mandatory Markings. Verify all function, instruction, caution and warning labels, markings or placards are clearly and properly marked including:

a. safe working load (SWL) and vehicle weight (VW) on both sides, in view of operator,

b. operator controls,

c. manufacturer’s nameplate/label,

d. accredited laboratory (UL, FM) certification, and

e. Aerial Work Platform (AWP) Safety Certification

f. Reject if the above markings are missing, illegible, expired or incorrect. All required markings that are rejected shall be recorded on the MHE Inspection Form, Enclosure 2.1.1, but is not a cause for removal from service.

7.6.29 Immediately report any adverse conditions noted to the OMT REP. The report is to include photographs of each MHE for record purposes.

7.7 Conduct maintenance on the MHE in accordance with the manufacturers design, installation, maintenance instructions and service bulletins using References 2.1.1 and 2.1.2 as guidance. The maintenance shall include/verify:

7.7.1 Replace the Hydraulic Oil

7.7.2 Replace the Hydraulic Oil Filter

7.7.3 Replace the Hydraulic Tank Breather

7.7.4 Replace the Coolant(**)

7.7.5 Replace the Engine Oil(**)

7.7.6 Replace the Engine Oil Filter(**)

7.7.7 Replace the Air Filter(**)

7.7.8 Replace the Fuel Water Separator, Diesel(**)

7.7.9 Replace the In-Line Fuel Filter(**)

7.7.10 Replace the Transmission Oil(**)

7.7.11 Replace the Transmission Oil Filter(**)

7.7.12 Replace the Brake Fluid

7.7.13 Replace Drive Hub oil

7.7.14 Adjust the brakes in accordance with the manufacturer's recommendations

7.7.15 Inspect fuse blocks, fuses and connectors for damage and corrosion

7.7.16 Verify and adjust all switches for proper operation (brake lights, seats, seat brake, parking brake, direction, start, e-stops, deadman, limits of motion, etc.)

7.7.17 Remove, clean and adjust the Fuel Injectors(**)

7.7.18 Visually examine all hydraulic hoses/fittings for damage, wear, age, cracking, leaks, and kinks

7.7.19 Clean battery and cable terminations

7.7.20 Clean and lubricate all moving parts where lubrication is specified using References 2.1.2 as guidance, including:

a. Boom Assembly

b. Turntable

c. Chassis

d. Platform

e. Brake Master Cylinder Rod End Pin

f. Steering

g. Pedals, Levers, Hinges, Linkages and Hood Latch

NOTE: (**) - Fuel powered MHE

7.8 Establish conditions for testing of the MHE.

7.8.1 Install barricades and warning signs to prevent access to areas of active testing. Erect and maintain control lines, warning lines, railings or similar barriers to mark the boundaries of the hazard areas. Brake test course shall be level, clean asphalt, brushed concrete, level non-skid, or equivalent surfaces, and of adequate length to permit safe conduct of the test.

7.8.2 Weight Test Loads. Only known values of test weights shall be used for testing. Dynamometers, or other recording equipment, are not permitted in lieu of dead weights. The assembled test load, including rigging, shall be accurate within +5/-0 percent of the nominal test load. The weights shall be measured using calibrated equipment with a minimum accuracy of ± 2 percent (of scale) traceable to the National Institute of Standards and Technology.

7.8.3 Marking of Test Loads. The test weight(s) shall be marked (e.g., steel stamping, etching, engraving, raised metal deposit, or stenciling) with a unique identification number and the weight in pounds.

7.8.4 Conduct a functional test, under no load conditions, evaluating the performance of each individual MHE. Record the results for each onto a MHE Safety Certification Form, Enclosure 2.2.3.

7.9 Conduct a load test using Reference 2.1.1 through 2.1.2 as guidance.

7.9.1 Position the lift truck at rest on a smooth, level surface with the mast vertical. Secure the truck as necessary to prevent overturn. Perform testing with the hydraulic system at normal operating temperatures.

7.9.2 Position a test load on the platform such that the center of mass of the test load is on the platform's centerline and at the rated load center. Unless restricted by the manufacturer, the test load shall be 150% of the MHE's SWL as a minimum; per NAVFAC P-300. Ensure the load is centered laterally. Extend the boom to its max. rated horizontal reach, raise the platform at least 12 inches off the deck/floor, and hold for 2 minutes. After 2 minutes, measure any downward and tilt drift, as applicable.

7.9.3 Swing the boom lateral thru its full range of motion monitoring its performance. Return the boom to centerline and the load to deck/floor.

7.9.4 With the boom centerline, platform centered and rated load on the floor, raise the platform to the maximum working height. After 2 minutes, measure any downward and tilt drift, as applicable.

7.9.5 Remove the load and restraints and visually inspect the platform, boom, and other structural components for cracks, elongation, permanent deformation, fractures, or failures. Record the results for each onto a MHE Safety Certification Form, Enclosure 2.2.3.

7.10 Conduct a drive speed and brake operation test using References 2.1.1 through 2.1.2 as guidance.

7.10.1 Load platform with a rated load (100% of the MHE's SWL). The test load shall bear against the floor and centered laterally.

7.10.2 Determine the manufacturer's maximum rated speed of the MHE (with boom stowed and raised/extended).

7.10.3 Test and record the Drive Speed with boom in the stowed position. This may be accomplished by determining the time it takes the MHE to pass through a 40-foot course as follows: Measure a 40-foot straight course. Transverse the course at maximum speed from a running start. Bring the machine to top drive speed before reaching the start line. Using an appropriate means to measure time (e.g., stop watch), note the time it takes for a reference point on the MHE to move from the start of the course to the end. Alternately, the maximum speed may be measured using a device such as a radar/laser gun.

7.10.4 Test and record the Drive Speed with boom raised or in the extended position similarly to 7.13.3.

7.10.5 With the MHE at maximum speed with boom stowed and after passing through the 40-foot course, release the drive joystick to apply the service brakes. Record the stopping distance measured from the point of brake application.

7.10.6 MHE shall be rejected if:

a. The recorded distance exceeds the maximum allowable stopping distance

b. The brakes do not gradually engage (grabbing) causing sudden or immediate stops.

c. Brake fluid is leaking, exceeding manufacturer’s specifications or standard industrial maintenance practices.

d. Any component fails.

7.11 Conduct a parking brake test using References 2.1.1 through 2.1.2 as guidance.

7.11.1 Test the parking brake by engaging the parking brake on the unloaded MHE. Attempt to drive the MHE forward by applying a moderate amount of power to the MHE. If the MHE moves, adjust the parking brake, and repeat the parking brake test. The MHE shall be rejected if proper adjustments cannot be achieved.

7.12 Testing is to be coordinated with the OMT Rep to allow for observation if deemed necessary. Demonstrate current calibration of all test equipment prior to any testing.

7.13 MHE Safety Certification Marking. Equipment passing periodic testing must have the previous weight test tags/markings removed and the following information either stenciled or labeled, per Reference 2.1.1, the testing activity's name and the expiration date (month/year). Expiration indicates the date the MHE should be scheduled for its next 12 month periodic test.

ANNUAL SAFETY INSPECTION / WEIGHT TEST ACCOMPLISHED BY:
IAW NAVFAC P300 (Section 4-1.11.5), MFG TECH MAN & ANSI A92.5 **EXPIRES: ________________________________**

7.14 Reports

7.14.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.14.2 Submit a report that includes “as released” condition of the sea growth and marine protection. Submit one electronic copy in Portable Document Format (PDF).

7.14.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.14.4 Prepare and submit Inspection, Test and Certification forms/reports (Enclosures 2.2.1 through 2.2.3) for each MHE detailing the results of their examination, test and certification. All reports are to be signed by the attending

Technical Representative who performed the exam and testing. Submit reports to the OMT REP in Adobe PDF format.

7.15 Painting/Lagging

7.15.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.15.2 Renew lagging where damaged or disturbed.

7.16 Manufacturer’s Representative

7.16.1 Provide the services of a Manufacturer’s Authorized Service Provider to inspect, test and certify the MHE listed in paragraph 3.2. Qualifications and certificates for the company and field service rep are to be submitted to the OMT REP for review and approval prior to the start of any work. Possible GENIE/JLG sources can be found in Enclosure 2.2.3.

7.17 Preparation of Drawings

7.17.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 - NAVSUP Publication 538, Fig. 8-12, MHE Initial Receipt Inspection Form

SHIPBOARD UNIT ☐	SHOREBASED UNIT ☐	INSPECTION DATE
MAKE	MHE TYPE	BLADE LENGTH
MODEL	MOTOR TYPE	FUEL TYPE
YEAR	TIRE TYPE	HOUR METER READING
SERIAL NO.	CAPACITY	EQUIPMENT COST CODE

ITEM No.	INSPECTION POINT	ACCEPT	REJECT	N/A
1	Manufacturer’s data/identification plate			
2	Manufacturer’s “STRUCTURALLY WEIGHT TESTED” marking (ammunition or shipboard handling)			
3	Accredited laboratory certification plate (e.g., UL, FM)			
4	Alphabetical designation (EE, DS, etc.)			
5	Operational controls correctly labeled/marked			
6	Battery identification plate or marking			
7	Fuel type marking (e.g., DIESEL FUEL ONLY)			
8	Warning decals and labels			
9	Safe working load marking (sides and mast) (e.g., SWL 6000 LBS)			
10	Vehicle weight marking (sides and mast) (e.g., VW 10,000 LBS)			
11	Registration number (e.g., 13-20000)			
12	SLEP data plate			
13	“LEAD AND CHROMATE FREE PAINT” marking			
14	“LIFT HERE” marking			
15	“TIEDOWN” marking			
16	Tire pressure marking (pneumatic only)			
17	Fuel cap (color coded)			
18	“SHIPBOARD USE APPROVED” marking			
19	Blue diagonal striping (type EX only)			
20	Instruction plates			
21	Grease fittings (on components) fully filled and accessible			

ENCLOSURE 2.2.1 - NAVSUP Publication 538, Fig. 8-12, MHE Initial Receipt Inspection Form

ITEM No.	INSPECTION POINT	ACCEPT	REJECT	N/A
22	Correct fluid levels (brake, radiator, transmission, hydraulic, etc.)			
23	Paint is smooth and adheres well			
24	No evidence of loose, missing or broken hardware			
25	No evidence of bent, cracked or worn accessories			
26	No evidence of missing covers, panels, loose or poor fit			
27	Overhead guard and load backrest (cargo guard) not damaged			
28	Forks are straight within 1% of length			
29	Hydraulic cylinders: no weld cracks, leaks or other damage			
30	Neoprene hydraulic lines: free of paint, routed correctly and secured			
31	Towing devices, hitch pins, tow chains, etc. are attached			
32	Battery cables: secure, no cuts/abrasions, and protected by non-conductive covers			
33	Battery mounts and hold downs are secure			
34	Wiring harness is correct with no frayed, brittle or crimped connections (soldered only)			
35	Tires are correct type, size and thread (no delamination shall be accepted)			
36	Wheel lugs are torqued as specified			
37	Check mast rollers and locks			
38	Fork heel pins and locks function			
39	Lift chains and anchor pins are secure and functional			
40	Operator pedals are equipped with rubber pads or non-slip coating			
41	Operator restraint system/seat belt equipped and functional			
42	Dash gauges are correct and marked			
43	Key ignition switch (verify key numbers match contract numbers)			
44	Start MHE and note any unusual noises			
45	Weight gauge is legible and operable			

ENCLOSURE 2.2.1 - NAVSUP Publication 538, Fig. 8-12, MHE Initial Receipt Inspection Form

ITEM No.	INSPECTION POINT	ACCEPT	REJECT	N/A
46	Steering wheel is smooth and little free play			
47	Verify all gauges function			
48	Verify hourmeter operates properly			
49	Verify horn operates			
50	Verify back-up alarm operates			
51	Verify all lights operate: brake, spot, battle/blackout, etc.			
52	No evidence of steering play or loose suspension			
53	Bring to full operating temperature (fuel power only) and check for leaks			
54	Check for arcing and operation of contactors, traction, steer and hydraulic pump motors (electric MHE only)			
55	Battery disconnect switch is operable			
56	Check accelerator and inching/declutching pedal operation			
57	Check brake operation			
58	Parking brake functions			
59	Check emergency stop operation			
60	Check for smoothness when raising mast to full height			
61	Verify hose reel functions smoothly			
62	Check tilt and side shift (rotation some models) operation and smoothness			
63	Check acceleration (all forward and reverse gears)			
64	Check braking (at speed) if smooth and straight			
65	Check transmission (forward and reverse)			
66	Clutch adjustment, engagement and operation			
67	Check for exhaust leaks			
68	Check applicable steering modes (2 wheel, 4 wheel, crab)			
69	Verify the turning radius/tracking			
70	For electric trucks, the presence of static conductive straps or tires			

ENCLOSURE 2.2.1 - NAVSUP Publication 538, Fig. 8-12, MHE Initial Receipt Inspection Form

INSPECTION ACTIVITY	INSPECTOR NAME	DATE (MMDDYY)
	PRINT:	
	SIGNATURE:	

USS Hershel Williams		
(ESB 4)		
HABITABILITY OUTFITTING AND FURNISHINGS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0666	CATEGORY "A"	2026-01-30
ESB-CCSI-INSPECT, TEST AND CERTIFY MATERIAL HANDLING EQUIPMENT - AERIAL WORK PLATFORM (1YR)		Manovich, David

ENCLOSURE 2.2.2 - JLG, Annual Machine Inspection Report

JLG An Oshkosh Corporation Company

JLG Industries, Inc.
1 JLG Drive
McConnellsburg, PA 17233-9533

BOOM LIFT and TRAILER MOUNTED BOOM LIFTS
Annual Machine Inspection Report

Serial Number: ____________________
Machine Model: ____________________
Hourmeter Reading: ____________________

Previous Inspection Date ____________________

JLG Account Holder Name & Address
Customer No. : ____________________

Product Owner/User Name & Address
☐ **Owner** ☐ **User**

ANNUAL MACHINE .... The Owner must perform an Annual Machine Inspection of this machine no later than 13 months from the date of the prior Annual Machine Inspection. This Annual Machine Inspection is to be performed by a mechanic qualified on the specific make and model of aerial work platform.

Check each item below. (Refer to Operators & Safety, Service & Maintenance Manuals for specific information regarding inspection procedures and criteria.) Indicate in the appropriate space as each item has been performed. If the item is found to be not acceptable, describe each discrepancy in the comments space at the bottom of the form. Use additional paper if necessary. Immediate action must be taken to correct all discrepancies. The Owner shall not place the machine in service until all discrepancies have been corrected.

Y=Yes (Passed) N=No (Failed) C=Corrected NA=Not Applicable	Y	N	C	NA
FUNCTIONS & CONTROLS				
1. All controls return to neutral/off position when released.				
2. Detents properly lock controls in neutral/off position. Check condition of control enclosures & protective boots/guards.				
3. Footswitch operates properly. *(shuts off function when released)*				
4. Emergency stop switches at the ground & platform control stations arrest all platform movement.				
5. All function & speed cut-outs operate properly.				
6. Manual descent system and/or auxilliary power system operates properly.				
7. Tower boom sychronization and sequence system operates properly. *(HA & A models)*				
8. Capacity indicator system operates properly.				
9. Brakes operate properly. *(swing & drive)*				
10. Machine controls operate properly at platform & ground control stations. *(lift, swing, drive, telescope, etc.)*				
PLATFORM ASSEMBLY				
1. Platform installed & secure. Mounting hardware free of damage & secure. Gate opens & latches properly.				
2. Platform guardrails and floor in place, secure and undamaged.				
3. Lanyard anchorage points secure, undamaged & labeled.				
BOOM ASSEMBLY				
1. Boom sections free of damage, distortion & excessive wear.				
2. All nuts, bolts, pins, shafts, shields, bearings, wear pads, locking devices, checked for proper installation, no excessive wear, cracks or distortion.				
3. Inspect all sheaves, sheave pins and bearings for proper installation, secure, no excessive wear, cracks or distortion.				
4. Boom chains and cables are free of damage, properly installed, properly torqued & properly lubricated.				
5. Powertrack is free of damage, distortion & excessive wear. Cables/hoses are properly routed, no chafing or leaks.				

Y=Yes (Passed) N=No (Failed) C=Corrected NA=Not Applicable	Y	N	C	NA
BOOM ASSEMBLY (continued)				
6. Inspect all cylinder pins, pivot pins, attaching & retention hardware for damage, distortion & excessive wear.				
TURNTABLE				
1. Hood doors open & latch properly.				
2. Hood props in place, secure and functioning properly.				
3. Turntable bearing, swing drive & gear secure and undamaged, properly lubricated. No missing bearing bolts or signs of looseness.				
4. Check swing bearing bolt tolerance per the Service & Maintenance Manual.				
5. Turntable lock secure, undamaged & operable. (If equipped)				
CHASSIS				
1. Wheel rim nuts torqued properly.				
2. Proper tires installed.				
3. Tires free of gouges and excessive wear, no cords showing. If pneumatic, properly inflated. Tire bead properly seated around rim.				
4. Oscillating axle operates properly, locks in place when turntable swings to side.				
5. Outrigger/extendible axles operate properly. Interlocks function properly.				
POWER SYSTEM				
1. Engine idle, throttle & RPM set properly.				
2. Battery fluid levels correct.				
3. Battery charger scrolls through diagnostics when plugged in.				
4. Batteries accept charge.				
5. Air & fuel filter clean.				
6. Fuel, coolant and engine oil level correct.				
7. Fuel cap tight & vent open.				
8. Exhaust system free of leaks.				
HYDRAULIC/ELECTRICAL SYSTEM				
1. All cylinders free of damage, no evidence of leaks.				

Y=Yes (Passed) N=No (Failed) C=Corrected NA=Not Applicable	Y	N	C	NA
HYDRAULIC/ELECTRICAL SYSTEM (continued)				
2. All areas around hydraulic components (pump, oil lines, reservoir) free of oil, no evidence of leaks.				
3. Hydraulic oil free of contaminants, hydraulic filter clean.				
4. Hydraulic level in tank & torque hubs correct.				
5. Hydraulic tank cap tight & vent open.				
6. All hydraulic fittings & lines secure, free of damage, chafing & leaks.				
7. All electrical connections tight, no corrosion or abrasions.				
8. Instruments, switches, gauges, horn & lights operate properly.				
9. Pump & motor secure, undamaged & free of leaks.				
10. All hydraulic pressures properly adjusted.				
MANUALS & DECALS				
1. ANSI/SIA Manual of Responsibilities in manual storage box.				
2. Operators & Safety Manual in manual storage box.				
3. AEM Handbook in manual storage box.				
4. Capacity decals installed & legible at both platform & ground stations.				
5. All safety & instructional decals installed, secure & legible.				
GENERAL				
1. Lift is free of unauthorized modifications or additions.				
2. Paint and overall appearance.				
3. Applicable Safety Bulletins completed.				
4. Inspect general structural condition including all welds.				
5. Grease and lubricate per Service & Maintenance Manual.				
6. Drive & operate machine to test all machine functions.				
7. Record inspection date on "IMPORTANT" decal on frame.				
8. If ownership has changed, complete attached Owner Update form and return to JLG.				
Comments:				

The undersigned certifies that this machine has been inspected, per each area of inspection, and any and all discrepancies have been brought to the attention of the Owner/User, and that all discrepancies have been corrected prior to any further use of this machine.

JLG Account Holder: ____________ / ____________ / ____________ **Owner/User:** ____________ / ____________ / ____________
Authorized Signature Printed Signature Date Authorized Signature Printed Signature Date

Copy to JLG Account Holder; Copy to Owner/User; Owner Update Form send to JLG Industries, Inc. as required

Form No.- 3124166 - CGF483 - 3/31/2005

USS Hershel Williams		
(ESB 4)		
HABITABILITY OUTFITTING AND FURNISHINGS		CONTRACT NO. N68171-26-RN-003
ITEM NO. 0666	CATEGORY "A"	2026-01-30
ESB-CCSI-INSPECT, TEST AND CERTIFY MATERIAL HANDLING EQUIPMENT - AERIAL WORK PLATFORM (1YR)		Manovich, David

ENCLOSURE 2.2.3 - MHE Safety Certification Form

USN/MSC REGISTRATION NO.	MHE CLASS/LIFT CODE/TYPE	SWL	HOUR READER METER	DATE

TEST	PARAGRAPH	PROCEDURE	RESULTS					
MARKINGS	6-6.1	VISUAL INSPECTION:	ACCEPT		REJECT			
FUEL SYSTEM	6-6.2	LEAK INSPECTION:	ACCEPT		REJECT		N/A	
NEUTRAL START SWITCH (Fueled MHE)	6-6.3	OPERATIONAL INSPECTION:	ACCEPT		REJECT		N/A	
WARNING DEVICES (Powered MHE)	6-6.4	OPERATIONAL INSPECTION:	ACCEPT		REJECT		N/A	
SPARK EMISSION (Powered MHE)	6-6.5	VISUAL INSPECTION:	ACCEPT		REJECT		N/A	
STATIC DISCHARGE (Electric and HS/Ordnance Pallet Truck)	6-6.6	CONDUCTOR TYPE:	TIRES		STRAPS		N/A	
		RESISTANCE READING (OHMS):	LEFT		RIGHT		N/A	
			ACCEPT		REJECT		N/A	
CARRIAGE REACH & ROLLER (Reach and Tier)	6-6.7	OPERATIONAL INSPECTION:	ACCEPT		REJECT		N/A	
DRIVE CONTROL (Powered Pallet Trucks)	6-6.8	OPERATIONAL INSPECTION:	ACCEPT		REJECT		N/A	
BRAKE (Except Straddle Carriers and Manual Pallet Trucks)	6-6.9.1	TRAVEL DISTANCE (TABLE 6-2): FEET	ACCEPT		REJECT		N/A	
		PEDAL FEEL:	ACCEPT		REJECT		N/A	
	6-6.9.2	PARKING BRAKE:	ACCEPT		REJECT		N/A	
	6-6.9.3	SAFETY BRAKING/DISCONNECT SYSTEM:	ACCEPT		REJECT		N/A	
BRAKE (Straddle Carrier)	6-6.10	OPERATIONAL INSPECTION:	ACCEPT		REJECT		N/A	
OPERATIONAL WEIGHT TEST (Forklift Trucks)	6-6.11	100% RATED LOAD HELD FOR 2 MINUTES:	ACCEPT		REJECT		N/A	

ENCLOSURE 2.2.3 - MHE Safety Certification Form

LOAD TEST (Aerial Work Platform)	8-7.6	150% RATED LOAD HELD FOR 2 MINUTES:	ACCEPT		REJECT		N/A	
KING PIN (Straddle Carrier)	6-6.13	VISUAL INSPECTION:	ACCEPT		REJECT		N/A	
LIFTING WEIGHT TEST (Straddle Carrier)	6-6.14	100% RATED LOAD:	ACCEPT		REJECT		N/A	
OVERLOAD WEIGHT TEST (Except Pallet Trucks and Straddle Carriers)	6-5.3	AFTER REPAIRS/MODIFICATIONS AFFECTING LOAD-BEARING COMPONENTS	YES		NO		N/A	
	6-7	100% LOAD HELD FOR 2 MINUTES:	ACCEPT		REJECT		N/A	
		150% LOAD HELD FOR 2 MINUTES:	ACCEPT		REJECT		N/A	
LIFT EYES (Except Straddle Carriers)	6-5.4	AFTER REPAIR/REPLACEMENT	YES		NO		N/A	
	6-8	150% WEIGHT OF MHE HELD FOR 2 MINUTES:	ACCEPT		REJECT		N/A	

THIS IS TO CERTIFY THAT THE ABOVE MHE WAS INSPECTED AND TESTED IN ACCORDANCE WITH NAVSEA SW023-AH-WHM-010.

MECHANIC/MAINTENANCE PROVIDER:

PRINT FULL NAME	SIGNATURE	DATE

CERTIFIED BY:

PRINT FULL NAME	SIGNATURE	DATE

THIS 18 MONTH CERTIFICATION EXPIRES ON: ____________

MONTH/YEAR

USSS Hershel Williams		
(ESB 3 4)		
HVAC		**CONTRACT NO. N68171-26-RN-003**
ITEM NO. 0851A	**CATEGORY "A"**	**2026-02-01**
ESB-CCSI-GALLEY VENTILATION SYSTEM AND GAYLORD HOOD CLEANING (2.5 YR)		**Port Engineer's Name**

1.0 ABSTRACT

The purpose of this item is to inspect, clean and test the Galley Ventilation Systems and Gaylord Hoods.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 MSC Drawing No. 512-8499376, HVAC Diagram FWD Accommodations

2.1.2 MSC Drawing No. 512-8468337, HVAC Diagram Accommodations, Aft House

2.1.3 Tech Manual T6161-EJ-FSE-010, Hood Assembly Galley Operating and Maintenance Instructions

2.1.4 Tech Manual T6161-LP-MMC-010, Vapor Hood, Exhaust Ventilator and Command Control Center; Installation, Operation and Maintenance Manual

2.1.5 Tech Manual T6161-LG-FSE-010, Galley Hood Fire Suppression System

2.1.6 NFPA 96 Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations (Available Commercially)

2.2 Enclosure

2.2.1 Depth Gauge Comb

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

3.1.2 Aft Galley (A-47-0)

3.1.3 Fwd Galley (A4-92-1)

3.1.4 Quantity: Ventilation Systems for spaces identified in 3.1.1 and two Gaylord hoods identified in paragraph 3.2.

3.2 Item Description/Manufacturer's Data: Gaylord Hood Manufacturer: Gaylord

3.2.1 One Gaylord Command Center, MFG: Gaylord, Model: C-5000

3.2.2 One Vapor Hood, MFG: Gaylord, Model: VH2-W-35.984

3.2.3 One Exhaust Ventilator, MFG: Gaylord, Model: NAD-48

3.2.4 One Exhaust Ventilator, MFG: Gaylord, Model: NBDL-DS-CL-CA-62.008

3.2.5 One Exhaust Ventilator, MFG: Gaylord, Model: NAC-24

3.2.6 One Exhaust Ventilator, MFG: Gaylord, Model: NAD-CA-48.032

3.2.7 One Exhaust Ventilator, MFG: Gaylord, Model: NAD-CA-6.937

3.2.8 One Fire Damper Control Switch, MFG: Gaylord, Model: N66L

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of

this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS

6.1 All items will be serviced and maintained in accordance with OEMs specifications and certification requirements as set forth by ABS and National Fire Protection Agency (NFPA).

7.0 STATEMENT OF WORK REQUIRED

7.1 Inspect, clean, and test the Galley Ventilation Systems and Gaylord Hoods identified in section 3.0 using References 2.1.1 through 2.1.6 and Enclosure 2.2.1 as guidance. Accomplish inspection, cleaning, and testing in accordance with IMO, SOLAS, and the Manufacturer’s requirements.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Drain and dispose of any liquid in accordance with all local, state, and federal laws.

7.4 Inspect overhead and bulkhead paneling required to be temporarily removed in the presence of the OMT REP. Submit an “as found” condition report.

7.5 Temporarily remove overhead and bulkhead paneling to gain access to the supply and exhaust ducting. Temporarily remove access covers and plates to the ventilation systems to allow inspection and cleaning. Existing openings will be used to the maximum extent possible. New access openings will be approved in advance by the OMT REP only. New openings in watertight ducting sections are not permitted. Upon completion and acceptance of all work, reinstall access covers and panels using new gaskets and sheet metal fasteners.

7.6 Conduct an annual inspection of the Galley ventilation and Gaylord Hoods in accordance with the manufacturer’s design, installation, maintenance instructions and service bulletins. The examination will include/verify:

7.6.1 The ventilation systems and Gaylord hoods have not been modified or relocated.

7.6.2 Record the position and settings of any dampers prior to cleaning.

7.6.3 With the assistance of ship’s force, conduct a hot detergent water wash of the Gaylord hoods.

Water Pressure:	40 psi minimum	Water Temperature:	140° F minimum
	80 psi maximum		180° F maximum

7.6.4 Demonstrate all features of the Gaylord Hoods using Reference 2.1.3 as guidance.

7.6.5 Inspect and test check Gaylord Hoods failsafe thermostats and damper control switches.

7.6.6 Prove satisfactory operation of gaylord hoods. Submit an “as found” condition report to document any discrepancies prior to cleaning.

7.7 Conduct annual maintenance on all Galley ventilation systems and Gaylord Hoods in accordance with the manufacturer’s design, installation, maintenance instructions, and service bulletins. The maintenance will include/verify:

7.7.1 Clean the range and fryer hoods, grease removal devices, fans, dampers, ducting and other appurtenances including troughs and drain lines integral to the hoods to remove all dirt, lint, loose scale, debris, grease and combustible contaminants.

7.7.2 Each system identified in section 3.0 is to be cleaned in its entirety, inclusive of fan, filters, cooling and heating coils, grease precipitators, diffusers, vents, hoods, all branches, exterior of fan motors, fan discharge to duct terminus and terminus screens. Assume that 1/3 of fasteners on fan access panels will require renewal.

7.7.3 Approved cleaning methods

a. Cryogenically clean ducting by means of dry ice blasting. Other modern methods of cleaning of the ducting may be considered upon review and approval by the OMT REP.

b. Where access allows, manual scraping and scrubbing followed by non-toxic solvent wash.

c. Use of pressure washers, steam cleaners and other wet wash methods are permissible as long as the ventilation ducting is proven watertight prior to washing and suitable means to collect wash water are identified and utilized.

d. Provide Safety Data Sheets (SDS) for any cleaning agents, degreaser solution or solvent proposed for use. These products will be non-toxic and suitable for the degreaser application.

7.7.4 Temporarily remove each re-useable type filter, blow clean with compressed air and wash with hot water containing an approved washing compound.

7.7.5 Where compressed air or other inert gas cleaning systems are utilized, downstream air/debris collectors will utilize HEPA filters and will be arranged to prevent contamination of the workspace.

7.7.6 Flammable solvents or cleaning aids will not be used.

7.7.7 Cleaning chemicals will not be used on fusible links or other detection devices.

7.7.8 Clean surfaces to remove all combustible contaminants to a minimum of .002” using Reference 2.1.6 and Enclosure 2.2.1 as guidance. Place a grease depth gauge comb on ducting to measure the depth of any remaining grease to verify cleanliness. Demonstrate final cleanliness to the OMT REP.

7.7.9 Final visual inspection must be conducted using a borescope or camera fitted to a probe. All sections of ducting that were cleaned must be inspected by OMT REP. Upon completion and acceptance of cleanliness by OMT REP, restore the fire extinguishing systems, ventilation system, cooking equipment, electrical switches and alarms, and vent access covers and plates to a ready for service condition.

7.8 Conduct annual testing of the Galley ventilation systems and Gaylord Hoods in accordance with the manufacturer’s design, installation, maintenance instructions and service bulletins. Coordinate testing with OMT REP. The testing will include/verify:

7.8.1 Proper operation of the Galley ventilation.

7.8.2 Check gaylord hood air velocity at throat of ventilator and check exhaust fans for belt tightness, belt alignment, and lubrication.

7.8.3 Demonstrate the airtight integrity of the access covers, openings and panels to the OMT REP. Upon approval by the OMT REP, reinstall duct insulation.

7.8.4 With assistance of ship’s force and OEM Rep, demonstrate proper operation of the Gaylord Hoods.

7.9 Maintain workspaces in a clean condition during and upon completion of the work requirements. Prevent contamination of any adjacent equipment and spaces.

7.10 Upon completion of all inspections, tests and repairs, return the Galley ventilation and Gaylord Hoods to a ready for service condition.

7.11 Reports

7.11.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.11.2 Submit a report that includes “as released” condition of the all Galley ventilation and Gaylord Hoods(s) affirming they are grease free and safe for use. Submit one electronic copy in Portable Document Format (PDF).

7.11.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.12 Painting/Lagging

7.12.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.12.2 Renew lagging where damaged or disturbed.

7.13 Manufacturer’s Representative

7.13.1 Persons performing all inspections, maintenance, service and testing of the Gaylord Hoods will be OEM authorized service technicians.

7.13.2 Possible Sources

EMS Ice, Inc.
PO Box 1671
Chesapeake, VA 23327
Office Phone - (757) 494-9903
Toll Free - (877) 236-7423

Innovative Vacuum Services, Inc.
20909 70th Ave West
Edmonds, WA 98026-7201
Phone: 206-783-3317
Toll-Free: 800-945-4081
www.innovac.com

7.13.3 Companies and persons performing maintenance and testing will have available the appropriate certificates, servicing manuals, service bulletins, correct tools, materials, and manufacturer’s replacement parts.

7.13.4 Provide the OMT REP with documentation of certification requested in paragraph 7.13.1 prior to the start of any work.

7.14 Preparation of Drawings

7.14.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 DEPTH GAUGE COMB

FOR GUIDANCE ONLY

ACCEPTABLE CLEANED SURFACE = .002” or less

DEPTH GAUGE COMB

1.0 ABSTRACT

The purpose of this item is to clean, polish, and inspect the propeller(s), rudder(s), sea chest grates, transducers and underwater hull while waterborne.

2.0 REFERENCES/ENCLOSURES

2.1 References

2.1.1 NAVSEA Drawing No. 830-8499361,Selected Record Drawing, Docking Plan

2.1.2 Tech Manual T9244-A1-MMC-010, Fixed Pitch Propeller and Shafting with Bearings (Volume I), Wartsila Drawing: F.P. Propeller, Page 1-29/30

2.1.3 Tech Manual S9243-CT-MMC-010, Shaft Seal, Stern Tube, Type MB

2.1.4 Becker Marine Systems Drawing No. A-0. 10058, Arrangement of Becker Rudder

2.1.5 Becker Marine Systems Drawing No. R-1. 630476, Mounting Instruction for Rudder

2.1.6 MSC Drawing. No. 801-8499371, General Arrangement

2.1.7 NAVSEA Tech Pub No. S9086-HP-STM-010, NSTM Chapter 245, Propellers and Propulsors

2.2 Enclosure

2.2.1 Blade and Hub Inspection Forms

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Location: Underwater Hull

a. LOA: 239.325 M

b. LBP: 233.215 M

c. Beam at DWL: 50 M

3.1.2 Quantity

a. Two Fixed-Pitch Propellers

b. Two Rudders

c. Nine Sea Chest Suction Grates

d. Two External Stern Tube Shaft Seals

e. Two Rope Guards

3.2 Item Description/Manufacturer’s Data

3.2.1 Propeller

a. Diameter: 7450 MM

b. Number of Blades: 4

c. Pitch: 5923 MM (mean)

d. Propeller Material: Nickel Aluminum Bronze

e. Propeller Weight: 24,013 KG

3.2.2 Rudder: See Reference 2.1.4 and 2.1.5 for details.

3.2.3 Sea Chest Suction Grates: See Reference 2.1.1 for details.

3.2.4 Stern Tube Shaft Seal

a. Stern Tube Shaft Seal Type MB, Wartsila Drawing H78690-01

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES

5.1 The contractor and all subcontractors, regardless of tier must consult the General Technical Requirements (GTR) to determine applicability to this work item. In performance of this work item, the contractor and all subcontractors regardless of tier must comply with the requirements of all applicable GTRs including but not limited to GTRs 1 through 7, 22, 23, 24, 28, and 29.

5.2 This ship contains high voltage (hv) electrical systems. Obey all posted and verbal instructions regarding safety and exclusion from high voltage areas. At no time approach, work on, or enter a high voltage area without proper authorization.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Clean, polish, and inspect the propeller and sea chest suction grates identified in paragraph 3.1, and inspect the underwater hull using References 2.1.1 through 2.1.7 and Enclosure 2.2.1 as guidance. Accomplish all tasks using IMO, SOLAS, and the Manufacturer’s requirements as guidance.

7.2 Coordinate with ship’s force to have associated equipment locked and/or tagged out per ship’s lock out/tag out policy. When the maintenance is complete, coordinate with ship’s force to have associated equipment unlocked and/or tagged in per ship’s lock out/tag out policy.

7.3 Conduct a Pre-Dive Planning Meeting with the Diver’s Representative, Master, Chief Engineer and OMT REP. The meeting agenda and discussion covers, at a minimum:

7.3.1 Confirm there is no record or indication of abnormal deterioration or damage to underwater body, propeller, rudder, stern frame, stern seal, underwater structure, or sea valves.

7.3.2 Discuss means of communication between the Bridge, Dive Team, and Engine Room.

7.3.3 Discuss procedures/plans for emergencies.

7.3.4 Discuss the water clarity (visibility) and the current cleanliness of the underwater hull. Determine if additional cleaning or spot cleaning will be necessary.

7.3.5 Discuss the vessel’s anticipated drafts, pier-side clearance, and under-keel clearance at the time of the cleaning.

7.3.6 Discuss impact of the tides and currents on divers and the examination.

7.3.7 Discuss anticipated duration of the underwater work.

NOTE: Clean all seal areas at the completion of work

7.4 Clean and polish the propeller and sea chest suction grates to meet the manufacturer’s design, installation, maintenance instructions, and service bulletin requirements. Use References 2.1.1 through 2.1.3 as guidance. Perform the following:

7.4.1 Remove all sea growth from the propeller and sea chest suction grates, including, but not limited to: calcium deposits, sea grass, and barnacles. Cleaning must not result in damage to the propeller or the removal of propeller material. Utilize the following approved cleaning methods:

a. Hand-cleaning methods such as scraping, and the use of brushes and pads. Plastic or hardwood scrapers, nylon, and polypropylene brushes such as A-1 and A-2, and hand-held plastic conditioning pads such as Scotch-Brite “Green” are acceptable for cleaning all propeller surfaces. It is preferred that the final cleaning operation of propeller blade edges (within 3 inches) be cleaned by hand with plastic surface conditioning pads. Do not attempt to round the edges of the blades with brushes and discs.

b. Powered cleaning processes. Only nylon and polypropylene brushes such as A-1 and A-2, and non-abrasive plastic surface conditioning discs such as Scotch-Brite “Green” may be used with power tools on less sensitive areas of painted propellers. Do not remove markings. Powered cleaning methods are not to be used within three inches of blade edges, tips, cusps, fillets (excluding hub to blade interface), areas of high curvature, or other sensitive areas of the propeller. When cleaning the outer periphery of the blades, the brushes and discs must be kept flat on the blade surface.

7.5 Conduct an inspection of the propeller, rope guards, rudder, shaft seals, sea chest suction grates and underwater hull to meet the manufacturer’s design, installation, maintenance instructions and service bulletin requirements. Use References 2.1.1 through 2.1.7 and Enclosure 2.2.1 as guidance. Perform the following:

7.5.1 Examine all visible parts of the propeller. Inspect the hub, blades, and blade edges for cracks, curling, porosity, cavitation erosion, nicks, dents, bends, cable marks, flat spots, ridges, punch marks, and gouges. Verify that the propeller is free of conditions that may cause the propeller to become out of balance.

7.5.2 Inspect Bow Thruster Propeller and Shafting as follows:

a. Inspect the following underwater equipment for damage, fouling, roughness, loose or missing parts; propeller blades, hubs, propeller caps, devices, rope guards and fairwaters.

b. Remove any wire, rope, hose, cable, or any foreign material which may cause vibration and/or noise or damage to equipment being inspected.

7.5.3 Visually examine stern bearing and rope guards for fishing nets, wire rope, and oil seal leaks.

7.5.4 Visually examine rudders for damage, missing hardware, missing inspection plates, and wire rope.

7.5.5 Visually examine all sea chest suction grates for damage, missing hardware, obstruction, and debris.

7.5.6 Visually examine the entire underwater hull for fouling or damage. Record the location and extent, dimensions, etc. of any hull damage observed. This examination is also to include bilge keels, thrusters, rudders, shafting, seals, propellers, rope guards, rope cutters, transducers and appendages.

7.5.7 Submit photographs and a completed Enclosure 2.2.1 to the OMT REP detailing any observed damage or deficient conditions.

7.6 Maintain workspaces in a clean condition during and upon completion of the work requirements. Prevent contamination of any adjacent equipment and spaces.

7.7 Upon completion of all inspections and maintenance, return the propeller and sea chest suction grates to a ready for service condition.

7.8 Coordinate with ship’s force to tag-in all required systems and equipment.

7.9 Reports

7.9.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the OMT REP with recommended repairs and parts required. Submit one electronic copy in Portable Document Format (PDF).

7.9.2 Submit a report that includes “as released” condition of the propeller(s), rudders, shaft seals, rope guard(s), rope cutters, transducers, thrusters, sea chest grates and underwater hull. Submit one electronic copy in Portable Document Format (PDF).

7.9.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company’s representative.

7.10 Painting

7.10.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

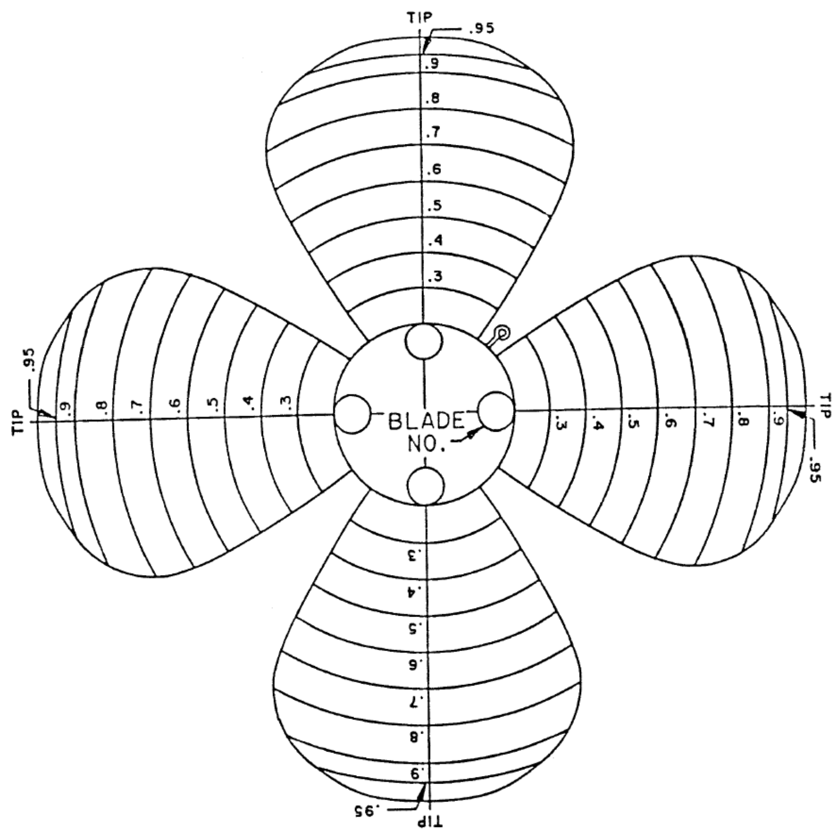
7.11 Manufacturer’s Representative: None

7.12 Preparation of Drawings

7.12.1 Provide electronic copies of all revised and/or updated technical documentation including drawings, technical manuals, service bulletins, parts lists, certification and test data associated with the work performed or parts supplied to the OMT REP.

8.0 GENERAL REQUIREMENTS: None additional

ENCLOSURE 2.2.1 BLADE AND HUB INSPECTION FORMS



☐ PRESSURE FACE ☐ SUCTION FACE

General Notes:____

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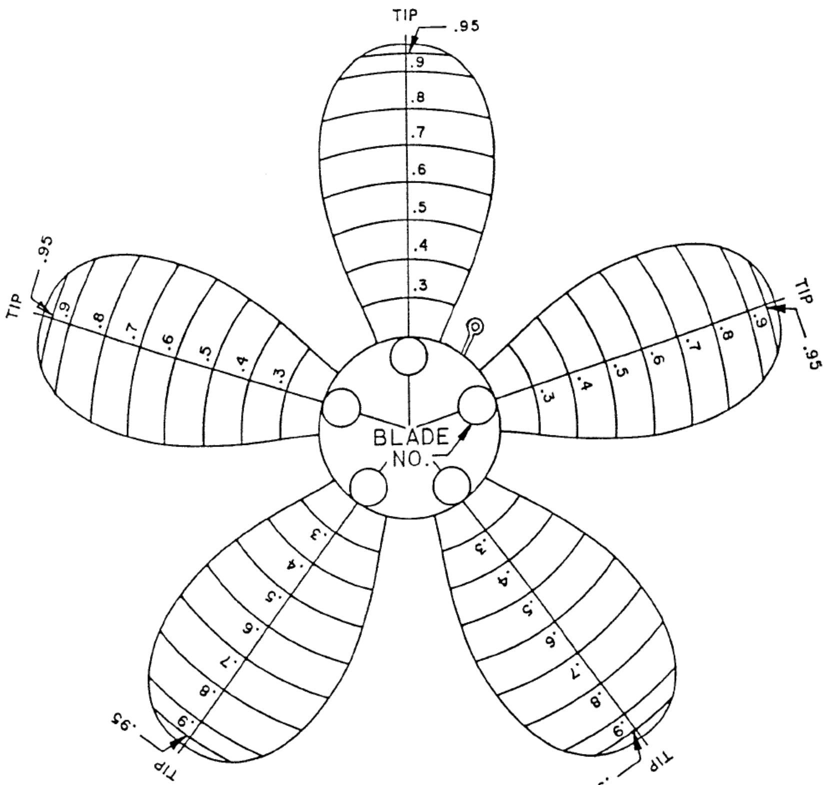
SERIAL NO.:____________________ PAGE _____ OF _____

NAVSEA 9245/3 (3/16) FORM 8

ENCLOSURE 2.2.1 BLADE AND HUB INSPECTION FORMS



☐ PRESSURE FACE ☐ SUCTION FACE

General Notes:____

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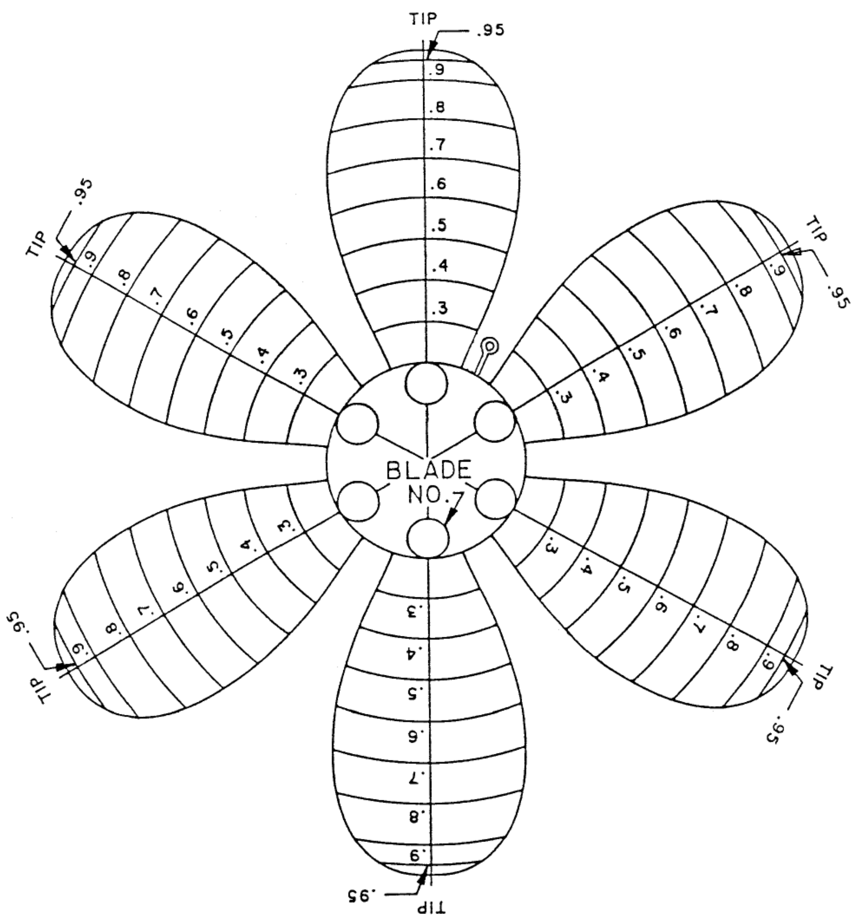
SERIAL NO.:____________________ PAGE _____ OF _____

NAVSEA 9245/3 (3/16) FORM 9

ENCLOSURE 2.2.1 BLADE AND HUB INSPECTION FORMS



☐ PRESSURE FACE ☐ SUCTION FACE

General Notes:____
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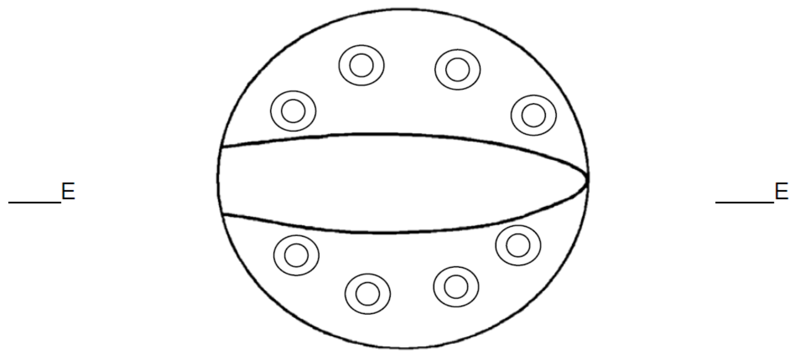
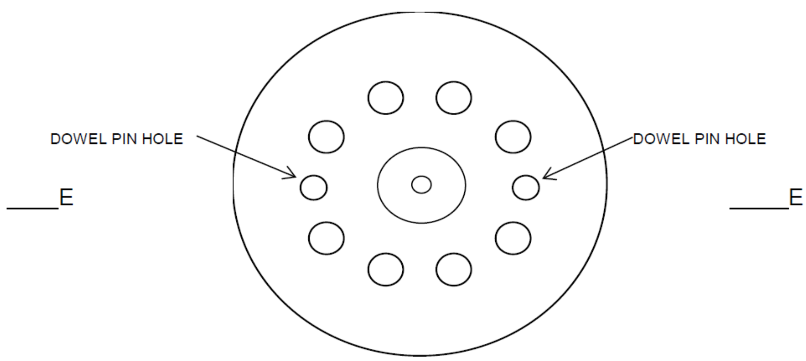
NAVSEA 9245/3 (3/16) FORM 10

ENCLOSURE 2.2.1 BLADE AND HUB INSPECTION FORMS

BLADE PALM





BLADE NO. __________

HEAT NO. __________

☐ LEFT HAND ☐ RIGHT HAND

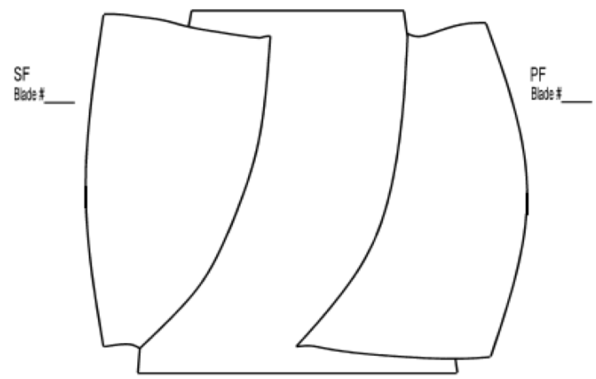
SERIAL NO.:______________________ PAGE _____ OF _____

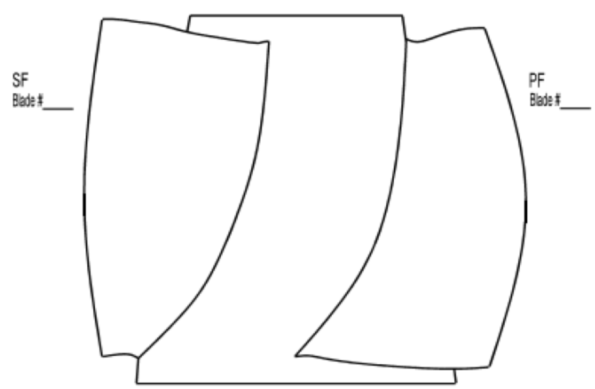
NAVSEA 9245/3 (3/16) FORM 27

ENCLOSURE 2.2.1 BLADE AND HUB INSPECTION FORMS

FILLET AREA & HUB O.D.

RH PROPELLER





General Notes:____
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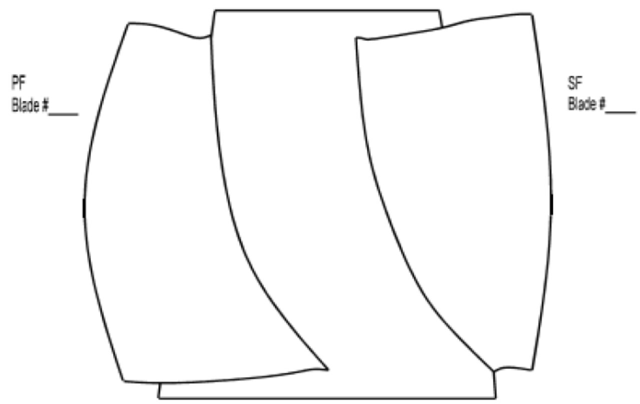
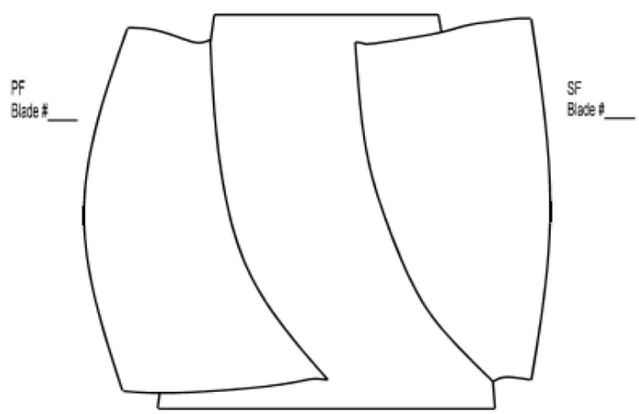
NAVSEA 9245/3 (3/16) FORM 31

ENCLOSURE 2.2.1 BLADE AND HUB INSPECTION FORMS

FILLET AREA & HUB O.D.
LH PROPELLER





General Notes:____
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NAVSEA 9245/3 (3/16) FORM 32